CHAPTER

21

AIR CONDITIONING

CHAPTER 21
AIR CONDITIONING

Subject/Page	Date	COC
21-EFFECTIVE PAGES		
1 thru 14	Sep 15/2026	
21-CONTENTS		
1	May 15/2026	
2	May 15/2024	
3	Jan 15/2025	
4	Jan 15/2025	
5	Jan 15/2025	
6	Jan 15/2025	
7	May 15/2024	
8	Sep 15/2025	
9	May 15/2024	
10	May 15/2026	
11	May 15/2026	
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13	May 15/2026	
14	May 15/2026	
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27	May 15/2026	
28	May 15/2026	
29	May 15/2026	
30	May 15/2026	
31	May 15/2026	
32	BLANK	
21-00-00		
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202	May 15/2026	
203	May 15/2026	
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205	May 15/2026	
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21-00-00		
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502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
21-00-00		
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903	May 15/2024	
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907	May 15/2024	
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909	May 15/2024	
910	May 15/2024	
911	May 15/2024	
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913	May 15/2024	
914	May 15/2024	
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916	May 15/2024	
917	May 15/2024	
918	May 15/2024	
919	May 15/2024	
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921	May 15/2024	
922	May 15/2024	
923	May 15/2024	
924	May 15/2024	
925	May 15/2024	
21-00-00 (cont)		
926	Jan 15/2025	
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929	Jan 15/2025	
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R 953	Sep 15/2026	
954	Jan 15/2025	
955	Jan 15/2025	
956	Jan 15/2025	
957	Jan 15/2025	
958	Jan 15/2025	
959	Jan 15/2025	
960	Jan 15/2025	
961	Jan 15/2025	

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21-EFFECTIVE PAGES

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Subject/Page	Date	COC
21-00-00 (cont)		
962	Jan 15/2025	
963	Jan 15/2025	
964	Jan 15/2025	
965	Jan 15/2025	
966	Jan 15/2025	
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981	Jan 15/2025	
982	Jan 15/2025	
983	Jan 15/2025	
R 984	Sep 15/2026	
O 985	Sep 15/2026	
986	Jan 15/2025	
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21-00-01		
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203	Sep 15/2025	
204	Sep 15/2025	
205	Sep 15/2025	
206	May 15/2026	
207	May 15/2026	
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21-00-05		
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R 202	Sep 15/2026	
R 203	Sep 15/2026	
R 204	Sep 15/2026	
R 205	Sep 15/2026	
R 206	Sep 15/2026	
R 207	Sep 15/2026	
208	Jan 15/2025	
209	Jan 15/2025	
210	Jan 15/2025	
21-20-02		
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803	May 15/2024	
804	May 15/2024	
805	May 15/2024	
806	May 15/2024	
807	May 15/2024	
808	May 15/2024	
809	May 15/2024	
810	Jan 15/2025	
811	May 15/2024	
812	May 15/2024	
813	May 15/2024	
814	May 15/2024	
815	May 15/2024	
21-20-02 (cont)		
816	May 15/2024	
817	May 15/2024	
818	May 15/2024	
819	May 15/2024	
820	May 15/2024	
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822	May 15/2024	
823	May 15/2024	
R 824	Sep 15/2026	
825	May 15/2024	
R 826	Sep 15/2026	
827	May 15/2024	
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833	May 15/2024	
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844	May 15/2024	
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847	May 15/2024	
848	May 15/2024	
849	May 15/2024	
850	May 15/2024	
851	May 15/2024	

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AIR CONDITIONING

Subject/Page	Date	COC
21-20-02 (cont)		
852	May 15/2024	
21-21-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2025	
408	Sep 15/2025	
409	May 15/2024	
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413	May 15/2024	
414	Sep 15/2025	
415	Sep 15/2025	
416	Jan 15/2026	
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419	Jan 15/2026	
420	Sep 15/2025	
421	Sep 15/2025	
422	Sep 15/2025	
423	Sep 15/2025	
424	BLANK	
21-22-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
21-22-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
21-22-02 (cont)		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
21-22-05		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
O 403	Sep 15/2026	
404	Sep 15/2024	
405	Sep 15/2024	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
R 408	Sep 15/2026	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
21-22-07		
701	May 15/2024	
702	May 15/2024	
703	May 15/2024	
704	May 15/2024	
705	May 15/2024	
706	May 15/2024	
707	May 15/2024	
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21-22-09		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
21-23-01		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
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207	May 15/2024	
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209	May 15/2024	
210	May 15/2024	
211	May 15/2024	
212	May 15/2024	
213	May 15/2024	
214	BLANK	
21-23-02		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	BLANK	
21-23-03		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	May 15/2024	
209	May 15/2024	
210	BLANK	
21-23-04		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	

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Subject/Page	Date	COC
21-23-04 (cont)		
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
21-23-05		
701	May 15/2026	
702	May 15/2026	
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21-24-04		
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402	May 15/2024	
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404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
21-25-00		
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202	May 15/2024	
203	May 15/2024	
204	May 15/2026	
205	May 15/2026	
206	BLANK	
21-25-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2026	
21-25-01		
401	Jan 15/2025	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	Jan 15/2025	

Subject/Page	Date	COC
21-25-01 (cont)		
409	Jan 15/2025	
410	BLANK	
21-25-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
21-25-03		
401	Sep 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
21-25-03		
601	Sep 15/2024	
602	Sep 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	BLANK	
21-26-01		
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403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
21-26-03		
201	May 15/2024	
202	May 15/2024	

Subject/Page	Date	COC
21-26-03 (cont)		
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	BLANK	
21-27-00		
201	May 15/2026	
202	May 15/2024	
203	May 15/2026	
204	May 15/2026	
205	May 15/2026	
206	BLANK	
21-27-00		
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502	May 15/2024	
503	May 15/2026	
504	May 15/2026	
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506	May 15/2026	
507	May 15/2026	
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521	May 15/2026	
522	May 15/2026	
523	May 15/2026	
524	May 15/2026	
525	May 15/2026	

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AIR CONDITIONING

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527	May 15/2026	
528	May 15/2026	
529	May 15/2026	
530	May 15/2026	
531	May 15/2026	
532	May 15/2026	
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534	May 15/2026	
535	May 15/2026	
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538	May 15/2026	
21-27-01		
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402	Sep 15/2025	
403	Sep 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2025	
408	BLANK	
21-27-02		
401	Sep 15/2024	
402	Sep 15/2024	
403	May 15/2024	
404	Sep 15/2024	
405	May 15/2026	
406	Sep 15/2024	
407	Sep 15/2024	
408	BLANK	
21-27-03		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	Jan 15/2026	

Subject/Page	Date	COC
21-27-03 (cont)		
206	Jan 15/2026	
207	Jan 15/2026	
208	Jan 15/2026	
21-27-04		
401	Sep 15/2025	
402	Sep 15/2025	
403	May 15/2024	
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405	Sep 15/2025	
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407	Sep 15/2025	
408	May 15/2024	
409	May 15/2024	
410	Jan 15/2025	
411	May 15/2024	
412	BLANK	
21-27-04		
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702	May 15/2024	
21-27-07		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
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21-27-10		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
21-31-00		
201	Sep 15/2025	
202	Sep 15/2025	
203	Sep 15/2025	
204	May 15/2024	
205	May 15/2024	

Subject/Page	Date	COC
21-31-00 (cont)		
206	May 15/2024	
207	May 15/2024	
208	May 15/2024	
21-31-00		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
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521	May 15/2024	
522	May 15/2024	
523	May 15/2024	
524	May 15/2024	
525	May 15/2024	
526	May 15/2024	

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AIR CONDITIONING

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21-31-00 (cont)		
527	May 15/2024	
528	May 15/2024	
529	May 15/2024	
530	May 15/2024	
21-31-01		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
21-31-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
21-31-03		
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402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	May 15/2024	
406	Jan 15/2025	
407	May 15/2024	
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413	May 15/2024	
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415	May 15/2024	
416	May 15/2024	
417	Jan 15/2025	
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21-32-01		
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402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
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21-32-01		
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203	Sep 15/2025	
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205	Jan 15/2025	
206	Jan 15/2025	
207	Jan 15/2025	
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21-32-05		
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403	May 15/2024	
21-32-05 (cont)		
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504	May 15/2024	
505	May 15/2024	
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507	May 15/2024	
508	May 15/2024	
21-33-01		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
21-33-02		
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21-33-04		
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21-33-07		
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402	May 15/2024	
403	May 15/2024	

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Subject/Page	Date	COC
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21-45-02		
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406	May 15/2024	
21-45-02 (cont)		
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411	Sep 15/2025	
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204	May 15/2026	
205	May 15/2026	
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207	May 15/2026	
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509	Sep 15/2025	
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21-51-00		
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602	Jan 15/2025	
21-51-00 (cont)		
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605	Jan 15/2025	
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607	Jan 15/2025	
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21-51-01		
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403	Jan 15/2026	
404	May 15/2024	
405	May 15/2024	
406	Jan 15/2026	
R 407	Sep 15/2026	
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R 417	Sep 15/2026	
O 418	Sep 15/2026	
O 419	Sep 15/2026	
O 420	Sep 15/2026	
D 421	Sep 15/2026	
D 422	BLANK	
21-51-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

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Subject/Page	Date	COC
21-51-02 (cont)		
406	May 15/2026	
21-51-02		
501	Jan 15/2025	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	Jan 15/2025	
506	Jan 15/2025	
507	Jan 15/2025	
508	Jan 15/2025	
509	Jan 15/2025	
510	Jan 15/2025	
511	Jan 15/2025	
512	BLANK	
21-51-03		
201	May 15/2026	
202	May 15/2026	
203	May 15/2026	
204	May 15/2026	
205	May 15/2026	
206	May 15/2026	
207	May 15/2026	
208	May 15/2026	
209	May 15/2026	
210	May 15/2026	
211	May 15/2026	
212	BLANK	
21-51-03		
401	May 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
407	Jan 15/2026	
408	May 15/2026	
409	Sep 15/2025	

Subject/Page	Date	COC
21-51-03 (cont)		
410	May 15/2026	
411	Sep 15/2025	
412	Sep 15/2025	
413	Sep 15/2025	
414	Sep 15/2025	
415	Sep 15/2025	
416	Sep 15/2025	
417	Sep 15/2025	
418	Sep 15/2025	
419	Sep 15/2025	
420	Sep 15/2025	
421	Sep 15/2025	
422	Sep 15/2025	
423	Sep 15/2025	
424	Sep 15/2025	
425	Sep 15/2025	
426	Sep 15/2025	
427	Sep 15/2025	
428	Jan 15/2026	
429	Jan 15/2026	
430	Sep 15/2025	
431	Jan 15/2026	
432	Sep 15/2025	
433	Sep 15/2025	
434	May 15/2026	
435	May 15/2026	
436	May 15/2026	
437	May 15/2026	
438	May 15/2026	
21-51-03		
601	Jan 15/2025	
602	May 15/2025	
603	May 15/2025	
604	Jan 15/2026	
605	Jan 15/2026	
606	Jan 15/2025	

Subject/Page	Date	COC
21-51-03 (cont)		
607	Jan 15/2025	
608	Jan 15/2025	
21-51-03		
701	May 15/2025	
702	Jan 15/2025	
703	Jan 15/2025	
704	Sep 15/2025	
705	Jan 15/2025	
706	Jan 15/2025	
707	Jan 15/2025	
708	Jan 15/2026	
709	Jan 15/2026	
710	Jan 15/2026	
711	Jan 15/2025	
712	Jan 15/2025	
713	Jan 15/2025	
714	Jan 15/2025	
715	May 15/2026	
716	May 15/2026	
R 717	Sep 15/2026	
O 718	Sep 15/2026	
O 719	Sep 15/2026	
O 720	Sep 15/2026	
O 721	Sep 15/2026	
O 722	Sep 15/2026	
723	May 15/2026	
724	May 15/2026	
725	May 15/2026	
726	May 15/2026	
21-51-04		
401	Jan 15/2025	
402	Jan 15/2026	
R 403	Sep 15/2026	
R 404	Sep 15/2026	
O 405	Sep 15/2026	
R 406	Sep 15/2026	
R 407	Sep 15/2026	

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Subject/Page	Date	COC
21-51-04 (cont)		
408	May 15/2024	
R 409	Sep 15/2026	
410	Jan 15/2026	
R 411	Sep 15/2026	
412	Jan 15/2025	
R 413	Sep 15/2026	
414	Jan 15/2026	
415	Jan 15/2026	
R 416	Sep 15/2026	
R 417	Sep 15/2026	
R 418	Sep 15/2026	
419	Jan 15/2026	
420	BLANK	
21-51-07		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
405	Jan 15/2025	
406	Jan 15/2025	
407	Jan 15/2025	
408	Jan 15/2025	
409	Jan 15/2025	
410	May 15/2024	
411	May 15/2024	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
21-51-08		
401	Jan 15/2025	
402	Sep 15/2025	
R 403	Sep 15/2026	
404	May 15/2026	
R 405	Sep 15/2026	
406	Sep 15/2025	
407	Jan 15/2025	
408	Sep 15/2025	
21-51-08 (cont)		
409	Sep 15/2025	
410	Sep 15/2025	
411	May 15/2026	
412	May 15/2026	
R 413	Sep 15/2026	
414	May 15/2026	
21-51-10		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
405	Jan 15/2025	
406	Jan 15/2026	
407	Jan 15/2026	
408	Jan 15/2026	
21-51-11		
401	Jan 15/2026	
402	Jan 15/2026	
R 403	Sep 15/2026	
404	Sep 15/2025	
405	Jan 15/2026	
406	Jan 15/2026	
407	May 15/2024	
408	Jan 15/2026	
409	Jan 15/2026	
410	Jan 15/2026	
411	Jan 15/2026	
412	Jan 15/2026	
21-51-11		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
21-51-12		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
21-51-13		
401	Sep 15/2025	
402	Sep 15/2025	
403	Sep 15/2025	
404	Sep 15/2025	
405	May 15/2026	
406	Sep 15/2025	
21-51-14		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
405	Jan 15/2025	
406	Jan 15/2025	
407	Jan 15/2025	
408	BLANK	
21-51-15		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
405	Jan 15/2025	
406	Jan 15/2026	
407	Jan 15/2026	
408	BLANK	
21-51-16		
401	Jan 15/2026	
402	Jan 15/2026	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	

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CHAPTER 21
AIR CONDITIONING

Subject/Page	Date	COC
21-51-16 (cont)		
406	Jan 15/2026	
407	Jan 15/2026	
408	Jan 15/2026	
21-51-16		
601	Jan 15/2026	
602	Jan 15/2026	
603	Jan 15/2026	
604	Jan 15/2026	
605	Jan 15/2026	
606	Jan 15/2025	
607	Jan 15/2025	
608	Jan 15/2025	
21-51-17		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
407	Jan 15/2025	
408	Jan 15/2026	
409	Jan 15/2026	
410	Jan 15/2026	
21-51-21		
401	May 15/2024	
402	May 15/2024	
403	Sep 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	Sep 15/2024	
409	May 15/2024	
410	May 15/2026	
411	May 15/2025	
412	May 15/2025	
413	Sep 15/2025	

Subject/Page	Date	COC
21-51-21 (cont)		
414	Jan 15/2026	
415	Jan 15/2026	
416	Jan 15/2026	
417	Jan 15/2026	
418	Jan 15/2026	
21-51-22		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
21-51-22		
501	Jan 15/2026	
502	Jan 15/2026	
503	Jan 15/2026	
R 504	Sep 15/2026	
R 505	Sep 15/2026	
506	Jan 15/2026	
507	Jan 15/2026	
508	Jan 15/2026	
509	Jan 15/2026	
510	Jan 15/2026	
511	Jan 15/2025	
512	Jan 15/2025	
513	Jan 15/2025	
514	Jan 15/2025	
515	Jan 15/2025	
516	BLANK	

Subject/Page	Date	COC
21-51-22		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	BLANK	
21-51-23		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Jan 15/2025	
408	Jan 15/2025	
409	Jan 15/2025	
410	Jan 15/2025	
411	May 15/2026	
412	May 15/2026	
21-51-24		
401	Jan 15/2026	
402	Jan 15/2026	
R 403	Sep 15/2026	
R 404	Sep 15/2026	
405	Sep 15/2025	
406	Sep 15/2025	
R 407	Sep 15/2026	
408	Sep 15/2025	
409	Sep 15/2025	
410	Sep 15/2025	
411	Sep 15/2025	
412	Sep 15/2025	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
R 417	Sep 15/2026	
R 418	Sep 15/2026	
R 419	Sep 15/2026	

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Subject/Page	Date	COC
21-51-24 (cont)		
O 420	Sep 15/2026	
D 421	Sep 15/2026	
D 422	BLANK	
21-51-24		
801	Jan 15/2025	
802	May 15/2024	
803	May 15/2024	
804	Jan 15/2025	
805	May 15/2024	
806	May 15/2024	
807	May 15/2024	
808	May 15/2024	
809	Jan 15/2025	
810	May 15/2024	
811	May 15/2024	
812	May 15/2024	
813	May 15/2024	
814	May 15/2024	
815	May 15/2024	
816	BLANK	
21-51-27		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
R 408	Sep 15/2026	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
O 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
414	BLANK	
21-51-40		
401	Jan 15/2026	
402	Jan 15/2026	
403	Jan 15/2026	
404	Jan 15/2025	
405	Jan 15/2025	
406	Jan 15/2026	
407	Sep 15/2025	
408	Sep 15/2025	
21-51-40		
501	May 15/2026	
502	May 15/2026	
503	May 15/2026	
504	May 15/2026	
21-51-41		
401	Jan 15/2026	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	Jan 15/2025	
R 406	Sep 15/2026	
407	Jan 15/2026	
408	BLANK	
21-51-41		
501	May 15/2026	
502	Jan 15/2026	
503	Jan 15/2026	
504	BLANK	
21-51-42		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	Jan 15/2026	
406	Jan 15/2026	
21-51-42		
501	Jan 15/2026	
502	Jan 15/2026	
21-51-50		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
407	Jan 15/2025	
408	Jan 15/2025	
21-51-51		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
21-51-51		
501	Sep 15/2025	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	BLANK	
21-51-52		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	Jan 15/2025	
405	Jan 15/2025	
406	BLANK	
21-51-52		
501	Sep 15/2025	
502	Sep 15/2025	
503	Sep 15/2025	
504	May 15/2024	
21-51-60		
401	Jan 15/2025	
402	Jan 15/2025	

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Subject/Page	Date	COC
21-51-60 (cont)		
403	May 15/2024	
404	May 15/2026	
405	May 15/2026	
406	BLANK	
21-51-65		
401	Sep 15/2025	
402	Sep 15/2025	
403	May 15/2026	
404	Sep 15/2025	
405	Sep 15/2025	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
O 408	Sep 15/2026	
21-61-00		
501	May 15/2026	
502	May 15/2026	
503	Sep 15/2025	
504	Sep 15/2025	
505	Sep 15/2025	
506	Sep 15/2025	
21-61-04		
401	May 15/2025	
402	May 15/2025	
403	May 15/2024	
404	May 15/2026	
405	May 15/2024	
406	May 15/2026	
407	May 15/2025	
408	May 15/2025	
21-61-04		
501	Sep 15/2025	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
21-61-04 (cont)		
508	May 15/2024	
21-61-05		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2026	
405	May 15/2024	
406	May 15/2026	
407	May 15/2024	
408	May 15/2024	
21-61-06		
201	Sep 15/2025	
202	Sep 15/2025	
203	Sep 15/2025	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	May 15/2024	
209	May 15/2025	
210	Sep 15/2025	
211	May 15/2025	
212	Sep 15/2025	
213	May 15/2025	
214	BLANK	
21-61-07		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
21-61-09		
401	Sep 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2024	
21-61-09 (cont)		
405	Sep 15/2025	
406	Sep 15/2025	
407	May 15/2026	
408	May 15/2026	
409	May 15/2026	
410	May 15/2026	
21-61-09		
601	May 15/2024	
602	May 15/2024	
21-61-10		
201	Jan 15/2026	
202	Jan 15/2026	
203	May 15/2024	
204	Jan 15/2025	
205	May 15/2024	
206	May 15/2024	
207	Sep 15/2025	
208	May 15/2024	
209	Sep 15/2025	
210	May 15/2024	
21-61-10		
601	Sep 15/2025	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
609	May 15/2024	
610	May 15/2024	
611	May 15/2024	
612	BLANK	
21-61-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	

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Subject/Page	Date	COC
21-61-11 (cont)		
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
R 409	Sep 15/2026	
410	May 15/2024	
411	May 15/2024	
412	BLANK	
21-61-12		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
407	Jan 15/2025	
408	Jan 15/2025	
409	Jan 15/2025	
410	Jan 15/2025	
411	Jan 15/2026	
412	Jan 15/2026	
21-61-14		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
R 207	Sep 15/2026	
208	May 15/2024	
209	May 15/2024	
210	May 15/2024	
R 211	Sep 15/2026	
R 212	Sep 15/2026	
213	May 15/2024	
214	May 15/2024	

Subject/Page	Date	COC
21-61-14 (cont)		
215	May 15/2024	
216	BLANK	
21-61-15		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
406	BLANK	
21-61-21		
401	May 15/2026	
402	Jan 15/2025	
403	May 15/2025	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
409	May 15/2026	
410	May 15/2026	
21-61-22		
201	Sep 15/2024	
202	BLANK	
21-61-22		
401	Jan 15/2026	
402	Jan 15/2026	
403	Jan 15/2026	
404	Jan 15/2026	
405	Jan 15/2026	
406	BLANK	
21-61-23		
401	Jan 15/2025	
R 402	Sep 15/2026	
R 403	Sep 15/2026	
404	Jan 15/2025	
R 405	Sep 15/2026	
R 406	Sep 15/2026	

Subject/Page	Date	COC
21-61-23 (cont)		
R 407	Sep 15/2026	
R 408	Sep 15/2026	
O 409	Sep 15/2026	
410	BLANK	
21-61-23		
701	May 15/2024	
702	May 15/2024	
703	May 15/2024	
704	May 15/2024	
705	May 15/2024	
706	BLANK	
21-61-24		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2026	
405	Jan 15/2025	
406	Jan 15/2025	
21-61-30		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
21-61-31		
401	May 15/2026	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
406	May 15/2026	
21-61-32		
401	May 15/2024	
402	May 15/2024	

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Subject/Page	Date	COC
21-61-32 (cont)		
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
21-62-00		
501	May 15/2026	
502	May 15/2026	
503	Sep 15/2024	
504	BLANK	
21-62-01		
401	Jan 15/2025	
402	Sep 15/2025	
403	Sep 15/2025	
404	May 15/2024	
405	Sep 15/2025	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
O 408	Sep 15/2026	
O 409	Sep 15/2026	
410	BLANK	
21-73-01		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
405	Jan 15/2026	
406	Jan 15/2026	
407	Jan 15/2025	
408	BLANK	

Subject/Page	Date	COC

Subject/Page	Date	COC

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CHAPTER 21
AIR CONDITIONING

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
AIR CONDITIONING - GENERAL - MAINTENANCE PRACTICES	21-00-00		201	PTF ALL
Supply Conditioned Air to the Airplane TASK 21-00-00-800-801			201	PTF ALL
Remove Conditioned Air from the Airplane TASK 21-00-00-800-802			201	PTF ALL
Supply Conditioned Air with a Cooling Pack TASK 21-00-00-800-803			202	PTF ALL
Remove Conditioned Air Supplied by a Cooling Pack TASK 21-00-00-800-804			203	PTF ALL
Supply Conditioned Air with a Ground Air Source TASK 21-00-00-800-805			203	PTF ALL
Remove Conditioned Air Supplied by a Ground Air Source TASK 21-00-00-800-806			204	PTF ALL
AIR CONDITIONING - GENERAL - ADJUSTMENT/TEST	21-00-00		501	PTF ALL
Air Conditioning, System Test, Air Conditioning IBIT TASK 21-00-00-730-801			501	PTF ALL
Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test TASK 21-00-00-730-802			502	PTF ALL
AIR CONDITIONING - DDG MAINTENANCE PROCEDURES	21-00-00		901	PTF ALL
DDG 21-21-01 Ground Conditioned Air Connector Check Valve Inoperative - Preparation TASK 21-00-00-040-844			902	PTF ALL
DDG 21-21-01 Ground Conditioned Air Connector Check Valve Inoperative - Restoration TASK 21-00-00-440-831			903	PTF ALL
DDG 21-23-01 Return Air Grille Broken or Missing - Preparation TASK 21-00-00-040-845			903	PTF ALL

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CHAPTER 21
AIR CONDITIONING

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 21-23-01 Return Air Grille Broken or Missing - Restoration TASK 21-00-00-440-832			904	PTF ALL
DDG 21-27-01 Equipment Cooling Supply and Exhaust Fans Inoperative - Preparation TASK 21-00-00-040-846			905	PTF ALL
DDG 21-27-02A Overboard Exhaust Valve Inoperative - Valve in SMOKE Position - Preparation TASK 21-00-00-040-847			907	PTF ALL
DDG 21-27-02A Overboard Exhaust Valve Inoperative - Valve in SMOKE Position - Restoration TASK 21-00-00-440-833			911	PTF ALL
DDG 21-27-02B Overboard Exhaust Valve Inoperative - Valve Fully Open - Preparation TASK 21-00-00-040-848			912	PTF ALL
DDG 21-27-02B Overboard Exhaust Valve Inoperative - Valve Fully Open - Restoration TASK 21-00-00-440-834			912	PTF ALL
DDG 21-27-03 Equipment Cooling Low Flow Detector System Inoperative - Preparation TASK 21-00-00-040-849			913	PTF ALL
DDG 21-27-03 Equipment Cooling Low Flow Detector System Inoperative - Restoration TASK 21-00-00-440-835			917	PTF ALL
DDG 21-27-04 Equipment Cooling Air Filter Removed - Preparation TASK 21-00-00-040-850			917	PTF ALL
DDG 21-27-04 Equipment Cooling Air Filter Removed - Restoration TASK 21-00-00-440-836			918	PTF ALL
DDG 21-27-05 Enhanced E6 Rack Equipment Cooling Exhaust Fan Inoperative - Preparation TASK 21-00-00-040-870			918	PTF ALL
DDG 21-27-05 Enhanced E6 Rack Equipment Cooling Exhaust Fan Inoperative - Restoration TASK 21-00-00-440-856			919	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 21-31-01-01A Cabin Pressure Control System - One Automatic Mode Inoperative - Preparation TASK 21-00-00-040-851			919	PTF ALL
DDG 21-31-01-01A Cabin Pressure Control System - One Automatic Mode Inoperative - Restoration TASK 21-00-00-440-837			920	PTF ALL
DDG 21-31-01-01B Cabin Pressure Control System - Both Automatic Modes Inoperative - Preparation TASK 21-00-00-040-852			921	PTF ALL
DDG 21-31-01-01B Cabin Pressure Control System - Both Automatic Modes Inoperative - Restoration TASK 21-00-00-440-838			921	PTF ALL
DDG 21-31-01-02 Cabin Pressure Control System - Manual Mode Inoperative - Preparation TASK 21-00-00-040-853			922	PTF ALL
DDG 21-31-01-02 Cabin Pressure Control System - Manual Mode Inoperative - Restoration TASK 21-00-00-440-839			923	PTF ALL
DDG 21-31-02 Main Outflow Valve Inoperative - Preparation TASK 21-00-00-040-854			924	PTF ALL
DDG 21-31-02 Main Outflow Valve Inoperative - Restoration TASK 21-00-00-440-840			925	PTF ALL
DDG 21-32-01A Positive Pressure Relief Valve Inoperative - Valve Closed - Preparation TASK 21-00-00-040-855			926	PTF ALL
DDG 21-32-01A Positive Pressure Relief Valve Inoperative - Valve Closed - Restoration TASK 21-00-00-440-841			930	PTF ALL
DDG 21-32-01B Positive Pressure Relief Valve Inoperative - Valve Open - Preparation TASK 21-00-00-040-856			931	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 21-32-01B Positive Pressure Relief Valve Inoperative - Valve Open - Restoration TASK 21-00-00-440-842			931	PTF ALL
DDG 21-33-01 Cabin Rate of Climb Indicator Inoperative - Preparation TASK 21-00-00-040-857			932	PTF ALL
DDG 21-33-01 Cabin Rate of Climb Indicator Inoperative - Restoration TASK 21-00-00-440-843			932	PTF ALL
DDG 21-33-02–02 Cabin Differential Pressure Indication Inoperative - Preparation TASK 21-00-00-040-858			933	PTF ALL
DDG 21-33-02-02 Cabin Differential Pressure Indication Inoperative - Restoration TASK 21-00-00-440-844			933	PTF ALL
DDG 21-33-03–02 Cabin Altitude Indication Inoperative - Preparation TASK 21-00-00-040-859			934	PTF ALL
DDG 21-33-03–02 Cabin Altitude Indication Inoperative - Restoration TASK 21-00-00-440-845			934	PTF ALL
DDG 21-33-05 Outflow Valve Position Indicator Inoperative - Preparation TASK 21-00-00-040-860			935	PTF ALL
DDG 21-45-01 Door Area Heater System Inoperative - Preparation TASK 21-00-00-040-861			936	PTF ALL
DDG 21-45-01 Door Area Heater System Inoperative - Restoration TASK 21-00-00-440-847			938	PTF ALL
DDG 21-51-01 Air Conditioning Packs Inoperative - Preparation TASK 21-00-00-040-862			939	PTF ALL
DDG 21-51-01 Air Conditioning Packs Inoperative - Restoration TASK 21-00-00-440-848			941	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 21-51-02 Pack Flow Control and Shutoff Valves Inoperative - Preparation TASK 21-00-00-040-863			943	PTF ALL
DDG 21-51-02 Pack Flow Control and Shutoff Valves Inoperative - Restoration TASK 21-00-00-440-849			950	PTF ALL
DDG 21-51-04 Pack Flow Sensor Inoperative - Preparation TASK 21-00-00-040-871			953	PTF ALL
DDG 21-51-05 Pack Temperature Control Valves Inoperative - Preparation TASK 21-00-00-040-869			953	PTF ALL
DDG 21-51-05 Pack Temperature Control Valves Inoperative - Restoration TASK 21-00-00-440-855			959	PTF ALL
DDG 21-51-09 Pack Ram Air Systems Inoperative - Preparation TASK 21-00-00-040-864			961	PTF ALL
DDG 21-51-09 Pack Ram Air Systems Inoperative - Restoration TASK 21-00-00-440-850			972	PTF ALL
DDG 21-61-01 Trim Air Modulating Valve Inoperative - Preparation TASK 21-00-00-040-865			976	PTF ALL
DDG 21-61-01 Trim Air Modulating Valve Inoperative - Restoration TASK 21-00-00-440-851			981	PTF ALL
DDG 21-61-04 Trim Air Check Valve Inoperative - Preparation TASK 21-00-00-040-866			982	PTF ALL
DDG 21-61-04 Trim Air Check Valve Inoperative - Restoration TASK 21-00-00-440-852			988	PTF ALL
DDG 21-62-01 Trim Air Pressure Regulating and Shutoff Valve Inoperative - Preparation TASK 21-00-00-040-867			990	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 21-62-01 Trim Air Pressure Regulating and Shutoff Valve Inoperative - Restoration TASK 21-00-00-440-853			995	PTF ALL
AIR CONDITIONING SYSTEM OIL CONTAMINATION REMOVAL - MAINTENANCE PRACTICES	21-00-01		201	PTF ALL
Oil Contamination Removal from Air Conditioning and Pneumatic Systems TASK 21-00-01-100-801			201	PTF ALL
CONFIDENCE CHECK OF AIRPLANE ABILITY TO MAINTAIN CABIN PRESSURE IN FLIGHT - MAINTENANCE PRACTICES	21-00-05		201	PTF ALL
Airplane Pack Operation - Confidence Check TASK 21-00-05-780-801			201	PTF ALL
AIR CONDITIONING SYSTEM DUCTS - REPAIRS	21-20-02		801	PTF ALL
Taped Joint Repair - Fiberglass/Kevlar Duct TASK 21-20-02-330-801			802	PTF ALL
External Patch - Fiberglass/Kevlar Duct TASK 21-20-02-330-802			808	PTF ALL
Structural Repair - Fiberglass/Kevlar Duct TASK 21-20-02-330-803			810	PTF ALL
Taped Joint Repair - Polyurethane Foam Duct TASK 21-20-02-330-804			815	PTF ALL
Duct End Repair - Polyurethane Foam Repair TASK 21-20-02-330-805			820	PTF ALL
External Patch - Polyurethane Foam Duct TASK 21-20-02-330-806			821	PTF ALL
Structural Repair - Polyurethane Foam Duct TASK 21-20-02-330-807			823	PTF ALL
Inner Lining Repair - Polyurethane Foam Duct TASK 21-20-02-330-808			826	PTF ALL
Duct Section Replacement - Polyurethane Foam Duct TASK 21-20-02-330-809			828	PTF ALL
Tape Replacement - Foam Wrapped Insulated Duct TASK 21-20-02-330-810			830	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Insulation Repair - Foam Wrapped Insulated Duct TASK 21-20-02-330-811			835	PTF ALL
Insulation Replacement - Foam Wrapped Insulated Duct TASK 21-20-02-330-812			837	PTF ALL
Tape Replacement - Fiberglass Insulated Duct TASK 21-20-02-330-813			839	PTF ALL
Insulation and Covering Repair - Fiberglass Insulated Duct TASK 21-20-02-330-814			842	PTF ALL
Insulation and Covering Replacement - Fiberglass Insulated Duct TASK 21-20-02-330-815			843	PTF ALL
External Patch - Reinforced Thermoplastic Laminate Duct TASK 21-20-02-330-816			844	PTF ALL
Structural Repair - Reinforced Thermoplastic Laminate Duct TASK 21-20-02-330-817			848	PTF ALL
Duct End Repair - Reinforced Thermoplastic Laminate Duct TASK 21-20-02-330-818			850	PTF ALL
GROUND CONDITIONED AIR CONNECTION DUCT AND CHECK VALVE - REMOVAL/INSTALLATION	21-21-01		401	PTF ALL
Ground Conditioned Air Connection Duct Removal TASK 21-21-01-000-801			401	PTF ALL
Ground Conditioned Air Connection Duct Installation TASK 21-21-01-400-801			405	PTF ALL
Ground Conditioned Air Check Valve Removal TASK 21-21-01-000-802			406	PTF ALL
Ground Conditioned Air Check Valve Installation TASK 21-21-01-400-802			414	PTF ALL
Cooling Exchange Check Valve Removal TASK 21-21-01-000-803			418	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Cooling Exchange Check Valve Installation TASK 21-21-01-400-803			422	PTF ALL
WINDSHIELD AND FOOT AIR OUTLET VALVE - REMOVAL/INSTALLATION	21-22-01		401	PTF ALL
Windshield and Foot Air Outlet Valve Removal TASK 21-22-01-000-801			401	PTF ALL
Windshield and Foot Air Outlet Installation TASK 21-22-01-400-801			405	PTF ALL
AIR OUTLET VALVE CONTROL CABLES - REMOVAL/INSTALLATION	21-22-02		401	PTF ALL
Air Outlet Valve Control Cables Removal TASK 21-22-02-000-801			401	PTF ALL
Air Outlet Valve Control Cables Installation TASK 21-22-02-400-801			404	PTF ALL
AIR OUTLET VALVE CONTROL CABLES - ADJUSTMENT/TEST	21-22-02		501	PTF ALL
Air Outlet Valve Control Cables Adjustment and Test TASK 21-22-02-820-801			501	PTF ALL
FLIGHT DECK AIR DISTRIBUTION SYSTEM MUFFLER/DUCT ASSEMBLY - REMOVAL/INSTALLATION	21-22-05		401	PTF ALL
Flight Deck Air Distribution System Muffler/Duct Assembly Removal TASK 21-22-05-000-801			401	PTF ALL
Flight Deck Air Distribution System Muffler/Duct Assembly Installation TASK 21-22-05-400-801			408	PTF ALL
FLIGHT DECK DUCT ASSEMBLY SCREEN - CLEANING/PAINTING	21-22-07		701	PTF ALL
Flight Deck Duct Assembly Removal TASK 21-22-07-020-801			701	PTF ALL
Flight Deck Duct Assembly Screen Cleaning TASK 21-22-07-100-801			706	PTF ALL
Flight Deck Duct Assembly Installation TASK 21-22-07-420-801			707	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FLIGHT COMPARTMENT GASPER AIR OUTLETS - REMOVAL/INSTALLATION	21-22-09		401	PTF ALL
Gasper Air Outlet Removal TASK 21-22-09-000-801			401	PTF ALL
Gasper Air Outlet Installation TASK 21-22-09-400-801			404	PTF ALL
OVERHEAD DISTRIBUTION DUCT - MAINTENANCE PRACTICES	21-23-01		201	PTF ALL
Overhead Distribution Duct Removal TASK 21-23-01-000-801			201	PTF ALL
Overhead Distribution Duct Installation TASK 21-23-01-400-801			204	PTF ALL
Overhead Distribution Duct Insulation Repair TASK 21-23-01-300-801			205	PTF ALL
Insulation Replacement/Repair - Overhead Distribution Duct Sidewall Risers Interface Segment TASK 21-23-01-300-802			207	PTF ALL
Overhead Distribution Duct Cleaning TASK 21-23-01-100-801			211	PTF ALL
SIDEWALL RISER DUCTS/TRANSITION DUCT - MAINTENANCE PRACTICES	21-23-02		201	PTF ALL
Sidewall Riser Ducts/Transition Duct Inspection TASK 21-23-02-212-801			201	PTF ALL
Sidewall Riser Ducts/Transition Duct Repair TASK 21-23-02-360-801			203	PTF ALL
RETURN AIR GRILLES (DECOMPRESSION VENTS) - MAINTENANCE PRACTICES	21-23-03		201	PTF ALL
Return Air Grille Removal TASK 21-23-03-000-801			201	PTF ALL
Return Air Grille Baffle Removal TASK 21-23-03-000-802			204	PTF ALL
Return Air Grille and Baffle Cleaning TASK 21-23-03-100-801			206	PTF ALL
Return Air Grille Baffle Installation TASK 21-23-03-400-801			206	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Return Air Grille Installation TASK 21-23-03-400-802			207	PTF ALL
Return Air Grille Baffle Inspection TASK 21-23-03-200-801			208	PTF ALL
SIDEWALL RISER DUCTS - REMOVAL/INSTALLATION	21-23-04		401	PTF ALL
Sidewall Riser Ducts Removal TASK 21-23-04-000-801			401	PTF ALL
Sidewall Riser Ducts Installation TASK 21-23-04-400-801			404	PTF ALL
SIDEWALL DIFFUSER ORIFICE - CLEANING/PAINTING	21-23-05		701	PTF ALL
Sidewall Diffuser Orifice Cleaning TASK 21-23-05-100-801			701	PTF ALL
PASSENGER CABIN GASPER AIR OUTLETS - REMOVAL/INSTALLATION	21-24-04		401	PTF ALL
Gasper Air Outlet (PSU) Removal TASK 21-24-04-000-801			401	PTF ALL
Gasper Air Outlet (PSU) Installation TASK 21-24-04-400-801			404	PTF ALL
Gasper Air Outlet (LSU) Removal TASK 21-24-04-000-802			404	PTF ALL
Gasper Air Outlet (LSU) Installation TASK 21-24-04-400-802			406	PTF ALL
RECIRCULATION SYSTEM - MAINTENANCE PRACTICES	21-25-00		201	PTF ALL
Recirculation System - Deactivation TASK 21-25-00-800-801			201	PTF ALL
Recirculation System - Activation TASK 21-25-00-800-802			205	PTF ALL
RECIRCULATION SYSTEM - ADJUSTMENT/TEST	21-25-00		501	PTF ALL
Recirculation System - Operational Test TASK 21-25-00-710-801			501	PTF ALL

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RECIRCULATION AIR FILTER - REMOVAL/INSTALLATION	21-25-01		401	PTF ALL
Recirculation Air Filter - Removal TASK 21-25-01-000-801			401	PTF ALL
Recirculation Air Filter - Installation TASK 21-25-01-400-801			408	PTF ALL
RECIRCULATION FAN - REMOVAL/INSTALLATION	21-25-02		401	PTF ALL
Recirculation Fan Removal TASK 21-25-02-000-801			401	PTF ALL
Recirculation Fan Installation TASK 21-25-02-400-801			405	PTF ALL
RECIRCULATION FAN CHECK VALVE - REMOVAL/INSTALLATION	21-25-03		401	PTF ALL
Recirculation Fan Check Valve Removal TASK 21-25-03-000-801			401	PTF ALL
Recirculation Fan Check Valve Installation TASK 21-25-03-400-801			405	PTF ALL
RECIRCULATION FAN CHECK VALVE - INSPECTION/CHECK	21-25-03		601	PTF ALL
Recirculation Fan Check Valve Inspection TASK 21-25-03-200-801			601	PTF ALL
GALLEY VENTILATION MUFFLER - REMOVAL/INSTALLATION	21-26-01		401	PTF ALL
Galley Ventilation Muffler Removal TASK 21-26-01-000-801			401	PTF ALL
Galley Ventilation Muffler Installation TASK 21-26-01-400-801			406	PTF ALL
GALLEY CEILING VENTILATION GRILL - MAINTENANCE PRACTICES	21-26-03		201	PTF ALL
Galley Ceiling Ventilation Grill Removal TASK 21-26-03-000-801			201	PTF ALL
Galley Ceiling Ventilation Grill Cleaning TASK 21-26-03-100-801			204	PTF ALL
Galley Ceiling Ventilation Grill Installation TASK 21-26-03-400-801			204	PTF ALL

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EQUIPMENT COOLING SYSTEM - MAINTENANCE PRACTICES	21-27-00		201	PTF ALL
Equipment Cooling System - Deactivation TASK 21-27-00-800-801			201	PTF ALL
Equipment Cooling System - Activation TASK 21-27-00-800-802			203	PTF ALL
EQUIPMENT COOLING SYSTEM - ADJUSTMENT/TEST	21-27-00		501	PTF ALL
Equipment Cooling Low Flow Sensors - Operational Test TASK 21-27-00-700-801			501	PTF ALL
Equipment Cooling Fans - Operational Test TASK 21-27-00-700-802			503	PTF ALL
Equipment Cooling Overboard Exhaust Valve - Functional Test TASK 21-27-00-700-803			506	PTF ALL
Flight Compartment Display Unit Air Flow Test TASK 21-27-00-700-804			517	PTF ALL
Alternate Equipment Cooling Supply Fan - Operational Test TASK 21-27-00-890-801			518	PTF ALL
Alternate Equipment Cooling Exhaust Fan - Operational Test TASK 21-27-00-700-805			520	PTF ALL
Equipment Cooling Overboard Exhaust Valve Smoke Clearance Mode - Operational Test TASK 21-27-00-700-806			522	PTF ALL
Smoke Clearance Mode - Operational Test TASK 21-27-00-700-807			525	PTF ALL
Aft Equipment Cooling Fan - Functional Test TASK 21-27-00-700-808			534	PTF ALL
EQUIPMENT COOLING AIR FILTER - REMOVAL/INSTALLATION	21-27-01		401	PTF ALL
Equipment Cooling Air Filter Removal TASK 21-27-01-000-801			401	PTF ALL

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Equipment Cooling Air Filter Installation TASK 21-27-01-400-801			407	PTF ALL
EQUIPMENT COOLING SUPPLY FAN - REMOVAL/INSTALLATION	21-27-02		401	PTF ALL
Forward Equipment Cooling Supply Fan Removal TASK 21-27-02-000-801			401	PTF ALL
Forward Equipment Cooling Supply Fan Installation TASK 21-27-02-400-801			405	PTF ALL
EQUIPMENT COOLING LOW FLOW SENSORS - MAINTENANCE PRACTICES	21-27-03		201	PTF ALL
Equipment Cooling Low Flow Sensors Removal TASK 21-27-03-000-801			201	PTF ALL
Equipment Cooling Low Flow Sensors Installation TASK 21-27-03-400-801			205	PTF ALL
Equipment Cooling Low Flow Sensors Cleaning TASK 21-27-03-400-802			208	PTF ALL
EQUIPMENT COOLING EXHAUST FAN - REMOVAL/INSTALLATION	21-27-04		401	PTF ALL
Equipment Cooling Exhaust Fan Removal TASK 21-27-04-000-801			401	PTF ALL
Equipment Cooling Exhaust Fan Installation TASK 21-27-04-400-801			405	PTF ALL
Aft Equipment Cooling Exhaust Fan Removal TASK 21-27-04-000-802			406	PTF ALL
Aft Equipment Cooling Exhaust Fan Installation TASK 21-27-04-400-802			410	PTF ALL
EQUIPMENT COOLING EXHAUST FAN - CLEANING/PAINTING	21-27-04		701	PTF ALL
Equipment Cooling Exhaust Fan Cleaning TASK 21-27-04-110-801			701	PTF ALL
EQUIPMENT COOLING MUFFLER - REMOVAL/INSTALLATION	21-27-07		401	PTF ALL
Equipment Cooling Muffler Removal TASK 21-27-07-000-801			401	PTF ALL

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Equipment Cooling Muffler Installation TASK 21-27-07-400-801			404	PTF ALL
EQUIPMENT COOLING EXHAUST DUCT - REMOVAL/INSTALLATION	21-27-10		401	PTF ALL
Equipment Cooling Exhaust Duct - Removal TASK 21-27-10-000-801			401	PTF ALL
Equipment Cooling Exhaust Duct - Installation TASK 21-27-10-400-801			404	PTF ALL
PRESSURIZATION CONTROL SYSTEM - MAINTENANCE PRACTICES	21-31-00		201	PTF ALL
Pressurization Control System - Deactivation TASK 21-31-00-800-801			201	PTF ALL
Pressurization Control System - Activation TASK 21-31-00-800-802			208	PTF ALL
PRESSURE EQUALIZATION VALVE - REMOVAL/INSTALLATION	21-31-00		401	PTF ALL
Pressure Equalization Valve Removal TASK 21-31-00-000-801			401	PTF ALL
Pressure Equalization Valve Installation TASK 21-31-00-400-801			406	PTF ALL
PRESSURIZATION CONTROL SYSTEM - ADJUSTMENT/TEST	21-31-00		501	PTF ALL
Pressurization System Manual Mode Test TASK 21-31-00-710-801			501	PTF ALL
750 Feet Per Minute Descent With Cargo Fire Mode Test TASK 21-31-00-710-802			507	PTF ALL
Master Caution Test TASK 21-31-00-700-801			509	PTF ALL
Pressurization System Ground Test TASK 21-31-00-700-802			511	PTF ALL
Cabin Pressure Controller (CPC) BITE Procedure TASK 21-31-00-740-801			518	PTF ALL

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CABIN PRESSURE CONTROLLER (CPC) - REMOVAL/INSTALLATION	21-31-01		401	PTF ALL
Cabin Pressure Controller (CPC) Removal TASK 21-31-01-000-801			401	PTF ALL
Cabin Pressure Controller (CPC) Installation TASK 21-31-01-400-801			405	PTF ALL
CABIN PRESSURE CONTROL MODULE - REMOVAL/INSTALLATION	21-31-02		401	PTF ALL
Cabin Pressure Control Module Removal TASK 21-31-02-000-801			401	PTF ALL
Cabin Pressure Control Module Installation TASK 21-31-02-400-801			403	PTF ALL
AFT OUTFLOW VALVE ASSEMBLY - REMOVAL/INSTALLATION	21-31-03		401	PTF ALL
Aft Outflow Valve Assembly Removal TASK 21-31-03-000-801			401	PTF ALL
Aft Outflow Valve Assembly Installation TASK 21-31-03-400-801			412	PTF ALL
POSITIVE PRESSURE RELIEF VALVE - REMOVAL/INSTALLATION	21-32-01		401	PTF ALL
Positive Pressure Relief Valve Removal TASK 21-32-01-000-801			401	PTF ALL
Positive Pressure Relief Valve Installation TASK 21-32-01-400-801			404	PTF ALL
POSITIVE PRESSURE RELIEF VALVE - ADJUSTMENT/TEST	21-32-01		501	PTF ALL
Positive Pressure Relief Valve - System Test with the Use of Hamilton Sundstrand Test Equipment TASK 21-32-01-700-801			501	PTF ALL
Positive Pressure Relief Valve - System Test with the Use of Boeing Test Equipment TASK 21-32-01-700-802			506	PTF ALL
POSITIVE PRESSURE RELIEF VALVE FILTER - REMOVAL/INSTALLATION	21-32-02		401	PTF ALL
Positive Pressure Relief Valve Filter Removal TASK 21-32-02-000-801			401	PTF ALL

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Positive Pressure Relief Valve Filter Installation TASK 21-32-02-400-801			406	PTF ALL
NEGATIVE PRESSURE RELIEF VALVE - MAINTENANCE PRACTICES	21-32-03		201	PTF ALL
Negative Pressure Relief Valve Removal TASK 21-32-03-000-801			201	PTF ALL
Negative Pressure Relief Valve Installation TASK 21-32-03-400-801			205	PTF ALL
Negative Pressure Relief Valve Adjustment/Test TASK 21-32-03-700-801			206	PTF ALL
CARGO COMPARTMENT BLOWOUT PANELS - REMOVAL/INSTALLATION	21-32-05		401	PTF ALL
Ceiling Blowout Panel Removal TASK 21-32-05-000-801			401	PTF ALL
Ceiling Blowout Panel Installation TASK 21-32-05-400-801			404	PTF ALL
Cargo Compartment Bulkhead Blowout Panel Removal TASK 21-32-05-000-802			405	PTF ALL
Cargo Compartment Bulkhead Blowout Panel Installation TASK 21-32-05-400-802			409	PTF ALL
CABIN PRESSURE ALTITUDE WARNING SYSTEM - ADJUSTMENT/TEST	21-33-00		501	PTF ALL
Cabin Altitude Warning Test TASK 21-33-00-700-802			501	PTF ALL
Cabin Rate of Climb Indicator Check TASK 21-33-00-700-801			507	PTF ALL
CABIN RATE OF CLIMB INDICATOR - REMOVAL/INSTALLATION	21-33-01		401	PTF ALL
Cabin Rate of Climb Indicator Removal TASK 21-33-01-000-801			401	PTF ALL
Cabin Rate of Climb Indicator Installation TASK 21-33-01-400-801			404	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
CABIN ALTITUDE AND DIFFERENTIAL PRESSURE INDICATOR - REMOVAL/INSTALLATION	21-33-02		401	PTF ALL
Cabin Altitude and Differential Pressure Indicator Removal TASK 21-33-02-000-801			401	PTF ALL
Cabin Altitude and Differential Pressure Indicator Installation TASK 21-33-02-400-801			405	PTF ALL
CABIN PRESSURE SWITCH - REMOVAL/INSTALLATION	21-33-04		401	PTF ALL
Cabin Pressure Warning Switch Removal TASK 21-33-04-000-801			401	PTF ALL
Cabin Pressure Warning Switch Installation TASK 21-33-04-400-801			403	PTF ALL
CABIN ALTIMETER AND RATE OF CLIMB MODULE - REMOVAL AND INSTALLATION	21-33-07		401	PTF ALL
Cabin Altimeter and Rate of Climb Module Removal TASK 21-33-07-000-801			401	PTF ALL
Cabin Altimeter and Rate of Climb Module Installation TASK 21-33-07-400-801			403	PTF ALL
OVERBOARD EXHAUST VALVE - REMOVAL/INSTALLATION	21-43-01		401	PTF ALL
Overboard Exhaust Valve Removal TASK 21-43-01-000-801			401	PTF ALL
Overboard Exhaust Valve Installation TASK 21-43-01-400-801			405	PTF ALL
DOOR AREA HEATING SYSTEM - ADJUSTMENT/TEST	21-45-00		501	PTF ALL
Door Area Heater Operational Test TASK 21-45-00-700-801			501	PTF ALL
Emergency Exit Hatch Heaters Operational Test TASK 21-45-00-700-802			502	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DOOR AREA HEATER - REMOVAL/INSTALLATION	21-45-01		401	PTF ALL
Door Area Heater Removal TASK 21-45-01-000-801			401	PTF ALL
Door Area Heater Installation TASK 21-45-01-400-801			408	PTF ALL
EMERGENCY EXIT DOOR HEATER SYSTEM - REMOVAL/INSTALLATION	21-45-02		401	PTF ALL
Emergency Exit Door Heater Blanket Removal TASK 21-45-02-000-801			401	PTF ALL
Emergency Exit Door Heater Blanket Installation TASK 21-45-02-400-801			404	PTF ALL
Emergency Exit Door Surround Panel Heater Blanket Removal TASK 21-45-02-000-802			405	PTF ALL
Emergency Exit Door Surround Panel Heater Blanket Installation TASK 21-45-02-400-802			409	PTF ALL
Emergency Exit Door Closeout Panel Heater Blanket Removal TASK 21-45-02-000-803			410	PTF ALL
Emergency Exit Door Closeout Panel Heater Blanket Installation TASK 21-45-02-400-803			414	PTF ALL
PACK FLOW CONTROL AND PACK COOLING SYSTEM - MAINTENANCE PRACTICES	21-51-00		201	PTF ALL
Air Cycle Air Conditioning and Temperature Control System - Deactivation TASK 21-51-00-800-801			201	PTF ALL
Air Cycle Air Conditioning and Temperature Control System - Activation TASK 21-51-00-800-802			203	PTF ALL
Ram Air System - Deactivation TASK 21-51-00-800-803			205	PTF ALL
Ram Air System - Activation TASK 21-51-00-800-804			206	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
PACK FLOW CONTROL AND PACK COOLING SYSTEM - ADJUSTMENT/TEST	21-51-00		501	PTF ALL
Pack Flow Control Valve Operational Test TASK 21-51-00-700-802			501	PTF ALL
Ram Air System Operational Test TASK 21-51-00-730-801			503	PTF ALL
Ground Pack Run Test - APU AUTO Flow Mode TASK 21-51-00-700-804			508	PTF ALL
PACK FLOW CONTROL AND PACK COOLING SYSTEM - INSPECTION/CHECK	21-51-00		601	PTF ALL
Air Cycle Machine Static Breakaway Torque Check TASK 21-51-00-700-803			601	PTF ALL
FLOW CONTROL AND SHUTOFF VALVE - REMOVAL/INSTALLATION	21-51-01		401	PTF ALL
Left Flow Control and Shutoff Valve Removal TASK 21-51-01-000-801			401	PTF ALL
Left Flow Control and Shutoff Valve Installation TASK 21-51-01-400-801			407	PTF ALL
Right Flow Control and Shutoff Valve Removal TASK 21-51-01-000-802			411	PTF ALL
Right Flow Control and Shutoff Valve Installation TASK 21-51-01-400-802			417	PTF ALL
AIR CONDITIONING ACCESSORY UNIT (ACAU) - REMOVAL/INSTALLATION	21-51-02		401	PTF ALL
Air Conditioning Accessory Unit (ACAU) Removal TASK 21-51-02-000-801			401	PTF ALL
Air Conditioning Accessory Unit (ACAU) Installation TASK 21-51-02-400-801			403	PTF ALL
AIR CONDITIONING ACCESSORY UNIT - ADJUSTMENT/TEST	21-51-02		501	PTF ALL
Air Conditioning Accessory Unit Operational Test TASK 21-51-02-710-801			501	PTF ALL
Pack Overheat Shutoff Operational Test TASK 21-51-02-710-802			508	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
HEAT EXCHANGER AND PLENUM/DIFFUSER ASSEMBLY - CLEARING DEBRIS/BLOCKAGES	21-51-03		201	PTF ALL
Heat Exchanger - Blowing Out Debris/Blockages with Nitrogen Gas/Compressed Air TASK 21-51-03-160-805			201	PTF ALL
HEAT EXCHANGER AND PLENUM/DIFFUSER ASSEMBLY - REMOVAL/INSTALLATION	21-51-03		401	PTF ALL
Heat Exchanger and Plenum/Diffuser Assembly - Removal TASK 21-51-03-000-803			401	PTF ALL
Heat Exchanger and Plenum/Diffuser Assembly - Installation TASK 21-51-03-400-802			426	PTF ALL
FAN BYPASS CHECK VALVE DOOR - INSPECTION/CHECK	21-51-03		601	PTF ALL
Fan Bypass Check Valve Door - Operational Check TASK 21-51-03-710-802			601	PTF ALL
HEAT EXCHANGER AND PLENUM/DIFFUSER ASSEMBLY - CLEANING/PAINTING	21-51-03		701	PTF ALL
Heat Exchanger and Plenum/Diffuser Assembly Cleaning TASK 21-51-03-000-804			701	PTF ALL
Heat Exchanger - Cleaning with Soap-Water-Air TASK 21-51-03-160-803			715	PTF ALL
AIR CYCLE MACHINE - REMOVAL/INSTALLATION	21-51-04		401	PTF ALL
Air Cycle Machine Removal TASK 21-51-04-000-801			401	PTF ALL
Air Cycle Machine Installation TASK 21-51-04-400-801			412	PTF ALL
CONDITIONED AIR CHECK VALVE - REMOVAL/INSTALLATION	21-51-07		401	PTF ALL
Conditioned Air Check Valve Removal TASK 21-51-07-000-801			401	PTF ALL
Conditioned Air Check Valve Installation TASK 21-51-07-400-801			405	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Left Pack Conditioned Air Check Valve Sense Line Removal TASK 21-51-07-000-802			408	PTF ALL
Left Pack Conditioned Air Check Valve Sense Line Installation TASK 21-51-07-400-802			412	PTF ALL
HIGH PRESSURE WATER SEPARATOR (HPWS) - REMOVAL/INSTALLATION	21-51-08		401	PTF ALL
High Pressure Water Separator Removal TASK 21-51-08-000-801			401	PTF ALL
High Pressure Water Separator Installation TASK 21-51-08-400-801			408	PTF ALL
TEMPERATURE CONTROL VALVE - REMOVAL/INSTALLATION	21-51-10		401	PTF ALL
Temperature Control Valve Removal TASK 21-51-10-000-801			401	PTF ALL
Temperature Control Valve Installation TASK 21-51-10-400-801			405	PTF ALL
STANDBY TEMPERATURE CONTROL VALVE - REMOVAL/INSTALLATION	21-51-11		401	PTF ALL
Standby Temperature Control Valve Removal TASK 21-51-11-000-801			401	PTF ALL
Standby Temperature Control Valve Installation TASK 21-51-11-400-801			408	PTF ALL
STANDBY TEMPERATURE CONTROL VALVE - ADJUSTMENT/TEST	21-51-11		501	PTF ALL
Standby Temperature Control Valve Primary Pneumatic Control - Operational Check TASK 21-51-11-710-801			501	PTF ALL
CONDENSER - REMOVAL/INSTALLATION	21-51-12		401	PTF ALL
Condenser Removal TASK 21-51-12-000-801			401	PTF ALL
Condenser Installation TASK 21-51-12-400-801			405	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
REHEATER - REMOVAL/INSTALLATION	21-51-13		401	PTF ALL
Reheater Removal TASK 21-51-13-000-801			401	PTF ALL
Reheater Installation TASK 21-51-13-400-801			404	PTF ALL
WATER EXTRACTOR - REMOVAL/INSTALLATION	21-51-14		401	PTF ALL
Water Extractor Removal TASK 21-51-14-000-801			401	PTF ALL
Water Extractor Installation TASK 21-51-14-400-801			405	PTF ALL
WATER EXTRACTOR DUCT - REMOVAL/INSTALLATION	21-51-15		401	PTF ALL
Water Extractor Duct Removal TASK 21-51-15-000-801			401	PTF ALL
Water Extractor Duct Installation TASK 21-51-15-400-801			405	PTF ALL
WATER SPRAY NOZZLE - REMOVAL/INSTALLATION	21-51-16		401	PTF ALL
Water Spray Nozzle Removal TASK 21-51-16-000-801			401	PTF ALL
Water Spray Nozzle Installation TASK 21-51-16-400-801			406	PTF ALL
WATER SPRAY NOZZLE - INSPECTION/CHECK	21-51-16		601	PTF ALL
Water Spray Nozzle Check TASK 21-51-16-200-801			601	PTF ALL
HIGH PRESSURE WATER SEPARATOR MIX MUFF - REMOVAL/INSTALLATION	21-51-17		401	PTF ALL
High Pressure Water Separator Mix Muff Removal TASK 21-51-17-000-801			401	PTF ALL
High Pressure Water Separator Mix Muff Installation TASK 21-51-17-400-801			407	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
RAM AIR INLET ACTUATOR - REMOVAL/INSTALLATION	21-51-21		401	PTF ALL
Ram Air Inlet Actuator Removal TASK 21-51-21-000-801			401	PTF ALL
Ram Air Inlet Actuator Installation TASK 21-51-21-400-801			410	PTF ALL
RAM AIR INLET DEFLECTOR DOOR AND SHAFT ASSEMBLY - REMOVAL/INSTALLATION	21-51-22		401	PTF ALL
Ram Air Inlet Deflector Door Removal TASK 21-51-22-000-801			401	PTF ALL
Ram Air Inlet Deflector Door Installation TASK 21-51-22-400-801			405	PTF ALL
Ram Air Inlet Deflector Door Shaft Assembly Removal TASK 21-51-22-000-802			407	PTF ALL
Ram Air Inlet Deflector Door Shaft Assembly Installation TASK 21-51-22-400-802			412	PTF ALL
RAM AIR INLET SYSTEM - ADJUSTMENT	21-51-22		501	PTF ALL
Ram Air Inlet Deflector Door and Modulation Panels Adjustment TASK 21-51-22-400-803			501	PTF ALL
RAM AIR SHAFT ASSEMBLY AND ARM ASSEMBLIES INSPECTION	21-51-22		601	PTF ALL
Ram Air Inlet Door Linkage Inspection TASK 21-51-22-200-801			601	PTF ALL
RAM AIR INLET MODULATION PANELS - REMOVAL/INSTALLATION	21-51-23		401	PTF ALL
Ram Air Inlet Modulation Panels Removal TASK 21-51-23-000-801			401	PTF ALL
Ram Air Inlet Modulation Panels Installation TASK 21-51-23-400-801			407	PTF ALL
RAM AIR DUCTS - REMOVAL/INSTALLATION	21-51-24		401	PTF ALL
Ram Air Ducts Removal TASK 21-51-24-000-801			401	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Ram Air Ducts Installation TASK 21-51-24-400-801			413	PTF ALL
RAM AIR DUCTS - REPAIRS	21-51-24		801	PTF ALL
Resin Damage - Fiberglass Ducts TASK 21-51-24-340-801			801	PTF ALL
External Patch - Fiberglass Ducts TASK 21-51-24-340-802			803	PTF ALL
Structural Repair - Fiberglass Ducts TASK 21-51-24-340-803			809	PTF ALL
RAM AIR EXHAUST ACTUATOR - REMOVAL/INSTALLATION	21-51-27		401	PTF ALL
Ram Air Exhaust Actuator - Removal TASK 21-51-27-000-801			401	PTF ALL
Ram Air Exhaust Actuator - Installation TASK 21-51-27-400-801			408	PTF ALL
COMPRESSOR DISCHARGE OVERHEAT SWITCH - REMOVAL/INSTALLATION	21-51-40		401	PTF ALL
Compressor Discharge Overheat Switch Removal TASK 21-51-40-000-801			401	PTF ALL
Compressor Discharge Overheat Switch Installation TASK 21-51-40-400-801			406	PTF ALL
COMPRESSOR DISCHARGE OVERHEAT SWITCH - ADJUSTMENT/TEST	21-51-40		501	PTF ALL
Compressor Discharge Overheat Switch Functional Test TASK 21-51-40-710-801			501	PTF ALL
TURBINE INLET OVERHEAT SWITCH - REMOVAL/INSTALLATION	21-51-41		401	PTF ALL
Turbine Inlet Overheat Switch Removal TASK 21-51-41-000-801			401	PTF ALL
Turbine Inlet Overheat Switch Installation TASK 21-51-41-400-801			406	PTF ALL

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AIR CONDITIONING

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
TURBINE INLET THERMAL SWITCH - ADJUSTMENT/TEST	21-51-41		501	PTF ALL
Turbine Inlet Thermal Switch - Functional Test TASK 21-51-41-710-801			501	PTF ALL
PACK DISCHARGE OVERHEAT SWITCH - REMOVAL/INSTALLATION	21-51-42		401	PTF ALL
Pack Discharge Overheat Switch Removal TASK 21-51-42-000-801			401	PTF ALL
Pack Discharge Overheat Switch Installation TASK 21-51-42-400-801			405	PTF ALL
PACK DISCHARGE OVERHEAT SWITCH - ADJUSTMENT/TEST	21-51-42		501	PTF ALL
Pack Discharge Overheat Switch - Functional Test TASK 21-51-42-710-801			501	PTF ALL
RAM AIR CONTROL TEMPERATURE SENSOR - REMOVAL/INSTALLATION	21-51-50		401	PTF ALL
Ram Air Control Temperature Sensor Removal TASK 21-51-50-000-801			401	PTF ALL
Ram Air Control Temperature Sensor Installation TASK 21-51-50-400-801			406	PTF ALL
MIX MANIFOLD TEMPERATURE SENSOR - MAINTENANCE PRACTICES	21-51-51		201	PTF ALL
Mix Manifold Temperature Sensor Removal TASK 21-51-51-000-801			201	PTF ALL
Mix Manifold Temperature Sensor Cleaning TASK 21-51-51-110-801			205	PTF ALL
Mix Manifold Temperature Sensor Installation TASK 21-51-51-400-801			205	PTF ALL
MIX MANIFOLD TEMPERATURE SENSOR - ADJUSTMENT/TEST	21-51-51		501	PTF ALL
Mix Manifold Temperature Sensor Test TASK 21-51-51-710-801			501	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
PACK TEMPERATURE SENSOR - REMOVAL/INSTALLATION	21-51-52		401	PTF ALL
Pack Temperature Sensor Removal TASK 21-51-52-000-801			401	PTF ALL
Pack Temperature Sensor Installation TASK 21-51-52-400-801			404	PTF ALL
PACK TEMPERATURE SENSOR - ADJUSTMENT/TEST	21-51-52		501	PTF ALL
Pack Temperature Sensor Test TASK 21-51-52-720-801			501	PTF ALL
PACK TEMPERATURE BULB - REMOVAL/INSTALLATION	21-51-60		401	PTF ALL
Pack Temperature Bulb Removal TASK 21-51-60-000-801			401	PTF ALL
Pack Temperature Bulb Installation TASK 21-51-60-400-801			404	PTF ALL
AIR CONDITIONING MODULE - REMOVAL/INSTALLATION	21-51-65		401	PTF ALL
Air Conditioning Module Removal TASK 21-51-65-000-801			401	PTF ALL
Air Conditioning Module Installation TASK 21-51-65-400-801			404	PTF ALL
ZONE TEMPERATURE CONTROL AND INDICATION - ADJUSTMENT/TEST	21-61-00		501	PTF ALL
Cabin Temperature Control and Indication - Operational Test TASK 21-61-00-700-801			501	PTF ALL
Duct Overheat Switch - Operational Test TASK 21-61-00-700-802			502	PTF ALL
DUCT TEMPERATURE LIMIT SENSOR - REMOVAL/INSTALLATION	21-61-04		401	PTF ALL
Duct Temperature Limit Sensor Removal TASK 21-61-04-000-801			401	PTF ALL
Duct Temperature Limit Sensor Installation TASK 21-61-04-400-801			407	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DUCT TEMPERATURE LIMIT SENSOR - ADJUSTMENT/TEST	21-61-04		501	PTF ALL
Duct Temperature Limit Sensor Test TASK 21-61-04-000-802			501	PTF ALL
DUCT OVERHEAT SWITCH - REMOVAL/INSTALLATION	21-61-05		401	PTF ALL
Duct Overheat Switch Removal TASK 21-61-05-000-801			401	PTF ALL
Duct Overheat Switch Installation TASK 21-61-05-400-801			407	PTF ALL
CABIN TEMPERATURE SENSOR ASSEMBLY FILTER - MAINTENANCE PRACTICES	21-61-06		201	PTF ALL
Cabin Temperature Sensor Assembly Filter Removal TASK 21-61-06-000-801			201	PTF ALL
Cabin Temperature Sensor Assembly Filter Cleaning TASK 21-61-06-100-801			209	PTF ALL
Cabin Temperature Sensor Assembly Filter Installation TASK 21-61-06-400-801			211	PTF ALL
CABIN TEMPERATURE SELECTOR - REMOVAL/INSTALLATION	21-61-07		401	PTF ALL
Cabin Temperature Selector Removal TASK 21-61-07-000-801			401	PTF ALL
Cabin Temperature Selector Installation TASK 21-61-07-400-801			405	PTF ALL
TRIM AIR CHECK VALVE - REMOVAL/INSTALLATION	21-61-09		401	PTF ALL
Trim Air Check Valve Removal TASK 21-61-09-000-801			401	PTF ALL
Trim Air Check Valve Installation TASK 21-61-09-400-801			407	PTF ALL
TRIM AIR CHECK VALVE - INSPECTION/CHECK	21-61-09		601	PTF ALL
Trim Air Check Valve Inspection TASK 21-61-09-200-801			601	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
CABIN TEMPERATURE SENSOR - MAINTENANCE PRACTICES	21-61-10		201	PTF ALL
Cabin Temperature Sensor Removal TASK 21-61-10-000-801			201	PTF ALL
Cabin Temperature Sensor Cleaning TASK 21-61-10-100-801			208	PTF ALL
Cabin Temperature Sensor Installation TASK 21-61-10-400-801			208	PTF ALL
CABIN TEMPERATURE SENSOR - INSPECTION/CHECK	21-61-10		601	PTF ALL
Cabin Temperature Sensor Resistance Check TASK 21-61-10-200-801			601	PTF ALL
CABIN TEMPERATURE SENSOR FAN - REMOVAL/INSTALLATION	21-61-11		401	PTF ALL
Cabin Temperature Sensor Fan Removal TASK 21-61-11-000-801			401	PTF ALL
Cabin Temperature Sensor Fan Installation TASK 21-61-11-400-801			409	PTF ALL
TRIM AIR DUCT PRESSURE SEAL - REMOVAL/INSTALLATION	21-61-12		401	PTF ALL
Trim Air Duct Pressure Seal Removal TASK 21-61-12-000-801			401	PTF ALL
Trim Air Duct Pressure Seal Installation TASK 21-61-12-400-801			409	PTF ALL
TEMPERATURE SENSOR ASSEMBLY DUCT/HOUSING - MAINTENANCE PRACTICES	21-61-14		201	PTF ALL
Flight Compartment Temperature Sensor Assembly Housing Removal TASK 21-61-14-000-801			201	PTF ALL
Flight Compartment Temperature Sensor Assembly Housing Installation TASK 21-61-14-400-801			205	PTF ALL
Passenger Cabin Temperature Sensor Assembly Duct Removal TASK 21-61-14-000-802			206	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Passenger Cabin Temperature Sensor Assembly Duct Installation TASK 21-61-14-400-802			212	PTF ALL
Flight Compartment Temperature Sensor Assembly Housing Cleaning TASK 21-61-14-100-801			213	PTF ALL
Passenger Cabin Temperature Sensor Assembly Duct Cleaning TASK 21-61-14-100-802			214	PTF ALL
TEMPERATURE CONTROL MODULE - REMOVAL/INSTALLATION	21-61-15		401	PTF ALL
Temperature Control Module Removal TASK 21-61-15-000-801			401	PTF ALL
Temperature Control Module Installation TASK 21-61-15-400-801			404	PTF ALL
ZONE TRIM AIR MODULATING VALVE - REMOVAL/INSTALLATION	21-61-21		401	PTF ALL
Zone Trim Air Modulating Valve Removal TASK 21-61-21-000-801			401	PTF ALL
Zone Trim Air Modulating Valve Installation TASK 21-61-21-400-801			406	PTF ALL
INTEGRATED AIR SYSTEMS CONTROLLER - MAINTENANCE PRACTICES	21-61-22		201	PTF ALL
Integrated Air System Controller Configuration Check TASK 21-61-22-700-801			201	PTF ALL
INTEGRATED AIR SYSTEMS CONTROLLER - REMOVAL/INSTALLATION	21-61-22		401	PTF ALL
Integrated Air Systems Controller (IASC) Removal TASK 21-61-22-000-801			401	PTF ALL
Integrated Air Systems Controller (IASC) Installation TASK 21-61-22-400-801			404	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
INLET PRESSURE SENSOR - REMOVAL/INSTALLATION	21-61-23		401	PTF ALL
Inlet Pressure Sensor Removal TASK 21-61-23-000-801			401	PTF ALL
Inlet Pressure Sensor Installation TASK 21-61-23-400-801			406	PTF ALL
PACK PRESSURE SENSOR SENSE LINE - CLEANING/PAINTING	21-61-23		701	PTF ALL
Pack Inlet Pressure Sensor Sense Line Cleaning TASK 21-61-23-100-801			701	PTF ALL
FLOW SENSOR - REMOVAL/INSTALLATION	21-61-24		401	PTF ALL
Flow Sensor Removal TASK 21-61-24-000-801			401	PTF ALL
Flow Sensor Installation TASK 21-61-24-400-801			404	PTF ALL
DUCT TEMPERATURE BULB - REMOVAL/INSTALLATION	21-61-30		401	PTF ALL
Duct Temperature Bulb Removal TASK 21-61-30-000-801			401	PTF ALL
Duct Temperature Bulb Installation TASK 21-61-30-400-801			407	PTF ALL
CABIN TEMPERATURE BULB - REMOVAL/INSTALLATION	21-61-31		401	PTF ALL
Cabin Temperature Bulb Removal TASK 21-61-31-000-801			401	PTF ALL
Cabin Temperature Bulb Installation TASK 21-61-31-400-801			405	PTF ALL
CABIN TEMPERATURE INDICATOR - REMOVAL/INSTALLATION	21-61-32		401	PTF ALL
Cabin Temperature Indicator Removal TASK 21-61-32-000-801			401	PTF ALL
Cabin Temperature Indicator Installation TASK 21-61-32-400-801			404	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
TRIM AIR PRESSURE REGULATION AND SHUTOFF CONTROL - ADJUSTMENT/TEST	21-62-00		501	PTF ALL
Trim Air Pressure Regulating and Shutoff Valve - Operational Test TASK 21-62-00-710-801			501	PTF ALL
TRIM AIR PRESSURE REGULATING AND SHUTOFF VALVE - REMOVAL/INSTALLATION	21-62-01		401	PTF ALL
Trim Air Pressure Regulating and Shutoff Valve Removal TASK 21-62-01-000-801			401	PTF ALL
Trim Air Pressure Regulating and Shutoff Valve Installation TASK 21-62-01-400-801			406	PTF ALL
OZONE CONVERTER - REMOVAL/INSTALLATION	21-73-01		401	PTF ALL
Ozone Converter Removal TASK 21-73-01-000-801			401	PTF ALL
Ozone Converter Installation TASK 21-73-01-400-801			405	PTF ALL

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1. General

A. This procedure has these tasks:

(1) Supply Conditioned Air to the Airplane

(2) Remove Conditioned Air from the Airplane

(3) Supply Conditioned Air with a Cooling Pack

(4) Remove Conditioned Air Supplied by a Cooling Pack

(5) Supply Conditioned Air with a Ground Air Source

(6) Remove Conditioned Air Supplied by a Ground Air Source

B. It is recommended that the conditioned air to cool the airplane on the ground be supplied from a ground air source, when practical, as an alternative to operating the cooling pack (SL 737-21-053).

TASK 21-00-00-800-801

2. Supply Conditioned Air to the Airplane

A. General

SUBTASK 21-00-00-860-001

(1) It is recommended that the conditioned air to cool the airplane on the ground be supplied from a ground air source, when practical, as an alternative to operating the cooling packs. To supply conditioned air, do this task: Supply Conditioned Air with a Ground Air Source, TASK 21-00-00-800-805.

B. Procedure

SUBTASK 21-00-00-860-002

(1) Do one of these tasks to supply conditioned air to the airplane:

(a) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

(b) Do this task: Supply Conditioned Air with a Ground Air Source, TASK 21-00-00-800-805.

——— END OF TASK ———

TASK 21-00-00-800-802

3. Remove Conditioned Air from the Airplane

A. Procedure

SUBTASK 21-00-00-860-003

(1) Do one of these tasks to remove conditioned air from the airplane:

(a) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, TASK 21-00-00-800-804.

(b) Do this task: Remove Conditioned Air Supplied by a Ground Air Source, TASK 21-00-00-800-806.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-800-803

4. Supply Conditioned Air with a Cooling Pack

A. References

Reference	Title
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)

B. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right
191	Lower Wing-To-Body Fairing - Forward of Wing Box
211	Flight Compartment - Left
212	Flight Compartment - Right

C. General

SUBTASK 21-00-00-860-004

(1) It is recommended that the conditioned air to cool the airplane on the ground be supplied from a ground air source, when practical, as an alternative to operating the cooling packs. To supply conditioned air, do this task: Supply Conditioned Air with a Ground Air Source, TASK 21-00-00-800-805.

SUBTASK 21-00-00-860-005

(2) When conditions permit, open the doors for the air conditioning packs when they operate.

D. Prepare to Supply Conditioned Air with a Cooling Pack

SUBTASK 21-00-00-860-007

(1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-00-00-860-008

WARNING MAKE SURE THAT THERE IS AN EXIT IN THE AIRPLANE FUSELAGE FOR THE CONDITIONED AIR. IF THERE IS NO EXIT FOR THE AIR, THE PRESSURE IN THE CABIN WILL INCREASE. THIS CAN CAUSE INJURIES TO PERSONNEL.

(2) Do one of these steps to make sure there is an exit for the conditioned air:

(a) Make sure the cabin pressure outflow valve is open.

(b) Make sure at least one passenger entry door is open.

SUBTASK 21-00-00-860-009

CAUTION DO NOT OPERATE A COOLING PACK AND A GROUND AIR SOURCE AT THE SAME TIME. THE COOLING PACK CAN NOT CONTROL THE AIR TEMPERATURE IF YOU ALSO OPERATE A GROUND AIR SOURCE. IF YOU OPERATE THEM AT THE SAME TIME, DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Make sure that the ground air source is disconnected from the ground conditioned air connector.

(a) If it is necessary, do this task: Remove Conditioned Air Supplied by a Ground Air Source, TASK 21-00-00-800-806.

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E. Supply Conditioned Air with a Cooling Pack

SUBTASK 21-00-00-860-010

(1) Turn these selectors on the P5-17 Cabin Temperature Control Panel to the AUTO position:

(a) CONT CAB

(b) FWD CAB

(c) AFT CAB

SUBTASK 21-00-00-860-011

(2) Set the applicable L PACK switch or the R PACK switch, on the P5-10 Air Conditioning Panel, to the AUTO position.

——— END OF TASK ———

TASK 21-00-00-800-804

5. Remove Conditioned Air Supplied by a Cooling Pack

A. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Remove Conditioned Air Supplied by a Cooling Pack

SUBTASK 21-00-00-860-012

(1) Set the applicable L PACK switch or the R PACK switch, on the P5-10 Air Conditioning Panel, to the OFF position.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-860-013

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

——— END OF TASK ———

TASK 21-00-00-800-805

6. Supply Conditioned Air with a Ground Air Source

A. Location Zones

Zone	Area
100	Lower Half of Fuselage

B. Access Panels

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

C. Supply Conditioned Air with a Ground Air Source

SUBTASK 21-00-00-010-001

(1) Get access to the ground conditioned air connector, do this step:

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Open this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

SUBTASK 21-00-00-210-001

(2) Make sure that the swing check valve in the ground conditioned air connector can move freely.

SUBTASK 21-00-00-860-015

(3) Do these steps to connect the duct from the ground air source to the ground conditioned air connector:

(a) Engage the fasteners on the duct with the slots in the ground conditioned air connector.

(b) Turn the duct until it is locked to the ground conditioned air connector.

SUBTASK 21-00-00-860-016

WARNING MAKE SURE THAT THERE IS AN EXIT IN THE AIRPLANE FUSELAGE FOR THE CONDITIONED AIR. IF THERE IS NO EXIT FOR THE AIR, THE PRESSURE IN THE CABIN WILL INCREASE. THIS CAN CAUSE INJURIES TO PERSONNEL.

(4) Do one of these steps to make sure that there is an exit for the conditioned air:

(a) Make sure that the cabin pressure outflow valve is open.

(b) Make sure that at least one passenger entry door is open.

SUBTASK 21-00-00-860-017

CAUTION DO NOT OPERATE A COOLING PACK AND A GROUND AIR SOURCE AT THE SAME TIME. THE COOLING PACK CAN NOT CONTROL THE AIR TEMPERATURE IF YOU ALSO OPERATE A GROUND AIR SOURCE. IF YOU OPERATE THEM AT THE SAME TIME, DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Make sure that the L PACK and R PACK switches, on the P5-10 air conditioning panel, are set to the OFF position.

NOTE: It is optional to make sure that the L RECIRC FAN and R RECIRC FAN switches are set to the OFF position.

SUBTASK 21-00-00-860-019

(6) Operate the ground air source to supply conditioned air to the airplane.

(a) Make sure that the conditioned air pressure does not exceed 15 in/H_2O (0.54 psi).

(b) Make sure that the conditioned air temperature does not exceed 160°F (71°C).

——— END OF TASK ———

TASK 21-00-00-800-806

7. Remove Conditioned Air Supplied by a Ground Air Source

A. Location Zones

Zone	Area
100	Lower Half of Fuselage

B. Access Panels

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Procedure

SUBTASK 21-00-00-860-020

(1) Stop the operation of the ground air source.

SUBTASK 21-00-00-020-001

(2) Remove the duct for the ground air source from the ground conditioned air connector.

SUBTASK 21-00-00-010-002

(3) Close this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR CONDITIONING - GENERAL - ADJUSTMENT/TEST

1. General

A. This procedure has these tests:

(1) 21 Air Conditioning, System Test, Air Conditioning IBIT - Left

(2) 21 Air Conditioning, System Test, Air Conditioning IBIT - Right

(3) 21 Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test - Left

(4) 21 Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test - Right

TASK 21-00-00-730-801

2. Air Conditioning, System Test, Air Conditioning IBIT

A. General

(1) This test makes sure that the air conditioning valves and ram air door actuators operate correctly.

(a) Air conditioning valves and ram air doors will open and close.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

D. Prepare for the Test

SUBTASK 21-00-00-860-021

(1) Make sure that the electrical power is on (TASK 24-22-00-860-801).

SUBTASK 21-00-00-860-022

(2) If the GND TEST switch, on the P61-4 panel, is in NORM position, set the GND TEST switch to ENABLE position.

SUBTASK 21-00-00-700-001

(3) Make sure that the airplane is in “ON GROUND” mode.

(a) If it is necessary, do this task to set the airplane to ON GROUND mode: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-00-00-860-023

(4) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L and R PACK switches to the AUTO position.

(b) Set the BLEED 1 and 2 switches to the OFF position.

(c) Set the BLEED APU switch to the OFF position.

(d) Set the L and R RECIRC FAN switches to the AUTO position.

(e) Set the ISOLATION VALVE switch to the OPEN position.

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SUBTASK 21-00-00-860-024

(5) Do these steps on the P5-17 cabin temperature panel (located on the P5 forward overhead panel):

(a) Set the CONT CAB, FWD CAB, and AFT CAB selectors to the AUTO position.

(b) Set the TRIM AIR switch to the ON position.

E. System Test - Air Conditioning IBIT

SUBTASK 21-00-00-730-001

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE AWAY FROM THE RAM AIR INLET DOOR. WHEN YOU PUSH THE VERIFY SWITCH, THE RAM AIR COMPONENTS WILL MOVE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Start the applicable ground test (TASK 46-13-02-710-801):

(a) 21 Air Conditioning, System Test, Air Conditioning IBIT - Left.

(b) 21 Air Conditioning, System Test, Air Conditioning IBIT - Right.

SUBTASK 21-00-00-730-004

(2) When the test is completed, make sure that PASS shows adjacent to TEST CONDITION.

SUBTASK 21-00-00-730-005

(3) If FAILED shows, refer to the IFIM and do the applicable procedure(s).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-860-025

(1) Put the switches back to the usual condition.

SUBTASK 21-00-00-860-026

(2) If all maintenance on the airplane is completed, set the GND TEST switch, on the P61-4 panel, to the NORM position.

——— END OF TASK ———

TASK 21-00-00-730-802

3. Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test

A. General

(1) This test makes sure that the air conditioning valves, ram air door actuators, and flow sensor operate correctly.

(a) Air conditioning valves and ram air doors will open and close.

(2) This test will take approximately 4-5 minutes to complete.

(3) If the Air Conditioning System has been running, prior to this test, or if it is a hot day, shut down the Air Conditioning System for at least 10 minutes before running this test. This will prevent setting a drift fault.

B. References

Reference	Title
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
31-65-00-970-801	System Maintenance Data Pages Function (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
46-13-02-710-801	Onboard Maintenance Function Ground Test (P/B 201)

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C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
221	Passenger Compartment - Aft of Control Compartment to Forward Entry Door - Left

D. Prepare for the Test

SUBTASK 21-00-00-710-012

(1) On the MAX Display System (MDS), go to the 21 AIR CONDITIONING display under the MAINT DATA PGS (TASK 31-65-00-970-801).

(a) Make sure that the PACK INLET TEMP is less than 150°F (66°C) before running the IBIT with Flow Sensor Test.

NOTE: If the PACK INLET TEMP is greater than 150°F (66°C), the test will cause a fault to occur.

1) If it is necessary, do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

SUBTASK 21-00-00-860-028

(2) If the GND TEST switch, on the P61-4 panel, is in NORM position, set the GND TEST switch to ENABLE position.

SUBTASK 21-00-00-860-029

(3) Make sure that the airplane is in “ON GROUND” mode.

(a) If it is necessary, do this task to set the airplane to ON GROUND mode: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-00-00-860-030

(4) Do these steps on the P5-10 air conditioning module:

(a) Set the L and R PACK switches to the AUTO position.

(b) Set the BLEED 1 and 2 switches to the OFF position.

(c) Set the BLEED APU switch to the OFF position.

(d) Set the L and R RECIRC FAN switches to the AUTO position.

(e) Set the ISOLATION VALVE switch to the OPEN position.

SUBTASK 21-00-00-860-136

(5) On the P5-11 engine and wing anti-ice panel, set the WING ANTI-ICE switch to the OFF position.

SUBTASK 21-00-00-860-031

(6) Do these steps on the P5-17 cabin temperature selector:

(a) Set the CONT CAB, FWD CAB, and AFT CAB selectors to the AUTO position.

(b) Set the TRIM AIR switch to the ON position.

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E. System Test - Air Conditioning IBIT

SUBTASK 21-00-00-730-002

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE AWAY FROM THE RAM AIR INLET DOOR. WHEN YOU PUSH THE VERIFY SWITCH, THE RAM AIR COMPONENTS WILL MOVE AND CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do this task and select the ground test: Onboard Maintenance Function Ground Test, TASK 46-13-02-710-801.

(a) 21 Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test - Left.

(b) 21 Air Conditioning, System Test, Air Conditioning IBIT with Flow Sensor Test - Right.

SUBTASK 21-00-00-730-006

(2) When the test is completed, make sure that PASS shows adjacent to TEST CONDITION.

SUBTASK 21-00-00-730-007

(3) If FAILED shows, refer to the IFIM and do the applicable procedure(s).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-860-032

(1) Put the switches back to the usual condition.

SUBTASK 21-00-00-860-033

(2) If all maintenance on the airplane is completed, set the GND TEST switch, on the P61-4 panel, to the NORM position.

——— END OF TASK ———

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AIR CONDITIONING - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has the maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks prepare the airplane for flight with certain systems/components inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. These are the tasks for the air conditioning components:

(1) DDG 21-21-01 Ground Conditioned Air Connector Check Valve Inoperative - Preparation

(2) DDG 21-21-01 Ground Conditioned Air Connector Check Valve Inoperative - Restoration

(3) DDG 21-23-01 Return Air Grille Broken or Missing - Preparation

(4) DDG 21-23-01 Return Air Grille Broken or Missing - Restoration

(5) DDG 21-27-01 Equipment Cooling Supply and Exhaust Fans Inoperative - Preparation

(6) DDG 21-27-02A Overboard Exhaust Valve Inoperative - Valve in SMOKE Position - Preparation

(7) DDG 21-27-02A Overboard Exhaust Valve Inoperative - Valve in SMOKE Position - Restoration

(8) DDG 21-27-02B Overboard Exhaust Valve Inoperative - Valve Fully Open - Preparation

(9) DDG 21-27-02B Overboard Exhaust Valve Inoperative - Valve Fully Open - Restoration

(10) DDG 21-27-03 Equipment Cooling Low Flow Detector System Inoperative - Preparation

(11) DDG 21-27-03 Equipment Cooling Low Flow Detector System Inoperative - Restoration

(12) DDG 21-27-04 Equipment Cooling Air Filter Removed - Preparation

(13) DDG 21-27-04 Equipment Cooling Air Filter Removed - Restoration

(14) DDG 21-27-05 Enhanced E6 Rack Equipment Cooling Exhaust Fan Inoperative - Preparation

(15) DDG 21-27-05 Enhanced E6 Rack Equipment Cooling Exhaust Fan Inoperative - Restoration

(16) DDG 21-31-01-01A Cabin Pressure Control System - One Automatic Mode Inoperative - Preparation

(17) DDG 21-31-01-01A Cabin Pressure Control System - One Automatic Mode Inoperative - Restoration

(18) DDG 21-31-01-01B Cabin Pressure Control System - Both Automatic Modes Inoperative - Preparation

(19) DDG 21-31-01-01B Cabin Pressure Control System - Both Automatic Modes Inoperative - Restoration

(20) DDG 21-31-01-02 Cabin Pressure Control System - Manual Mode Inoperative - Preparation

(21) DDG 21-31-01-02 Cabin Pressure Control System - Manual Mode Inoperative - Restoration

(22) DDG 21-31-02 Main Outflow Valve Inoperative - Preparation

(23) DDG 21-31-02 Main Outflow Valve Inoperative - Restoration

(24) DDG 21-32-01A Positive Pressure Relief Valve Inoperative - Valve Closed - Preparation

(25) DDG 21-32-01A Positive Pressure Relief Valve Inoperative - Valve Closed - Restoration

(26) DDG 21-32-01B Positive Pressure Relief Valve Inoperative - Valve Open - Preparation

(27) DDG 21-32-01B Positive Pressure Relief Valve Inoperative - Valve Open - Restoration

(28) DDG 21-33-01 Cabin Rate of Climb Indicator Inoperative - Preparation

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(29) DDG 21-33-01 Cabin Rate of Climb Indicator Inoperative - Restoration
(30) DDG 21-33-02–02 Cabin Differential Pressure Indication Inoperative - Preparation
(31) DDG 21-33-02–02 Cabin Differential Pressure Indication Inoperative - Restoration
(32) DDG 21-33-03–02 Cabin Altitude Indication Inoperative - Preparation
(33) DDG 21-33-03–02 Cabin Altitude Indication Inoperative - Restoration
(34) DDG 21-33-05 Outflow Valve Position Indicator Inoperative - Preparation
(35) DDG 21-45-01 Door Area Heater System Inoperative - Preparation
(36) DDG 21-45-01 Door Area Heater System Inoperative - Restoration
(37) DDG 21-51-01 Air Conditioning Packs Inoperative - Preparation
(38) DDG 21-51-01 Air Conditioning Packs Inoperative - Restoration
(39) DDG 21-51-02 Pack Flow Control and Shutoff Valves Inoperative - Preparation
(40) DDG 21-51-02 Pack Flow Control and Shutoff Valves Inoperative - Restoration
(41) DDG 21-51-04 Pack Flow Sensor Inoperative - Preparation
(42) DDG 21–51–05 Pack Temperature Control Valves Inoperative - Preparation
(43) DDG 21–51–05 Pack Temperature Control Valves Inoperative - Restoration
(44) DDG 21-51-09 Pack Ram Air Systems Inoperative - Preparation
(45) DDG 21-51-09 Pack Ram Air Systems Inoperative - Restoration
(46) DDG 21-61-01 Trim Air Modulating Valve Inoperative - Preparation
(47) DDG 21-61-01 Trim Air Modulating Valve Inoperative - Restoration
(48) DDG 21-61-04 Trim Air Check Valve Inoperative - Preparation
(49) DDG 21-61-04 Trim Air Check Valve Inoperative - Restoration
(50) DDG 21-62-01 Trim Air Pressure Regulating and Shutoff Valve Inoperative - Preparation
(51) DDG 21-62-01 Trim Air Pressure Regulating and Shutoff Valve Inoperative - Restoration

TASK 21-00-00-040-844

2. DDG 21-21-01 Ground Conditioned Air Connector Check Valve Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the ground conditioned air connector check valve inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right
212	Flight Compartment - Right

C. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-040-001

(1) For dispatch with the ground preconditioned air connection check valve inoperative in the open position, do these steps:

(a) On the P5 overhead panel, do these steps:

1) Put the pressurization mode selector in the MAN position.

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2) Hold the outflow valve toggle switch in the OPEN position until the outflow valve position indicator shows FULL OPEN.

(b) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

——— END OF TASK ———

TASK 21-00-00-440-831

3. DDG 21-21-01 Ground Conditioned Air Connector Check Valve Inoperative - Restoration

A. General

(1) This task puts the airplane to its usual condition after operation with the ground conditioned air connector check valve inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right
212	Flight Compartment - Right

C. Ground Preconditioned Air Connection Check Valve Restoration.

SUBTASK 21-00-00-760-001

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-00-00-710-001

(2) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

——— END OF TASK ———

TASK 21-00-00-040-845

4. DDG 21-23-01 Return Air Grille Broken or Missing - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the return air grille broken or missing.

B. References

Reference	Title
21-23-03 P/B 201	RETURN AIR GRILLES (DECOMPRESSION VENTS) - MAINTENANCE PRACTICES
25	EQUIPMENT/FURNISHINGS

C. Location Zones

Zone	Area
231	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Left
232	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Right

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(Continued)

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Return Air Grille Preparation

SUBTASK 21-00-00-010-003

(1) Find the damaged return air grille.

SUBTASK 21-00-00-200-001

(2) Determine if the damaged return air grille is in a location that is permitted to have a damaged or missing return air grille.

(a) It is permitted for a return air grille to be missing or damaged in these locations:

1) Station 887-907.

NOTE: For a damaged return air grille not located in the listed locations, it is permitted to replace the damaged return air grille with one in the allowed location if the two return air grille assemblies have the same part number.

SUBTASK 21-00-00-010-004

(3) Remove interior furnishings that block access to the damaged return air grille (CHAPTER 25).

SUBTASK 21-00-00-020-002

(4) Remove the damaged return air grille (PAGEBLOCK 21-23-03/201).

SUBTASK 21-00-00-400-001

(5) Install a solid panel that fully covers the opening caused by removing the damaged return air grille.

NOTE: Only one return air grille can be blocked at a time.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-420-001

(1) Install the interior furnishings removed for access to the damaged return air grille (CHAPTER 25).

——— END OF TASK ———

TASK 21-00-00-440-832

5. DDG 21-23-01 Return Air Grille Broken or Missing - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with a broken or missing return air grille.

B. References

Reference	Title
21-23-03 P/B 201	RETURN AIR GRILLES (DECOMPRESSION VENTS) - MAINTENANCE PRACTICES
25	EQUIPMENT/FURNISHINGS

C. Location Zones

Zone	Area
231	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Left
232	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Right

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(Continued)

Zone	Area
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Procedure

SUBTASK 21-00-00-010-005

(1) Remove the interior furnishings necessary to access the blocked return air grille location (CHAPTER 25).

SUBTASK 21-00-00-020-003

(2) Remove the solid panel used in the location of the damaged return air grille (PAGEBLOCK 21-23-03/201).

SUBTASK 21-00-00-400-002

(3) Install a new return air grille (PAGEBLOCK 21-23-03/201).

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-410-001

(1) Install the interior furnishings removed for access to the damaged return air grille (CHAPTER 25).

——— END OF TASK ———

TASK 21-00-00-040-846

6. DDG 21-27-01 Equipment Cooling Supply and Exhaust Fans Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with an electronic equipment cooling supply or exhaust fan inoperative.

NOTE: A fan may be inoperative as long as all remaining fans and both low flow detectors are inspected to operate normally.

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Equipment Cooling Supply And Exhaust Fan Inspection

SUBTASK 21-00-00-720-001

(1) On the P5 Equipment Cooling Panel, put the applicable SUPPLY or EXHAUST switch to the operative fan position (NORM or ALTN).

(a) Make sure that the applicable cooling supply or exhaust OFF light goes off.

SUBTASK 21-00-00-720-002

(2) When the supply system has two fans operative, do these steps:

(a) Put the EQUIP COOLING SUPPLY switch to the NORM position.

(b) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

(c) Make sure the equipment cooling supply OFF light comes on.

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(d) Put the EQUIP COOLING SUPPLY switch to the ALTN position.

(e) Make sure the equipment cooling supply OFF light goes off.

(f) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

(g) Make sure the equipment cooling supply OFF light comes on.

(h) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

(i) Make sure that the equipment cooling supply OFF light goes off.

(j) Put the EQUIPMENT COOLING SUPPLY switch to the NORM position.

SUBTASK 21-00-00-720-003

(3) When the exhaust system has the two fans operative, do these steps:

(a) Put the EQUIP COOLING EXHAUST switch to the NORM position.

(b) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

(c) Make sure the equipment cooling exhaust OFF light comes on.

(d) Put the EQUIP COOLING EXHAUST switch to the ALTN position.

(e) Make sure the equipment cooling exhaust OFF light goes off.

(f) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

(g) Make sure the equipment cooling exhaust OFF light comes on.

(h) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

(i) Make sure that the equipment cooling exhaust OFF light goes off.

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(j) Put the EQUIPMENT COOLING EXHAUST switch to the NORM position.

——— END OF TASK ———

TASK 21-00-00-040-847

7. DDG 21-27-02A Overboard Exhaust Valve Inoperative - Valve in SMOKE Position - Preparation

(Figure 901)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the overboard exhaust valve inoperative with the valve in the SMOKE position.

B. References

Reference	Title
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

Reference	Description
STD-3907	Mirror - Dental

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Equipment Cooling Overboard Exhaust Valve Inspection

SUBTASK 21-00-00-010-006

(1) To get access to the equipment cooling overboard exhaust valve, do these steps:

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Remove the equipment cooling cover panel [27].

1) Remove the screws [28].

2) Remove the equipment cooling cover panel [27].

SUBTASK 21-00-00-210-002

(2) Use a dental mirror, STD-3907, to make sure the position indicator shows that the valve is in the SMOKE position.

G. If the overboard exhaust valve has not failed in the SMOKE position, do these steps to put and deactivate the valve in the SMOKE position:

SUBTASK 21-00-00-860-034

(1) Do this task to put the airplane in the air mode: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-00-00-860-035

(2) Put the applicable PACK switches to the HIGH position.

(3) Put the R RECIRC FAN switch to the OFF position.

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(4) Make sure that the overboard exhaust valve is in the SMOKE position.

NOTE: This will take approximately 10 seconds.

(5) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(6) Return the applicable PACK switches to the AUTO position.

(7) Return the R RECIRC FAN switch to the AUTO position.

H. Equipment Cooling Overboard Exhaust Valve Deactivation

SUBTASK 21-00-00-860-036

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(2) Disconnect, cap and stow the electrical connector [34] D11880 on the overboard exhaust valve.

(3) Make sure the equipment cooling overboard exhaust valve is in the SMOKE position.

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-00-00-410-018

(5) Install the equipment cooling cover panel [27].

(a) Put the equipment cooling cover panel [27] in place.

(b) Install the screws [28].

SUBTASK 21-00-00-410-017

(6) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

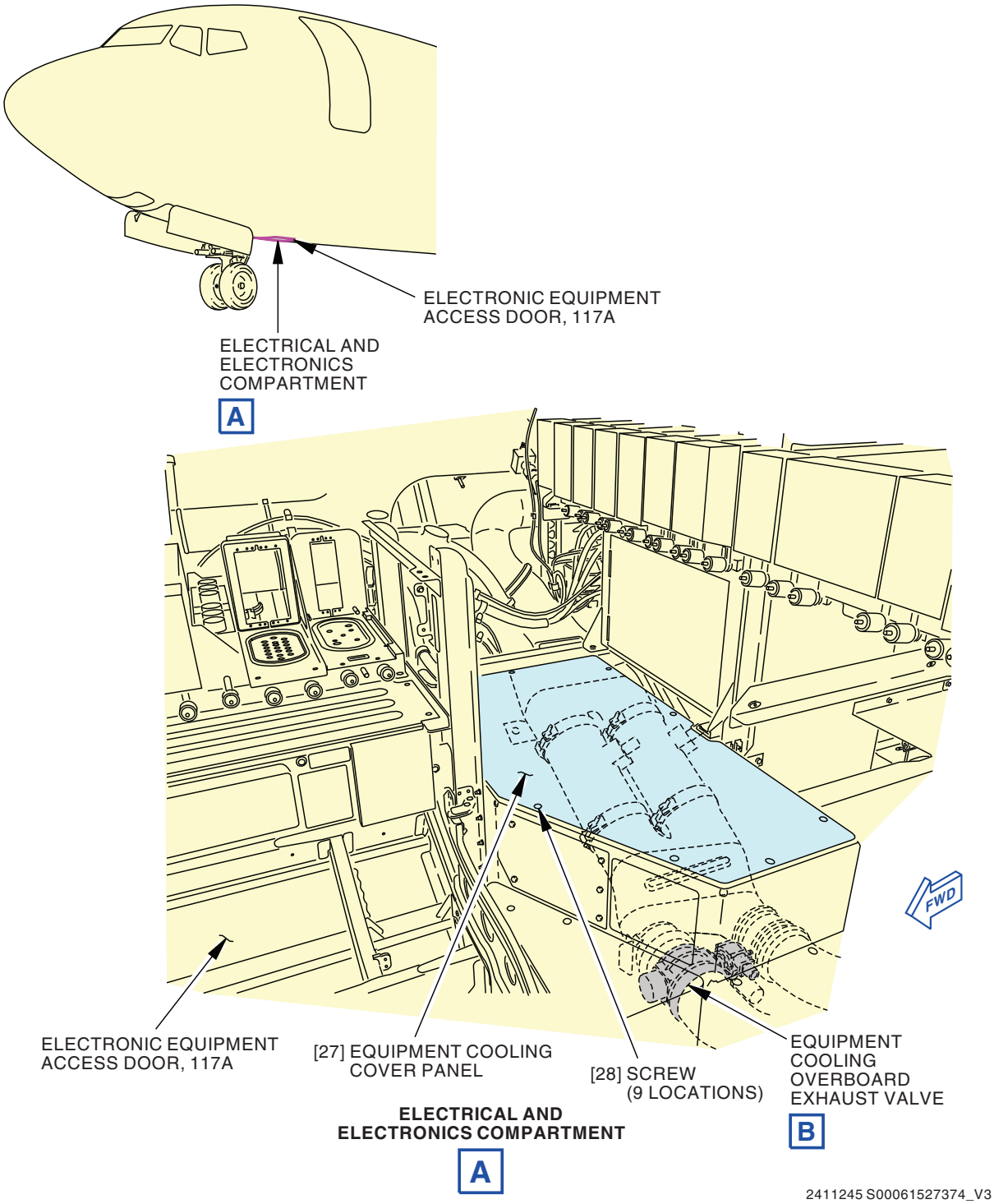
SUBTASK 21-00-00-860-037

(7) Make sure that the airplane is in “ON GROUND” mode.

(a) If it is necessary, do this task to set the airplane to ON GROUND mode: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

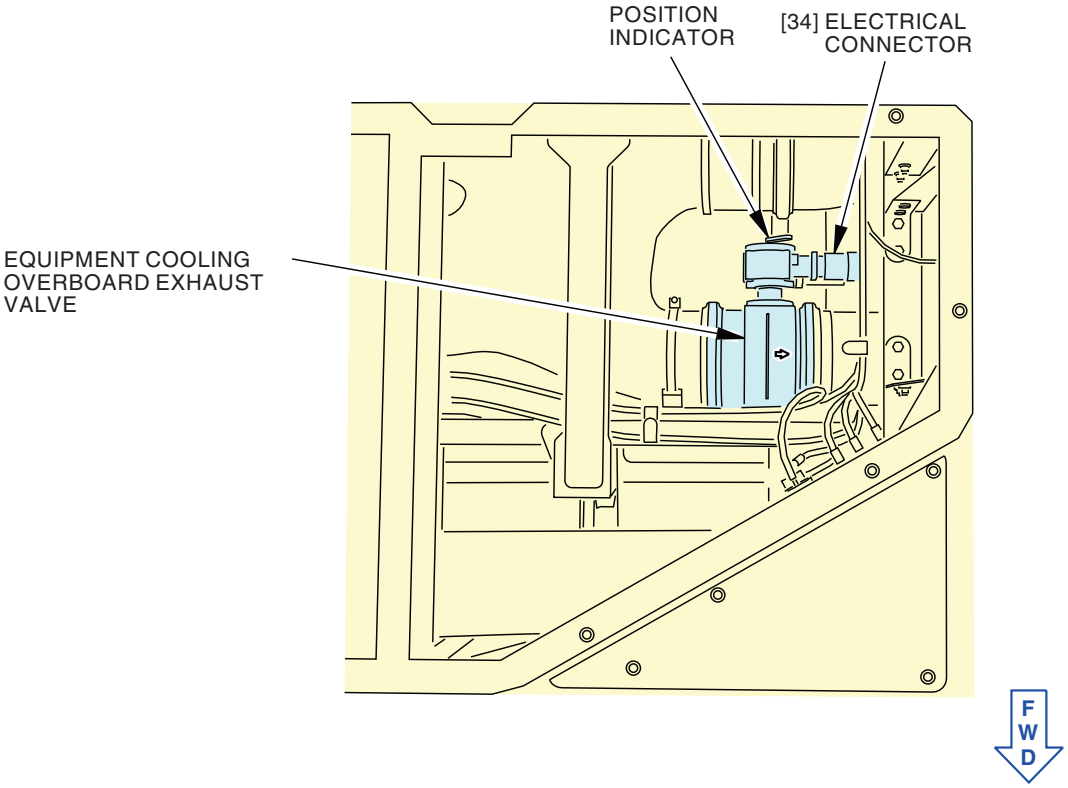
——— END OF TASK ———

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Equipment Cooling Overboard Exhaust Valve Deactivation
Figure 901/21-00-00-990-809 (Sheet 1 of 2)

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EQUIPMENT COOLING OVERBOARD EXHAUST VALVE
(OXYGEN BOTTLE INSTALLATION NOT SHOWN FOR CLARITY)
(VIEW IN THE DOWN DIRECTION)

B

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Equipment Cooling Overboard Exhaust Valve Deactivation
Figure 901/21-00-00-990-809 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-833

8. DDG 21-27-02A Overboard Exhaust Valve Inoperative - Valve in SMOKE Position - Restoration

(Figure 901)

A. General

(1) This task puts the airplane back to its usual condition after operation with an equipment cooling overboard exhaust valve inoperative with the valve in the SMOKE position.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Procedure

SUBTASK 21-00-00-010-038

(1) To get access to the equipment cooling overboard exhaust valve, do these steps:

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Remove the equipment cooling cover panel [27].

1) Remove the screws [28].

2) Remove the equipment cooling cover panel [27].

SUBTASK 21-00-00-860-102

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-00-00-860-103

(3) Connect the electrical connector [34] D11880 on the overboard exhaust valve.

SUBTASK 21-00-00-860-104

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-00-00-410-019

(5) Install the equipment cooling cover panel [27].

(a) Put the equipment cooling cover panel [27] in place.

(b) Install the screws [28].

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SUBTASK 21-00-00-410-016

(6) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 21-00-00-040-848

9. DDG 21-27-02B Overboard Exhaust Valve Inoperative - Valve Fully Open - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the overboard exhaust valve inoperative with the valve in the fully open position.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

C. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-040-002

(1) On the P5 overhead panel, do these steps:

(a) Put the pressurization mode selector in the MAN position.

(b) Hold the outflow valve toggle switch in the OPEN position until the outflow valve position indicator shows FULL OPEN.

(2) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

——— END OF TASK ———

TASK 21-00-00-440-834

10. DDG 21-27-02B Overboard Exhaust Valve Inoperative - Valve Fully Open - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with an equipment cooling overboard exhaust valve inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

C. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-960-002

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-710-013

(2) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

——— END OF TASK ———

TASK 21-00-00-040-849

11. DDG 21-27-03 Equipment Cooling Low Flow Detector System Inoperative - Preparation
(Figure 902)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with an equipment cooling low flow detector system inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Equipment Cooling Supply and Exhaust Fan Inspection

SUBTASK 21-00-00-010-008

(1) To get access to the equipment cooling overboard exhaust valve, do these steps:

Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-00-00-710-002

(2) If the supply system low flow detector does not operate, then do these steps:

(a) Put the SUPPLY switch to the NORMAL position on the Equipment Cooling Panel.

(b) Make sure the normal equipment cooling supply fan operates.

1) Put your hand on the fan to feel for vibration.

NOTE: If you feel vibration, the fan is on.

(c) Put the SUPPLY switch to the ALTERNATE position.

(d) Make sure the alternate equipment cooling supply fan operates.

1) Put your hand on the fan to feel for vibration.

NOTE: If you feel vibration, the fan is on.

SUBTASK 21-00-00-710-003

(3) If the exhaust system low flow detector does not operate, then do these steps:

(a) Put the EXHAUST switch to the NORMAL position on the Equipment Cooling Panel.

(b) Remove the equipment cooling cover panel [35].

1) Remove the nine screws [36].

2) Remove the equipment cooling cover panel [35].

(c) Make sure that the normal equipment cooling exhaust fan operates.

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1) Put your hand on the fan to feel for vibration or feel for air exiting the overboard exhaust vent at the bottom of the airplane near the centerline at STA 370.

NOTE: If you feel vibration from the fan or air from the vent, the fan is on.

(d) Put the EXHAUST switch to the ALTERNATE position.

(e) Make sure that the alternate equipment cooling exhaust fan operates.

1) Put your hand on the fan to feel for vibration or feel for air exiting the overboard exhaust vent at the bottom of the airplane near the centerline at STA 370.

NOTE: If you feel vibration from the fan or air from the vent, the fan is on.

(f) Install the equipment cooling cover panel [35].

1) Put the nine screws [36] in place.

2) Install the equipment cooling cover panel [35].

SUBTASK 21-00-00-010-010

(4) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-00-00-930-001

(5) To make sure that the remaining detector system operates normally, do this task: DDG 21-27-01 Equipment Cooling Supply and Exhaust Fans Inoperative - Preparation, TASK 21-00-00-040-846.

SUBTASK 21-00-00-860-038

(6) Put the SUPPLY and the EXHAUST switches on the Equipment Cooling Panel to their usual positions.

SUBTASK 21-00-00-040-003

(7) For the ground crew call horn sounding on the ground with a low flow detector system inoperative (high alarm condition), open the applicable circuit breaker:

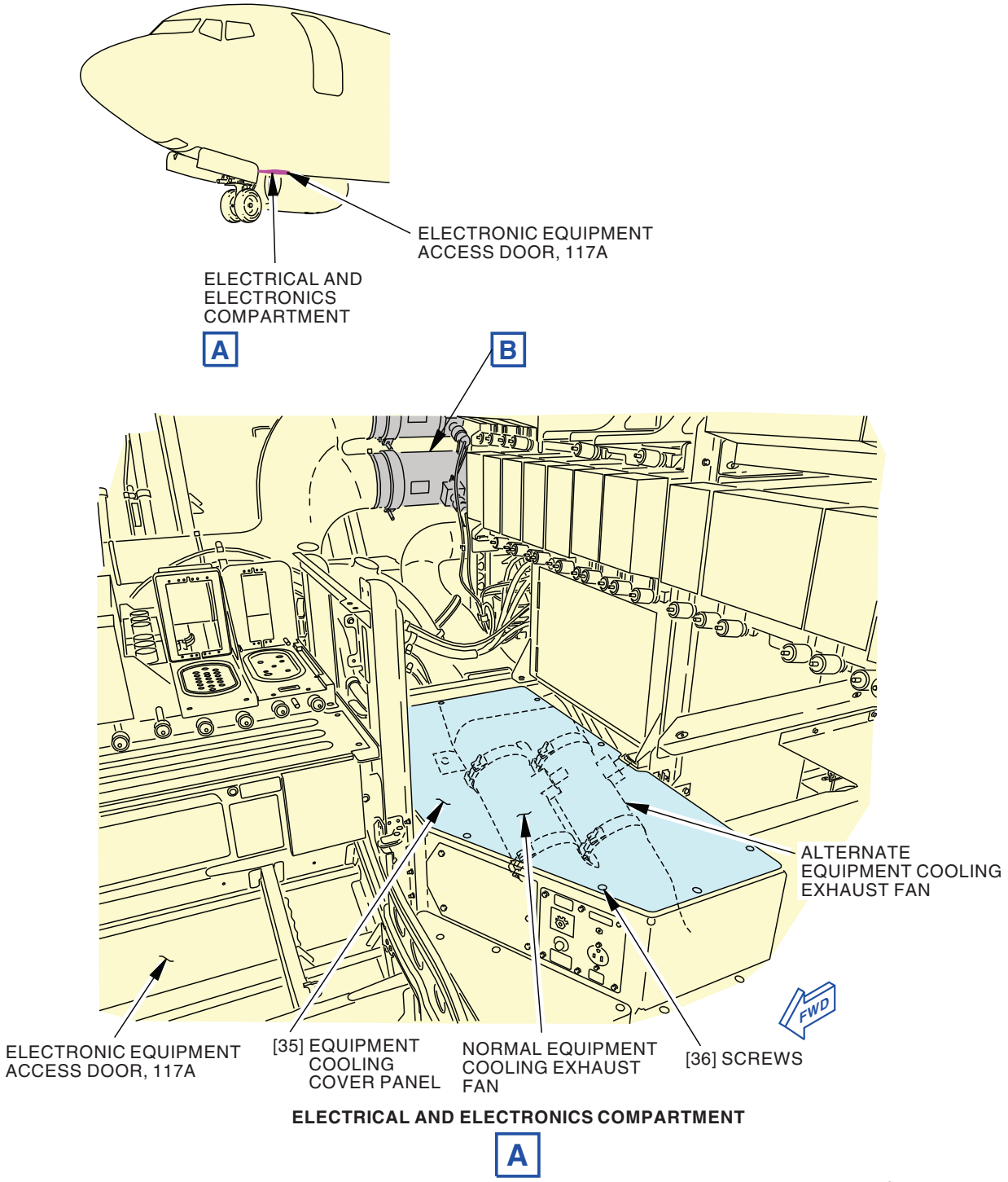
(a) Open these circuit breakers and install safety locks:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

——— END OF TASK ———

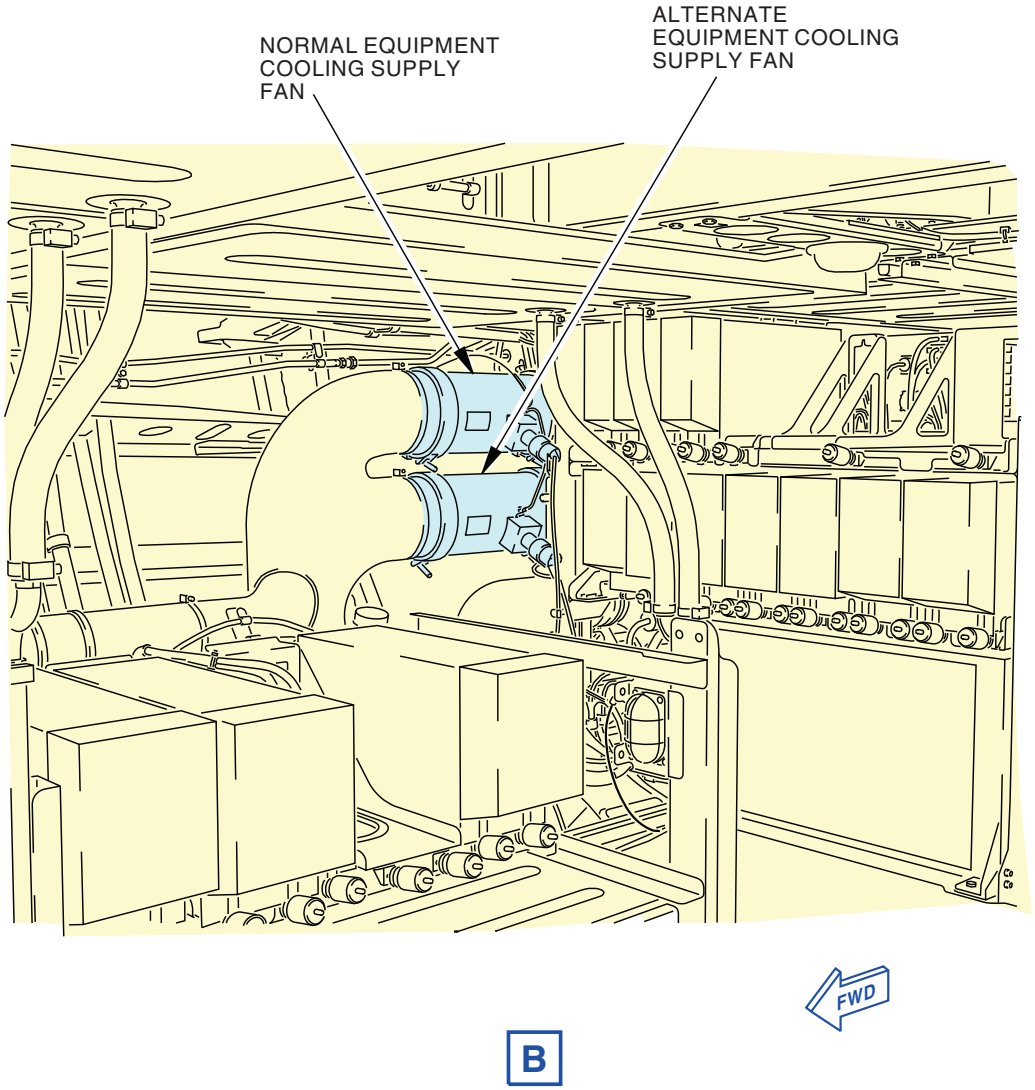
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Equipment Cooling Low Flow Detector System Deactivation
Figure 902/21-00-00-990-810 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Equipment Cooling Low Flow Detector System Deactivation
Figure 902/21-00-00-990-810 (Sheet 2 of 2)

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TASK 21-00-00-440-835

12. DDG 21-27-03 Equipment Cooling Low Flow Detector System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with an equipment cooling low flow detector system inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

C. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-860-039

(1) Close the applicable circuit breaker

(a) Remove the safety locks and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

——— END OF TASK ———

TASK 21-00-00-040-850

13. DDG 21-27-04 Equipment Cooling Air Filter Removed - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the equipment cooling air filter removed.

B. References

Reference	Title
21-27-01-000-801	Equipment Cooling Air Filter Removal (P/B 401)
50-11-01-400-801	Forward Cargo Compartment Forward Bulkhead Liner Installation (P/B 401)

C. Location Zones

Zone	Area
122	Forward Cargo Compartment - Right

D. Equipment Cooling Air Filter Removal

SUBTASK 21-00-00-020-004

(1) Do this task: Equipment Cooling Air Filter Removal, TASK 21-27-01-000-801.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-410-002

(1) Install the forward right bulkhead liner in the forward cargo compartment (TASK 50-11-01-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-836

14. DDG 21-27-04 Equipment Cooling Air Filter Removed - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the equipment cooling air filter removed.

B. References

Reference	Title
21-27-01-400-801	Equipment Cooling Air Filter Installation (P/B 401)
24-22-00-860-802	Remove Electrical Power (P/B 201)
50-11-01-000-801	Forward Cargo Compartment Forward Bulkhead Liner Removal (P/B 401)

C. Location Zones

Zone	Area
122	Forward Cargo Compartment - Right

D. Prepare for the Installation

SUBTASK 21-00-00-860-040

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802

SUBTASK 21-00-00-410-003

(2) Remove the forward right bulkhead liner in the forward cargo compartment (TASK 50-11-01-000-801).

E. Equipment Cooling Air Filter - Installation

SUBTASK 21-00-00-020-005

(1) Do this task: Equipment Cooling Air Filter Installation, TASK 21-27-01-400-801.

——— END OF TASK ———

TASK 21-00-00-040-870

15. DDG 21-27-05 Enhanced E6 Rack Equipment Cooling Exhaust Fan Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the enhanced E6 rack equipment cooling exhaust fan inoperative.

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
212	Flight Compartment - Right

C. Enhanced E6 Rack Equipment Cooling Exhaust Fan Deactivation

SUBTASK 21-00-00-860-132

(1) Open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	8	C04069	EQUIPMENT COOLING E6 RACK FAN POWER
D	10	C04070	EQUIPMENT COOLING E6 RACK FAN CONTROL

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SUBTASK 21-00-00-020-022

(2) Do one of these steps:

(a) Remove all actively and passively cooled equipment found on the E6-1 and E6-3 rack.

(b) Open and install safety lock on the applicable circuit breakers for all actively and passively cooled equipment found on the E6-1 and E6-3 rack.

——— END OF TASK ———

TASK 21-00-00-440-856

16. DDG 21-27-05 Enhanced E6 Rack Equipment Cooling Exhaust Fan Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the enhanced E6 rack equipment cooling exhaust fan inoperative.

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
212	Flight Compartment - Right

C. Enhanced E6 Rack Equipment Cooling Exhaust Fan Activation

SUBTASK 21-00-00-860-133

(1) Remove the safety locks and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	8	C04069	EQUIPMENT COOLING E6 RACK FAN POWER
D	10	C04070	EQUIPMENT COOLING E6 RACK FAN CONTROL

SUBTASK 21-00-00-420-022

(2) Do the applicable step:

(a) Install all actively and passively cooled equipment found on the E6-1 and E6-3 rack.

(b) Remove the safety lock and close the applicable circuit breakers for all actively and passively cooled equipment found on the E6-1 and E6-3 rack.

——— END OF TASK ———

TASK 21-00-00-040-851

17. DDG 21-31-01-01A Cabin Pressure Control System - One Automatic Mode Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with one automatic mode of the cabin pressure control system inoperative.

B. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. One Automatic Cabin Pressure Controller Inoperative - Deactivation Procedure

SUBTASK 21-00-00-860-041

(1) When the manual mode operates and one of the two automatic modes does not operate, then deactivate the controller that is Inoperative.

(2) Open the applicable circuit breaker to deactivate the inoperative Cabin Pressure Controller:

(a) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

(b) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

——— END OF TASK ———

TASK 21-00-00-440-837

18. DDG 21-31-01-01A Cabin Pressure Control System - One Automatic Mode Inoperative - Restoration

A. General

(1) This task puts the airplane to its usual condition after operation with one automatic mode of the cabin pressure control system inoperative.

B. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

C. Cabin Pressure Controller Reactivation.

SUBTASK 21-00-00-960-004

(1) Close the applicable circuit breaker:

(a) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

(b) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-040-852

19. DDG 21-31-01-01B Cabin Pressure Control System - Both Automatic Modes Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with two modes of the cabin pressure control system inoperative.

B. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

C. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-860-042

(1) Do these steps to prepare for unpressurized flight:

(a) Put the pressurization mode selector on the P5-6 pressurization control panel in the MAN position.

(b) Hold the outflow valve toggle switch in the OPEN position until the valve position indicator indicates FULL OPEN.

(c) Open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

——— END OF TASK ———

TASK 21-00-00-440-838

20. DDG 21-31-01-01B Cabin Pressure Control System - Both Automatic Modes Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with two modes of the cabin pressure control system inoperative.

B. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

C. Cabin Pressure Control System Reactivation

SUBTASK 21-00-00-860-043

(1) Remove the safety locks and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

SUBTASK 21-00-00-710-014

(2) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

——— END OF TASK ———

TASK 21-00-00-040-853

21. DDG 21-31-01-02 Cabin Pressure Control System - Manual Mode Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the manual modes of the cabin pressure control system inoperative.

B. References

Reference	Title
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)
71-00-00-910-801-G00	Prepare the Engine for Operation (P/B 201)
71-00-00-910-802-G00	Start the Engine (Selection) (P/B 201)
71-00-00-910-806-G00	Stop the Engine (Usual Engine Stop) (P/B 201)

C. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

D. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-860-044

(1) Do these steps to put the outflow valve in the FULL OPEN position:

(a) Put the airplane in a position where it is safe to do high power engine run and set the parking brake.

(b) Do this task: Prepare the Engine for Operation, TASK 71-00-00-910-801-G00

(c) To start the engines do this task: TASK 71-00-00-910-802-G00

(d) Let the engines become stable at the idle-power position for two minutes.

(e) Slowly move the forward thrust levers until N1 is greater than 60% and N2 is greater than 89%

(f) Put the pressurization mode selector on the P5-6 pressurization control panel in the MAN position.

(g) Hold the outflow valve toggle switch in the OPEN position until the valve position indicator indicates FULL OPEN.

(h) Open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

(i) Slowly move the forward thrust levers to the idle-power position.

(j) To stop the engines, do this task: TASK 71-00-00-910-806-G00.

SUBTASK 21-00-00-000-006

(2) If the outflow valve can not be put to the FULL OPEN position, then remove the outflow valve.

(a) Do this task: Aft Outflow Valve Assembly Removal, TASK 21-31-03-000-801.

(b) Cap and stow the electrical connectors.

SUBTASK 21-00-00-040-046

(3) Open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

——— END OF TASK ———

TASK 21-00-00-440-839

22. DDG 21-31-01-02 Cabin Pressure Control System - Manual Mode Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the manual mode of the cabin pressure control system inoperative.

B. References

Reference	Title
21-31-03-400-801	Aft Outflow Valve Assembly Installation (P/B 401)

C. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

D. Cabin Pressure Control System Reactivation

SUBTASK 21-00-00-440-018

(1) Remove the safety locks and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

SUBTASK 21-00-00-420-002

(2) If the outflow valve was removed, do these steps:

(a) Remove the caps from the electrical connectors.

(b) Do this task to install the outflow valve: TASK 21-31-03-400-801.

SUBTASK 21-00-00-860-119

(3) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

——— END OF TASK ———

TASK 21-00-00-040-854

23. DDG 21-31-02 Main Outflow Valve Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the main outflow valve inoperative.

B. References

Reference	Title
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)

C. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

D. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-860-046

(1) Do these steps to put the outflow valve in the FULL OPEN position:

(a) Put the pressurization mode selector, on the P5-6 pressurization control panel, in the MAN position.

(b) Hold the outflow valve toggle switch in the OPEN position until the valve position indicator indicates FULL OPEN.

(c) Open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

SUBTASK 21-00-00-000-001

(2) If the outflow valve can not be put to the FULL OPEN position, then remove the outflow valve.

(a) Do this task: Aft Outflow Valve Assembly Removal, TASK 21-31-03-000-801.

(b) Cap and stow the electrical connectors.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-040-004

(3) Open these circuit breakers and install safety locks:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

END OF TASK

TASK 21-00-00-440-840

24. DDG 21-31-02 Main Outflow Valve Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the main outflow valve inoperative.

B. References

Reference	Title
21-31-03-400-801	Aft Outflow Valve Assembly Installation (P/B 401)

C. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

D. Cabin Pressure Control System Reactivation

SUBTASK 21-00-00-440-002

(1) Remove the safety locks and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

SUBTASK 21-00-00-420-003

(2) If the outflow valve was removed, do these steps:

(a) Remove the caps from the electrical connectors.

(b) Do this task to install the outflow valve: TASK 21-31-03-400-801.

SUBTASK 21-00-00-710-015

(3) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-040-855

25. DDG 21-32-01A Positive Pressure Relief Valve Inoperative - Valve Closed - Preparation

(Figure 903)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with one of the Positive Pressure Relief Valves Inoperative.

B. References

Reference	Title
21-32-01-000-801	Positive Pressure Relief Valve Removal (P/B 401)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00095	Lubricant - Dry Film - MS122	

D. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left

E. Positive Pressure Relief Valve Deactivation

SUBTASK 21-00-00-010-011

(1) Remove the aft bulkhead liner in the aft cargo compartment, do this task: TASK 50-12-02-000-801.

SUBTASK 21-00-00-010-051

(2) Remove the E6 electrical equipment rack, aft access panel.

SUBTASK 21-00-00-020-006

(3) To remove the positive pressure relief valve that does not operate, do this task: TASK 21-32-01-000-801

(a) Keep the clamp [23] and gasket [21].

SUBTASK 21-00-00-350-001

(4) Make a plate from 0.1250 +0.0625 / -0.0000 in. (3.1750 +1.5875 / -0.0000 mm) 2024 T3 aluminum to seal the pedestal:

(a) Use the gasket [21] as a template to calculate the circumference of the plate [22].

(b) Drill three holes in the plate [22] with dimensions and locations the same as the gasket [21].

(c) Set the plate [22] on the pedestal, make sure that the fitment is correct.

SUBTASK 21-00-00-640-001

(5) Apply the MS122 lubricant, D00095, to both sides of the gasket [21].

SUBTASK 21-00-00-420-004

(6) Install the gasket [21], plate [22], and clamp [23] on the pedestal.

SUBTASK 21-00-00-020-007

(7) Put an airtight cap on the ambient sense tube.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-410-004

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

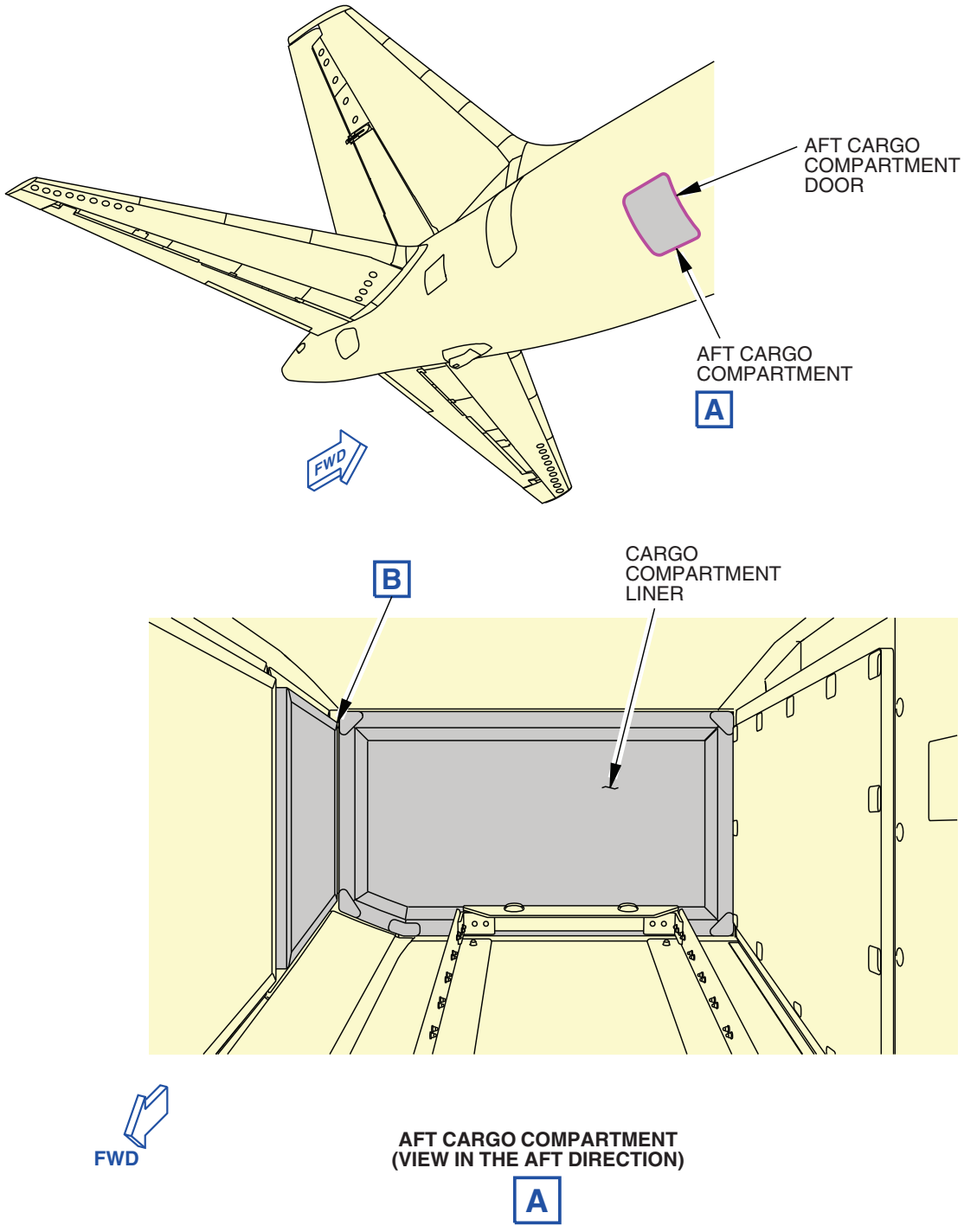
(8) Install the aft bulkhead liner of the aft cargo compartment, do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 21-00-00-410-034

(9) Install the electrical equipment rack, E6, aft access panel.

END OF TASK

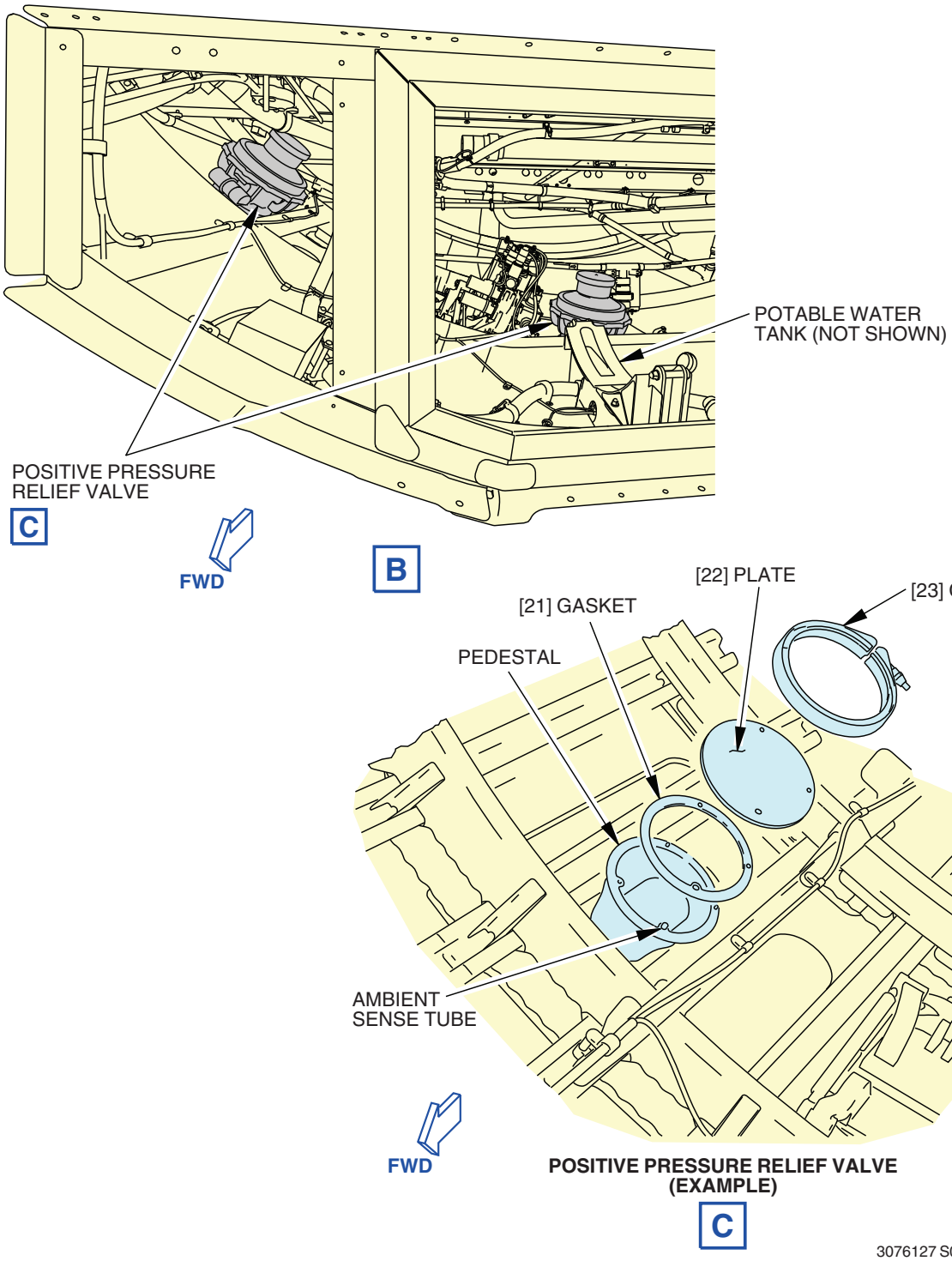
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Positive Pressure Relief Valve Deactivation
Figure 903/21-00-00-990-805 (Sheet 1 of 2)

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Positive Pressure Relief Valve Deactivation
Figure 903/21-00-00-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-841

26. DDG 21-32-01A Positive Pressure Relief Valve Inoperative - Valve Closed - Restoration

(Figure 903)

A. General

(1) This task puts the airplane to its usual condition after operation with one of the positive pressure relief valves inoperative.

B. References

Reference	Title
21-32-01-400-801	Positive Pressure Relief Valve Installation (P/B 401)
21-32-01-700-802	Positive Pressure Relief Valve - System Test with the Use of Boeing Test Equipment (P/B 501)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left

D. Positive Pressure Relief Valve Reactivation

SUBTASK 21-00-00-010-052

(1) Remove the E6 electrical equipment rack, aft access panel.

SUBTASK 21-00-00-010-012

(2) Remove the aft bulkhead liner in the aft cargo compartment, do this task: TASK 50-12-02-000-801.

SUBTASK 21-00-00-020-008

(3) Remove the clamp [23] and plate [22].

SUBTASK 21-00-00-020-009

(4) Remove the cap from the ambient sense tube.

SUBTASK 21-00-00-420-005

(5) Install the positive pressure relief valve, do this task: TASK 21-32-01-400-801.

SUBTASK 21-00-00-720-004

(6) Do this task: Positive Pressure Relief Valve - System Test with the Use of Boeing Test Equipment, TASK 21-32-01-700-802.

NOTE: If a new positive pressure relief valve was installed you do not need to do this test.

SUBTASK 21-00-00-010-013

(7) Install the aft bulkhead liner in the aft cargo compartment, do this task: TASK 50-12-02-400-801.

SUBTASK 21-00-00-410-038

(8) Install the electrical equipment rack, E6, aft access panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-040-856

27. DDG 21-32-01B Positive Pressure Relief Valve Inoperative - Valve Open - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the positive pressure relief valves inoperative.

B. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

C. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-040-047

(1) Do these steps to prepare for unpressurized flight:

(a) Put the pressurization mode selector on the P5-6 pressurization control panel in the MAN position.

(b) Hold the outflow valve toggle switch in the OPEN position until the valve position indicator indicates FULL OPEN.

(c) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

——— END OF TASK ———

TASK 21-00-00-440-842

28. DDG 21-32-01B Positive Pressure Relief Valve Inoperative - Valve Open - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the positive pressure relief valves inoperative.

B. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

C. Cabin Pressure Control System Reactivation

SUBTASK 21-00-00-860-048

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-00-00-710-016

(2) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-040-857

29. DDG 21-33-01 Cabin Rate of Climb Indicator Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the cabin rate of climb indicator inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right
212	Flight Compartment - Right

C. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-860-050

(1) Do these steps to prepare for unpressurized flight:

(a) Put the pressurization mode selector on the P5-6 pressurization control panel in the MAN position.

(b) Hold the outflow valve toggle switch in the OPEN position until the valve position indicator indicates FULL OPEN.

(c) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

——— END OF TASK ———

TASK 21-00-00-440-843

30. DDG 21-33-01 Cabin Rate of Climb Indicator Inoperative - Restoration

A. General

(1) This task puts the airplane to its usual condition after operation with the cabin rate of climb indicator inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right
212	Flight Compartment - Right

C. Cabin Rate of Climb Indicator Restoration.

SUBTASK 21-00-00-760-002

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-710-017

(2) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

——— END OF TASK ———

TASK 21-00-00-040-858

31. DDG 21-33-02–02 Cabin Differential Pressure Indication Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the cabin differential pressure indication inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right
212	Flight Compartment - Right

C. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-040-048

(1) Do these steps to prepare for unpressurized flight:

(a) Put the pressurization mode selector on the P5-6 pressurization control panel in the MAN position.

(b) Hold the outflow valve toggle switch in the OPEN position until the valve position indicator indicates FULL OPEN.

(c) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

——— END OF TASK ———

TASK 21-00-00-440-844

32. DDG 21-33-02-02 Cabin Differential Pressure Indication Inoperative - Restoration

A. General

(1) This task puts the airplane to its usual condition after operation with the cabin differential pressure indicator inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Cabin Differential Pressure Indicator Restoration.

SUBTASK 21-00-00-440-003

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-00-00-710-018

(2) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

——— END OF TASK ———

TASK 21-00-00-040-859

33. DDG 21-33-03–02 Cabin Altitude Indication Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the cabin altitude indication inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right
212	Flight Compartment - Right

C. Prepare for Unpressurized Flight.

SUBTASK 21-00-00-040-049

(1) Do these steps to prepare for unpressurized flight:

(a) Put the pressurization mode selector on the P5-6 pressurization control panel in the MAN position.

(b) Hold the outflow valve toggle switch in the OPEN position until the valve position indicator indicates FULL OPEN.

(c) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

——— END OF TASK ———

TASK 21-00-00-440-845

34. DDG 21-33-03–02 Cabin Altitude Indication Inoperative - Restoration

A. General

(1) This task puts the airplane to its usual condition after operation with the cabin altitude indication inoperative.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
122	Forward Cargo Compartment - Right
212	Flight Compartment - Right

C. Cabin Altitude Indication Restoration.

SUBTASK 21-00-00-860-054

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-00-00-710-019

(2) Put the pressurization mode selector on the P5-6 pressurization control panel in the AUTO position.

——— END OF TASK ———

TASK 21-00-00-040-860

35. DDG 21-33-05 Outflow Valve Position Indicator Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the outflow valve position indicator inoperative.

B. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left
212	Flight Compartment - Right

C. Preparation - Outflow Valve Operational Test

SUBTASK 21-00-00-860-057

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

SUBTASK 21-00-00-210-003

(2) Look at the aft outflow valve to make sure it moved to the fully open position.

D. Outflow Valve Operational Test

SUBTASK 21-00-00-710-005

(1) On the P5-6 Pressurization Control Panel, put the mode selector to MAN.

SUBTASK 21-00-00-710-006

(2) Make sure the MANUAL light goes on.

SUBTASK 21-00-00-710-007

(3) Hold the VALVE switch in the CLOSED position for 30 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-710-008

(4) Look at the aft outflow valve to make sure it moved to the fully closed position.

SUBTASK 21-00-00-710-009

(5) On the P5-6 Pressurization Control Panel, put the mode selector to AUTO.

SUBTASK 21-00-00-210-004

(6) Make sure the MANUAL light goes out.

SUBTASK 21-00-00-710-010

(7) Wait 30 seconds and then look at the aft outflow valve to make sure it moved to the fully open position.

——— END OF TASK ———

TASK 21-00-00-040-861

36. DDG 21-45-01 Door Area Heater System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with any of these door area heater systems inoperative:

(a) Forward Door Area Heater System

(b) Overwing Emergency Exit Door Heater Systems

(c) Aft Door Area Heater Systems

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
830	Subzone - Passenger Compartment Doors, Left
840	Subzone - Passenger Compartment Doors, Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Door Area Heater System Deactivation

SUBTASK 21-00-00-010-047

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-00-00-865-001

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) To deactivate the forward door area or the overwing emergency exit door heater systems:

Open this circuit breaker and install safety lock:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

NOTE: Opening the circuit breaker deactivates both the forward entry door area heater and all overwing exit door heaters.

SUBTASK 21-00-00-865-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) To deactivate the aft door area heater system:

Open this circuit breaker and install safety lock:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	7	C01279	DOOR AREA HTR-AFT

SUBTASK 21-00-00-410-032

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-847

37. DDG 21-45-01 Door Area Heater System Inoperative - Restoration

A. General

(1) This task puts back the airplane to its usual condition after operation with any of these door area heater systems inoperative.

(a) Forward Door Area Heater System

(b) Overwing Emergency Exit Door Heater Systems

(c) Aft Door Area Heater Systems

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
830	Subzone - Passenger Compartment Doors, Left
840	Subzone - Passenger Compartment Doors, Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Door Area Heater System Activation

SUBTASK 21-00-00-010-048

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-00-00-440-010

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) To activate the forward door area or the overwing emergency exit door heater systems:

Remove the safety lock and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-440-008

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) To activate the aft door area heater system:

Remove the safety lock and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	7	C01279	DOOR AREA HTR-AFT

SUBTASK 21-00-00-410-031

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

TASK 21-00-00-040-862

38. DDG 21-51-01 Air Conditioning Packs Inoperative - Preparation

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with both air conditioning packs inoperative.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for Deactivation

SUBTASK 21-00-00-630-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-860-137

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-860-138

(3) To get access to the flow control and shutoff valve for the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-860-139

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the flow control and shutoff valve for the right cooling pack do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Both Air Conditioning Packs Inoperative Deactivation

SUBTASK 21-00-00-040-006

(1) Manually position the outflow valve in the FULL OPEN position.

(a) Put the Pressurization Mode selector to MAN.

(b) Hold the valve toggle switch in the OPEN position until valve position indicator indicates FULL OPEN.

SUBTASK 21-00-00-040-050

(2) Deactivate (locked closed) the flow control and shutoff valve for the associated pack.

(a) For the left pack flow control and shutoff valve disconnect and stow the D15830 electrical connector.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) For the right pack flow control and shutoff valve disconnect and stow the D15834 electrical connector.

(c) Do these steps to lock the applicable pack flow control and shutoff valve in the closed position:

1) Push in the manual override.

2) If it is already in, pull it out and push it back in to verify that the valve is in the closed position.

(d) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(e) Put one of the L or R PACK switches in the HIGH position and attach a DO NOT OPERATE tag, STD-858.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-410-039

(1) To close the left ECS access doors, do this step:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-040

(2) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

TASK 21-00-00-440-848

39. DDG 21-51-01 Air Conditioning Packs Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with both air conditioning packs inoperative.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for Restoration

SUBTASK 21-00-00-860-140

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-860-141

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-010-053

(3) To get access to the flow control and shutoff valve for the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-010-054

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the flow control and shutoff valve for the right cooling pack do these steps: (TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Both Air Conditioning Packs Inoperative Restoration

SUBTASK 21-00-00-020-023

(1) For the left pack flow control and shutoff valve connect the D15830 electrical connector.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-020-024

(2) For the right pack flow control and shutoff valve connect the D15834 electrical connector.

SUBTASK 21-00-00-860-142

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-860-143

(4) Put the pressurization mode selector to the AUTO position.

SUBTASK 21-00-00-860-144

(5) Put the PACK switches to the AUTO position.

(a) Remove the DO NOT OPERATE tag.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-410-041

(1) To close the left ECS access doors, do this step:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-042

(2) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

TASK 21-00-00-040-863

40. DDG 21-51-02 Pack Flow Control and Shutoff Valves Inoperative - Preparation

(Figure 904)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with a pack flow control and shutoff valve inoperative.

NOTE: When the pack flow control and shutoff valve is locked closed you will not be able to operate the related pack.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-15135	Cable - Jumper (Make from 0.5 inch length 20 AWG wire, crimp or solder both end to pin contact size 16 P/N M39029/31-228)

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for Deactivation

SUBTASK 21-00-00-860-059

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-860-060

(2) Do this step to deactivate the pack flow control and shutoff valve for the left cooling pack:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-860-061

(3) Do this step to deactivate the pack flow control and shutoff valve for the right cooling pack:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-00-00-010-014

(4) To get access to the flow control and shutoff valve for the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-010-015

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) To get access to the flow control and shutoff valve for the right cooling pack, do these steps:
(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Flow Control and Shutoff Valve Deactivation (Locked Closed)

SUBTASK 21-00-00-020-010

(1) For the left pack flow control and shutoff valve disconnect and stow the D15830 electrical connector.

SUBTASK 21-00-00-020-011

(2) For the right pack flow control and shutoff valve disconnect and stow the D15834 electrical connector.

SUBTASK 21-00-00-020-012

(3) Disconnect and stow both D15830 and D15834 electrical connectors to deactivate the left and right flow control and shutoff valves. This will prevent the operation of both packs and prevent the airplane from being pressurized.

SUBTASK 21-00-00-040-007

(4) If the left pack airflow shut-off valve limit switch has failed in the open position, do these steps at connector D15832 at the left pack flow control and shutoff valve:

(a) Disconnect connector D15832 from the left pack flow control and shutoff valve.

(b) Install a jumper cable, STD-15135, between pins 1 and 3 of connector D15832.

(c) Make sure there is continuity to airplane ground at pin 3 of connector D15832.

(d) Cap and stow the electrical connector D15832 with the jumper cable, STD-15135, installed between pins 1 and 3.

SUBTASK 21-00-00-040-008

(5) If the right pack airflow shut-off valve limit switch has failed in the open position, do these steps at connector D15836 at the right pack flow control and shutoff valve:

(a) Disconnect connector D15836 from the right flow control and shutoff valve.

(b) Install a jumper cable, STD-15135, between pins 1 and 3 of connector D15836.

(c) Make sure there is continuity to airplane ground at pin 3 of connector D15836.

(d) Cap and stow the electrical connector D15836 with the jumper cable, STD-15135, installed between pins 1 and 3.

SUBTASK 21-00-00-040-009

(6) Do these steps to lock the applicable pack flow control and shutoff valve in the closed position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Push in the manual override. If it is already in, pull it out and push it back in to verify that the valve is in the closed position.

SUBTASK 21-00-00-860-062

(7) If you deactivated the flow control and shutoff valve for the left cooling pack, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-860-063

(8) If you deactivated the flow control and shutoff valve for the right cooling pack, do this step:

Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-00-00-040-010

(9) If both pack airflow shutoff valves are inoperative, do these steps:

(a) Manually position the outflow valve in the FULL OPEN position:

1) Put the Pressurization Mode selector to MAN.

2) Hold the valve toggle switch in the OPEN position until valve position indicator indicates FULL OPEN.

(b) Do this step to put the recirculation fan in the INOP ON condition:

1) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(c) Put one of the L or R PACK switches in the HIGH position and attach a DO NOT OPERATE tag, STD-858.

(d) If installed, shut off power to the air/ground public telephone system.

SUBTASK 21-00-00-410-005

(10) To close the left ECS access doors, do this step:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-006

(11) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

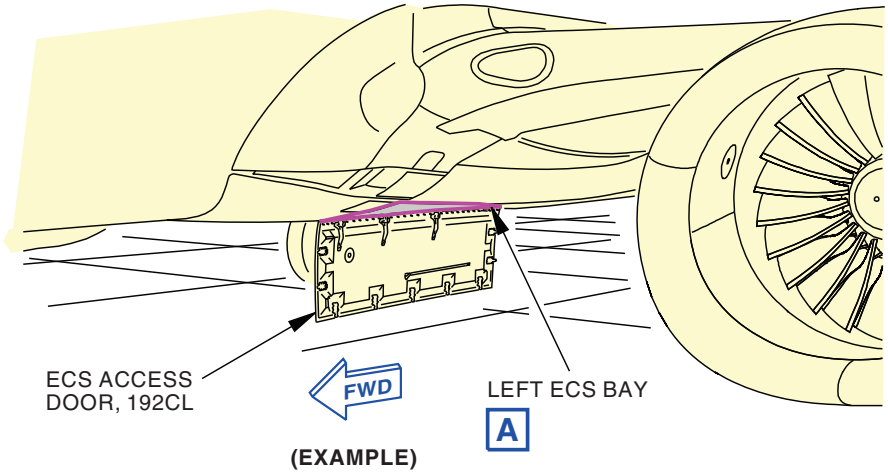
NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

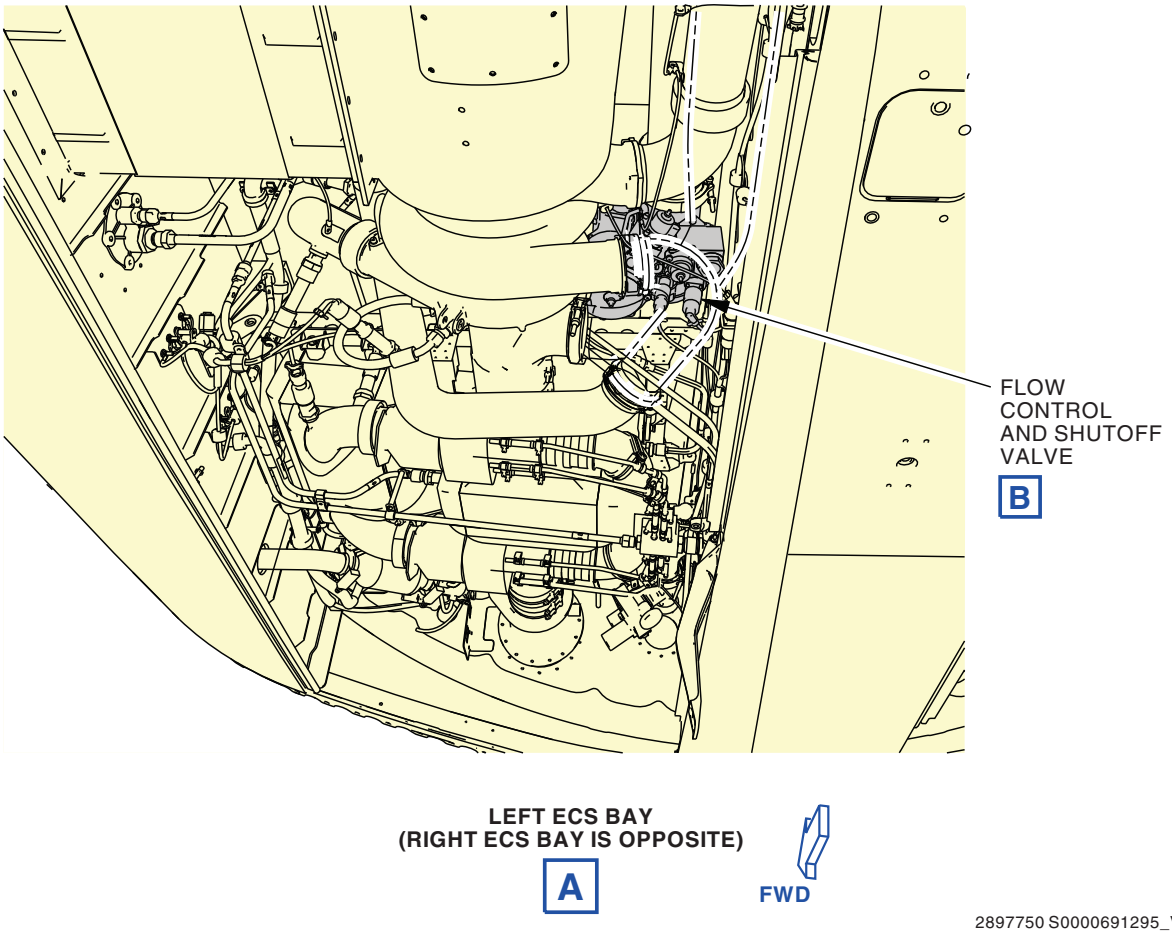
SUBTASK 21-00-00-810-006

(12) If necessary, extinguish the MAINT light on the P5-1 panel. Do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

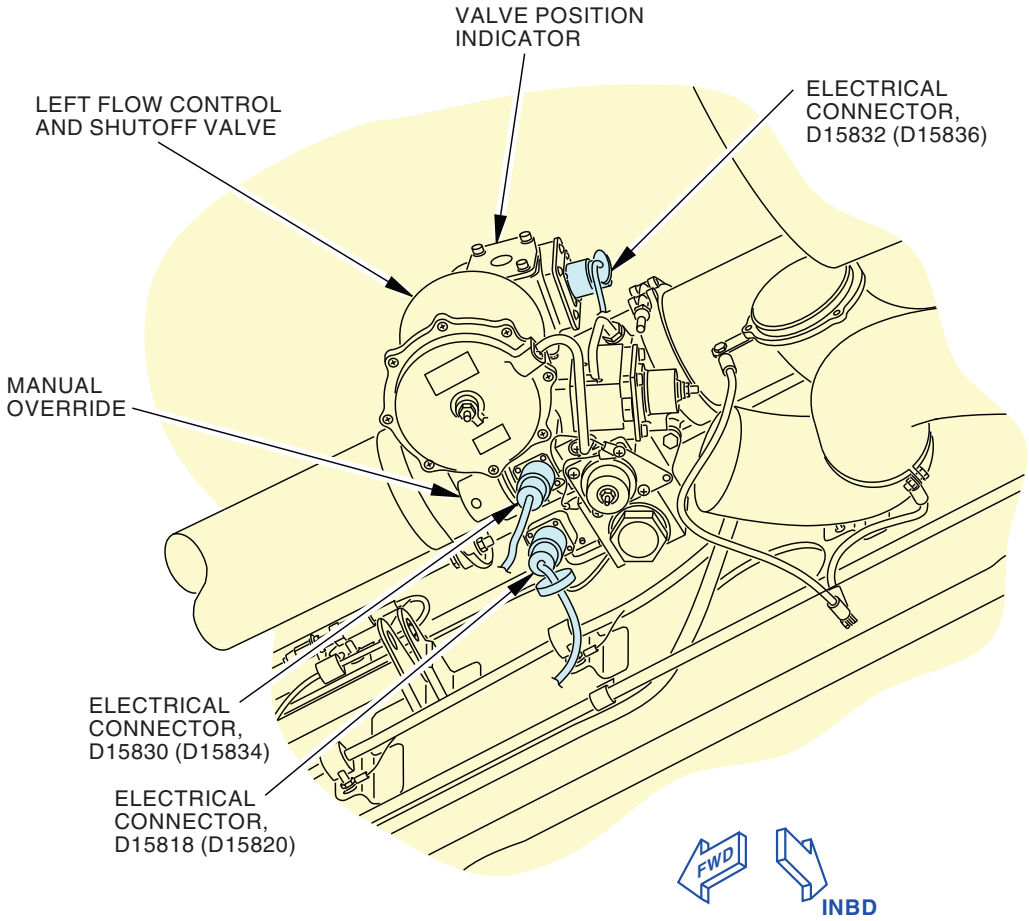




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Flow Control and Shutoff Valve Deactivation
Figure 904/21-00-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT ECS BAY
(RIGHT SIDE IS EQUIVALENT)

B

2411233 S00061527324_V2

Flow Control and Shutoff Valve Deactivation
Figure 904/21-00-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-849

41. DDG 21-51-02 Pack Flow Control and Shutoff Valves Inoperative - Restoration

(Figure 904)

A. General

(1) This task puts the airplane back to its usual condition after operation with a pack flow control and shutoff valves inoperative.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-15135	Cable - Jumper (Make from 0.5 inch length 20 AWG wire, crimp or solder both end to pin contact size 16 P/N M39029/31-228)

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for Reactivation

SUBTASK 21-00-00-860-064

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-860-065

(2) If you deactivated the pack flow control and shutoff valve for the left cooling pack, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-860-066

(3) If you deactivated the pack flow control and shutoff valve for the right cooling pack, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-010-016

(4) To get access to the flow control and shutoff valve for the left cooling pack, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-010-017

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) To get access to the flow control and shutoff valve for the right cooling pack, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Flow Control Valve and Shutoff Valve Reactivation

SUBTASK 21-00-00-440-011

(1) For the left pack flow control and shutoff valve connect the D15830 electrical connector.

SUBTASK 21-00-00-440-012

(2) For the right pack flow control and shutoff valve connect the D15834 electrical connector.

SUBTASK 21-00-00-440-013

(3) Connect both D15830 and D15834 electrical connectors to activate the left and right flow control and shutoff valves if both pack flow control and shutoff valves were inoperative.

SUBTASK 21-00-00-440-005

(4) Do these steps at connector D15832 at the left flow control and shutoff valve if capped and stowed:

(a) Remove the cap from connector D15832.

(b) If installed, remove the jumper cable, STD-15135, between pins 1 and 3 of connector D15832.

SUBTASK 21-00-00-440-006

(5) Do these steps at connector D15836 at the right flow control and shutoff valve if capped and stowed:

(a) Remove the cap from connector D15836.

(b) If installed, remove the jumper cable, STD-15135, between pins 1 and 3 of connector D15836.

SUBTASK 21-00-00-860-067

(6) If you put the flow control and shutoff valve for the left cooling pack in its usual condition, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-860-068

(7) If you put the flow control and shutoff valve for the right cooling pack in its usual condition, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-00-00-860-069

(8) If both pack flow control and shutoff valves where inoperative:

(a) Do this step to put the recirculation fan in the usual condition:

1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(b) Put the pressurization mode selector to the AUTO position.

(c) Put the PACK switches in the AUTO position and remove the DO NOT OPERATE tag, STD-858.

(d) If installed, supply power to the air/ground public telephone system.

SUBTASK 21-00-00-410-027

(9) To close the left ECS access doors, do this step:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-028

(10) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. MAINT Light Status Message

SUBTASK 21-00-00-750-001

(1) Uncheck the applicable MAINT light Status Message(s) on the MAINT LIGHT menu (TASK 31-65-00-750-801).

——— END OF TASK ———

TASK 21-00-00-040-871

42. DDG 21-51-04 Pack Flow Sensor Inoperative - Preparation

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the pack flow sensors inoperative.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
31-65-00-860-802	Scheduled Maintenance Task (SMT) Messages Check (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Dispatch the Airplane with the Pack Flow Sensor Inoperative

SUBTASK 21-00-00-200-004

(1) Make sure that scheduled maintenance is not necessary for the applicable pack inlet pressure sensors (Scheduled Maintenance Task (SMT) Messages Check, TASK 31-65-00-860-802):

(a) For the left pack flow sensor:

1) Make sure that PACK CONTROL CHANNEL L scheduled maintenance task does not show.

(b) For the right pack flow sensor:

1) Make sure that PACK CONTROL CHANNEL R scheduled maintenance task does not show.

SUBTASK 21-00-00-420-025

(2) Disconnect, cap, and stow the electrical connector from the applicable pack flow sensor.

SUBTASK 21-00-00-860-135

(3) If it is necessary, extinguish the MAINT light (TASK 31-65-00-750-801).

——— END OF TASK ———

TASK 21-00-00-040-869

43. DDG 21-51-05 Pack Temperature Control Valves Inoperative - Preparation

(Figure 905)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the pack temperature control valves inoperative in closed position provided that the associated standby pack temperature control valve is verified to operate normally.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for Deactivation

SUBTASK 21-00-00-860-070

(1) Make sure that the pneumatic pressure is removed from the pneumatic system (TASK 36-00-00-860-806).

SUBTASK 21-00-00-860-071

(2) Make sure that the L PACK and R PACK switches, on the P5-10 air conditioning panel, are on the OFF position.

SUBTASK 21-00-00-860-128

(3) To deactivate the left temperature control valve, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-00-00-860-129

(4) To deactivate the right temperature control valve, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-00-00-010-018

(5) To get access to the left temperature control valve, do this step:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-010-019

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(6) To get access to the right temperature control valve, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-00-00-040-013

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(7) Deactivate the applicable temperature control valve in closed position:

(a) Disconnect and stow the electrical connector from temperature control valve.

(b) Turn the manual override on the valve until the visual position indicator shows that the valve is fully closed.

SUBTASK 21-00-00-860-131

(8) If the left pack Temperature Control Valve was deactivated, do this step:

(a) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-00-00-860-130

(9) If the right pack Temperature Control Valve was deactivated, do this step:

(a) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-00-00-410-009

(10) To close the left ECS access doors, do this step:

(TASK 06-41-00-800-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-010

(11) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-00-00-750-004

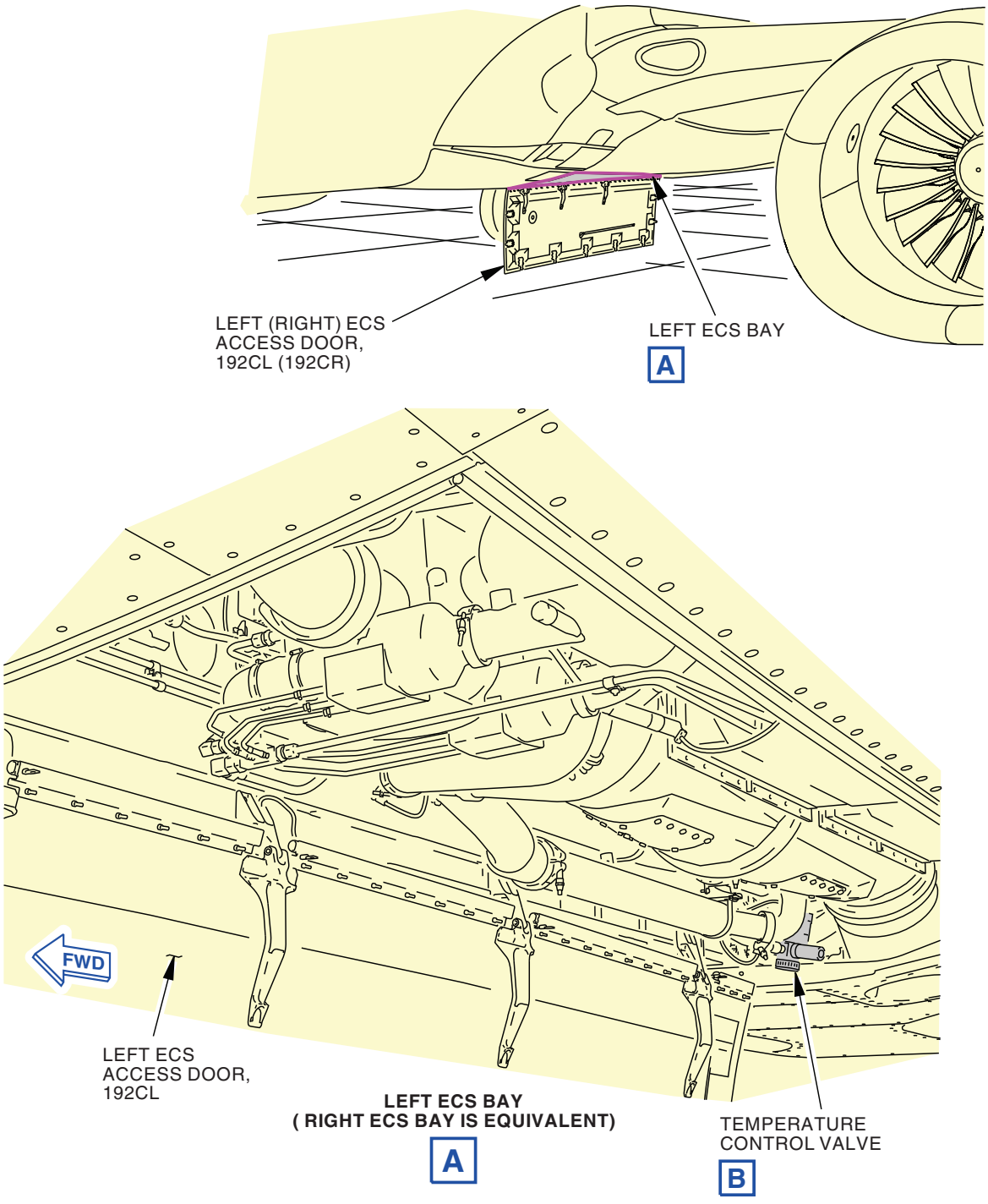
(12) Make sure that the MAINT light does not turn on for the associated active status message (TASK 31-65-00-750-801).

SUBTASK 21-00-00-750-005

(13) If it is necessary, extinguish the MAINT light, on the P5-1 panel, do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

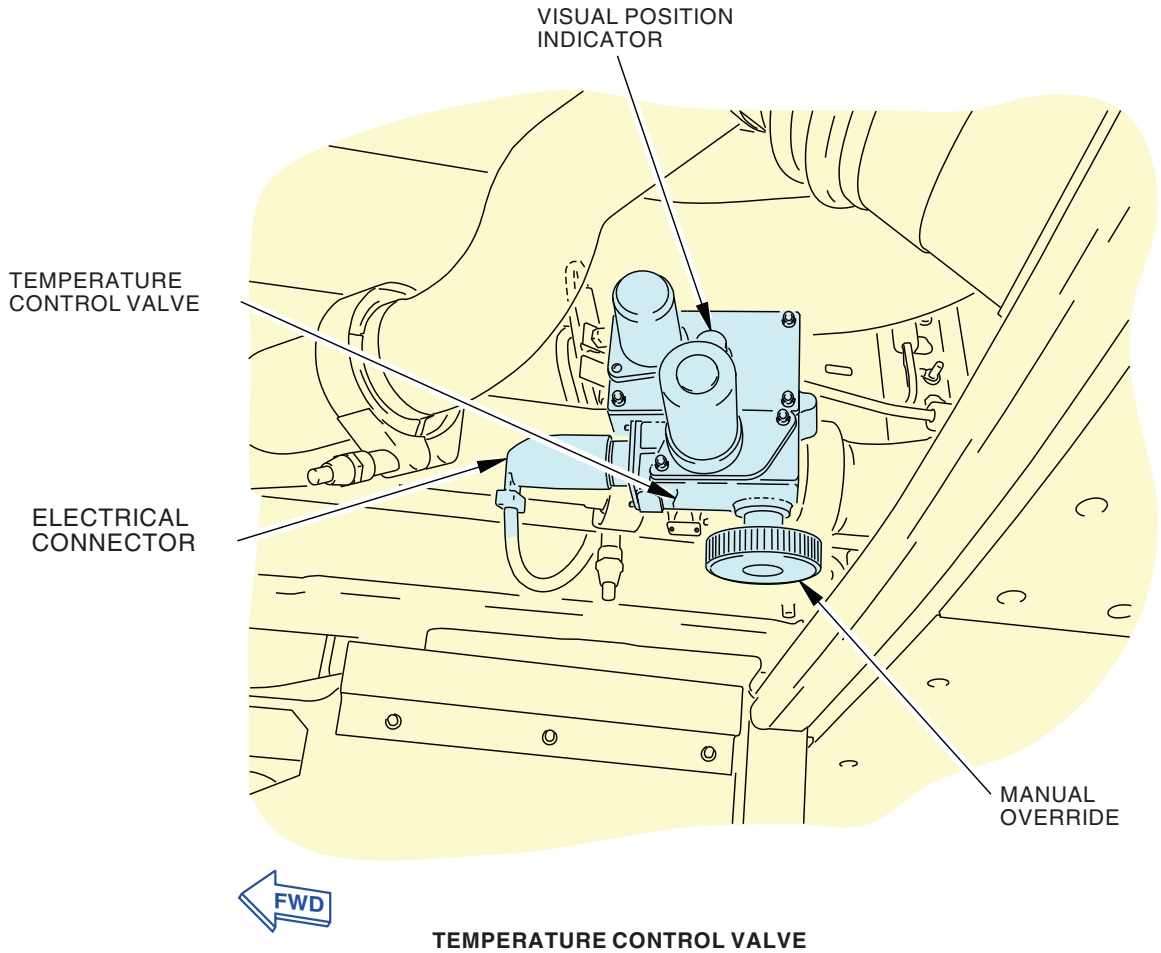
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Temperature Control Valve
Figure 905/21-00-00-990-811 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2482219 S0000583632_V1

Temperature Control Valve
Figure 905/21-00-00-990-811 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-855

44. DDG 21-51-05 Pack Temperature Control Valves Inoperative - Restoration

(Figure 905)

A. General

(1) This task puts the airplane to its usual condition after operation with the pack temperature control valves inoperative.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for Reactivation

SUBTASK 21-00-00-860-120

(1) Make sure that the pneumatic pressure is removed from the pneumatic system (TASK 36-00-00-860-806).

SUBTASK 21-00-00-860-121

(2) Make sure that the L PACK and R PACK switches, on the P5-10 air conditioning panel, are on the OFF position.

SUBTASK 21-00-00-860-126

(3) To restore the left temperature control valve, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-00-00-860-127

(4) To restore the right temperature control valve, do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-010-045

(5) To get access to the left deactivated temperature control valve, do this step:

(a) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-010-046

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(6) To get access to the right deactivated temperature control valve, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-00-00-440-019

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(7) Restore the applicable temperature control valve as follows:

(a) Turn the manual override on the valve until the visual position indicator shows that the valve is between the OPEN and CLOSED positions.

(b) Connect the electrical connector to the temperature control valve.

SUBTASK 21-00-00-860-122

(8) If the left pack Temperature Control Valve was restored, do this step:

(a) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-00-00-860-123

(9) If the right pack Temperature Control Valve was restored, do this step:

(a) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-00-00-410-029

(10) To close the left ECS access doors, do this step:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-030

(11) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

TASK 21-00-00-040-864

45. DDG 21-51-09 Pack Ram Air Systems Inoperative - Preparation

(Figure 906)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the pack ram air systems inoperative

NOTE: If the ram air inlet panels are put into an inoperative flight open condition, the ram air modulated exhaust duct louvers must also be in an open condition. Conversely, the ram air inlet panels must be into an inoperative flight open condition if the fault condition is with the ram air modulated exhaust system.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-27-000-801	Ram Air Exhaust Actuator - Removal (P/B 401)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14767	Link - Stowage SRADA Part #: C21009-1 Supplier: 81205

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

F. Prepare for Deactivation

SUBTASK 21-00-00-020-013

(1) When the ram air inlet door and ram air exhaust duct louvers for the left cooling pack will be deactivated, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT

SUBTASK 21-00-00-020-014

(2) When the ram air inlet door and ram air exhaust duct louvers for the right cooling pack will be deactivated, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT

SUBTASK 21-00-00-010-020

(3) To get access to the ram air inlet door actuator for the left cooling pack, do this step:

Open this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-010-021

(4) To get access to the ram air inlet door actuator for the right cooling pack, do this step:

Open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

SUBTASK 21-00-00-010-022

(5) To get access to the ram air exhaust door actuator for the left cooling pack, do this step:

(a) Open these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

SUBTASK 21-00-00-010-023

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(6) To get access to the ram air exhaust door actuator for the right cooling pack, do this step:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

(b) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(c) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Ram Air Inlet Modulating System Deactivation (Secured Open)

Figure 906

SUBTASK 21-00-00-020-015

(1) Disconnect and stow the electrical connectors [1].

SUBTASK 21-00-00-020-016

(2) Remove the bolt [2], washer [3], washer [7], bushing [8], and nut [6] that connects the actuator rod [9] end to the bellcrank [4].

SUBTASK 21-00-00-420-006

(3) Remove the actuator rod [9] end from the bellcrank [4] and reinstall the bolt [2], washer [3], washer [7], bushing [8], and nut [6] back to the bellcrank [4].

SUBTASK 21-00-00-980-001

(4) Do these steps to secure the door in the flight open position:

(a) Turn the bellcrank [4] to align the rigging pin holes on the shaft assembly with the holes on the bearing housing support [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Install the bolt [10] through the rigging pin holes.

1) Use a BACB30UU3P36D bolt or an equivalent bolt with a 0.1875 in. (4.7625 mm) shank diameter and 2.2 in. (55.9 mm) shank length.

2) If necessary, drill the shank end of bolt [10] for a cotter pin [12].

(c) Install the nut [11] and cotter pin [12] on the bolt [10].

1) Use a BACN10JD3 nut or an equivalent nut.

2) Use a BACP18BC01D06P cotter pin or an equivalent cotter pin with a 0.047 in. (1.194 mm) diameter and a 0.75 in. (19.05 mm) length.

SUBTASK 21-00-00-420-007

(5) Use lockwire [13] to connect the actuator rod [9] end to the bolt [2].

SUBTASK 21-00-00-860-072

(6) If you deactivated the ram air inlet door for the left cooling pack, close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

SUBTASK 21-00-00-860-073

(7) If you deactivated the ram air inlet door for the right cooling pack, close this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

H. Ram Air Exit Modulating System Deactivation (Secured Open)

Figure 907

SUBTASK 21-00-00-000-002

(1) Disconnect the electrical connectors [26] for the ram air exhaust actuator [25].

(2) If necessary, remove the ram air exhaust actuator [25]. Do this task: Ram Air Exhaust Actuator - Removal, TASK 21-51-27-000-801.

NOTE: If you will keep the ram air exhaust actuator [25] installed you must install the Link - Stowage SRADA, SPL-14767 tool.

SUBTASK 21-00-00-040-015

(3) Cap and stow the electrical connectors [26] for the ram air exhaust actuator [25].

SUBTASK 21-00-00-000-003

(4) Remove the nut [20], washer [19], washer [17], and bolt [16] that attach the most forward louver drive arm [14] to the link assembly [15].

SUBTASK 21-00-00-820-001

(5) Move the link assembly [15] to open the exit louvers until the most forward hole in the link assembly [15] lines up with the hole in the most forward louver drive arm [14].

SUBTASK 21-00-00-420-008

(6) In the same order that they were removed, install the bolt [16], washer [17], washer [19], and nut [20] through the most forward hole in the link assembly [15] and the most forward hole in the louver drive arm [14].

NOTE: The louvers should be open approximately 56°arc (1 rad) relative to the airplane skin.

SUBTASK 21-00-00-420-020

(7) If the ram air exhaust actuator [25] will stay installed, then install the Link - Stowage SRADA, SPL-14767 tool:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Attach the SRADA stowage link [31] to the actuator rod [24] end using the bolt [29], countersunk washer [30], washer [33], nut [32] that are included in Link - Stowage SRADA, SPL-14767 tool kit.

(b) Attach the other end of the SRADA stowage link [31] to the actuator drive arm [18] using the nut, washer, washer, and bolt, which had been used to attach the actuator rod [24] end to the actuator drive arm [18].

SUBTASK 21-00-00-410-011

(8) If you deactivated the ram air exhaust louvers for the left cooling pack, do this step:

(a) Close these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

SUBTASK 21-00-00-410-036

(9) If you deactivated the ram air exhaust louvers for the right cooling pack, do this step:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(c) Close this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

SUBTASK 21-00-00-420-009

(10) If you deactivated the ram air exhaust louvers for the left cooling pack, do this step:

(a) Remove the safety tags and close these circuit breakers:

Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-420-010

(11) If you deactivated the ram air exhaust louvers for the right cooling pack, do this step:

(a) Remove the safety tags and close these circuit breakers:

Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

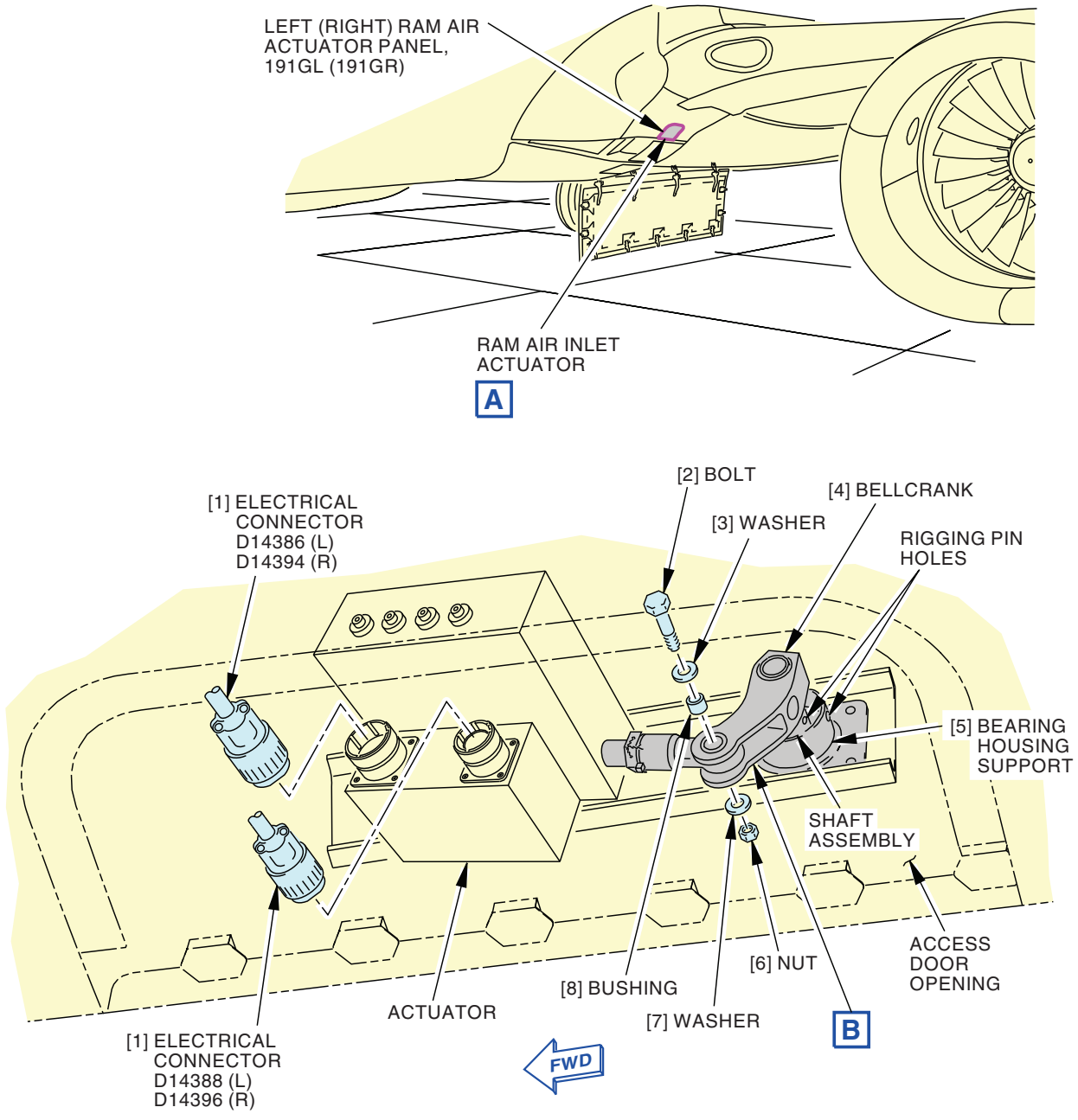
Row	Col	Number	Name
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-00-00-810-007

(12) If necessary, extinguish the MAINT light on the P5-1 panel. Do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



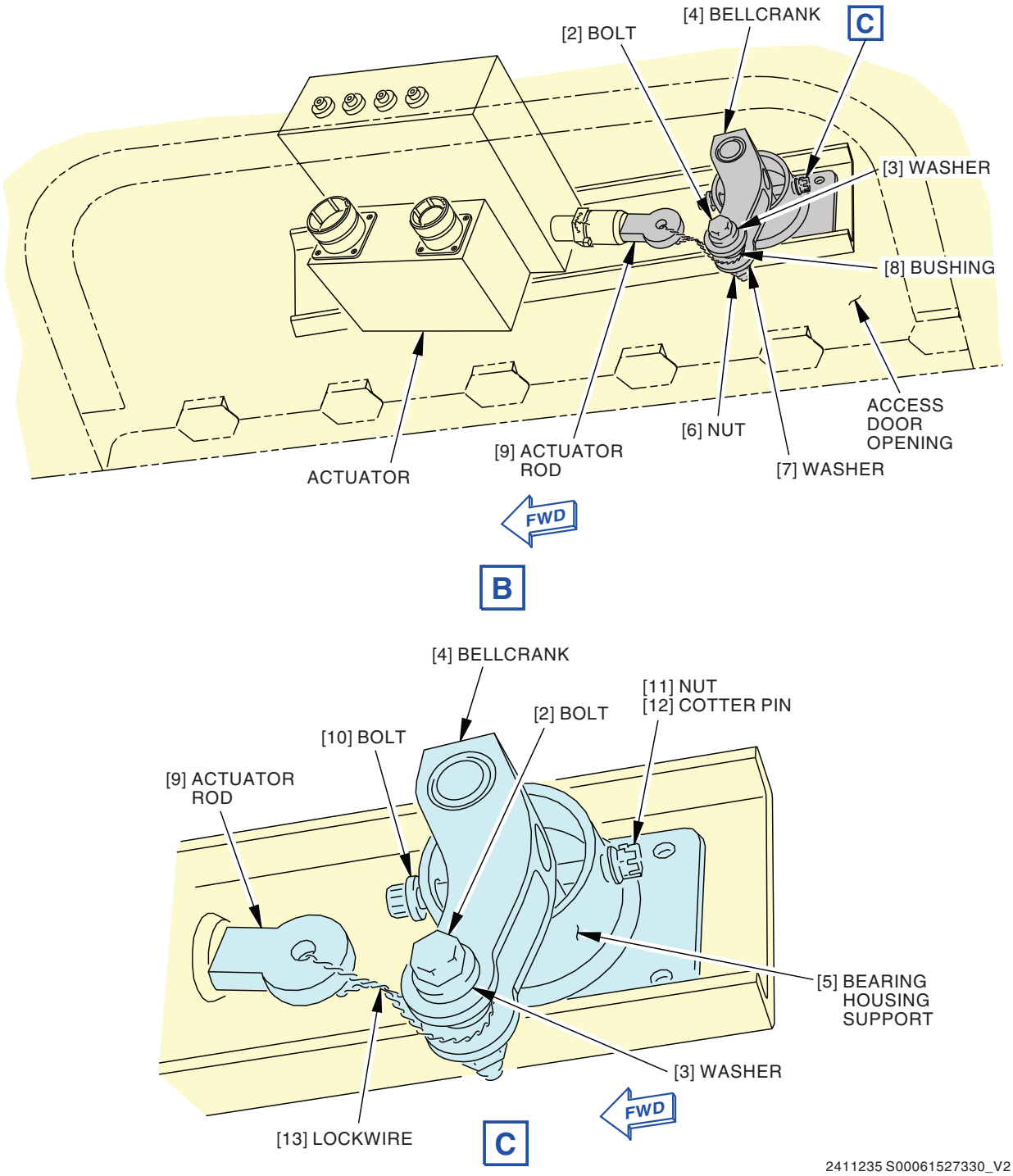
LEFT RAM AIR INLET ACTUATOR
(RIGHT RAM AIR INLET ACTUATOR IS OPPOSITE)

A

2411234 S00061527329_V2

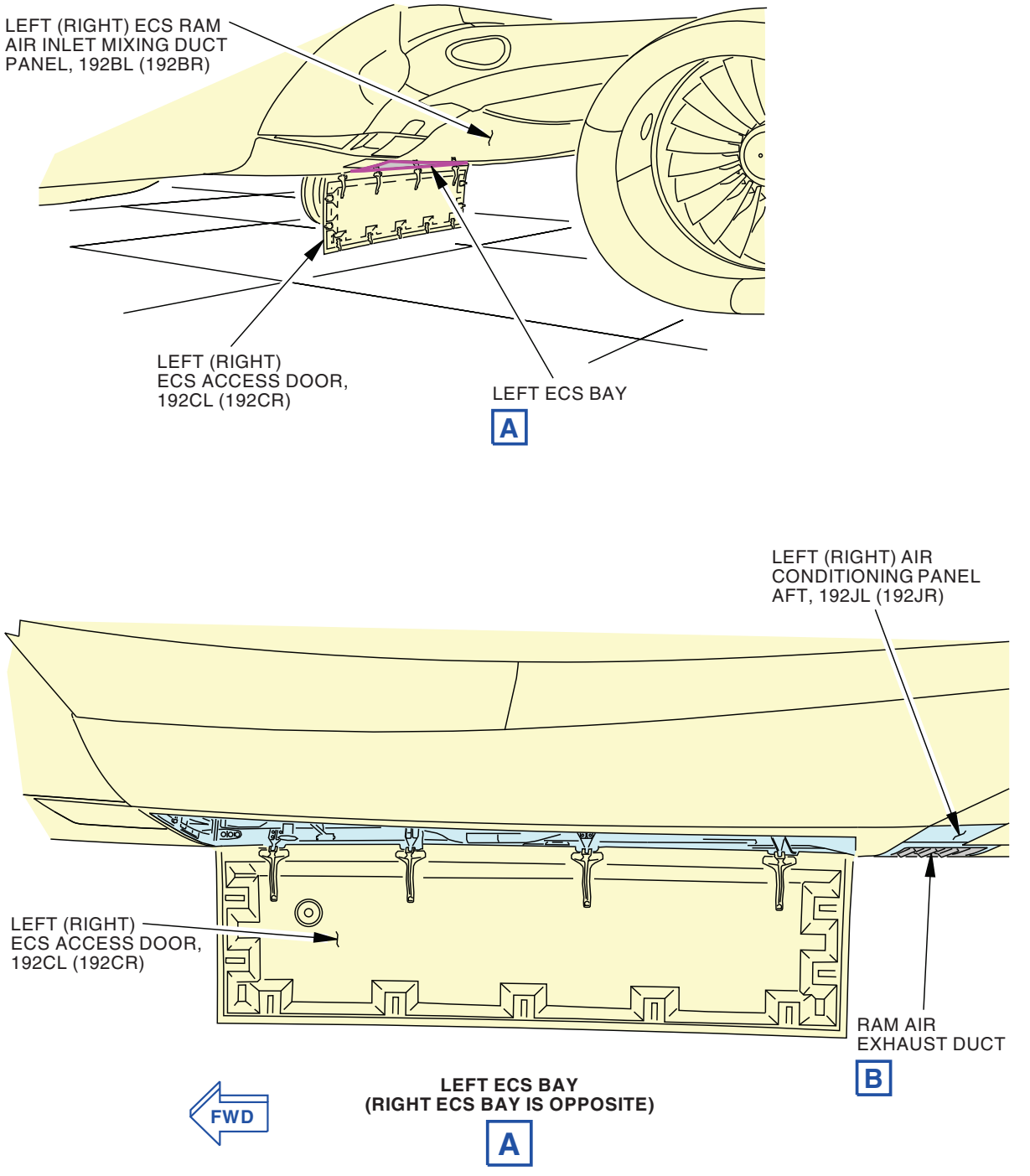
Ram Air Modulation System Deactivation
Figure 906/21-00-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



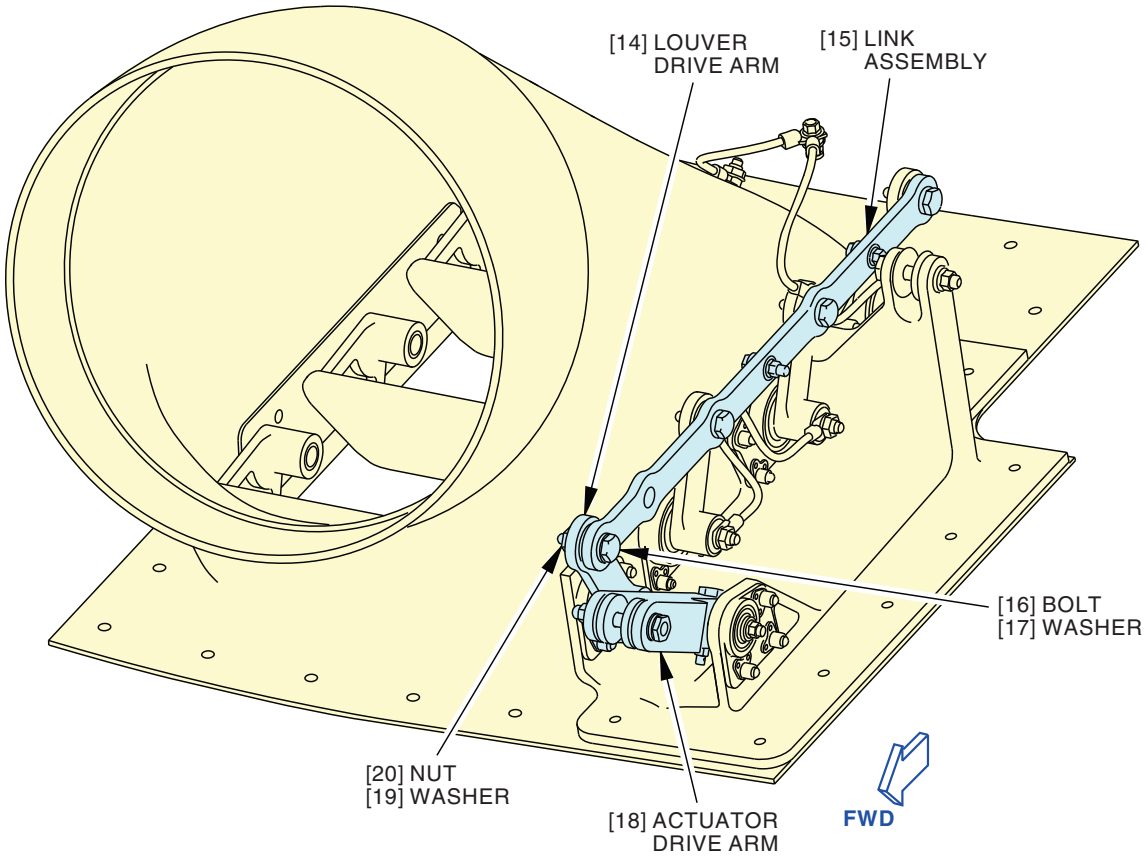
Ram Air Modulation System Deactivation
Figure 906/21-00-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ram Air Exit Louvers Modulation System Deactivation
Figure 907/21-00-00-990-804 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



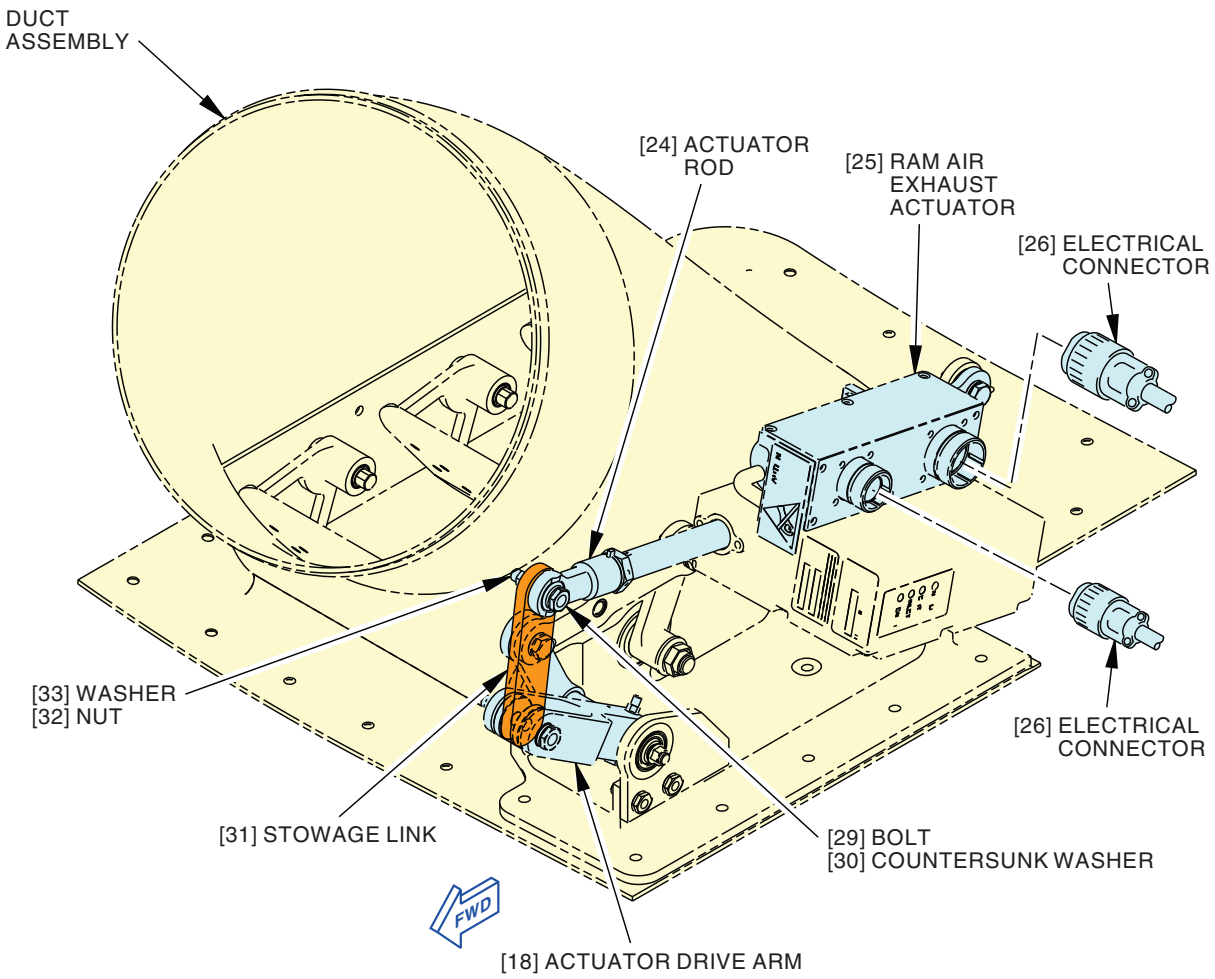
RAM AIR EXHAUST DUCT
(RAM AIR EXHAUST ACTUATOR REMOVED)

B

2411237 S00061527332_V3

Ram Air Exit Louvers Modulation System Deactivation
Figure 907/21-00-00-990-804 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RAM AIR EXHAUST DUCT
(RAM AIR EXHAUST ACTUATOR INSTALLED WITH STOWAGE LINK)

B

2502308 S0000588651_V3

Ram Air Exit Louvers Modulation System Deactivation
Figure 907/21-00-00-990-804 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-850

46. DDG 21-51-09 Pack Ram Air Systems Inoperative - Restoration

(Figure 906)

A. General

(1) This task restores the airplane to its usual condition after operation with the pack ram air systems inoperative

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
21-51-27-400-801	Ram Air Exhaust Actuator - Installation (P/B 401)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14767	Link - Stowage SRADA Part #: C21009-1 Supplier: 81205

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward
191GR	Ram Air Actuator Panel - Forward
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

F. Prepare for Reactivation

SUBTASK 21-00-00-860-074

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-020-017

(2) If the ram air inlet door and ram air exhaust duct louvers for the left cooling pack were deactivated, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-420-011

(3) If the ram air inlet door and ram air exhaust duct louvers for the right cooling pack were deactivated, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

SUBTASK 21-00-00-010-024

(4) To get access to the ram air inlet door actuator for the left cooling pack, open this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

SUBTASK 21-00-00-010-025

(5) To get access to the ram air inlet door actuator for the right cooling pack, open this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

SUBTASK 21-00-00-010-026

(6) To get access to the ram air exhaust duct louvers for the left cooling pack, open these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

SUBTASK 21-00-00-010-027

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(7) To get access to the ram air exhaust door actuator for the right cooling pack, do this step:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(c) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Ram Air Inlet Modulating System Reactivation

Figure 906

SUBTASK 21-00-00-420-012

(1) Remove the lockwire [13] from the actuator rod end and the bolt [2].

SUBTASK 21-00-00-020-018

(2) Remove the cotter pin [12], nut [11], and bolt [10].

SUBTASK 21-00-00-020-021

(3) Remove and retain the bolt [2], washer [3], washer [7], bushing [8] and nut [6] from the bellcrank [4].

SUBTASK 21-00-00-820-002

(4) Turn the bellcrank [4] to align with the actuator rod [9] end.

SUBTASK 21-00-00-420-013

(5) Install the bolt [2], washer [3], washer [7], bushing [8] and nut [6] that connect the actuator rod [9] end to the bellcrank [4].

SUBTASK 21-00-00-420-014

(6) Install the electrical connectors [1] on the actuator.

SUBTASK 21-00-00-860-075

(7) If you put the ram air inlet door for the left cooling pack to its usual condition, close this access panel:

Number	Name/Location
191GL	Ram Air Actuator Panel - Forward

SUBTASK 21-00-00-860-076

(8) If you put the ram air inlet door for the right cooling pack to its usual condition, close this access panel:

Number	Name/Location
191GR	Ram Air Actuator Panel - Forward

H. Ram Air Exit Modulating System Reactivation

Figure 907

SUBTASK 21-00-00-000-004

(1) If the ram air exhaust actuator [25] was still installed and deactivated, then remove the Link - Stowage SRADA, SPL-14767 tool:

(a) Disconnect the SRADA stowage link [31] from the actuator rod [24] end by removing the bolt [29], countersunk washer [30], washer [33], and nut [32] that are part of the Link - Stowage SRADA, SPL-14767 tool kit.

(b) Disconnect the other end of the SRADA stowage link [31] from the actuator drive arm [18] by removing the bolt, washers, and nut which will be used to attach the actuator rod [24] end to the actuator drive arm [18].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-820-003

(2) Remove the nut [20], washer [19], washer [17], and bolt [16] that are attaching the most forward louver drive arm [14] to the most forward hole in the link assembly [15].

SUBTASK 21-00-00-820-004

(3) Move the link assembly [15] until the hole in the most forward louver drive arm [14] aligns with the second most forward hole in the link assembly [15].

SUBTASK 21-00-00-420-015

(4) Install the bolt [16], washer [17], washer [19], and nut [20] through the hole in the most forward louver drive arm [14]and the second most forward hole in the link assembly [15].

SUBTASK 21-00-00-420-023

(5) If necessary, install the ram air exhaust actuator [25]. Do this task: Ram Air Exhaust Actuator - Installation, TASK 21-51-27-400-801.

SUBTASK 21-00-00-420-024

(6) Connect the electrical connectors [26] to the ram air exhaust actuator [25].

SUBTASK 21-00-00-410-013

(7) If the ram air exhaust duct louvers for the left cooling pack were put back to their usual condition, do this step:

(a) Close these access panels:

Number	Name/Location
192CL	ECS Access Door
192JL	Air Conditioning Panel - Aft

SUBTASK 21-00-00-410-014

(8) If the ram air exhaust duct louvers for the right cooling pack were put back to their usual condition, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

(c) Close this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

SUBTASK 21-00-00-420-017

(9) If the ram air exhaust duct louvers for the left cooling pack were put back to their usual condition, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-00-00-420-018

(10) If the ram air exhaust duct louvers for the right cooling pack were put back to their usual condition, do this step:

Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R

I. Ram Air Modulating System Check

SUBTASK 21-00-00-730-003

(1) Do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

J. MAINT Light Status Message

SUBTASK 21-00-00-750-002

(1) Uncheck the applicable MAINT light Status Message(s) on the MAINT LIGHT menu (TASK 31-65-00-750-801).

——— END OF TASK ———

TASK 21-00-00-040-865

47. DDG 21-61-01 Trim Air Modulating Valve Inoperative - Preparation

(Figure 908)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with a Zone Trim Air Modulating Valve Inoperative.

NOTE: There are three zone trim air modulating valves. The valve for the control cabin is in the left ECS bay. The valves for the forward and the aft passenger cabin zones are in the right ECS bay.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for Deactivation

SUBTASK 21-00-00-860-077

(1) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

SUBTASK 21-00-00-860-078

(2) Set the CONT CAB, FWD CAB and/or AFT CAB temperature selectors on the P5-17 temperature control panel to OFF.

SUBTASK 21-00-00-010-028

(3) To get access to the control compartment trim air modulating valve, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-010-029

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the aft and forward passenger compartment trim air modulating valves, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Zone Trim Air Modulating Valve Deactivation (Locked Closed)

SUBTASK 21-00-00-040-017

(1) Turn the manual override on the inoperative trim air modulating valve until the visual position indicator shows that it is in the full closed position.

SUBTASK 21-00-00-040-016

WARNING

DO NOT TOUCH AIR CONDITIONING PARTS UNTIL THEY BECOME COOL. AFTER THE AIR CONDITIONING SYSTEM OPERATES, SOME AIR CONDITIONING PARTS CAN STAY HOT FOR ALMOST ONE HOUR. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(2) Disconnect and stow the electrical connector for the inoperative valve.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-410-020

(1) To close the left ECS access doors, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-021

(2) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-00-00-860-109

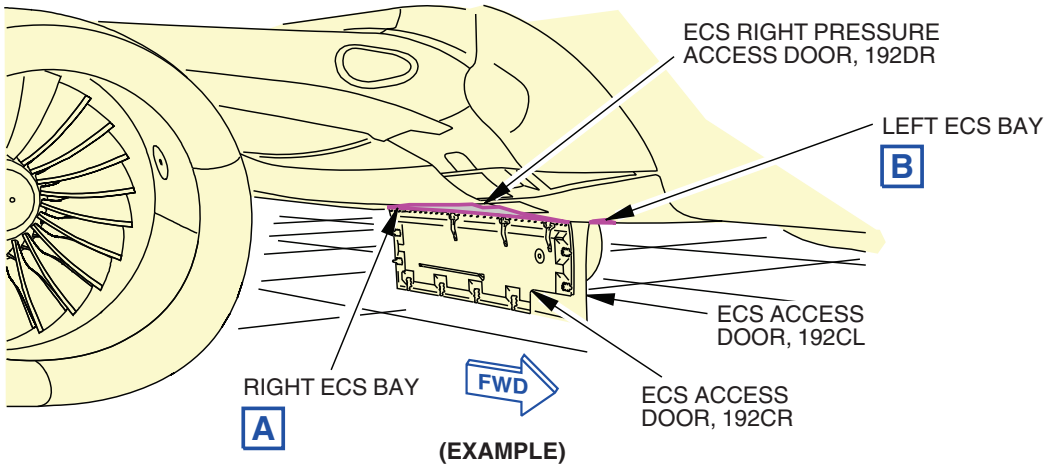
(3) Set the temperature selector associated with the deactivated trim air modulating valve to AUTO.

SUBTASK 21-00-00-810-008

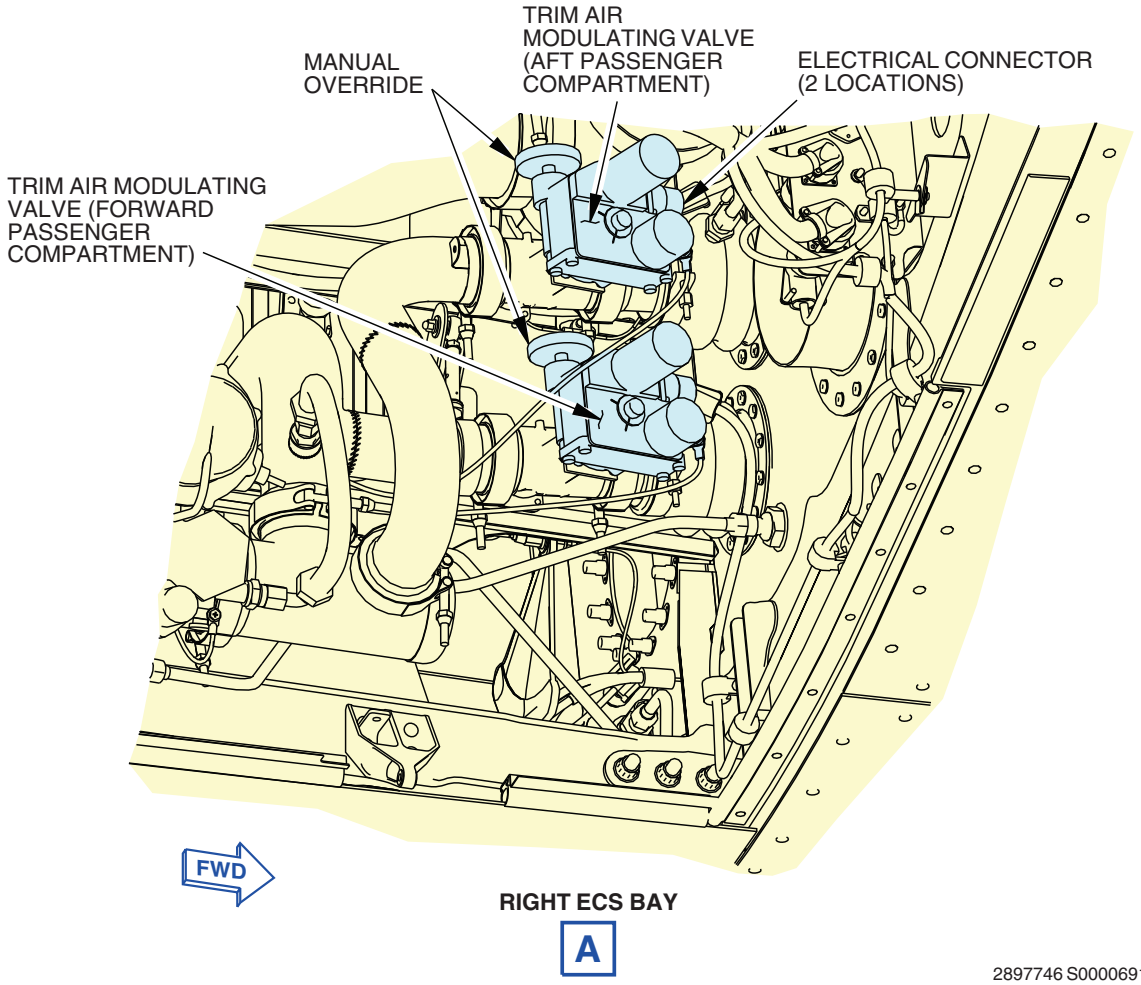
(4) If necessary, extinguish the MAINT light on the P5-1 panel. Do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



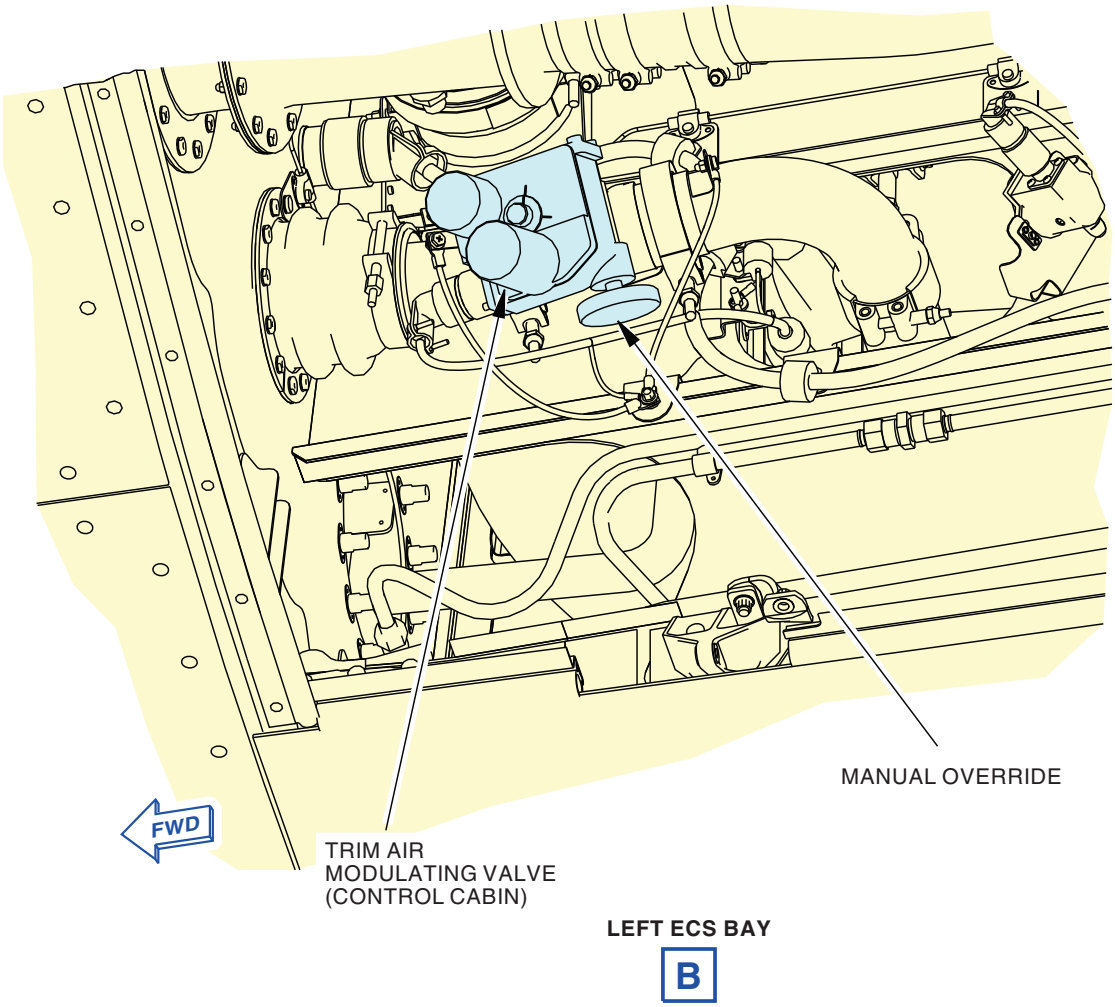
(EXAMPLE)



2897746 S0000691296_V2

Zone Trim Air Modulating Valve Deactivation
Figure 908/21-00-00-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411242 S00061527365_V1

Zone Trim Air Modulating Valve Deactivation
Figure 908/21-00-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-851

48. DDG 21-61-01 Trim Air Modulating Valve Inoperative - Restoration

(Figure 908)

A. General

(1) This task puts the airplane to its usual condition after operation with a Zone Trim Air Modulating Valve Inoperative.

NOTE: There are three Zone Trim Air Modulating Valves. The valve for the control cabin is in the left ECS bay. The valves for the forward and the aft passenger cabin zones are in the right ECS bay.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Zone Trim Air Modulating Valve Restoration

SUBTASK 21-00-00-860-110

(1) Set the CONT CAB, FWD CAB and/or AFT CAB temperature selectors on the P5-17 temperature control panel to OFF.

SUBTASK 21-00-00-010-039

(2) To get access to the control compartment trim air modulating valve, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-010-040

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(3) To get access to the aft and forward passenger compartment trim air modulating valves, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-00-00-400-003

WARNING DO NOT TOUCH AIR CONDITIONING PARTS UNTIL THEY BECOME COOL. AFTER THE AIR CONDITIONING SYSTEM OPERATES, SOME AIR CONDITIONING PARTS CAN STAY HOT FOR ALMOST ONE HOUR. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(4) Connect the electrical connector for the inoperative valve.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-410-022

(1) To close the left ECS access doors, do this step:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-023

(2) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-00-00-860-111

(3) Set the temperature selector associated with the deactivated trim air modulating valve to AUTO.

G. MAINT Light Status Message

SUBTASK 21-00-00-750-003

(1) Uncheck the applicable MAINT light Status Message(s) on the MAINT LIGHT menu (TASK 31-65-00-750-801).

——— END OF TASK ———

TASK 21-00-00-040-866

49. DDG 21-61-04 Trim Air Check Valve Inoperative - Preparation

(Figure 909)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with a trim air check valve inoperative.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-24-400-801	Ram Air Ducts Installation (P/B 401)
21-61-09-000-801	Trim Air Check Valve Removal (P/B 401)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11158	Blank-Off Plate, Trim Air Check Valve (Dispatch) Part #: A21015-8 Supplier: 81205

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for Deactivation

SUBTASK 21-00-00-040-019

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-040-020

(2) Do these steps on the P5-10 air conditioning panel:

(a) Set the BLEED 1 and 2 switches to the OFF position and attach DO-NOT-OPERATE tags.

(b) Set the BLEED APU switch to the OFF position and attach a DO-NOT-OPERATE tag.

SUBTASK 21-00-00-010-031

(3) To get access to the trim air check valve in the left ECS bay, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-010-032

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the trim air check valve in the right ECS bay, do these steps:

(TASK 06-41-00-800-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Trim Air Check Valve Deactivation

SUBTASK 21-00-00-040-021

(1) Do these steps to deactivate the trim air check valve:

(a) Do the steps in this task to remove the duct connection clamp on the check valve: Trim Air Check Valve Removal, TASK 21-61-09-000-801.

(b) Install the blank-off plate, SPL-11158, between the duct assembly and the inoperative trim air check valve.

NOTE: If blank-off plate, SPL-11158 is not available, then fabricate a blanking plate which has these specifications:

Allowable materials:

1. Titanium 6AL-4V per MIL-T-9046 Type 3 Comp C

2. Inconel 718 per AMS 5596

3. 301 CRES per MIL-S-5059 1/2, 3/4, or Fullhard

Dimensions:

Diameter; 3.10 +0.10 / -0.00 in. (78.74 +2.54 / -0.00 mm)

Thickness; 0.040 in. (1.016 mm)

1) Install the blanking plate on the upstream side of the inoperative trim air check valve.

NOTE: If the inoperative trim air check valve flow body is cracked or leaking, dispatch is not allowed.

(c) Do the steps in this task to install the clamp on the check valve and duct connection: Ram Air Ducts Installation, TASK 21-51-24-400-801.

SUBTASK 21-00-00-700-002

(2) Perform a leakage test:

(a) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

1) Set the BLEED 1 and 2 switches to the ON position and remove the DO-NOT-OPERATE tags.

2) Set the BLEED APU switch to the ON position and remove the DO-NOT-OPERATE tag.

(b) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

(c) Put the applicable PACK switch to the AUTO position.

(d) Put the TRIM AIR switch to the OFF position.

(e) Check valve flange for leakage.

1) Diffused leakage is satisfactory.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Jet blast leakage must be repaired by joint or clamp realignment.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-010-034

(1) To close the left ECS access doors, do this step:

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-010-035

(2) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-00-00-860-080

(3) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-440-016

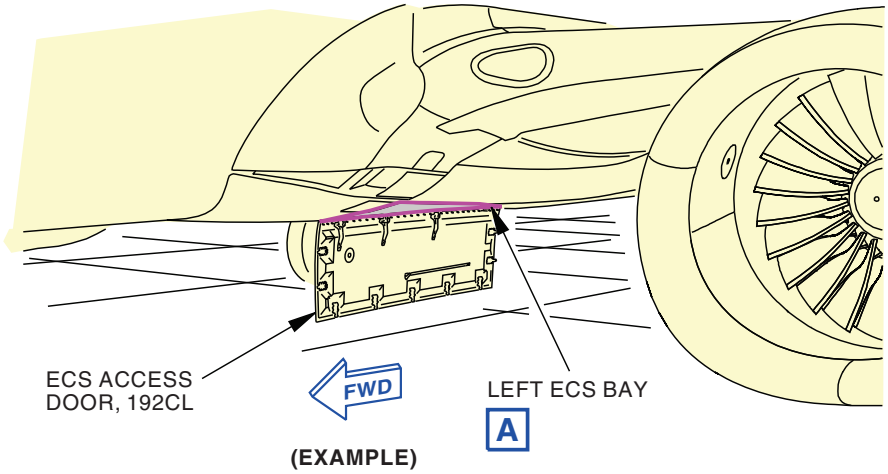
(4) Do these steps on the P5-10 air conditioning panel:

(a) Set the BLEED 1 and 2 switches to the ON position and remove the DO-NOT-OPERATE tags.

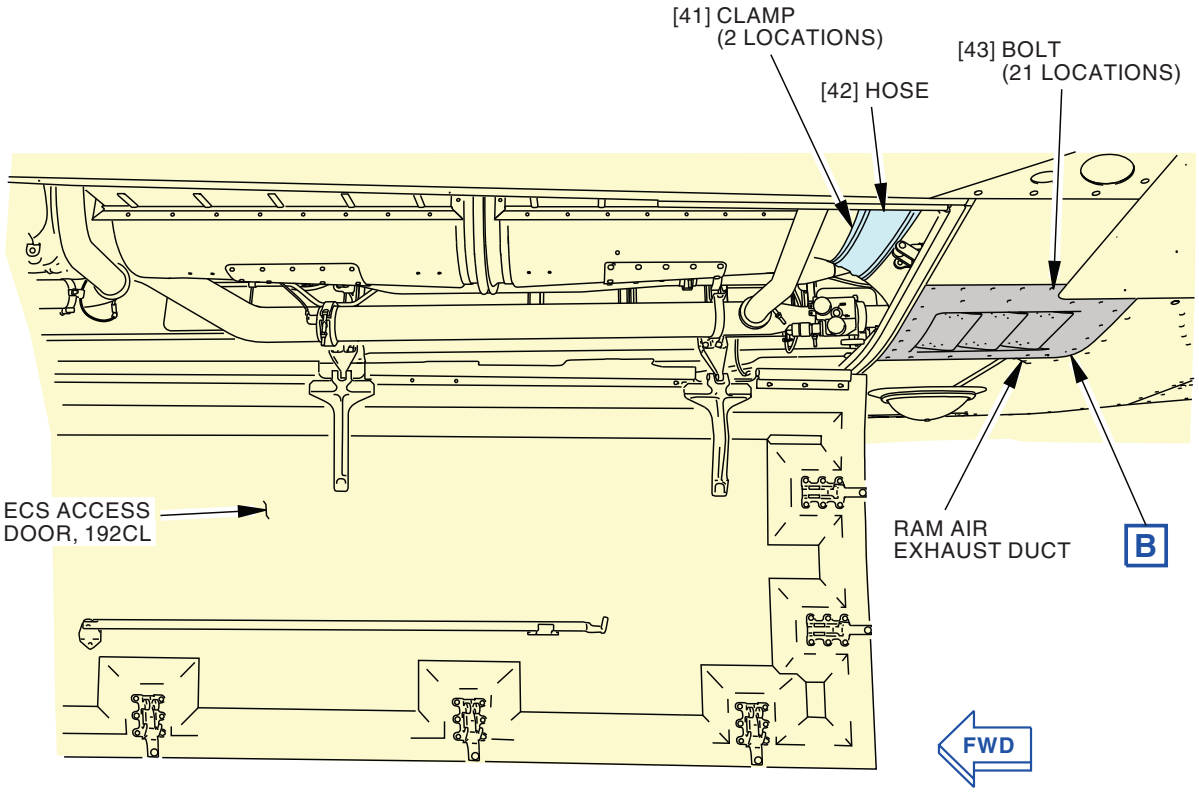
(b) Set the BLEED APU switch to the ON position and remove the DO-NOT-OPERATE tag.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)



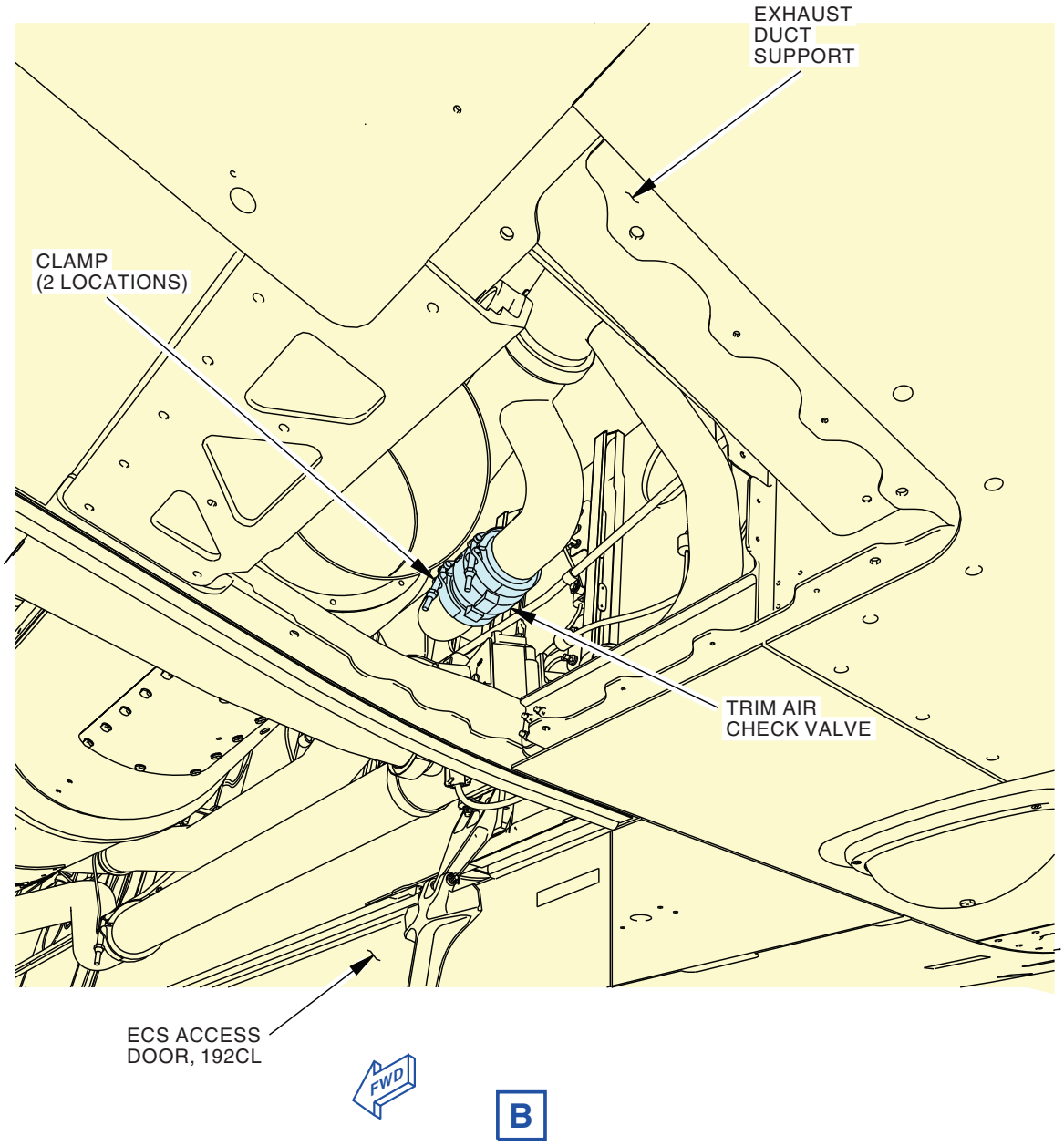
LEFT ECS BAY
(RIGHT ECS BAY IS OPPOSITE)

A

2897755 S0000691297_V1

Trim Air Check Valve Deactivation
Figure 909/21-00-00-990-808 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Trim Air Check Valve Deactivation
Figure 909/21-00-00-990-808 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-852

50. DDG 21-61-04 Trim Air Check Valve Inoperative - Restoration

(Figure 909)

A. General

(1) This task puts the airplane to its usual condition after operation with a trim air check valve inoperative.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-51-24-400-801	Ram Air Ducts Installation (P/B 401)
21-61-09-000-801	Trim Air Check Valve Removal (P/B 401)
21-61-09-400-801	Trim Air Check Valve Installation (P/B 401)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Prepare for Reactivation

SUBTASK 21-00-00-040-044

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-040-045

(2) Do these steps on the P5-10 air conditioning panel:

(a) Set the BLEED 1 and 2 switches to the OFF position and attach DO-NOT-OPERATE tags.

(b) Set the BLEED APU switch to the OFF position and attach a DO-NOT-OPERATE tag.

SUBTASK 21-00-00-010-041

(3) To get access to the trim air check valve in the left ECS bay, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-010-042

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the trim air check valve in the right ECS bay, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

F. Trim Air Check Valve Reactivation

SUBTASK 21-00-00-960-018

(1) Remove the blanking plate installed between the trim air check valve and duct flange.

(a) Do the steps in this task to remove the blanking plate: Trim Air Check Valve Removal, TASK 21-61-09-000-801

1) Be sure the blanking plate used between the duct assembly and the inoperative trim air check valve is removed at the completion of task: Trim Air Check Valve Installation, TASK 21-61-09-400-801.

NOTE: Blanking plate material specifications are as follows: Titanium 6AL-4V per MIL-T-9046 Type 3 Comp C, Inconel 718 per AMS 5596, and 301 CRES per MIL-S-5059 1/2, 3/4, or Fullhard. Blanking plate dimension specifications are as follows: Diameter is 3.10 +0.10 / -0.00 in. (78.74 +2.54 / -0.00 mm) and Thickness is 0.040 in. (1.016 mm).

(b) Do the steps in this task to install the clamp on the check valve and duct connection: Ram Air Ducts Installation, TASK 21-51-24-400-801.

SUBTASK 21-00-00-700-003

(2) Perform a leakage test:

(a) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

1) Set the BLEED 1 and 2 switches to the ON position and remove the DO-NOT-OPERATE tags.

2) Set the BLEED APU switch to the ON position and remove the DO-NOT-OPERATE tag.

(b) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801

(c) Put the applicable PACK switch to the AUTO position.

(d) Put the TRIM AIR switch to the OFF position.

(e) Check valve flange for leakage.

1) Diffused leakage is satisfactory.

2) Jet blast leakage must be repaired by joint or clamp realignment.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-00-00-410-024

(1) To close the left ECS access doors, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-00-410-025

(2) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-00-00-860-112

(3) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-00-00-440-017

(4) Do these steps on the P5-10 air conditioning panel:

(a) Set the BLEED 1 and 2 switches to the ON position and remove the DO-NOT-OPERATE tags.

(b) Set the BLEED APU switch to the ON position and remove the DO-NOT-OPERATE tag.

——— END OF TASK ———

TASK 21-00-00-040-867

51. DDG 21-62-01 Trim Air Pressure Regulating and Shutoff Valve Inoperative - Preparation

(Figure 910)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with a Trim Air Pressure Regulating and Shutoff Valve Inoperative.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for Deactivation

SUBTASK 21-00-00-860-082

(1) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

SUBTASK 21-00-00-860-084

(2) On the P5-17 Cabin Temperature Panel, put the TRIM AIR switch to the OFF position and attach a DO NOT OPERATE tag, STD-858.

SUBTASK 21-00-00-860-083

(3) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

SUBTASK 21-00-00-010-036

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the trim air pressure regulating and shutoff valve, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Trim Air Pressure Regulating and Shutoff Valve Deactivation (Locked Closed)

SUBTASK 21-00-00-040-022

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH AIR CONDITIONING PARTS UNTIL THEY BECOME COOL. AFTER THE AIR CONDITIONING SYSTEM OPERATES, SOME AIR CONDITIONING PARTS CAN STAY HOT FOR ALMOST ONE HOUR. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(1) Disconnect, cap and stow the electrical connector from the shutoff valve.

SUBTASK 21-00-00-040-023

(2) Lock the shutoff valve in the closed position.

(a) Remove the lockwire from the manual override hex (TASK 20-10-44-000-801).

(b) Turn the manual override hex on the shutoff valve until LOCKED shows.

(c) Install lockwire on the manual override hex (TASK 20-10-44-400-801).

SUBTASK 21-00-00-010-037

(3) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

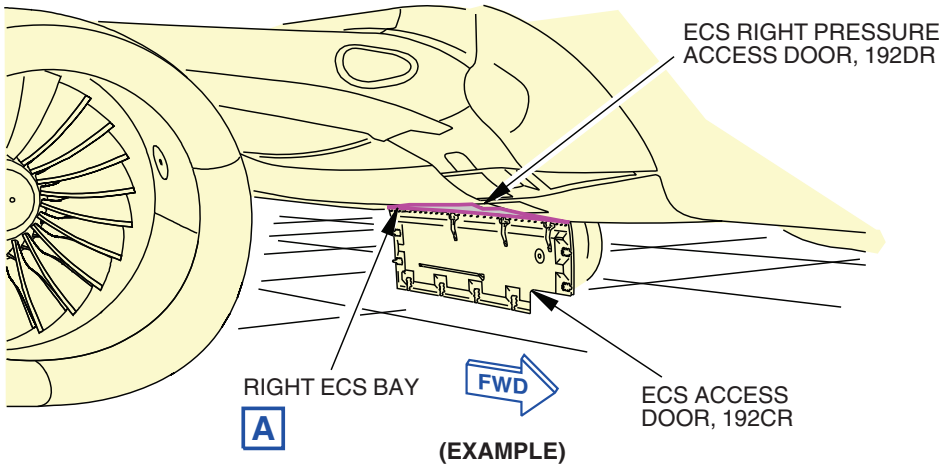
(b) Close this access panel:

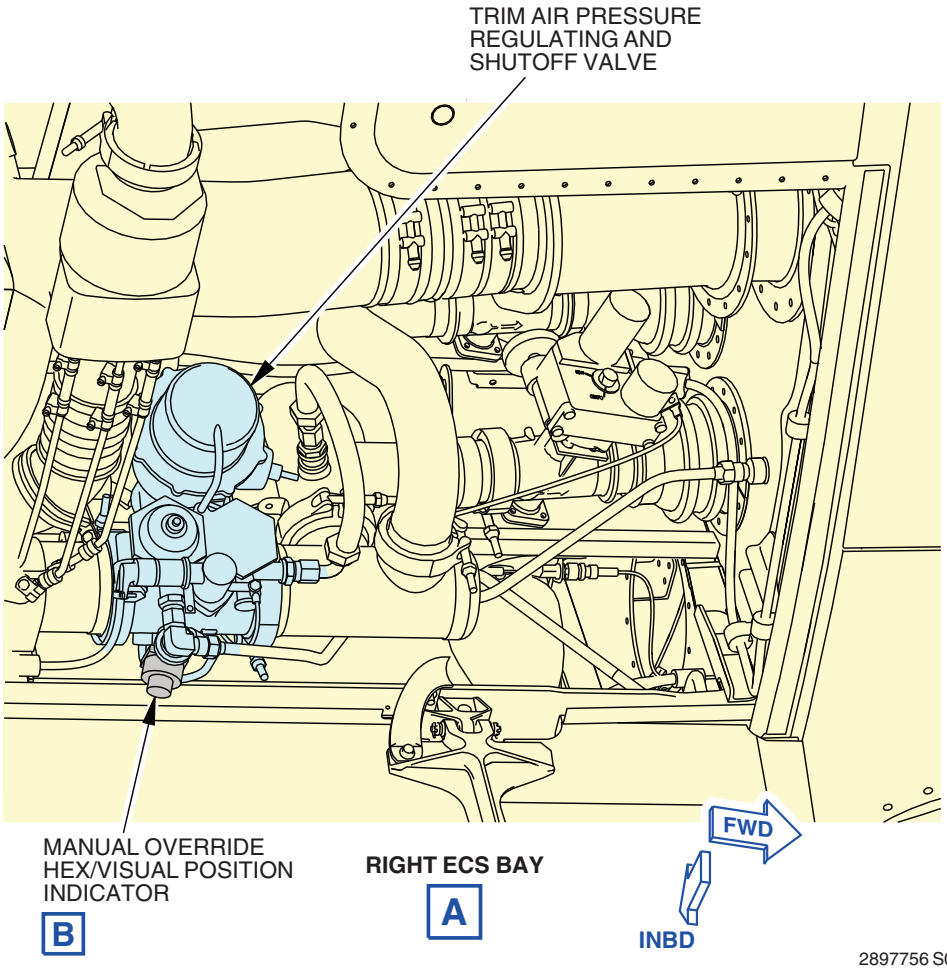
Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

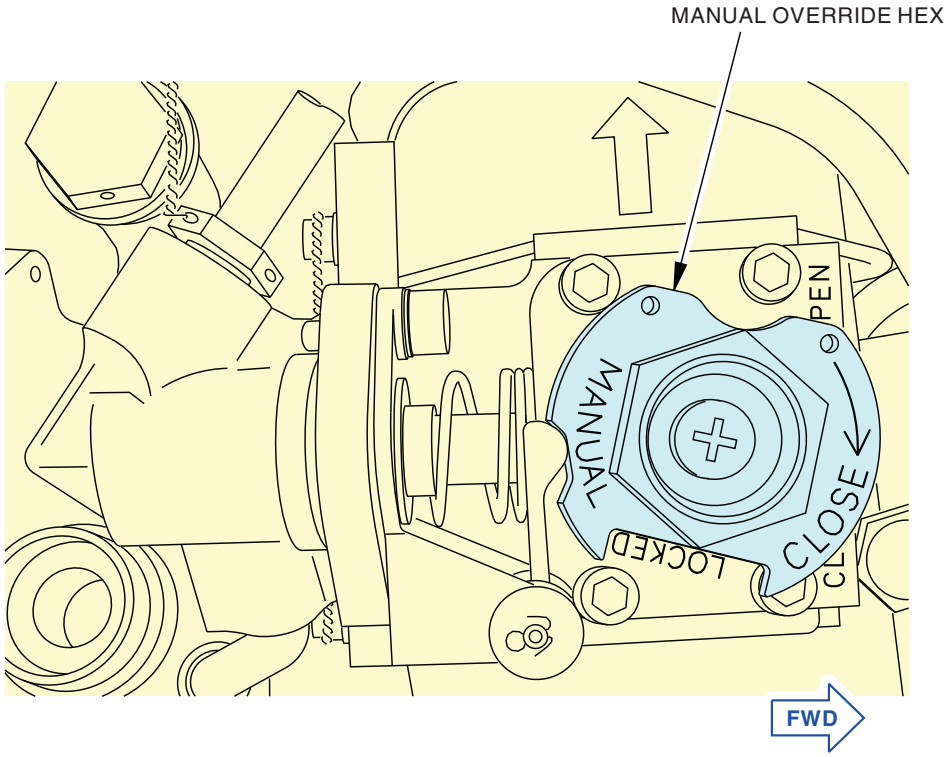
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Trim Air Pressure Regulating and Shutoff Valve Deactivation
Figure 910/21-00-00-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MANUAL OVERRIDE HEX/VISUAL POSITION INDICATOR

B

2561541 S0000611430_V1

Trim Air Pressure Regulating and Shutoff Valve Deactivation
Figure 910/21-00-00-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-00-00-440-853

52. DDG 21-62-01 Trim Air Pressure Regulating and Shutoff Valve Inoperative - Restoration

(Figure 910)

A. General

(1) This task gives the maintenance steps which return the airplane to its usual condition after flight with the Trim Air Pressure Regulating and Shutoff Valve Inoperative.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-44-000-801	Lockwire, Cotter Pins, and Lockrings - Removal (P/B 401)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for Reactivation

SUBTASK 21-00-00-860-113

(1) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

SUBTASK 21-00-00-860-114

(2) Make sure that the TRIM AIR switch, on the P5-17 cabin temperature panel, is on the OFF position and has a DO NOT OPERATE tag, STD-858.

SUBTASK 21-00-00-860-117

(3) Make sure that this circuit breaker is open and has a safety lock:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

SUBTASK 21-00-00-010-044

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(4) To get access to the trim air pressure regulating and shutoff valve, do these steps:

(TASK 06-41-00-800-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

G. Trim Air Pressure Regulating and Shutoff Valve Restoration

SUBTASK 21-00-00-440-015

(1) Unlock the shutoff valve.

(a) Remove the lockwire from the manual override hex (TASK 20-10-44-000-801).

(b) Turn the manual override hex on the shutoff valve until LOCKED does not show.

(c) Install lockwire on the manual override hex (TASK 20-10-44-400-801).

SUBTASK 21-00-00-440-014

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT TOUCH AIR CONDITIONING PARTS UNTIL THEY BECOME COOL. AFTER THE AIR CONDITIONING SYSTEM OPERATES, SOME AIR CONDITIONING PARTS CAN STAY HOT FOR ALMOST ONE HOUR. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(2) Remove the cap and connect the electrical connector to the shutoff valve.

SUBTASK 21-00-00-410-026

(3) To close the right ECS access doors, do these steps:
(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-00-00-860-134

(4) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-00-00-860-115

(5) On the P5-17 Cabin Temperature Panel, put the TRIM AIR switch to the ON position and remove the DO NOT OPERATE tag, STD-858.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR CONDITIONING SYSTEM OIL CONTAMINATION REMOVAL - MAINTENANCE PRACTICES

1. General

A. The oil fumes and the smoke from an APU/engine failure can get into the airplane cabin and cause contamination of the conditioned air. This procedure gives instructions to remove the oil contamination from the air conditioning and pneumatic systems. You must first isolate the cause of the oil contamination and repair the problem before you do this procedure.

B. The APU is the most likely source of the smoke or odors in the airplane. The smoke or odors can be caused by either an internal APU failure which can release oil into the pneumatic and/or air conditioning systems. Another source of smoke or odor is oil, glycol or hydraulic fluid being ingested into the inlet of the APU or engines where it then can enter the pneumatic and/or air conditioning systems and vaporize at higher temperatures.

C. Do not do this procedure with the crew and passengers on the airplane, since this procedure can generate smoke in the airplane.

TASK 21-00-01-100-801

2. Oil Contamination Removal from Air Conditioning and Pneumatic Systems

A. References

Reference	Title
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
21-25-01-000-801	Recirculation Air Filter - Removal (P/B 401)
21-25-01-400-801	Recirculation Air Filter - Installation (P/B 401)
21-51-03-000-804	Heat Exchanger and Plenum/Diffuser Assembly Cleaning (P/B 701)
21-51-04-000-801	Air Cycle Machine Removal (P/B 401)
21-51-04-400-801	Air Cycle Machine Installation (P/B 401)
21-51-08-000-801	High Pressure Water Separator Removal (P/B 401)
21-51-08-400-801	High Pressure Water Separator Installation (P/B 401)
21-51-12-000-801	Condenser Removal (P/B 401)
21-51-12-400-801	Condenser Installation (P/B 401)
21-73-01-000-801	Ozone Converter Removal (P/B 401)
21-73-01-400-801	Ozone Converter Installation (P/B 401)
49-11-00-860-801	APU Starting and Operation (P/B 201)
71-00-00-800-801-G00	Engine Ground Safety Precautions (P/B 201)
71-00-00-910-802-G00	Start the Engine (Selection) (P/B 201)
71-00-00-910-806-G00	Stop the Engine (Usual Engine Stop) (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2462	Deactivator - Check Valve Part #: C21006-7 Supplier: 81205 Opt Part #: C21006-1 Supplier: 81205 Opt Part #: PF80-012-500 Supplier: 053H3

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
SPL-15817	Adapter - Hose, Exhaust Part #: C21011-1 Supplier: 81205

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Oil Contamination Removal from Air Conditioning and Pneumatic System

SUBTASK 21-00-01-010-001

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(1) To get access to the cooling packs, open these access doors in the sequence that follows:

(a) Open this access door:

Number	Name/Location
192CL	ECS Access Door

(b) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(c) Open this access door:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-00-01-020-001

WARNING

PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(2) Install the deactivator, SPL-2462, in the ground air connector as follows:

(a) Put the deactivator, SPL-2462, with the exhaust hose adapter, SPL-15817, in the ground air connector so that the check valve flappers are open.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Install the coupling on the supply hose from the ground air cart onto the nipple of the ground air connector check valve assembly so that the deactivator, SPL-2462, is held in position.

NOTE: The supply hose should not be connected to the ground cart as the exhaust from the hose should be vented to ambient.

SUBTASK 21-00-01-860-002

(3) Make sure that these switches, on the P5 overhead panel, are set to the indicated positions:

(a) On the P5-10 air conditioning panel:

1) L PACK and R PACK switches are set to OFF.

2) L RECIRC FAN and R RECIRC FAN switches are set to OFF.

(b) On the P5-11 thermal anti-ice panel:

1) WING, ENGINE 1, and ENGINE 2, ANTI-ICE switches are set to OFF.

SUBTASK 21-00-01-860-003

(4) Start the Auxiliary Power Unit (APU) and let the APU operation become stable (TASK 49-11-00-860-801).

SUBTASK 21-00-01-860-004

(5) Do the steps which follow to remove the oil contamination from the APU pneumatic ducts:

NOTE: This procedure will remove the oil contamination from the APU bleed air. The APU bleed air will go out through the check valve in the pneumatic ground air connector.

(a) Make sure that the ISOLATION VALVE switch, on the P5-10 air conditioning panel is set to the OPEN position.

(b) Make sure that the APU BLEED switch, on the P5-10 air conditioning panel, is set to the ON position.

WARNING DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(c) Let the APU operate until the air that comes out of the pneumatic ground connector has no smell of the oil contamination.

SUBTASK 21-00-01-860-005

(6) Set the ISOLATION VALVE switch, on the P5-10 air conditioning panel to the CLOSE position.

SUBTASK 21-00-01-860-006

(7) Start the left engine and let the engine operate at ground idle speed (TASK 71-00-00-910-802-G00).

(a) Use the APU to start the engine as it is still in operation.

(b) Set the APU BLEED switch, on the P5-10 air conditioning panel, to OFF.

(c) Set the ISOLATION VALVE switch, on the P5-10 air conditioning panel to OPEN.

SUBTASK 21-00-01-860-007

(8) Do the steps which follow to remove the oil contamination from the left engine pneumatic ducts:

(a) Set the engine pneumatic BLEED 1 switch, on the P5-10 air conditioning panel, to ON.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBSERVE PROPER SAFETY PRECAUTIONS AROUND RUNNING ENGINE. WEAR EAR PROTECTORS AND STAY CLEAR OF ENGINE HAZARD AREAS. SEE "ENGINE GROUND SAFETY PRECAUTIONS" FOR ENGINE HAZARD DESCRIPTION.

(b) Make sure that you obey all safety precautions when you are near an engine that is in operation (TASK 71-00-00-800-801-G00).

WARNING DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(c) Let the engine 1 pneumatic air flow until the air that comes out of the pneumatic ground connector has no smell of the oil contamination.

(d) Set the engine pneumatic BLEED 1 switch to the OFF position.

SUBTASK 21-00-01-860-008

(9) Stop the operation of the left engine (TASK 71-00-00-910-806-G00).

SUBTASK 21-00-01-080-001

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(10) Remove the ground cart supply hose coupling from the nipple on the ground air connector check valve assembly.

SUBTASK 21-00-01-090-001

(11) Remove the deactivator, SPL-2462, from the pneumatic ground air connector.

SUBTASK 21-00-01-860-009

(12) Set the APU BLEED switch, on the P5-10 air conditioning panel, to the ON position.

SUBTASK 21-00-01-860-010

(13) Start the right engine and let the engine operate at ground idle speed (TASK 71-00-00-910-802-G00).

(a) Use the APU to start the engine as it is still in operation.

(b) Set the APU BLEED switch, on the P5-10 air conditioning panel, to the OFF position.

WARNING OBSERVE PROPER SAFETY PRECAUTIONS AROUND RUNNING ENGINE. WEAR EAR PROTECTORS AND STAY CLEAR OF ENGINE HAZARD AREAS. SEE "ENGINE GROUND SAFETY PRECAUTIONS" FOR ENGINE HAZARD DESCRIPTION.

(c) Make sure that you obey all safety precautions when you are near an engine that is in operation (TASK 71-00-00-800-801-G00).

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(d) Install the deactivator, SPL-2462, in the ground air connector:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put the deactivator, SPL-2462, in the ground air connector so that the check valve flappers are open.

2) Install the coupling on the supply hose from the ground air cart onto the nipple of the ground air connector check valve assembly so that the deactivator, SPL-2462, is held in position.

NOTE: The supply hose should not be connected to the ground cart as the exhaust from the hose should be vented to ambient.

(e) Set the ISOLATION VALVE switch, on the P5-10 air conditioning panel to the CLOSE position.

SUBTASK 21-00-01-860-011

(14) Do the steps which follow to remove the oil contamination from the right engine pneumatic ducts:

(a) Set the engine pneumatic BLEED 2 switch, on the P5-10 air conditioning panel, to the ON position.

WARNING OBSERVE PROPER SAFETY PRECAUTIONS AROUND RUNNING ENGINE. WEAR EAR PROTECTORS AND STAY CLEAR OF ENGINE HAZARD AREAS. SEE "ENGINE GROUND SAFETY PRECAUTIONS" FOR ENGINE HAZARD DESCRIPTION.

(b) Make sure that you obey all safety precautions when you are near an engine that is in operation (TASK 71-00-00-800-801-G00).

WARNING DO NOT PUT YOUR HAND OR BODY IN THE AIR THAT COMES OUT OF THE PNEUMATIC GROUND CONNECTOR. THE HOT AIR WILL CAUSE INJURIES.

(c) Let the engine 2 pneumatic air flow until the air that comes out of the pneumatic ground connector has no smell of the oil contamination.

(d) Set the engine pneumatic BLEED 2 switch to the OFF position.

SUBTASK 21-00-01-860-012

(15) Stop the operation of the right engine (TASK 71-00-00-910-806-G00).

SUBTASK 21-00-01-800-001

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(16) Remove the ground cart supply hose and deactivator, SPL-2462, from the pneumatic ground connector.

SUBTASK 21-00-01-860-013

(17) Do the steps which follow to remove the oil contamination from the air conditioning packs and the ducts that are downstream of the packs:

(a) Supply conditioned air with the left and right cooling packs (TASK 21-00-00-800-803).

(b) Put the TRIM AIR switch to the ON position.

(c) Turn the CONT CAB, FWD CAB, AND AFT CAB temperature selectors, on the P5-17 cabin temperature control panel, to the full WARM position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Let the packs operate until the air that goes into the control cabin and the passenger cabin has no smell of the oil contamination.

(e) Shut down the cooling packs (TASK 21-00-00-800-804).

SUBTASK 21-00-01-200-001

(18) Do this inspection of the Air Cycle Machine (ACM):

(a) Remove the ACM (TASK 21-51-04-000-801).

(b) Make sure that you can turn the rotating components of the ACM with 20 in-lb (2 N·m) of torque applied on the axial nut just aft of the fan impeller.

(c) If the ACM does not rotate with 20 in-lb (2 N·m) of torque, replace the ACM with a serviceable unit.

(d) Do this task: Air Cycle Machine Installation, TASK 21-51-04-400-801.

SUBTASK 21-00-01-100-001

(19) Clean the heat exchanger and plenum/diffuser assembly, if contamination is found (TASK 21-51-03-000-804).

NOTE: Cleaning of the heat exchanger and plenum/diffuser assembly is recommended once the contamination has been rectified.

SUBTASK 21-00-01-210-001

(20) Examine the high pressure water separator and condenser for contamination.

NOTE: Removal and cleaning of these components is recommended once the contamination has been rectified.

(a) Do this task: High Pressure Water Separator Removal, TASK 21-51-08-000-801.

(b) Do this task: Condenser Removal, TASK 21-51-12-000-801.

(c) Clean the components that have contamination, replace the components if they have too much contamination.

(d) Do this task: Condenser Installation, TASK 21-51-12-400-801.

(e) Do this task: High Pressure Water Separator Installation, TASK 21-51-08-400-801.

SUBTASK 21-00-01-160-001

(21) Clean the pneumatic ducts and the components where you can see the oil contamination.

SUBTASK 21-00-01-960-001

(22) Replace the components if they have too much contamination.

SUBTASK 21-00-01-210-002

(23) Examine the recirculation air filters for contamination:

(a) Replace the filters if they have contamination. These are the tasks:

- Recirculation Air Filter - Removal, TASK 21-25-01-000-801,
- Recirculation Air Filter - Installation, TASK 21-25-01-400-801.

SUBTASK 21-00-01-900-001

(24) Replace the ozone converter. These are the tasks:

- Ozone Converter Removal, TASK 21-73-01-000-801,
- Ozone Converter Installation, TASK 21-73-01-400-801.

F. Put the Airplane Back to its Usual Condition

SUBTASK 21-00-01-010-002

(1) Close these access doors in the sequence that follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access door:

Number	Name/Location
192CL	ECS Access Door

(b) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(c) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONFIDENCE CHECK OF AIRPLANE ABILITY TO MAINTAIN CABIN PRESSURE IN FLIGHT - MAINTENANCE PRACTICES

1. General

A. The function of this procedure is to do a ground check of the capability of the air conditioning pack(s) to pressurize the airplane when it is in flight. This test can be accomplished prior to dispatch at an interval best determined by the operator.

TASK 21-00-05-780-801

2. Airplane Pack Operation - Confidence Check

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This confidence check consists primarily of these tests:

(a) Phase I Test - pressurize cabin from 0 psid (0 kPa) to 4 psid (28 kPa) using the Auxiliary Power Unit (APU), and the packs.

(b) Phase II Test - perform a cabin leak down test from 4 psid (28 kPa) cabin pressure and note the cabin differential pressure for 120 seconds at 10 second intervals.

(2) Figure 201, Figure 202 provide graphical representation of the confidence check tests.

(3) Table 201 and Table 202 use to record the Phase I and II test results.

(4) Figure 203 provides supplemental troubleshooting information.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
49-11-00-860-801	APU Starting and Operation (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
200	Upper Half of Fuselage

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Phase I Test. Airplane Pressurization

SUBTASK 21-00-05-860-002

(1) Prepare for airplane pressurization as follows:

(a) Make sure that the nose and main landing gear are pinned.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) To install the downlock pins, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

(b) Inspect the condition of the following items for evidence of wear and tear:

1) The forward and aft cargo door seals
2) The main entry door seals
3) The service door seals
4) The duct pressure seals.

NOTE: The duct pressure seals are the seals on the ducts that go from an unpressurized compartment such as the Environmental Control System (ECS) bays to the pressurized distribution bay compartment or aft cargo compartment.

(c) Record the condition of the door seals and the duct seals on the Table 201.

Table 201/21-00-05-993-802 Initial Cabin Pressurization (0.0 to 4.0 psid) - Confidence Check

Airplane _____	**Hours** _____	**Cycles**_____	**Date**_____	
Phase I - Initial Cabin Pressurization (0.0 to 4.0 psid)				
Condition of cargo and main entry door seals				
Flow control valve indicator modulation		Left valve _____	Right valve_____	
Any obvious air leak noise from inside cabin				
Any obvious leaks outside airplane				
Estimated time to pressurize from 0.0 to 4.0 psid				
Bleed source(s) used to achieve 4.0 psid		Left engine_____	Right engine____	APU_____

(d) Open this access panel:

Number	**Name/Location**
192CL	ECS Access Door

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(e) To open the right ECS access door do these steps:

(TASK 06-41-00-800-801)

1) Open this access door:

Number	**Name/Location**
192DR	ECS High Pressure Access Door

2) Open this access door:

Number	**Name/Location**
192CR	ECS Access Door

SUBTASK 21-00-05-480-001

(2) Make sure that there is audio communication between the flight compartment personnel and ground personnel.

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SUBTASK 21-00-05-860-003

(3) Do these steps on the Flight Management Computer (FMC) to display the FMCS ANALOG page 1/4:

NOTE: Pack and isolation valve operation/position information can be obtained from the FMC Maintenance Discrete page.

(a) Press the INIT REF function key on the Control Display Unit (CDU) front panel.

(b) Push the Line Select Key (LSK) adjacent to <INDEX (LSK 6L) to show the INIT REF INDEX 1/1 page.

(c) Push the LSK adjacent to MAINT> (LSK 6R) to show the MAINT BITE INDEX page.

(d) Push the LSK adjacent to <FMCS (LSK 1L) to show the FMCS BITE 1/1 page.

(e) Push the LSK adjacent to <DISCRETES (LSK 4L) to show the FMCS ANALOG DISC 1/4 page.

(f) Use the ECS PACK H/L (high flow, low flow) line entry on the ANALOG DISC 1/4 page.

(g) The following information can be displayed on the FMC:

1) ECS PACK ON/OFF (LEFT and RIGHT)

a) Indicates if the left or right pack is on or off.

2) ECS PACK H/L LO HI (LEFT and RIGHT)

a) Indicates the pack is in the low flow mode or if it has been commanded to the high flow mode. A pack can be commanded to the high flow mode even if it is off.

3) ISOLATION VALVE OPEN (CLOSED)

a) Indicates if the valve is open or closed.

SUBTASK 21-00-05-860-004

(4) Do this task: APU Starting and Operation, TASK 49-11-00-860-801.

(a) Let the APU become stable at the governed speed.

SUBTASK 21-00-05-860-005

(5) Position the ISOLATION VALVE switch to OPEN.

SUBTASK 21-00-05-860-006

(6) Set the BLEED 1 and BLEED 2 switches on the P5-10 air conditioning panel to the ON position.

NOTE: The cabin is not being pressurized at this point.

SUBTASK 21-00-05-710-001

(7) Do these steps to make sure that the pack flow control and shutoff valves operate correctly:

(a) Pressurize the pneumatic system with the APU (TASK 36-00-00-860-803).

(b) Set the APU BLEED switch, on the P5-10 air conditioning panel, to ON position.

(c) Set the L and R PACK switches, on the P5-10 air conditioning panel, to AUTO position.

(d) Make sure that the ECS PACK H/L indication on the FMC display shows LO for both packs.

(e) Set the L PACK switch to HIGH.

(f) Make sure that the ECS PACK H/L indication on the FMC display shows HI for the left pack and LO for the right pack.

(g) Set the R PACK switch to HIGH.

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(h) Make sure that the ECS PACK H/L indication on the FMC display shows HI for both packs.

(i) Put the L and R PACK switches, on the P5-10 air conditioning panel, to AUTO position.

(j) Make sure that the ECS PACK H/L indication on the FMC display shows LO for both packs.

(k) Put the L and R PACK switches to OFF.

SUBTASK 21-00-05-780-001

(8) Pressurize the airplane with both packs as follows:

(a) Pressurize the airplane to 4 psid (28 kPa) as follows:

1) Make sure that the APU BLEED switch is set to ON.

2) Make sure that the ISOLATION VALVE switch is set to OPEN.

3) Make sure that the engine BLEED1 and BLEED 2 switches are set to OFF.

4) Make sure that the L PACK and R PACK switches are set to AUTO.

(b) Manually operate the outflow valve MANUAL VALVE CLOSE/OPEN switch to maintain the cabin pressure of 4 psid (28 kPa) as follows:

NOTE: Pressurizing the airplane to 4 psid (28 kPa) for the short duration of the test will not harm the airplane.

1) Make sure that the pressurization mode selector is set to MAN.

2) Move the MANUAL VALVE CLOSE/OPEN toggle switch as necessary to maintain 4 psid (28 kPa) on the DIFF PRESS gauge.

(c) If a differential pressure of 4 psid (28 kPa) cannot be maintained, either a pack supply or airplane leakage problem exists.

1) To repair leakage, do the applicable fault isolation tasks.

F. Phase II Test. Airplane Leak Down Check

SUBTASK 21-00-05-200-001

(1) Do this check of the leakage of the cabin pressure.

(a) Make a record of the cabin temperature.

SUBTASK 21-00-05-210-001

(2) Inspect these components for any sign of leakage:

NOTE: This step is optional.

(a) Doors

(b) Hatches

(c) Equipment cooling valve

(d) Bilge drains

(e) Flight deck windows

(f) Safety relief valves

(g) APU duct seals at the pressure bulkhead penetrations at Sta. 727 and Sta. 1016

(h) Duct seals in the air conditioning bays at Sta. 540

(i) Outflow valve seal.

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SUBTASK 21-00-05-860-007

(3) When 4.2 psid (29.0 kPa)±0.1 psid (0.7 kPa) on the DIFF PRESS gauge has been achieved and stabilized, fully close (manually) the aft outflow valve.

(a) Make sure that the indication on the gauge above the CLOSE/OPEN switch, on the P5-17 pressurization control panel, shows that the valve is fully closed.

SUBTASK 21-00-05-040-001

(4) Put the L PACK and R PACK switches to OFF.

SUBTASK 21-00-05-780-004

CAUTION

KEEP THE APU IN OPERATION. IF YOU DO NOT KEEP THE APU IN OPERATION, EQUIPMENT DAMAGE COULD OCCUR.

(5) Continue to operate the APU.

SUBTASK 21-00-05-700-001

(6) Let the cabin pressure leak down to 4 psid (28 kPa) (time zero) then start the timer.

(a) Record of the pressure differential indications on the Table 202 at 10 second intervals until the cabin pressure reduces to 2.5 psid (17.2 kPa).

Table 202/21-00-05-993-803 Airplane Leak Down (No Pack Operation) - Confidence Check

Airplane_____ Hours_____ Cycles_____ Date_____		
Phase II - Airplane Leak Down (No Pack Operation)		
Cabin temperature _____		
Time (seconds)	Cabin (psid)	Cabin (rate of change)
10		
20		
30		
40		
50		
60		
70		
80		
90		
100		
110		
120		

1) See the table in Performance Summary for system performance evaluation.

SUBTASK 21-00-05-860-008

(7) Depressurize the airplane and restore it to its unpressurized condition as follows:

(a) Turn on either pack by putting the appropriate PACK switch to ON.

(b) Modulate the outflow valve to depressurize the airplane at approximately 500 feet per minute.

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SUBTASK 21-00-05-860-009

(8) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

(9) Make sure that the mode selector switch, on the P5-6 cabin pressurization control panel, is returned to the AUTO position.

SUBTASK 21-00-05-080-001

(10) Remove the nose and main landing gear safety locks (TASK 32-00-01-080-801).

G. Performance Summary

SUBTASK 21-00-05-780-005

(1) Best performance of the air conditioning packs and airplane structure is as follows:

Table 203/21-00-05-993-801

SYSTEM	BEST PERFORMANCE
For -7/-8/-8200/-9: Cabin Leakage Rate from 4 psid (28 kPa) to 2.5 psid (17.2 kPa)	120-130 seconds is optimum; Airplane will maintain pressure in flight even with slightly degraded inflow. 110-120 seconds is acceptable; Airplane will maintain pressure in flight with normal inflow. 100-110 seconds is marginally acceptable; Airplane will maintain pressure in flight with normal inflow. Less than 100 seconds is unacceptable; Airplane will not maintain pressure in flight. Perform maintenance.
For -10: Cabin Leakage Rate from 4 psid (28 kPa) to 2.5 psid (17.2 kPa)	Above 180 seconds is optimum; Airplane will maintain pressure in flight even with slightly degraded inflow. 170-180 seconds is acceptable; Airplane will maintain pressure in flight with normal inflow. 160-170 seconds is marginally acceptable; Airplane will maintain pressure in flight with normal inflow. Less than 160 seconds is unacceptable; Airplane will not maintain pressure in flight. Perform maintenance.

SUBTASK 21-00-05-780-006

(2) If the test data agrees with the best performance data for the packs (inflow) and the airplane structure (outflow), you can be very sure that cabin pressure can be maintained at cruise (FL250) on a single pack.

SUBTASK 21-00-05-780-007

(3) Refer to the data in Figure 202, Table 201, and Table 202 and the troubleshooting information in Figure 203 to better determine the condition of the packs and the airplane and what action if any is recommended.

H. Put the Airplane Back to Its Usual Conditions

SUBTASK 21-00-05-410-002

(1) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-00-05-410-001

(2) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

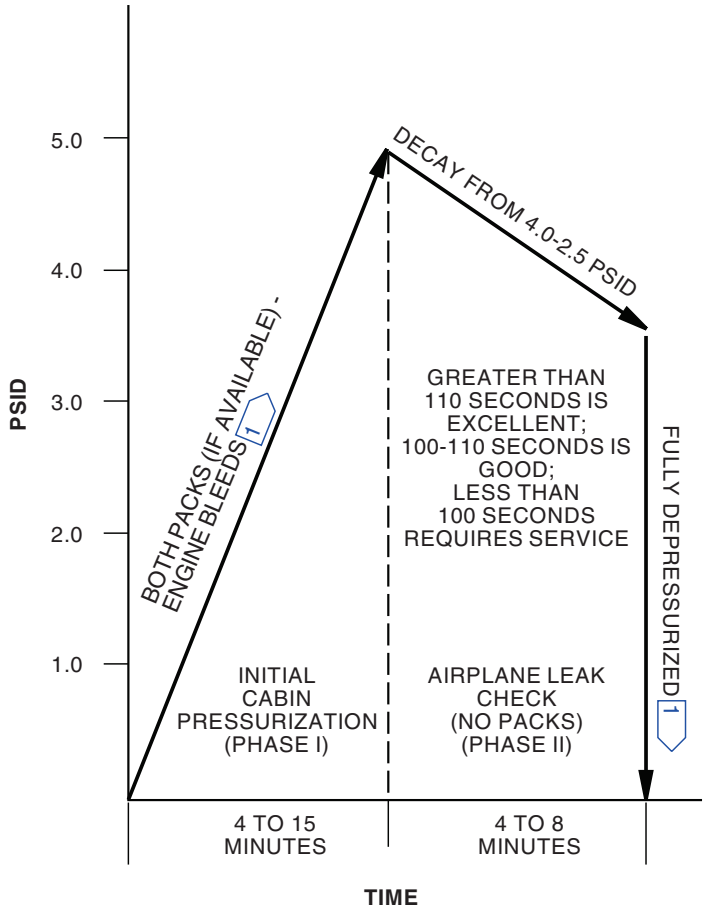
(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

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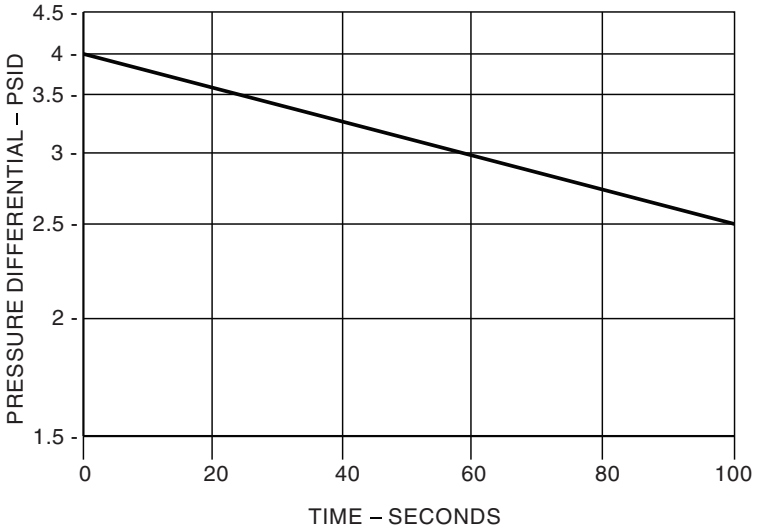
NOTE:

1 OUTFLOW VALVE IN MANUAL MODE

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Airplane Dispatch with Single Pack Operation - Confidence Check
Figure 201/21-00-05-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411251 S00061527395_V3

Airplane Dispatch with Single Pack Operation - Confidence Check
Figure 202/21-00-05-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SYSTEM	MINIMUM REQUIREMENT	REQUIREMENT SATISFIED	
		YES	NO
AIRPLANE/ STRUCTURE AIR LEAKAGE	LEAK DOWN RATE FROM 4.0 TO 2.5 PSID IS GREATER THAN OR EQUAL TO A MINIMUM OF 100 SECONDS (110 TO 120 SECONDS IS THE TARGET)	CABIN LEAKAGE IS ACCEPTABLE. OPERATORS MAY WISH TO CONSIDER CONDUCTING AMM 05-51-91/201 LEAKAGE TEST AT THE NEXT HEAVY MAINTENANCE CHECK BASIS AND MONITOR PERFORMANCE.	INSPECT AND REPAIR ANY AIR LEAKAGE AS NOTED ON THE DATA SHEET. SPECIFICALLY: DOORS AND HATCHES – MISSING OR TORN SEALS, POOR CONTACT BETWEEN DOOR SEALS AND AIRPLANE DUE TO DIRT, ETC., INSPECT SEALING COMPOUND AREAS AND REPAIR DOOR SEALS (AMM 52-09-10/801). OVERBOARD EXHAUST VALVE – P/N 20798-01 OEV SHOULD CLOSE AT 0.7 TO 1.1 PSID. **NOTE:** ALL BILGE DRAINS OPEN IS EQUIVALENT TO A 3.75 INCH (95.25 mm) DIAMETER HOLE IN THE PRESSURE VESSEL. BILGE DRAINS – SHOULD CLOSE AT APPROXIMATELY 2.0 PSID: – CABIN PRESSURE SAFETY RELIEF VALVE – OUTFLOW VALVE SEAL – NEGATIVE PRESSURE RELIEF VALVE SEAL – WATER SERVICE PANEL – FLIGHT DECK WINDOWS – PRESSURE BULKHEAD PENETRATIONS SHOULD ALL HAVE MINIMAL LEAKAGE – SIGNIFICANT LEAKS SHOULD BE RESOLVED AS REQUIRED.

2411253 S00061527397_V2

Airplane Dispatch with Single Pack Operation - Troubleshooting
Figure 203/21-00-05-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR CONDITIONING SYSTEM DUCTS - REPAIRS

1. General

A. This procedure gives instructions for the repair of air conditioning ducts that are made of these materials:

(1) Fiberglass
(2) Kevlar
(3) Polyurethane foam
(4) Ducts wrapped with Foam Insulation
(5) Ducts with Fiberglass Insulation/Flame-Resistant Covering

B. This procedure has these tasks:

(1) Fiberglass or Kevlar Ducts
  (a) Taped Joint Repair
  (b) External Patch Repair
  (c) Structural Repair
(2) Polyurethane Foam Duct
  (a) Taped Joint Repair
  (b) Duct End Repair
  (c) External Patch Repair
  (d) Structural Repair
  (e) Inner Lining Repair
  (f) Duct Section Replacement
(3) Ducts wrapped with Foam Insulation
  (a) Tape Replacement
  (b) Insulation Repair
  (c) Insulation Replacement
(4) Ducts with Fiberglass Insulation/Flame-Resistant Covering
  (a) Tape Replacement
  (b) Insulation and Covering Repair
  (c) Insulation and Covering Replacement

C. This procedure also uses these materials

(1) Tape - pressure sensitive glass cloth (Scotch No. 361)
(2) Polyurethane foam - rigid (BMS8-131 Type I Grade 20 Form A)
(3) Polyurethane foam duct (to match damaged duct)
(4) Plastic sheet - polyethylene or vinyl (commercially available)
(5) Potting compound - BMS 5-28, Type 19 (Epocast 1619A/B)
(6) Sandpaper - 320 grit (commercially available)

D. Some of the ducts in the air distribution system are wrapped with insulation material. CFR 14 Part 121.312 mandates that insulation material installed in the fuselage as a replacement after September 2, 2005 must meet the flame propagation requirements of CFR Part 25.856, effective September 2, 2003.

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(1) Existing insulation material can be removed and re-installed if it is not damaged. Replacement insulation material must meet the flame propagation requirement.

(2) BMS 8-385 Type 4 is the preferred foam insulation material that meets the flame propagation requirement. BMS 8-385 Type 6 and BMS 8-300 Type VIII insulation are new, qualified materials optional to BMS 8-385 Type 4.

(3) If BMS 8-300, Type II, Grade 5 foam insulation material is used, it must be completely covered with BMS 5-157, Type I, Class 1, Grade B, composition MPVF tape or BMS 5-157, Type I, Class 2, Grade C, composition MPEEK that meets the flame propagation requirement.

(4) Do not use BMS8-39 polyurethane foam insulation for repairs due to the degradation in flammability properties over time.

(5) When insulation material is removed and re-installed, any existing tape that does not meet the flame propagation requirement must be removed or completely covered with tape that does meet the requirement.

(a) Tape specifications BMS 5-157, Type I, Class 1, Grade B, composition MPVF and BMS 5-157, Type I, Class 2, Grade C, composition MPEEK are qualified tapes used to secure insulation to ducts.

TASK 21-20-02-330-801

2. Taped Joint Repair - Fiberglass/Kevlar Duct

(Figure 801)

A. General

(1) This procedure is for the repair of a taped joint that connects two sections of Fiberglass or Kevlar air conditioning ducts.

(2) This procedure may also be used to make a temporary repair to a damaged Fiberglass or Kevlar air conditioning duct when time is not available to make a permanent repair or if the materials are not available to make a permanent repair. If a temporary repair is made, make the permanent repair as soon as possible.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00066	Sealant - RTV - Dow Corning 90-006	
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
C00580	Primer - Adhesive Bonding, RTV - PR-1200 (Formerly DC 1200)	
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
G50449	Foam - Flexible Melamine	BMS8-385 Type 4 Grade 1
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51169	Foam - Flexible Melamine,	BMS8-385 Type 6
G51170	Foam - Flexible Polyimide	BMS8-300 Type VIII
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

D. Prepare for the Repair

SUBTASK 21-20-02-840-001

(1) Do this task:Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-002

(2) Gain access to the damaged air conditioning duct joint.

SUBTASK 21-20-02-840-003

(3) If insulation is installed on the duct joint, then remove the insulation.

SUBTASK 21-20-02-840-004

(4) Do these steps to remove loose tape and unwanted adhesive from the duct joint.

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Use the solvent, B00148 and a cotton wiper, G00034 to clean the duct joint area.

(b) Sand the duct joint area to remove any adhesive that remains on the duct joint.

(c) Use the solvent, B00148 and a cotton wiper, G00034 to clean the duct joint area.

SUBTASK 21-20-02-840-005

(5) Cut a piece of the glass fabric, G00031 that is 3 inches wide and is long enough to go around the duct a minimum of two times.

E. Repair the Taped Joint

SUBTASK 21-20-02-340-001

(1) Apply a thin, smooth layer of the PR-1200 primer, C00580 to the duct joint area and let it dry at room temperature for a minimum of 1 hour.

SUBTASK 21-20-02-340-002

(2) Apply the Dow Corning 90-006 sealant, A00066 to one side of the Fiberglass glass fabric, G00031.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-20-02-340-003

(3) Wrap the Fiberglass glass fabric, G00031 around the duct joint a minimum of two times.

NOTE: Make sure that the sealant on the Fiberglass glass fabric, G00031 touches the PR-1200 primer, C00580 on the duct.

SUBTASK 21-20-02-340-004

(4) Smooth the Fiberglass glass fabric, G00031 to remove any bubbles or any wrinkles.

SUBTASK 21-20-02-340-005

(5) Wrap the glass cloth tape around the two edges of the Fiberglass fabric repair. Make sure the tape lies flat with no wrinkles or gaps.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-006

(1) If insulation was removed from the duct joint, install new insulation:

(a) You must install insulation that meets one of these specifications:

1) BMS 8-385 Type 4 foam insulation, G50449

2) BMS 8-385 Type 6 foam insulation, G51169

3) BMS 8-300 Type VIII foam insulation, G51170.

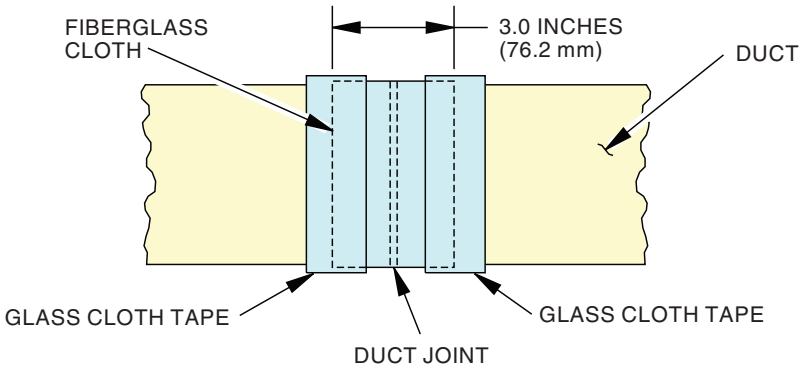
(b) Use BMS 5-157 MPVF tape, G50327, tape, G51202 or BMS 5-157 MPEEK tape, G50789 to install the insulation.

SUBTASK 21-20-02-840-007

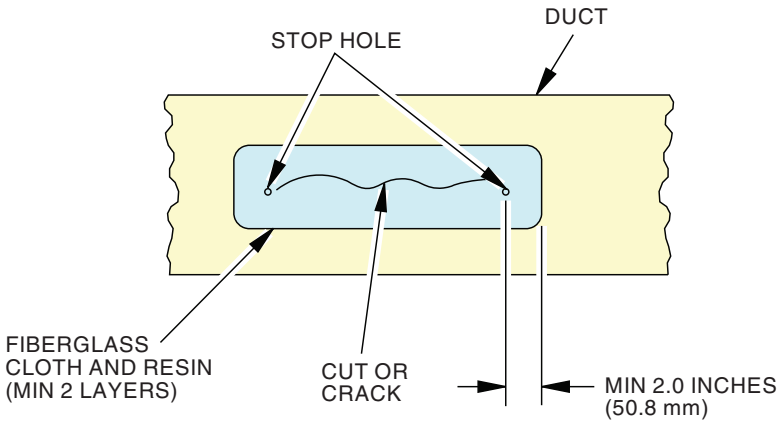
(2) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TAPED JOINT REPAIR - FIBERGLASS/KEVLAR DUCT

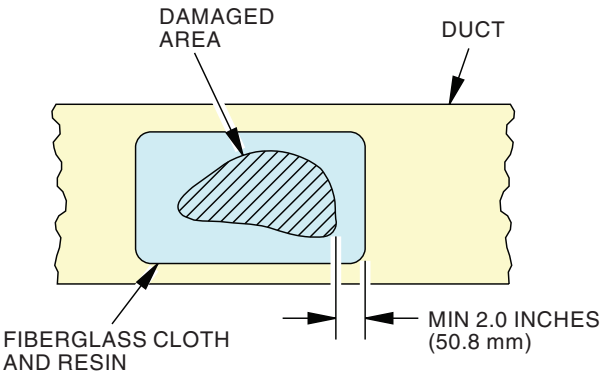


EXTERNAL PATCH - FIBERGLASS/KEVLAR DUCT

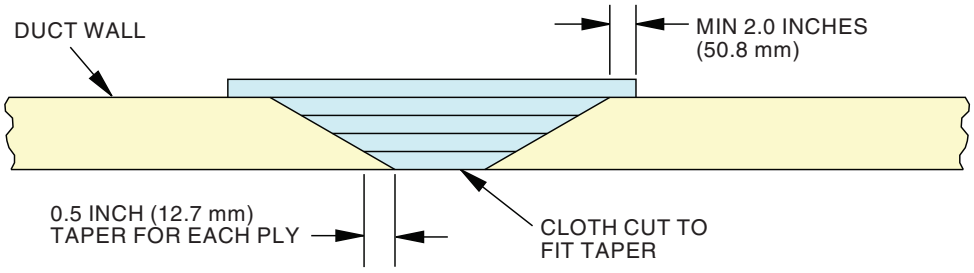
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Air Conditioning System Ducts Repair
Figure 801/21-20-02-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STRUCTURAL REPAIR - FIBERGLASS/KEVLAR DUCT - OVERHEAD VIEW

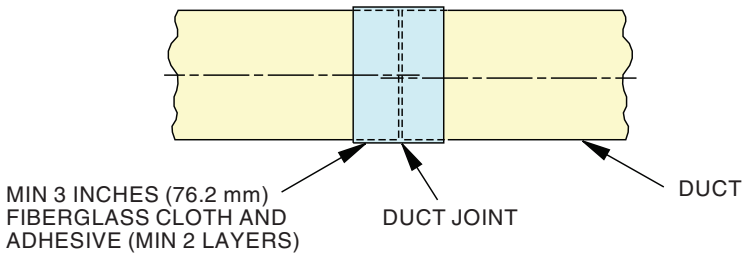


STRUCTURAL REPAIR - FIBERGLASS/KEVLAR DUCT - CROSS-SECTION VIEW

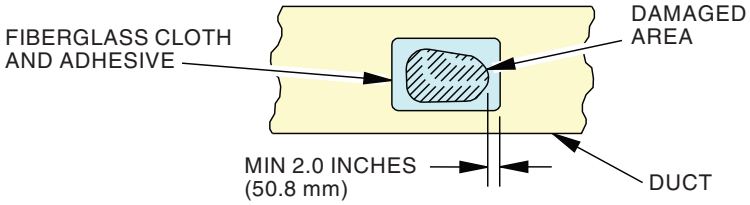
2411256 S00061527404_V1

Air Conditioning System Ducts Repair
Figure 801/21-20-02-990-801 (Sheet 2 of 3)

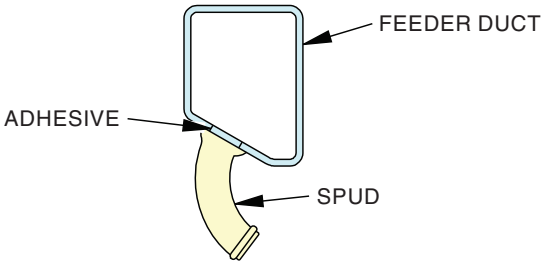
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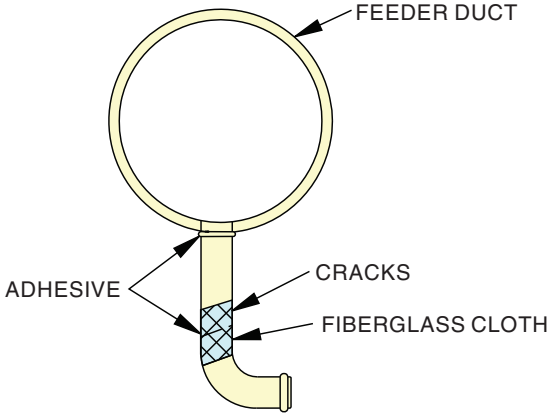
REPAIR FOR THE TAPE CONNECTION AT THE DUCT JOINT AREA



REPAIR OF A DUCT WITHOUT INNER LINING DAMAGE



FEEDER DUCT REPAIR



FEEDER DUCT REPAIR

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Air Conditioning System Ducts Repair
Figure 801/21-20-02-990-801 (Sheet 3 of 3)

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TASK 21-20-02-330-802

3. External Patch - Fiberglass/Kevlar Duct

(Figure 801)

A. General

(1) This procedure gives instructions for the repair of a crack or cut in a Fiberglass or Kevlar air conditioning duct with an external patch.

(2) An external patch is applicable when the cut or the crack is less than 1/4 inch wide and is less than 3 inches long.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00040	Resin - Self-Extinguishing Laminating - Epocast 50-A1 with Epocast 9816	BMS8-201 Type III (Long work life)
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02137	Resin - Polyester - Hetron 92	
G50449	Foam - Flexible Melamine	BMS8-385 Type 4 Grade 1
G51169	Foam - Flexible Melamine,	BMS8-385 Type 6
G51170	Foam - Flexible Polyimide	BMS8-300 Type VIII

D. Prepare for the Repair

SUBTASK 21-20-02-840-008

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-009

(2) Gain access to the damaged air conditioning duct joint.

SUBTASK 21-20-02-840-010

(3) If insulation is installed on the duct, then remove the insulation.

SUBTASK 21-20-02-840-011

(4) Do these steps to prepare the damaged area of the duct.

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Use the solvent, B00148 and a cotton wiper, G00034 to clean the duct joint area.

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(b) Drill a hole at each end of the cut or crack to prevent the cut or crack from getting larger. The holes should be a maximum of 1/16 inch diameter.

(c) Sand the damaged area to remove any unwanted contamination from the duct.

(d) Use the solvent, B00148 and a cotton wiper, G00034 to clean the duct joint area.

SUBTASK 21-20-02-840-012

(5) Cut sufficient pieces of the Fiberglass glass fabric, G00031 to equal the thickness of the duct. Cut the Fiberglass glass fabric, G00031 pieces 1/2 inch larger on all sides than the damage. Round the corners of the pieces of the Fiberglass fabric.

SUBTASK 21-20-02-840-013

(6) Prepare the applicable resin:

(a) Fiberglass duct repair - Hetron 92 resin, G02137

(b) Kevlar duct repair - resin, A00040

E. Repair the Air Conditioning Duct

SUBTASK 21-20-02-340-006

(1) Apply a thin, smooth layer of the applicable Hetron 92 resin, G02137 or resin, A00040 to the duct where the Fiberglass fabric will be applied. Make sure the area of the resin is a minimum of the size and shape of the Fiberglass fabric pieces.

SUBTASK 21-20-02-340-007

(2) Place a piece of the Fiberglass fabric on the resin so that it extends beyond the cut or crack a minimum of 1/2 inch on all sides.

SUBTASK 21-20-02-340-008

(3) Soak the Fiberglass fabric with the applicable resin.

SUBTASK 21-20-02-340-009

(4) Put a piece of plastic sheet on the Fiberglass fabric and press lightly to remove any bubbles or any wrinkles and to remove unwanted resin.

SUBTASK 21-20-02-340-010

(5) Remove the plastic sheet from the Fiberglass fabric piece and remove any unwanted resin with a wiping cloth.

SUBTASK 21-20-02-340-011

(6) Place another piece of the Fiberglass fabric on the first piece, soak it with resin, and smooth it. Continue using the same procedure until the thickness of the patch equals the thickness of the duct.

SUBTASK 21-20-02-340-012

(7) Allow the resin to cure for 24 hours at room temperature.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-014

(1) If insulation was removed from the duct joint, install new insulation:

(a) You must install insulation that meets one of these specifications:

1) BMS 8-385 Type 4, foam insulation, G50449

2) BMS 8-385 Type 6, foam insulation, G51169

3) BMS 8-300 Type VIII, foam insulation, G51170I.

(b) Use BMS 5-157 MPVF or BMS 5-157 MPEEK tape to install the insulation.

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SUBTASK 21-20-02-840-015

(2) Replace the panels or other airplane structure removed to gain access to the duct joint.

END OF TASK

TASK 21-20-02-330-803

4. Structural Repair - Fiberglass/Kevlar Duct

(Figure 801)

A. General

(1) This procedure gives instructions for making a structural repair to a fiberglass or Kevlar air conditioning duct.

(2) A structural repair is applicable when the damage to the air conditioning duct is a cut or a hole that is more than 0.25 in. (6.35 mm) wide and more than 3 in. (76.2 mm) long but less than 5 percent of the total surface area of the duct. If the damage is more than 5 percent of the total surface area of the duct, then replace the duct section.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-94	Bag - Vacuum
STD-122	Brush
STD-821	Squeegee - Plastic

D. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02137	Resin - Polyester - Hetron 92	
G50104	Film, FEP or TFE Solid Parting Film, 0.001 inch Thick Minimum	
G50173	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type H-3 Style 7781
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50399	Resin - Fiberglass Layup, Long Worklife, Non-Brominated	BMS8-201 Type III (Supersedes BMS8-201 Type I)
G50400	Resin - Fiberglass Layup, Short Worklife, Non-Brominated	BMS8-201 Type IV (Supersedes BMS8-201 Type II)

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(Continued)

Reference	Description	Specification
G50449	Foam - Flexible Melamine	BMS8-385 Type 4 Grade 1
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G50797	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type D Style 120
G50798	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type D-1 Style 220
G51169	Foam - Flexible Melamine,	BMS8-385 Type 6
G51170	Foam - Flexible Polyimide	BMS8-300 Type VIII
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF
G51266	Fabric - Glass Fabric Reinforcements For Laminated Plastics Products	BMS9-3 Type H-2 Style 1581

E. Prepare for the Repair

SUBTASK 21-20-02-840-016

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-017

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-018

(3) If insulation is installed on the duct, then remove the insulation.

SUBTASK 21-20-02-840-019

(4) Do these steps to prepare the damaged area of the duct.

(a) Cut away the damaged area to make a rounded opening with a smooth shape.

1) Cut the sides of the opening at an angle to make a taper of 0.5 in. (12.7 mm) between each ply of material.

(b) Sand the edge of the opening to make a smooth surface.

(c) Sand the surface of the duct around the opening a distance of 0.5 in. (12.7 mm) for each ply in the duct.

NOTE: For example, if there are 6 plies, sand 3 in. (76.2 mm) from the opening on all sides (1/2 x 6 = 3 inches) or (12.7 x 6 = 76.2 millimeters).

WARNING

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(d) Use the solvent, B00148, and a cotton wiper, G00034, to clean the edge of the opening and the sanded surface of the duct.

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SUBTASK 21-20-02-840-020

(5) Cut sufficient pieces of the glass fabric, G00031, fabric, G50173, or fabric, G51266, to be equal to the thickness of the duct.

(a) Cut each piece of the fiberglass fabric 0.5 in. (12.7 mm) larger to fit the taper of the opening.

(b) Cut one piece of the fiberglass fabric a minimum of 2 in. (50.8 mm) larger on all sides than the opening in the duct.

SUBTASK 21-20-02-840-021

(6) Cut a piece of solid parting film, G50104, 2 in. (50.8 mm) to 3 in. (76.2 mm) larger on all sides than the opening.

SUBTASK 21-20-02-840-103

(7) Cut a piece of cardboard or aluminum 2 in. (50.8 mm) to 3 in. (76.2 mm) larger than the opening on all sides.

SUBTASK 21-20-02-840-022

(8) Attach the solid parting film, G50104, to the inside of the opening with adhesive tape so that it covers the opening.

SUBTASK 21-20-02-840-104

(9) Attach the cardboard or aluminum to the inside of the opening with adhesive tape so that it covers the opening and provides support for the solid parting film, G50104.

SUBTASK 21-20-02-840-023

(10) Prepare the applicable resin:

(a) Fiberglass duct repair - Hetron 92 resin, G02137, or resin, G50399

(b) Kevlar duct repair - resin, G50399, or resin, G50400

F. Repair the Air Conditioning Duct

SUBTASK 21-20-02-340-013

(1) Apply a thin, smooth layer of resin to the solid parting film, G50104, in the opening in the duct.

SUBTASK 21-20-02-340-014

(2) Put a piece of the fiberglass fabric in the opening over the solid parting film, G50104, so that it covers the opening completely.

SUBTASK 21-20-02-340-015

(3) Soak the fiberglass fabric with the applicable resin.

SUBTASK 21-20-02-340-016

(4) Put a piece of solid parting film, G50104, on the fiberglass fabric.

(a) Apply light pressure with a plastic squeegee, STD-821, or with your hand.

(b) From the center of the fabric to the edges, remove all bubbles, wrinkles, and unwanted resin.

SUBTASK 21-20-02-340-017

(5) Remove the solid parting film, G50104, from the fiberglass fabric piece.

SUBTASK 21-20-02-340-102

(6) Remove unwanted resin with a cotton wiper, G00034.

SUBTASK 21-20-02-340-018

(7) Put the next piece of the fiberglass fabric on the first piece of fabric.

(a) Make sure that the fabric is on all of the opening.

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(b) If it is necessary, cut the fiberglass fabric.

(c) Soak the fiberglass fabric with resin.

1) Make sure that the resin is smooth.

(d) Continue with the same procedure until the thickness of the patch is equal to the thickness of the duct.

SUBTASK 21-20-02-340-019

(8) Put the last piece of the fiberglass fabric on top of the patch so it extends beyond the opening a minimum of 2 in. (50.8 mm) on all sides.

(a) Make the fiberglass fabric smooth to remove bubbles or wrinkles.

SUBTASK 21-20-02-340-020

WARNING

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(9) Wipe the duct around the patch with a cotton wiper, G00034, and solvent, B00148, to remove unwanted resin.

SUBTASK 21-20-02-340-104

(10) To cure resin, G50399, or resin, G50400, do these steps (Structural Repair Manual (SRM) 51-30-10):

(a) Attach a vacuum bag, STD-94, to hold the repair layers and to keep the duct shape.

1) Allow the resin to cure for 24 hours at 75 ±5°F (24 ±3°C).

2) Cure for a minimum of 2 - 2.5 hours under a minimum of 20 in/Hg (68 kPa) at 170 ±10°F (77 ±6°C).

NOTE: Moisture protection is not necessary during 170 ±10°F (77 ±6°C) cure.

SUBTASK 21-20-02-340-021

(11) To cure Hetron 92 resin, G02137, allow the resin to cure for 24 hours at room temperature.

G. Repair the Beaded End

NOTE: More repair plies are necessary on the external and internal surfaces. If no damage is found on the duct plies (the plies below the bead), do the bead repairs and not the duct plies repairs. Only repair the ply that attaches to the bead.

SUBTASK 21-20-02-340-105

(1) Prepare the applicable resin:

(a) Fiberglass duct repair - Hetron 92 resin, G02137, or resin, G50399.

(b) Kevlar duct repair - resin, G50399, or resin, G50400.

SUBTASK 21-20-02-340-090

(2) Apply a thin, smooth layer of resin to the solid parting film, G50104, in the opening in the duct.

SUBTASK 21-20-02-340-091

(3) Put a piece of fabric in the opening on the solid parting film, G50104.

(a) Use fabric, G50797, fabric, G50798, fabric, G50173, glass fabric, G00031, fabric, G50173, or fabric, G51266, to repair the beaded end.

(b) Make sure that the fabric is on all of the opening.

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SUBTASK 21-20-02-340-092

(4) Soak the fabric with resin.

SUBTASK 21-20-02-340-093

(5) Put a piece of the solid parting film, G50104, on the fabric.

(a) Apply light pressure to the solid parting film, G50104, with a plastic squeegee, STD-821, or with your hand.

(b) From the center of the fabric to the edges, remove all bubbles, wrinkles, and unwanted resin.

SUBTASK 21-20-02-340-094

(6) Remove the solid parting film, G50104, from the fiberglass fabric piece.

SUBTASK 21-20-02-340-103

(7) Remove unwanted resin with a cotton wiper, G00034.

SUBTASK 21-20-02-340-095

(8) Put the next piece of fabric on the first piece of fabric.

(a) Make sure that the fabric is on all of the opening.

(b) If it is necessary, cut the fabric.

(c) Soak the fabric with resin.

1) Make sure that the resin is smooth.

(d) Continue with the same procedure with fabric until the bead agrees with the necessary duct end dimensions.

SUBTASK 21-20-02-340-096

(9) Cut the last piece of fabric to the necessary dimensions:

NOTE: The dimensions can include the repaired bead and the internal and external surfaces of the full diameter of the duct.

(a) Make sure that the external layer of the Outer Mold Line (OML) is on all of the repaired bead and is around the part end.

(b) Make sure that the external layer extends more than 0.5 in. (12.7 mm) from the repair layer of the Inner Mold Line (IML).

(c) From the center of the fabric to the edges, remove all bubbles or wrinkles.

SUBTASK 21-20-02-340-097

WARNING

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(10) Clean the duct around the patch with a cotton wiper, G00034, and solvent, B00184, to remove unwanted resin.

SUBTASK 21-20-02-340-098

(11) To cure the resin, G50399, or resin, G50400 (SRM 51-30-10):

(a) Attach a vacuum bag, STD-94, to hold the repair layers and to keep the duct shape.

1) Allow the resin to cure for 24 hours at 75 ±5°F (24 ±3°C).

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2) Cure for a minimum of 2 - 2.5 hours under a minimum of 20 in/Hg (68 kPa) at 170 ±10°F (77 ±6°C).

NOTE: Moisture protection is not necessary during 170 ±10°F (77 ±6°C) cure.

SUBTASK 21-20-02-340-106

(12) To cure Hetron 92 resin, G02137, allow the resin to cure for 24 hours at room temperature.

SUBTASK 21-20-02-340-099

(13) If it is necessary, sand the patch on the outer duct surface to get a correct fit.

(a) Keep a wall thickness of 0.045 in. (1.143 mm) - 0.065 in. (1.651 mm) in this area.

NOTE: Wrinkles, creases, and unwanted resin are not permitted in this area. Sand these items flush where necessary.

SUBTASK 21-20-02-340-100

(14) Use a brush, STD-122, to seal exposed fabric with a layer of resin.

SUBTASK 21-20-02-340-101

(15) Let the resin cure.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-102

(1) Remove the solid parting film, G50104, the cardboard or aluminum, and the adhesive tape you used to support the repair.

SUBTASK 21-20-02-840-024

(2) If insulation was removed from the duct joint, install new insulation:

(a) You must install insulation that meets one of these specifications:

1) BMS 8-385 Type 4 foam insulation, G50449

2) BMS 8-385 Type 6 foam insulation, G51169

3) BMS 8-300 Type VIII foam insulation, G51170.

(b) Use BMS 5-157 MPVF tape, G50327, tape, G51202, or BMS 5-157 MPEEK tape, G50789, to install the insulation.

SUBTASK 21-20-02-840-025

(3) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

TASK 21-20-02-330-804

5. Taped Joint Repair - Polyurethane Foam Duct

(Figure 802)

A. General

(1) This procedure gives instructions for the repair of the taped joint that connects sections of polyurethane foam air conditioning duct.

(2) This procedure may also be used to make a temporary repair to a damaged polyurethane foam air conditioning duct when time is not available to make a permanent repair or if the materials are not available to make a permanent repair. If a temporary repair is made, make the permanent repair as soon as possible.

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B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Prepare for the Repair

SUBTASK 21-20-02-840-026

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-027

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-028

(3) Do these steps to remove loose tape and unwanted adhesive from the duct joint.

WARNING

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(a) Use the solvent, B00148 and a cotton wiper, G00034 to clean the duct joint area.

(b) Sand the duct joint area to remove any adhesive that remains on the duct joint.

(c) Use the solvent, B00148 and a cotton wiper, G00034 to clean any dust from the duct joint area.

SUBTASK 21-20-02-840-029

(4) Examine the end of each air conditioning duct.

SUBTASK 21-20-02-840-030

(5) If the duct lining is damaged so that the foam core of the duct shows, then do the Duct End Repair - Polyurethane Foam Duct procedure below.

SUBTASK 21-20-02-840-031

(6) Cut a piece of the Fiberglass glass fabric, G00031 that is 3 inches wide and is long enough to go around the duct a minimum of two times.

SUBTASK 21-20-02-840-032

(7) Prepare the applicable resin.

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E. Repair the Taped Joint

SUBTASK 21-20-02-340-022

(1) Apply resin to the duct ends. Allow the resin to partially dry until it is sticky. Press the duct ends together by aligning the installation checkmarks.

SUBTASK 21-20-02-340-023

(2) Apply a thin, smooth layer of resin to the joint area a minimum of 2 inches wider than the duct joint.

SUBTASK 21-20-02-340-024

(3) Apply resin to one side of the Fiberglass glass fabric, G00031.

SUBTASK 21-20-02-340-025

(4) Wrap the Fiberglass fabric around the duct joint a minimum of two times. Make sure the resin on the Fiberglass fabric is in contact with the duct.

SUBTASK 21-20-02-340-026

(5) Smooth the Fiberglass fabric to remove bubbles or wrinkles.

SUBTASK 21-20-02-340-027

(6) Allow the resin to cure for 24 hours at room temperature.

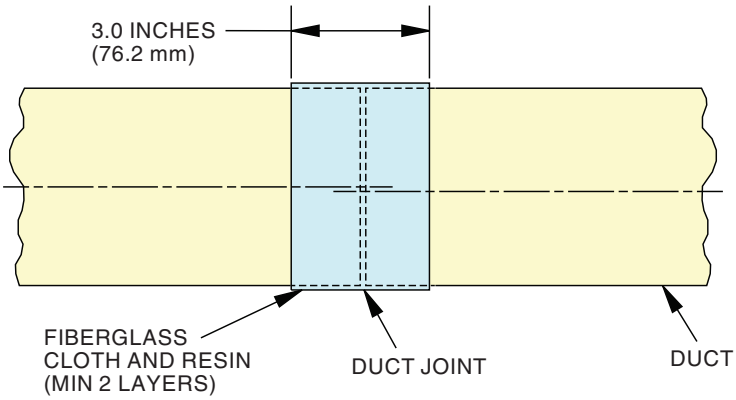
F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-033

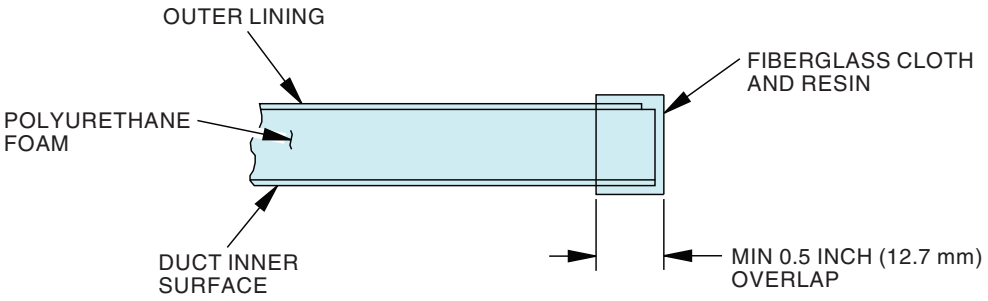
(1) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

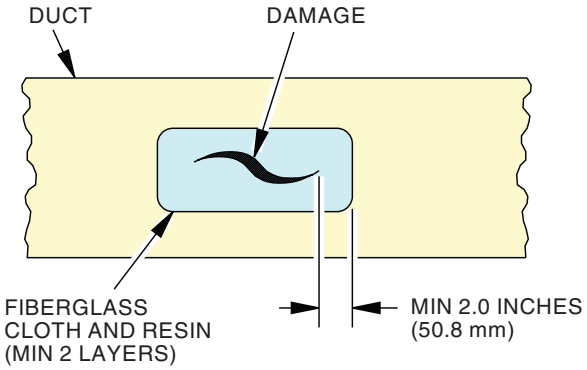
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TAPED JOINT REPAIR - POLYURETHANE FOAM DUCT



DUCT END REPAIR - POLYURETHANE FOAM DUCT

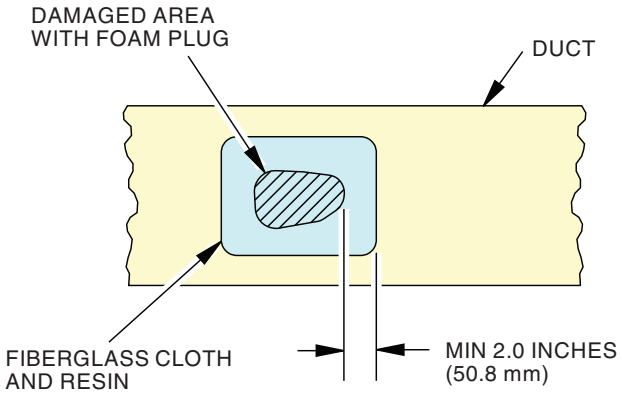


EXTERNAL PATCH - POLYURETHANE FOAM DUCT

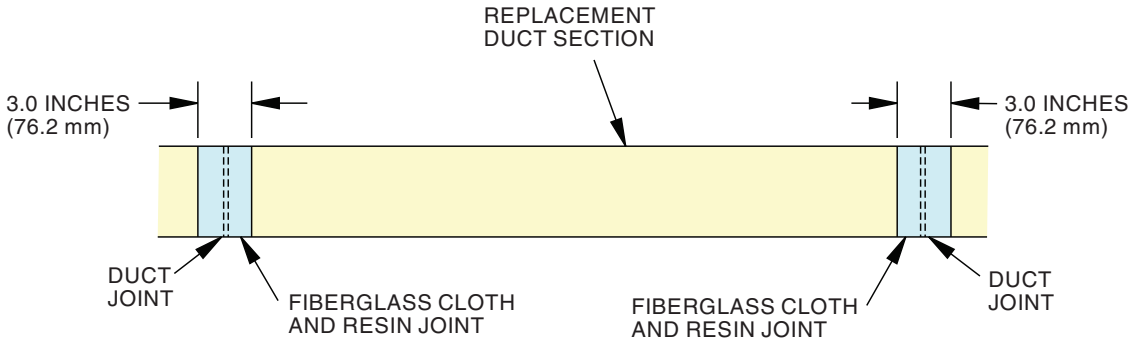
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Air Conditioning System Ducts Repair
Figure 802/21-20-02-990-802 (Sheet 1 of 2)

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STRUCTURAL REPAIR - POLYURETHANE FOAM DUCT



DUCT SECTION REPLACEMENT - POLYURETHANE FOAM DUCT

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Air Conditioning System Ducts Repair
Figure 802/21-20-02-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-20-02-330-805

6. Duct End Repair - Polyurethane Foam Repair

(Figure 802)

A. General

(1) This procedure gives instructions for the repair of the end of a polyurethane foam air conditioning duct. This procedure is done if the lining that covers the polyurethane foam at the duct end is damaged and the polyurethane foam is exposed.

(2) This procedure may be required when doing the repair of a taped duct joint.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00834	Cloth - Lint-free Cotton	

D. Prepare for the Repair

SUBTASK 21-20-02-840-034

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-035

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-036

(3) Do these steps to remove loose tape and unwanted adhesive from the duct end area.

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Use solvent, B00148 and a lint-free cloth, G00834 to clean the duct end area.

NOTE: Clean the inner surface and the outer surface of the duct a minimum of 1 inch from the duct end.

(b) Sand the duct end area to remove any adhesive that remains.

(c) Use the solvent, B00148 and a lint-free cloth, G00834 to clean any dust from the duct end area.

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SUBTASK 21-20-02-840-037

(4) Cut a piece of the Fiberglass glass fabric, G00031 the length of the outside circumference of the duct. The Fiberglass fabric should have the width needed to overlap the duct end and the inside and outside of the duct a minimum of 1/2 inch.

SUBTASK 21-20-02-840-038

(5) Prepare the resin.

E. Repair the Duct End

SUBTASK 21-20-02-340-028

(1) Apply a thin, smooth layer of resin on the duct end and on the surface of the duct a minimum of 1/2 inch from the duct end.

SUBTASK 21-20-02-340-029

(2) Wrap the Fiberglass fabric around the duct end so that it overlaps a minimum of 1/2 inch on both sides of the end.

SUBTASK 21-20-02-340-030

(3) Soak the Fiberglass fabric with resin.

SUBTASK 21-20-02-340-031

(4) Put a piece of plastic sheet on the Fiberglass fabric and press lightly with your hand to smooth the edges of the Fiberglass fabric and to remove bubbles or wrinkles.

SUBTASK 21-20-02-340-032

(5) Allow the resin to cure for 24 hours at room temperature.

SUBTASK 21-20-02-340-033

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(6) Clean the area with a cotton wiper, G00034 and solvent, B00148 to remove unwanted resin.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-039

(1) Replace the panels or other airplane structure removed to gain access to the duct joint.

END OF TASK

TASK 21-20-02-330-806

7. External Patch - Polyurethane Foam Duct

(Figure 802)

A. General

(1) This procedure gives instructions for the repair of a polyurethane foam air conditioning duct if the damage is less than 1/4 inch deep and less than 1 inch long and the inner lining is not damaged. If the damage is larger, do the Structural Repair - Polyurethane Foam Duct procedure below.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Prepare for the Repair

SUBTASK 21-20-02-840-040

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-041

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-210-001

(3) Examine the inner surface of the duct.

SUBTASK 21-20-02-210-002

(4) If the inner lining is damaged, then do the Duct Section Replacement - Polyurethane Foam Duct procedure below.

SUBTASK 21-20-02-210-003

(5) Examine the damage to the duct. If the damage is more than 3 inches long or more than 5 square inches in area, then do the Duct Section Replacement - Polyurethane Foam Duct procedure below.

SUBTASK 21-20-02-840-042

(6) Do these steps to prepare the air conditioning duct:

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Use the solvent, B00148 and a cotton wiper, G00034 to clean the damaged area.

(b) Sand the damaged area to remove any unwanted contamination.

(c) Use the solvent, B00148 and a cotton wiper, G00034 to clean any dust from the damaged area.

SUBTASK 21-20-02-840-043

(7) Cut two pieces of the Fiberglass glass fabric, G00031 1/2 inch larger on all sides than the damaged area. Round the corners of the Fiberglass fabric pieces.

SUBTASK 21-20-02-840-044

(8) Prepare the applicable resin.

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E. Repair the Duct

SUBTASK 21-20-02-340-034

(1) Apply a thin, smooth layer of resin to the duct where the Fiberglass fabric will be applied. Make sure the area of the resin is a minimum of the size and shape of the Fiberglass fabric.

SUBTASK 21-20-02-340-035

(2) Place a piece of the Fiberglass glass fabric, G00031 on the resin so that it extends beyond the damaged area a minimum of 1/2 inch to all sides.

SUBTASK 21-20-02-340-036

(3) Soak the Fiberglass fabric with resin.

SUBTASK 21-20-02-340-037

(4) Put a piece of plastic sheet on the Fiberglass fabric and press lightly with a squeegee or with your hand. Smooth from the center to the edges of the Fiberglass fabric to remove bubbles or wrinkles and to remove unwanted resin.

SUBTASK 21-20-02-340-038

(5) Remove the plastic sheet from the Fiberglass fabric and remove unwanted resin with a cotton wiper, G00034.

SUBTASK 21-20-02-340-039

(6) Follow the same procedure to apply the second piece of the Fiberglass fabric.

SUBTASK 21-20-02-340-040

(7) Allow the resin to cure for 24 hours at room temperature.

SUBTASK 21-20-02-340-041

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(8) Remove unwanted resin from the edges of the Fiberglass fabric with a cotton wiper, G00034 soaked with solvent, B00148.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-045

(1) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

TASK 21-20-02-330-807

8. Structural Repair - Polyurethane Foam Duct

(Figure 802)

A. General

(1) This procedure gives instructions for making a structural repair to a polyurethane foam air conditioning duct. A structural repair replaces the polyurethane foam in the surface of the duct if the damage or deterioration to the polyurethane foam is greater than 1/4 inch deep and 1 inch long but the inner lining is not damaged.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00040	Resin - Self-Extinguishing Laminating - Epocast 50-A1 with Epocast 9816	BMS8-201 Type III (Long work life)
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
B01027	Solvent - Butyl Carbitol (Dow Chemical - E-Series Glycol Ethers)	BAC5750
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00077	Foam - Flame Retardant Rigid Urethane	BMS8-133
G02137	Resin - Polyester - Hetron 92	
G50305	Sheeting - Plastic, Polyethylene	
G50399	Resin - Fiberglass Layup, Long Worklife, Non-Brominated	BMS8-201 Type III (Supersedes BMS8-201 Type I)
G50400	Resin - Fiberglass Layup, Short Worklife, Non-Brominated	BMS8-201 Type IV (Supersedes BMS8-201 Type II)

D. Prepare for the Repair

SUBTASK 21-20-02-840-046

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-047

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-210-004

(3) Examine the inner surface of the duct.

SUBTASK 21-20-02-210-005

(4) If the duct lining is damaged, then do the Duct Section Replacement - Polyurethane Foam Duct procedure below.

SUBTASK 21-20-02-210-006

(5) Examine the damage to the duct. If the damage is more than half the thickness of the foam duct material, then do the Duct Section Replacement - Polyurethane Foam Duct procedure below.

SUBTASK 21-20-02-840-048

(6) Do these steps to prepare the air conditioning duct.

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(a) Remove the damaged portion of the outer lining and the damaged foam material.

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Use the solvent, B00148 and a cotton wiper, G00034 to clean the damaged area.

(c) Sand the duct outer surface a minimum of 2 inches from the edges of the damaged area to remove any unwanted contamination.

(d) Use the solvent, B00148 and a cotton wiper, G00034 to clean any dust from the damaged area.

SUBTASK 21-20-02-840-049

(7) Cut a piece of the polyurethane foam foam, G00077 to fit the area where the damaged foam was removed.

(a) Put the foam in the opening to make sure it is the correct size and shape.

SUBTASK 21-20-02-840-050

(8) Prepare glass fabric, G00031 to cover the damaged area:

(a) Cut two pieces of the glass fabric, G00031 two inches larger on all sides than the damaged area.

(b) Round the corners of the two pieces of the glass fabric, G00031.

SUBTASK 21-20-02-840-051

(9) Prepare the applicable resin, G50400, or resin, G50399, or resin, A00040 or Hetron 92 resin, G02137.

E. Repair the Duct

SUBTASK 21-20-02-340-042

(1) Use the resin mixture to bond the piece of foam, G00077 (plug) to the damaged area on the duct:

(a) Make sure you fill all gaps and voids with the resin mixture.

SUBTASK 21-20-02-340-043

(2) Put the glass fabric, G00031 onto a piece of plastic sheeting, G50305.

SUBTASK 21-20-02-340-044

(3) Pour the resin mixture onto the glass fabric, G00031.

SUBTASK 21-20-02-300-001

(4) Put a second piece of plastic sheeting, G50305 over the glass fabric, G00031.

SUBTASK 21-20-02-300-002

(5) Use a plastic squeegee or your hand to remove unwanted resin, bubbles and wrinkles from the fiberglass fabric as follows:

(a) Smooth from the center of the fiberglass fabric to the edges of the fiberglass fabric.

SUBTASK 21-20-02-300-003

(6) Remove the top piece of the plastic sheeting from the fiberglass fabric.

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SUBTASK 21-20-02-300-004

(7) Put the glass fabric coated with the resin over the foam plug so that it extends beyond the damaged area a minimum of two inches on all sides.

SUBTASK 21-20-02-300-005

(8) Rub the glass fabric over the foam plug with a squeegee until the surface is smooth and unwanted resin is removed.

SUBTASK 21-20-02-340-045

(9) Follow the same steps above to apply a second piece of the glass fabric, G00031 over the damaged area.

SUBTASK 21-20-02-340-046

(10) Allow the resin to cure for 24 hours at room temperature.

SUBTASK 21-20-02-300-006

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(11) Remove unwanted resin from the edges of the repaired area with a cotton wiper, G00034 soaked with Butyl Carbitol solvent, B01027, solvent, B00184 or solvent, B00062.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-052

(1) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

TASK 21-20-02-330-808

9. Inner Lining Repair - Polyurethane Foam Duct

A. General

(1) This procedure gives instructions for the repair of the inner lining on a polyurethane foam air conditioning duct. Use this procedure if the inner lining has peeled away from the duct but the polyurethane foam is not damaged.

(2) If the polyurethane foam is damaged, do the procedure to replace a section of polyurethane foam air conditioning duct.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB

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(Continued)

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Prepare for the Repair

SUBTASK 21-20-02-840-053

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-054

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-055

(3) Do these steps to prepare the air conditioning duct.

(a) Cut away the inner lining that has peeled away from the foam. Do not damage the polyurethane foam.

(b) Examine the polyurethane foam. If the foam is damaged, do the Duct Section Replacement - Polyurethane Foam Duct procedure below.

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(c) Use the solvent, B00148 and a cotton wiper, G00034 to clean the damaged area.

(d) Sand the duct joint area to remove any unwanted contamination on the duct.

(e) Use the solvent, B00148 and a cotton wiper, G00034 to remove any dust from the damaged area.

SUBTASK 21-20-02-840-056

(4) Cut two pieces of the Fiberglass glass fabric, G00031 1/2 inch larger on all sides than the damaged area. Round the corners of the Fiberglass fabric pieces.

SUBTASK 21-20-02-840-057

(5) Prepare the applicable resin.

E. Repair the Inner Lining

SUBTASK 21-20-02-340-047

(1) Apply a thin, smooth layer of the applicable resin to the damaged area of the duct. Make sure the area of the resin is a minimum of the size and shape of the Fiberglass fabric.

SUBTASK 21-20-02-340-048

(2) Place a piece of the Fiberglass fabric on the resin so that it extends beyond the damaged area a minimum of 1/2 inch on all sides.

SUBTASK 21-20-02-340-049

(3) Soak the Fiberglass fabric with the applicable resin.

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SUBTASK 21-20-02-340-050

(4) Put a piece of the plastic sheet on the Fiberglass fabric and press lightly with a squeegee or with your hand. Smooth from the center to the edges of the Fiberglass fabric to remove any bubbles or any wrinkles.

SUBTASK 21-20-02-340-051

(5) Remove the plastic sheet from the patch and remove unwanted resin with a cotton wiper, G00034.

SUBTASK 21-20-02-340-052

(6) Follow the same procedure to apply the second piece of Fiberglass fabric.

SUBTASK 21-20-02-340-053

(7) Allow the resin to cure for 24 hours at room temperature.

SUBTASK 21-20-02-340-054

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(8) Remove unwanted resin from the edges of the repaired area with a cotton wiper, G00034 soaked with solvent, B00148.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-058

(1) Use the applicable procedure to return the duct to its usual condition.

SUBTASK 21-20-02-840-059

(2) Replace the panels or other airplane structure removed to gain access to the duct.

——— END OF TASK ———

TASK 21-20-02-330-809

10. Duct Section Replacement - Polyurethane Foam Duct

(Figure 802)

A. General

(1) This procedure gives instructions to replace a section of polyurethane foam air conditioning duct.

(2) If time or materials are not available to do this procedure, use the procedure to repair a taped duct joint to make a temporary repair. If a temporary repair is made, do this procedure as soon as possible to make a permanent repair.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740

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(Continued)

Reference	Description	Specification
G00031	Fabric - Glass, Finished, For Resin Laminates	AMS-C-9084 Type VIIIA or VIIIB
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Prepare for the Repair

SUBTASK 21-20-02-840-060

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-061

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-062

(3) Cut and remove the damaged section of polyurethane foam air conditioning duct. Cut the duct so the edges are straight and perpendicular to the circumference of the duct.

SUBTASK 21-20-02-840-063

(4) Do these steps to prepare the air conditioning duct.

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Use the solvent, B00148 and a cotton wiper, G00034 to clean the duct ends.

(b) Sand the duct ends to remove any contamination from the duct ends.

(c) Use the solvent, B00148 and a cotton wiper, G00034 to remove any dust from the duct ends.

SUBTASK 21-20-02-840-064

(5) Prepare a piece of the polyurethane foam duct that matches the damaged duct and is the correct length to replace the damaged section of the duct.

NOTE: Cut the replacement section with clearance for the material that will be added by the duct end repair procedure.

SUBTASK 21-20-02-840-065

(6) Do the Duct End Repair - Polyurethane Foam Duct procedure above for each end of the replacement duct section and for each end of the damaged duct.

SUBTASK 21-20-02-840-066

(7) Cut two pieces of the Fiberglass glass fabric, G00031 3 inches wide and long enough to go around the duct a minimum of four times.

SUBTASK 21-20-02-840-067

(8) Prepare the applicable resin.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Repair the Taped Joint

SUBTASK 21-20-02-340-055

(1) Apply the applicable resin to the duct ends. Allow the resin to partially dry until it is sticky. Press the duct ends together by aligning the installation checkmarks.

SUBTASK 21-20-02-340-056

(2) Apply the applicable resin to each joint area a minimum of 2 inches on both sides of the joint.

SUBTASK 21-20-02-340-057

(3) Apply the applicable resin to one side of the two pieces of the Fiberglass fabric.

SUBTASK 21-20-02-340-058

(4) Wrap one piece of the Fiberglass fabric around each duct joint a minimum of four times. Make sure the resin on the Fiberglass fabric is in contact with the duct.

SUBTASK 21-20-02-340-059

(5) Put a piece of plastic sheet on the joint and press on it to smooth the Fiberglass fabric to remove any bubbles or any wrinkles.

SUBTASK 21-20-02-340-060

(6) Allow the resin to cure for 24 hours at room temperature.

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(7) Clean any unwanted resin from the joint areas with a cotton wiper, G00034 and solvent, B00148.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-068

(1) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

TASK 21-20-02-330-810

11. Tape Replacement - Foam Wrapped Insulated Duct

(Figure 803)

A. General

(1) This procedure has instructions to repair those air conditioning ducts insulated with Melamine or Polyimide Foam. The Foam Insulation is wrapped around the duct and held in place with an adhesive tape.

(2) CFR 14 Part 121.312 mandates that insulation material installed in the fuselage as a replacement after September 2, 2005 must meet the flame propagation requirements of CFR Part 25.856, effective September 2, 2003.

(a) Existing insulation material can be removed and re-installed if it is not damaged. Replacement insulation material must meet the flame propagation requirement.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Melamine foam insulation, G50449, BMS8-385, is the preferred foam insulation material that meets the flame propagation requirement. Melamine BMS8-385 Type 4 is the preferred foam insulation material that meets the flame propagation requirement. Melamine BMS 8-385 Type 6 and polyimide BMS 8-300 Type VIII insulation are optional to BMS8-385 Type 4.

(c) Do not use BMS8-39 polyurethane foam insulation for repairs due to the degradation in flammability properties over time.

(d) When insulation material is removed and re-installed, any existing tape that does not meet the flame propagation requirement must be removed or completely covered with tape that does meet the requirement.

(e) Air conditioning ducts with polyimide foam insulation, G02470 must be wrapped with tape, G50327, tape, G51202 or tape, G50789 to meet the flammability/flame propagation requirement. Wrap the tape, G50327, tape, G51202 or tape, G50789 across 100% of the brown hypalon insulation surface of the polyimide foam insulation, G02470.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G02470	Foam - Flexible Polyimide	BMS8-300 Type I
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50449	Foam - Flexible Melamine	BMS8-385 Type 4 Grade 1
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

D. Prepare for the Repair

SUBTASK 21-20-02-840-069

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-070

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-071

(3) If necessary, remove the air conditioning duct.

SUBTASK 21-20-02-840-072

(4) Do these steps to remove loose tape and unwanted adhesive from the repair area:

(a) Carefully remove loose tape from the repair area.

(b) Use the solvent, B00184 or alcohol, B00130 and a wiping fabric to clean the repair area.

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E. Replace the Tape

SUBTASK 21-20-02-420-001

(1) Install the tape, G50327, tape, G51202 or tape, G50789 to the repair area (Figure 803).

(a) Make sure the tape, G50327, tape, G51202 or tape, G50789 has an overlap of all the edges and seams of the insulation by a minimum of 1.00 inch (25.4 mm).

F. Rework Ducts Insulated with Polyimide Foam Insulation

SUBTASK 21-20-02-420-002

(1) Install tape, G50327, tape, G51202 or tape, G50789 to completely cover all exposed surfaces of foam insulation as shown in (Figure 803).

(a) The tape, G50327, tape, G51202 or tape, G50789 must cover 100% of the brown colored hypalon insulation surface.

(b) Cut the tape, G50327, tape, G51202 or tape, G50789 as required to fit around air outlets, branches and insulation vent holes.

(c) The tape, G50327, tape, G51202 or tape, G50789 may be installed spiral wrapped or lengthwise down the duct depending on installers preference.

(d) The tape, G50327, tape, G51202 or tape, G50789 may be installed in any configuration as long as 100% complete coverage of the foam insulation, G02470 insulation is obtained.

(e) Maintain a minimum of 1/8 inch overlap of tape, G50327, tape, G51202 or tape, G50789.

(f) Avoid compression of the foam insulation, G02470 insulation.

(g) Wrinkles in the tape, G50327, tape, G51202 or tape, G50789 are acceptable if pinched together and contain no voids.

(h) The tape, G50327, tape, G51202 or tape, G50789 may be installed to the foam insulation, G02470 insulation prior to installation on the duct if desired by the installer.

(i) For foam insulation, G02470 insulation installed on duct joints, tape, G50327, tape, G51202 or tape, G50789 may be applied to the foam insulation, G02470 insulation before or after the foam insulation, G02470 insulation is installed over the duct joint.

(j) The tape, G50327, tape, G51202 or tape, G50789 orientation is optional.

SUBTASK 21-20-02-420-003

(2) If necessary, install the air conditioning duct.

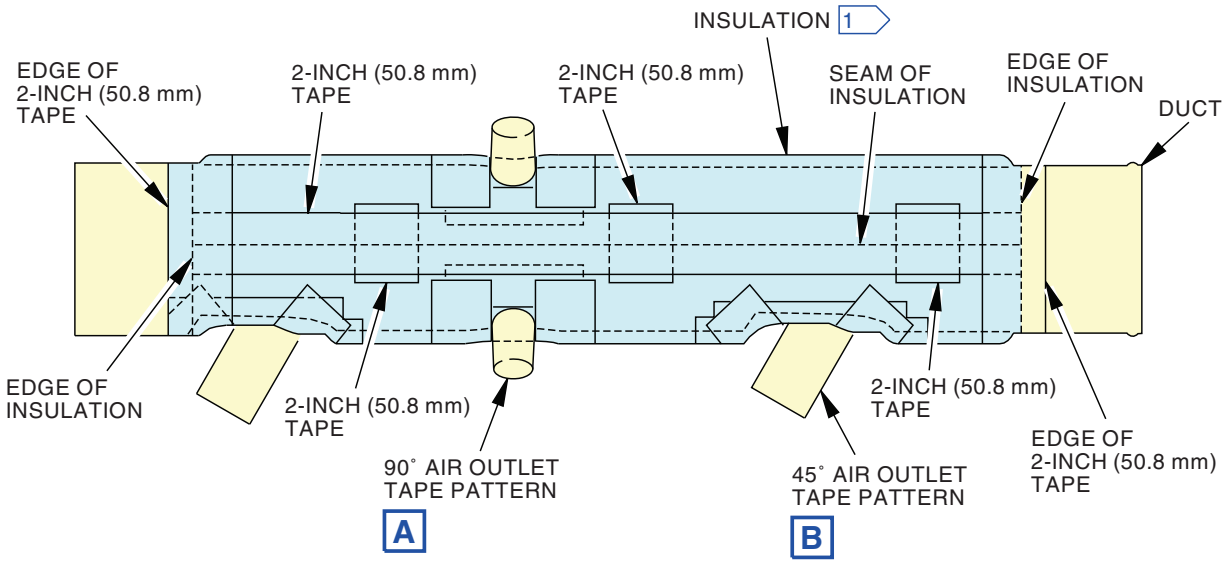
G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-073

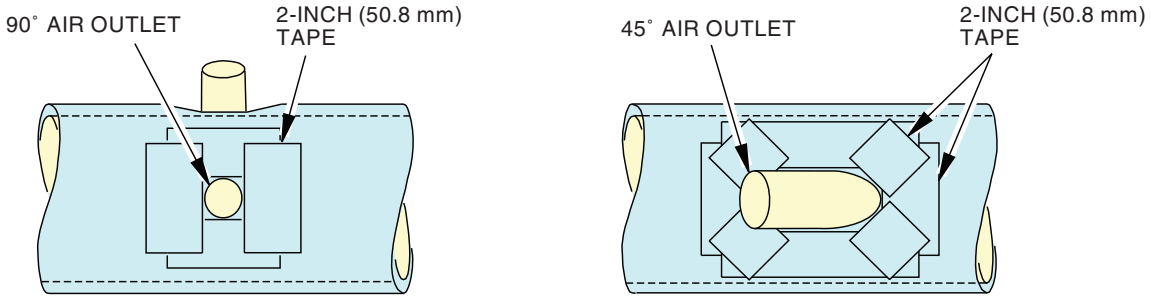
(1) Replace the panels or other airplane structure removed to gain access to the duct.

——— END OF TASK ———

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FOAM WRAPPED INSULATED DUCT (EXAMPLE)



90˚ AIR OUTLET TAPE PATTERN

A

45˚ AIR OUTLET TAPE PATTERN

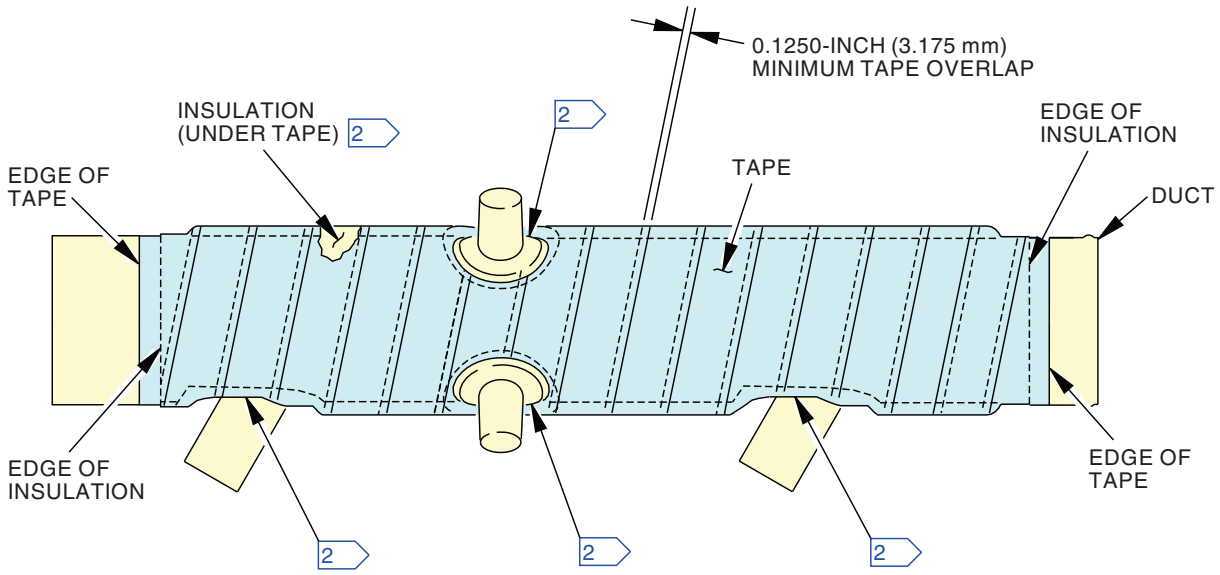
B

1 FOR INSULATION DAMAGE APPLY A MINIMUM 2-INCH (50.8 mm) PATCH OF PRESSURE SENSITIVE ADHESIVE TAPE TO THE DAMAGED AREA. MAKE SURE YOU HAVE FULL COVERAGE OF THE DAMAGED OR TORN AREA WITH NO LESS THAN 0.5-INCH (12.7 mm) OVERLAP.

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Air Conditioning System Insulated Duct Repair (Foam Wrapped)
Figure 803/21-20-02-990-803 (Sheet 1 of 2)

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**REWORK POLYIMIDE FOAM INSULATED DUCT
(EXAMPLE)**

NOTE:

- 2-INCH (50.8 mm) WIDE, 3-INCH (76.2 mm) WIDE OR 12-INCH (304.8 mm)WIDE TAPE (BMS 5-157 TYPE I, CLASS 1, GRADE B) MUST COVER 100% OF THE BROWN HYPALON INSULATION SURFACE.
- TAPE MAY BE INSTALLED SPIRAL WRAPPED AS SHOWN OR LENGTHWISE DOWN THE DUCT DEPENDING ON THE INSTALLERS PREFERENCE.
- SHOWN PATTERN IS FOR REFERENCE ONLY. TAPE MAY BE APPLIED IN ANY CONFIGURATION AS LONG AS COMPLETE BMS 8-300 COVERAGE IS OBTAINED.
- WIDER TAPE MAY BE CUT INTO NARROWER STRIPS OR ANY UNIQUE SHAPE IF DESIRED TO ASSIST IN OBTAINING COMPLETE BMS 8-300 COVERAGE.
- MAINTAIN MINIMUN 0.1250-INCH (3.175 mm) OVERLAP OF TAPE.
- AVOID COMPRESSION OF BMS 8-300 INSULATION.
- WRINKLES IN TAPE ARE ACCEPTABLE IF PINCHED TOGETHER AND CONTAIN NO VOIDS.
- COVERING OF EXISTING BMS 5-157 TAPE IS NOT REQUIRED.
- TAPE MAY BE APPLIED TO THE BLANKET PRIOR TO THE BLANKET INSTALLATION ON THE DUCT IF DESIRED BY THE INSTALLER.
- FOR BLANKETS INSTALLED ON DUCT JOINTS, TAPE MAY BE APPLIED TO THE BLANKET BEFORE OR AFTER THE BLANKET IS INSTALLED OVER THE DUCT JOINT. TAPE ORIENTATION IS OPTIONAL.

2 COVER ALL EXPOSED INSULATION (BMS 8-300 TYPE II, GRADE 5) WITH TAPE (BMS 5-157 TYPE I, CLASS 1, GRADE B). CUT TAPE AS REQUIRED TO FIT AROUND AIR OUTLETS, BRANCHES AND INSULATION VENT HOLES.

2411261 S00061527418_V1

**Air Conditioning System Insulated Duct Repair (Foam Wrapped)
Figure 803/21-20-02-990-803 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-20-02-330-811

12. Insulation Repair - Foam Wrapped Insulated Duct

(Figure 803)

A. General

(1) This procedure has instructions to repair those air conditioning ducts insulated with Melamine or Polyimide Foam. The Foam Insulation is wrapped around the duct and held in place with an adhesive tape.

(2) CFR 14 Part 121.312 mandates that insulation material installed in the fuselage as a replacement after September 2, 2005 must meet the flame propagation requirements of CFR Part 25.856, effective September 2, 2003.

(a) Existing insulation material can be removed and re-installed if it is not damaged. Replacement insulation material must meet the flame propagation requirement.

(b) Melamine foam insulation, G50449, BMS8-385, Type 4 is the preferred foam insulation material that meets the flame propagation requirement. Melamine foam insulation, G51169, BMS 8-385, Type 6 and polyimide foam insulation, G51170, BMS 8-300, Type VIII insulation are new, qualified materials optional to BMS 8-385, Type 4.

(c) Do not use BMS8-39 polyurethane foam insulation for repairs due to the degradation in flammability properties over time.

(d) When insulation material is removed and re-installed, any existing tape that does not meet the flame propagation requirement must be removed or completely covered with tape that does meet the requirement.

(e) Air conditioning ducts with polyimide foam insulation, G02470 must be wrapped with tape, G50327 or tape, G51202 to meet the flammability/flame propagation requirement. Wrap the tape, G50327 or tape, G51202 across 100% of the brown hypalon insulation surface of the polyimide foam insulation, G02470. The tape, G50789 is optional to the tape, G50327.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G02470	Foam - Flexible Polyimide	BMS8-300 Type I
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50449	Foam - Flexible Melamine	BMS8-385 Type 4 Grade 1
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51169	Foam - Flexible Melamine,	BMS8-385 Type 6
G51170	Foam - Flexible Polyimide	BMS8-300 Type VIII

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(Continued)

Reference	Description	Specification
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

D. Prepare for the Repair

SUBTASK 21-20-02-840-074

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-075

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-076

(3) If necessary, remove the air conditioning duct.

SUBTASK 21-20-02-840-077

(4) Do these steps to remove loose tape and unwanted adhesive from the repair area:

(a) Use the solvent, B00184 or alcohol, B00130 and a wiping cloth to clean the repair area.

E. Repair the Insulation

SUBTASK 21-20-02-420-004

(1) Install a minimum 2.0 inch (50.8 mm) wide patch of the tape, G50327, tape, G51202 or tape, G50789 to completely cover the damaged area of the insulation (Figure 803).

(a) Make sure the tape, G50327, tape, G51202 or tape, G50789 has an overlap of all the edges of the damaged area by a minimum of 0.50 inch (12.7 mm).

F. Rework Ducts Insulated with Polymide Foam Insulation

SUBTASK 21-20-02-420-005

(1) Install tape, G50327, tape, G51202 to completely cover all exposed surfaces of foam insulation as shown in (Figure 803).

(a) The tape, G50327, tape, G51202 or tape, G50789 must cover 100% of the brown colored hypalon insulation surface.

(b) Cut the tape, G50327, tape, G51202 or tape, G50789 as required to fit around air outlets, branches and insulation vent holes.

(c) The tape, G50327, tape, G51202 or tape, G50789 may be installed spiral wrapped or lengthwise down the duct depending on installers preference.

(d) The tape, G50327, tape, G51202 or tape, G50789 may be installed in any configuration as long as 100% complete coverage of the foam insulation, G02470 insulation is obtained.

(e) Maintain a minimum of 1/8 inch overlap of tape, G50327, tape, G51202 or tape, G50789.

(f) Avoid compression of the foam insulation, G02470 insulation.

(g) Wrinkles in tape, G50327, tape, G51202 or tape, G50789 are acceptable if pinched together and contain no voids.

(h) The tape, G50327, tape, G51202 or tape, G50789 may be installed to the foam insulation, G02470 insulation prior to installation on the duct if desired by the installer.

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(i) For foam insulation, G02470 insulation installed on duct joints, tape, G50327, tape, G51202 or tape, G50789 may be applied to the foam insulation, G02470 insulation before or after the foam insulation, G02470 insulation is installed over the duct joint.

(j) The tape, G50327, tape, G51202 or tape, G50789 orientation is optional.

SUBTASK 21-20-02-420-006

(2) If necessary, install the air conditioning duct.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-078

(1) Replace the panels or other airplane structure removed to gain access to the duct.

——— END OF TASK ———

TASK 21-20-02-330-812

13. Insulation Replacement - Foam Wrapped Insulated Duct

(Figure 803)

A. General

(1) This procedure has instructions to repair those air conditioning ducts insulated with Melamine or Polyimide Foam. The Foam Insulation is wrapped around the duct and held in place with an adhesive tape.

(2) CFR 14 Part 121.312 mandates that insulation material installed in the fuselage as a replacement after September 2, 2005 must meet the flame propagation requirements of CFR Part 25.856, effective September 2, 2003.

(a) Existing insulation material can be removed and re-installed if it is not damaged. Replacement insulation material must meet the flame propagation requirement.

(b) Melamine foam insulation, G50449, BMS8-385, Type 4 is the preferred foam insulation material that meets the flame propagation requirement. BMS 8-385 Type 6 flexible melamine foam and BMS 8-300 Type VIII flexible polyimide foam insulation are optional to BMS8-385 Type 4.

(c) Do not use BMS8-39 polyurethane foam insulation for repairs due to the degradation in flammability properties over time.

(d) When insulation material is removed and re-installed, any existing tape that does not meet the flame propagation requirement must be removed or completely covered with tape that does meet the requirement.

(e) Air conditioning ducts with polyimide foam insulation, G02470 must be wrapped with tape, G50327, tape, G51202 or tape, G50789 to meet the flammability/flame propagation requirement. Wrap the tape, G50327, tape, G51202 or tape, G50789 across 100% of the brown hypalon insulation surface of the polyimide foam insulation, G02470.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G02470	Foam - Flexible Polyimide	BMS8-300 Type I

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(Continued)

Reference	Description	Specification
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50449	Foam - Flexible Melamine	BMS8-385 Type 4 Grade 1
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51169	Foam - Flexible Melamine,	BMS8-385 Type 6
G51170	Foam - Flexible Polyimide	BMS8-300 Type VIII
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

D. Prepare for the Repair

SUBTASK 21-20-02-840-079

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-080

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-081

(3) Remove the damaged air conditioning duct to permit removal of the foam insulation.

SUBTASK 21-20-02-840-082

(4) Do these steps to remove the tape and the foam insulation from the duct:

(a) Remove the tape from the duct.

(b) Remove the insulation from the repair duct.

(c) Use the solvent, B00184 or alcohol, B00130 and a wiping cloth to clean the repair area.

E. Replace the Insulation

SUBTASK 21-20-02-420-007

(1) Do these steps to install one of these foam insulations: foam insulation, G50449, foam insulation, G51169, foam insulation, G51170 or foam insulation, G02470 on the duct:

(a) Use the solvent, B00184 or alcohol, B00130 and a wiping cloth to clean the outer surface of the foam insulation.

(b) Wrap the foam insulation around the duct.

(c) Install strips of 2 inch wide (50.8 mm) tape, G50327, tape, G51202 or tape, G50789 to the edges and seams of the foam insulation (Figure 803)

(d) Make sure the tape has an overlap of all the edges and seams of the foam insulation by a minimum of 1.00 inch (25.4 mm).

(e) Install the tape, G50327, tape, G51202 or tape, G50789 as shown in (Figure 803)

F. Rework Ducts insulated with Polyimide Foam Insulation

SUBTASK 21-20-02-420-008

(1) Install tape, G50327, tape, G51202 or tape, G50789 to completely cover all exposed surfaces of foam insulation as shown in (Figure 803).

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(a) The tape, G50327, tape, G51202 or tape, G50789 must cover 100% of the brown colored hypalon insulation surface.

(b) Cut the tape, G50327, tape, G51202 or tape, G50789 as required to fit around air outlets, branches and insulation vent holes.

(c) The tape, G50327, tape, G51202 or tape, G50789 may be installed spiral wrapped or lengthwise down the duct depending on installers preference.

(d) The tape, G50327, tape, G51202 or tape, G50789 may be installed in any configuration as long as 100% complete coverage of the foam insulation, G02470 insulation is obtained.

(e) Maintain a minimum of 1/8 inch overlap of tape, G50327, tape, G51202 or tape, G50789.

(f) Avoid compression of the foam insulation, G02470 insulation.

(g) Wrinkles in the tape, G50327, tape, G51202 or tape, G50789 are acceptable if pinched together and contain no voids.

(h) The tape, G50327, tape, G51202 or tape, G50789 may be installed to the foam insulation, G02470 insulation prior to installation on the duct if desired by the installer.

(i) For foam insulation, G02470 insulation installed on duct joints, tape, G50327, tape, G51202 or tape, G50789 may be applied to the foam insulation, G02470 insulation before or after the foam insulation, G02470 insulation is installed over the duct joint.

(j) The tape orientation is optional.

SUBTASK 21-20-02-420-009

(2) Install the air conditioning duct you removed for the repair.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-083

(1) Replace the panels or other airplane structure removed to gain access to the duct.

——— END OF TASK ———

TASK 21-20-02-330-813

14. Tape Replacement - Fiberglass Insulated Duct

(Figure 803, Figure 804)

A. General

(1) This procedure has instructions to repair those air conditioning ducts insulated with Fiberglass Insulation (BMS 8-48) and a flame-resistant covering (BMS 8-142) which have been overwrapped with cargo liner joint sealing tape (BMS 5-146).

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF

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(Continued)

Reference	Description	Specification
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

D. Prepare for the Repair

SUBTASK 21-20-02-840-084

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-085

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-086

(3) If necessary, remove the air conditioning duct.

SUBTASK 21-20-02-840-087

(4) Do these steps to remove loose tape and unwanted adhesive from the repair area:

(a) Carefully remove loose tape from the repair area.

(b) Use the solvent, B00184 or alcohol, B00130 and a wiping cloth to clean the repair area.

SUBTASK 21-20-02-840-088

(5) Install the tape, G50327, tape, G51202 or tape, G50789 to the repair area (Figure 803, Figure 804).

(a) Use a continuous strip of tape to wrap around the insulation/covering and the duct.

(b) Make sure the tape has an overlap of the edges of the adjacent tape by a minimum of 0.12 inch (3.17 mm).

(c) Make sure the new tape has an overlap onto the ends of the duct by a minimum of 1.00 inch (25.4 mm).

SUBTASK 21-20-02-840-089

(6) If necessary, install the air conditioning duct.

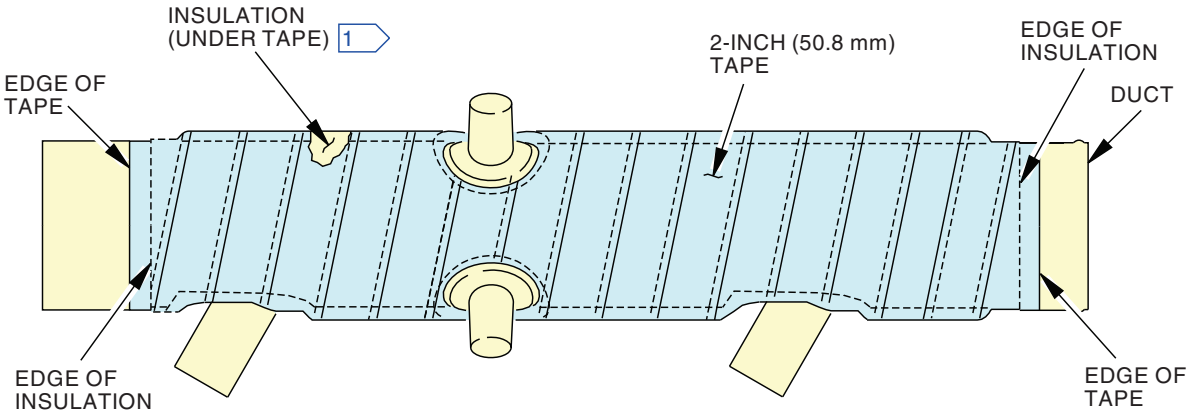
E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-090

(1) Replace the panels or other airplane structure removed to gain access to the duct.

——— END OF TASK ———

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**FIBERGLASS INSULATED DUCT
(EXAMPLE)**

1 COVER ALL EXPOSED INSULATION WITH CARGO LINER JOINT SEALER TAPE. CUT TAPE AS REQUIRED TO FIT AROUND AIR OUTLETS.

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**Air Conditioning System Insulated Duct Repair (Fiberglass)
Figure 804/21-20-02-990-804**

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TASK 21-20-02-330-814

15. Insulation and Covering Repair - Fiberglass Insulated Duct

(Figure 804)

A. General

(1) This procedure has instructions to repair those air conditioning ducts insulated with Fiberglass Insulation (BMS 8-48) and a flame-resistant covering (BMS 8-142) which have been overwrapped with cargo liner joint sealing tape (BMS 5-146).

(2) This procedure has these repair instructions:

(a) Replacement of loose or old tape (Repair Method 1)

(b) Repair of cuts or tears in the flame-resistant covering and Fiberglass insulation (Repair Method 2)

(c) Replacement of the Fiberglass insulation (Repair Method 3).

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

D. Tape Replacement (Repair Method 1)

SUBTASK 21-20-02-840-091

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-092

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-093

(3) If necessary, remove the air conditioning duct.

SUBTASK 21-20-02-840-094

(4) Remove all loose or old cargo liner joint sealing tape from the insulation/covering.

SUBTASK 21-20-02-110-001

(5) Use the solvent, B00184 or alcohol, B00130 and a wiping cloth to clean the repair area.

SUBTASK 21-20-02-420-010

(6) Install new tape, G50327, tape, G51202 or tape, G50789 to completely cover all surfaces of insulation and existing tape as shown in (Figure 803):

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(a) Use a continuous strip of tape to wrap around the insulation/covering and the duct.

1) Make sure the new tape overlaps the edges of the adjacent tape by a minimum of 1/8-inch.

2) Make sure the new tape overlaps onto the ends of the duct by a minimum of 1-inch.

SUBTASK 21-20-02-840-095

(7) If removed, install the air conditioning duct.

E. Cuts or Tears in Fiberglass Insulation/Covering (Repair Method 2)

SUBTASK 21-20-02-020-001

(1) Remove the duct if needed for better access.

SUBTASK 21-20-02-110-002

(2) Use the solvent, B00184 or alcohol, B00130 and a wiping cloth to clean the repair area.

SUBTASK 21-20-02-340-061

(3) Apply a minimum 2-inch wide patch of tape, G50327, tape, G51202 or tape, G50789 to completely cover the damaged area on the insulation/covering.

(a) Make sure to apply a large enough patch to assure full coverage of the damaged area with a minimum 2-inch overlap.

SUBTASK 21-20-02-420-011

(4) If removed, install the air conditioning duct.

F. Fiberglass Insulation Replacement (Repair Method 3)

SUBTASK 21-20-02-020-002

(1) Remove the fiberglass insulation from the duct.

SUBTASK 21-20-02-420-012

(2) Install new melamine foam insulation. To install the insulation, do this task: Insulation Replacement - Foam Wrapped Insulated Duct, TASK 21-20-02-330-812.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-096

(1) Replace the panels or other airplane structure removed to gain access to the duct.

——— END OF TASK ———

TASK 21-20-02-330-815

16. Insulation and Covering Replacement - Fiberglass Insulated Duct

(Figure 803)

A. General

(1) This procedure has instructions to repair those air conditioning ducts insulated with Fiberglass Insulation (BMS 8-48) and a flame-resistant covering (BMS 8-142) which have been overwrapped with cargo liner joint sealing tape (BMS 5-146).

(2) Air conditioning ducts with polyimide foam insulation, G02470 must be wrapped with tape, G50327, tape, G51202 or tape, G50789 to meet new flammability/flame propagation requirements. Wrap the tape, G50327, tape, G51202 or tape, G50789 across 100% of the brown hypalon insulation surface on the polyimide foam insulation, G02470.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Consumable Materials

Reference	Description	Specification
G02470	Foam - Flexible Polyimide	BMS8-300 Type I
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

D. Prepare for the Repair

SUBTASK 21-20-02-840-097

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-840-098

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-840-099

(3) Remove the damaged air conditioning duct.

SUBTASK 21-20-02-840-100

(4) Install a new air conditioning duct with polyimide foam insulation, G02470 (Tape Replacement - Foam Wrapped Insulated Duct, TASK 21-20-02-330-810).

(5) Rework the duct insulated with polyimide (Tape Replacement - Foam Wrapped Insulated Duct, TASK 21-20-02-330-810).

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-840-101

(1) Replace the panels or other airplane structure removed to gain access to the duct.

——— END OF TASK ———

TASK 21-20-02-330-816

17. External Patch - Reinforced Thermoplastic Laminate Duct

(Figure 805)

A. General

(1) This procedure gives instructions for the repair of a puncture, crack or cut in a Reinforced Thermoplastic Laminate air conditioning duct with an external patch.

(2) An external patch is applicable when the cut or the crack is less than 0.25 inches (6.35 mm) wide and is less than 3.0 inches (76.2 mm) long.

(3) An external patch is applicable when the puncture is less than 0.25 inches (6.35 mm) diamter.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00316	Fabric - Woven Glass Reinforcements For Laminating Plastics	BMS9-3
G50400	Resin - Fiberglass Layup, Short Worklife, Non-Brominated	BMS8-201 Type IV (Supersedes BMS8-201 Type II)

D. Prepare for the Repair

SUBTASK 21-20-02-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-010-001

(2) Gain access to the damaged air conditioning duct joint.

SUBTASK 21-20-02-010-002

(3) If insulation is installed on the duct, then remove the insulation.

SUBTASK 21-20-02-120-001

(4) Do these steps to prepare the damaged area of the duct.

(a) Use sandpaper on the damaged area to remove the gloss from the finish of the duct.

(b) Use the solvent, B00062 and a cotton wiper, G00034 to clean the duct joint area.

(c) Make sure fibers from the damaged area of the duct do not extend more than 0.125 inches (3.175 mm) into the inside of the duct.

SUBTASK 21-20-02-340-062

(5) Do these steps to prepare the materials for the duct repair:

(a) Cut sufficient pieces of the Fiberglass fabric, G00316 to equal the number of plies of the damaged duct. Cut the Fiberglass fabric, G00316 pieces 0.50 inches (12.7 mm) larger on all sides than the damage. Round the corners of the pieces of the Fiberglass fabric.

(b) Prepare the Fiberglass duct repair resin, G50400.

E. Repair the Air Conditioning Duct

SUBTASK 21-20-02-340-063

(1) Apply a thin, smooth layer of the resin, G50400 to the duct where the Fiberglass fabric will be applied. Make sure the area of the resin is a minimum of the size and shape of the Fiberglass fabric pieces.

SUBTASK 21-20-02-340-064

(2) Place a piece of the Fiberglass fabric on the resin so that it extends beyond the area of damage a minimum of 1/2 inch on all sides.

SUBTASK 21-20-02-340-065

(3) Soak the Fiberglass fabric with the applicable resin.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-20-02-340-066

(4) Put a piece of plastic sheet on the Fiberglass fabric and press lightly to remove any bubbles or any wrinkles and to remove unwanted resin.

SUBTASK 21-20-02-340-067

(5) Remove the plastic sheet from the Fiberglass fabric piece and remove any unwanted resin with a wiping cloth.

SUBTASK 21-20-02-340-068

(6) Place another piece of the Fiberglass fabric on the first piece, soak it with resin, and smooth it. Continue using the same procedure for each new ply of the fabric.

SUBTASK 21-20-02-340-069

(7) Allow the resin to cure for 24 hours at room temperature.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-410-001

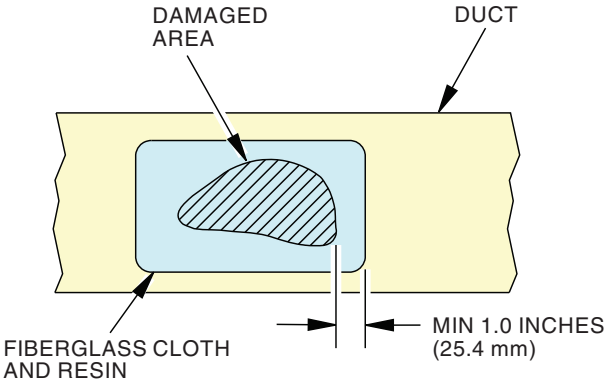
(1) If insulation was removed from the duct joint, install new insulation.

SUBTASK 21-20-02-410-002

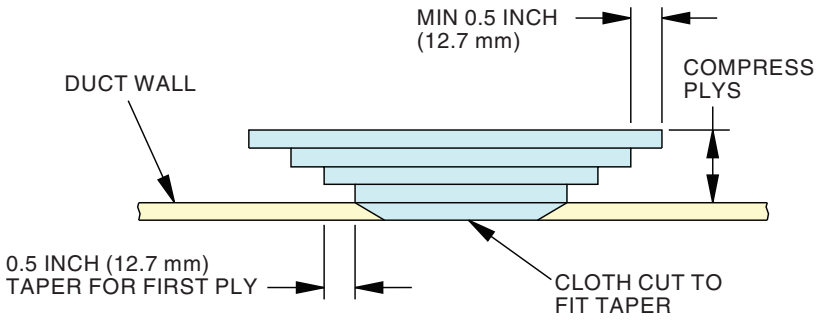
(2) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STRUCTURAL REPAIR - REINFORCED THERMOPLASTIC LAMINATE - OVERHEAD VIEW



STRUCTURAL REPAIR - REINFORCED THERMOPLASTIC LAMINATE - CROSS-SECTION VIEW

2411263 S00061527426_V1

Strutural Repair - Reinforced Thermoplastic Laminate Duct
Figure 805/21-20-02-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-20-02-330-817

18. Structural Repair - Reinforced Thermoplastic Laminate Duct

(Figure 805)

A. General

(1) This procedure gives instructions for making a structural repair to a Reinforced Thermoplastic Laminate air conditioning duct.

(2) A structural repair is applicable when the damage to the air conditioning duct is a cut or a hole that is more than 0.25 inches (6.35 mm) wide and more than 3.0 inches (76.2 mm) long but less than 5 percent of the total surface area of the duct. If the damage is more than 5 percent of the total surface area of the duct, then replace the duct section.

(3) A structural repair is applicable when the damage to the air conditioning duct is a puncture that is more than 0.25 inches (6.35 mm) in diameter.

(4) It is not permitted to overlap structural repairs to the air conditioning duct.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00316	Fabric - Woven Glass Reinforcements For Laminating Plastics	BMS9-3
G50400	Resin - Fiberglass Layup, Short Worklife, Non-Brominated	BMS8-201 Type IV (Supersedes BMS8-201 Type II)

D. Prepare for the Repair

SUBTASK 21-20-02-010-003

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-010-004

(2) Gain access to the damaged air conditioning duct.

SUBTASK 21-20-02-010-005

(3) If insulation is installed on the duct, then remove the insulation.

SUBTASK 21-20-02-120-002

(4) Do these steps to prepare the damaged area of the duct.

(a) Cut away the damaged area to make a rounded opening with a smooth shape.

(b) Use sandpaper to prepare the surface of the duct around the opening a distance of 0.50 inches (12.7 mm) for each ply in the duct. For example, if there are 6 plies, sand 3 inches from the opening on all sides (0.50 x 6.0 = 3.0 inches)(12.7 x 152.4 = 76.2 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Use sandpaper to prepare the surface of the duct an additional 0.50 inches (12.7 mm) from the repair area.

(d) Make sure fibers from the damaged area of the duct do not extend more than 0.125 inches (3.175 mm) into the inside of the duct.

(e) Use the solvent, B00062 and a cotton wiper, G00034 to clean the duct joint area.

SUBTASK 21-20-02-350-001

(5) Cut a piece of plastic sheet 2.0 to 3.0 inches (50.8 to 76.2 mm) larger on all sides than the opening. Cut a piece of cardboard or aluminum 2.0 to 3.0 inches (50.8 to 76.2 mm) larger than the opening on all sides.

SUBTASK 21-20-02-350-002

(6) Attach the plastic sheet to the inside of the opening with adhesive tape so that it covers the opening. Attach the cardboard or aluminum to the inside of the opening with adhesive tape so that it covers the opening and provides support for the plastic sheet.

SUBTASK 21-20-02-340-070

(7) Do these steps to prepare the materials for the duct repair:

(a) Cut sufficient pieces of the Fiberglass fabric, G00316 to equal the number of plies of damaged the duct. Cut an additional piece of fabric, G00316

(b) Cut the first piece of the Fiberglass fabric, G00316 pieces 0.50 inches (12.7 mm) larger on all sides than the damage. Round the corners of the pieces of the Fiberglass fabric.

(c) Cut each piece of the Fiberglass fabric, G00316 0.50 inches (12.7 mm) larger on all sides than the piece before.

(d) Prepare the Fiberglass duct repair resin, G50400.

E. Repair the Air Conditioning Duct

SUBTASK 21-20-02-340-071

(1) Apply a thin, smooth layer of resin, G50400 to the plastic sheet in the opening in the duct.

SUBTASK 21-20-02-340-072

(2) Place a piece of the Fiberglass fabric in the opening over the plastic sheet so that it covers the opening completely.

SUBTASK 21-20-02-340-073

(3) Soak the Fiberglass fabric with the applicable resin.

SUBTASK 21-20-02-340-074

(4) Put a piece of plastic sheet on the Fiberglass fabric and press lightly with a squeegee or with your hand. Smooth from the center to the edges of the Fiberglass fabric to remove bubbles or wrinkles and to remove unwanted resin.

SUBTASK 21-20-02-340-075

(5) Remove the plastic sheet from the patch and remove unwanted resin with a cotton wiper, G00034.

SUBTASK 21-20-02-340-076

(6) Place the next piece of the Fiberglass fabric on the first piece so that it covers the opening. Trim the Fiberglass fabric if necessary. Soak the Fiberglass fabric with resin and smooth it. Continue using the same procedure until the thickness of the patch equals the thickness of the duct.

SUBTASK 21-20-02-340-077

(7) Round the corners of the last piece of the Fiberglass fabric. Smooth the Fiberglass fabric to remove bubbles or winkles.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-20-02-340-078

(8) Wipe the duct around the patch with a cotton wiper, G00034 and solvent, B00148 to remove unwanted resin.

SUBTASK 21-20-02-340-079

(9) Allow the resin to cure for 24 hours at room temperature.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-410-003

(1) If insulation was removed from the duct joint, install new insulation.

SUBTASK 21-20-02-410-004

(2) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

TASK 21-20-02-330-818

19. Duct End Repair - Reinforced Thermoplastic Laminate Duct

(Figure 805)

A. General

(1) This procedure gives instructions for making a repair to the end of a Reinforced Thermoplastic Laminate air conditioning duct.

(2) A duct end repair replaces a full section of a damaged duct. A repair section to a duct end can not extend more than 4.0 inches (101.6 mm) from the end of the duct.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00316	Fabric - Woven Glass Reinforcements For Laminating Plastics	BMS9-3
G50400	Resin - Fiberglass Layup, Short Worklife, Non-Brominated	BMS8-201 Type IV (Supersedes BMS8-201 Type II)

D. Prepare for the Repair

SUBTASK 21-20-02-040-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

NOTE: This will make sure that there is no airflow in the ducts when you do the repair.

SUBTASK 21-20-02-010-006

(2) Gain access to the damaged air conditioning duct.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-20-02-010-007

(3) If insulation is installed on the duct, then remove the insulation.

SUBTASK 21-20-02-120-003

(4) Do these steps to prepare the damaged area of the duct.

(a) Remove the damaged area of the duct end.

(b) Cut the duct end straight and perpendicular to the circumference of the duct.

(c) Use sandpaper to taper the surface of the duct around the opening a distance of 0.50 inches (12.7 mm).

(d) Use sandpaper to prepare the surface of the duct a distance of 0.50 inches (12.7 mm) for each ply in the duct. For example, if there are 6 plies, sand 3 inches from the duct end on all sides (0.50 x 6.0 = 3.0 inches)(12.7 x 152.4 = 76.2 mm).

(e) Use sandpaper to remove the gloss of the duct finish an additional of 0.50 inches (12.7 mm) from the repair area.

(f) Make sure that fibers from the cut end of the duct do not extend more than 0.125 inches (3.175 mm) into the inner surface of the duct.

(g) Use the solvent, B00062 and a cotton wiper, G00034 to clean the duct joint area.

SUBTASK 21-20-02-350-003

(5) Do these steps to prepare a parting surface to hold the new duct section:

(a) Get a cardboard tube of an equal diameter to the damaged duct. Make sure that the cardboard tube is of a sufficient length to give support to the new duct suction.

(b) Cut a piece of a plastic sheet of a sufficient size to cover the cardboard tube.

(c) Attach the plastic sheet to the outer surface of the cardboard tube with adhesive tape.

SUBTASK 21-20-02-340-080

(6) Do these steps to prepare the materials for the duct repair:

(a) Do the steps that follow to cut the pieces of the Fiberglass fabric, G00316 to the correct size:

1) Cut the same number pieces of the Fiberglass fabric, G00316 as the number of plies of the damaged duct. Cut an additional piece of fabric, G00316

2) Cut the pieces of the Fiberglass fabric, G00316 to a length sufficient to go around the circumference of the duct, plus 0.50 inches (12.7 mm) to make an overlap.

3) Cut each piece of the Fiberglass fabric, G003160.50 inches (12.7 mm) wider than the piece before.

4) Round the corners of the last piece of the Fiberglass fabric.

(b) Prepare the Fiberglass duct repair resin, G50400.

E. Repair the Air Conditioning Duct

SUBTASK 21-20-02-340-081

(1) Apply a thin, smooth layer of resin, G50400 to the plastic sheet in the opening in the duct.

SUBTASK 21-20-02-340-082

(2) Place a piece of the Fiberglass fabric in the opening over the plastic sheet so that it covers the opening completely.

SUBTASK 21-20-02-340-083

(3) Soak the Fiberglass fabric with the applicable resin.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-20-02-340-084

(4) Put a piece of plastic sheet on the Fiberglass fabric and press lightly with a squeegee or with your hand. Smooth from the center to the edges of the Fiberglass fabric to remove bubbles or wrinkles and to remove unwanted resin.

SUBTASK 21-20-02-340-085

(5) Remove the plastic sheet from the patch and remove unwanted resin with a cotton wiper, G00034.

SUBTASK 21-20-02-340-086

(6) Place the next piece of the Fiberglass fabric on the first piece so that it covers the opening. Trim the Fiberglass fabric if necessary. Soak the Fiberglass fabric with resin and smooth it. Continue using the same procedure to apply each new ply of fabric.

SUBTASK 21-20-02-340-087

(7) Place the last piece of the Fiberglass fabric on top of the patch so it extends beyond the damage a minimum of 2.0 inches (50.8 mm) on all sides. Smooth the Fiberglass fabric to remove bubbles or winkles.

SUBTASK 21-20-02-340-088

(8) Wipe the duct around the patch with a cotton wiper, G00034 and solvent, B00148 to remove unwanted resin.

SUBTASK 21-20-02-340-089

(9) Allow the resin to cure for 24 hours at room temperature.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-20-02-010-008

(1) If insulation was removed from the duct joint, install new insulation.

SUBTASK 21-20-02-010-009

(2) Replace the panels or other airplane structure removed to gain access to the duct joint.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GROUND CONDITIONED AIR CONNECTION DUCT AND CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the ground conditioned air connection duct.

(2) An installation of the ground conditioned air connection duct.

(3) A removal of the ground conditioned air check valve.

(4) An installation of the ground conditioned air check valve.

(5) A removal of the cooling exchange check valve.

(6) An installation of the cooling exchange check valve.

B. The ground conditioned air connection duct is installed below the mix bay. The function of the duct is to provide a connection for a supply of external conditioned air. To gain access to the duct, do this step:

Open this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

C. The check valve is installed in the mix bay just forward of the air conditioning bays. The function of the check valve is to let conditioned air from an external source to be put into the distribution system and also to seal the distribution system when external conditioned air is not being used.

D. The check valve is a component of the lower mix manifold. It is necessary to remove the mix manifold to get access to the check valve.

TASK 21-21-01-000-801

2. Ground Conditioned Air Connection Duct Removal

(Figure 401)

A. References

Reference	Title
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
191E	Low Pressure ECS Panel - Forward
821	Forward Cargo Door

D. Prepare for the Removal

SUBTASK 21-21-01-860-002

(1) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-21-01-010-001

(2) Open this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 21-21-01-010-002

(3) Remove the center aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

E. Ground Conditioned Air Connection Duct Removal

SUBTASK 21-21-01-010-003

(1) Remove the 15 bolts [4] and washers [5] that attach the ground conditioned air connection duct [1] to the airplane skin.

SUBTASK 21-21-01-010-004

(2) Remove the bolts [7] and washers [5] that attach the clamps [6] to the ground conditioned air connection duct [1].

(a) Make a record of the locations for the bolts [7], washers [5] and clamps [6] to make sure they are installed in the same positions.

SUBTASK 21-21-01-010-005

(3) To get access to the ground conditioned air connection duct [1], do this step:

(a) Open this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

SUBTASK 21-21-01-970-001

(4) Make a record of the correct position that the ground conditioned air connection duct [1] is installed so that it can be reinstalled the same way.

SUBTASK 21-21-01-020-001

(5) Remove the ground conditioned air connection duct [1].

SUBTASK 21-21-01-410-001

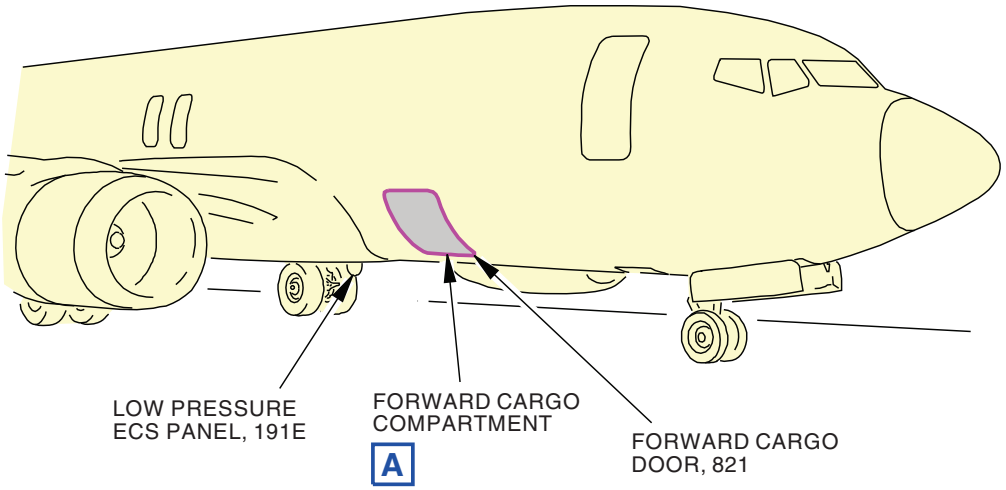
(6) To prevent the entry of unwanted materials into the check valve duct, do this step:

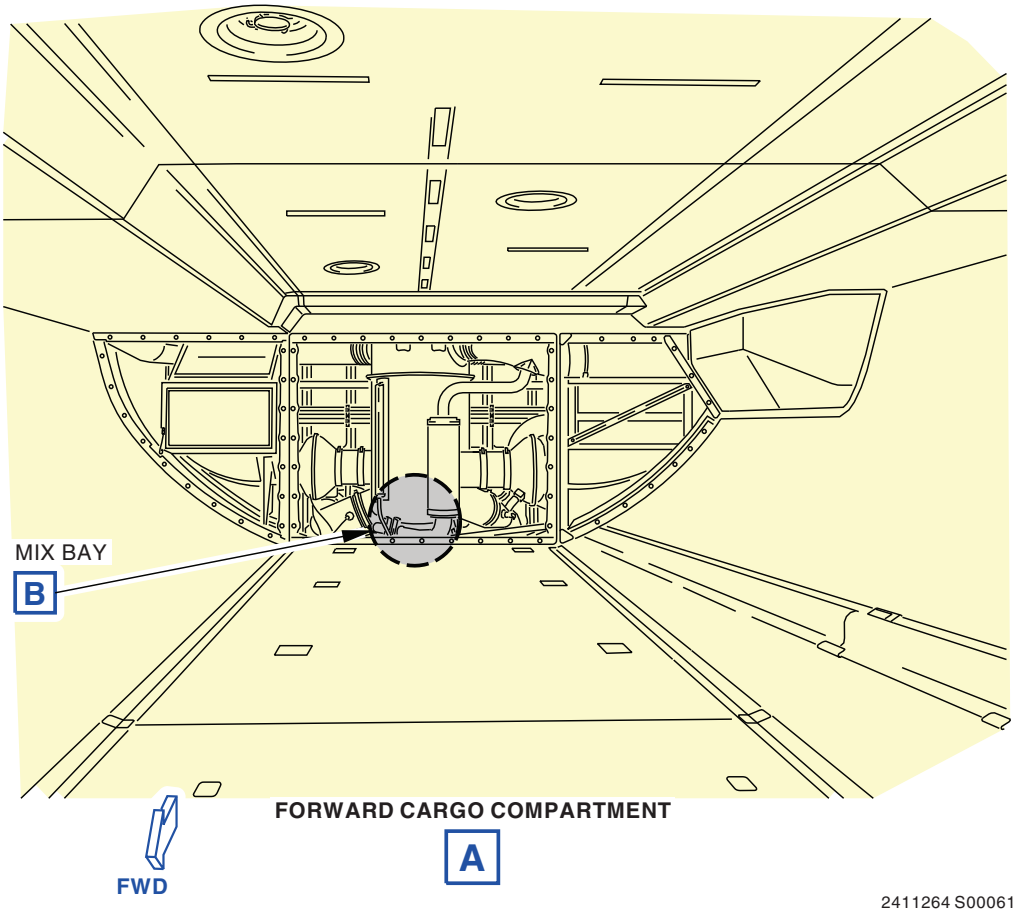
(a) Close this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

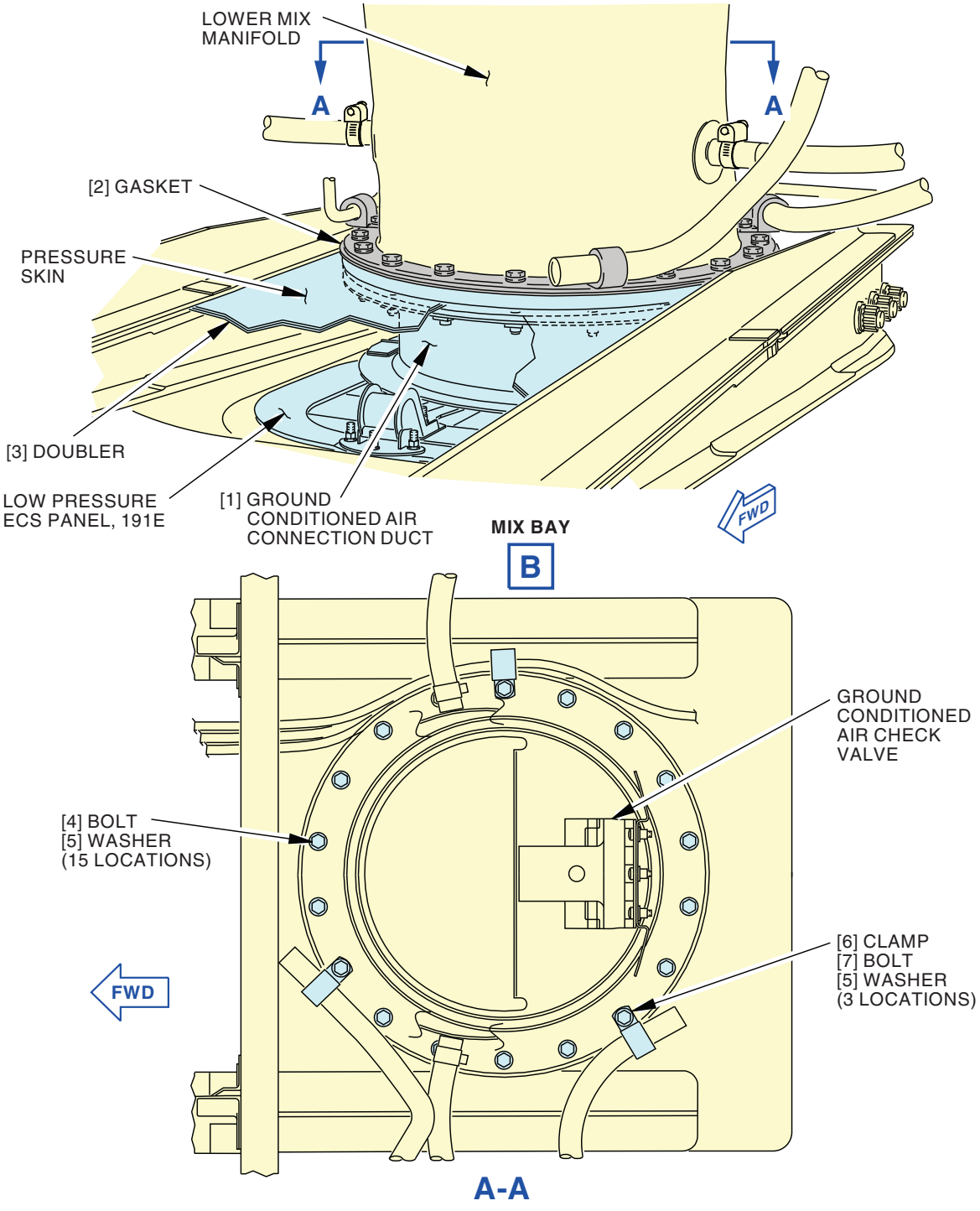




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Ground Conditioned Air Connection Duct Installation
Figure 401/21-21-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ground Conditioned Air Connection Duct Installation
Figure 401/21-21-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-21-01-400-801

3. Ground Conditioned Air Connection Duct Installation

(Figure 401)

A. References

Reference	Title
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
51-22-00-160-801	Prepare For Sealing (P/B 201)
51-22-00-390-801	Non-Removable Faying (Mated) Surface Seal Application (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Air connection duct	21-21-01-02-470	PTF ALL

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
191E	Low Pressure ECS Panel - Forward
821	Forward Cargo Door

F. Prepare for the Installation

SUBTASK 21-21-01-010-006

(1) Open this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

SUBTASK 21-21-01-140-001

(2) Prepare the surfaces for the application of sealant (TASK 51-22-00-160-801).

SUBTASK 21-21-01-390-001

(3) Apply a layer of the sealant, A00247, to the faying surface between the doubler [3] on the airplane skin and the ground conditioned air connection duct [1] (TASK 51-22-00-390-801).

G. Ground Conditioned Air Connection Duct Installation

SUBTASK 21-21-01-420-001

(1) Install the ground conditioned air connection duct [1] in its position:

(a) Make sure the ground conditioned air connection duct [1] is installed in the same position as when it was removed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-21-01-420-002

(2) Install the bolts [7] and washers [5] for the clamps [6].

(a) Make sure they are installed in the positions they were before the removal.

SUBTASK 21-21-01-420-003

(3) Install the 15 bolts [4] and washers [5].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-21-01-410-002

(1) Install the center aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 21-21-01-010-007

(2) Close this access panel:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 21-21-01-410-003

(3) Close this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

——— END OF TASK ———

TASK 21-21-01-000-802

4. Ground Conditioned Air Check Valve Removal

(Figure 402)

A. General

(1) The check valve is a component of the lower mix manifold. It is necessary to remove the mix manifold to get access to the check valve.

B. References

Reference	Title
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
126	Air Conditioning Distribution Bay - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Access Panels

Number	Name/Location
821	Forward Cargo Door

F. Prepare for the Removal

SUBTASK 21-21-01-860-005

(1) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

SUBTASK 21-21-01-860-006

(2) Put these switches on the P5-10 Air Conditioning Panel, to the OFF position and attach DO NOT OPERATE tags, STD-858:

(a) L and R RECIRC FAN

(b) L and R PACK

(c) BLEED 1, 2 and APU.

SUBTASK 21-21-01-860-013

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

SUBTASK 21-21-01-010-008

(4) Open this access door:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 21-21-01-010-009

(5) Remove the center aft bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

G. Mix Manifold Removal

SUBTASK 21-21-01-010-010

(1) To remove the check valves [39] for the two recirculation fans, do these steps:

(a) Hold the check valves [39] while you remove the couplings [37] and the sleeves [38].

(b) Remove the check valves [39].

SUBTASK 21-21-01-010-011

(2) To remove the forward muffler [41] in the mix bay, do these steps:

(a) Remove the clamp [46] that is installed at the rear bulkhead of the mix bay.

(b) Remove the nut [47], washer [48], and spacer [49] that are below the bracket.

(c) Remove the screw [51], washer [50] and washer [48] from the clamp [52].

(d) Remove the coupling [35] and sleeve [36] from the upper duct of the forward muffler.

(e) Carefully remove the forward muffler [41].

SUBTASK 21-21-01-010-012

(3) To remove the upper mix manifold [31], do these steps:

(a) Disconnect the electrical connectors [32] from the mix manifold temperature sensors.

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(b) Remove the clamps [33] and sleeves [34].

(c) Remove the screws [56] and washers [55].

(d) Carefully remove the upper mix manifold [31].

SUBTASK 21-21-01-010-013

(4) Remove the collector ring [54] and the two gaskets [53].

SUBTASK 21-21-01-010-014

(5) Remove the couplings [40] from the lower ducts of the lower mix manifold [43].

SUBTASK 21-21-01-020-002

(6) To disconnect the drain tubes from the lower mix manifold [43]:

(a) Remove the four clamps [42].

(b) Pull the four drain tubes off the lower mix manifold [43].

(c) Make a record of the locations of the drain tubes and the clamps [42] to make sure that they are installed in the same positions.

SUBTASK 21-21-01-020-003

(7) To remove the lower mix manifold [43] from the mix bay, do these steps:

(a) Remove the screws [44] and washers [45] that attach the support brackets of the lower mix manifold [43] to the airplane structure.

(b) Remove the bolts [61] and the washers [62] that attach the clamps [60] to the lower mix manifold [43].

1) Make a record of the locations of the bolts [61], washers [62] and clamps [60] to make sure that they are installed in the same positions.

(c) Remove the bolts [65] and washers [62] that attach the lower mix manifold [43] to the spacer on the airplane skin.

(d) Carefully remove the lower mix manifold [43] from the mix bay.

(e) Remove the gasket [58] that was below the rear duct of the lower mix manifold [43].

H. Ground Conditioned Air Check Valve Removal

SUBTASK 21-21-01-020-004

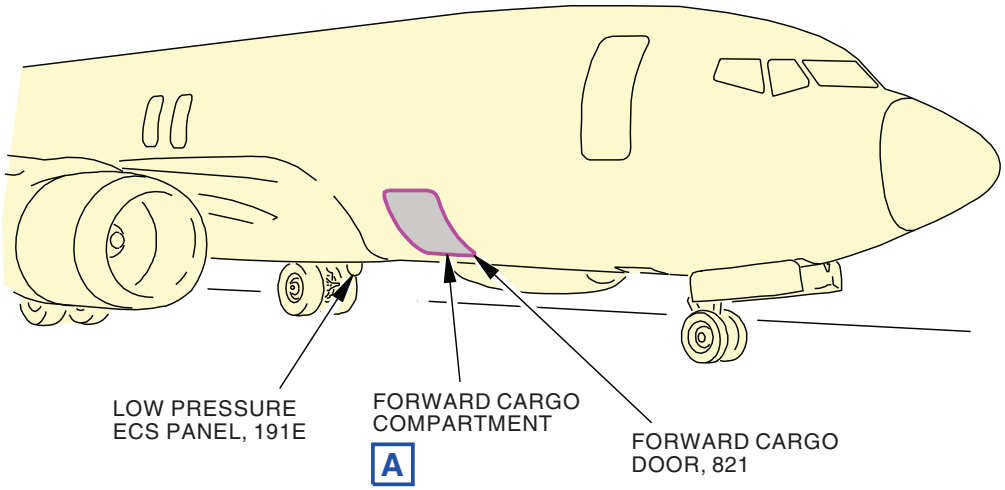
(1) To remove the ground conditioned air check valve [59] from the lower mix manifold [43], do these steps:

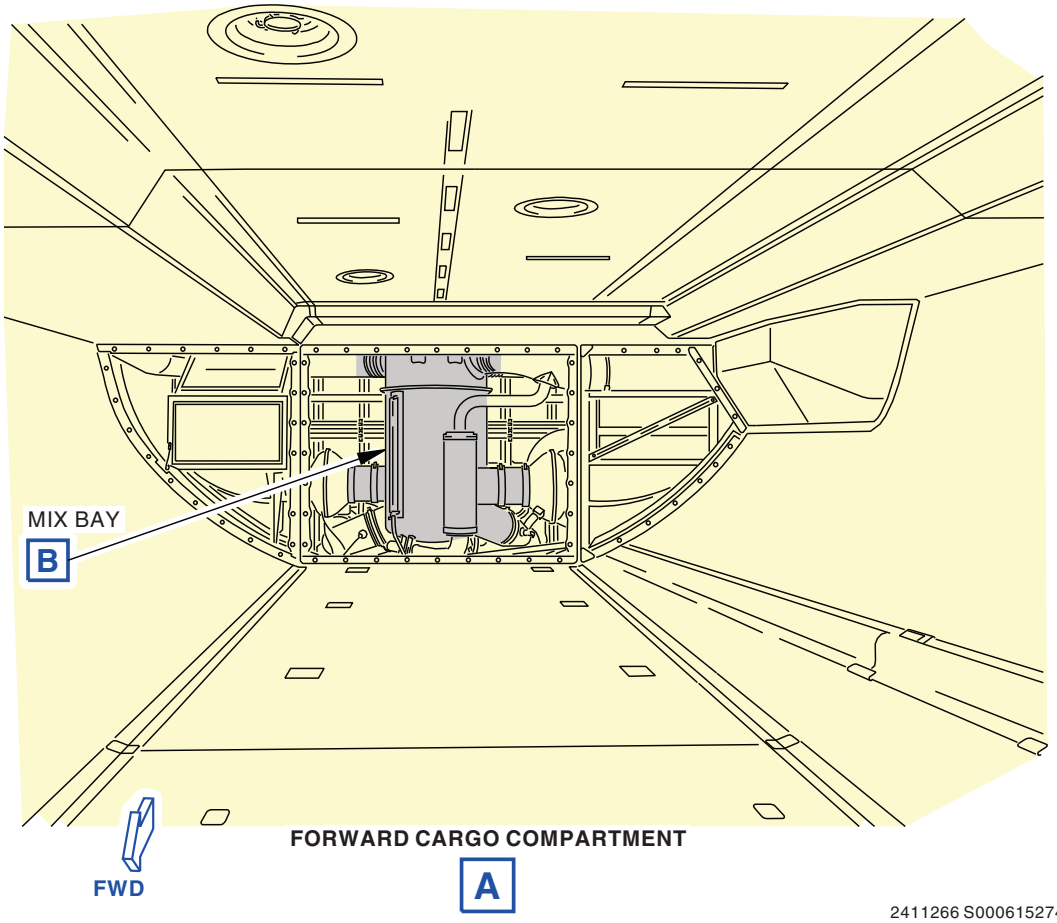
(a) Remove the bolts [63] and the washers [64].

(b) Remove the ground conditioned air check valve [59] from its position in the lower mix manifold [43].

——— END OF TASK ———

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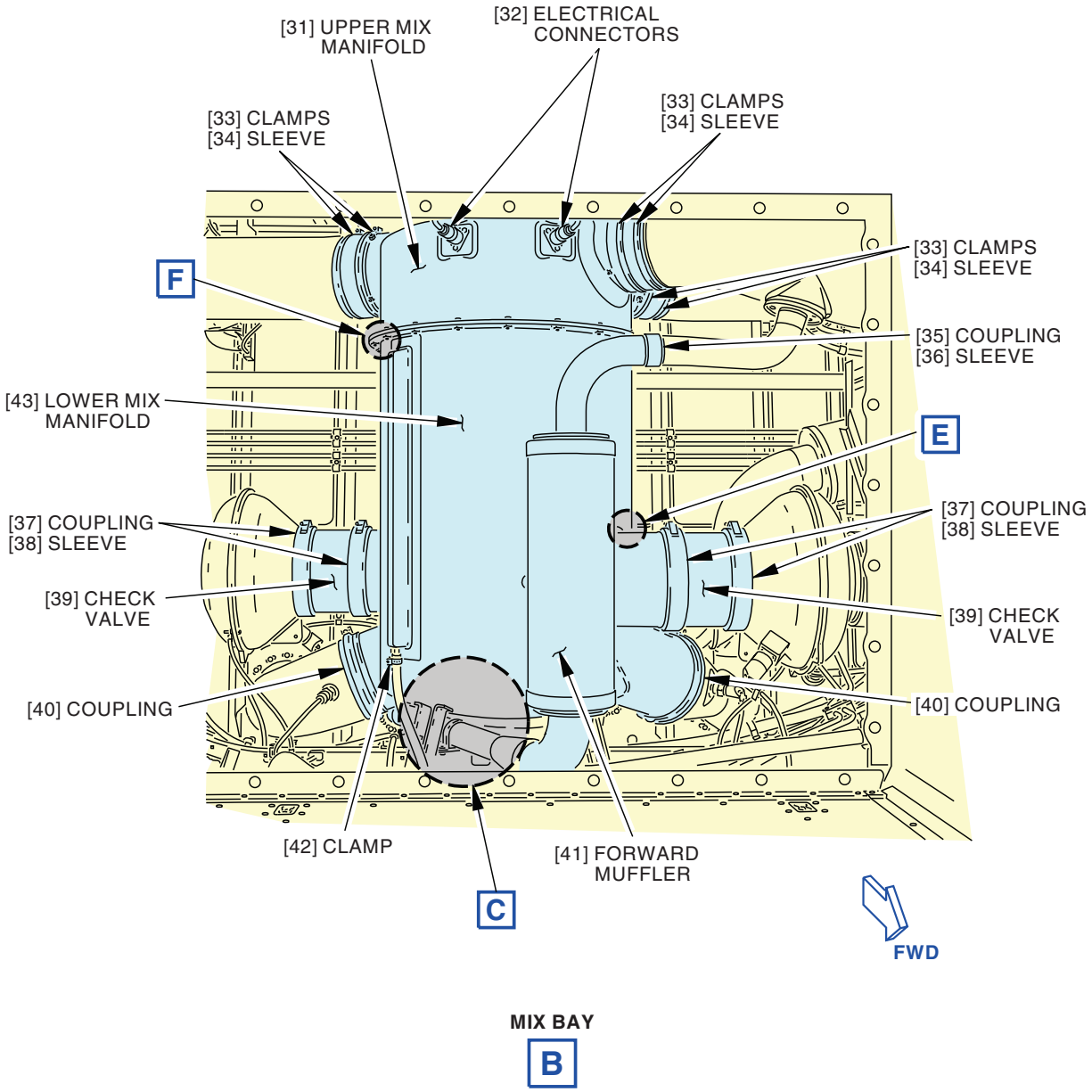




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Ground Conditioned Air Check Valve Installation
Figure 402/21-21-01-990-802 (Sheet 1 of 5)

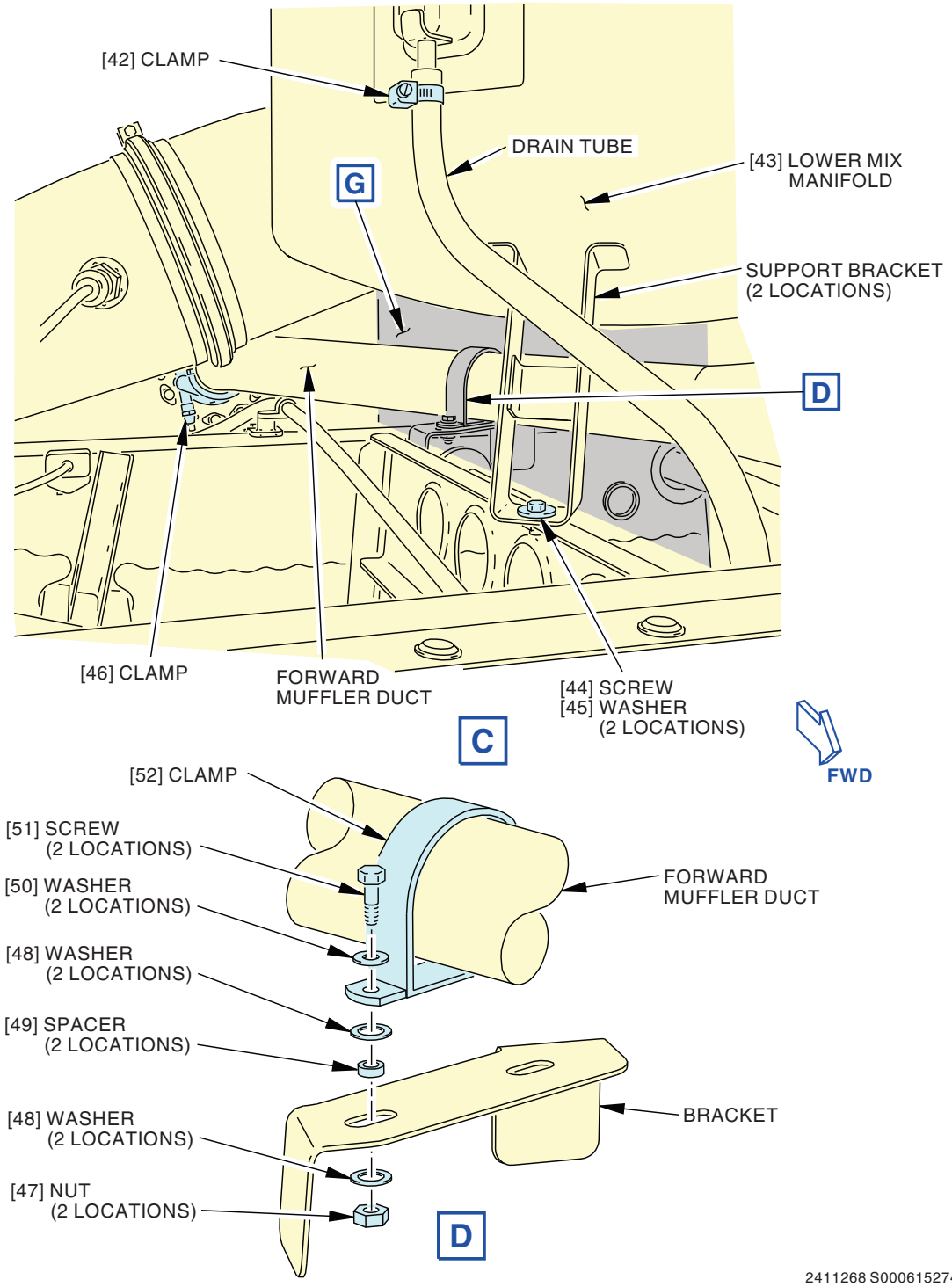
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411267 S00061527438_V1

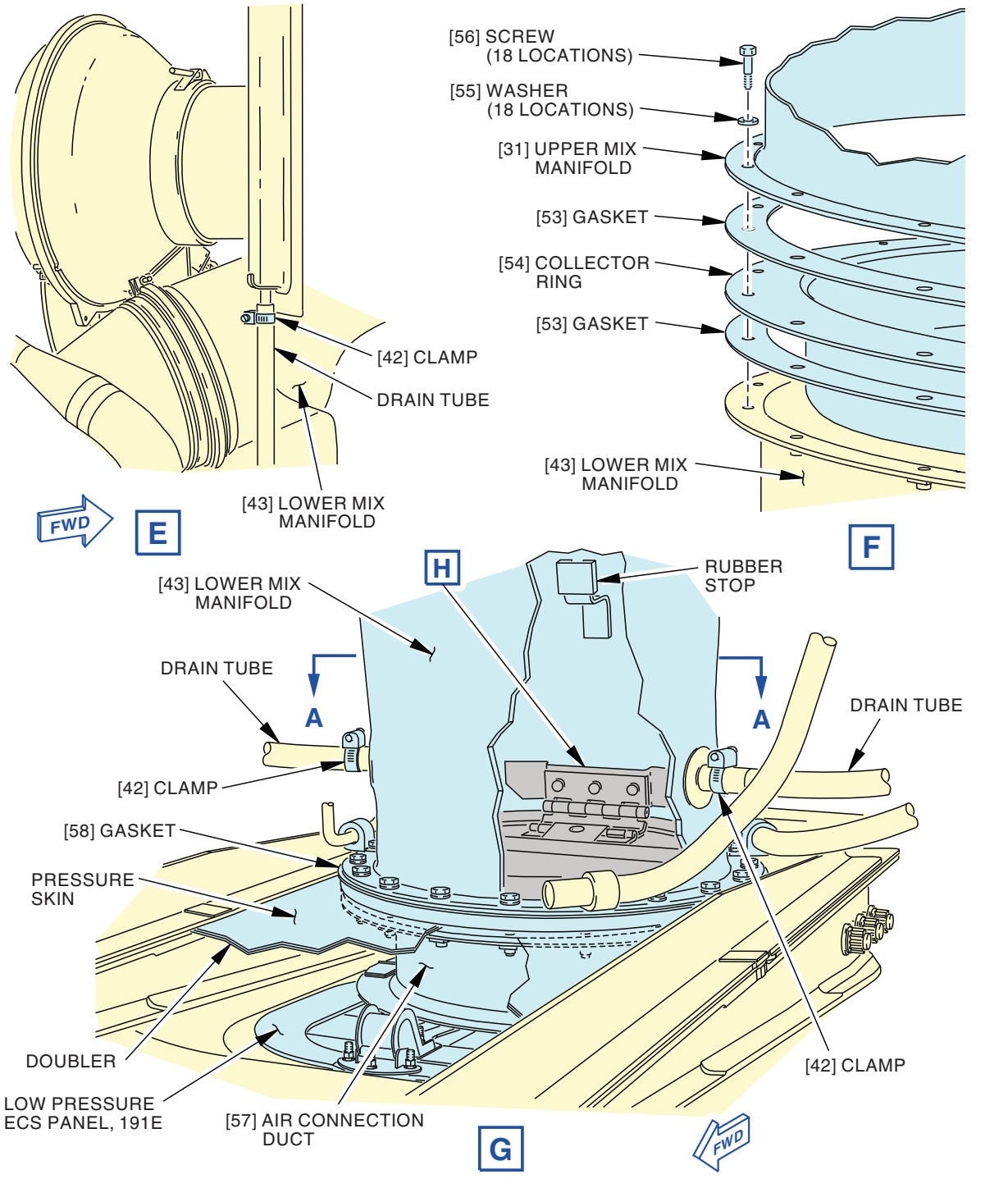
Ground Conditioned Air Check Valve Installation
Figure 402/21-21-01-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Ground Conditioned Air Check Valve Installation
Figure 402/21-21-01-990-802 (Sheet 3 of 5)

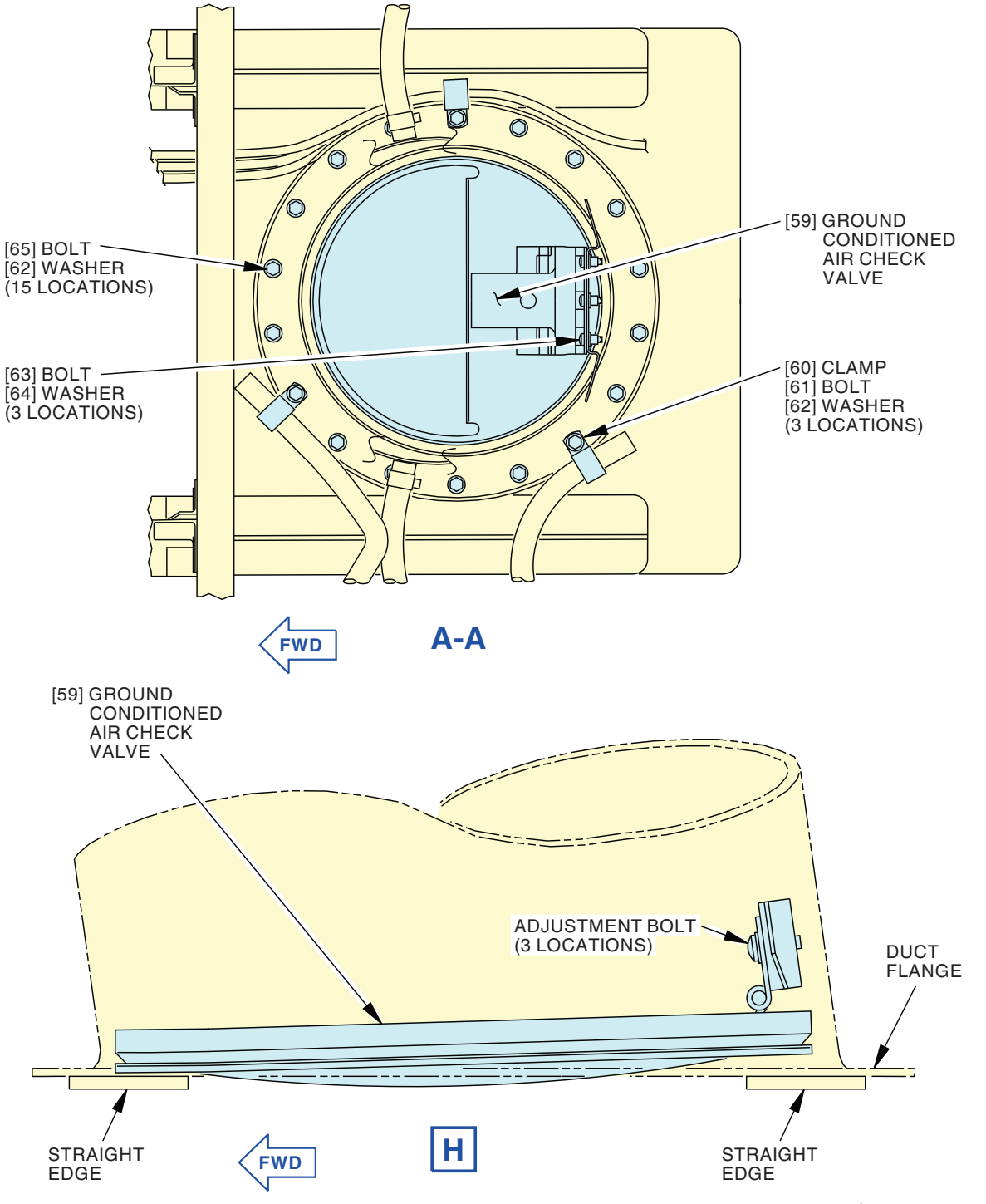
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411269 S00061527440_V1

Ground Conditioned Air Check Valve Installation
Figure 402/21-21-01-990-802 (Sheet 4 of 5)

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Ground Conditioned Air Check Valve Installation
Figure 402/21-21-01-990-802 (Sheet 5 of 5)

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TASK 21-21-01-400-802

5. Ground Conditioned Air Check Valve Installation

(Figure 402)

A. General

(1) The hinge on the check valve has a spring which holds the check valve almost open when the air conditioning system is off. When the air conditioning system is on, the pressure in the mix manifold causes the swing check valve to close. Before you install the lower mix manifold, you must do a check of the check valve and adjust the hinge if it is necessary. The check valve must open and close correctly.

B. References

Reference	Title
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
53	Gasket	21-21-01-02-530	PTF ALL
58	Gasket	21-21-01-02-545	PTF ALL
59	Ground conditioned air check valve	21-21-05-01-145	PTF ALL

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
126	Air Conditioning Distribution Bay - Right
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
191E	Low Pressure ECS Panel - Forward
821	Forward Cargo Door

F. Ground Conditioned Air Check Valve Installation

SUBTASK 21-21-01-420-004

(1) Do these steps to install the ground conditioned air check valve [59] in the lower mix manifold [43]:

(a) Put the ground conditioned air check valve [59] in its position in the lower mix manifold [43].

(b) Loosely install the bolts [63] and the washers [64].

(c) Adjust the ground conditioned air check valve [59] so that the bolts [63] are in the center of the adjustment slots.

(d) Fully tighten the bolts [63] and the washers [64].

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G. Ground Conditioned Air Check Valve Adjustment

SUBTASK 21-21-01-710-001

(1) Do this check of the ground conditioned air check valve [59]:

(a) Put a straight edge on the flange of the rear duct of the lower mix manifold [43].

(b) Move the straight edge to the end of the ground conditioned air check valve [59] opposite the hinge.

(c) Make sure that the end of the ground conditioned air check valve [59] touches the straight edge.

(d) Move the straight edge to the end of the ground conditioned air check valve [59] below the hinge.

(e) Make sure that there is a small clearance between the edge of the ground conditioned air check valve [59] and the straight edge.

(f) If there is no clearance, adjust the hinge:

1) Loosen the hinge adjustment bolts [63].

2) Move the hinge up as necessary.

3) Tighten the hinge adjustment bolts [63].

(g) Push down on the ground conditioned air check valve [59] at the hinge.

(h) Make sure that the clearance in the above step closes without too much downward force.

(i) Release the pressure on the ground conditioned air check valve [59].

(j) Make sure that there is clearance again between the edge of ground conditioned air check valve [59] and the straight edge.

(k) If too much force is necessary to remove the clearance when you push down on the ground conditioned air check valve [59], adjust the hinge:

1) Loosen the hinge adjustment bolts [63].

2) Move the hinge down as necessary.

3) Tighten the hinge adjustment bolts [63].

(l) Do this check again to make sure that the ground conditioned air check valve [59] operates correctly.

H. Mix Manifold Installation

SUBTASK 21-21-01-420-005

(1) To install the gasket [58] that was below the rear duct of the lower mix manifold [43], do these steps:

(a) Make sure that the gasket [58] is not damaged.

1) Replace the gasket [58], if it is damaged.

(b) When you install the gasket [58], align the slot in the gasket [58] with the slot on the spacer on the airplane skin.

SUBTASK 21-21-01-420-006

(2) Install the lower mix manifold [43] in the mix bay:

(a) Carefully move the lower mix manifold [43] to its position.

(b) Make sure that the ducts on the lower mix manifold [43] are aligned with the other ducts in the mix bay.

(c) Install the bolts [61] and washers [62] for the clamps [60].

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1) Make sure that the bolts [61] and washers [62] are installed in the positions they were before the removal.

(d) Install the bolts [65] and the washers [62] that attach the lower mix manifold [43] to the spacer on the airplane skin.

(e) Install the screws [44] and washers [45] that attach support brackets of the lower mix manifold [43] to the airplane structure.

SUBTASK 21-21-01-010-015

(3) Install the couplings [40] for the lower ducts of the lower mix manifold [43].

(a) Tighten the nut on the couplings [40] to 20 in-lb (2.3 N·m) to 25 in-lb (2.8 N·m).

SUBTASK 21-21-01-010-016

(4) Install the collector ring [54] and the gaskets [53]:

(a) Make sure that the gaskets [53] are not damaged.

1) Replace the gaskets [53], if they are damaged.

(b) Put one of the gaskets [53] on the lower mix manifold [43].

(c) Put the collector ring [54] on the lower mix manifold [43].

(d) Put the other gasket [53] on the collector ring [54].

1) Make sure that the holes in the gaskets [53], collector ring [54], and the lower mix manifold [43] are aligned.

SUBTASK 21-21-01-010-017

(5) To install the upper mix manifold [31], do these steps:

(a) Put the upper mix manifold [31] on the lower mix manifold [43].

1) Make sure that the holes in the upper mix manifold [31], lower mix manifold [43], gaskets [53], and collector ring [54] are aligned.

(b) Install the screws [56] and washers [55].

(c) Install the electrical connectors [32] on the mix manifold temperature sensors.

(d) Install the clamps [33] and sleeves [34].

SUBTASK 21-21-01-020-005

(6) To connect the drain tubes to the lower mix manifold [43]:

(a) Push the four drain tubes on lower mix manifold [43].

1) Make sure that the drain tubes and the clamps [42] are installed in the positions they were before the removal.

(b) Install the clamps [42] on the drain tubes.

SUBTASK 21-21-01-010-018

(7) To install the forward muffler [41] in the mix bay, do these steps:

(a) Carefully move the forward muffler [41] to its location in the mix bay.

(b) Install the clamp [46] at the rear bulkhead of the mix bay.

1) Tighten the nut on the clamp to 55 in-lb (6.2 N·m) - 60 in-lb (6.8 N·m).

2) Make sure that the loop bands do not touch.

3) Make sure that the clearance between the retainers is 0.050 in. (1.270 mm) - 0.310 in. (7.874 mm).

(c) Move the clamp [52] to its position on the bracket.

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(d) Put the washers [48] under the clamp [52].

(e) Install the screw [51] and the washer [48].

(f) Install the spacer [49], washer [48] and nut [47] below the bracket.

1) Make sure that the spacer [49] is in the slot on the bracket.

(g) Install the coupling [35] and the sleeve [36] for the top duct of the forward muffler [41].

SUBTASK 21-21-01-010-019

(8) To install the check valve [39] for the recirculation fans, do these steps for each of the two check valves [39]:

(a) Hold the check valve [39] in its position.

1) Make sure that the flow arrow on the check valve [39] points inboard.

(b) Install the couplings [37] and the sleeves [38].

SUBTASK 21-21-01-860-014

(9) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

I. Check Valve Duct Installation Test

SUBTASK 21-21-01-740-001

(1) Do a Built-In-Test Equipment (BITE) test on the left Integrated Air Systems Controller (M2939), and the right Integrated Air Systems Controller (M2957):

NOTE: This will clear any fault introduced during maintenance when power is interrupted.

(a) Do the applicable ground test in this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

SUBTASK 21-21-01-860-007

(2) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-21-01-860-008

(3) Supply conditioned air to the air conditioning system (TASK 21-00-00-800-803).

SUBTASK 21-21-01-790-001

(4) Make sure that there are no leaks at any of the ducts or hoses you disconnected for this installation.

SUBTASK 21-21-01-010-020

(5) Open this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

SUBTASK 21-21-01-710-002

(6) Make sure that air flows out the drain hole.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 21-21-01-860-009

(1) Remove the conditioned air from the air conditioning system (TASK 21-00-00-800-802)

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SUBTASK 21-21-01-710-003

(2) Make sure that the swing check valve is not completely closed.

SUBTASK 21-21-01-410-004

(3) Install the center liner in the aft bulkhead in the forward cargo compartment (TASK 50-11-02-400-801)

SUBTASK 21-21-01-410-005

(4) Close this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

SUBTASK 21-21-01-410-006

(5) Close this access door:

Number	Name/Location
821	Forward Cargo Door

——— END OF TASK ———

TASK 21-21-01-000-803

6. Cooling Exchange Check Valve Removal

(Figure 403)

A. General

(1) This task includes the steps to remove the Cooling Exchange Check Valve.

(2) You must remove electrical power from the airplane before you remove the cooling exchange check valve. This will make sure the electrical/electronic equipment does not receive electrical power when the equipment cooling system is not in operation.

B. References

Reference	Title
12-15-21-600-804	Crew Oxygen Cylinder Replacement (P/B 301)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 21-21-01-010-021

(1) To get access to the cooling exchange check valve, do this step:

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-21-01-010-022

(2) Remove the access panel on the top of the raised platform just aft and right of the aircraft center line:

(a) Remove the 9 screws that attach the access panel to the top of the raised platform.

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(b) Remove the raised platform.

SUBTASK 21-21-01-010-023

(3) Remove the crew oxygen system cylinder (TASK 12-15-21-600-804).

F. Cooling Exchange Check Valve Removal

SUBTASK 21-21-01-020-006

(1) Remove the strap [4].

SUBTASK 21-21-01-020-007

(2) Remove the sleeve [3].

SUBTASK 21-21-01-020-008

(3) Loosen the coupling [2].

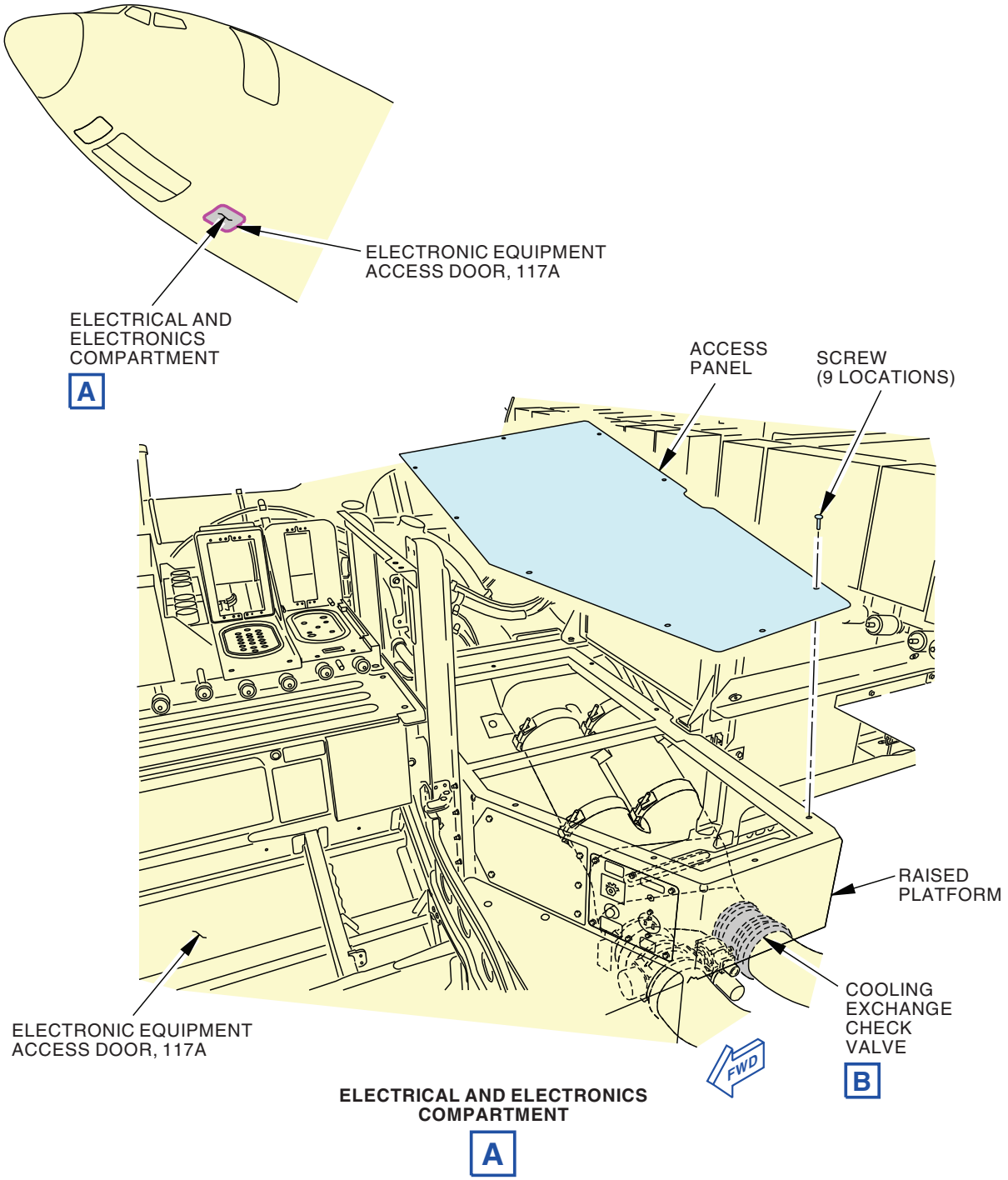
SUBTASK 21-21-01-020-009

(4) Remove the cooling exchange check valve [1]:

(a) Put covers on the duct openings to prevent the entry of unwanted material.

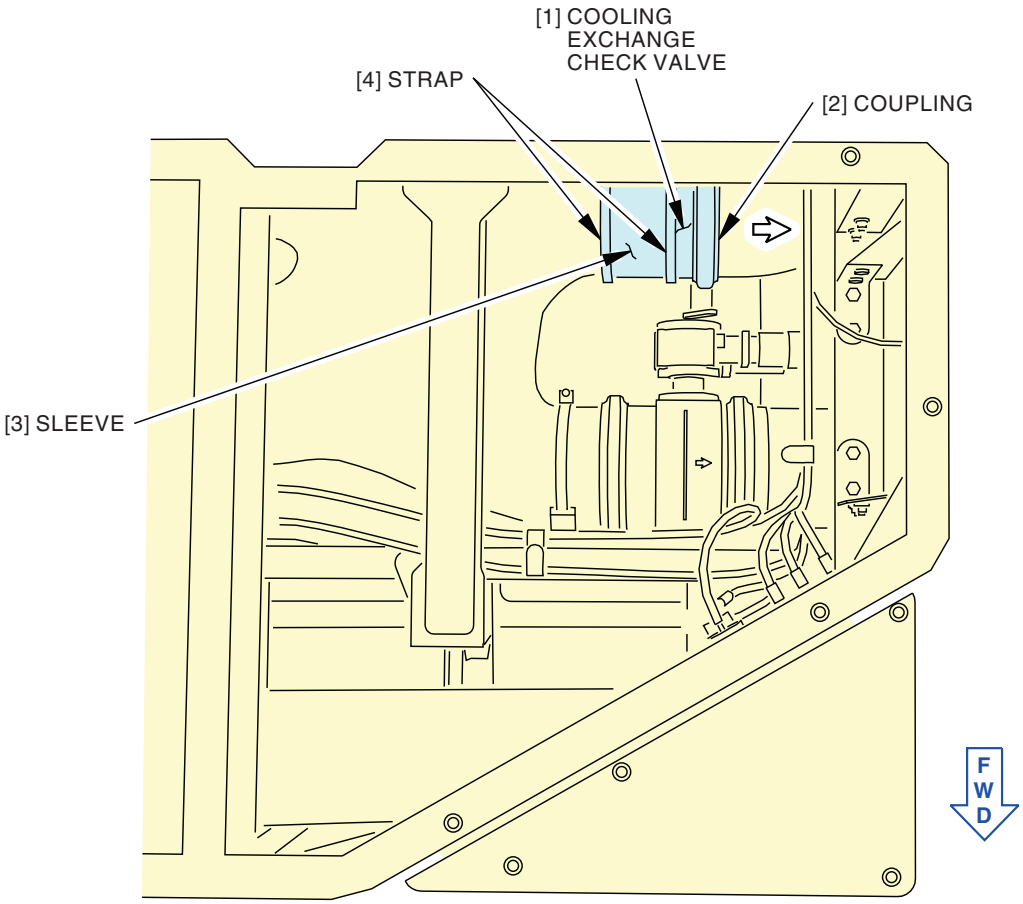
——— END OF TASK ———

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Cooling Exchange Check Valve Installation
Figure 403/21-21-01-990-803 (Sheet 1 of 2)

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COOLING EXCHANGE CHECK VALVE
(OXYGEN BOTTLE INSTALLATION IS NOT SHOWN)

B

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Cooling Exchange Check Valve Installation
Figure 403/21-21-01-990-803 (Sheet 2 of 2)

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TASK 21-21-01-400-803

7. Cooling Exchange Check Valve Installation

(Figure 403)

A. General

(1) This task includes the steps to install the Cooling Exchange Check Valve.

B. References

Reference	Title
50-10-04-000-801	Cargo Compartment Floor Panel Removal (P/B 401)
50-10-04-400-801	Cargo Compartment Floor Panel Installation (P/B 401)
52-31-00-580-802	Close the Cargo Door (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-3923	Tool - Installation, Tie Strap Part #: GS2B-E Supplier: 06383 Part #: GS4H-E Supplier: 06383 Part #: GTS-E Supplier: 06383 Part #: PPTS Supplier: 06383 Opt Part #: GS2B Supplier: 06383 Opt Part #: GS2BL Supplier: 06383 Opt Part #: GS4H Supplier: 06383 Opt Part #: GTS Supplier: 06383

D. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

E. Cooling Exchange Check Valve Installation

SUBTASK 21-21-01-420-007

(1) Install the Cooling Exchange Check Valve:

(a) Remove the duct covers.

(b) Put the cooling exchange check valve [1] in its position.

1) Install the cooling exchange check valve [1] so the direction of the flow is from the E/E compartment to the forward cargo compartment.

(c) Install the coupling [2].

(d) Tighten the coupling [2].

(e) Move the sleeve [3] on to the cooling exchange check valve [1] and the duct flanges.

(f) Put the strap [4] around the sleeve [3].

(g) Tighten the strap [4].

1) To apply the correct tension, set the selector knob to heavy and the tension indicator to 8.

a) Use the panduit tie strap installation tool, COM-3923 (with blue handle).

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F. Cooling Exchange Check Valve Installation Test

SUBTASK 21-21-01-860-011

(1) Make sure these switches, on the P5 Overhead Panel, are in the NORMAL position:

(a) EQUIP COOLING, EXHAUST

(b) EQUIP COOLING, SUPPLY

SUBTASK 21-21-01-710-004

(2) Make sure there is no air leakage at the coupling [2] and strap [4].

SUBTASK 21-21-01-010-024

(3) Remove the forward cargo compartment forward floor panel (TASK 50-10-04-000-801).

SUBTASK 21-21-01-710-005

(4) Make sure there is air flow out of the forward cargo heat duct port.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-21-01-410-007

(1) Install the forward cargo compartment forward cargo floor panel (TASK 50-10-04-400-801).

SUBTASK 21-21-01-860-012

(2) Put these switches in the AUTO position:

(a) L PACK

(b) R PACK

(c) L RECIRC FAN

(d) R RECIRC FAN

SUBTASK 21-21-01-410-008

(3) Close the forward cargo door (TASK 52-31-00-580-802).

——— END OF TASK ———

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WINDSHIELD AND FOOT AIR OUTLET VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the windshield and foot air outlet valve.

(2) An installation of the windshield and foot air outlet valve.

B. There is one windshield and foot air outlet valve for the captain and one for the first officer. The valves control the flow of air to the foot air outlets and the windshield defoggers. The valves are installed behind the display units on the captains instrument panel, P1, and the first officers instrument panel, P3.

TASK 21-22-01-000-801

2. Windshield and Foot Air Outlet Valve Removal

(Figure 401)

A. References

Reference	Title
31-65-11-000-801	MAX Display Unit Removal (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 21-22-01-860-001

(1) Make sure that the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, are set to the OFF position.

SUBTASK 21-22-01-860-002

(2) Make sure that the L and R PACK switches, on the P5-10 air conditioning panel, are set to the OFF position.

SUBTASK 21-22-01-010-001

(3) To get access to the captain's windshield and foot air outlet valve [4], do this step:

(a) Remove the display units on the captain's instrument panel (P1) (TASK 31-65-11-000-801).

SUBTASK 21-22-01-010-002

(4) To get access to the first officer's windshield and foot air outlet valve [4], do this step:

(a) Remove the display units on the first officer's instrument panel (P3) (TASK 31-65-11-000-801).

D. Windshield and Foot Air Outlet Valve Removal

SUBTASK 21-22-01-020-001

(1) Disconnect the cable assemblies [9] from the windshield and foot air outlet valve [4] as follows:

(a) Remove the cotter pins [2], the washers [3], and the pins [1].

SUBTASK 21-22-01-010-003

(2) Remove the strap [8] that holds the inlet flex duct to the valve [4].

SUBTASK 21-22-01-010-004

(3) Remove the strap [5] that holds the flex duct to the windshield and foot air outlet valve [4].

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SUBTASK 21-22-01-010-005

(4) Slide the inlet duct and the two outlets ducts away from the windshield and foot air outlet valve [4].

SUBTASK 21-22-01-010-006

(5) Remove the windshield and foot air outlet valve [4] from the wire tray bracket, as follows:

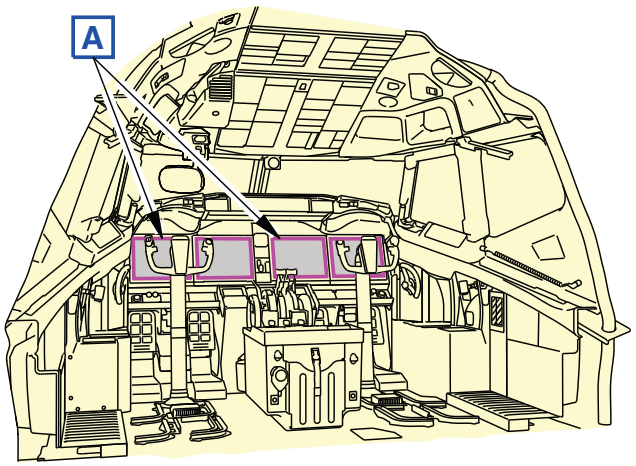
(a) Remove the screws [6] and the washers [7].

SUBTASK 21-22-01-020-002

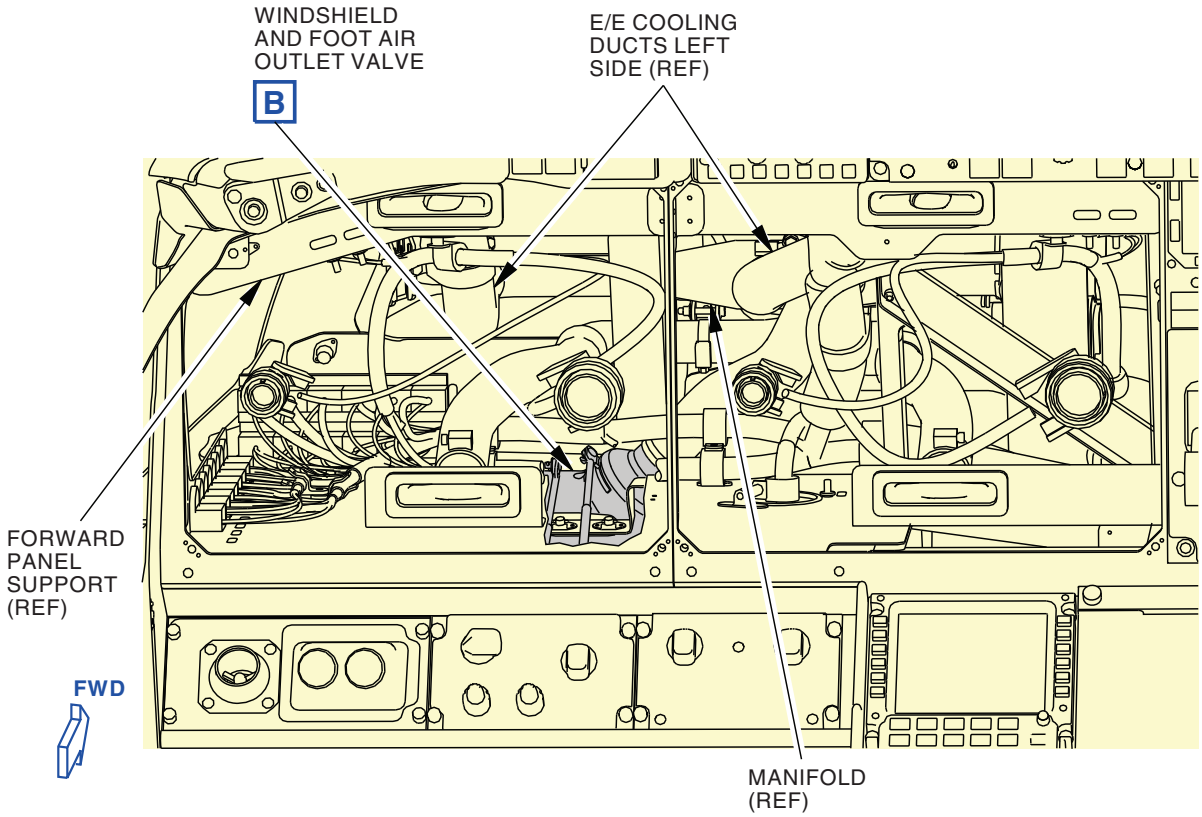
(6) Remove the windshield and foot air outlet valve [4].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

NOTE:

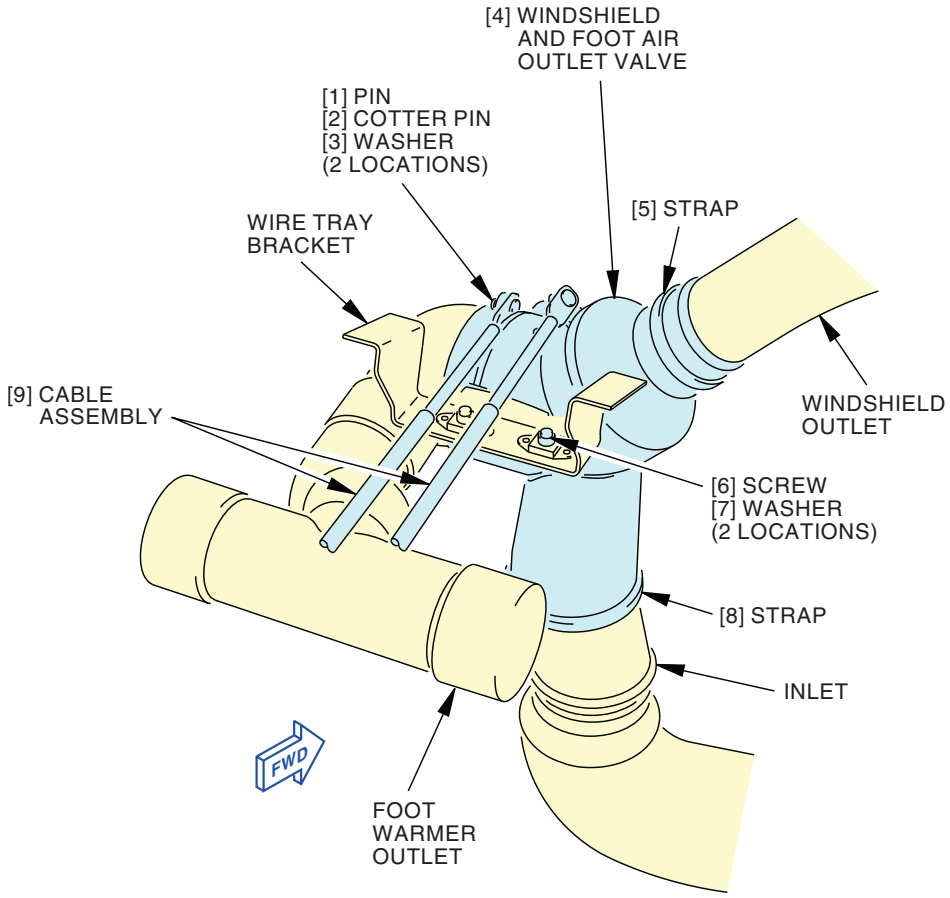
MAX DISPLAY UNITS REMOVED

A

2585568 S0000617654_V1

Windshield and Foot Air Outlet Valve Installation
Figure 401/21-22-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WINDSHIELD AND FOOT AIR OUTLET VALVE

B

2411274 S00061527454_V2

Windshield and Foot Air Outlet Valve Installation
Figure 401/21-22-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-22-01-400-801

3. Windshield and Foot Air Outlet Installation

(Figure 401)

A. References

Reference	Title
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
21-22-02-820-801	Air Outlet Valve Control Cables Adjustment and Test (P/B 501)
31-65-11-400-801	MAX Display Unit Installation (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Windshield and Foot Air Outlet Valve Installation

SUBTASK 21-22-01-210-001

(1) Make sure that there is no unwanted material or contamination in the inlet duct to the windshield and foot air outlet valve [4] or the outlet ducts from the windshield and foot air outlet valve [4].

SUBTASK 21-22-01-420-001

(2) Install the windshield and foot air outlet valve [4] on the wire tray bracket as follows (Figure 401):

(a) Position the windshield and foot air outlet valve [4] in its correct location.

(b) Install the washers [7] and the screws [6].

SUBTASK 21-22-01-410-001

(3) Install the inlet duct and the two outlet ducts on the windshield and foot air outlet valve [4] as follows:

(a) Slide the ends of the inlet duct and the outlet ducts onto the windshield and foot air outlet valve [4].

(b) Install the strap [5], and the strap [8] to hold the ducts onto the valve [4].

SUBTASK 21-22-01-420-002

(4) Attach the cable assemblies [9] to the windshield and foot air outlet valve [4] as follows:

(a) Put the control cable for the windshield air on the applicable valve body of the windshield and foot air outlet valve [4].

(b) Put the control cable for the foot air on the applicable valve body of the windshield and foot air outlet valve [4].

(c) Install the pins [1], the washers [3], and the cotter pins [2].

D. Air Outlet Valve Installation Test

SUBTASK 21-22-01-710-001

(1) Do a check of the windshield and foot air outlet valve [4] as follows:

(a) Do this task: Supply Conditioned Air to the Airplane, TASK 21-00-00-800-801.

(b) Operate the FOOT AIR or WINDSHIELD AIR knob, as applicable, to close the valve [4].

(c) Make sure that no air flow is detected at the windshield defogger outlet or at the foot air outlet, as applicable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) If air flow is detected, adjust the applicable cable assembly (TASK 21-22-02-820-801).

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-22-01-410-002

(1) Install the display units for the captains instrument panel (P1) or the first officers instrument panel (P3), as applicable (TASK 31-65-11-400-801).

SUBTASK 21-22-01-860-003

(2) Make sure that the L and R RECIRC FAN switches on the P5-10 air conditioning panel are set to the AUTO position.

SUBTASK 21-22-01-410-003

(3) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR OUTLET VALVE CONTROL CABLES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Air outlet valve control cables removal.

(2) Air outlet valve control cables installation.

B. There are two control cables that go to each windshield and foot air outlet valve. One of the cables controls the flow of air to the foot air outlet. The other cable controls the flow of air to the windshield defogger.

TASK 21-22-02-000-801

2. Air Outlet Valve Control Cables Removal

(Figure 401)

A. References

Reference	Title
31-65-11-000-801	MAX Display Unit Removal (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 21-22-02-010-001

(1) To get access the control cables for the outlet valve, do this step:

(a) Remove the display units on the captain's instrument panel (P1) or the first officer's instrument panel (P3), as applicable (TASK 31-65-11-000-801).

D. Air Outlet Valve Control Cables Removal

SUBTASK 21-22-02-020-002

(1) Remove the nut [9] from the end of the cable assembly [1] or the cable assembly [2].

SUBTASK 21-22-02-020-003

(2) Remove the cotter pin [6], the washer [7], and the pin [5] that attach the end of cable assembly [1] or cable assembly [2] to the tab on the applicable windshield and foot air outlet valve.

SUBTASK 21-22-02-020-004

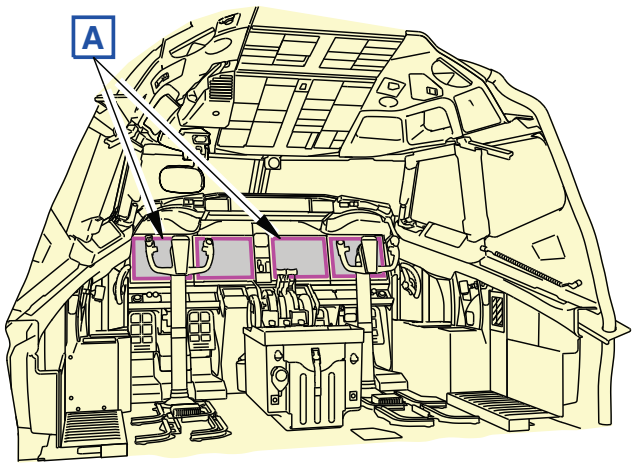
(3) Loosen the nut [3] or the nut [8] to release the cable assembly [1] or the cable assembly [2] from the bracket.

SUBTASK 21-22-02-020-005

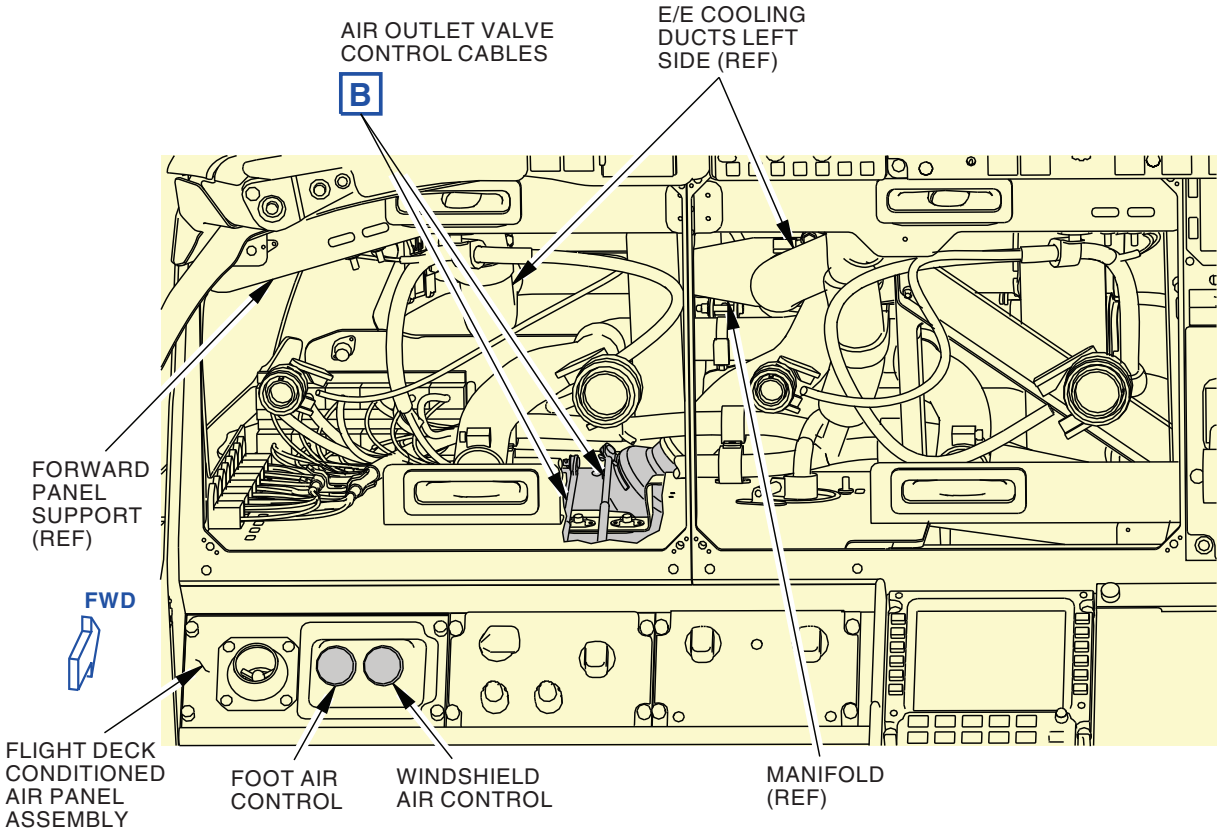
(4) Remove the cable assembly [1] or the cable assembly [2].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

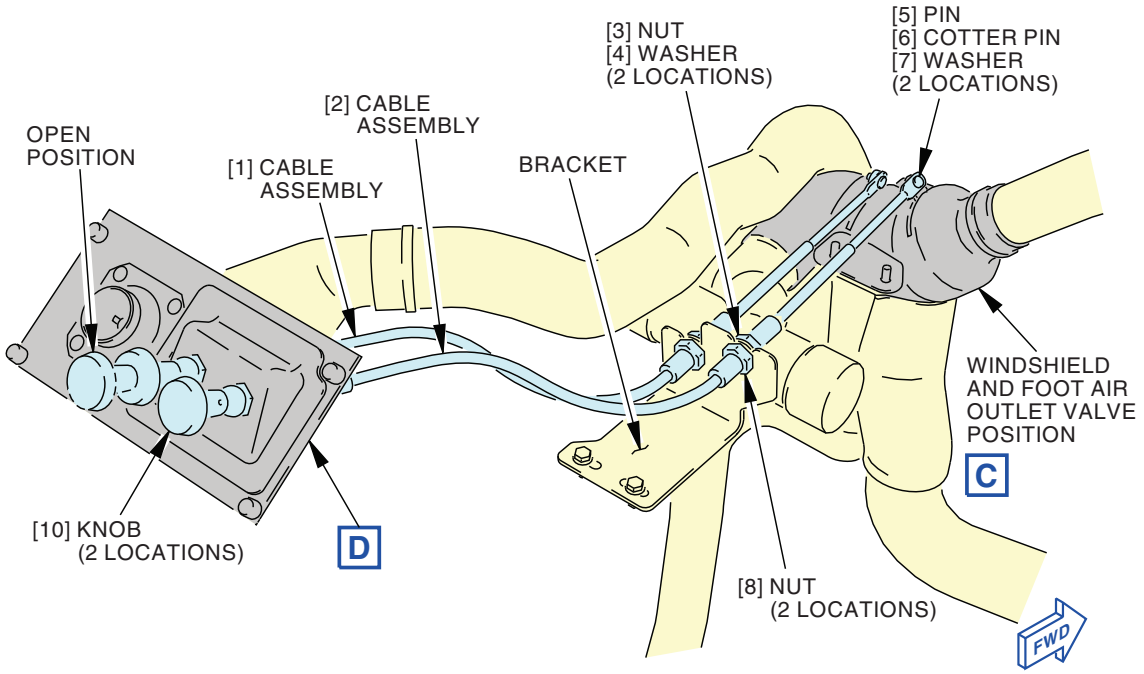
NOTE:

MAX DISPLAY UNITS REMOVED

2411275 S00061527459_V2

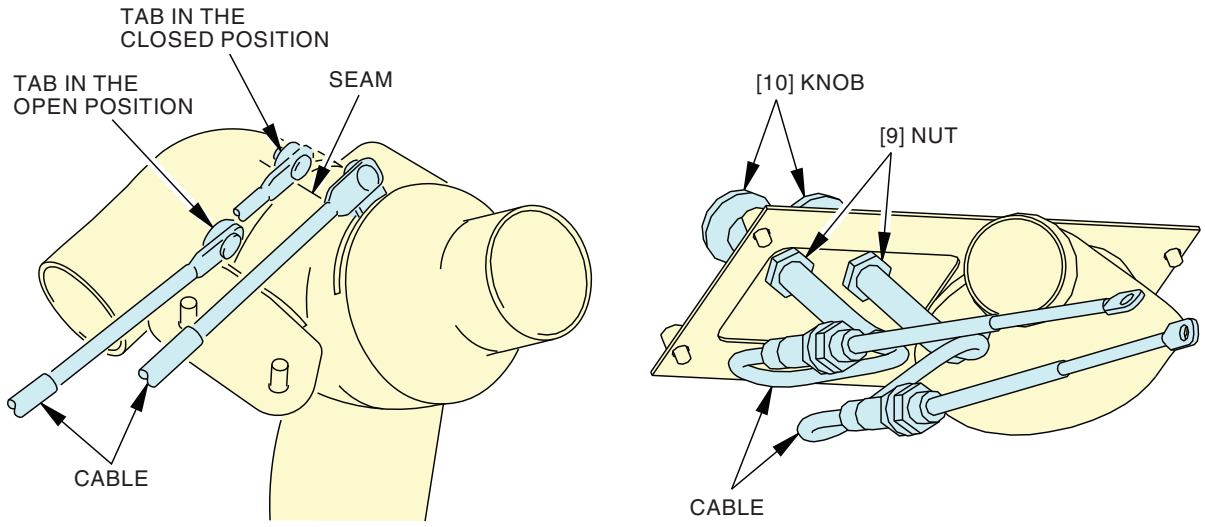
Air Outlet Valve Control Cables Installation
Figure 401/21-22-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AIR OUTLET VALVE CONTROL CABLES

B



WINDSHIELD AND FOOT AIR OUTLET VALVE POSITION

C

D

2411276 S00061527460_V1

Air Outlet Valve Control Cables Installation
Figure 401/21-22-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-22-02-400-801

3. Air Outlet Valve Control Cables Installation

(Figure 401)

A. References

Reference	Title
21-22-02-820-801	Air Outlet Valve Control Cables Adjustment and Test (P/B 501)
31-65-11-400-801	MAX Display Unit Installation (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Air Outlet Valve Control Cables Installation

SUBTASK 21-22-02-210-001

(1) Position cable assembly [1] or cable assembly [2] in its correct position (Figure 401).

SUBTASK 21-22-02-420-001

(2) Install the nut [9] from the end of cable assembly [1] or cable assembly [2] and tighten firmly to the plate.

SUBTASK 21-22-02-410-001

(3) Tighten the nut [3] or the nut [8] to firmly hold cable assembly [1] or cable assembly [2] in the bracket.

SUBTASK 21-22-02-860-001

(4) Make sure that the windshield and foot air outlet valve ports are fully closed.

NOTE: The tabs on the valve will be centered on the seam on the valve between the two halves of the valve when the outlet ports are fully closed.

SUBTASK 21-22-02-860-002

(5) Push the knob [10] fully forward to put the cable assembly [1] or the cable assembly [2] in the closed position.

SUBTASK 21-22-02-820-001

(6) Adjust the nut [3] or the nut [8] until the hole in the end of cable assembly [1] or cable assembly [2] aligns with the hole in the tab on the windshield and foot air outlet valve.

SUBTASK 21-22-02-420-003

(7) Install the pin [5], the washer [7], and a new cotter pin [6].

SUBTASK 21-22-02-700-001

(8) Perform the air outlet valve control cables test (TASK 21-22-02-820-801).

D. Put the Airplane Back to Its Usual Condition

SUBTASK 21-22-02-410-005

(1) Install the display units for the captain's instrument panel (P1) or the first officer's instrument panel (P3), as applicable (TASK 31-65-11-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR OUTLET VALVE CONTROL CABLES - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) An adjustment of the control cables for the outlet valve for the windshield and foot air.

B. There are two control cables for each outlet valve for the windshield and foot air. The adjustment nuts on the control cables are installed behind the display units on the P1 captain's instrument panel or the P3 first officer's instrument panel.

TASK 21-22-02-820-801

2. Air Outlet Valve Control Cables Adjustment and Test

(Figure 501)

A. References

Reference	Title
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
31-65-11-000-801	MAX Display Unit Removal (P/B 401)
31-65-11-400-801	MAX Display Unit Installation (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Adjustment

SUBTASK 21-22-02-010-004

(1) To get access to the control cables for the captain's foot outlet and windshield outlet, do this step:

(a) Remove the display units on the P1 captain's instrument panel (TASK 31-65-11-000-801).

SUBTASK 21-22-02-010-005

(2) To get access to the control cables for the first officer's foot outlet and windshield outlet, do this step:

(a) Remove the display units on the P3 first officer's instrument panel (TASK 31-65-11-000-801).

D. Air Outlet Valve Control Cables Adjustment

SUBTASK 21-22-02-020-006

(1) Disconnect the cable assembly [1] or cable assembly [2] from the outlet valve [8] as follows:

(a) Remove the cotter pin [6], washer [7], and pin [5].

SUBTASK 21-22-02-860-004

(2) Make sure that the outlet port on the outlet valve [8] that the cable controls is fully closed.

NOTE: The port on the valve is fully closed when the tab on the valve for the control cable is centered on the seam of the valve (View C, Figure 501).

SUBTASK 21-22-02-860-005

(3) Push the knob [10] on the cable assembly [1] or cable assembly [2] fully forward.

NOTE: When the knob is pushed fully forward, the valve is in the fully closed position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-22-02-820-002

(4) Adjust the nut [3] and nut [9] until the hole in the end of the cable assembly [1] or cable assembly [2] aligns with the hole in the tab on the outlet valve [8].

SUBTASK 21-22-02-420-006

(5) Install the pin [5], washer [7], and cotter pin [6].

E. Air Outlet Valve Control Cables Test

SUBTASK 21-22-02-710-001

(1) Do this task: Supply Conditioned Air to the Airplane, TASK 21-00-00-800-801.

SUBTASK 21-22-02-860-006

(2) Pull out the knob [10] for the cable assembly that was adjusted.

SUBTASK 21-22-02-710-002

(3) Make sure that air comes out of the foot air outlet or at the base of the windshield, as applicable.

SUBTASK 21-22-02-710-003

(4) Push the knob [10] fully in for the cable that was adjusted.

SUBTASK 21-22-02-710-004

(5) Make sure that the flow of air stops.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-22-02-860-007

(1) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

SUBTASK 21-22-02-010-006

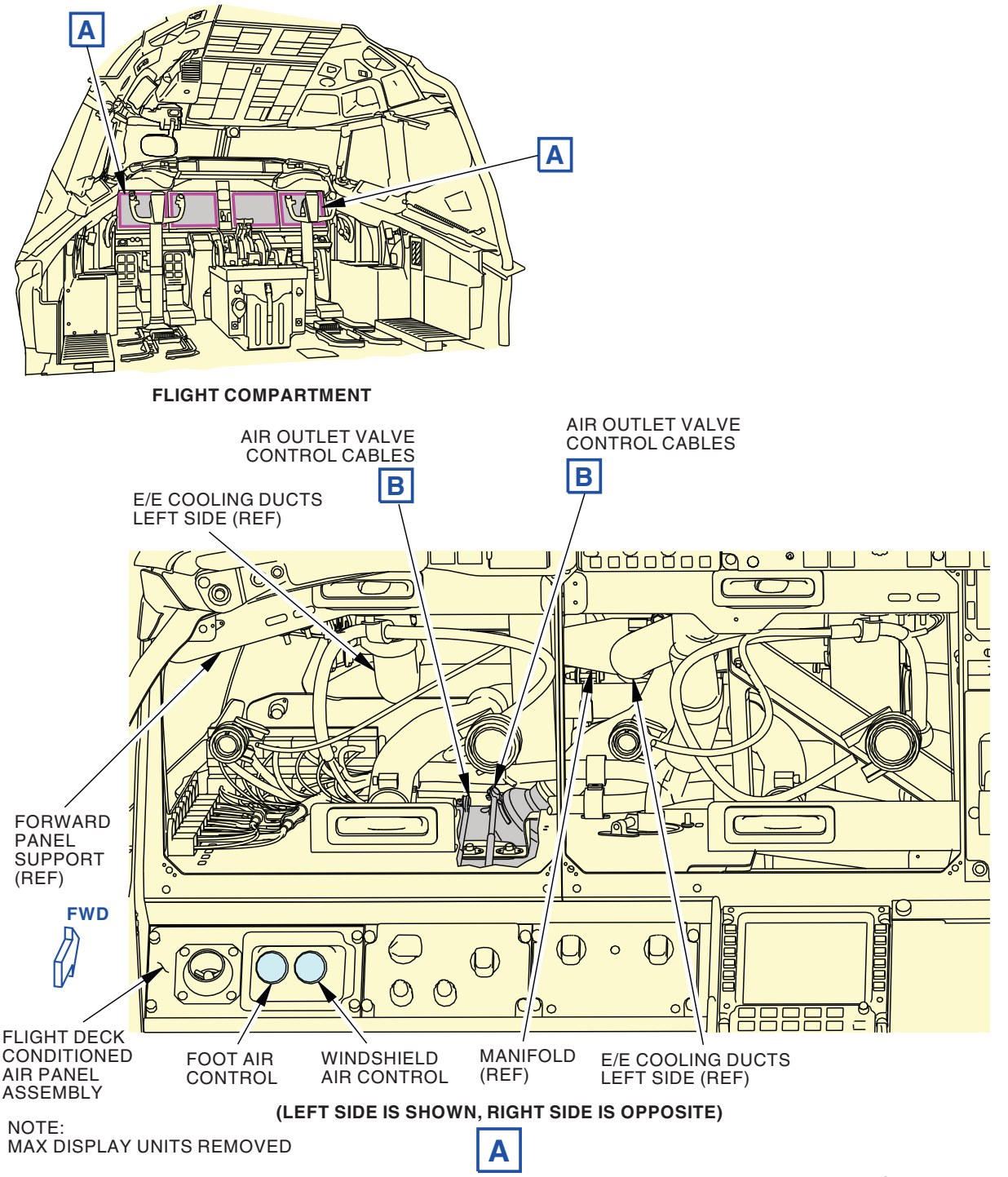
(2) Install the display units on the P1 captains's instrument panel if applicable (TASK 31-65-11-400-801).

SUBTASK 21-22-02-010-007

(3) Install the display units on the P3 first officer's instrument panel if applicable (TASK 31-65-11-400-801).

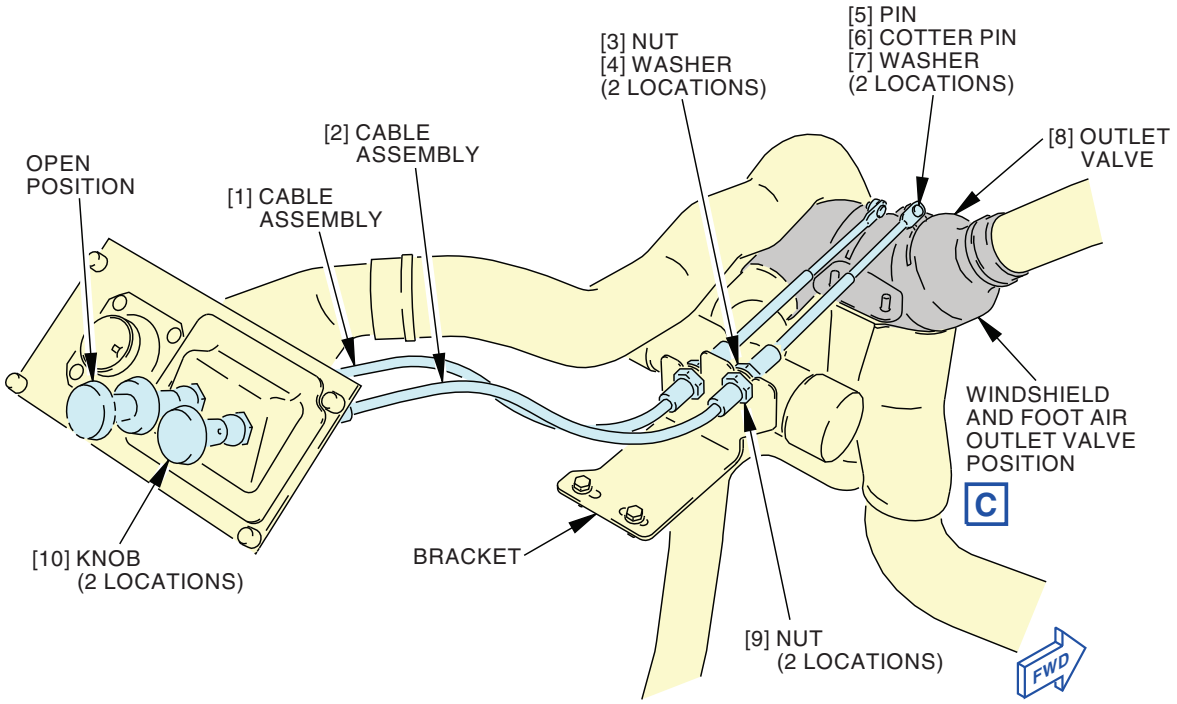
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



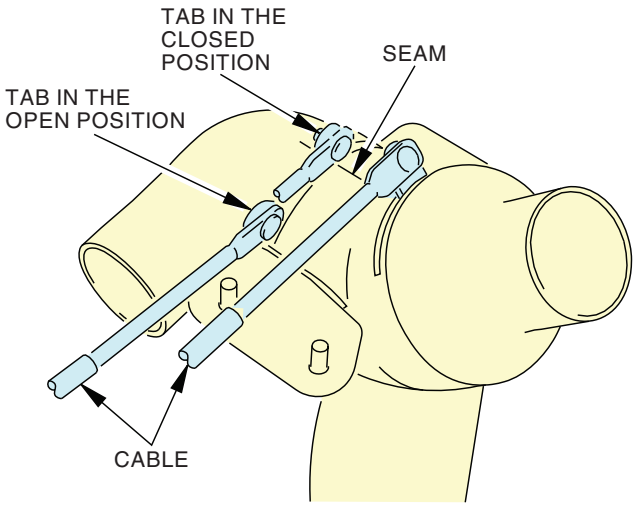
Air Outlet Valve Control Cables Adjustment
Figure 501/21-22-02-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AIR OUTLET VALVE CONTROL CABLES

B



WINDSHIELD AND FOOT AIR OUTLET VALVE POSITION

C

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Air Outlet Valve Control Cables Adjustment
Figure 501/21-22-02-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT DECK AIR DISTRIBUTION SYSTEM MUFFLER/DUCT ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the flight deck air distribution system muffler/duct assembly.

(2) An installation of the flight deck air distribution system muffler/duct assembly.

B. The flight deck air distribution system muffler/duct assembly is located in the left portion of the air distribution mix bay.

C. The muffler is bonded to two elbow ducts to form the muffler/duct assembly.

TASK 21-22-05-000-801

2. Flight Deck Air Distribution System Muffler/Duct Assembly Removal

(Figure 401)

A. References

Reference	Title
21-25-02-000-801	Recirculation Fan Removal (P/B 401)
50-10-03-000-801	Cargo Compartment Ceiling Liner Removal (P/B 401)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented
STD-6439	Coveralls - Hooded

C. Consumable Materials

Reference	Description	Specification
G50140	Gloves - Protective, Latex or Nitrile	

D. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
125	Air Conditioning Distribution Bay - Left
212	Flight Compartment - Right

E. Prepare for the Removal

SUBTASK 21-22-05-860-001

(1) Make sure that the L and R PACK switches, on the P5-10 air conditioning panel, are set to OFF and install the DO NOT OPERATE tag, STD-858.

SUBTASK 21-22-05-860-002

(2) Make sure that the L and R RECIRC FAN switches are set to OFF and install the DO NOT OPERATE tag, STD-858.

SUBTASK 21-22-05-010-001

(3) Open the forward cargo compartment door.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-22-05-010-002

(4) Remove the center and left aft bulkhead liners from the forward cargo compartment (TASK 50-11-02-000-801).

SUBTASK 21-22-05-010-003

(5) Remove the left, aft ceiling liner from the forward cargo compartment (TASK 50-10-03-000-801).

SUBTASK 21-22-05-940-001

WARNING PUT ON THE PERSONAL PROTECTIVE EQUIPMENT BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES AND INJURIES TO PERSONS.

(6) Before you touch the muffler/duct assembly, put on the personal protective equipment listed below:

(a) hooded coveralls, STD-6439

(b) safety goggles, STD-6437

(c) particulate respirator, STD-6435

(d) protective gloves, G50140.

F. Flight Deck Air Distribution System Muffler/Duct Assembly Removal

SUBTASK 21-22-05-020-001

(1) Remove the muffler/duct assembly [9]:

(a) Disconnect the drain tube [14] from the muffler/duct assembly [9].

(b) Remove the left recirculation fan (TASK 21-25-02-000-801).

(c) Remove the fasteners that attach the duct support [7] to the filter housing.

(d) Remove the fasteners that attach the bracket assembly [2] and bracket assembly [8] to the filter housing.

(e) Remove and discard the strap [13] that connects the muffler/duct assembly [9] to the filter housing.

PTF 101 PRE SB 737-21-1262

(f) Remove and discard the straps [11], strap [12], and straps [15] that connect the muffler/duct assembly [9] to the ducts that are forward and aft of the muffler/duct assembly [9].

PTF 102-999; PTF 101 POST SB 737-21-1262

(g) Remove the straps [11], strap [12], clamp [17], and clamp [18] that connect the muffler/duct assembly [9] to the ducts that are forward and aft of the muffler/duct assembly [9].

1) Discard the straps [11] and strap [12].

PTF ALL

(h) Remove the screw [5] and the washer [6].

(i) Remove the screws [3] and the washers [4].

(j) Remove the filter housing from the mix bay.

(k) Move the sleeve [10] and sleeve [16] away from the ends of the elbow ducts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

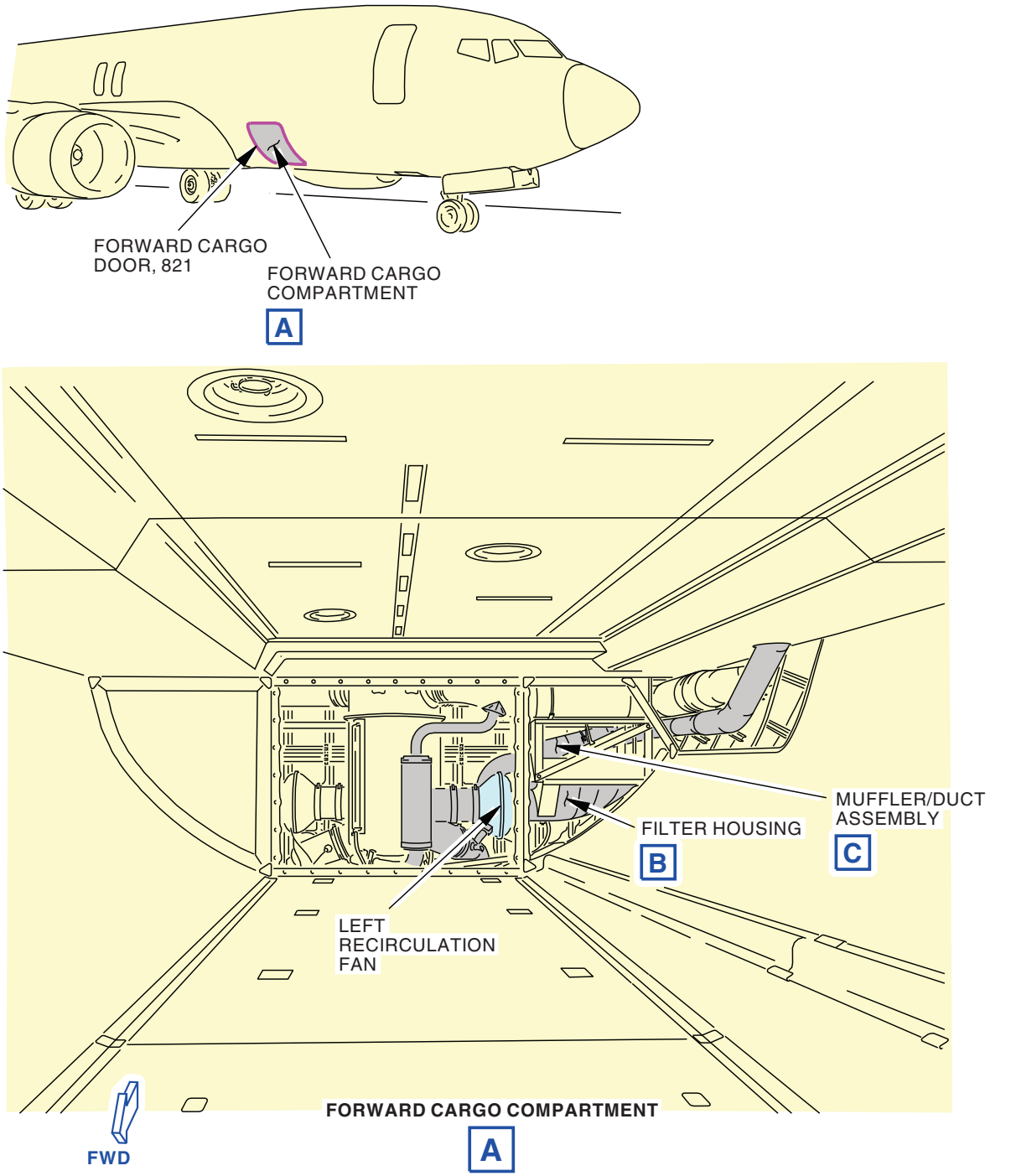
(l) Remove the muffler/duct assembly [9] from the mix bay.

SUBTASK 21-22-05-400-001

(2) Put covers over the open ducts to prevent the entry of unwanted materials.

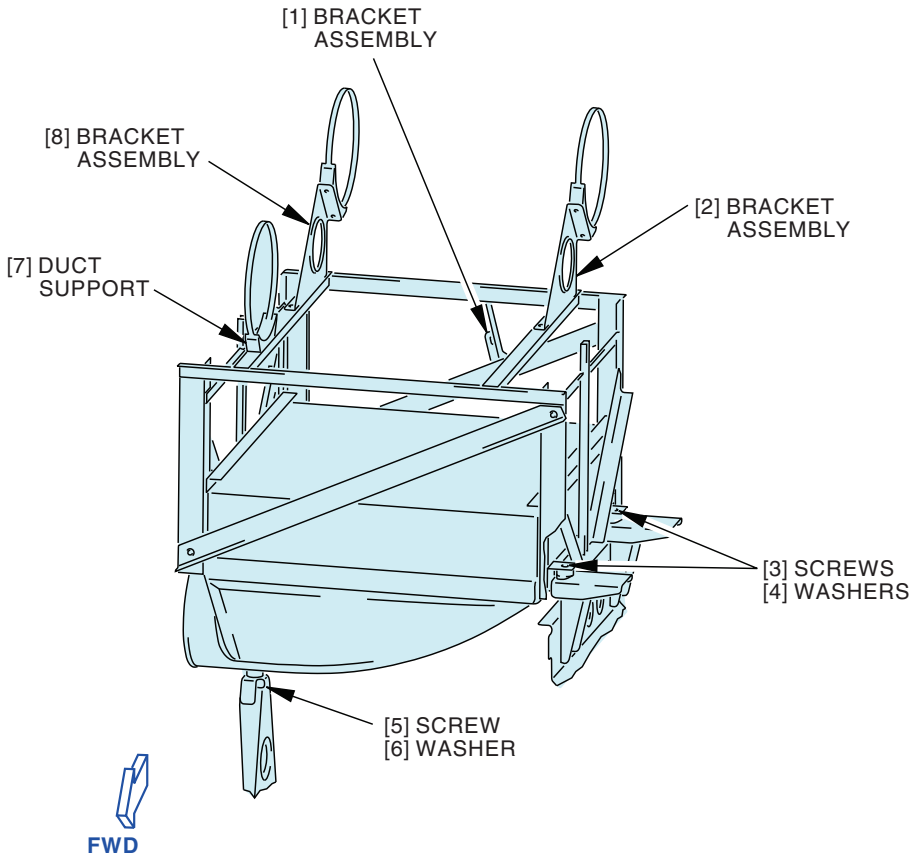
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Deck Air Distribution System Muffler/Duct Assembly Installation
Figure 401/21-22-05-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



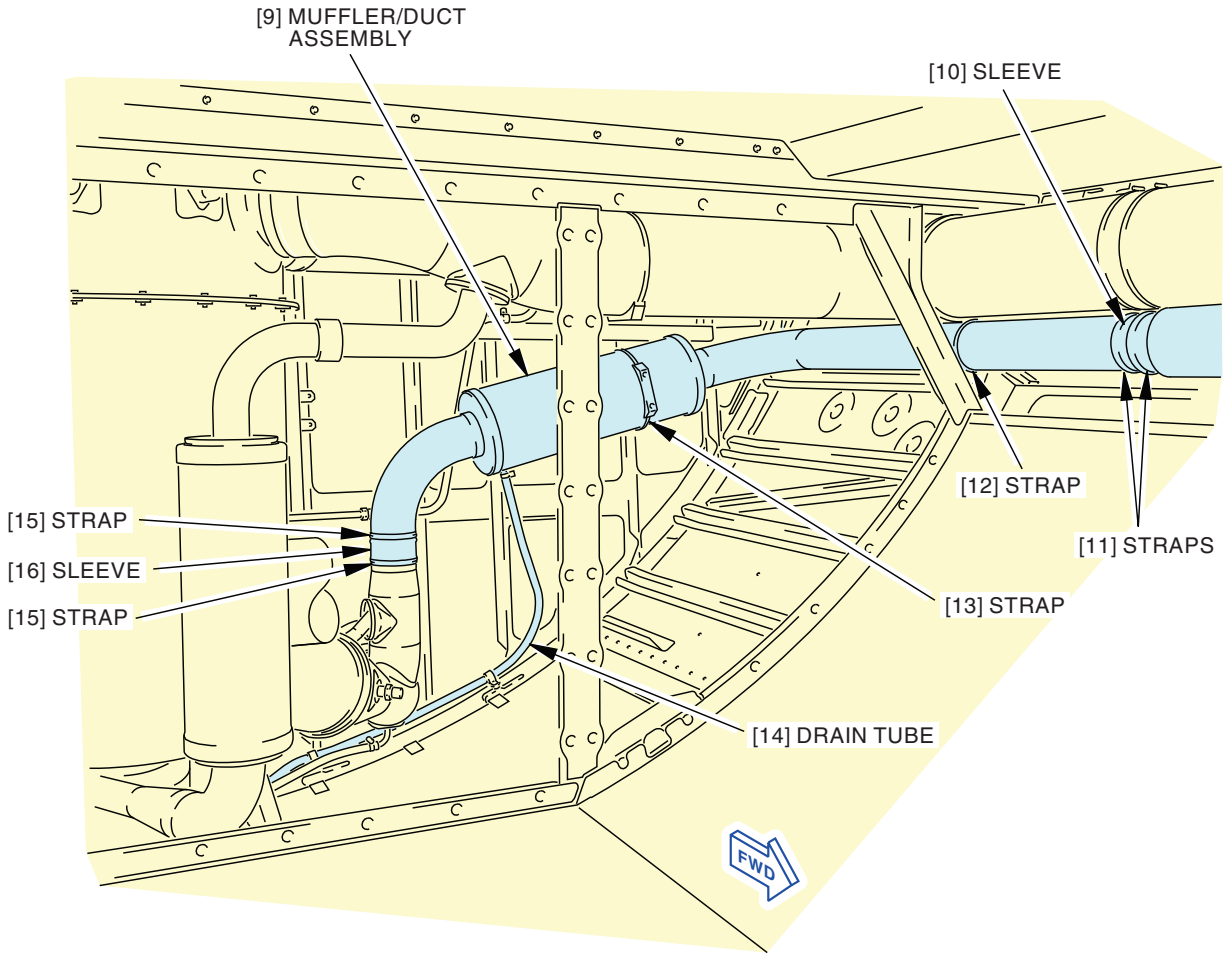
FILTER HOUSING
(DUCTS NOT SHOWN)

B

2411281 S00061527471_V1

Flight Deck Air Distribution System Muffler/Duct Assembly Installation
Figure 401/21-22-05-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



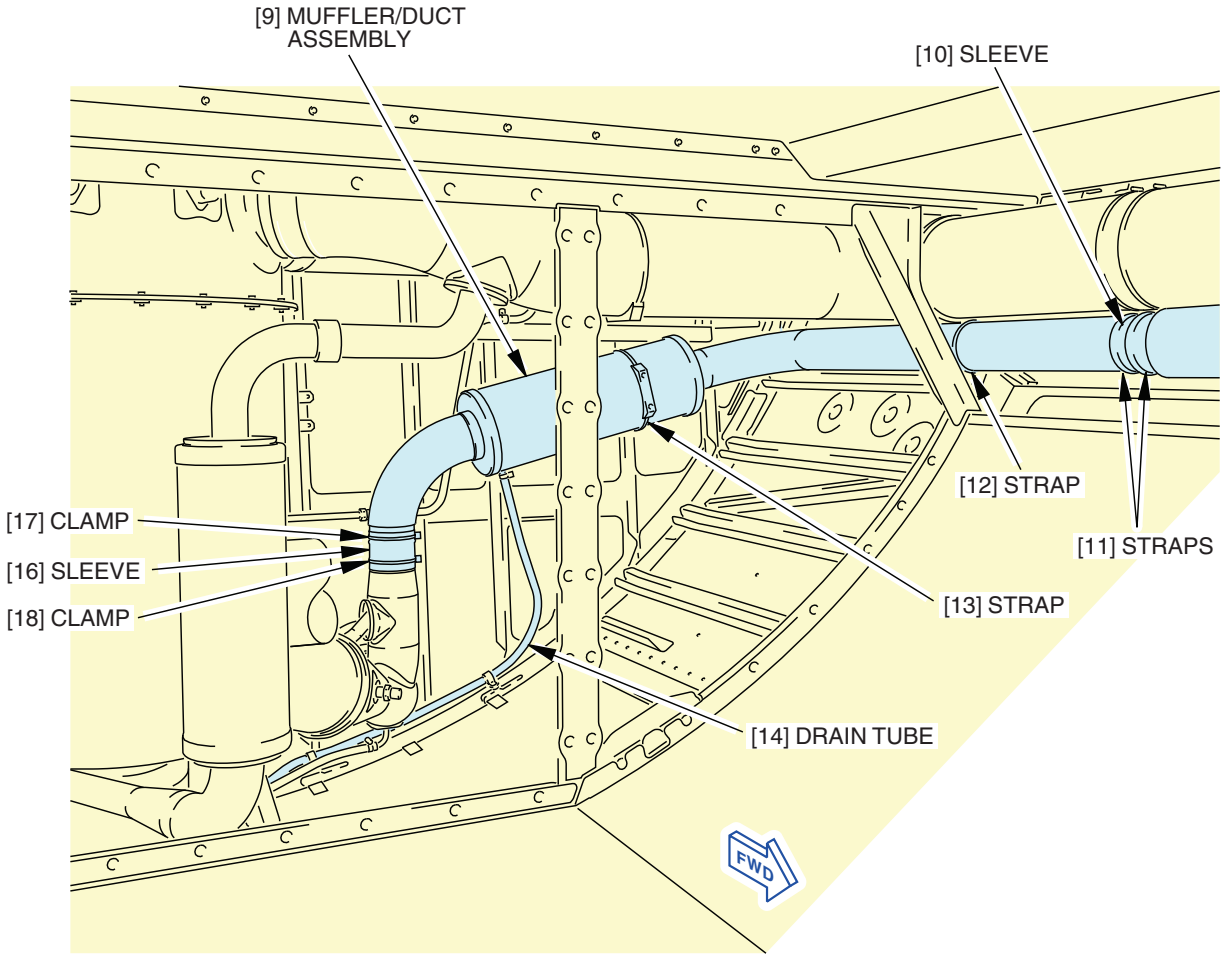
MUFFLER/DUCT ASSEMBLY

C

2411282 S00061527472_V1

Flight Deck Air Distribution System Muffler/Duct Assembly Installation
Figure 401/21-22-05-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MUFFLER/DUCT ASSEMBLY

C

3070446 S0000831166_V1

Flight Deck Air Distribution System Muffler/Duct Assembly Installation
Figure 401/21-22-05-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-22-05-400-801

3. Flight Deck Air Distribution System Muffler/Duct Assembly Installation

(Figure 401)

A. References

Reference	Title
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
21-25-02-400-801	Recirculation Fan Installation (P/B 401)
50-10-03-400-801	Cargo Compartment Ceiling Liner Installation (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-3923	Tool - Installation, Tie Strap Part #: GS2B-E Supplier: 06383 Part #: GS4H-E Supplier: 06383 Part #: GTS-E Supplier: 06383 Part #: PPTS Supplier: 06383 Opt Part #: GS2B Supplier: 06383 Opt Part #: GS2BL Supplier: 06383 Opt Part #: GS4H Supplier: 06383 Opt Part #: GTS Supplier: 06383

C. Consumable Materials

Reference	Description	Specification
G00624	Bag - Plastic, General Purpose	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Muffler/duct assembly	Not Specified	
11	Strap	Not Specified	
12	Strap	Not Specified	

E. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
125	Air Conditioning Distribution Bay - Left
212	Flight Compartment - Right

F. Flight Deck Air Distribution System Muffler/Duct Assembly Installation

SUBTASK 21-22-05-000-001

(1) Remove the duct covers from the connecting ducts.

SUBTASK 21-22-05-080-001

(2) Install the muffler/duct assembly [9]:

NOTE: The muffler/duct assembly comes with two elbow ducts bonded to the muffler.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put the muffler/duct assembly [9] in its position in the left side of the mix bay.

(b) Connect the sleeve [10] and sleeve [16] to the muffler/duct assembly [9].

PTF 101 PRE SB 737-21-1262

(c) Install the new straps [11] and new straps [15] that attach to the sleeve [10] and sleeve [16] with a tie strap installation tool, COM-3923.

PTF 102-999; PTF 101 POST SB 737-21-1262

(d) Install the new straps [11] that attach to the sleeve [10] with a tie strap installation tool, COM-3923.

(e) Install the clamp [17] that attaches to the sleeve [16].

1) Tighten the clamp to 5 in-lb (0.56 N·m) - 9 in-lb (1.02 N·m).

(f) Install the clamp [18] that attaches to the sleeve [16].

1) Tighten the clamp to 13 in-lb (1.47 N·m) - 17 in-lb (1.92 N·m).

PTF ALL

(g) Install the new strap [12] and attach it with the tie strap installation tool, COM-3923.

(h) Install the filter housing:

1) Put the filter housing into the left side of the mix bay.
2) Install the washer [6] and screw [5].
3) Install the washers [4] and screws [3].
4) Attach the duct support [7] to the filter housing.
5) Attach the bracket assembly [8] and bracket assembly [2] to the filter housing.
6) Install the strap [13] and secure it with the tie strap installation tool, COM-3923, to the filter housing.
7) Install the left recirculation fan (TASK 21-25-02-400-801).

(i) Connect the drain tube [14] to the muffler/duct assembly [9].

SUBTASK 21-22-05-420-001

(3) Supply conditioned air to the flight deck (TASK 21-00-00-800-803).

SUBTASK 21-22-05-790-001

(4) Make sure that there are no leaks at the interfaces at the ends of the muffler/duct assembly.

(a) Repair all leaks that you find.

SUBTASK 21-22-05-420-002

(5) Operate the flight deck air conditioning outlets to make sure that an adequate supply of conditioned air is available to the flight deck.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-22-05-020-002

WARNING DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(1) Remove all personal protective equipment and put them in a plastic bag, G00624. Seal the bag closed with tape.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(a) Discard the bag with the personal protective equipment. Refer to the airline, local health, safety and regulatory procedures for disposal of material.

SUBTASK 21-22-05-110-001

WARNING CLEAN YOUR HANDS WITH SOAP AND FLUSH WITH HOT WATER. DIRTY HANDS WITH CONTAMINATION CAN CAUSE DISEASE AND ILLNESSES TO PERSONS.

(2) Clean your hands with soap and running hot water.

SUBTASK 21-22-05-860-003

(3) Remove the conditioned air to the flight deck (TASK 21-00-00-800-804).

SUBTASK 21-22-05-860-004

(4) Install the center and left aft bulkhead panels in the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 21-22-05-410-001

(5) Install the forward cargo compartment ceiling liner that was removed (TASK 50-10-03-400-801).

SUBTASK 21-22-05-860-005

(6) Close the forward cargo compartment door.

SUBTASK 21-22-05-860-006

(7) Put the L and R PACK switches, on the P5-10 air conditioning panel, to the AUTO position and remove the DO-NOT-OPERATE tag.

SUBTASK 21-22-05-860-007

I (8) Put the L and R RECIRC FAN switches, on the P5-10 panel, to the AUTO position and remove the DO-NOT-OPERATE tag.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT DECK DUCT ASSEMBLY SCREEN - CLEANING/PAINTING

1. General

A. This procedure has these tasks:

(1) Flight Deck Duct Assembly Removal

(2) Flight Deck Duct Assembly Screen Cleaning

(3) Flight Deck Duct Assembly Installation.

B. The duct assembly is located along the lower left side of the flight deck compartment to the left of the nose landing gear wheel well.

C. The duct assembly with the wire mesh screen must be removed from the ducts that are connected to it and then the screen can be cleaned.

TASK 21-22-07-020-801

2. Flight Deck Duct Assembly Removal

(Figure 701)

A. Tools/Equipment

Reference	Description
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented
STD-6439	Coveralls - Hooded

B. Consumable Materials

Reference	Description	Specification
G50140	Gloves - Protective, Latex or Nitrile	

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 21-22-07-864-001

(1) Make sure that the L and R PACK switches on the P5-10 air conditioning panel are set to OFF.

SUBTASK 21-22-07-864-002

(2) Make sure the L and R RECIRC FAN switches on the P5-10 air conditioning panel are set to OFF.

SUBTASK 21-22-07-010-001

(3) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-22-07-940-001

WARNING WEAR APPLICABLE PERSONAL PROTECTIVE EQUIPMENT (PPE) WHEN YOU DO THIS TASK. BEFORE YOU USE PPE, READ, UNDERSTAND AND OBEY ALL INSTRUCTIONS FOR ITS USE. YOU CAN BE INJURED IF YOU DO NOT USE PPE AND USE IT CORRECTLY.

(4) Before you touch the duct assembly, put on the personal protective equipment listed below:

(a) hooded coveralls, STD-6439

(b) safety goggles, STD-6437

(c) particulate respirator, STD-6435

(d) protective gloves, G50140

F. Flight Deck Duct Assembly Removal

SUBTASK 21-22-07-010-002

(1) Get access to the duct assembly [1] with the wire mesh cone assembly in the compartment to the left of the nose landing gear wheel well.

SUBTASK 21-22-07-020-001

(2) Remove the duct assembly [1] as follows:

(a) Remove the plastic tie straps [15] at the location where the duct assembly [1] connects to the duct assembly [3].

(b) Remove the plastic tie straps [15] at two locations where the duct assembly [1] connects to the duct assembly [5] and duct assembly [6].

(c) Do these steps to remove the clamp [4]:

1) Remove the nut [12], washer [13] and screw [14] that attach the clamp [4] to the structure.

2) Remove the clamp [4].

(d) Try to remove duct assembly [1].

(e) If you find that there is not enough clearance in the installation of the duct assemblies to permit removal of the duct assembly [1], then continue with the steps that follow:

NOTE: The ends of the duct assemblies overlap and clearance must be obtained so that duct assembly [1] can be removed.

1) Remove the plastic tie straps [15] that keep the flex sleeve hose [10] between the duct assembly [5] and duct assembly [8].

2) If it is necessary, remove the plastic tie strap [15] that attaches the duct assembly [8] and duct support [7].

3) Remove the plastic tie straps [15] that keep the flex sleeve hose [10] between the duct assembly [6] and duct assembly [11].

4) Loosen the screws [16] on the clamp [9] that keeps the duct assembly [8].

5) Slide the duct assembly [5] and duct assembly [6] as far aft as possible so that the duct assembly [1] can be removed.

(f) Remove duct assembly [1].

(g) Examine the duct assembly [1].

1) If the duct assembly [1] or the wire mesh screen is damaged, get a serviceable duct assembly [1].

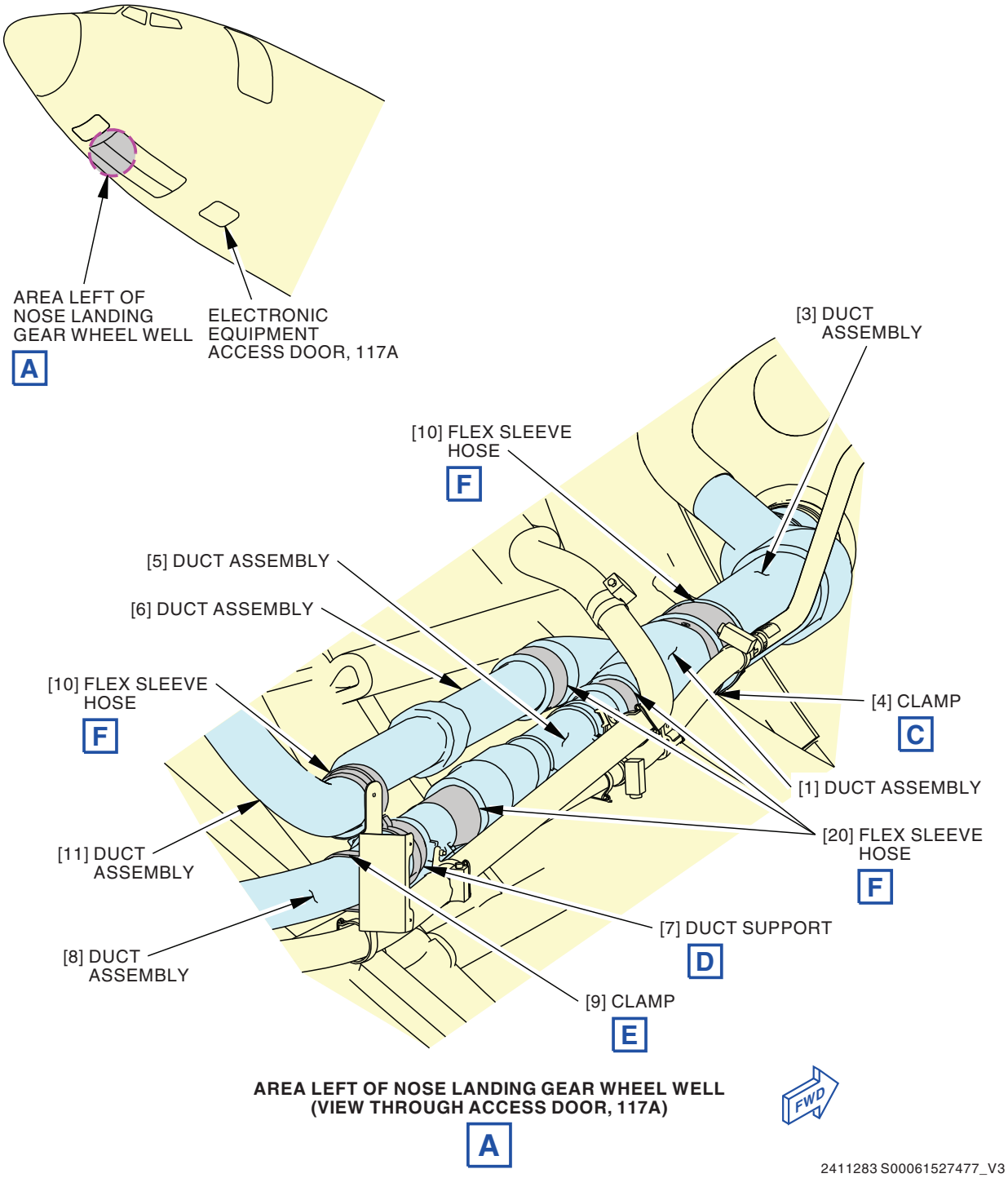
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-22-07-913-001

(3) Put protective covers on the open ends of the duct assembly [5], duct assembly [6] and duct assembly [3].

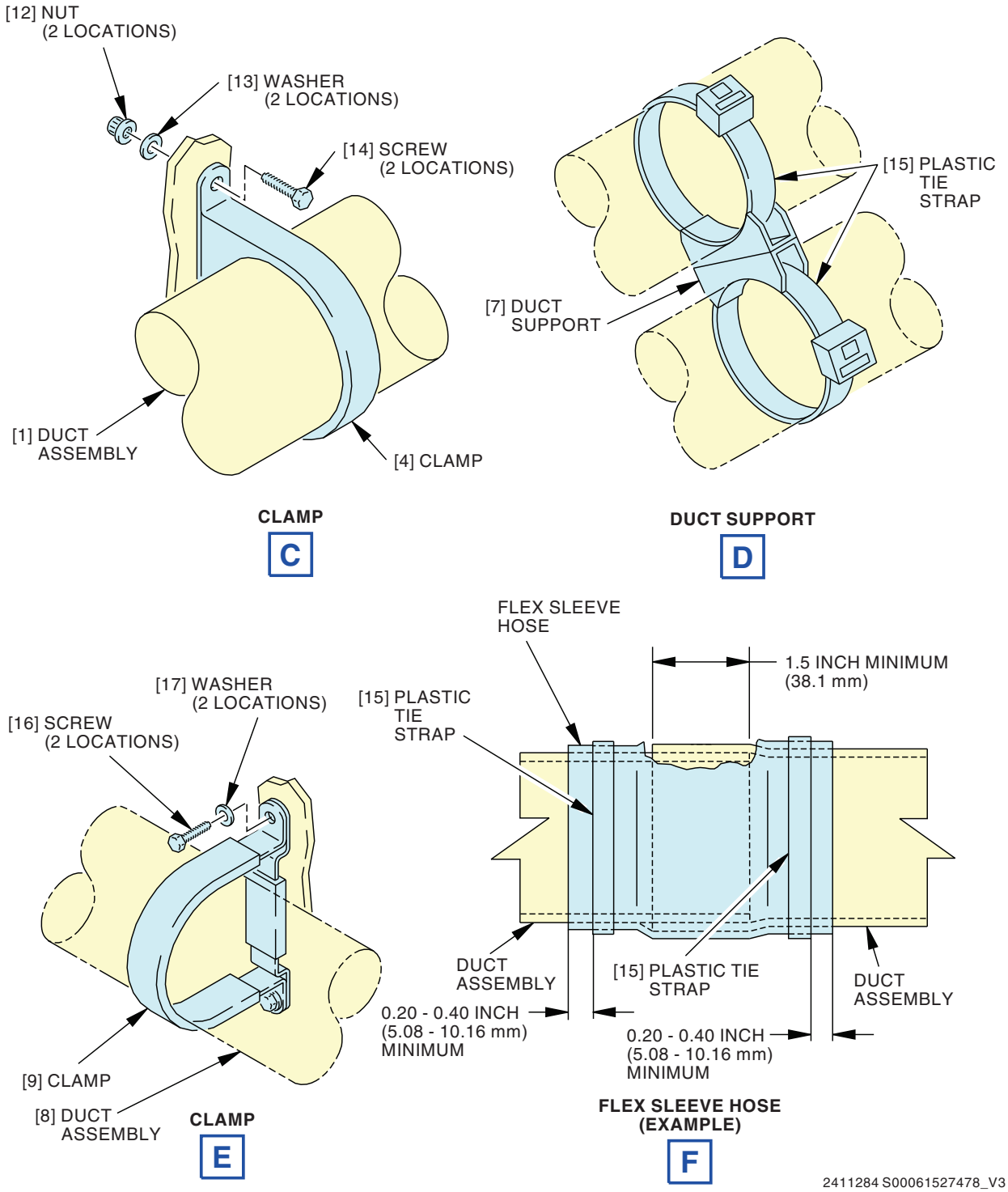
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Deck Duct Assembly Screen Cleaning
Figure 701/21-22-07-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Deck Duct Assembly Screen Cleaning
Figure 701/21-22-07-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-22-07-100-801

3. Flight Deck Duct Assembly Screen Cleaning

(Figure 701)

A. Tools/Equipment

Reference	Description
STD-1068	Vacuum - High Capacity

B. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Flight Deck Duct Assembly Screen Cleaning

SUBTASK 21-22-07-020-002

(1) If it is not already done, remove the duct assembly [1] with the wire mesh cone assembly (TASK 21-22-07-020-801).

SUBTASK 21-22-07-560-001

(2) Move the duct assembly [1] to a location where there is a source of high capacity vacuum, STD-1068.

SUBTASK 21-22-07-160-001

(3) Remove all debris that is accumulated on the wire mesh screen inside the duct assembly [1] with the high capacity vacuum, STD-1068.

SUBTASK 21-22-07-210-001

(4) Visually inspect and make sure that all debris has been removed from the wire mesh cone assembly.

SUBTASK 21-22-07-670-001

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(5) Disinfect the wire mesh screen with a solution of 70% alcohol, B00130.

(a) Use pre-moistened cotton wiper, G00034.

NOTE: Follow the manufacture's instruction and refer to the local health authority guidance for required contact/dwell time.

SUBTASK 21-22-07-420-001

(6) Install the duct assembly [1] with the screen (TASK 21-22-07-420-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-22-07-420-801

4. Flight Deck Duct Assembly Installation

(Figure 701)

A. References

Reference	Title
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-3923	Tool - Installation, Tie Strap Part #: GS2B-E Supplier: 06383 Part #: GS4H-E Supplier: 06383 Part #: GTS-E Supplier: 06383 Part #: PPTS Supplier: 06383 Opt Part #: GS2B Supplier: 06383 Opt Part #: GS2BL Supplier: 06383 Opt Part #: GS4H Supplier: 06383 Opt Part #: GTS Supplier: 06383

C. Consumable Materials

Reference	Description	Specification
G00624	Bag - Plastic, General Purpose	
G50218	Strap - Plastic, Adjustable, Self-locking, 27.50 Inches (698.50MM) Long	BACS38K6
G51390	Strap - Plastic, Adjustable, Self-locking, 11.40 Inches (289.56MM) Long	BACS38K7

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Flight Deck Duct Assembly Installation

SUBTASK 21-22-07-420-002

(1) Install the duct assembly [1] as follows:

(a) Install the flex sleeve hose [10] and make sure to position a minimum of 1.5 in. (38.1 mm) of overlap between duct walls of duct assembly [1] and duct assembly [3] (Figure 701, View F).

(b) Install the flex sleeve hose [20] and make sure to position a minimum of 1.5 in. (38.1 mm) of overlap between duct walls of (Figure 701, View F):

1) duct assembly [1] and duct assembly [5]

2) duct assembly [1] and duct assembly [6].

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(c) If it is necessary, install the flex sleeve hose [20] and make sure to position a minimum of 1.5 in. (38.1 mm) of overlap between duct walls of duct assembly [5] and duct assembly [8] (Figure 701, View F):

(d) If it is necessary, install the flex sleeve hose [10] and make sure to position a minimum of 1.5 in. (38.1 mm) of overlap between duct walls of duct assembly [6] and duct assembly [11] (Figure 701, View F):

(e) Install the plastic tie straps [15] to the flex sleeve hose with the tie strap installation tool, COM-3923:

1) Use the adjustable plastic straps, G50218, between the duct assembly [1] and duct assembly [3].

2) Use the adjustable plastic straps, G51390, at these locations:

a) duct assembly [1] connects to duct assembly [5] and duct assembly [6]

b) between duct assembly [5] and duct assembly [8]

c) between duct assembly [6] and duct assembly [11].

3) Put the plastic tie straps [15] in position 0.20 in. (5.08 mm) - 0.40 in. (10.16 mm) from the ends of the flex sleeve hose.

(f) Install the clamp [4] with the screws [14], washers [13] and nuts [12].

(g) If it is necessary, install the plastic tie straps [15] (adjustable plastic strap, G51390) that attaches the duct assembly [8] and duct support [7] with the tie strap installation tool, COM-3923.

(h) Tighten the screws [16] for the clamp [9] that holds the duct assembly [8].

SUBTASK 21-22-07-863-001

(2) Supply conditioned air to the flight compartment (TASK 21-00-00-800-801).

SUBTASK 21-22-07-710-001

(3) Make sure that the conditioned air outflow from the flight compartment outlets is satisfactory.

SUBTASK 21-22-07-790-001

(4) Make sure that there are no leaks at the joints of the duct assemblies that were disturbed.

(a) If there are leaks, repair the duct assembly joint to stop the leak.

SUBTASK 21-22-07-020-003

WARNING DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(5) Remove all personal protective equipment and put them in a plastic bag, G00624.

(6) Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(7) Discard the bag with the personal protective equipment. Refer to the airline, local health, safety and regulatory procedures for disposal of material.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-22-07-110-001

WARNING CLEAN YOUR HANDS WITH SOAP AND FLUSH WITH HOT WATER. DIRTY HANDS WITH CONTAMINATION CAN CAUSE DISEASE AND ILLNESSES TO PERSONS.

(8) Clean your hands with soap and running hot water.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-22-07-410-001

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-22-07-864-003

(2) Remove conditioned air from the airplane (TASK 21-00-00-800-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT COMPARTMENT GASPER AIR OUTLETS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the gasper air outlets installed in the control cabin.

(2) An installation of the gasper air outlets installed in the control cabin.

B. The gasper air outlets are referred to as captain/first officer's gasper air outlet and overhead gasper air outlet in this procedure.

TASK 21-22-09-000-801

2. Gasper Air Outlet Removal

(Figure 401)

A. References

Reference	Title
25-11-21 P/B 201	FLIGHT COMPARTMENT PANELS - MAINTENANCE PRACTICES

B. Consumable Materials

Reference	Description	Specification
G51390	Strap - Plastic, Adjustable, Self-locking, 11.40 Inches (289.56MM) Long	BACS38K7
G51414	Strap - Plastic, Adjustable, Self-locking, 7.30 Inches (185.42 mm) Long	BACS38K1

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 21-22-09-010-001

(1) Remove the flight compartment panel for the applicable gasper air oulet [1] (PAGEBLOCK 25-11-21/201).

E. Gasper Air Outlet Removal

SUBTASK 21-22-09-030-001

(1) For the captain/first officers gasper air oulet [1] removal, do these steps:

(a) Remove the strap [2] (adjustable plastic strap, G51414) that holds the hose [3] to the gasper air oulet [1].

(b) Remove the hose [3] from the gasper air oulet [1].

(c) Discard the strap [2] (adjustable plastic strap, G51414).

(d) Remove the screws [4] that attach the gasper air oulet [1] to the panel.

(e) Remove the gasper air oulet [1].

SUBTASK 21-22-09-020-002

(2) For the overhead gasper air oulet [1] removal, do these steps:

(a) Remove the strap [6] (adjustable plastic strap, G51390) that holds the hose [7] to the gasper air oulet [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the hose [7] from the gasper air oulet [5].

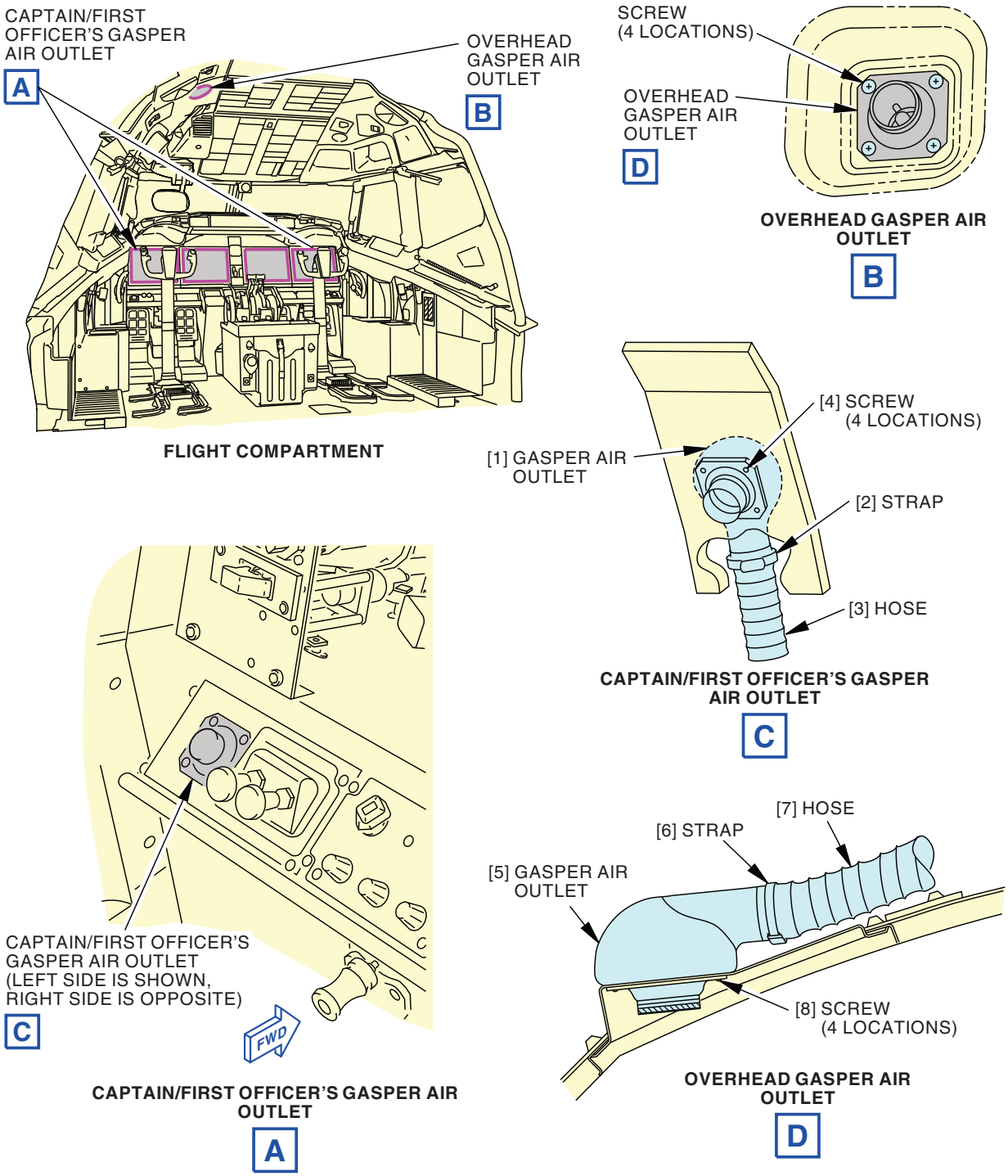
(c) Discard the strap [6] (adjustable plastic strap, G51390).

(d) Remove the screws [8] that attach the gasper air oulet [5] to the panel.

(e) Remove the gasper air oulet [5].

——— END OF TASK ———

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Gasper Air Outlet Installation
Figure 401/21-22-09-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-22-09-400-801

3. Gasper Air Outlet Installation

(Figure 401)

A. References

Reference	Title
25-11-21 P/B 201	FLIGHT COMPARTMENT PANELS - MAINTENANCE PRACTICES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-3923	Tool - Installation, Tie Strap Part #: GS2B-E Supplier: 06383 Part #: GS4H-E Supplier: 06383 Part #: GTS-E Supplier: 06383 Part #: PPTS Supplier: 06383 Opt Part #: GS2B Supplier: 06383 Opt Part #: GS2BL Supplier: 06383 Opt Part #: GS4H Supplier: 06383 Opt Part #: GTS Supplier: 06383

C. Consumable Materials

Reference	Description	Specification
G51390	Strap - Plastic, Adjustable, Self-locking, 11.40 Inches (289.56MM) Long	BACS38K7
G51414	Strap - Plastic, Adjustable, Self-locking, 7.30 Inches (185.42 mm) Long	BACS38K1

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Gasper Air Outlet Installation

SUBTASK 21-22-09-400-001

(1) For the captain/first officers gasper air oulet [1]gasper air outlet installation, do these steps:

(a) Put the gasper air oulet [1] in the panel.

(b) Install the screws [4] that attach the gasper air oulet [1] to the panel.

(c) Install the hose [3] on the gasper air oulet [1]:

1) Loosely put a strap [2] (adjustable plastic strap, G51414) around the hose [3].

2) Install the hose [3] onto the gasper air oulet [1].

3) Put the head of the strap [2] (adjustable plastic strap, G51414) towards the panel assembly to prevent head contact with wiring or wire bundles.

4) Tighten the strap [2] (adjustable plastic strap, G51414) with a tie strap installation tool, COM-3923.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-22-09-420-003

(2) For the overhead gasper air outlet installation, do these steps:

(a) Put the gasper air oulet [5] in the panel.

(b) Install the screws [8] that attach the gasper air oulet [5] to the panel.

(c) Install the hose [7] on the gasper air oulet [5]:

1) Loosely put a strap [6] (adjustable plastic strap, G51390) around the hose [7].

2) Install the hose [7] onto the gasper air oulet [5].

3) Put the head of the strap [6] (adjustable plastic strap, G51390) towards the panel assembly to prevent head contact with wiring or wire bundles.

4) Tighten the strap [6] (adjustable plastic strap, G51390) with a tie strap installation tool, COM-3923.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-22-09-410-001

(1) Install the flight compartment panel for the applicable gasper (PAGEBLOCK 25-11-21/201).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OVERHEAD DISTRIBUTION DUCT - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of a section of the overhead distribution duct

(2) An installation of a section of the overhead distribution duct

(3) A repair of the insulation for the overhead distribution duct

(4) A cleaning of the passenger overhead distribution duct.

B. Some of the ducts in the air distribution system are wrapped with insulation material. CFR 14 Part 121.312 mandates that insulation material installed in the fuselage as a replacement after September 2, 2005 must meet the flame propagation requirements of CFR Part 25.856, effective September 2, 2003.

C. Existing insulation material can be removed and re-installed if it is not damaged. Replacement insulation material must meet the flame propagation requirement.

D. BMS8-385 Type 4 is the preferred foam insulation material that meets the flame propagation requirement. BMS 8-385 Type 6 and BMS 8-300 Type VIII insulation are new, qualified materials optional to BMS 8-385 Type 4.

E. If BMS8-300, Type 1 foam insulation material is used, it must be completely covered with tape that meets the flame propagation requirement.

F. Do not use BMS8-39 polyurethane foam insulation for repairs due to the degradation in flammability properties over time.

G. When insulation material is removed and re-installed, any existing tape that does not meet the flame propagation requirement must be removed or completely covered with tape that does meet the requirement.

(1) Tape specifications BMS 5-157 MPVF and BMS 5-157 MPEEK are qualified tapes used to secure insulation to ducts.

TASK 21-23-01-000-801

2. Overhead Distribution Duct Removal

(Figure 201)

A. Consumable Materials

Reference	Description	Specification
G50344	Tape - Nitto P-212HD WHT (Formerly Permacel) Glass Cloth	
G50625	Strap - Plastic, Adjustable, Self-locking, 14.40 Inches (365.76 mm) Long	BACS38K3

B. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Prepare for the Removal

SUBTASK 21-23-01-860-001

(1) Make sure that these switches, on the P5-10 air conditioning panel, are set to the OFF position:

(a) L PACK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) R PACK

(c) L RECIRC FAN

(d) R RECIRC FAN.

SUBTASK 21-23-01-010-001

(2) Remove the ceiling panels as required to get access to the duct.

SUBTASK 21-23-01-010-002

(3) If it is necessary, remove the interior furnishings.

D. Overhead Distribution Duct Removal

SUBTASK 21-23-01-030-001

(1) Remove the self locking strap [14] (adjustable plastic straps, G50625) that are on the tape [3] at the joints between the diffuser/hose assembly [2] and the spuds on the distribution duct [1].

SUBTASK 21-23-01-020-002

(2) Remove the tape [3] (Nitto P-212HD WHT tape, G50344) that connects the diffuser/hose assembly [2] to the distribution duct [1].

SUBTASK 21-23-01-020-003

(3) Remove the strap [4] that hold the distribution duct [1] to the bracket assemblies.

SUBTASK 21-23-01-020-004

(4) Loosen the strap [5] at each end of the distribution duct [1] that secure the flex sleeves [6] to the duct.

SUBTASK 21-23-01-020-006

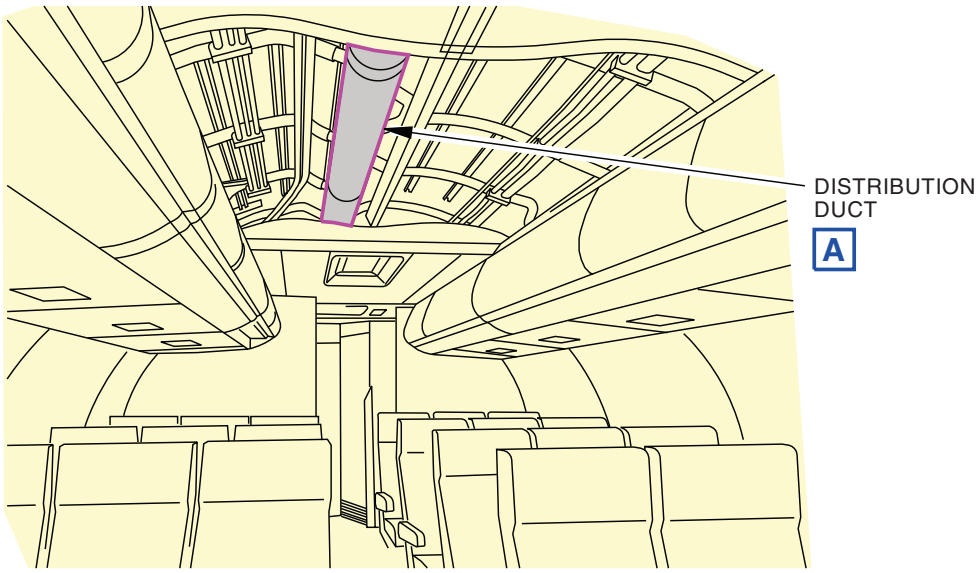
(5) Support the distribution duct [1] and slide the flex sleeves [6] away from the ends of the distribution duct [1].

SUBTASK 21-23-01-020-007

(6) Remove the distribution duct [1].

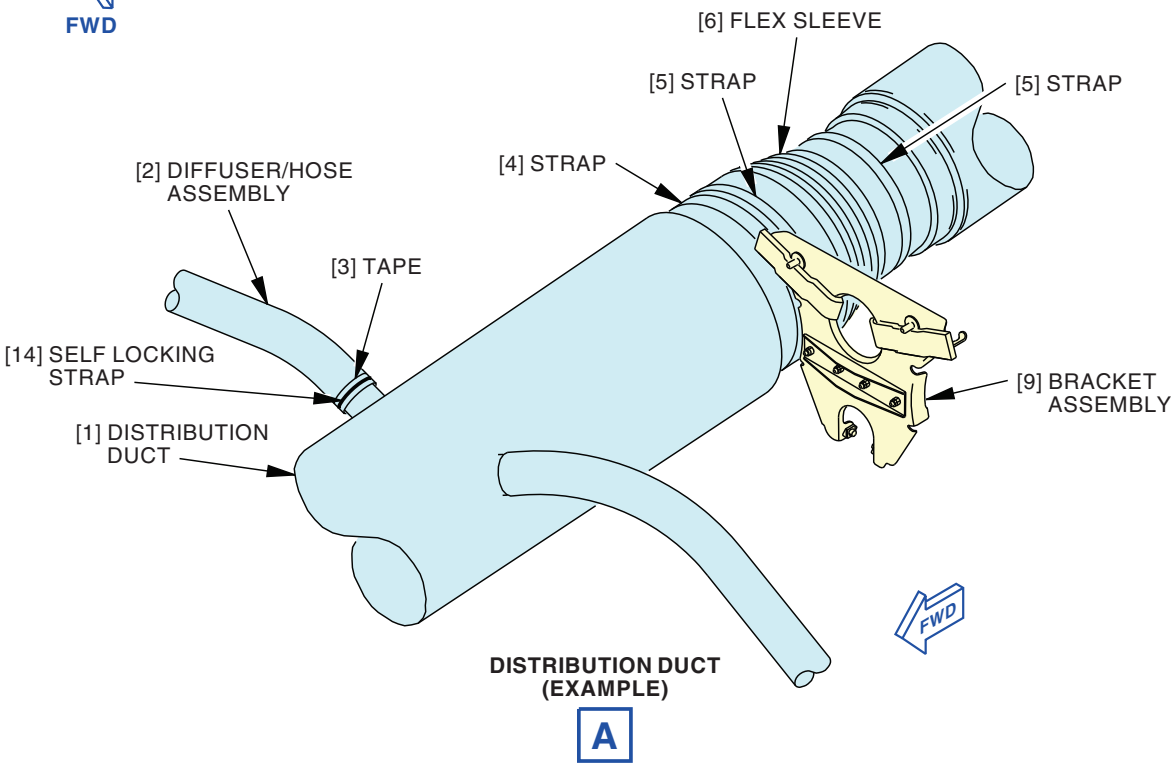
——— END OF TASK ———

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PASSENGER COMPARTMENT
(EXAMPLE)

FWD



DISTRIBUTION DUCT
(EXAMPLE)

A

2411286 S00061527490_V3

Passenger Cabin Overhead Air Distribution Duct Installation
Figure 201/21-23-01-990-801

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TASK 21-23-01-400-801

3. Overhead Distribution Duct Installation

(Figure 201)

A. References

Reference	Title
21-00-00-800-805	Supply Conditioned Air with a Ground Air Source (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)

B. Consumable Materials

Reference	Description	Specification
B00068	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002, MIL-E-51454 Type II
G50344	Tape - Nitto P-212HD WHT (Formerly Permacel) Glass Cloth	
G50625	Strap - Plastic, Adjustable, Self-locking, 14.40 Inches (365.76 mm) Long	BACS38K3

C. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

D. Overhead Distribution Duct Installation

SUBTASK 21-23-01-980-001

(1) Put the distribution duct [1] in its position above the ceiling panels.

SUBTASK 21-23-01-420-001

(2) Slide the flex sleeves [6] onto the ends of the distribution duct [1] as follows:

(a) Make sure that each flex sleeve [6] overlaps the duct end by 1.63 ±0.13 in. (41.40 ±3.30 mm).

(b) Loosely position the duct strap [5] from the ends of the flex sleeves [6].

SUBTASK 21-23-01-020-009

(3) Connect the diffuser/hose assembly [2] to the distribution duct [1] as follows:

(a) Install the diffuser/hose assembly [2] over the outlets on the distribution duct.

1) Make sure that there is at least 1/2-inch of overlap between each diffuser/hose assembly and the outlet on the distribution duct.

2) Use alcohol, B00068, to clean the area on the outlet on the distribution duct and the area on the diffuser/hose assembly where you will apply the tape [3].

(b) Install tape [3] (two to four laps of 2-inch wide Nitto P-212HD WHT tape, G50344) to the center of the joint over the ends of the diffuser/hose assembly [2] and the outlets on the distribution duct as follows:

1) Keep the tape taut and smooth as you apply the two to four laps.

(c) Install adjustable plastic strap, G50625, around the center portion of the tape to hold the diffuser/hose assembly tightly to the outlet on the distribution duct.

SUBTASK 21-23-01-420-003

(4) Install the strap [4] to secure the distribution duct to the bracket assemblies [9].

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SUBTASK 21-23-01-420-004

(5) Tighten the straps [5] that hold the flex sleeves [6] to the ends of the distribution duct.

E. Overhead Distribution Duct Test

SUBTASK 21-23-01-790-001

(1) Do this check of the overhead distribution duct:

(a) If the packs are operational, do the steps that follow for the test of the overhead distribution test:

1) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

2) Put the L and R PACK switches, on the P5-10 air conditioning panel, to the AUTO position

3) Make sure that there is not excessive leakage of air at the connections on the distribution duct that was replaced.

4) Repair any connection in the overhead distribution ducts where airflow can be felt on a hand a minimum of 2 inches away from the surface of a joint.

(b) If the packs are not operational and a test of the overhead distribution ducts is desired, do these steps:

1) Supply ground conditioned air to the distribution system, do this task: Supply Conditioned Air with a Ground Air Source, TASK 21-00-00-800-805.

2) Make sure that there is not excessive leakage of air at the connections on the distribution duct that was replaced.

a) Excessive leakage is airflow that can be felt on a hand a minimum of 2 inches away from the surface of a joint.

3) Repair any connection in the overhead distribution ducts where airflow can be felt on a hand a minimum of 2 inches away from the surface of a joint.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-23-01-410-001

(1) Install any interior furnishings that were removed for access.

SUBTASK 21-23-01-410-002

(2) Install all ceiling panels that were removed for access.

——— END OF TASK ———

TASK 21-23-01-300-801

4. Overhead Distribution Duct Insulation Repair

(Figure 201)

A. General

(1) The repair of the duct insulation consists of replacement of the tape that holds the insulation to the duct.

(2) CFR 14 Part 121.312 mandates that insulation material installed in the fuselage as a replacement after September 2, 2005 must meet the flame propagation requirements of CFR Part 25.856, effective September 2, 2003.

(3) Existing insulation material can be removed and re-installed if it is not damaged. Replacement insulation material must meet the flame propagation requirement.

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(4) Melamine foam insulation, G50449 is the preferred foam insulation material that meets the flame propagation requirement. Melamine foam insulation, G51169, BMS 8-385, Type 6 and foam insulation, G51170, BMS 8-300, Type VIII insulation is optional to BMS8-385 Type 4.

(5) Air conditioning ducts with polyimide foam insulation, G02470 must be wrapped with tape, G50327, tape, G51202 or tape, G50789 to meet the flammability/flame propagation requirements. Wrap tape, G50327, tape, G51202 or tape, G50789 across 100% of the brown hypalon insulation surface on the polyimide foam insulation, G02470.

(6) Do not use BMS8-39 polyurethane foam insulation for repairs due to the degradation in flammability properties over time.

(7) When insulation material is removed and re-installed, any existing tape that does not meet the flame propagation requirement must be removed or completely covered with tape that does meet the requirement.

B. Consumable Materials

Reference	Description	Specification
G02470	Foam - Flexible Polyimide	BMS8-300 Type I
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50449	Foam - Flexible Melamine	BMS8-385 Type 4 Grade 1
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51169	Foam - Flexible Melamine,	BMS8-385 Type 6
G51170	Foam - Flexible Polyimide	BMS8-300 Type VIII
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

C. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

D. Insulation Repair for Overhead Distribution Duct

SUBTASK 21-23-01-010-003

(1) Remove ceiling panels as required to get access to the distribution duct.

SUBTASK 21-23-01-010-004

(2) As required, remove the interior furnishings for access.

SUBTASK 21-23-01-020-010

(3) Remove all loose tape from the ends or seams of the insulation blanket on the distribution duct.

SUBTASK 21-23-01-420-005

(4) Apply tape to the seams or ends of the insulation blanket as follows:

(a) Put the edges of the insulation together and apply two-inch wide tape, G50327, tape, G51202 or tape, G50789 to the seam of the insulation blanket so that no insulation foam edge can be seen.

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(b) Apply a circumferential wrap of two-inch wide tape, G50327, tape, G51202 or tape, G50789 on the insulation at the ends of the duct so that the tape extends one inch beyond the end of the insulation blanket.

(c) Apply circumferential wraps of one-inch wide tape, G50327, tape, G51202 or tape, G50789 along the length of the duct to secure the insulation blanket to the duct.

SUBTASK 21-23-01-410-003

(5) Install all ceiling panels that were removed for access.

SUBTASK 21-23-01-410-004

(6) Install any interior furnishings removed for access.

——— END OF TASK ———

TASK 21-23-01-300-802

5. Insulation Replacement/Repair - Overhead Distribution Duct Sidewall Risers Interface Segment

Figure 202

A. References

Reference	Title
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

B. Consumable Materials

Reference	Description	Specification
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00834	Cloth - Lint-free Cotton	
G02470	Foam - Flexible Polyimide	BMS8-300 Type I
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

C. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

D. General

SUBTASK 21-23-01-350-001

(1) This procedure provides instructions for either the replacement or repair of the insulation on the duct segments of the overhead distributgfhe vbyenion system that connect to the sidewall risers. Some airplanes did not have insulation installed on the production line in the area of the spuds that connect to the sidewall risers. For those airplanes, refer to SB 737-21A1139 for instructions to install this insulation.

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E. Preparation for the Replacement/Repair of the Insulation

SUBTASK 21-23-01-864-001

(1) Make sure that the L and R PACK switches on the P5-10 air conditioning panel are set to OFF.

SUBTASK 21-23-01-864-002

(2) Remove the pressure from the pneumatic system. To remove the pressure, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-23-01-010-005

(3) Remove the ceiling panels as required to get access to the duct segment of the overhead distribution system that connects to the sidewall risers.

SUBTASK 21-23-01-020-011

(4) If necessary, remove the overhead distribution system duct segment that connects to the sidewall risers ducts. To remove the duct segment, do this task: Overhead Distribution Duct Removal, TASK 21-23-01-000-801.

SUBTASK 21-23-01-020-012

(5) If applicable, remove all tape and insulation from the area around the riser spuds on the duct segment to be repaired.

SUBTASK 21-23-01-110-001

(6) Clean the area around the riser spuds without the insulation as follows:

WARNING

DO NOT GET METHYL ETHYL KETONE (MEK) IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM MEK. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE MEK. KEEP MEK AWAY FROM SPARKS, FLAME AND HEAT. MEK IS A POISONOUS AND FLAMMABLE SOLVENT WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Wet a lint-free cloth, G00834 with solvent, B00062 or solvent, B00184.

(b) Wipe the area around the riser spuds on the duct segment until all traces of dirt, debris, or tape is removed.

(c) Use a lint-free cloth, G00834 and remove all traces of the solvent.

SUBTASK 21-23-01-350-002

(7) Cut the foam insulation, G02470 insulation into four sections to the approximate size necessary to cover the bare surfaces around the riser spuds.

(a) Refer to Figure 202that illustrates the four sections of foam insulation.

SUBTASK 21-23-01-350-003

(8) Position each section in its place and mark the location on the section where the riser spuds will intersect the section.

SUBTASK 21-23-01-350-004

(9) Make small cuts in the areas of the insulation where the riser spuds will fit so that the insulation will fit around the spuds.

(a) Cut from the center of the semicircle that you marked to the circumference of the marked semicircle.

(b) Make the cuts on each of the four sections of insulation.

SUBTASK 21-23-01-420-006

(10) Position the four sections of foam insulation around the riser spuds on the duct segment.

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SUBTASK 21-23-01-420-007

(11) Install tape, G50327, tape, G51202 or tape, G50789 on the foam insulation as shown on Figure 202and in accordance with these guidelines:

(a) One-inch wide or 2-inch wide advanced insulation blanket tape, G50327, tape, G51202 or tape, G50789 must cover 100% of the brown colored “Hypalon” insulation surface.

(b) Cut the tape, G50327, tape, G51202 or tape, G50789 as required to fit around the spuds.

(c) Tape may be installed spiral wrapped or lengthwise depending on the installers' preference.

(d) Tape may be installed in any configuration as long as 100% coverage of the foam insulation is obtained.

(e) The tape must overlap all edges and seams of the insulation by a minimum of 1.00 inch (25.4mm).

(f) Wider tape may be cut into narrower strips or any unique shape if desired to assist in complete foam insulation coverage.

(g) Maintain a minimum of 1/8-inch overlap of the tape.

(h) Avoid compression of the foam insulation.

(i) Wrinkles in the tape are acceptable if the wrinkles are pinched together to remove all voids.

(j) Covering of existing insulation tape is not required.

(k) Tape may be installed to the foam insulation prior to installation of the duct if desired by the installer.

(l) Tape orientation is optional.

SUBTASK 21-23-01-420-008

(12) Install the repaired overhead distribution system duct segment. To install the duct segment, do this task: Overhead Distribution Duct Installation, TASK 21-23-01-400-801.

SUBTASK 21-23-01-780-001

(13) Do this test of the overhead distribution system:

(a) Supply conditioned air to the overhead distribution system. To supply conditioned air, do this task: Supply Conditioned Air to the Airplane, TASK 21-00-00-800-801.

(b) Make sure that there are no leaks at the joints of the repaired duct segment of the overhead distribution assembly.

(c) Repair any leak that you can feel with your hand when it is 2 in. (51 mm) from the duct.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-23-01-864-003

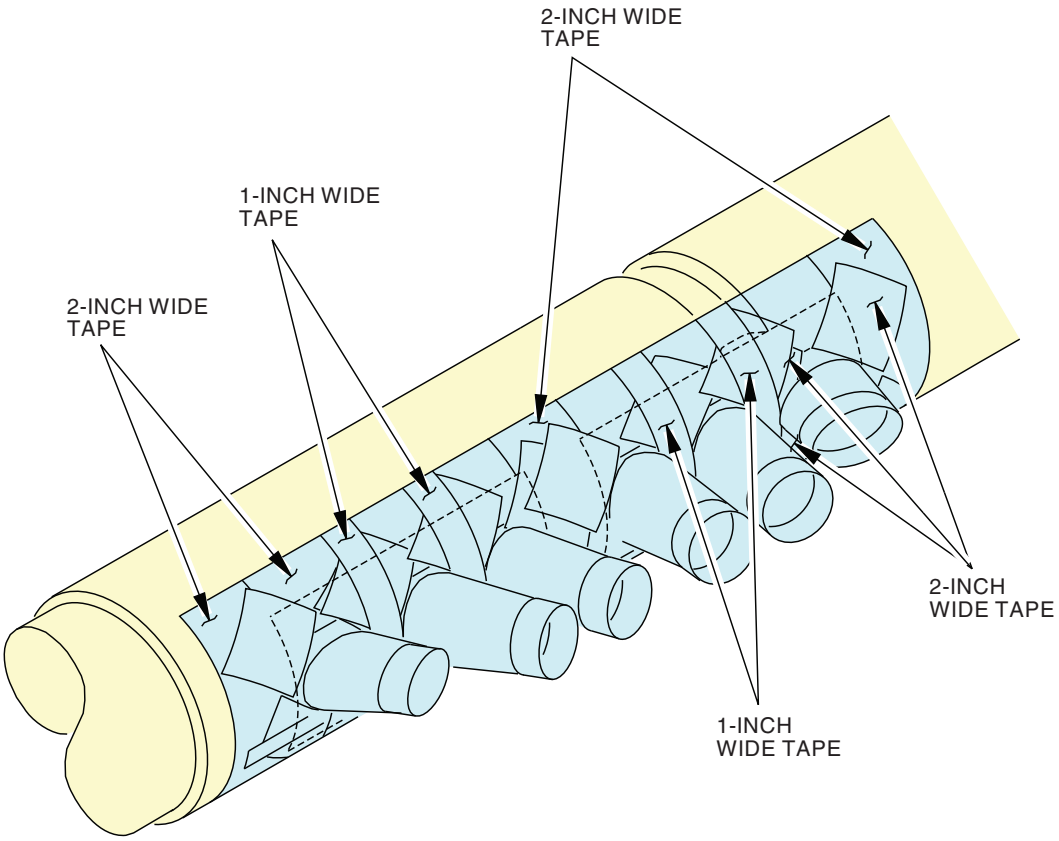
(1) Remove the conditioned air from the airplane if it is not necessary. To remove conditioned air, do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

SUBTASK 21-23-01-410-005

(2) Install all ceiling panels that were removed for access to the overhead distribution system duct segment.

——— END OF TASK ———

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TAPE INSTALLATION
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE) (TYPICAL)

C

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Insulation Repair/Replacement - Overhead Distribution Sidewall Risers Connecting Section
Figure 202/21-23-01-990-802

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TASK 21-23-01-100-801

6. Overhead Distribution Duct Cleaning

A. General

(1) This task has these procedures:

(a) Light check cleaning.

(b) Heavy check cleaning.

NOTE: Heavy check cleaning should be done during the crown corrosion inspection.

B. References

Reference	Title
21-22-09 P/B 401	FLIGHT COMPARTMENT GASPER AIR OUTLETS - REMOVAL/INSTALLATION
21-24-04 P/B 401	PASSENGER CABIN GASPER AIR OUTLETS - REMOVAL/INSTALLATION

C. Tools/Equipment

Reference	Description
STD-133	Brush - Stiff Bristle, Non-metallic
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

E. Prepare for the Cleaning

NOTE: The overhead distribution duct consists of 13 duct segments. Duct segments are connected in the two separated lines. First line has 3 duct segments which supply air to the forward passenger zone, and second line has 10 duct segments which supply air to the aft passenger zone. The overhead distribution duct starts at body station 403 and end at body station 962.

SUBTASK 21-23-01-860-003

(1) Put the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, to the OFF position and install the DO NOT OPERATE tags, STD-858.

SUBTASK 21-23-01-860-004

(2) Put the L and R PACK switches, on the P5-10 air conditioning panel, to the OFF position and install the DO NOT OPERATE tag, STD-858.

SUBTASK 21-23-01-860-005

(3) Close all gasper valves in cockpit (PAGEBLOCK 21-22-09/401).

SUBTASK 21-23-01-860-017

(4) Close all gasper valves in galley and cabin (PAGEBLOCK 21-24-04/401).

F. Light Check Cleaning

SUBTASK 21-23-01-020-014

(1) Remove the FWD galley air supply hose from the round manifold duct at STA 360 and connect to a hose routed overboard.

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SUBTASK 21-23-01-860-006

(2) Remove the DO NOT OPERATE tags, STD-858, and put the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, to the ON position.

SUBTASK 21-23-01-860-007

(3) Remove the DO NOT OPERATE tags, STD-858, and put the L and R PACK switches, on the P5-10 air conditioning panel, to the ON position.

(a) Wait for 5 minutes until no debris come out from the hose.

NOTE: The airflow will dislodge the debris through the hose routed overboard.

SUBTASK 21-23-01-860-008

(4) Put the L and R PACK switches, on the P5-10 air conditioning panel, to the OFF position and install the DO NOT OPERATE tags, STD-858.

SUBTASK 21-23-01-860-009

(5) Put the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, to the OFF position and install the DO NOT OPERATE tags, STD-858.

SUBTASK 21-23-01-020-015

(6) Disconnect the hose that is routed overboard and install the FWD galley air supply hose that you removed from the round manifold duct.

SUBTASK 21-23-01-420-009

(7) Remove the AFT galley air supply hose from the round manifold duct at STA 967 and connect to a hose routed overboard.

SUBTASK 21-23-01-860-010

(8) Remove the DO NOT OPERATE tags, STD-858, and put the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, to the ON position.

SUBTASK 21-23-01-860-011

(9) Remove the DO NOT OPERATE tags, STD-858, and put the L and R PACK switches, on the P5-10 air conditioning panel, to the ON position.

(a) Wait for 5 minutes until no debris come out from the hose.

NOTE: The airflow will dislodge the debris through the hose routed overboard.

SUBTASK 21-23-01-860-012

(10) Put the L and R PACK switches, on the P5-10 air conditioning panel, to the OFF position and install the DO NOT OPERATE tags, STD-858.

SUBTASK 21-23-01-860-013

(11) Put the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, to the OFF position and install the DO NOT OPERATE tags, STD-858.

SUBTASK 21-23-01-420-010

(12) Disconnect the hose that is routed overboard and install the AFT galley air supply hose that you removed from the round manifold duct.

SUBTASK 21-23-01-211-001

(13) Inspect the duct distribution system to see if the airflow distribution is equally distributed throughout the cabin and cargo compartment.

G. Heavy Check Cleaning

SUBTASK 21-23-01-860-014

(1) Do this task: Overhead Distribution Duct Removal, TASK 21-23-01-000-801.

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SUBTASK 21-23-01-170-001

(2) Clean the overhead distribution duct [1] as follow:

(a) Fill the overhead distribution duct [1] with water and rinse it.

1) Do not use the soap water.

(b) Use long stick stiff bristle brush, STD-133, to scrub the duct.

(c) Repeat the cleaning until no dirty water shows from the overhead distribution duct [1].

SUBTASK 21-23-01-160-001

(3) Let the ducts dry before reinstalling.

SUBTASK 21-23-01-860-015

(4) Do this task: Overhead Distribution Duct Installation, TASK 21-23-01-400-801.

SUBTASK 21-23-01-210-001

(5) Inspect the duct distribution system to see if the airflow distribution is equally distributed throughout the cabin and cargo compartment.

SUBTASK 21-23-01-860-016

(6) Put the airplane back to its usual condition.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SIDEWALL RISER DUCTS/TRANSITION DUCT - MAINTENANCE PRACTICES

1. General

A. This procedure consists of these tasks:

(1) Inspection of the sidewall risers/transition duct installation

(2) Repair of the sidewall risers/transition duct installation

B. There are two configurations of the installation where the sidewall riser ducts connect to the spuds on the transition duct. One configuration consists of the sidewall riser ducts directly connected to the spuds on the transition duct with a clamp to secure each riser duct and spud together. The other configuration uses a flex sleeve between each riser duct and spud with two clamps to secure the connection.

TASK 21-23-02-212-801

2. Sidewall Riser Ducts/Transition Duct Inspection

(Figure 201)

A. References

Reference	Title
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)

B. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

C. Sidewall Riser Ducts/Transition Duct Inspection

SUBTASK 21-23-02-010-001

(1) If it is not already done, remove the applicable cabin sidewall panels that are adjacent to the forward wing-to-body fairing.

SUBTASK 21-23-02-780-001

(2) Supply conditioned air to the passenger cabin (TASK 21-00-00-800-801).

SUBTASK 21-23-02-212-001

(3) Make sure the sidewall riser ducts [3]are securely connected to the spuds [6] on the transition duct [7].

SUBTASK 21-23-02-790-001

(4) Make sure that there is no excessive conditioned air leakage from the sidewall riser ducts/transition duct connections.

NOTE: Excessive leakage is leakage that can be felt with the hand 2 in. (51 mm) away from the joint.

SUBTASK 21-23-02-840-001

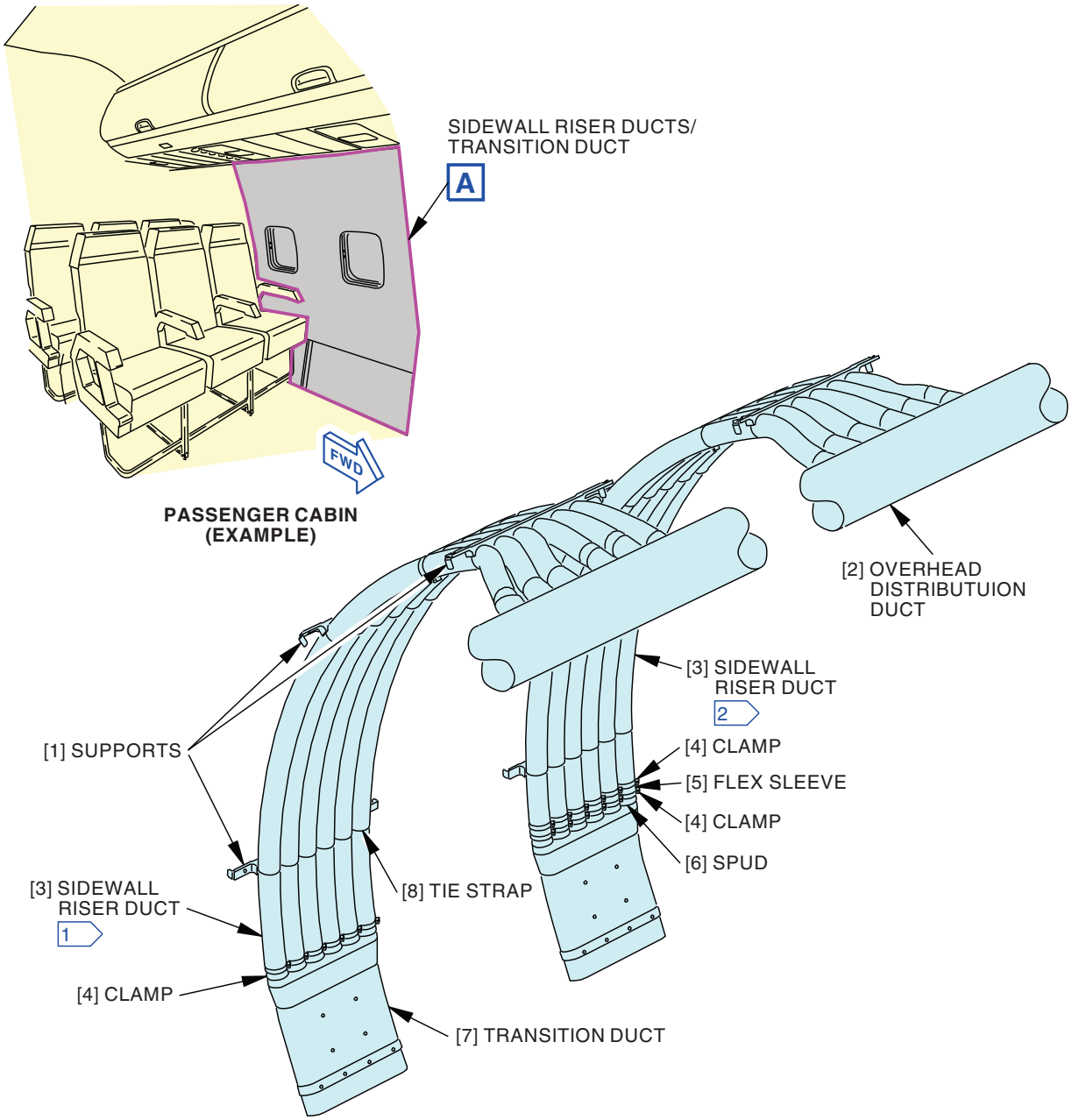
(5) Remove the conditioned air from the airplane (TASK 21-00-00-800-802).

SUBTASK 21-23-02-360-001

(6) If the sidewall riser ducts [3] have separated from the transition duct [7] or there is excessive leakage, do this task : Sidewall Riser Ducts/Transition Duct Repair, TASK 21-23-02-360-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Sidewall Riser Ducts/Transition Duct Installation
Figure 201/21-23-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-23-02-360-801

3. Sidewall Riser Ducts/Transition Duct Repair

(Figure 201)

A. References

Reference	Title
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)

B. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00130	Alcohol - Isopropyl	TT-I-735
G00093	Tape - Fiberglass, Silicone Pressure Sensitive Adhesive - G565	
G50218	Strap - Plastic, Adjustable, Self-locking, 27.50 Inches (698.50MM) Long	BACS38K6
G50262	Wiper - Cleaning	BMS15-5

C. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

D. Sidewall Riser Ducts/Transition Duct Repair

SUBTASK 21-23-02-210-001

(1) Make sure that the bottom of the transition duct [7] is approximately 8 in. (203 mm) above the passenger compartment floor.

SUBTASK 21-23-02-211-001

(2) If it is necessary, reposition the transition duct [7] so that the bottom of the transition duct [7] is correctly positioned 8 in. (203 mm) above the passenger compartment floor.

SUBTASK 21-23-02-430-001

(3) If the sidewall riser ducts [3] need to be moved down to be installed on the spuds [6], do these steps:

(a) Cut the plastic tie straps [8] from the three supports [1] that hold the sidewall riser ducts [3].

(b) Put the sidewall riser ducts [3] in position so that they can be installed on the spuds [6]].

SUBTASK 21-23-02-360-002

(4) If the sidewall riser ducts [3] are installed without a flex sleeve, then do these steps for the repair:

(a) Make sure there is no contamination in the sidewall riser ducts [3] or the spuds [6].

(b) Remove all tape from the spuds [6]on the transition duct [7].

(c) Remove all remaining parts of tape from the sidewall riser ducts [3] and the spuds [6] with a wiper, G50262, that is wet with alcohol, B00130, or solvent, B00083.

(d) Use a dry wiper, G50262, to remove the remaining alcohol, B00130, or solvent, B00083, from the sidewall riser ducts [3] and the spuds [6].

1) Do not allow the alcohol, B00130 or solvent, B00083 to dry on the sidewall riser ducts [3] and the spuds [6].

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(e) Wrap the ends of the spuds [6] on the transition duct [7] with G565 tape, G00093, to build-up the diameter of the spud [6] to match the diameter of the sidewall riser duct [3] so that there is a tight fit between the two.

(f) Slide the sidewall riser duct [3] over the spud [6] so that there is a 1 in. (25 mm) overlap between the spud [6] and the sidewall riser duct [3].

1) Shape the round sidewall riser duct [3] as necessary to fit over the oblong spud [6].

(g) Install the clamps [4] that hold the sidewall riser ducts [3] and spuds [6] together:

1) Put the clamps [4] in position equally between the end of the sidewall riser duct [3] and the transition duct [7] bead (0.25 in. (6.35 mm) to 0.45 in. (11.43 mm) from the end of the sidewall riser duct [3]).

2) Put the clamps [4] in position so that the screw head is 45 degrees forward or aft from the station line.

NOTE: The screw head can interfere with the side panel if it is put on the station line.

3) Tighten the clamp [4] to 15 ±2 in-lb (2 ±1 N·m).

SUBTASK 21-23-02-360-003

(5) If the sidewall riser ducts [3] are installed with a flex sleeve [5], then do these steps for the repair:

(a) Make sure there is no contamination in the sidewall riser ducts [3], flex sleeves [5], or spuds [6].

(b) If it is necessary, clean the sidewall riser ducts [3], flex sleeves [5], and spuds [6] with a wiper, G50262, that is wet with alcohol, B00130, or solvent, B00083.

(c) If it is necessary, use a dry wiper, G50262, to remove the remaining alcohol, B00130, or solvent, B00083, from the sidewall riser ducts [3], flex sleeves [5], and spuds [6].

1) Do not let the alcohol, B00130, or solvent, B00083, dry on the sidewall riser ducts [3], flex sleeves [5], and spuds [6].

(d) Slide the flex sleeve [5] over the sidewall riser duct [3] or spud [6] so that there is a 1 in. (25 mm) overlap between the sidewall riser duct [3] or spud [6] and flex sleeve [5].

1) Shape the round flex sleeve [5] as necessary to fit over the oblong spud [6].

(e) Install the clamps [4] that hold the sidewall riser ducts [3], flex sleeves [5], and spuds [6] together:

1) Put the clamps [4] in position equally between the end of the flex sleeve [5] and the sidewall riser duct [3] bead (0.25 in. (6.35 mm) to 0.45 in. (11.43 mm) from the end of the flex sleeve [5]).

2) Put the clamps [4] in position so that the screw head is 45 degrees forward or aft from the station line.

NOTE: The screw head can interfere with the side panel if it is positioned on the station line.

3) Tighten the clamp [4] to 15 ±2 in-lb (2 ±1 N·m).

SUBTASK 21-23-02-430-002

(6) If the sidewall riser ducts [3] were moved down and the plastic tie straps [8] removed, install a new adjustable plastic strap, G50218, to hold each sidewall riser duct [3] to the supports [1], if it is necessary.

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E. Post-Repair Check of the Sidewall Riser Ducts/Transition Duct

SUBTASK 21-23-02-710-001

(1) Supply conditioned air to the passenger cabin (TASK 21-00-00-800-801).

SUBTASK 21-23-02-790-002

(2) Do a check for leaks in the sidewall riser ducts/transition duct installation.

(a) Repair all leaks with the installation of tape on the spud [6].

SUBTASK 21-23-02-840-002

(3) Remove the conditioned air from the airplane (TASK 21-00-00-800-802).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-23-02-410-001

(1) Install the applicable cabin sidewall panels that were removed for this repair.

——— END OF TASK ———

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RETURN AIR GRILLES (DECOMPRESSION VENTS) - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Return air grille removal.

(2) Return air grille installation.

(3) Return air grille baffle removal.

(4) Return air grille baffle installation.

(5) Return air grille and baffle cleaning.

(6) Return air grille baffle inspection.

B. The return air grilles, which are also called decompression vents. They are near the floor, in the passenger compartment sidewall. The return air grilles let the air in the passenger cabin to flow into the lower fuselage areas. This will prevent damage to the airplane if a decompression condition occurs. The return air grille will be called the grille throughout this procedure.

TASK 21-23-03-000-801

2. Return Air Grille Removal

(Figure 201)

A. References

Reference	Title
25-22-00-000-801	Passenger Seat Removal (P/B 401)

B. Tools/Equipment

Reference	Description
STD-726	Punch - Awl

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for the Procedure

SUBTASK 21-23-03-000-001

(1) If necessary to gain access to the panels, do this task to remove the passenger seats: TASK 25-22-00-000-801.

E. Return Air Grille Removal

SUBTASK 21-23-03-020-001

(1) Do the steps that follow to remove the return air grille [1] with latch assembly:

(a) Insert awl punch, STD-726 in the release notch of the latch head.

1) Push the handle of awl punch, STD-726, up to release the latch.

2) Make sure that the mounting latches move freely.

(b) Move the return air grille [1] sideway sufficient to disengage the tongue and groove channel along vertical edges of the return air grille [1].

NOTE: Interference within channel can still exist at this point in removal process, it is necessary to unlatch the latches on the adjacent return air grille to allow sideways movement sufficient to remove the return air grille.

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(c) Move the bottom edge of the return air grille [1] away from the crease beam to clear interferences.

(d) Fully disengage the tongue and groove channel from the adjacent return air grille [1].

1) If it is necessary, press gently and deform the panel to fit into position.

(e) Pull the lower edge of the return air grille [1] down until clear of the window panel.

(f) Remove the return air grille [1].

SUBTASK 21-23-03-020-002

(2) Do the steps that follow to remove the return air grille [2] with window panel clip assembly:

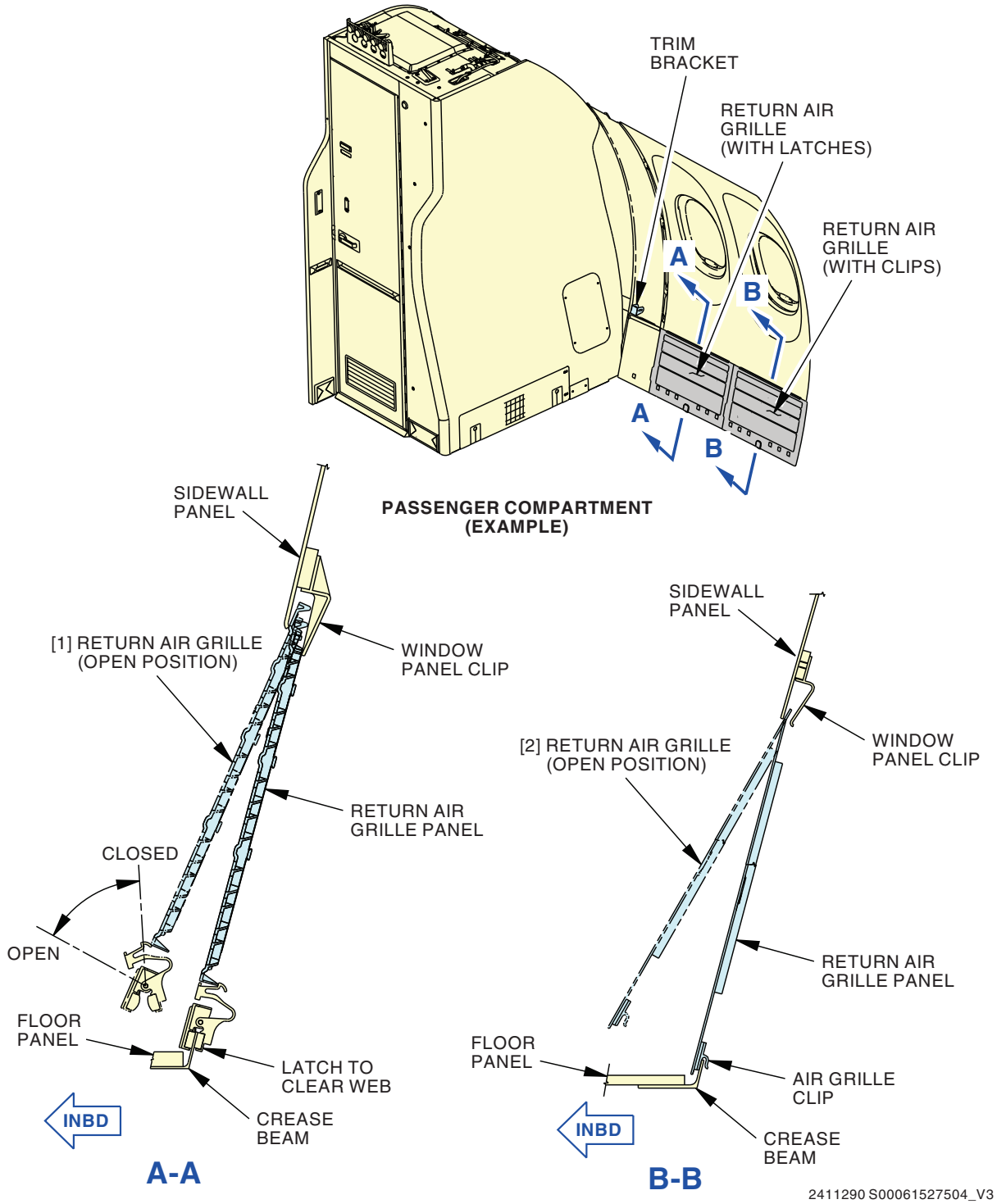
(a) Move the return air grille [2] up until the window panel clip disengages from the crease beam.

(b) Move inboard until the return air grille [2] clears the crease beam then lower to clear the window panel.

(c) Remove the return air grille [2].

——— END OF TASK ———

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Return Air Grille Installation
Figure 201/21-23-03-990-801

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TASK 21-23-03-000-802

3. Return Air Grille Baffle Removal

Figure 202

A. Location Zones

Zone	Area
200	Upper Half of Fuselage

B. Procedure

SUBTASK 21-23-03-020-003

(1) Do the steps that follow to remove the baffle [3] from the return air grille [1]:

(a) Remove the return air grill and baffle assembly, (TASK 21-23-03-000-801).

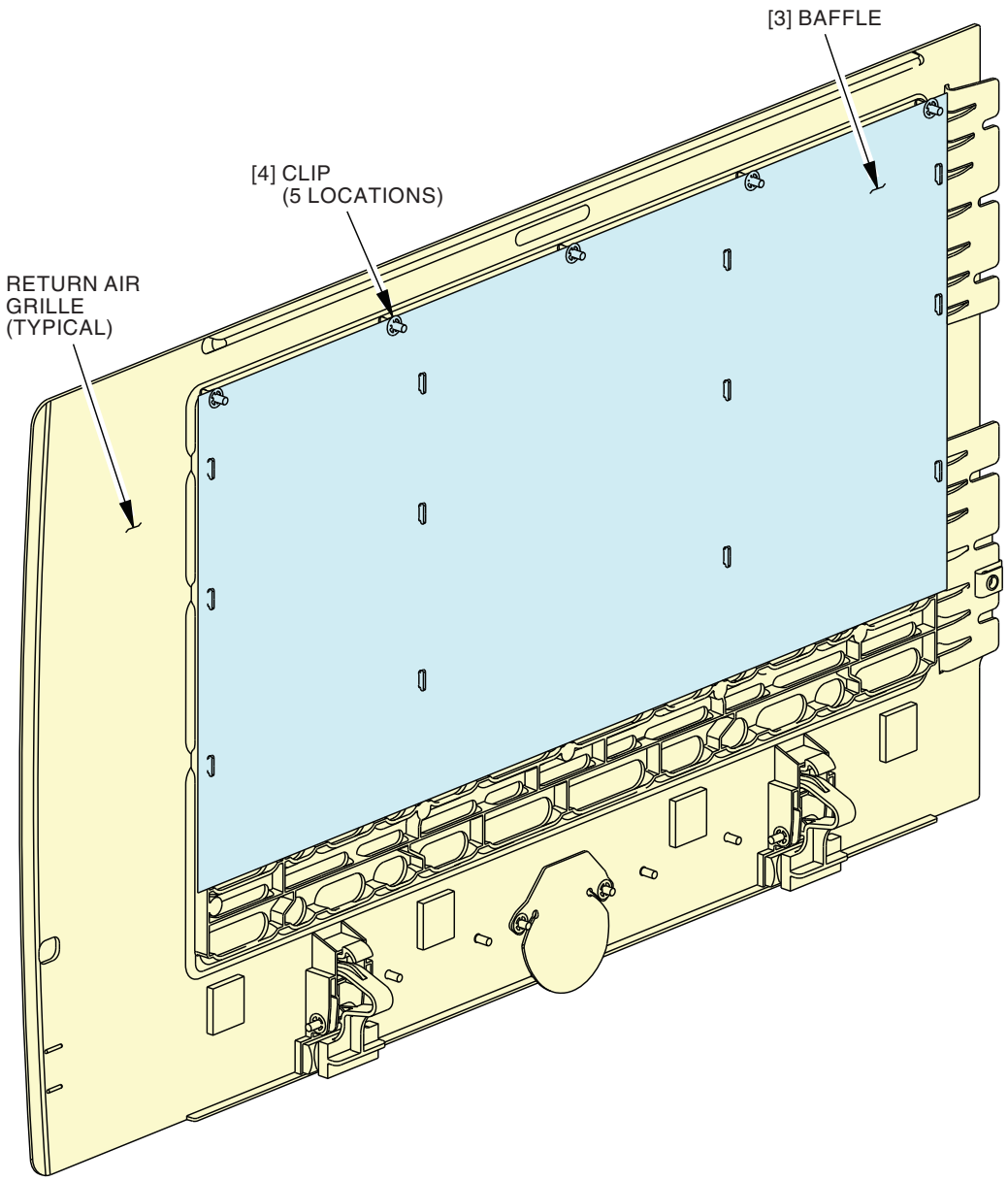
(b) Remove the retainer clips [4], at 5 locations, which attach the top of the baffle [3] to the return air grille [1].

(c) Slide the baffle [3] over the tabs on the return air grille [1].

(d) Remove the baffle [3].

——— END OF TASK ———

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Return Air Grille Baffle Installation
Figure 202/21-23-03-990-802

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TASK 21-23-03-100-801

4. Return Air Grille and Baffle Cleaning

A. Tools/Equipment

Reference	Description
STD-1068	Vacuum - High Capacity

B. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Procedure

SUBTASK 21-23-03-160-001

(1) Do the steps that follow to clean the return air grille and baffle assembly:

(a) Do this task to remove the return air grille and baffle assembly: TASK 21-23-03-000-801.

(b) Use the high capacity vacuum, STD-1068 to remove the dust, lint, or unwanted material from the return air grille and baffle assembly.

(c) Use the high capacity vacuum, STD-1068 to remove the dust, lint, or unwanted material from the airplane structure behind the return air grill and baffle assembly.

(d) Wipe the return air grille and baffle assembly with a cotton wiper, G00034 that is moist with water.

(e) Do this task to install the return air grille and baffle assembly: TASK 21-23-03-400-802.

——— END OF TASK ———

TASK 21-23-03-400-801

5. Return Air Grille Baffle Installation

Figure 202

A. Location Zones

Zone	Area
200	Upper Half of Fuselage

B. Procedure

SUBTASK 21-23-03-420-001

(1) Do the steps that follow to attach the baffle [3] to the return air grille [1]:

(a) Slide the baffle [3] over the tabs on the return air grille [1].

1) Make sure that all the edges of the baffle [3] are in their correct positions to the tabs.

(b) Install the retainer clips [4], at 5 locations, which attach the top of the baffle [3] to the return air grille [1]

(c) Install the return air grille and baffle assembly (TASK 21-23-03-400-802).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-23-03-400-802

6. Return Air Grille Installation

(Figure 201)

A. References

Reference	Title
25-22-00 P/B 401	PASSENGER SEATS - REMOVAL/INSTALLATION

B. Tools/Equipment

Reference	Description
STD-726	Punch - Awl

C. Consumable Materials

Reference	Description	Specification
G50156	Tape - Carpet, Double-sided, 16 oz/sq. yd., Black. Storage Life 18 Months.	BMS5-133 Type II Class 2

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Return Air Grille Installation

SUBTASK 21-23-03-420-002

(1) Do the steps that follow to install the return air grille [1] with latch assembly:

(a) Apply tape, G50156, is in 0.5 in. (12.7 mm) by 12 in. (304.8 mm) strips on the outboard side of the return air grille [1].

1) Make sure that tape, G50156, aligns with the trim bracket.

(b) Make sure that the mounting latches on the return air grille [1] are unlatched and move freely.

1) If it is necessary, push the handle of awl punch, STD-726, to release the latches.

(c) Move the top edge of the return air grille [1] behind the sidewall panel.

1) Align the mating edges and insert into the window panel clips.

(d) Position the bottom edge of the return air grille [1] to rest on the floor structure and inboard side of the crease beam.

1) Make sure that the panel pads are positioned to fully engage against inboard side of the crease beam and clamp pads mate to the outboard side of crease beam when closed.

NOTE: The fasteners attached to the crease beam can be considered part of the crease beam.

(e) Adjust position sideways to ensure proper engagement of the tongue and groove channel along the mating vertical edges.

1) If it is necessary, press gently and deform the panel to fit into position.

NOTE: To ease the instillation issues specific to engagement of the edge channel, release the clamp latches to unsecure adjacent panels. This will allow sufficient side-to-side movement to position the panel being installed. Once positioned, all panels should be adjusted for the proper channel engagement.

(f) Make sure that the latches are filly engaged in the locked position.

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(g) Close latches over the crease beam.

SUBTASK 21-23-03-420-003

(2) Do the steps that follow to install the return air grille [2] with clip assembly:

(a) Move the top of the return air grille [2] into the window panel clip as high as possible.

(b) Move the return air grille [2] outboard until it is above the crease beam.

(c) Move the return air grille [2] down until the clip engages the crease beam.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-23-03-410-001

(1) If necessary, install the applicable passenger seats (PAGEBLOCK 25-22-00/401).

——— END OF TASK ———

TASK 21-23-03-200-801

7. Return Air Grille Baffle Inspection

A. General

(1) This task has instructions to inspect, remove, and install the baffles on the backside of the return air grilles.

(2) In the event of a rapid decompression of the airplane fuselage, the baffles are designed to detach from the backside of the return air grilles to permit airflow movement from the main cabin to the lower lobe compartments. After a decompression event, it is necessary to inspect the baffles and reinstall them to the backside of the return air grilles.

B. References

Reference	Title
25-22-00-000-801	Passenger Seat Removal (P/B 401)
25-22-00-400-801	Passenger Seat Installation (P/B 401)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for Inspection

SUBTASK 21-23-03-020-006

(1) Do this task to remove the passenger seats as needed to get access to the return air grille [1]: TASK 25-22-00-000-801.

SUBTASK 21-23-03-020-004

(2) Do this task to remove the return air grille [1]: TASK 21-23-03-000-801.

E. Baffle Inspection

SUBTASK 21-23-03-212-001

(1) If the baffle [3] is missing or has damage (tears, punctures, rips, cuts), install a new baffle [3].

SUBTASK 21-23-03-212-002

(2) If the baffle [3] has become partially or fully detached from the backside of the return air grille [1], reinstall the baffle [3] with the attachment hardware.

SUBTASK 21-23-03-212-003

(3) If the baffle [3] has missing or damaged attachment hardware, install new attachment hardware.

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SUBTASK 21-23-03-212-004

(4) Inspect the lower sidewall areas for unwanted materials or missing baffle [3] hardware, and clean the areas as applicable before you reinstall the return air grille [1].

F. Baffle Removal

SUBTASK 21-23-03-020-005

(1) Do this task to remove the baffle on the return air grille [1]: TASK 21-23-03-000-802.

G. Baffle Installation

SUBTASK 21-23-03-420-004

(1) Do this task to install the baffle on the return air grille [1]: TASK 21-23-03-400-801.

H. Put the airplane back to its usual condition

SUBTASK 21-23-03-420-005

(1) Do this task to install the return air grille [1]: TASK 21-23-03-400-802.

SUBTASK 21-23-03-420-006

(2) Do this task to install the passenger seats if you removed them for access: TASK 25-22-00-400-801.

——— END OF TASK ———

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SIDEWALL RISER DUCTS - REMOVAL/INSTALLATION

1. General

A. This task removes the sidewall riser ducts between the transition duct in the mix bay and the main deck overhead distribution duct.

TASK 21-23-04-000-801

2. Sidewall Riser Ducts Removal

(Figure 401)

A. General

(1) This task removes the sidewall riser ducts between the distribution bay and the overhead distribution ducts.

(a) The sidewall riser duct installation consists of these components:

1) Six ducts
2) Three support assemblies
3) Clamps and plastic ties
4) Six flexible hoses
5) Six flexible sleeves
6) A transition duct assembly.

(2) There are three sets of riser ducts, two on the left side of the passenger cabin and one set on the right side.

(3) Access to the sidewall riser ducts is through the sidewall panels directly above the distribution bay in the main passenger cabin.

(4) There are two configurations of the installation where the sidewall riser ducts connect to the spuds on the transition duct. One configuration consists of the sidewall riser ducts directly connected to the spuds on the transition duct with a clamp to attach each riser duct and spud together. The other configuration uses a flex sleeve between each riser duct and transition duct spud with two clamps to attach the connections.

B. References

Reference	Title
25-22-00-000-801	Passenger Seat Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Location Zones

Zone	Area
231	Forward Passenger Compartment - Forward Entry Door to STA 663.75 - Left

D. Preparation for the Removal

SUBTASK 21-23-04-860-001

(1) Remove pressure from the pneumatic system (TASK 36-00-00-860-806).

SUBTASK 21-23-04-860-002

(2) Make sure that the L PACK and R PACK switches on the P5-10 air conditioning panel are set to OFF.

SUBTASK 21-23-04-020-001

(3) If it is necessary, remove passenger seats in the area of the sidewall riser ducts (TASK 25-22-00-000-801).

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SUBTASK 21-23-04-020-002

(4) Remove the cabin sidewall panels that are directly above main distribution outlets in the main distribution manifold to the sidewall riser ducts.

SUBTASK 21-23-04-020-003

(5) If it is necessary, remove the ceiling panels in the passenger cabin above the sidewall panels that you removed.

E. Sidewall Riser Ducts Removal

SUBTASK 21-23-04-020-004

(1) Do all of the applicable steps that follow to disconnect the six flex hoses [1] or six flex hoses [15] from the overhead distribution duct [3]:

(a) Remove the tie straps [2] and the tapes [16] at six locations that connect the flex hoses [1] and the spuds [7] on the overhead distribution duct [3].

SUBTASK 21-23-04-020-005

(2) Do one of the steps that follow to loosen the sidewall riser ducts [4] at the three support [14]locations:

(a) Cut the plastic tie straps [11] from the three supports [14] that hold the sidewall riser ducts [4].

1) Discard the plastic tie straps [11].

(b) Disconnect the supports [14] from structure.

SUBTASK 21-23-04-020-006

(3) Do one of these steps that follows to loosen the sidewall riser ducts [4] at the transition duct [12]:

(a) Loosen the twelve clamps [5] or six clamps [13] that attach the flex sleeves [6] and the spuds [7] to the sidewall riser ducts [4].

SUBTASK 21-23-04-020-007

(4) Do these steps to remove the transition duct [12] from the elbow duct [10] from the mix bay:

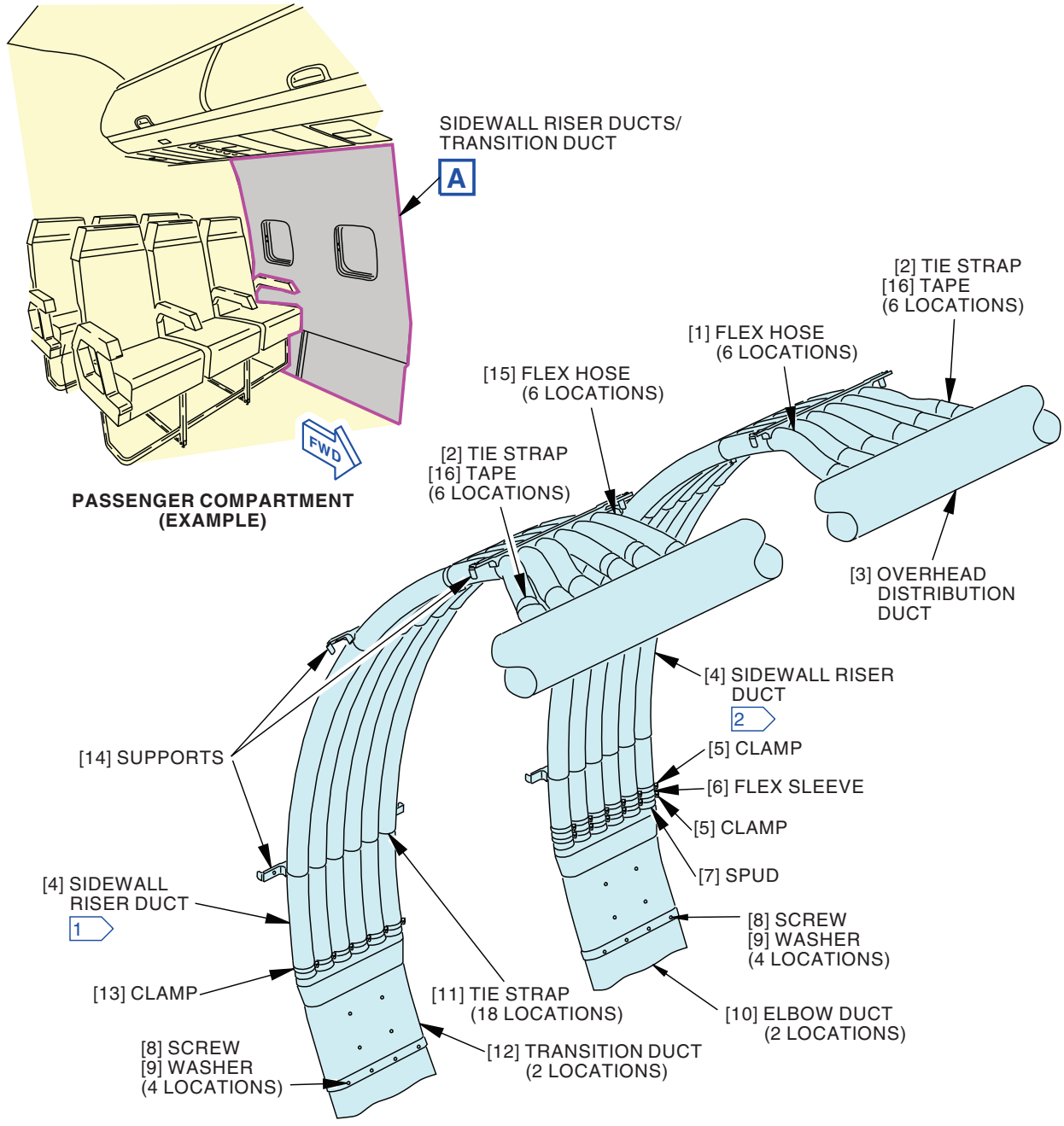
NOTE: This step is not necessary if the intent of maintenance is to replace just the round ducts as an assembly or an individual duct.

(a) Remove the screws [8] and washers [9].

(b) Remove the transition duct [12] from the elbow duct [10].

——— END OF TASK ———

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Sidewall Riser Ducts Installation
Figure 401/21-23-04-990-801

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TASK 21-23-04-400-801

3. Sidewall Riser Ducts Installation

(Figure 401)

A. References

Reference	Title
21-00-00-800-801	Supply Conditioned Air to the Airplane (P/B 201)
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
25-22-00-400-801	Passenger Seat Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
A00027	Adhesive - Silicone Rubber, 1 Part, RTV	BAC5010 Type 60
A50126	Primer - Adhesive Bonding (Formerly Dow Corning 1204 Clear)	
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00102	Abrasive - Aluminum Oxide Coated Cloth	
B00130	Alcohol - Isopropyl	TT-I-735
C00580	Primer - Adhesive Bonding, RTV - PR-1200 (Formerly DC 1200)	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50140	Gloves - Protective, Latex or Nitrile	
G50262	Wiper - Cleaning	BMS15-5
G50344	Tape - Nitto P-212HD WHT (Formerly Permacel) Glass Cloth	
G50625	Strap - Plastic, Adjustable, Self-locking, 14.40 Inches (365.76 mm) Long	BACS38K3

C. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

D. Sidewall Riser Ducts Installation

SUBTASK 21-23-04-160-001

(1) If you removed the transition duct [12] with the sidewall riser ducts [4], then remove all the sealant between the faying surfaces of the transition duct [12] and the elbow duct [10]:

(a) Lightly abrade the surfaces with old adhesive with 320 grit or finer abrasive cloth, B00102, until the old adhesive is removed.

(b) Wet a cotton wiper, G00034, with solvent, B00083, or alcohol, B00130, and wipe all faying surfaces until all traces of the old adhesive are gone.

1) Wet the cotton wiper, G00034, by pouring the solvent onto the wiper, not by dipping the wiper into the solvent source.

(c) Remove excess solvent with a clean cotton wiper, G00034.

(d) Do a final cleaning of the surfaces with a cotton wiper, G00034, wet with alcohol, B00130.

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(e) Immediately wipe all cleaned surfaces with a new cotton wiper, G00034, until all alcohol, B00130, is removed from the surfaces.

(f) Put on clean protective gloves, G50140, and avoid contact with the cleaned surfaces.

SUBTASK 21-23-04-390-001

(2) Apply the sealant to the faying surfaces on the elbow duct [10] and the transition duct [12]:

(a) Prime the surfaces:

1) Do not thin the primer or use primer that is cloudy or milky.

2) Brush or spray the surfaces with a thin coat of red or clear PR-1200 primer, C00580, or PR 1204 primer, A50126.

3) Let the primer dry at 65°F (18°C) - 100°F (38°C) for a minimum of 30 minutes and 4 hours maximum.

(b) Apply a continuous layer of adhesive, A00027, over both surfaces to be bonded.

NOTE: The inside bottom surface of the transition duct [12] and the outside top surface of the elbow duct [10] are the bonding surfaces.

(c) Install the transition duct [12] over the elbow duct [10] as soon as possible after the application of the adhesive.

(d) Install the four washers [9] and screws [8].

(e) Let the adhesive cure for a minimum of 24 hours at 65°F (18°C) - 100°F (38°C).

SUBTASK 21-23-04-210-001

(3) Make sure that the bottom of the transition duct [12] is approximately 8 in. (203 mm) above the passenger compartment floor.

SUBTASK 21-23-04-210-002

(4) If it is necessary, reposition the transition duct [12] so that the bottom of the transition duct [12] is approximately 8 in. (203 mm) above the passenger compartment floor.

SUBTASK 21-23-04-420-001

(5) Put the sidewall riser ducts [4] in their installed position.

SUBTASK 21-23-04-420-002

(6) Do these steps for the sidewall riser duct [4] installations without the flex sleeves [6]:

(a) Make sure that there is no contamination in the sidewall riser ducts [4] or the spuds [7].

(b) Remove all tape from the spuds [7] on the transition duct [12].

(c) If it is necessary, remove all remaining pieces of tape from the sidewall riser ducts [4] and the spuds [7] with a wiper, G50262, that is wet with alcohol, B00130, or solvent, B00083.

(d) Use a dry wiper, G50262, to remove the remaining alcohol, B00130, or solvent, B00083.

(e) Do not let the alcohol, B00130, or solvent, B00083, dry on the sidewall riser ducts [4] and the spuds [7].

(f) Wrap the ends of the spuds [7] on the with 2 in. (51 mm) Nitto P-212HD WHT tape, G50344, to build-up the diameter of the spuds [7] to match the diameter of the sidewall riser ducts [4] so that there is a tight fit between the two.

(g) Put the sidewall riser ducts [4] over the spuds [7] on the transition duct [12] until there is a 1 in. (25 mm) overlap between the ends of the sidewall riser ducts [4] and the spuds [7] on the transition duct [12].

1) Shape the round sidewall riser ducts [4] as necessary to fit over the oblong spuds [7].

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(h) Install the clamps [13] that hold the sidewall riser ducts [4] and spuds [7] together:

1) Put the clamps [13] in position equally between the end of the sidewall riser duct [4] and the transition duct [12] bead (0.25 in. (6.35 mm) to 0.45 in. (11.43 mm) from the end of the sidewall riser duct [4]).

2) Put the clamps [13] in position so that the screw head is 45 degrees forward or aft from the station line.

NOTE: The screw head can interfere with the sidewall panel if it is positioned on the station line.

3) Tighten the clamps [13] to 15 ±2 in-lb (2 ±1 N·m).

SUBTASK 21-23-04-420-003

(7) Do these steps for the sidewall riser duct [4] installations with the flex sleeves [6]:

(a) Make sure that there is no contamination in the sidewall riser ducts [4], flex sleeves [6], or the spuds [7].

(b) If it is necessary, clean the sidewall riser ducts [4], flex sleeves [6], and the spuds [7] with a wiper, G50262, that is wet with alcohol, B00130, or solvent,solvent, B00083.

(c) If it is necessary, use a dry wiper, G50262, to remove the remaining alcohol, B00130, or solvent, B00083, from the sidewall riser ducts [4], flex sleeves [6], and the spuds [7].

1) Do not let the alcohol, B00130 or solvent, B00083, dry on the sidewall riser ducts [4], flex sleeves [6], and the spuds [7].

(d) Slide the flex sleeves [6] over the sidewall riser ducts [4] or spuds [7] so that there is a 1 in. (25 mm) overlap between the sidewall riser duct [4], or spud [7] and the flex sleeve [6].

1) Shape the round flex sleeves [6] as necessary to fit over the oblong spuds [7].

(e) Install the clamps [5] that hold the sidewall riser ducts [4], flex sleeves [6], and the spuds [7] together:

1) Put the clamps [5] in position equally between the end of the flex sleeve [6] and the sidewall riser duct [4] bead (0.25 in. (6.35 mm) to 0.45 in. (11.43 mm) from the end of the flex sleeve [6]).

2) Put the clamps [5] in position so that the screw head is 45 degrees forward or aft from the station line.

NOTE: The screw head can interfere with the sidewall panel if it is positioned on the station line.

3) Tighten the clamps [5] to 15 ±2 in-lb (2 ±1 N·m).

SUBTASK 21-23-04-420-004

(8) If they are removed, install the supports [14].

SUBTASK 21-23-04-410-001

(9) Put the sidewall riser ducts [4] in position so that they are in the supports [14].

SUBTASK 21-23-04-420-005

(10) Install the adjustable plastic strap, G50625, and tie straps [11] to hold the sidewall riser ducts [4] in position in the supports [14].

SUBTASK 21-23-04-420-006

(11) To attach the flex hose [1] and flex hoses [15] to the overhead distribution duct [3], do these steps:

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(a) Make sure that there is no contamination in the flex hoses [15] or the overhead distribution duct [3].

(b) Remove all remaining pieces of tape from the flex hoses [15] and the overhead distribution duct spuds with a wiper, G50262, that is wet with alcohol, B00130, or solvent, B00083.

(c) If it is necessary, use a dry wiper, G50262, to remove the remaining alcohol, B00130, or solvent, B00083, from the flex hoses [1] and flex hoses [15] and the overhead distribution duct spuds.

(d) Install the Nitto P-212HD WHT tape, G50344:

1) Install the flex hoses [1] and flex hoses [15] on the spuds on the overhead distribution duct [3] with a minimum 0.5 in. (12.7 mm) of overlap.

2) Apply 2-4 laps of the tapes [16] to the center of the joint with these considerations:

a) Keep the tapes [16] taut as you wrap the joint.

b) Smooth out any wrinkles in the tapes [16] as you wrap the joint.

(e) Install six adjustable plastic straps, G50625 and tie straps [2] around the tapes [16] on the flex hoses [1]s.

(f) Make sure that the joints are tight and do not slip.

E. Sidewall Riser Ducts Installation Test

SUBTASK 21-23-04-860-003

(1) Supply conditioned air to the passenger cabin (TASK 21-00-00-800-801).

SUBTASK 21-23-04-790-001

(2) Make sure that there are no leaks in the sidewall riser ducts [4]:

(a) Do a check for leaks at all the joints in the sidewall riser ducts [4].

NOTE: A leak exists if airflow can be felt on a hand a minimum of 2 in. (51 mm) away from a joint or surface.

(b) Make sure that there is no whistling, vibration or other excessive air noise from the sidewall riser ducts [4].

(c) Repair all leaks that are found.

SUBTASK 21-23-04-860-004

(3) Remove the conditioned air from the airplane (TASK 21-00-00-800-802).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-23-04-410-002

(1) Install the sidewall panels that were removed for access.

SUBTASK 21-23-04-410-003

(2) Install all ceiling panels that were removed.

SUBTASK 21-23-04-410-004

(3) If it is necessary, install passenger seats in the area of the sidewall riser ducts (TASK 25-22-00-400-801).

——— END OF TASK ———

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SIDEWALL DIFFUSER ORIFICE - CLEANING/PAINTING

1. General

A. This procedure has this task:

(1) A cleaning of the sidewall diffuser orifice.

TASK 21-23-05-100-801

2. Sidewall Diffuser Orifice Cleaning

(Figure 701)

A. General

(1) This task provides the steps to clean the sidewall diffuser orifice that is blocked with debris.

B. References

Reference	Title
25-24-31-000-801	Overhead Stowage Bin - Removal (P/B 401)
25-24-31-000-802	Overhead Stowage Bin Module - Removal (P/B 401)
25-24-31-400-801	Overhead Stowage Bin - Installation (P/B 401)
25-24-31-400-802	Overhead Stowage Bin Module - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-133	Brush - Stiff Bristle, Non-metallic
STD-183	Cleaner - Vacuum
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G51500	Strap - Plastic, Adjustable, Self-locking, 7.95 Inches (201.93 MM) Long	BACS38K10

E. Location Zones

Zone	Area
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

F. Prepare for the Cleaning

SUBTASK 21-23-05-860-001

(1) Put the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, to the OFF position and install the DO NOT OPERATE tags, STD-858.

SUBTASK 21-23-05-860-002

(2) Put the L and R PACK switches, on the P5-10 air conditioning panel, to the OFF position and install the DO NOT OPERATE tag, STD-858.

SUBTASK 21-23-05-020-001

(3) Remove the overhead stowage bin, do this task: Overhead Stowage Bin Module - Removal, TASK 25-24-31-000-802 or Overhead Stowage Bin - Removal, TASK 25-24-31-000-801.

G. Sidewall Diffuser Orifice Cleaning

SUBTASK 21-23-05-100-001

(1) Clean the sidewall diffuser [3] as follows:

(a) Disconnect the flexible duct [2] from the sidewall diffuser [3].

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1) Remove the adjustable plastic strap.

(b) Clean the orifice [1] with the stiff bristle brush, STD-133.

1) If it is necessary, vacuum the debris with vacuum cleaner, STD-183.

(c) Connect the flexible duct [2] to the sidewall diffuser [3].

1) Install the adjustable plastic strap, G51500, on the flexible duct [2].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-23-05-420-001

(1) Install the overhead stowage bin, do this task: Overhead Stowage Bin - Installation, TASK 25-24-31-400-801 or Overhead Stowage Bin Module - Installation, TASK 25-24-31-400-802.

SUBTASK 21-23-05-020-002

(2) Remove the DO NOT OPERATE tags, STD-858, and put the L and R RECIRC FAN switches, on the P5-10 air conditioning panel, to the ON position.

SUBTASK 21-23-05-020-003

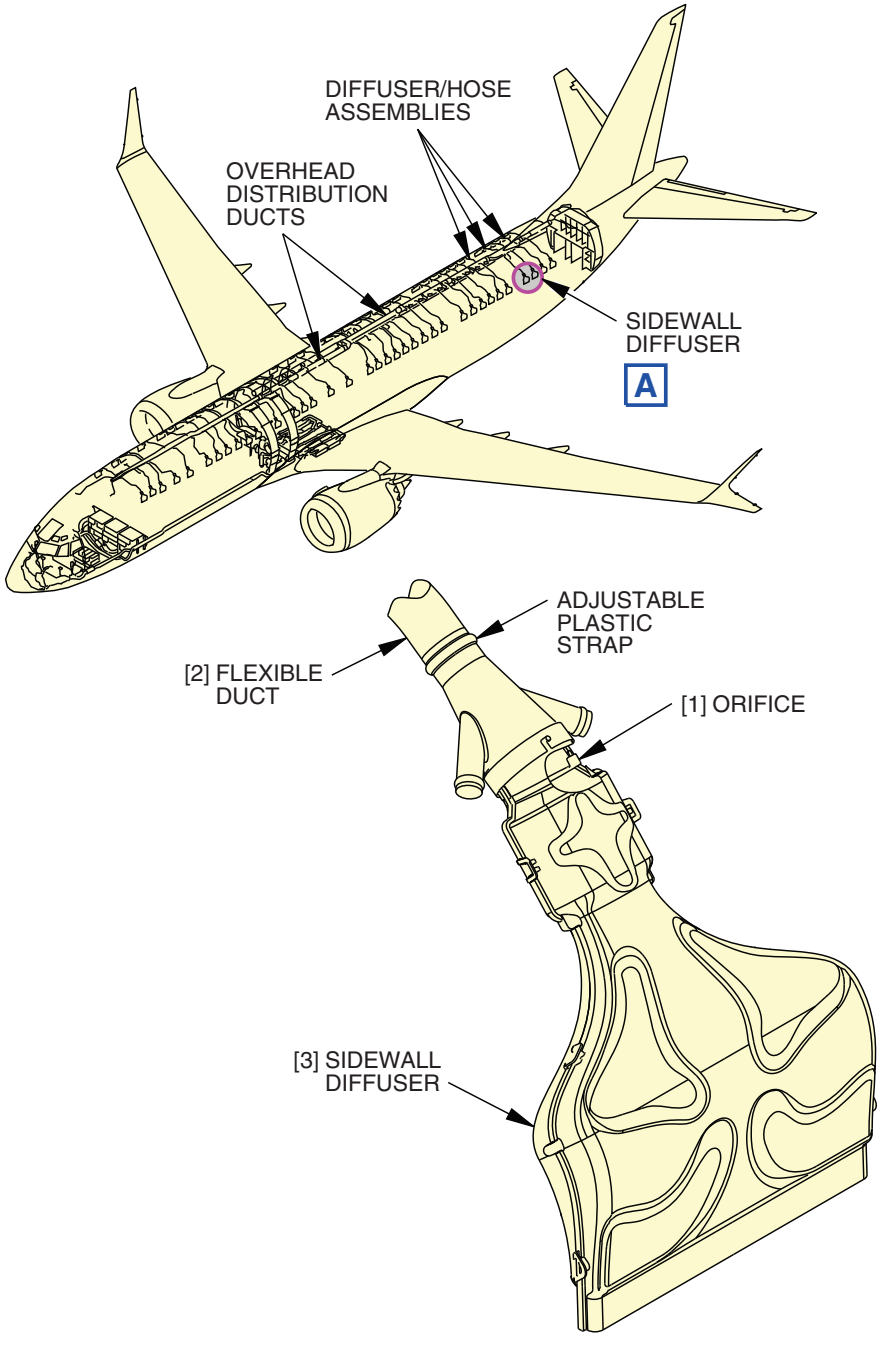
(3) Remove the DO NOT OPERATE tags, STD-858, and put the L and R PACK switches, on the P5-10 air conditioning panel, to the ON position.

SUBTASK 21-23-05-860-003

(4) Make sure that the airflow from the sidewall diffuser [3] is satisfactory.

——— END OF TASK ———

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Sidewall Diffuser Orifice Cleaning
Figure 701/21-23-05-990-801

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PASSENGER CABIN GASPER AIR OUTLETS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the gasper air outlets installed in the Passenger Service Units (PSU).

(2) An installation of the gasper air outlets installed in the Passenger Service Units (PSU).

(3) A removal of the gasper air outlets installed in the Lavatory Service Units (LSU).

(4) An installation of the gasper air outlets installed in the Lavatory Service Units (LSU).

B. The gasper air outlet is referred to as the gasper in this procedure.

TASK 21-24-04-000-801

2. Gasper Air Outlet (PSU) Removal

(Figure 401)

A. Location Zones

Zone	Area
200	Upper Half of Fuselage

B. Gasper Air Outlet Removal

SUBTASK 21-24-04-860-001

(1) Get a different gasper and turn the finger grips clockwise to fully retract the air guide.

SUBTASK 21-24-04-860-002

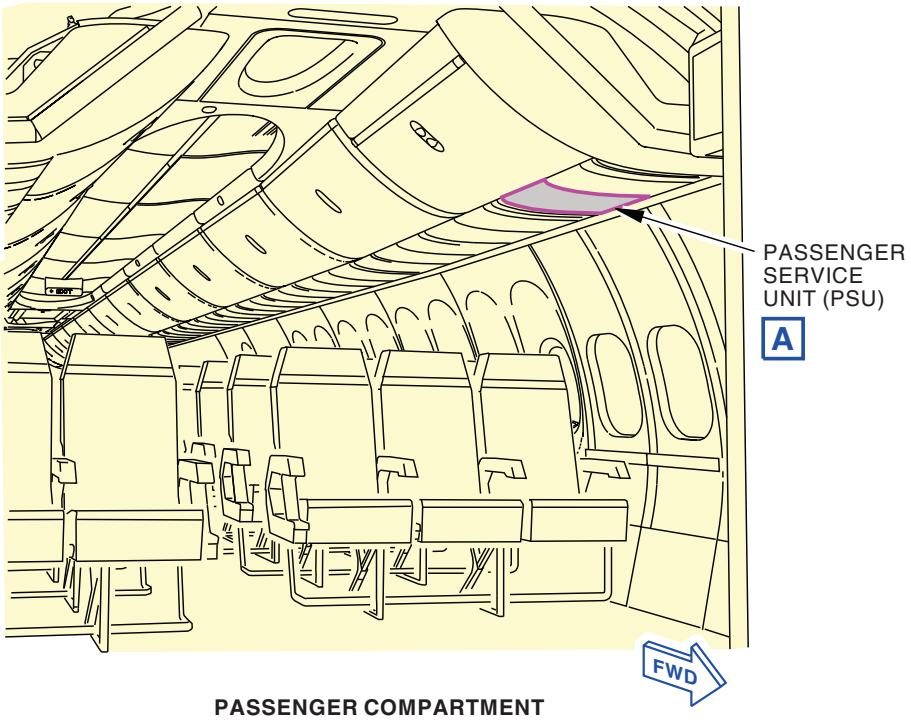
(2) Put the four tabs of the retracted gasper in the four holes of the air outlet gasper [1] being removed.

SUBTASK 21-24-04-020-001

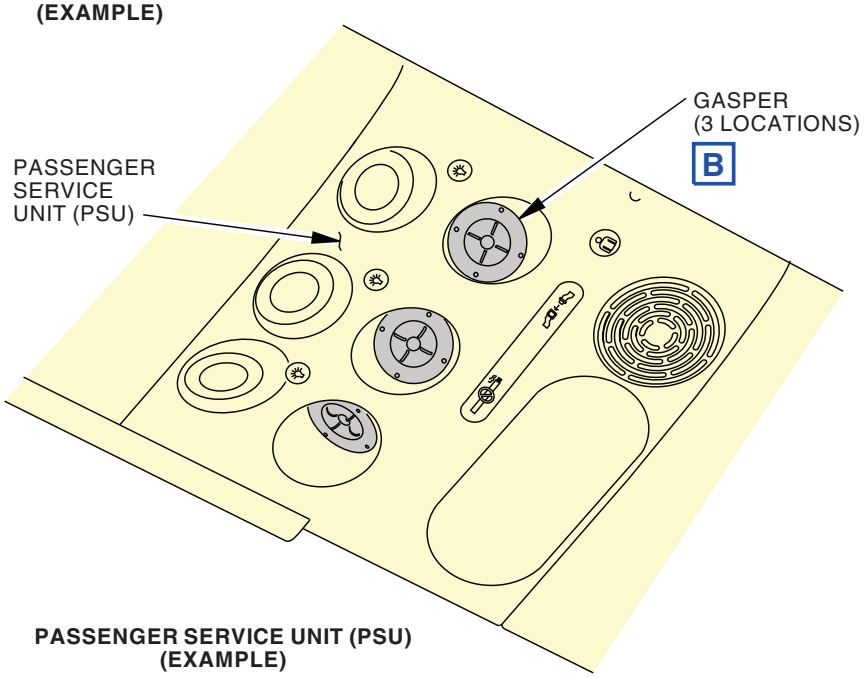
(3) Turn the retracted gasper counterclockwise to remove the air outlet gasper [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**PASSENGER COMPARTMENT
(EXAMPLE)**



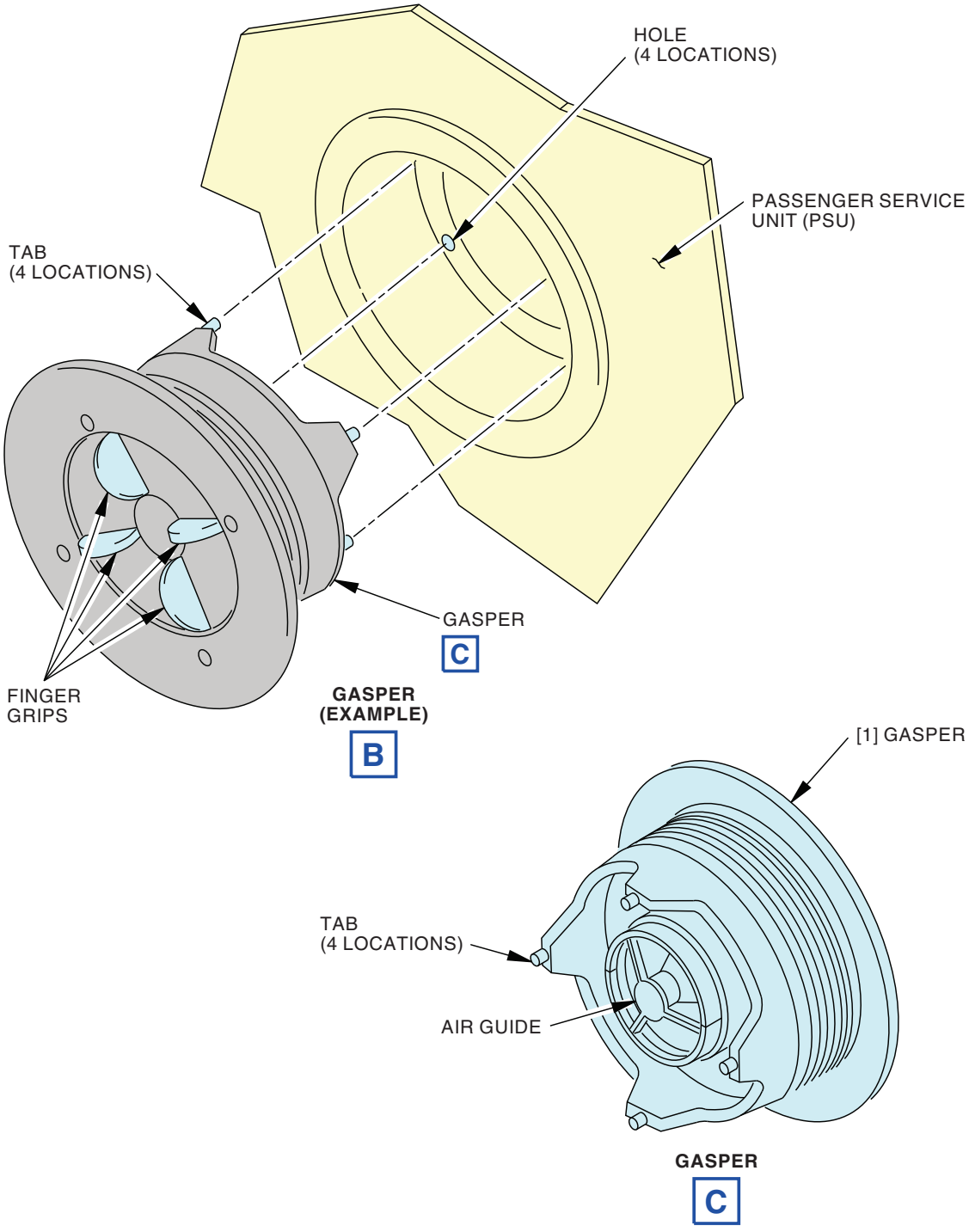
**PASSENGER SERVICE UNIT (PSU)
(EXAMPLE)**

A

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**Gasper Air Outlets (PSU) Installation
Figure 401/21-24-04-990-801 (Sheet 1 of 2)**

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Gasper Air Outlets (PSU) Installation
Figure 401/21-24-04-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-24-04-400-801

3. Gasper Air Outlet (PSU) Installation

(Figure 401)

A. Location Zones

Zone	Area
200	Upper Half of Fuselage

B. Gasper Air Outlet Installation

SUBTASK 21-24-04-420-001

(1) Loosely install the air outlet gasper [1] into the PSU gasper panel.

SUBTASK 21-24-04-860-003

(2) Get a different gasper and turn the finger grips in the clockwise direction to fully retract the air guide.

SUBTASK 21-24-04-860-004

(3) Put the four tabs of the retracted gasper in the four holes of the new air outlet gasper [1].

SUBTASK 21-24-04-420-002

(4) Use your hand to turn the air outlet gasper [1] in the clockwise direction and tighten.

——— END OF TASK ———

TASK 21-24-04-000-802

4. Gasper Air Outlet (LSU) Removal

(Figure 402)

A. References

Reference	Title
25-23-13-000-801	Attendant/Lavatory Service Unit Removal (P/B 401)

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Prepare for the Removal

SUBTASK 21-24-04-010-001

(1) Lower the applicable (LSU) gasper panel from the lavatory ceiling (TASK 25-23-13-000-801).

D. Gasper Air Outlet Removal

SUBTASK 21-24-04-020-002

(1) Do these steps to remove the air outlet gasper [2] from the strip:

(a) Loosen the strap that holds the A/C tubing to the air outlet gasper [2].

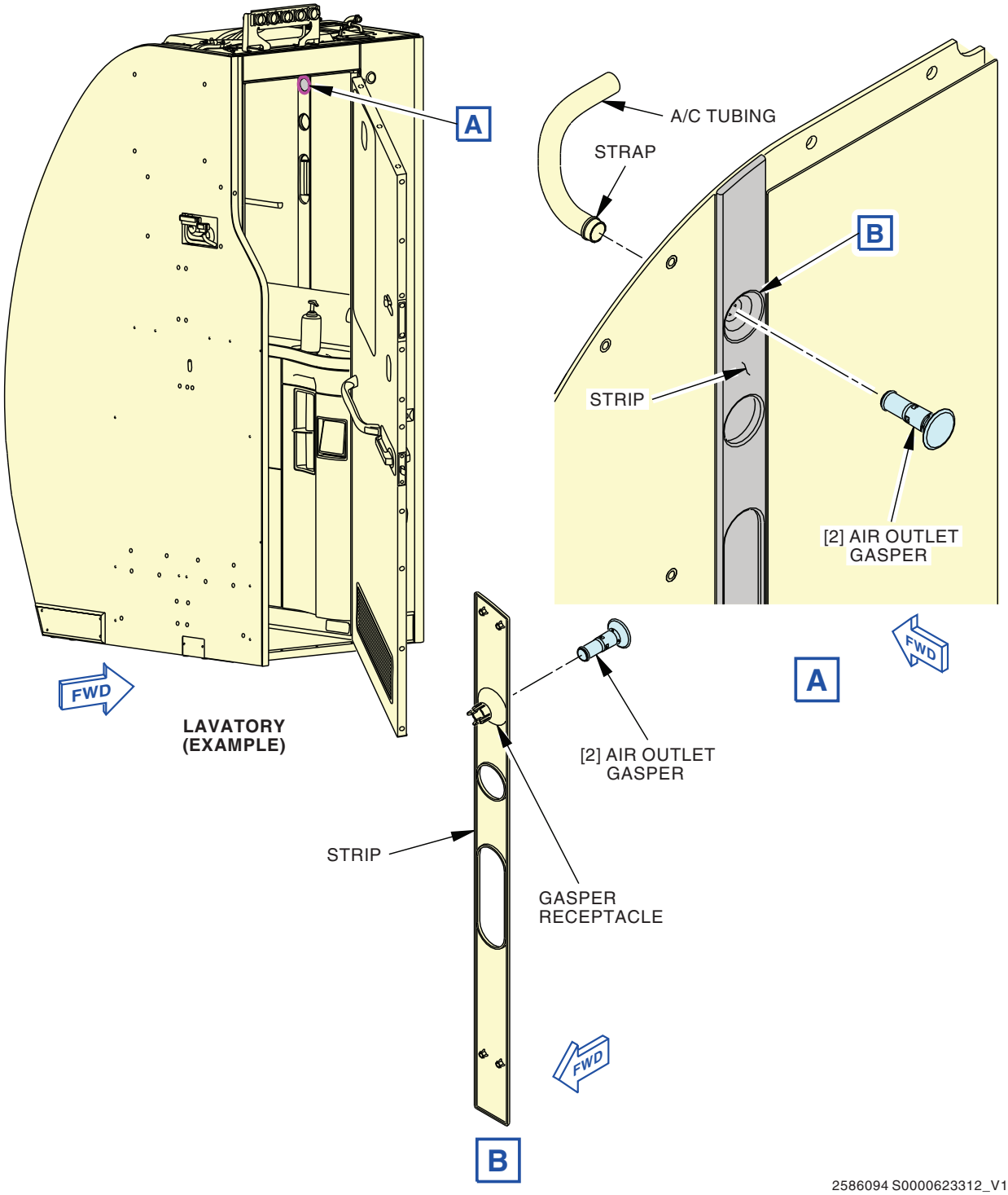
(b) Remove the A/C tubing from the air outlet gasper [2].

(c) Disengage the air outlet gasper [2] from the gasper receptacle clips on the strip.

(d) Remove the air outlet gasper [2] from the strip.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Gasper Air Outlet Installation
Figure 402/21-24-04-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-24-04-400-802

5. Gasper Air Outlet (LSU) Installation

(Figure 402)

A. References

Reference	Title
25-23-13-400-801	Attendant/Lavatory Service Unit Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Air outlet gasper	25-40-11-18E-195	PTF ALL

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Gasper Air Outlet Installation

SUBTASK 21-24-04-420-003

(1) Install the air outlet gasper [2] to the strip as follows (Figure 402):

(a) Put the air outlet gasper [2] into the gasper receptacle on the strip.

(b) Engage the air outlet gasper [2] with the gasper receptacle clips.

(c) Position the strap on the A/C tubing.

(d) Connect the A/C tubing to the air outlet gasper [2].

(e) Tighten the A/C tubing strap.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-24-04-410-001

(1) Install the applicable (LSU) gasper panel (TASK 25-23-13-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RECIRCULATION SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Recirculation System - Deactivation

(2) Recirculation System - Activation

TASK 21-25-00-800-801

2. Recirculation System - Deactivation

(Figure 201)

A. General

(1) This task will deactivate this component in the recirculation system:

(a) Recirculation Fan.

B. References

Reference	Title
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
821	Forward Cargo Door

E. Recirculation System Deactivation

SUBTASK 21-25-00-010-001

(1) Open this door:

Number	Name/Location
821	Forward Cargo Door

SUBTASK 21-25-00-010-002

(2) Remove the aft center bulkhead liner in the forward cargo compartment, do this task: Forward Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-11-02-000-801.

SUBTASK 21-25-00-860-001

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR
E	9	C01156	A/C RECIRC FAN LEFT CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	2	C01023	A/C RECIRC FAN RIGHT CONT.
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

F. Tryout

NOTE: This tryout is to make sure that the recirculation system is in a zero energy state.

SUBTASK 21-25-00-860-003

(1) Make sure that the CAB/UTIL switch, on the P5-13 electrical meters, battery and galley power panel, is set at the ON position.

SUBTASK 21-25-00-860-004

(2) Set these switches, on the P5-10 Air Conditioning Panel, to the OFF position:

(a) L PACK

(b) R PACK

(c) L RECIRC FAN

(d) R RECIRC FAN.

SUBTASK 21-25-00-860-005

(3) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-25-00-860-006

(4) Make sure that the BLEED 1 switch, on the P5-10 air conditioning panel, is set to the ON position.

SUBTASK 21-25-00-860-007

(5) Set the L and R RECIRC FAN switches to the AUTO position.

SUBTASK 21-25-00-860-008

(6) Set the L PACK switch and R PACK switch to the positions shown in Table 201 and make sure that the recirculation fans are off.

NOTE: Put your hand on the fan to feel for vibration. If you do not feel vibration, the fan is off.

Table 201/21-25-00-993-801

L PACK Switch Position	R PACK Switch Position	L Fan Condition	R Fan Condition
OFF	OFF	OFF	OFF
OFF	AUTO	OFF	OFF
OFF	HIGH	OFF	OFF
AUTO	OFF	OFF	OFF
AUTO	AUTO	OFF	OFF
AUTO	HIGH	OFF	OFF
HIGH	OFF	OFF	OFF
HIGH	AUTO	OFF	OFF
HIGH	HIGH	OFF	OFF

SUBTASK 21-25-00-860-009

(7) Set the L and R PACK switches to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-25-00-860-010

(8) Make sure that the L RECIRC FAN and R RECIRC FAN switches are in the AUTO position.

SUBTASK 21-25-00-860-011

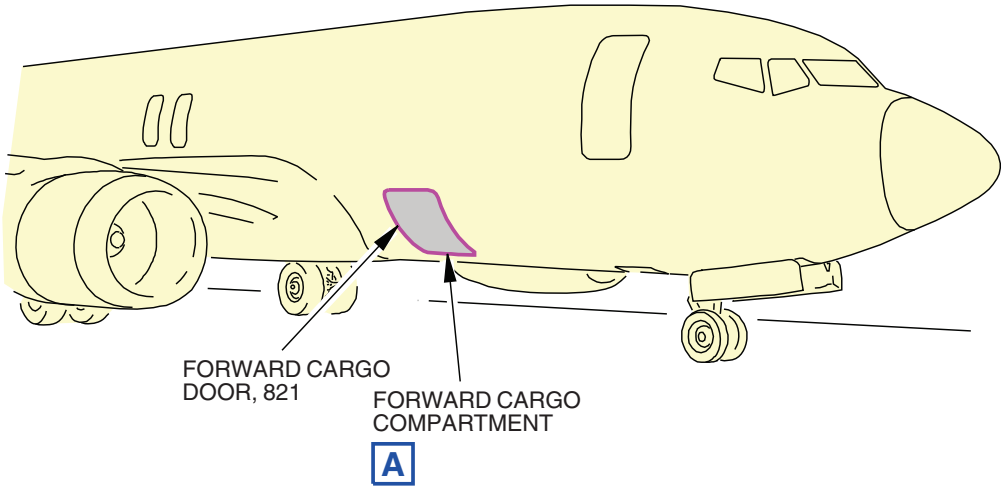
(9) Make sure that the BLEED 1 switch on the P5-10 air conditioning panel is set to the OFF position.

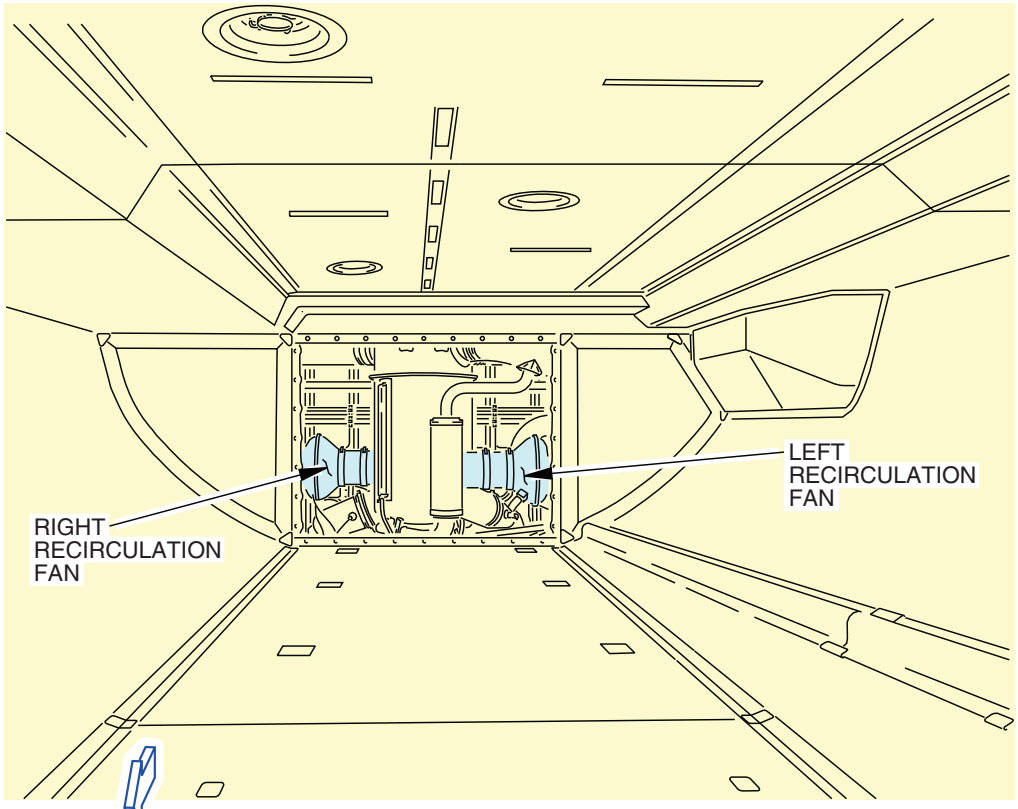
SUBTASK 21-25-00-860-012

(10) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Recirculation System
Figure 201/21-25-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-25-00-800-802

3. Recirculation System - Activation

(Figure 201)

A. General

(1) This task will activate this component in the recirculation system:

(a) Recirculation Fan.

B. References

Reference	Title
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
821	Forward Cargo Door

E. Recirculation System Activation

SUBTASK 21-25-00-860-014

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR
E	9	C01156	A/C RECIRC FAN LEFT CONT

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	2	C01023	A/C RECIRC FAN RIGHT CONT.
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

SUBTASK 21-25-00-410-001

(2) Install the aft center bulkhead liner in the forward cargo compartment, do this task: Forward Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-11-02-400-801.

SUBTASK 21-25-00-410-002

(3) Close this door:

Number	Name/Location
821	Forward Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RECIRCULATION SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure does an operational test of the recirculation system.

TASK 21-25-00-710-801

2. Recirculation System - Operational Test

(Figure 501)

A. References

Reference	Title
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Location Zones

Zone	Area
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

C. Prepare for the Operational Test

SUBTASK 21-25-00-860-016

(1) Make sure that the CAB/UTIL switch, on the P5-13 electrical meters, battery and galley power panel, is set at the ON position.

SUBTASK 21-25-00-860-017

(2) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR
E	9	C01156	A/C RECIRC FAN LEFT CONT

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
E	2	C01023	A/C RECIRC FAN RIGHT CONT.
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

SUBTASK 21-25-00-010-003

(3) Open the forward cargo door.

SUBTASK 21-25-00-010-004

(4) Remove the aft center bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

D. Recirculation System Operational Test

SUBTASK 21-25-00-860-018

(1) Set these switches, on the P5-10 air conditioning panel, to the OFF position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) L PACK

(b) R PACK

(c) L RECIRC FAN

(d) R RECIRC FAN.

SUBTASK 21-25-00-210-001

(2) Put your hand on the fan to feel for vibration.

(a) Make sure that the recirculation fans are off.

NOTE: If you do not feel vibration, the fan is off.

SUBTASK 21-25-00-910-001

(3) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-25-00-860-019

(4) Make sure that the BLEED 1 switch, on the P5-10 air conditioning panel, is set to the ON position.

SUBTASK 21-25-00-210-002

(5) Set the L and R RECIRC FAN switches to the AUTO position.

(a) Set the L PACK switch and the R PACK switch to the positions shown in Table 501 and make sure that the applicable fan condition occurs.

1) Put your hand on the fan to feel for vibration.

a) Make sure that the recirculation fans are off.

NOTE: If you do not feel vibration, the fan is off.

Table 501/21-25-00-993-803

L PACK Switch Position	R PACK Switch Position	L Fan Condition	R Fan Condition
OFF	OFF	ON	ON
OFF	AUTO	ON	ON
OFF	HIGH	ON	ON
AUTO	OFF	ON	ON
AUTO	AUTO	ON	ON
AUTO	HIGH	ON	ON
HIGH	OFF	ON	ON
HIGH	AUTO	ON	ON
HIGH	HIGH	OFF	ON

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-25-00-860-020

(1) Set the L and R PACK switches to the OFF position.

SUBTASK 21-25-00-860-021

(2) Set the L and R RECIRC FAN switches to the AUTO position.

SUBTASK 21-25-00-410-003

(3) Install the aft center bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-25-00-410-004

(4) Close the forward cargo door.

SUBTASK 21-25-00-860-022

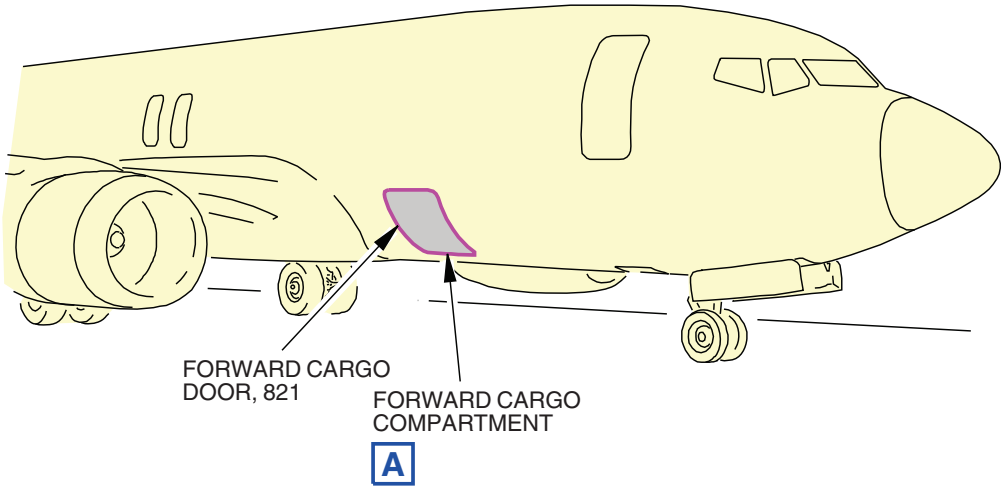
(5) Make sure that the BLEED 1 switch, on the P5-10 air conditioning panel, is set to the OFF position.

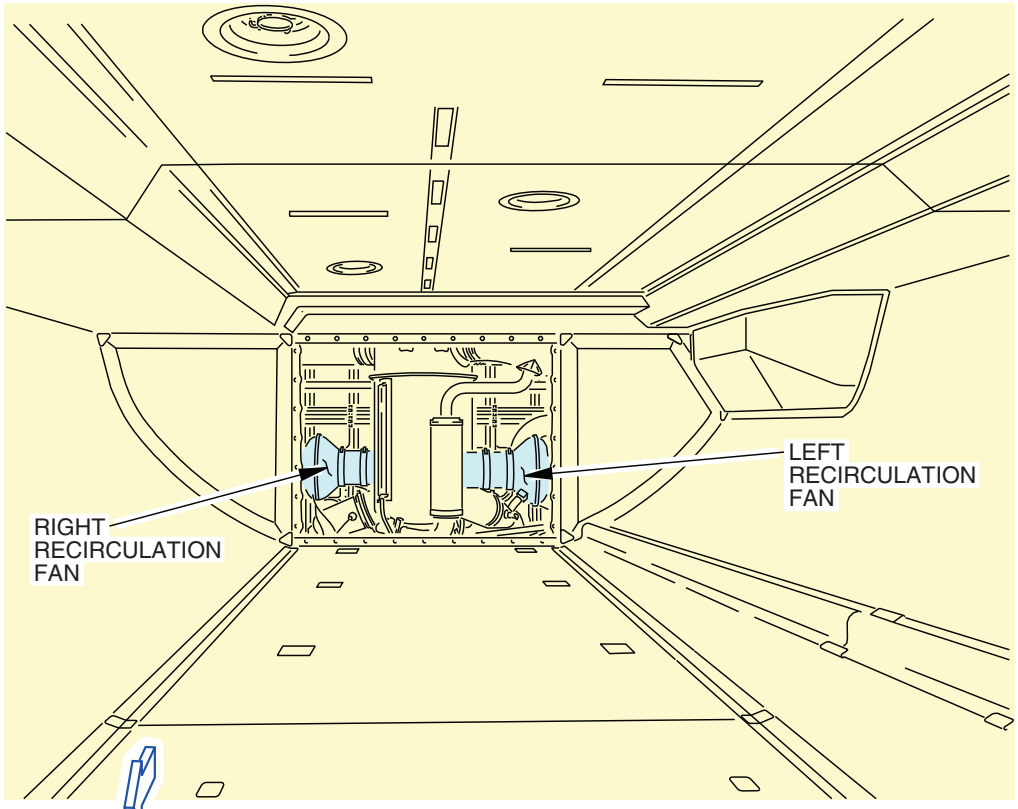
SUBTASK 21-25-00-910-002

(6) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2411297 S00061527531_V3

Recirculation System Test
Figure 501/21-25-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RECIRCULATION AIR FILTER - REMOVAL/INSTALLATION

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has these tasks:

(1) A removal of the recirculation air filter

(2) An installation of the recirculation air filter.

C. There are two recirculation air filters installed behind the end wall lining at the aft end of the forward cargo compartment. The air filters are in filter housings on the left and the right side of the mix manifold bay.

TASK 21-25-01-000-801

2. Recirculation Air Filter - Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented
STD-6439	Coveralls - Hooded

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B50280	Cleaner - Disinfectant - Zip-Chem Products Calla 1452	BAC5150
B50364	Cleaner - Disinfectant - Matrix #3	BAC5150
G00624	Bag - Plastic, General Purpose	
G50140	Gloves - Protective, Latex or Nitrile	

D. Location Zones

Zone	Area
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

E. Prepare for the Removal

SUBTASK 21-25-01-860-001

(1) Put the L RECIRC FAN and the R RECIRC FAN switches, on the P5-10 panel, to the OFF position and attach a DO NOT OPERATE tag, STD-858.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-25-01-865-001

(2) To prevent accidental recirculation fan operation during the removal of the recirculation air filters, open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR
E	9	C01156	A/C RECIRC FAN LEFT CONT

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	2	C01023	A/C RECIRC FAN RIGHT CONT.
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

SUBTASK 21-25-01-010-001

(3) Remove the left and right aft bulkhead liners in the forward cargo compartment (TASK 50-11-02-000-801).

SUBTASK 21-25-01-941-001

WARNING PUT ON THE PERSONAL PROTECTIVE EQUIPMENT BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES AND INJURIES TO PERSONS.

(4) Before you touch the recirculation air filter [3], put on the personal protective equipment listed below:

(a) hooded coveralls, STD-6439

(b) safety goggles, STD-6437

(c) particulate respirator, STD-6435

(d) protective gloves, G50140.

SUBTASK 21-25-01-500-001

(5) The plastic bag, G00624, must be larger than the filter dimensions of 20 in. (51 cm) in length, 20 in. (51 cm) in width, 5.5 in. (14 cm) in height and tape suitable for sealing the bag closed.

F. Recirculation Air Filter Removal

SUBTASK 21-25-01-020-001

(1) Remove the left recirculation air filter [3].

(a) Remove the two screws [6] from the retainer [7] on the filter housing [8].

(b) Remove the retainer [7] from the filter housing [8].

WARNING DO NOT LET THE FILTER TOUCH YOUR SKIN. DO NOT SHAKE OR HIT THE FILTER. DO NOT LET THE FILTER FALL. DO NOT USE COMPRESSED AIR TO CLEAN THE FILTER OR FILTER HOUSING. THIS CAN CAUSE THE INFECTIOUS MATERIAL TO BECOME AIRBORNE. DISCARD THE FILTER IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET THE RECIRCULATION FILTER TOUCH THE WIRE BUNDLE THAT IS IMMEDIATELY FORWARD OF THE FILTER. THE FORWARD AND AFT LOWER OUTBOARD CORNERS OF THE FILTER HAVE SHARP EDGES. THE SHARP EDGES CAN CUT THE INSULATION ON THE WIRE BUNDLE.

(c) If possible, slide the plastic bag, G00624, over the filter when removing the recirculation air filter [3] from the filter housing [8] to minimize contamination.

(d) Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(e) Discard the bag with the filter. Refer to the airline, local health, safety and regulatory procedures for the disposal of material.

SUBTASK 21-25-01-020-002

(2) Remove the right recirculation air filter [3].

(a) Release the quarter-turn fastener [1] which attach the access panel [2] to the filter housing [4].

(b) Remove the access panel [2].

WARNING DO NOT LET THE FILTER TOUCH YOUR SKIN. DO NOT SHAKE OR HIT THE FILTER. DO NOT LET THE FILTER FALL. DO NOT USE COMPRESSED AIR TO CLEAN THE FILTER OR FILTER HOUSING. THIS CAN CAUSE THE INFECTIOUS MATERIAL TO BECOME AIRBORNE. DISCARD THE FILTER IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(c) If possible, slide the plastic bag, G00624, over the filter when removing the recirculation air filter [3] from the filter housing [4] to minimize contamination.

(d) Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(e) Discard the bag with the filter. Refer to the airline, local health, safety and regulatory procedures for disposal of material.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Post-Filter Removal Cleanup

SUBTASK 21-25-01-670-003

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(1) Disinfect hard non-porous surfaces near the filter with a solution of 70% alcohol, B00130, Calla 1452 cleaner, B50280, or Matrix #3 cleaner, B50364.

NOTE: Follow the disinfectant manufacturer's instruction and refer to the EPA List N for applicable contact/dwell time.

NOTE: Refer to the EPA List N or local health authority guidance for required contact/dwell time for Isopropyl Alcohol to be used as a disinfectant.

SUBTASK 21-25-01-670-002

(2) Disinfect all tools used for filter replacement. Refer to the airline procedures and polices for the disinfecting of tools.

SUBTASK 21-25-01-020-003

WARNING DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(3) Remove the personal protective equipment and put them in a plastic bag, G00624.

(a) Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(b) Discard the bag with the personal protective equipment. Refer to the airline, local health, safety and regulatory procedures for disposal of material.

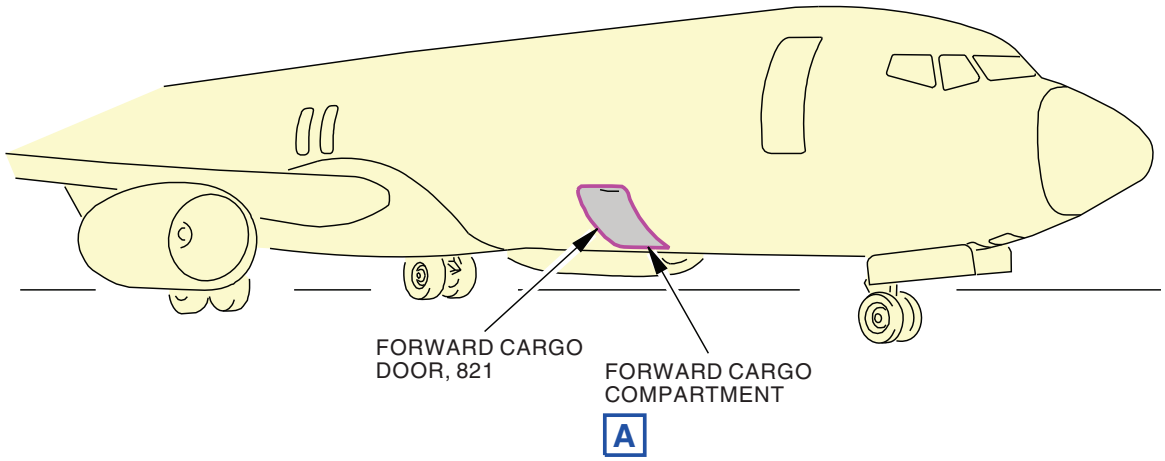
SUBTASK 21-25-01-110-001

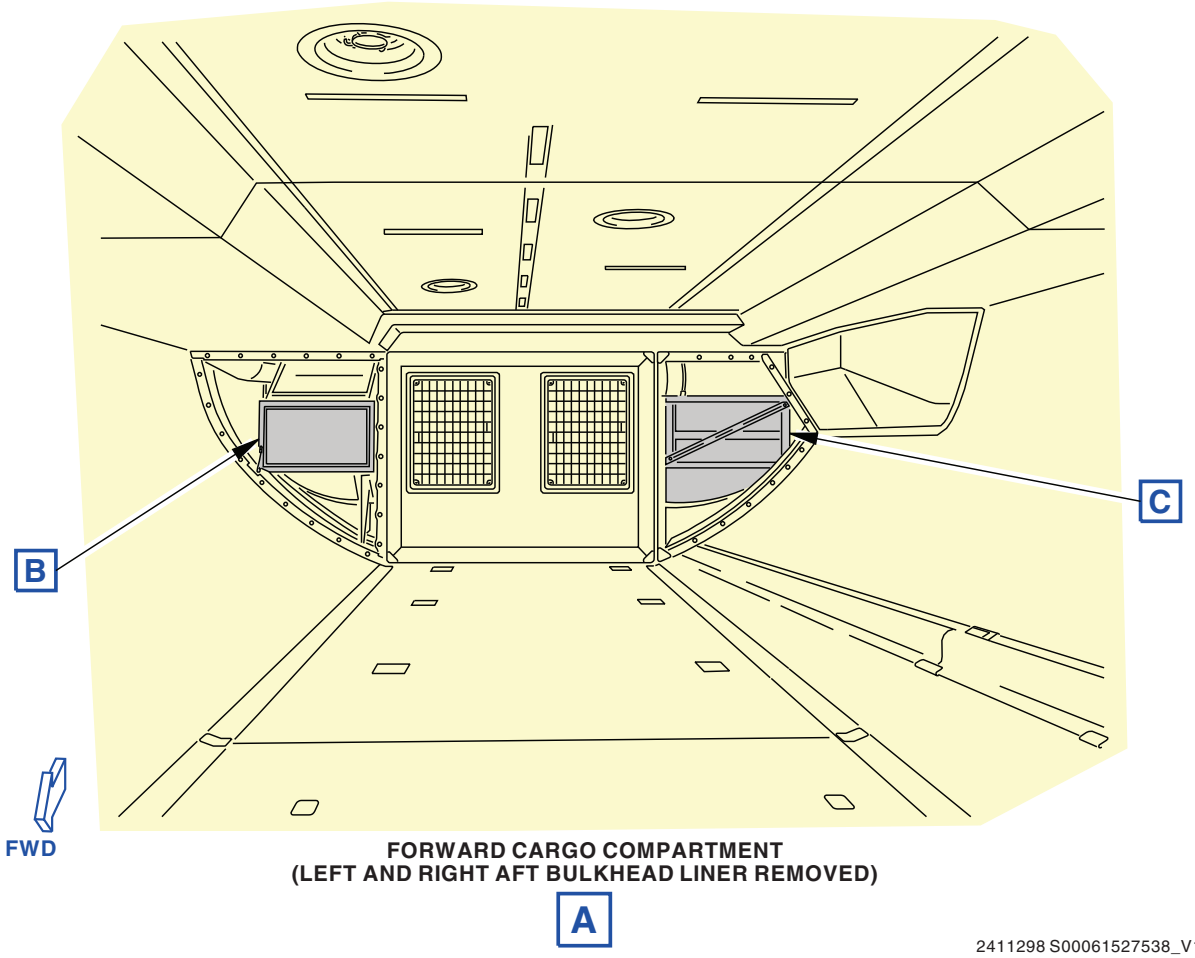
WARNING CLEAN YOUR HANDS WITH SOAP AND FLUSH WITH HOT WATER. DIRTY HANDS WITH CONTAMINATION CAN CAUSE DISEASE AND ILLNESSES TO PERSONS.

(4) Clean your hands with soap and running hot water.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





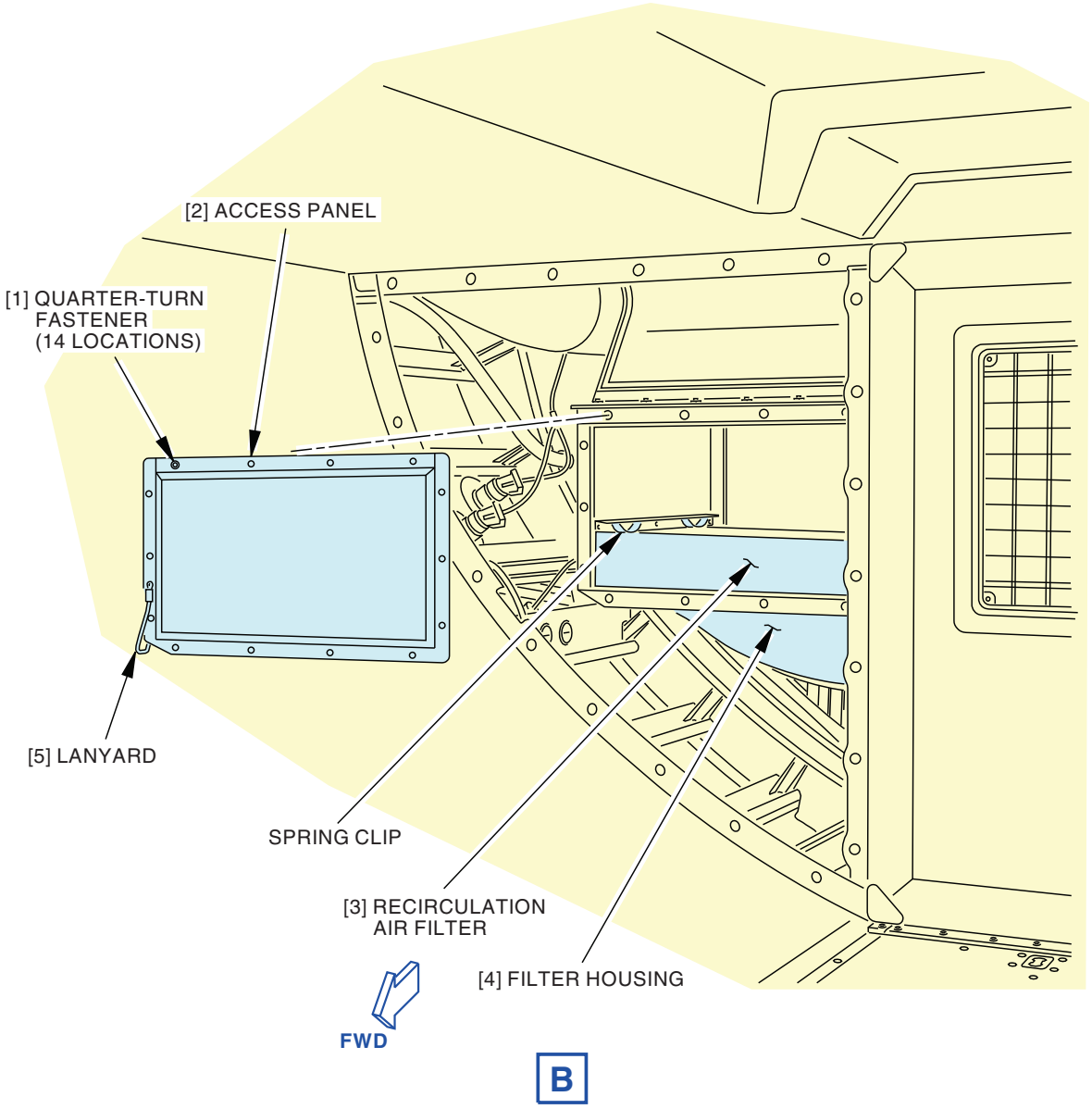
FORWARD CARGO COMPARTMENT
(LEFT AND RIGHT AFT BULKHEAD LINER REMOVED)

A

2411298 S00061527538_V1

Recirculation Air Filter Installation
Figure 401/21-25-01-990-801 (Sheet 1 of 3)

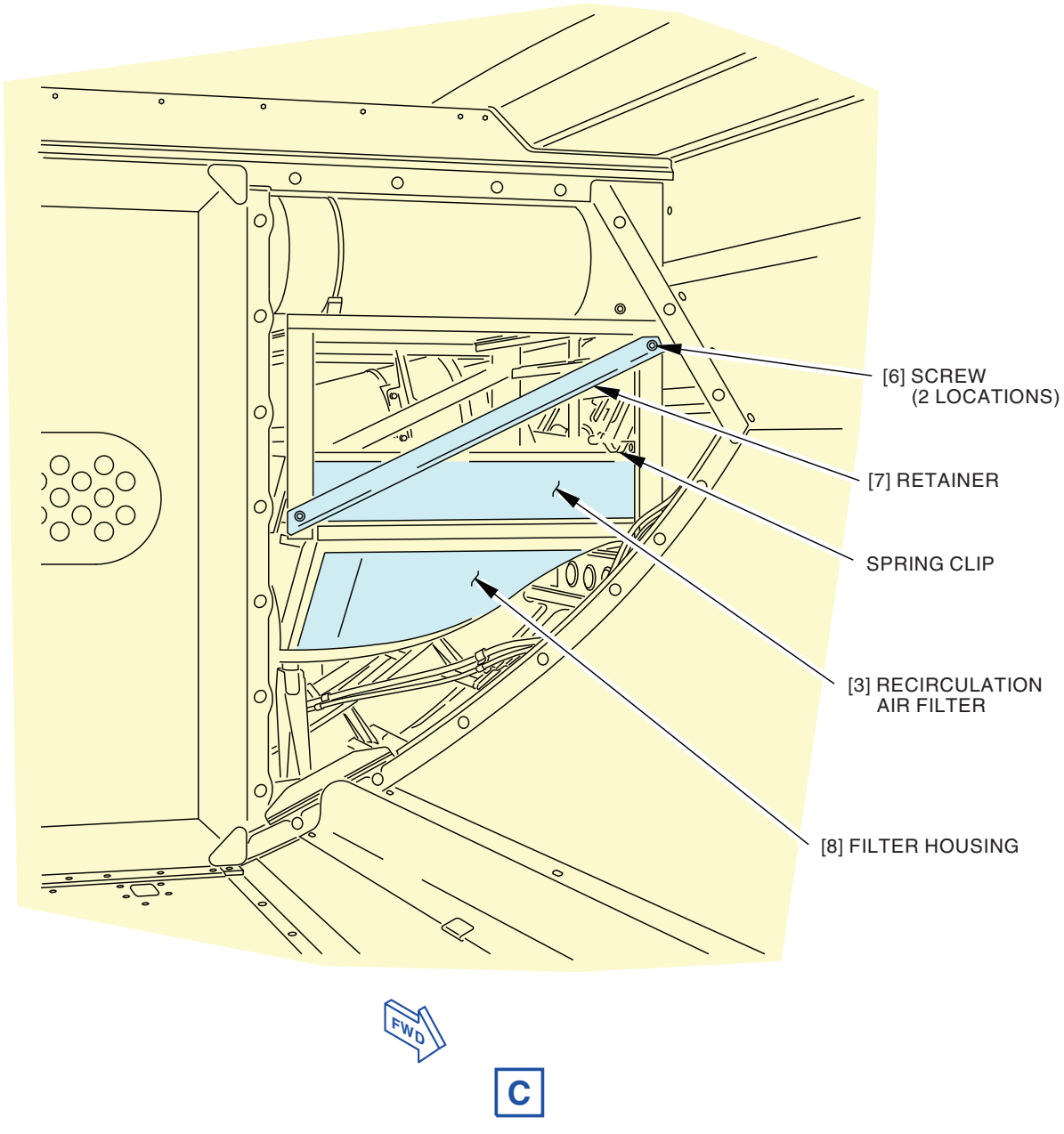
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411299 S00061527539_V1

Recirculation Air Filter Installation
Figure 401/21-25-01-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Recirculation Air Filter Installation
Figure 401/21-25-01-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-25-01-400-801

3. Recirculation Air Filter - Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Recirculation air filter	21-25-01-01-465	PTF ALL

C. Location Zones

Zone	Area
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

D. Recirculation Air Filter Installation

SUBTASK 21-25-01-420-001

(1) Install the left recirculation air filter [3].

(a) Make sure that there is no unwanted material or contamination in the filter housing [8] which could cause damage to the recirculation fan.

CAUTION

DO NOT LET THE RECIRCULATION FILTER TOUCH THE WIRE BUNDLE THAT IS IMMEDIATELY FORWARD OF THE FILTER. THE FORWARD AND AFT LOWER OUTBOARD CORNERS OF THE FILTER HAVE SHARP EDGES. THE SHARP EDGES CAN CUT THE INSULATION ON THE WIRE BUNDLE.

(b) Put the new recirculation air filter [3] in the filter housing [8].

(c) Check the spring clips.

1) Make sure that the spring clips apply pressure to keep the recirculation air filter [3] in position.

2) Make sure that all spring clips are installed and not damaged.

(d) Hold the retainer [7] in its position on the filter housing [8].

(e) Install the two screws [6] for the retainer [7] on the filter housing [8].

SUBTASK 21-25-01-420-002

(2) Install the right recirculation air filter [3].

(a) Make sure that there is no unwanted material in the filter housing [4] which could cause damage to the recirculation air filter [3].

CAUTION

MAKE SURE THAT THE AFT SPRING CLIPS OF THE RECIRCULATION AIR FILTER ARE IN TIGHT CONTACT WITH THE FILTER RAIL. IF THE AFT SPRING CLIPS ARE NOT IN THE CORRECT POSITION, THEN THE FLOATING SPRING CLIPS CAN CAUSE DAMAGE TO THE RECIRCULATION FAN AND FILTER HOUSING DURING INSTALLATION.

(b) Put the new recirculation air filter [3] in the filter housing [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Check the spring clips.

1) Make sure that the spring clips apply pressure to keep the recirculation air filter [3] in position.

2) Make sure that all spring clips are installed and not damaged.

(d) Put the access panel [2] in its position and turn the quarter-turn fasteners [1] to attach the access panel [2] to the filter housing [4].

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-25-01-010-002

(1) Install the left and right aft bulkhead liners in the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 21-25-01-865-002

(2) To restore recirculation fan operation after installation of the recirculation filter, remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR
E	9	C01156	A/C RECIRC FAN LEFT CONT

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	2	C01023	A/C RECIRC FAN RIGHT CONT.
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

SUBTASK 21-25-01-860-002

(3) Put the L RECIRC FAN and R RECIRC FAN switches, on the P5–10 panel, to the AUTO position and remove the DO-NOT-OPERATE tag.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RECIRCULATION FAN - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the recirculation fan.

(2) An installation of the recirculation fan.

B. There are two recirculation fans installed behind the liner at the aft end of the forward cargo compartment.

TASK 21-25-02-000-801

2. Recirculation Fan Removal

(Figure 401)

A. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 21-25-02-860-001

(1) To remove the left recirculation fan [1], do this step:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR
E	9	C01156	A/C RECIRC FAN LEFT CONT

SUBTASK 21-25-02-860-002

(2) To remove the right recirculation fan [1], do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	2	C01023	A/C RECIRC FAN RIGHT CONT.
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

SUBTASK 21-25-02-010-001

(3) Remove the aft center bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

D. Recirculation Fan Removal

SUBTASK 21-25-02-020-001

(1) Disconnect the electrical connector [4] from the recirculation fan [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-25-02-020-002

(2) Remove the nut [10], washer [9], and washer [8].

SUBTASK 21-25-02-020-003

(3) Remove the bonding jumper [5] from the recirculation fan [1].

SUBTASK 21-25-02-020-004

(4) Remove the coupling [2] and sleeve [3].

SUBTASK 21-25-02-020-005

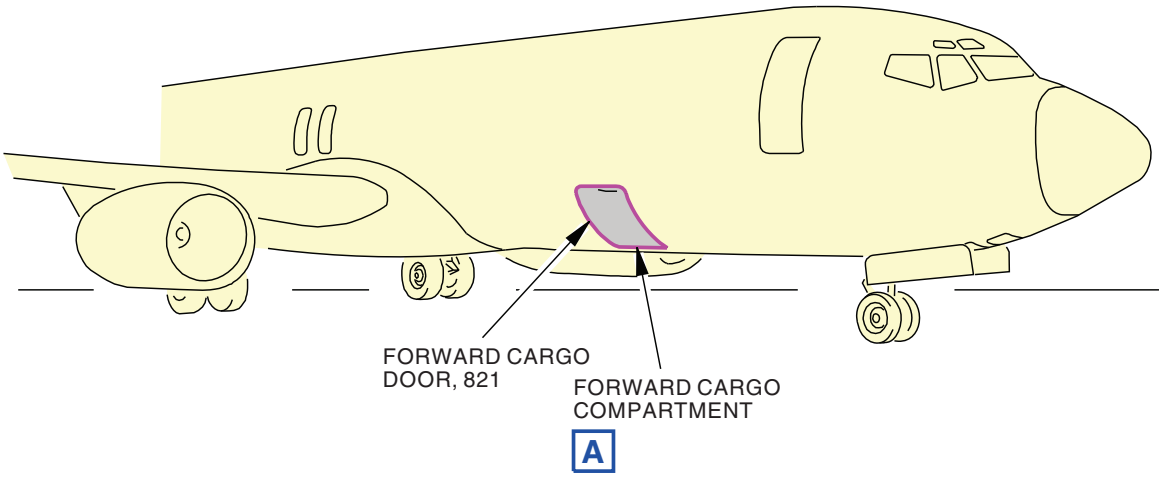
(5) Remove the coupling [6] and sleeve [7].

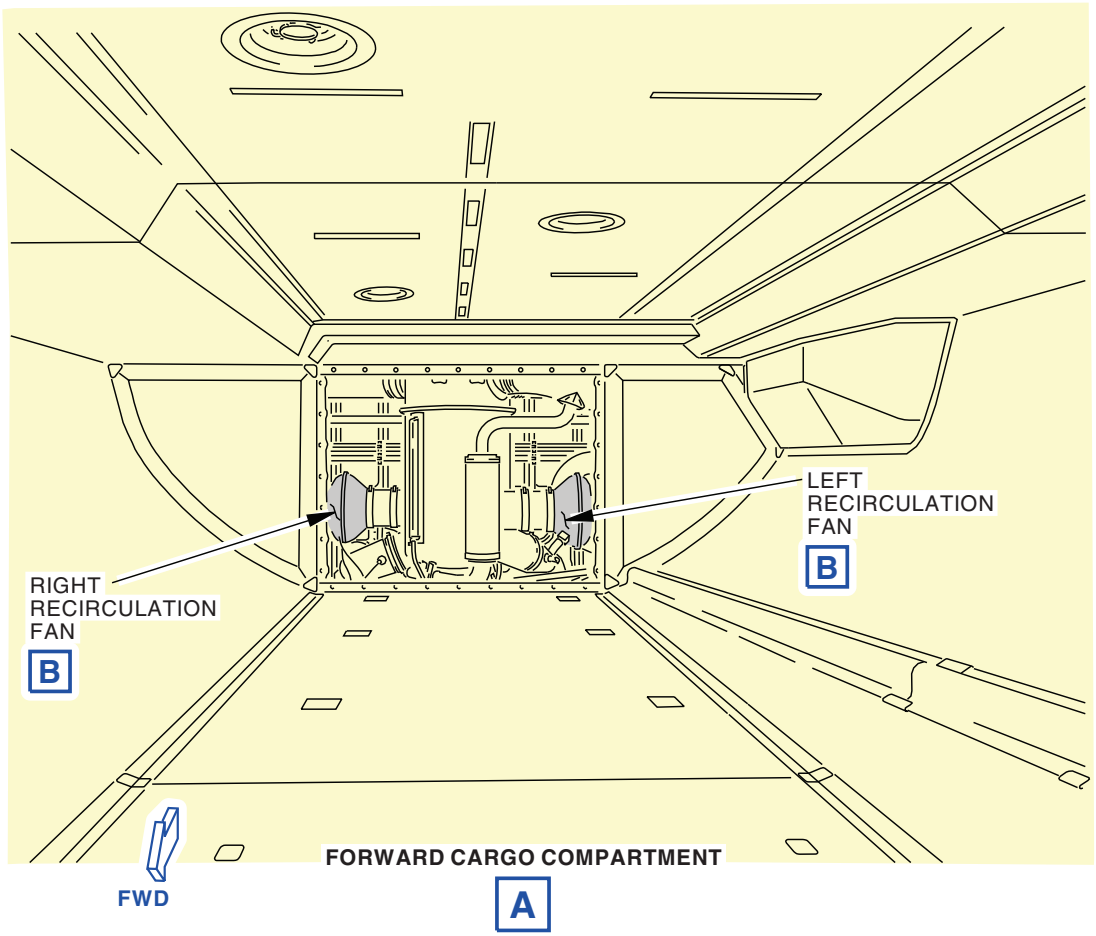
SUBTASK 21-25-02-020-006

(6) Remove the recirculation fan [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

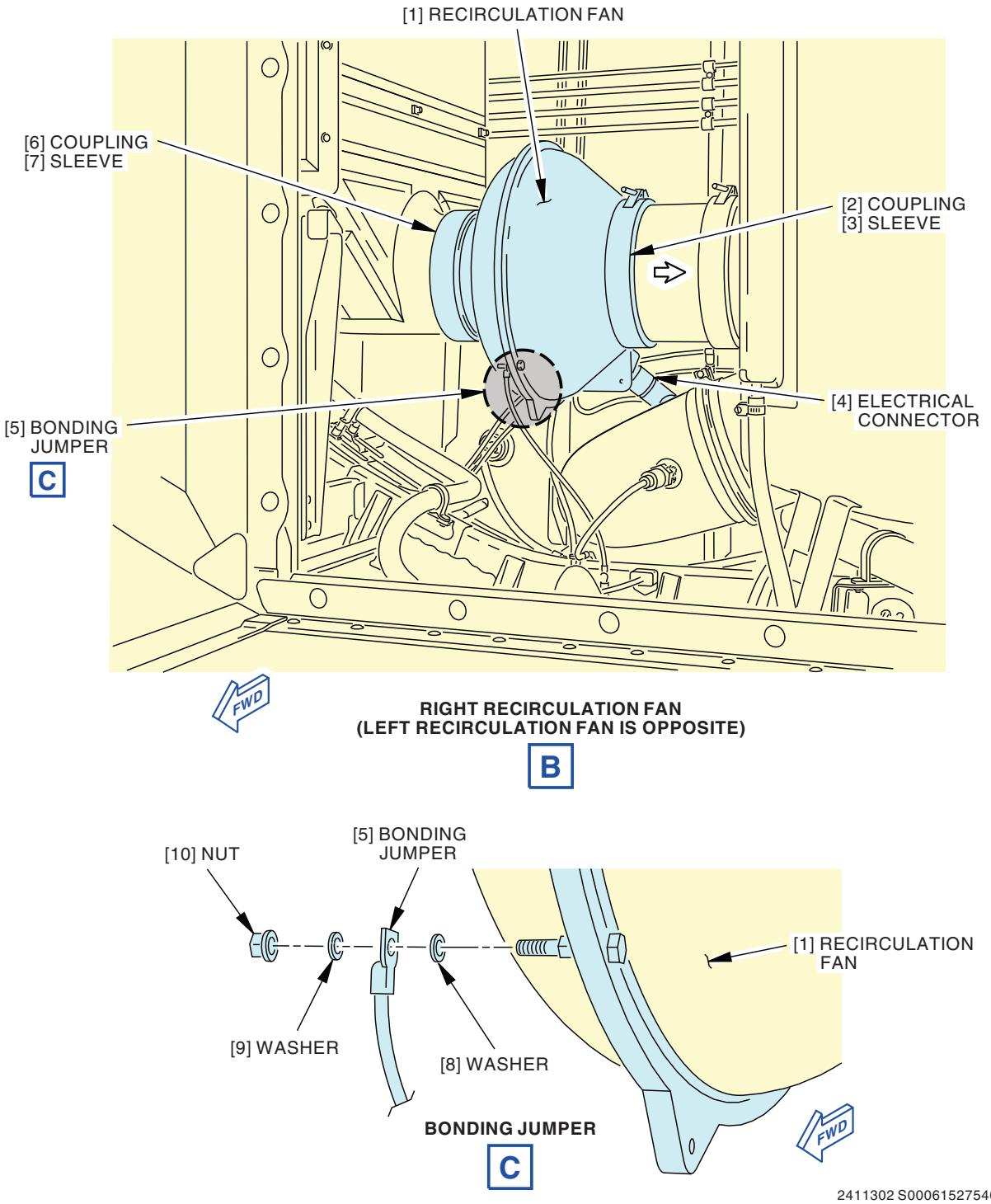




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Recirculation Fan Installation
Figure 401/21-25-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Recirculation Fan Installation
Figure 401/21-25-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-25-02-400-801

3. Recirculation Fan Installation

(Figure 401)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Recirculation fan	21-25-02-08-100	PTF ALL

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

E. Recirculation Fan Installation

SUBTASK 21-25-02-980-001

(1) Put the recirculation fan [1] in its position:

(a) Put the recirculation fan [1] in position so the flow arrow points inboard.

(b) Turn the recirculation fan [1] so you can install the electrical connector [4] and bonding jumper [5].

SUBTASK 21-25-02-420-001

(2) Install the sleeve [7] and coupling [6].

(a) Tighten the nut on the coupling [6] to 45 in-lb (5.1 N·m) - 55 in-lb (6.2 N·m).

SUBTASK 21-25-02-420-002

(3) Install the sleeve [3] and coupling [2].

(a) Tighten the nut on the coupling [2] to 45 in-lb (5.1 N·m) - 55 in-lb (6.2 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-25-02-420-003

(4) Do these steps to install the bonding jumper [5] on the recirculation fan [1]:

(a) Use the solvent cleaning method to clean these parts:

1) The bonding post

2) The terminal of the bonding jumper [5]

3) The washer [8]

4) The washer [9]

5) The nut [10].

(b) Put washer [8] and bonding jumper [5] in its position on the bonding post of the recirculation fan [1].

(c) Install the washer [9] and nut [10].

SUBTASK 21-25-02-760-001

(5) Measure the electrical bonding resistance between the bonding jumper [5] and recirculation fan [1] (SWPM 20-20-00).

(a) Use a non-intrinsically safe bonding meter, COM-614.

(b) Make sure that the electrical bonding resistance is 0.001 ohm (1 milliohm) or less.

SUBTASK 21-25-02-420-004

(6) Install the electrical connector [4] on the recirculation fan [1].

F. Recirculation Fan Installation Test

SUBTASK 21-25-02-860-003

(1) If it is necessary, do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-25-02-860-004

(2) Make sure that the CAB/UTIL switch, on the P5-13 electrical meters, battery and galley power panel, is set at the ON position.

SUBTASK 21-25-02-860-005

(3) If the left recirculation fan [1] was replaced, do this step:

(a) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR
E	9	C01156	A/C RECIRC FAN LEFT CONT

SUBTASK 21-25-02-860-006

(4) If the right recirculation fan [1] was replaced, do this step:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	2	C01023	A/C RECIRC FAN RIGHT CONT.
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

SUBTASK 21-25-02-860-007

(5) Make sure that the L PACK and R PACK switches, on the P5-10 air conditioning panel, are in the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-25-02-860-008

(6) Put the applicable RECIRC FAN switch, on the P5-10 air conditioning panel, to the AUTO position.

SUBTASK 21-25-02-710-001

(7) Make sure that the recirculation fan [1] operates.

(a) Put your hand on the fan to feel for vibration.

NOTE: If you feel vibration, the fan is on.

SUBTASK 21-25-02-790-001

(8) Make sure that there are no leaks around the fan connections.

(a) If you find a leak, repair it.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-25-02-410-001

(1) Install the aft center bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RECIRCULATION FAN CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the recirculation fan check valve

(2) An installation of the recirculation fan check valve.

B. There are two recirculation fan check valves installed behind the liner at the aft end of the forward cargo compartment.

TASK 21-25-03-000-801

2. Recirculation Fan Check Valve Removal

(Figure 401)

A. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 21-25-03-860-001

(1) Put the L RECIRC FAN and the R RECIRC FAN switches, on the P5-10 air conditioning panel, to the OFF position.

SUBTASK 21-25-03-860-002

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

SUBTASK 21-25-03-860-003

(3) Make sure that the L PACK and R PACK switches, on the P5-10 air conditioning panel, are in the OFF position.

SUBTASK 21-25-03-010-001

(4) Open the forward cargo door.

SUBTASK 21-25-03-010-002

(5) Remove the aft center bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Recirculation Fan Check Valve Removal

SUBTASK 21-25-03-020-001

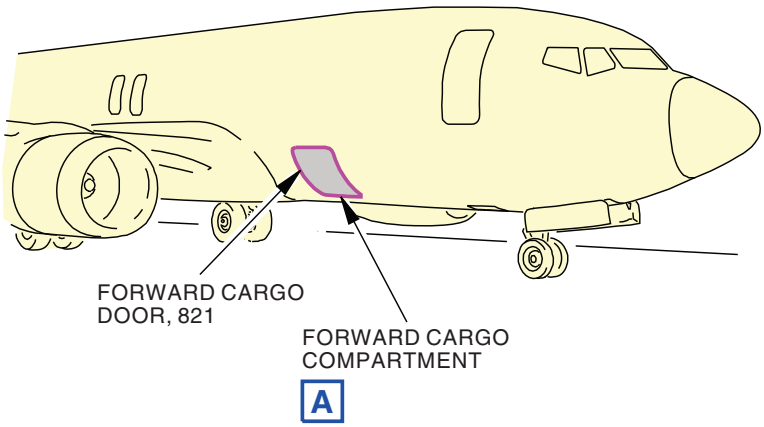
(1) Remove the coupling [2] and sleeve [3] at two locations.

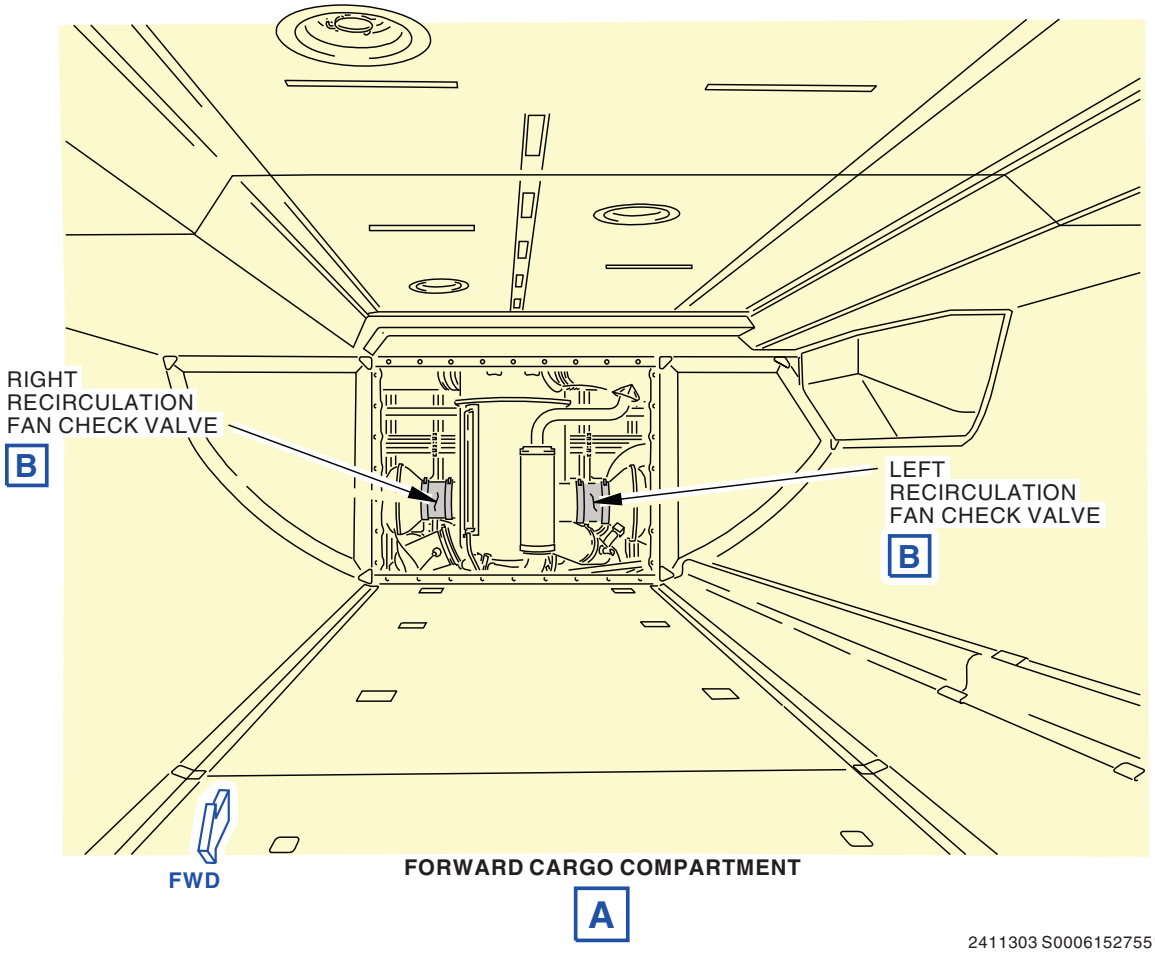
SUBTASK 21-25-03-020-002

(2) Remove the recirculation fan check valve [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

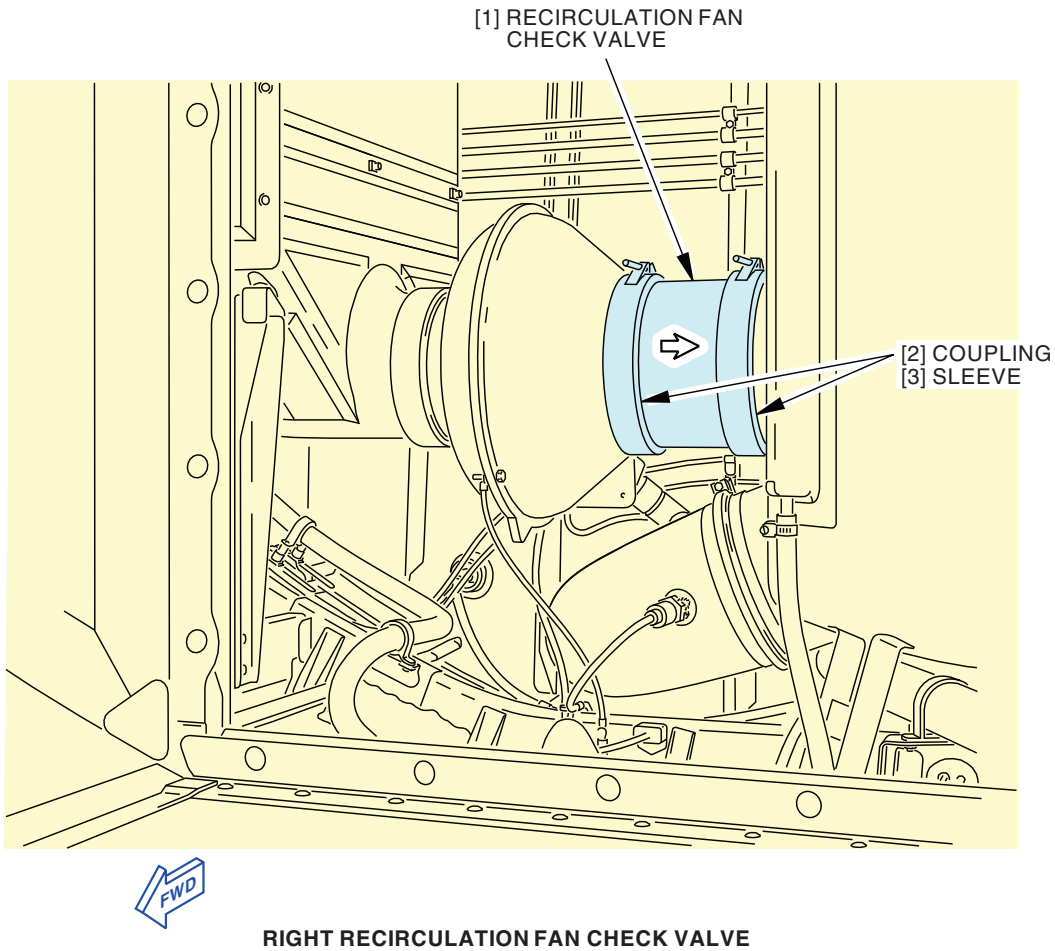




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Recirculation Fan Check Valve Installation
Figure 401/21-25-03-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIGHT RECIRCULATION FAN CHECK VALVE
(LEFT RECIRCULATION FAN CHECK VALVE IS OPPOSITE)

B

2411304 S00061527552_V1

Recirculation Fan Check Valve Installation
Figure 401/21-25-03-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-25-03-400-801

3. Recirculation Fan Check Valve Installation

(Figure 401)

A. References

Reference	Title
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right
212	Flight Compartment - Right

C. Recirculation Fan Check Valve Installation

SUBTASK 21-25-03-420-001

(1) Put the recirculation fan check valve [1] in its position.

(a) Make sure that the flapper valve shaft is in the vertical position.

NOTE: Installation with the flapper valve shaft in horizontal position reduces the ease of operation of the check valve.

(b) Make sure that the flow arrow points inboard.

SUBTASK 21-25-03-420-002

(2) Install the coupling [2] and sleeve [3] at two locations.

(a) Tighten the nut on the coupling [2] to 45 in-lb (5 N·m) - 55 in-lb (6 N·m).

D. Recirculation Fan Check Valve Leakage Test

SUBTASK 21-25-03-860-005

(1) Make sure that the CAB/UTIL switch, on the P5-13 electrical meters, battery and galley power panel, is set at the ON position.

SUBTASK 21-25-03-860-006

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	7	C00911	A/C RECIRC FAN LEFT CABIN AIR

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	4	C00884	AC RECIRC RIGHT FAN CABIN AIR

SUBTASK 21-25-03-860-007

(3) Make sure that the L PACK and R PACK switches, on the P5-10 air conditioning panel, are in the OFF position.

SUBTASK 21-25-03-860-008

(4) Put the applicable L or R RECIRC FAN switch, on the P5-10 air conditioning panel, to the AUTO position.

SUBTASK 21-25-03-790-001

(5) Make sure there are no leaks around the check valve connections.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If you find a leak, repair it.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-25-03-410-001

(1) Install the aft center bulkhead liner in the forward cargo compartment (TASK 50-11-02-400-801).

SUBTASK 21-25-03-010-003

(2) Close the forward cargo door.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RECIRCULATION FAN CHECK VALVE - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) An inspection of the recirculation fan check valve.

TASK 21-25-03-200-801

2. Recirculation Fan Check Valve Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has the steps to inspect the recirculation fan check valve.

B. References

Reference	Title
21-25-03-000-801	Recirculation Fan Check Valve Removal (P/B 401)
21-25-03-400-801	Recirculation Fan Check Valve Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-595	Micrometer
STD-625	Pen
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
125	Air Conditioning Distribution Bay - Left
126	Air Conditioning Distribution Bay - Right

E. Prepare for the Inspection

SUBTASK 21-25-03-020-003

(1) Do this task: Recirculation Fan Check Valve Removal, TASK 21-25-03-000-801.

F. Recirculation Fan Check Valve Inspection

SUBTASK 21-25-03-210-001

(1) Examine the recirculation fan check valve:

(a) Use your finger or a pen, STD-625, to push on each flapper to open them.

(b) Make sure that each flapper opens freely and moves smoothly to the stop tube.

(c) Make sure that each flapper closes smoothly and touches fully on the face of the check valve body.

(d) Make sure that there is no damage to the flappers:

1) Measure the thickness of the area on the flappers where the flappers contact the stop tube.

2) Make sure that the stop tube contact area on the flappers is a minimum of 0.020 in. (0.508 mm) thick.

a) Measure with a micrometer, STD-595, or vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097, or an equivalent tool.

3) Repair or replace check valves that do not meet the minimum thickness.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Make sure that there is no corrosion or lint on the flappers or the hinge pin.

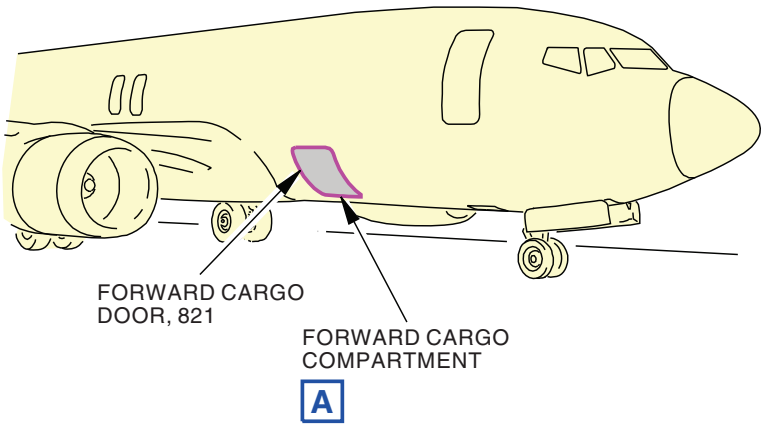
G. Put the Airplane Back to Its Usual Condition

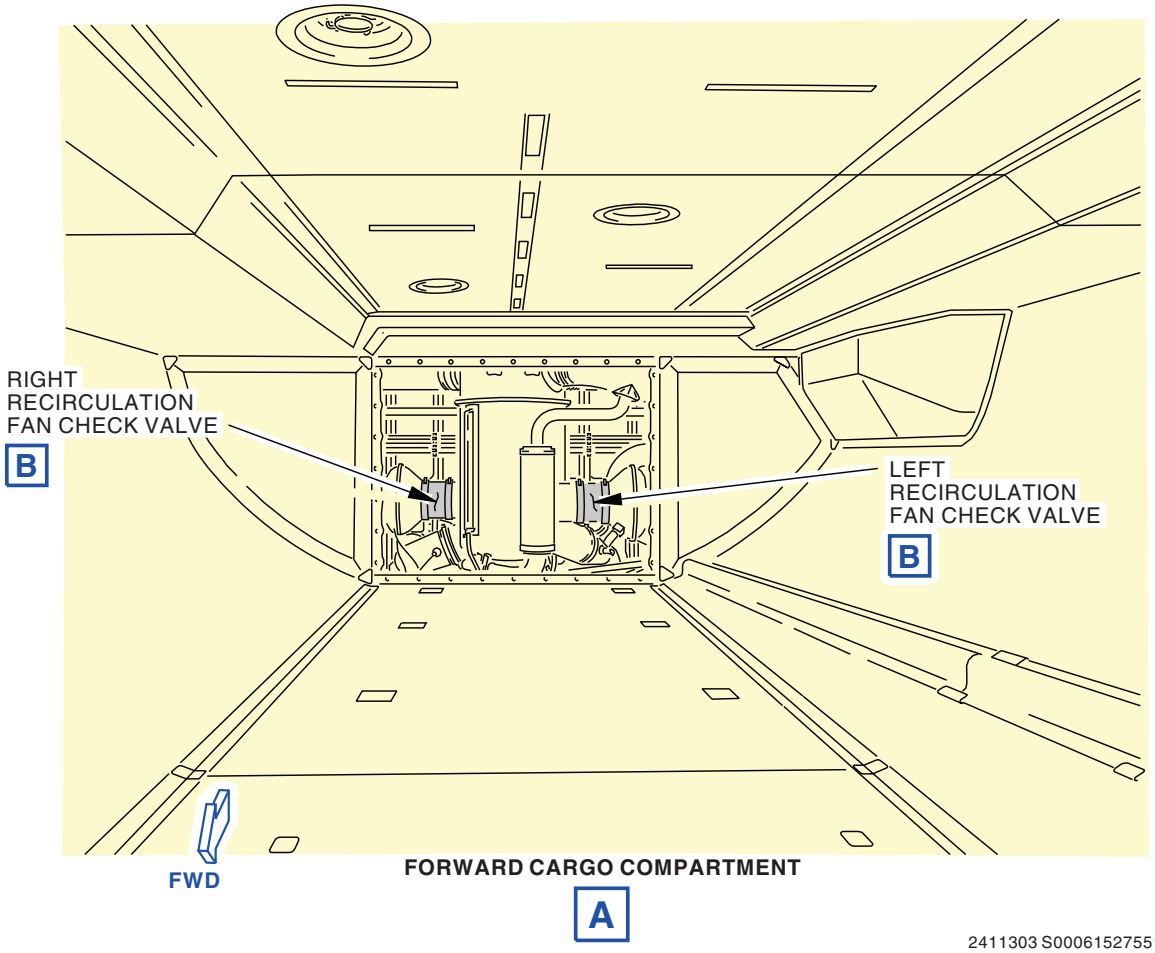
SUBTASK 21-25-03-420-003

(1) Do this task: Recirculation Fan Check Valve Installation, TASK 21-25-03-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

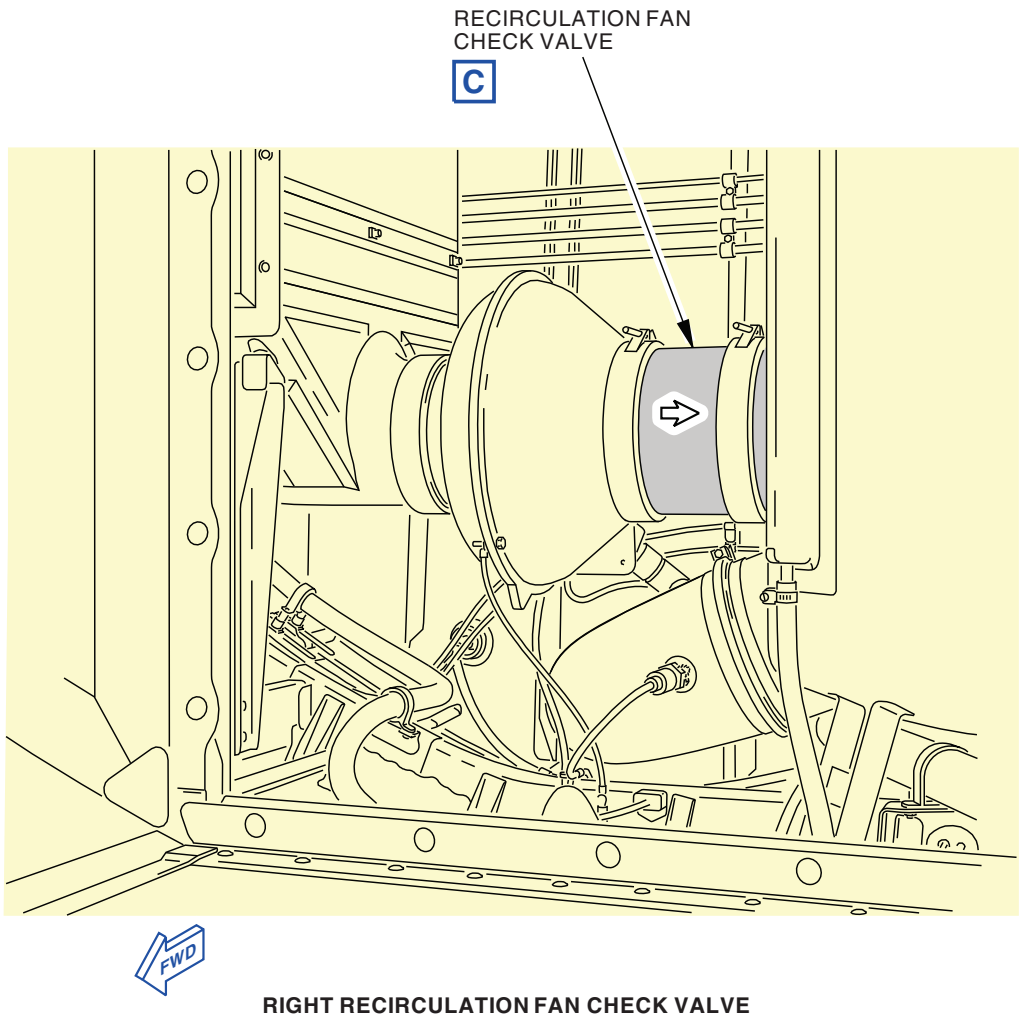




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Recirculation Fan Check Valve Inspection
Figure 601/21-25-03-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



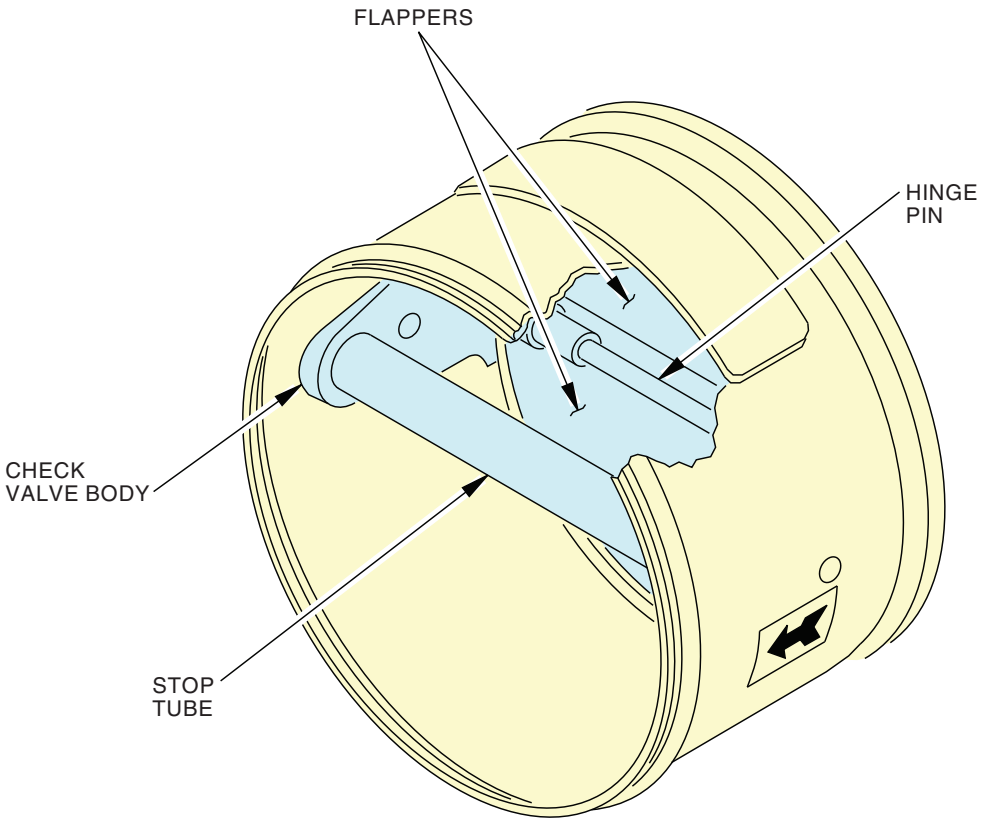
RIGHT RECIRCULATION FAN CHECK VALVE
(LEFT RECIRCULATION FAN CHECK VALVE IS OPPOSITE)

B

2536798 S0000602222_V2

Recirculation Fan Check Valve Inspection
Figure 601/21-25-03-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RECIRCULATION FAN CHECK VALVE

C

2411305 S00061527556_V1

Recirculation Fan Check Valve Inspection
Figure 601/21-25-03-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GALLEY VENTILATION MUFFLER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the forward and aft galley ventilation mufflers.

(2) An installation of the forward and aft galley ventilation mufflers.

B. There is a galley ventilation muffler installed above the lowered ceiling panels in the forward and aft entry ways.

TASK 21-26-01-000-801

2. Galley Ventilation Muffler Removal

(Figure 401, Figure 402)

A. Tools/Equipment

Reference	Description
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented
STD-6439	Coveralls - Hooded

B. Consumable Materials

Reference	Description	Specification
G50140	Gloves - Protective, Latex or Nitrile	

C. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

D. Prepare for the Removal

SUBTASK 21-26-01-940-001

WARNING PUT ON THE PERSONAL PROTECTIVE EQUIPMENT BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES AND INJURIES TO PERSONS.

(1) Before you touch the muffler/duct assembly, put on the personal protective equipment listed below:

(a) hooded coveralls, STD-6439

(b) safety goggles, STD-6437

(c) particulate respirator, STD-6435

(d) protective gloves, G50140

SUBTASK 21-26-01-010-001

(2) Lower the forward or aft entry light assembly to get access to the forward or aft galley ventilation muffler, as applicable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Galley Ventilation Muffler Removal

SUBTASK 21-26-01-010-002

(1) Do these steps to remove the forward galley ventilation muffler [1] from above the forward lowered ceiling:

(a) Loosen the clamp [3].

(b) Remove the strap [6].

(c) Move the flex hose [2] and flex hose [5] away from the ends of the forward galley ventilation muffler [1].

(d) Remove the clamps [4].

(e) Remove the forward galley ventilation muffler [1].

SUBTASK 21-26-01-020-001

(2) Do these steps to remove the aft galley ventilation muffler [23] from above the aft lowered ceiling:

(a) Remove the strap [22] that attaches the flex hose [21] to the aft galley ventilation muffler [23].

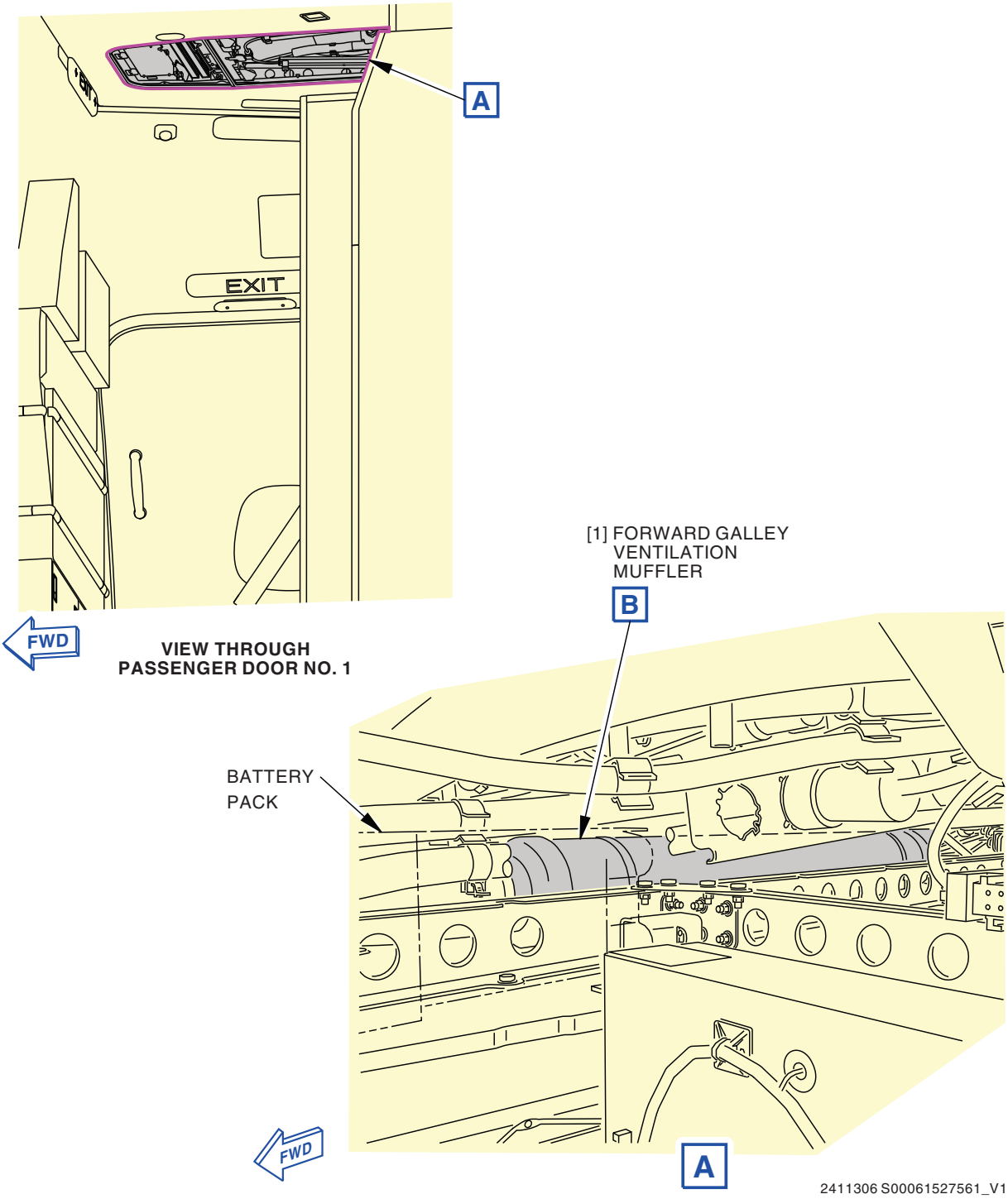
(b) Remove the strap [25] that attaches the flex sleeve [24] to the aft galley ventilation muffler [23].

(c) Remove the straps [22] that attach the aft galley ventilation muffler [23] to the support structure.

(d) Remove the aft galley ventilation muffler [23].

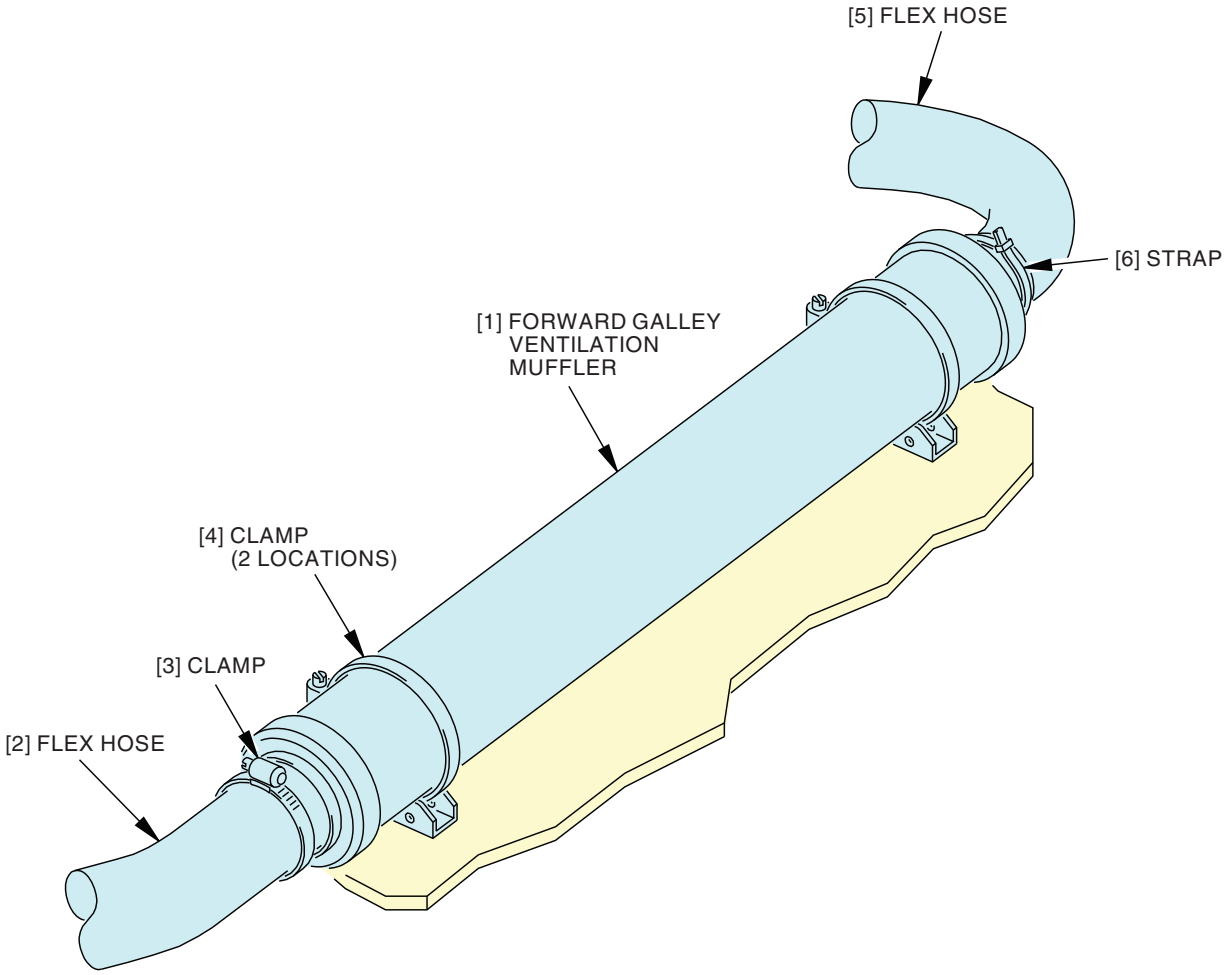
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Galley Ventilation Muffler Installation
Figure 401/21-26-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



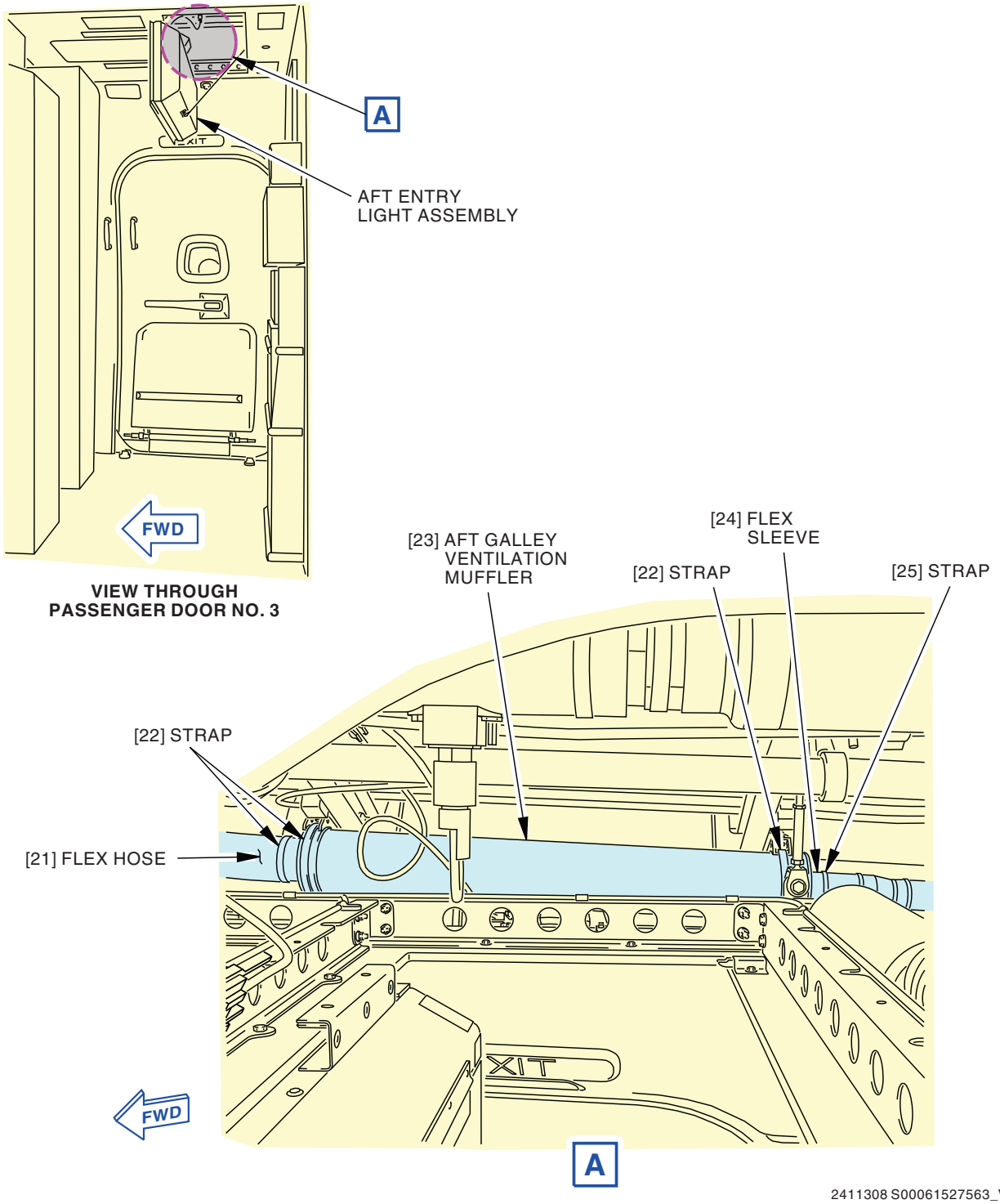
FORWARD GALLEY VENTILATION MUFFLER

B

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Forward Galley Ventilation Muffler Installation
Figure 401/21-26-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Galley Ventilation Muffler Installation
Figure 402/21-26-01-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-26-01-400-801

3. Galley Ventilation Muffler Installation

(Figure 401, Figure 402)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward galley ventilation muffler	21-26-01-10-010	PTF ALL
6	Strap	21-26-51-03-035	PTF ALL
22	Strap	21-26-51-03-035	PTF ALL
23	Aft galley ventilation muffler	21-26-51-03-145	PTF ALL
25	Strap	21-26-51-03-030	PTF ALL

B. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

C. Galley Ventilation Muffler Installation

SUBTASK 21-26-01-420-001

(1) Do these steps to install the forward galley ventilation muffler [1]:

(a) Put the forward galley ventilation muffler [1] in position on the lowered ceiling.

(b) Install the clamps [4].

(c) Move the flex hose [2] and flex hose [5] onto the ends of the forward galley ventilation muffler [1].

(d) Install the strap [6].

(e) Position the clamp [3] and tighten.

SUBTASK 21-26-01-420-002

(2) Do these steps to install the aft galley ventilation muffler [23]:

(a) Put the aft galley ventilation muffler [23] into its position on the support structure.

(b) Install the straps [22] to attach the aft galley ventilation muffler [23] to the support structure.

(c) Move the flex hose [21] and the flex sleeve [24] onto the ends of the aft galley ventilation muffler [23].

(d) Install the strap [25] to attach the flex sleeve [24] to the aft galley ventilation muffler [23].

(e) Install the strap [22] to attach the flex hose [21] to the aft galley ventilation muffler [23].

SUBTASK 21-26-01-410-001

(3) Close the forward or aft entry light assembly, as applicable.

SUBTASK 21-26-01-020-002

WARNING DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(4) Remove all personal protective equipment and put them in a plastic bag, G00624. Seal the bag closed with tape.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(a) Discard the bag with the personal protective equipment. Refer to the airline, local health, safety and regulatory procedures for disposal of material.

SUBTASK 21-26-01-110-001

WARNING CLEAN YOUR HANDS WITH SOAP AND FLUSH WITH HOT WATER. DIRTY HANDS WITH CONTAMINATION CAN CAUSE DISEASE AND ILLNESSES TO PERSONS.

(5) Clean your hands with soap and running hot water.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GALLEY CEILING VENTILATION GRILL - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the galley ceiling ventilation grill [1].

(2) A cleaning of the galley ceiling ventilation grill [1].

(3) An installation of the galley ceiling ventilation grill [1].

TASK 21-26-03-000-801

2. Galley Ceiling Ventilation Grill Removal

(Figure 201)

A. Location Zones

Zone	Area
200	Upper Half of Fuselage

B. Prepare for the Removal

SUBTASK 21-26-03-210-001

(1) Locate the galley ventilation grill [1] in the ceiling of either the forward-passenger door entry-way or the aft-passenger door entry-way.

C. Galley Ceiling Ventilation Grill Removal

SUBTASK 21-26-03-030-001

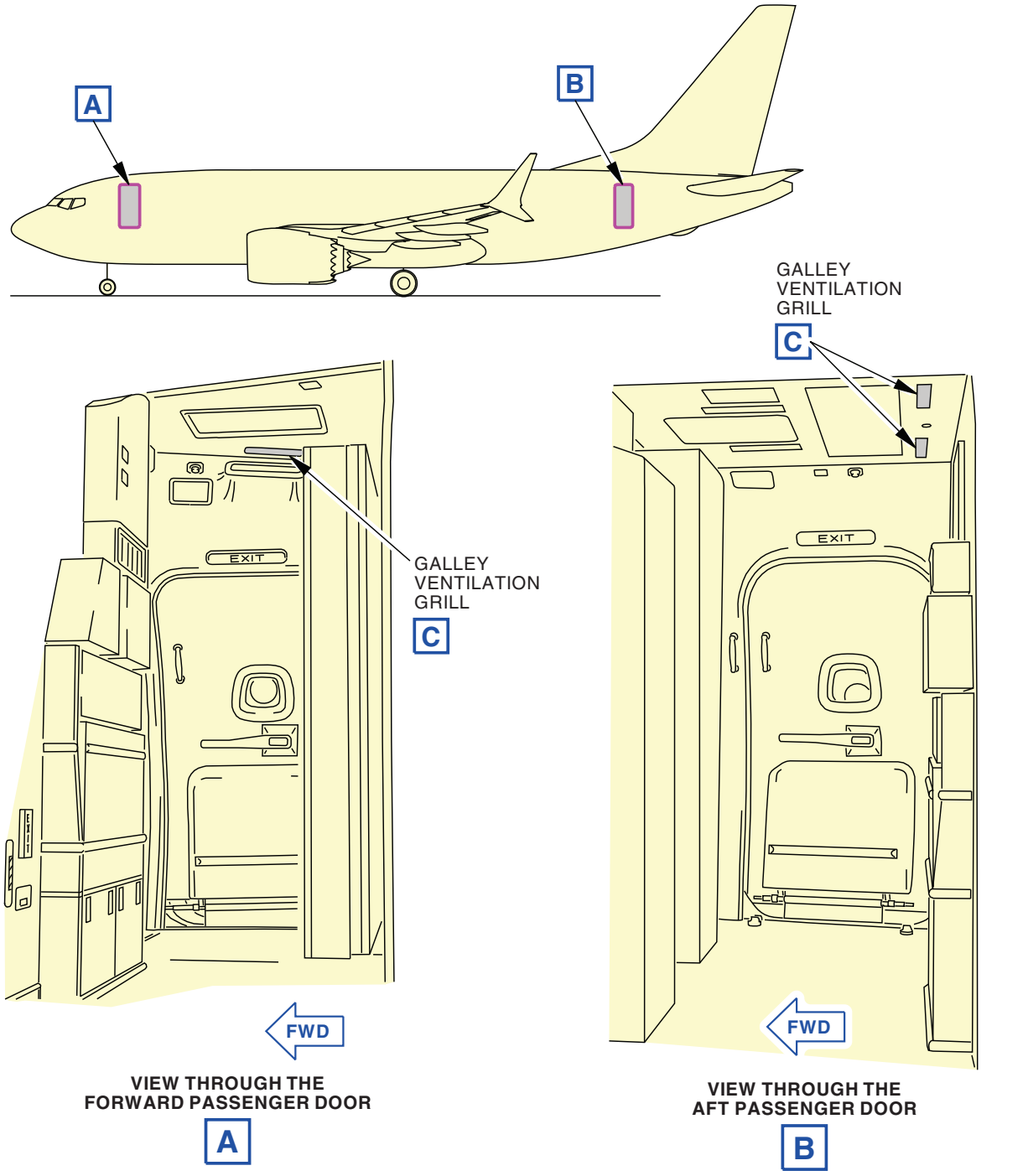
(1) Remove the two screws [2] that retain the ventilation grill [1] to the plenum above the ceiling.

SUBTASK 21-26-03-020-001

(2) Remove the ventilation grill [1] from the ceiling.

——— END OF TASK ———

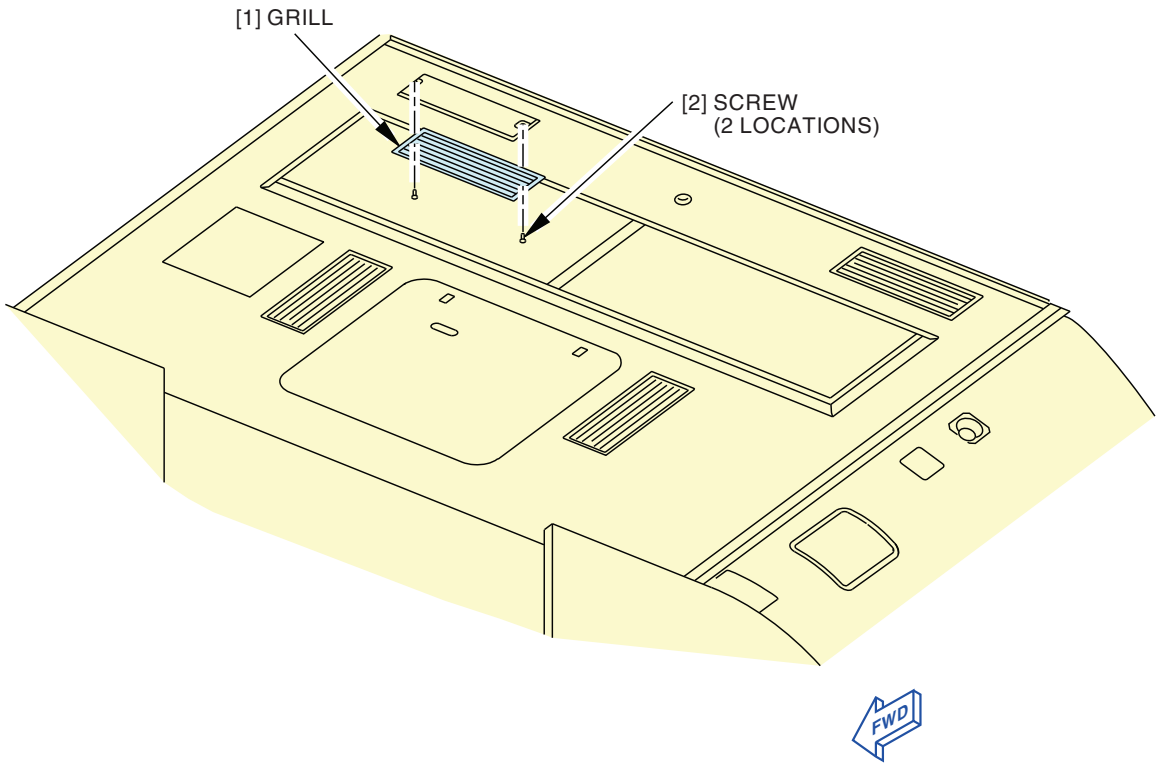
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Galley Ceiling Ventilation Grill Removal
Figure 201/21-26-03-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GALLEY VENTILATION GRILL
(EXAMPLE)

C

2411310 S00061527569_V1

Galley Ceiling Ventilation Grill Removal
Figure 201/21-26-03-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-26-03-100-801

3. Galley Ceiling Ventilation Grill Cleaning

(Figure 201)

A. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B00541	Cleaner - General Purpose Household Detergent	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Prepare for the Cleaning

SUBTASK 21-26-03-000-001

(1) Remove the ventilation grill [1] to be cleaned from the ceiling of the forward-passenger entry-way or the aft-passenger entry-way (TASK 21-26-03-000-801).

D. Galley Ceiling Ventilation Grill Cleaning

SUBTASK 21-26-03-100-001

(1) Do these steps to clean the ventilation grill [1]:

(a) Prepare a cleaning solution of a general purpose household detergent cleaner, B00541, diluted with water in accordance with the instructions on the detergent container.

(b) Clean the grill [1] with a cotton wiper, G00034, that is moist with the cleaning solution.

(c) If the cotton wiper, G00034, with the cleaning solution, did not remove all dirt deposits or contamination from the grill [1], clean the grill [1] with a cotton wiper, G00034, that is moist with alcohol, B00130.

(d) Flush the grill [1] with clean water until all traces of the detergent solution or alcohol are removed.

(e) Dry the grill [1] with a dry, clean cotton wiper, G00034.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-26-03-420-001

(1) Install the ventilation grill [1] in the entry-way ceiling (TASK 21-26-03-400-801).

——— END OF TASK ———

TASK 21-26-03-400-801

4. Galley Ceiling Ventilation Grill Installation

(Figure 201)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Grill	21-26-03-01-015	PTF ALL
		21-26-03-10-005	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Galley Ceiling Ventilation Grill Installation

SUBTASK 21-26-03-420-002

(1) Put the ventilation grill [1] in its position in the ceiling in either the forward or aft passenger entry-way.

SUBTASK 21-26-03-420-003

(2) Install the two screws [2] to retain the ventilation grill [1] to the plenum above the ceiling.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Equipment Cooling System - Deactivation

(2) Equipment Cooling System - Activation

TASK 21-27-00-800-801

2. Equipment Cooling System - Deactivation

A. General

(1) This task will deactivate these components in the equipment cooling system.

- Equipment Cooling Supply Fan
- Equipment Cooling Exhaust Fan
- Equipment Cooling Overboard Exhaust Valve

B. References

Reference	Title
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Equipment Cooling System - Deactivation

SUBTASK 21-27-00-860-177

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-27-00-010-001

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

SUBTASK 21-27-00-410-015

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Tryout

NOTE: This tryout is to make sure these components in the equipment cooling system are in a zero energy state.

- Equipment Cooling Supply Fan
- Equipment Cooling Exhaust Fan
- Equipment Cooling Overboard Exhaust Valve

NOTE: Wait a minimum of 60 seconds after deactivation before you do the tryout steps.

SUBTASK 21-27-00-860-002

(1) For the equipment cooling supply and exhaust fans, do these steps:

(a) Make sure that the EQUIP COOLING SUPPLY switch on the Equipment Cooling Panel is in the NORM position.

(b) Make sure that the EQUIP COOLING EXHAUST switch on the Equipment Cooling Panel is in the NORM position.

(c) Make sure that the EQUIP COOLING SUPPLY OFF light is on.

(d) Make sure that the EQUIP COOLING EXHAUST OFF light is on.

(e) Put the EQUIP COOLING SUPPLY switch in the ALTN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Put the EQUIP COOLING EXHAUST switch in the ALTN position.

(g) Wait a minimum of 60 seconds.

(h) Make sure that the EQUIP COOLING SUPPLY OFF light is on.

(i) Make sure that the EQUIP COOLING EXHAUST OFF light is on.

SUBTASK 21-27-00-860-003

(2) For the equipment cooling overboard exhaust valve, do these steps:

(a) Do this task: (Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803).

(b) Make sure that these switches, on the P5-10 Air Conditioning Panel, are in the AUTO position:

1) L RECIRC FAN

2) R RECIRC FAN

(c) Make sure that these switches, on the Equipment Cooling Panel, are in the NORM position:

1) EQUIP COOLING, EXHAUST

2) EQUIP COOLING, SUPPLY

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(d) Do this task: (Put the Airplane in the Air Mode, TASK 32-09-00-860-801).

(e) Put these switches, on the air conditioning module, to the HIGH position:

1) L PACK

2) R PACK

(f) Put the L RECIRC FAN switch to the OFF position.

(g) Put the R RECIRC FAN switch to the OFF position.

(h) Make sure that the equipment cooling overboard exhaust valve is in the NORMAL position.

NOTE: Look at the indicator on the valve actuator to see the position of the valve.

(i) Make sure that the equipment cooling overboard exhaust valve is NOT in the SMOKE position.

NOTE: Look at the indicator on the valve actuator to see the position of the valve.

END OF TASK

TASK 21-27-00-800-802

3. Equipment Cooling System - Activation

A. General

(1) This task will activate these components in the equipment cooling system.

- Equipment Cooling Supply Fan
- Equipment Cooling Exhaust Fan
- Equipment Cooling Overboard Exhaust Valve

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B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Equipment Cooling System - Activation

SUBTASK 21-27-00-860-178

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-27-00-010-011

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-00-860-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
A	8	C00935	EQPT CLG SPLY FAN PWR-ALTN
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM
D	12	C00837	EQPT CLG EXH FAN PWR-ALTN

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-410-001

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Equipment Cooling Operational Tests

NOTE: You need to do these tasks to do a complete system test of the equipment cooling system.

(a) Equipment Cooling Low Flow Sensors Operational Test

(b) Equipment Cooling Fans Operational Test

(c) Equipment Cooling Overboard Exhaust Valve Operational Test.

(d) Flight Compartment Display Unit Air Flow Test

(2) Equipment Cooling Scheduled Maintenance Tests for Task Cards

NOTE: You do not need to do the scheduled maintenance tests for task cards to do a complete system test of the equipment cooling system.

(a) Alternate Equipment Cooling Supply Fan Operational Test

(b) Alternate Equipment Cooling Exhaust Fan Operational Test

(c) Equipment Cooling Overboard Exhaust Valve Smoke Clearance Mode Operational Test.

TASK 21-27-00-700-801

2. Equipment Cooling Low Flow Sensors - Operational Test

A. General

(1) This task does a test of the equipment cooling supply and exhaust low flow sensors.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Equipment Cooling Low Flow Sensors - Operational Test

SUBTASK 21-27-00-860-133

(1) Make sure that the EQUIP COOLING SUPPLY switch on the Equipment Cooling Panel is in the NORM position.

SUBTASK 21-27-00-860-134

(2) Make sure that the EQUIP COOLING EXHAUST switch on the Equipment Cooling Panel is in the NORM position.

SUBTASK 21-27-00-860-006

(3) Make sure that the EQUIP COOLING SUPPLY and EXHAUST OFF lights on the Equipment Cooling Panel are not on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-710-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Do these steps to do a check of the equipment cooling supply low flow sensor:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Open this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

(c) Make sure that the EQUIP COOLING SUPPLY OFF light comes on after 20 to 60 seconds.

(d) Close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

(e) Make sure that the EQUIP COOLING SUPPLY OFF light goes out within 20 seconds.

SUBTASK 21-27-00-710-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Do these steps to do a check of the equipment cooling exhaust low flow sensor:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

(b) Make sure that the EQUIP COOLING EXHAUST OFF light comes on after 20 to 60 seconds.

(c) Close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

(d) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(e) Make sure that the EQUIP COOLING EXHAUST OFF light goes out within 20 seconds.

(f) If the MASTER CAUTION or AIR COND annunciator lights are on, push and release one of the two MASTER CAUTION lights.

——— END OF TASK ———

TASK 21-27-00-700-802

3. Equipment Cooling Fans - Operational Test

A. General

(1) This task does a test of these fans:

(a) Equipment cooling normal supply fan

(b) Equipment cooling alternate supply fan

(c) Equipment cooling normal exhaust fan

(d) Equipment cooling alternate exhaust fan

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Equipment Cooling Fans - Operational Test

SUBTASK 21-27-00-860-010

(1) Make sure that the EQUIP COOLING SUPPLY switch on the Equipment Cooling Panel is in the NORM position.

SUBTASK 21-27-00-860-011

(2) Make sure that the EQUIP COOLING EXHAUST switch on the Equipment Cooling Panel is in the NORM position.

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SUBTASK 21-27-00-860-135

(3) Make sure that the EQUIP COOLING SUPPLY and EXHAUST OFF lights on the Equipment Cooling Panel are not on.

SUBTASK 21-27-00-010-008

(4) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-00-860-012

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

SUBTASK 21-27-00-710-003

(6) Make sure that the EQUIP COOLING SUPPLY OFF light comes on after 20 to 60 seconds.

SUBTASK 21-27-00-860-013

(7) Put the EQUIP COOLING SUPPLY switch in the ALTN position.

SUBTASK 21-27-00-710-004

(8) Make sure that the EQUIP COOLING SUPPLY OFF light goes out within 20 seconds.

SUBTASK 21-27-00-860-014

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

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(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(9) Close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

SUBTASK 21-27-00-860-015

(10) Put the EQUIP COOLING SUPPLY switch in the NORM position.

SUBTASK 21-27-00-860-016

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(11) Open this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

SUBTASK 21-27-00-710-005

(12) Make sure that the EQUIP COOLING EXHAUST OFF light comes on after 20 to 60 seconds.

SUBTASK 21-27-00-860-017

(13) Put the EQUIP COOLING EXHAUST switch in the ALTN position.

SUBTASK 21-27-00-710-006

(14) Make sure that the EQUIP COOLING EXHAUST OFF light goes out within 20 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-860-018

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(15) Close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

SUBTASK 21-27-00-410-012

(16) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-00-860-019

(17) Put the EQUIP COOLING EXHAUST switch in the NORM position.

SUBTASK 21-27-00-860-020

(18) If the MASTER CAUTION or AIR COND annunciator lights are on, push and release one of the two MASTER CAUTION lights.

——— END OF TASK ———

TASK 21-27-00-700-803

4. Equipment Cooling Overboard Exhaust Valve - Functional Test

A. General

(1) This task does a functional test of the equipment cooling overboard exhaust valve. The interfaces of the exhaust fans, supply fans and recirculation fans with the overboard exhaust valve system are also tested.

B. References

Reference	Title
12-15-21-600-804	Crew Oxygen Cylinder Replacement (P/B 301)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

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C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
821	Forward Cargo Door

E. Prepare for the Functional Test

SUBTASK 21-27-00-010-002

(1) Get access to the equipment cooling overboard exhaust valve, the equipment cooling supply fans and the equipment cooling exhaust fans:

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

(b) Remove the access panel on the top of the raised platform that is just aft and outboard of the electronic equipment access door.

(c) Remove the crew oxygen system cylinder (TASK 12-15-21-600-804).

SUBTASK 21-27-00-010-003

(2) Get access to the recirculation fan in the mix bay at the aft end of the forward cargo compartment:

(a) Open this access door:

Number	Name/Location
821	Forward Cargo Door

(b) Remove the aft center bulkhead liner in the forward cargo compartment (TASK 50-11-02-000-801).

F. Equipment Cooling Overboard Exhaust Valve Functional Test

SUBTASK 21-27-00-860-136

(1) Make sure that the CAB/UTIL switch on the P5-13 Electrical meter, Battery and Galley Power Module is set to the ON position.

SUBTASK 21-27-00-860-023

(2) Make sure that the airplane is in the ground mode (TASK 32-09-00-860-802).

SUBTASK 21-27-00-860-024

(3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-27-00-860-025

(4) Set the EQUIP COOLING SUPPLY and EXHAUST switches on the Equipment Cooling Panel to the NORM positions.

SUBTASK 21-27-00-860-026

(5) Set these switches on the P5-10 Air Conditioning Panel to the AUTO position:

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(a) L PACK

(b) R PACK

(c) R RECIRC FAN.

SUBTASK 21-27-00-720-001

(6) Make sure that the position indicator on the overboard exhaust valve actuator is in the NORMAL position.

SUBTASK 21-27-00-860-027

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-27-00-860-028

(8) Set the R RECIRC FAN switch on the P5-10 Air Conditioning Panel to the OFF position.

SUBTASK 21-27-00-720-002

(9) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-860-029

(10) Set the R RECIRC FAN switch on the P5-10 Air Conditioning Panel to the AUTO position.

SUBTASK 21-27-00-860-030

(11) Set the L PACK switch on the P5-10 Air Conditioning Panel to the HIGH position.

SUBTASK 21-27-00-720-003

(12) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-860-031

(13) Set the L PACK switch on the P5-10 Air Conditioning Panel to the AUTO position.

SUBTASK 21-27-00-860-032

(14) Set the R PACK switch on the P5-10 Air Conditioning Panel to the HIGH position.

SUBTASK 21-27-00-720-004

(15) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-860-033

(16) Set the L PACK switch on the P5-10 Air Conditioning Panel to the HIGH position.

SUBTASK 21-27-00-720-005

(17) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-720-006

(18) Make sure that equipment cooling normal exhaust fan, M98, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-007

(19) Make sure that equipment cooling normal supply fan, M1322, is operating.

(a) Put your hand on the fan case to feel that it is operating.

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SUBTASK 21-27-00-860-034

(20) Put the EQUIP COOLING SUPPLY and EXHAUST fan switches to the ALTN positions.

SUBTASK 21-27-00-720-008

(21) Make sure that equipment cooling alternate exhaust fan, M99, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-009

(22) Make sure that the equipment cooling alternate supply fan, M1323, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-010

(23) Do these steps to send an OEV close enable signal to the overboard exhaust valve:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

(b) Unlatch and lower the P5 forward overhead panel to get access to the back of the P5-6 Cabin Pressurization Control Panel.

(c) Disconnect the electrical connector, D10726, from the P5-6 Cabin Pressurization Control Panel.

(d) Connect a jumper between pins 26 and 27 of electrical connector, D10726, on wire bundle, W2510.

SUBTASK 21-27-00-720-011

(24) Make sure that the position indicator on the overboard exhaust valve actuator is in the NORMAL position.

SUBTASK 21-27-00-720-012

(25) Do these steps to send an OEV open enable signal (OEV position indicator at the SMOKE position) to the overboard exhaust valve:

(a) Make a record of the time immediately after you do the next step.

(b) Remove the jumper between pins 26 and 27 of electrical connector, D10726, on wire bundle, W2510.

(c) Reconnect the electrical connector, D10726, to the P5-6 Cabin Pressurization Control Panel.

(d) Raise and latch the P5 forward overhead panel.

(e) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

(f) Make sure that after 4-1/2 to 5-1/2 minutes that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

NOTE: This does a check of the 5 minute delay on relay R649.

SUBTASK 21-27-00-860-035

(26) Set the R RECIRC FAN switch on the P5-10 Air Conditioning Panel to the OFF position.

(a) Make a record of the time when you do this step.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-720-013

(27) Make sure the equipment cooling alternate supply fan, M1323, is not operating.

SUBTASK 21-27-00-720-014

(28) Make sure the equipment cooling alternate exhaust fan, M99, is operating.

SUBTASK 21-27-00-720-015

(29) Make sure that the EQUIP COOLING SUPPLY OFF light is not on after 30 seconds since the R RECIRC FAN switch was set to OFF.

SUBTASK 21-27-00-720-016

(30) Make sure that the equipment cooling alternate supply fan, M1323, is operating within 4-1/2 to 5-1/2 minutes of when the R RECIRC FAN switch was set to OFF.

SUBTASK 21-27-00-720-017

(31) Make sure that the equipment cooling alternate exhaust fan, M99, is operating.

SUBTASK 21-27-00-720-069

(32) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-720-018

(33) Make sure that the EQUIP COOLING SUPPLY and EXHAUST OFF lights are not on.

SUBTASK 21-27-00-860-036

(34) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B
C	19	C01524	CARGO FIRE AFT DET A

SUBTASK 21-27-00-720-019

(35) Push and hold the TEST switch, on the P8 Cargo Fire Control Panel, until the FWD and AFT red cargo fire lights come on.

(a) Make a record of the time when you do this step.

SUBTASK 21-27-00-860-037

(36) Set the R RECIRC FAN switch on the P5-10 Air Conditioning Panel to the AUTO position.

SUBTASK 21-27-00-860-038

(37) Set the R RECIRC FAN switch on the P5-10 Air Conditioning Panel to the OFF position.

SUBTASK 21-27-00-720-021

(38) Make sure the equipment cooling alternate supply fan, M1323, is not operating.

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-720-022

(39) Make sure the equipment cooling alternate exhaust fan, M99, is not operating.

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-860-039

(40) Put the EQUIP COOLING EXHAUST switch to the NORM position.

SUBTASK 21-27-00-720-023

(41) Make sure the equipment cooling normal exhaust fan, M98, is not operating.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-720-070

(42) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-720-024

(43) Make sure that the EQUIP COOLING SUPPLY OFF light is not on after 30 seconds since the recirculation fan switch was set to OFF.

SUBTASK 21-27-00-720-071

(44) Make sure that the EQUIP COOLING EXHAUST OFF light is not on after 30 seconds since the R RECIRC FAN was set to OFF.

SUBTASK 21-27-00-720-026

(45) Make sure that after 4-1/2 to 5-1/2 minutes that the equipment cooling alternate supply fan, M1323, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-860-040

(46) Open this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

SUBTASK 21-27-00-860-041

(47) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(a) Make a record of the time when you do this step.

SUBTASK 21-27-00-720-027

(48) Make sure that the equipment cooling alternate supply fan, M1323, is not operating.

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-720-028

(49) Make sure that the equipment cooling normal exhaust fan, M98, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-072

(50) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-720-029

(51) Make sure that 30 seconds from when circuit breaker, C01444, was closed that the EQUIP COOLING SUPPLY OFF light is not on.

SUBTASK 21-27-00-720-030

(52) Make sure that 4-1/2 to 5-1/2 minutes from when circuit breaker, C01444, was closed that the equipment cooling alternate supply fan, M1323, is operating.

SUBTASK 21-27-00-860-042

(53) Set the EQUIP COOLING SUPPLY switch to the NORMAL position.

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SUBTASK 21-27-00-720-031

(54) Make sure the equipment cooling normal supply fan, M1322, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-860-043

(55) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A

SUBTASK 21-27-00-860-044

(56) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the AFT cargo compartment smoke detector light comes on.

SUBTASK 21-27-00-720-032

(57) Make sure that the equipment cooling normal supply fan, M1322, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-033

(58) Make sure that the equipment cooling normal exhaust fan, M98, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-034

(59) Make sure that the overboard exhaust valve position indicator is in the NORMAL position.

SUBTASK 21-27-00-860-045

(60) Open this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

SUBTASK 21-27-00-860-046

(61) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(a) Make a record of the time when you do this step.

SUBTASK 21-27-00-720-035

(62) Make sure that the equipment cooling normal supply fan, M1322, is not operating.

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-720-036

(63) Make sure that the equipment cooling normal exhaust fan, M98, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-037

(64) Make sure that the overboard exhaust valve position indicator is in the SMOKE position

SUBTASK 21-27-00-720-038

(65) Make sure that after 30 seconds the EQUIP COOLING SUPPLY OFF light is not on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-720-039

(66) Make sure that after 4-1/2 to 5-1/2 minutes that the equipment cooling normal supply fan, M1322, is operating.

(a) Put your hand on the fan case to feel that the fan is operating.

SUBTASK 21-27-00-720-040

(67) Do these checks of the recirculation fan:

(a) Make sure the right recirculation fan is not operating.

(b) Set the R RECIRC FAN switch to the AUTO position.

(c) Make sure that the right recirculation fan is operating.

1) Put your hand on the fan case to feel that it is operating.

(d) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

(e) Set the R RECIRC FAN switch to the OFF position.

1) Make a record of the time when you do this step.

(f) Make sure that the right recirculation fan, M1441, is operating.

1) Put your hand on the fan case to feel that it is operating.

(g) Make sure that the equipment cooling normal supply fan, M1322, is not operating.

1) Put your hand on the fan case to feel that it is not operating.

(h) Make sure that the equipment cooling normal exhaust fan, M98, is operating.

1) Put your hand on the fan case to feel that it is operating.

(i) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

(j) Make sure that after 30 seconds the EQUIP COOLING SUPPLY OFF light is not on.

(k) Make sure that after 4-1/2 to 5-1/2 minutes the normal equipment cooling supply fan, M1322, is operating.

1) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-860-047

(68) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A

SUBTASK 21-27-00-720-041

(69) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the FWD and AFT cargo compartment smoke detector lights come on.

SUBTASK 21-27-00-720-042

(70) Make sure that the equipment cooling normal supply fan, M1322, is operating.

(a) Put your hand on the fan case to feel that it is operating.

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SUBTASK 21-27-00-720-043

(71) Make sure that the equipment cooling normal exhaust fan, M98, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-044

(72) Make sure that the position indicator on the overboard exhaust valve actuator is in the NORMAL position.

SUBTASK 21-27-00-720-073

(73) Make sure that the right recirculation fan, M1441, is not operating.

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-860-048

(74) Open this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

SUBTASK 21-27-00-860-049

(75) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(a) Make a record of the time when you do this step.

SUBTASK 21-27-00-720-047

(76) Make sure that the equipment cooling normal supply fan, M1322, is not operating.

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-720-048

(77) Make sure that the equipment cooling normal exhaust fan, M98, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-049

(78) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-720-050

(79) Make sure that the right recirculation fan, M1441, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-051

(80) Make sure that 30 seconds from when circuit breaker, C01444, was closed that the EQUIP COOLING SUPPLY OFF light is not on.

SUBTASK 21-27-00-720-052

(81) Make sure that 4-1/2 to 5-1/2 minutes after circuit breaker, C01444, was closed that the equipment cooling normal supply fan, M1322, is operating.

(a) Put your hand on the fan case to feel that it is operating.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-860-050

(82) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A

SUBTASK 21-27-00-720-053

(83) Push and hold the TEST switch on the P8 Cargo Fire Control Panel until the AFT cargo compartment smoke detector light comes on.

SUBTASK 21-27-00-720-054

(84) Make sure that the equipment cooling normal supply fan, M1322, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-055

(85) Make sure that the equipment cooling normal exhaust fan, M98, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-056

(86) Make sure that the position indicator on the overboard exhaust valve actuator is in the NORMAL position.

SUBTASK 21-27-00-720-057

(87) Make sure that the right recirculation fan, M1441, is not operating.

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-860-051

(88) Open this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

SUBTASK 21-27-00-860-052

(89) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

(a) Make a record of the time when you do this step.

SUBTASK 21-27-00-720-058

(90) Make sure that the equipment cooling normal supply fan, M1322, is not operating.

(a) Put your hand on the fan case to feel that it is not operating.

SUBTASK 21-27-00-720-059

(91) Make sure that the equipment cooling normal exhaust fan, M98, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-060

(92) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-720-061

(93) Make sure that the right recirculation fan, M1441, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-062

(94) Make sure that 30 seconds after circuit breaker, C01444, was closed that the EQUIP COOLING SUPPLY OFF light is not on.

SUBTASK 21-27-00-720-063

(95) Make sure that 4-1/2 to 5-1/2 minutes after circuit breaker, C01444, was closed that the equipment cooling normal supply fan, M1322, is operating.

(a) Put your hand on the fan case to feel that it is operating.

SUBTASK 21-27-00-720-064

(96) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-27-00-720-065

(97) Make sure that the position indicator on the overboard exhaust valve actuator is in the SMOKE position.

SUBTASK 21-27-00-860-053

(98) Set the R RECIRC FAN switch to the AUTO position.

SUBTASK 21-27-00-720-066

(99) Make sure that the position indicator on the overboard exhaust valve actuator is in the NORMAL position.

SUBTASK 21-27-00-860-144

(100) Set the R RECIRC FAN switch to the OFF position.

SUBTASK 21-27-00-720-077

(101) Make sure that the position indicator on the overboard exhaust valve actuator is in the NORMAL position.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-00-860-054

(1) Put these switches on the P5-10 Air Conditioning Module to the AUTO position:

(a) L PACK

(b) R PACK

(c) R RECIRC FAN.

SUBTASK 21-27-00-860-055

(2) Make sure that the airplane is in the ground mode (TASK 32-09-00-860-802).

SUBTASK 21-27-00-410-003

(3) Install the crew oxygen system cylinder (TASK 12-15-21-600-804).

SUBTASK 21-27-00-860-162

(4) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A
C	18	C01525	CARGO FIRE AFT DET B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	19	C01524	CARGO FIRE AFT DET A

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-27-00-410-008

(5) Install the access panel in the electronics equipment bay on the top of the raised platform that you removed.

SUBTASK 21-27-00-410-004

(6) Install the forward cargo compartment aft bulkhead liner (TASK 50-11-02-400-801).

SUBTASK 21-27-00-410-005

(7) Close these access doors:

Number	Name/Location
117A	Electronic Equipment Access Door
821	Forward Cargo Door

——— END OF TASK ———

TASK 21-27-00-700-804

5. Flight Compartment Display Unit Air Flow Test

A. General

(1) This task makes sure that these display units (DU) have sufficient cooling air flow:

(a) Captain's Inboard DU

(b) Captain's Outboard DU

(c) First Officer's Inboard DU

(d) First Officer's Outboard DU

B. References

Reference	Title
31-65-11-000-801	MAX Display Unit Removal (P/B 401)
31-65-11-400-801	MAX Display Unit Installation (P/B 401)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Flight Compartment Display Unit Air Flow Test

SUBTASK 21-27-00-860-059

(1) Make sure the EQUIP COOLING SUPPLY and EXHAUST OFF lights on the Equipment Cooling Panel are not on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-860-060

(2) Make sure the EQUIP COOLING SUPPLY switch on the Equipment Cooling Panel is in the NORMAL position.

SUBTASK 21-27-00-860-061

(3) Make sure the EQUIP COOLING EXHAUST switch on the Equipment Cooling Panel is in the NORM position.

SUBTASK 21-27-00-750-001

(4) Do these steps to make sure the display units are receiving sufficient cooling airflow:

(a) Remove the four display units in the flight compartment (TASK 31-65-11-000-801).

(b) Make sure that you can feel airflow going into the air inlet at the first officer's inboard DU in the P3 panel.

NOTE: There is sufficient cooling airflow if you can feel air movement with your hand 3.0 in. (7.6 cm) from the air inlet or outlet.

(c) Make sure that you can feel airflow going into the air inlet at the first officer's outboard DU in the P3 panel.

(d) Make sure that you can feel airflow going into the air inlet at the captain's control display unit (CDU) in the P9 panel.

(e) Make sure that you can feel airflow going into the air inlet at the first officer's CDU in the P9 panel.

(f) Make sure that you can feel airflow going out of the lower air outlet at the captain's inboard DU in the P1 panel.

(g) Make sure that you can feel airflow going out of the lower air outlet at the captain's outboard DU in the P1 panel.

(h) Make sure that the passive exhaust duct connected to the captain's display units is not capped or bagged.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-00-420-001

(1) Install the display units (TASK 31-65-11-400-801).

——— END OF TASK ———

TASK 21-27-00-890-801

6. Alternate Equipment Cooling Supply Fan - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task does a test of the equipment cooling alternate supply fan.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Alternate Equipment Cooling Supply Fan - Operational Test

SUBTASK 21-27-00-860-064

(1) Make sure that the EQUIP COOLING SUPPLY OFF light on the Equipment Cooling Panel is not on.

SUBTASK 21-27-00-860-065

(2) Make sure that the EQUIP COOLING SUPPLY switch on the Equipment Cooling Panel is in the NORM position.

SUBTASK 21-27-00-010-009

(3) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-00-860-066

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

SUBTASK 21-27-00-710-007

(5) Make sure that the EQUIP COOLING SUPPLY OFF light comes on after 20 to 60 seconds.

SUBTASK 21-27-00-860-067

(6) Put the EQUIP COOLING SUPPLY switch in the ALTN position.

SUBTASK 21-27-00-710-008

(7) Make sure that the EQUIP COOLING SUPPLY OFF light goes out within 20 seconds.

SUBTASK 21-27-00-860-068

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(8) Close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

SUBTASK 21-27-00-410-013

(9) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-00-860-069

(10) Put the EQUIP COOLING SUPPLY switch in the NORM position.

SUBTASK 21-27-00-860-070

(11) If the MASTER CAUTION or AIR COND annunciator lights are on, push and release one of the two MASTER CAUTION lights.

——— END OF TASK ———

TASK 21-27-00-700-805

7. Alternate Equipment Cooling Exhaust Fan - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task does a test of the equipment cooling alternate exhaust fan.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Alternate Equipment Cooling Exhaust Fan - Operational Test

SUBTASK 21-27-00-860-073

(1) Make sure that the EQUIP COOLING EXHAUST OFF light on the Equipment Cooling Panel is not on.

SUBTASK 21-27-00-860-074

(2) Make sure that the EQUIP COOLING EXHAUST switch on the Equipment Cooling Panel is in the NORM position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-010-010

(3) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-00-860-075

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

SUBTASK 21-27-00-710-009

(5) Make sure that the EQUIP COOLING EXHAUST OFF light comes on after 20 to 60 seconds.

SUBTASK 21-27-00-860-076

(6) Put the EQUIP COOLING EXHAUST switch in the ALTN position.

SUBTASK 21-27-00-710-010

(7) Make sure that the EQUIP COOLING EXHAUST OFF light goes out within 20 seconds.

SUBTASK 21-27-00-860-077

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(8) Close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

SUBTASK 21-27-00-410-014

(9) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-00-860-078

(10) Put the EQUIP COOLING EXHAUST switch in the NORM position.

SUBTASK 21-27-00-860-079

(11) If the MASTER CAUTION or AIR COND annunciator lights are on, push and release one of the two MASTER CAUTION lights.

——— END OF TASK ———

TASK 21-27-00-700-806

8. Equipment Cooling Overboard Exhaust Valve Smoke Clearance Mode - Operational Test

A. General

(1) This task does a test of the equipment cooling overboard exhaust valve in the smoke removal mode.

(2) When persons are in the airplane during this test, they must be in good physical condition. When a person feels pain during a pressure change, you must make the pressure stable immediately or it must be lowered. Do this until the person can make the pressure equal in their ears, or be removed from the airplane.

B. References

Reference	Title
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Procedure

SUBTASK 21-27-00-860-082

(1) Extend the trailing edge flaps to position 1. To extend the flaps, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 21-27-00-860-083

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-27-00-860-084

WARNING MAKE SURE YOU OBEY THE PROPER PROCEDURES FOR COMPRESSION AND DECOMPRESSION WHEN YOU USE PERSONS IN A PRESSURIZED AREA. PRESSURE CHANGES THAT CAUSE PAIN MUST NOT BE DONE. IF YOU DO NOT OBEY THE PRECAUTIONS, INJURY TO PERSONS CAN OCCUR.

(3) Do these steps to pressurize the airplane cabin to 2.00 psi (13.79 kPa) differential pressure:

(a) Close all exterior doors.

(b) Make sure the flight compartment sliding windows are closed.

(c) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

(d) Set the R RECIRC FAN switch on the P5-10 Air Conditioning Panel to the AUTO position.

(e) Put the cabin pressurization switch on the P5–6 Cabin Pressure Control Panel to the MAN position.

(f) Begin to close the cabin pressurization outflow valve with the cabin pressurization valve MANUAL OPEN-CLOSE toggle switch on the P5–6 Cabin Pressure Control Panel:

1) Look at the VALVE position indicator to see the position of the valve.

WARNING DO NOT ALLOW THE INDICATION ON THE CABIN CLIMB GAUGE TO SHOW MORE THAN 1000 FEET PER MINUTE. WHEN YOU ALLOW A RATE OF CABIN CLIMB OF MORE THAN 1000 FEET PER MINUTE, DAMAGE TO THE AIRPLANE STRUCTURE OR INJURY TO PERSONNEL CAN OCCUR.

(g) Slowly increase the cabin pressure with the MANUAL OPEN-CLOSE toggle switch on the P5-6 panel so that the indication on the CABIN CLIMB gauge on the P5–16 panel shows a rate of approximately 300 feet per minute.

1) Do not change the cabin pressure at a rate of more than 1000 feet per minute.

(h) Increase the cabin pressure until the cabin DIFF PRESS indicator on the P5–16 Cabin Altitude and Rate of Climb Panel shows a differential pressure of 2.00 psi (13.79 kPa).

SUBTASK 21-27-00-860-085

(4) Make sure that the L PACK and R PACK switches on the P5-10 Air Conditioning Panel are in the AUTO position.

SUBTASK 21-27-00-860-086

(5) Make sure the R RECIRC FAN switch, on the P5-10 Air Conditioning Panel, is in the AUTO position.

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SUBTASK 21-27-00-710-011

(6) Make sure that there is no air flow out of the equipment cooling overboard exhaust valve port:

(a) Put your hand at the valve exhaust port on the bottom of the airplane to feel for airflow.

NOTE: The exhaust port is just aft of the E/E bay access door.

SUBTASK 21-27-00-860-087

WARNING DO NOT ALLOW THE INDICATION ON THE CABIN CLIMB GAUGE TO SHOW MORE THAN 1000 FEET PER MINUTE. WHEN YOU ALLOW A RATE OF CABIN CLIMB OF MORE THAN 1000 FEET PER MINUTE, DAMAGE TO THE AIRPLANE STRUCTURE OR INJURY TO PERSONNEL CAN OCCUR.

(7) Put the L PACK switch to HIGH.

NOTE: The cabin pressure can increase at more than 1000 feet per minute when the pack is in the high flow mode. Manually opening the outflow valve will reduce the rate of cabin pressurization.

(a) If necessary, operate the MANUAL OPEN-CLOSE toggle switch on the P5-6 Pressurization Control Panel so that the CABIN CLIMB indication on the P5-16 Cabin Altitude and Rate of Climb Panel does not exceed 1000 feet per minute.

SUBTASK 21-27-00-860-088

(8) Put the R PACK switch to OFF.

SUBTASK 21-27-00-860-089

(9) Put the R RECIRC FAN switch to the OFF position.

SUBTASK 21-27-00-710-012

(10) Make sure that there is air flow out of the equipment cooling overboard exhaust valve port.

NOTE: The CABIN CLIMB indication on the P5-16 Cabin Altitude and Rate of Climb Panel will indicate that the cabin pressure is climbing.

SUBTASK 21-27-00-860-090

WARNING DO NOT ALLOW THE INDICATION ON THE CABIN CLIMB GAUGE TO SHOW MORE THAN 1000 FEET PER MINUTE. WHEN YOU ALLOW A RATE OF CABIN CLIMB OF MORE THAN 1000 FEET PER MINUTE, DAMAGE TO THE AIRPLANE STRUCTURE OR INJURY TO PERSONNEL CAN OCCUR.

(11) Put the R PACK switch to HIGH.

NOTE: The cabin pressure can increase at more than 1000 feet per minute when the pack is in the high flow mode. Manually opening the outflow valve will reduce the rate of cabin pressurization.

(a) If necessary, operate the MANUAL OPEN-CLOSE switch on the P5-6 Pressurization Control Panel so that the CABIN CLIMB indication on the P5-16 Cabin Altitude and Rate of Climb Panel does not exceed 1000 feet per minute.

SUBTASK 21-27-00-860-091

(12) Put the L PACK switch to OFF.

SUBTASK 21-27-00-710-013

(13) Make sure that there is airflow out of the equipment cooling overboard exhaust valve port.

NOTE: The CABIN CLIMB indication on the P5-16 Cabin Altitude and Rate of Climb Panel will indicate that the cabin pressure is climbing.

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E. Put the Airplane Back to its Usual Condition

SUBTASK 21-27-00-860-092

(1) Put these switches to the AUTO position:

(a) L PACK

(b) R PACK

(c) R RECIRC FAN

SUBTASK 21-27-00-860-093

(2) Slowly open the outflow valve to decrease the cabin pressure until the cabin DIFF PRESS indicator shows a differential pressure of 0 psi (0 kPa).

SUBTASK 21-27-00-860-094

(3) Put the Cabin Pressurization Switch to the AUTO position.

SUBTASK 21-27-00-866-001

(4) Retract the trailing edge flaps to the full up position. To retract the flaps, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

SUBTASK 21-27-00-860-095

(5) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-27-00-860-096

(6) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, TASK 21-00-00-800-804.

——— END OF TASK ———

21-CMR-01

TASK 21-27-00-700-807

9. Smoke Clearance Mode - Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task does an operational test of the equipment cooling overboard exhaust valve system components that follow:

- The interfaces of the exhaust fans
- Supply fans
- Recirculation fans
- Smoke detection system.

(2) The equipment cooling overboard exhaust valve will be referred to by the acronym OEV in this test.

(3) In the event of a forward cargo fire, the cargo smoke detectors will detect smoke and annunciate to the flight crew of the forward cargo fire. The flight crew will then take the appropriate action for smoke clearance from the flight deck which results in the following:

(a) Power to the E/E cooling supply normal and alternate fans is removed for 5 minutes

(b) The E/E cooling supply low flow sensor warning is inhibited for 5 minutes

(c) The OEV opens to the SMOKE position for the duration of the flight

(d) Power to the E/E cooling exhaust normal and alternate fans is removed for the duration of the flight

(e) The E/E cooling exhaust low flow sensor warning is inhibited for the duration of the flight.

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B. References

Reference	Title
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
125	Air Conditioning Distribution Bay - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
191E	Low Pressure ECS Panel - Forward

E. Prepare for the Operational Test

SUBTASK 21-27-00-010-004

(1) Get access to the Cabin Pressure Controller (CPC) No. 2 on the E1-1 shelf in the E/E compartment:

NOTE: The CPC No. 2 will be used to determine if the OEV is in the NORMAL or SMOKE position.

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-00-860-099

(2) Make sure that the CAB/UTIL switch, on the P5-13 electrical meters, battery and galley power panel, is set at ON.

SUBTASK 21-27-00-210-001

(3) Make sure that the needle on the VALVE position indicator, on the P5-6 pressurization control panel, shows that the aft outflow valve is fully open.

(a) If it is necessary, do these steps:

1) Put the pressurization control switch to MAN position.

2) Put and hold the VALVE switch to OPEN position until the VALVE indicator is fully open.

SUBTASK 21-27-00-860-100

(4) Make sure that the airplane is in the ground mode (TASK 32-09-00-860-802).

SUBTASK 21-27-00-860-158

(5) Open these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

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CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC
E	6	C01205	SMYD-1 SNSR EXC AC

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

SUBTASK 21-27-00-860-160

(6) Open these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-27-00-860-181

(7) Close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-27-00-860-104

(8) Set the EQUIP COOLING SUPPLY and EXHAUST switches, on the equipment cooling panel in the middle of the P5 forward overhead panel, to the NORM positions.

SUBTASK 21-27-00-210-002

(9) Make sure that pneumatic power is off.

(a) Make sure that the L and R needle indicators on the DUCT PRESS gauge, on the P5-10 air conditioning panel, show 0 psi.

SUBTASK 21-27-00-860-105

(10) Set these switches, on the P5-10 air conditioning panel, to the AUTO position:

- L PACK
- R PACK
- L RECIRC FAN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-860-168

(11) Set the R RECIRC FAN switch, on the P5-10 air conditioning panel, to the OFF position.

SUBTASK 21-27-00-860-166

WARNING MAKE SURE THAT YOU DO THE STEPS TO PREPARE THE SYSTEMS FOR THE AIR MODE CORRECTLY. IF YOU DO NOT FOLLOW THESE INSTRUCTIONS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(12) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

NOTE: This is not to put the airplane in the air mode but to prepare the airplane for the air mode.

SUBTASK 21-27-00-860-167

(13) Open these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1

F. Smoke Clearance Mode Operational Test

SUBTASK 21-27-00-710-014

(1) Check that the left recirculation fan, M1440, is operating:

(a) If air is flowing out of the gaspers just outboard of the captain's and first officer's windshield and foot air controls, the left recirculation fan is operating.

(b) If the airflow out of the gaspers is not sufficient to positively determine left recirculation fan operation, do the steps that follow:

1) Open this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

2) Push up on the swing check valve.

3) If air flows out through the swing check valve, the left recirculation fan is operating.

SUBTASK 21-27-00-860-107

(2) Set the L RECIRC FAN switch, on the P5-10 air conditioning panel, to the OFF position.

SUBTASK 21-27-00-710-015

(3) Do this check that the recirculation fans are not operating:

(a) If air is not flowing out of the gaspers just outboard of the windshield and foot air controls, the recirculation fans are not operating.

SUBTASK 21-27-00-860-108

(4) Set the R RECIRC FAN switch, on the P5-10 air conditioning panel, to the AUTO position.

SUBTASK 21-27-00-710-016

(5) Do this check that the right recirculation fan, M1441, is operating:

(a) If air is flowing out of the gaspers just outboard of the captain's and first officer's windshield and foot air controls, the right recirculation fan is operating.

(b) If the airflow out of the gaspers is not sufficient to positively determine right recirculation fan operation, do the steps that follow:

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1) Open this access panel:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

2) Push up on the swing check valve.

3) If air flows out through the swing check valve, the right recirculation fan is operating.

SUBTASK 21-27-00-860-109

(6) Prepare the cabin pressure controller to display the OEV position indication:

(a) Set the pressurization mode selector switch, on the P5-6 pressurization control panel, to the AUTO position.

NOTE: The AUTO FAIL light may momentarily flash when AUTO is selected.

(b) Push the ON/OFF button on the front of the cabin pressure controller No.2 (M1654) on the E1-1 shelf in the E/E compartment.

(c) Push the MENU button.

(d) Continue to push the ↓ button until the display shows SYSTEM STATUS.

(e) Push the YES button.

(f) Push the ↓ button until the display shows SYSTEM CONFIG.

(g) Push the YES button.

(h) Push the ↓ button until the display shows CARGO HT VLV LOW.

NOTE: This display indicates that the OEV is in the NORMAL position.

SUBTASK 21-27-00-710-017

(7) Make sure that the CPC display shows CARGO HT VLV LOW.

SUBTASK 21-27-00-860-110

(8) Set the R PACK switch to the HIGH position.

SUBTASK 21-27-00-710-018

(9) Make sure that the CPC display shows CARGO HT VLV HIGH.

SUBTASK 21-27-00-860-111

(10) Set the R PACK switch to the AUTO position.

SUBTASK 21-27-00-710-019

(11) Make sure that the CPC display shows CARGO HT VLV LOW.

SUBTASK 21-27-00-860-112

(12) Set the L PACK switch to the HIGH position.

SUBTASK 21-27-00-710-020

(13) Make sure that the CPC display shows CARGO HT VLV HIGH.

SUBTASK 21-27-00-710-021

(14) Do an audible check that the equipment cooling normal supply fan, M1322, is operating.

SUBTASK 21-27-00-860-113

(15) Set the R RECIRC FAN switch, on the P5-10 air conditioning panel, to the OFF position:

(a) Make a note of the time when the switch is placed in the OFF position.

SUBTASK 21-27-00-710-022

(16) Make sure that the CPC display shows CARGO HT VLV HIGH.

NOTE: This display indicates that the OEV is in the SMOKE position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-00-710-023

(17) Do an audible check that the equipment cooling normal supply fan, M1322, is not operating.

SUBTASK 21-27-00-860-114

(18) Set the EQUIP COOLING SUPPLY and EXHAUST switches, on the equipment cooling panel in the middle of the P5 forward overhead panel, to the ALTN positions.

SUBTASK 21-27-00-710-024

(19) Do an audible check that the equipment cooling alternate supply fan, M1323, is not operating.

SUBTASK 21-27-00-710-025

(20) Do an audible check that the equipment cooling alternate supply fan, M1323, is operating within 4 1/2 to 5 1/2 minutes of when the R RECIRC FAN switch was set to OFF.

SUBTASK 21-27-00-860-115

(21) Open these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-27-00-860-116

(22) Open and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

SUBTASK 21-27-00-710-026

(23) Do an audible check that the equipment cooling alternate supply fan, M1323, is not operating.

SUBTASK 21-27-00-710-027

(24) Do this check to make sure that the right recirculation fan, M1441, is operating:

(a) If air is flowing out of the gaspers just outboard of the captain's and first officer's windshield and foot air controls, the right recirculation fan is operating.

(b) If the air flow out of the gaspers is not sufficient to positively determine recirculation fan operation, do the steps that follow:

1) Make sure that this access panel is open:

Number	Name/Location
191E	Low Pressure ECS Panel - Forward

2) Push up on the swing check valve.

3) If air is flowing out through the swing check valve, the right recirculation fan is operating.

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SUBTASK 21-27-00-860-151

(25) Close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A

SUBTASK 21-27-00-860-117

(26) Push and hold the TEST switch, on the P8 cargo fire control panel, until the FWD and AFT red cargo fire lights come on.

SUBTASK 21-27-00-860-118

(27) If it is necessary, push the BELL CUTOUT switch, on the engine and Auxiliary Power Unit (APU) fire control panel, to silence the flight compartment fire bell.

SUBTASK 21-27-00-860-182

(28) Release the TEST switch on the P8 cargo fire control panel.

SUBTASK 21-27-00-710-028

(29) Make sure that the right recirculation fan, M1441, is not operating:

(a) If air is not flowing out of the gaspers just outboard of the captain's and first officer's windshield and foot air controls, the right recirculation fan is not operating.

SUBTASK 21-27-00-860-119

CAUTION

MAKE SURE THAT YOU CLOSE THE FAN CIRCUIT BREAKER IN LESS THAN 5 MINUTES AFTER YOU OPEN IT. IF YOU KEEP THE CIRCUIT BREAKER OPEN MORE THAN 5 MINUTES, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(30) Open these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

SUBTASK 21-27-00-860-120

(31) Close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	18	C01434	EQPT COOLING LOW FLOW DETECT EXHAUST

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	14	C01435	EQPT COOLING EXHAUST FAN CONT NORMAL
C	15	C01436	EQPT COOLING EXHAUST FAN CONT ALTN

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(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	2	C01445	A/C OVERBOARD EXH VALVE RECONFIG CONT

SUBTASK 21-27-00-860-169

(32) Open and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

SUBTASK 21-27-00-710-029

(33) Do an audible check that the equipment cooling alternate exhaust fan, M99, is operating.

SUBTASK 21-27-00-860-121

(34) Set the EQUIP COOLING SUPPLY and EXHAUST switches, on the equipment cooling panel in the middle of the P5 forward overhead panel, to the NORM positions.

SUBTASK 21-27-00-710-030

(35) Do an audible check that the equipment cooling normal exhaust fan, M98, is operating.

SUBTASK 21-27-00-860-122

(36) Push and hold the TEST switch, on the P8 cargo fire control panel, until the FWD and AFT red cargo fire lights come on.

SUBTASK 21-27-00-860-179

(37) If it is necessary, push the BELL CUTOUT switch, on the engine and APU fire control panel, to silence the flight compartment fire bell.

SUBTASK 21-27-00-860-183

(38) Release the TEST switch on the P8 cargo fire control panel.

SUBTASK 21-27-00-710-031

(39) Do an audible check that the equipment cooling normal exhaust fan, M98, is not operating.

SUBTASK 21-27-00-860-123

(40) Set the EQUIP COOLING SUPPLY and EXHAUST switches, on the equipment cooling panel in the middle of the P5 forward overhead panel, to the ALTN positions.

SUBTASK 21-27-00-710-032

(41) Do an audible check that the equipment cooling alternate exhaust fan, M99, is not operating.

SUBTASK 21-27-00-710-033

(42) Make sure that the CPC display shows CARGO HT VLV HIGH.

NOTE: This display shows that the OEV is in the SMOKE position.

SUBTASK 21-27-00-860-125

(43) Open and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

NOTE: The FWD and AFT cargo fire alarm relays R945 and R946 are not latched.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-00-860-175

(1) Close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1

SUBTASK 21-27-00-860-161

(2) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-27-00-860-127

(3) Put these switches to the AUTO position:

- R RECIRC FAN
- L RECIRC FAN

SUBTASK 21-27-00-860-128

(4) Put these switches to the OFF position:

- L PACK
- R PACK

SUBTASK 21-27-00-860-130

(5) Put these switches, on the center panel of the P5 forward overhead panel, to the NORM position:

- EQUIP COOLING SUPPLY
- EQUIP COOLING EXHAUST.

SUBTASK 21-27-00-860-131

(6) Close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	5	C01204	SMYD-1 CMPTR DC
E	6	C01205	SMYD-1 SNSR EXC AC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

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F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	12	C01116	EQPT COOLING SUPPLY FAN CONT-NORMAL
C	13	C01117	EQPT COOLING SUPPLY FAN CONTROL-ALTN

SUBTASK 21-27-00-860-176

(7) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	17	C01433	EQPT COOLING LOW FLOW DETECT SUPPLY

SUBTASK 21-27-00-410-007

(8) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
191E	Low Pressure ECS Panel - Forward

——— END OF TASK ———

TASK 21-27-00-700-808

10. Aft Equipment Cooling Fan - Functional Test

(Figure 501)

A. General

(1) This task gives the instructions to test the aft equipment, E6 rack cooling exhaust fan.

B. References

Reference	Title
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

Reference	Description
STD-11325	Manometer - Digital (-5 to 5 PSI, Accuracy +/- 0.5%)

D. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
822	Aft Cargo Door

F. Prepare for Functional Test

SUBTASK 21-27-00-010-005

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 21-27-00-010-006

(2) Open the access panels on the E6 rack to get access to the electrical equipment.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Aft Equipment Cooling Fan Functional Test

SUBTASK 21-27-00-720-078

(1) Make sure that all required avionics boxes are installed in electrical equipment E6-1 shelf and E6-3 shelf.

(2) Make sure that the visible metering tray holes are plugged with red or black plugs.

(3) Make sure that the pressure equalization holes underneath the rack E6-3 are plugged.

SUBTASK 21-27-00-720-079

(4) Set the IFE/PASS SEAT switch on the P5-13 panel to ON.

SUBTASK 21-27-00-720-080

(5) Measure rack E6-1 shelf pressure at the test port on the front of the rack using digital manometer, STD-11325. (Refer to Figure 501 for location of test port).

(a) Make sure that pressure is at or above (more negative than) the acceptable minimum -2.0 inches of water.

NOTE: For example, a pressure reading of -2.2 inches of water is acceptable and a reading of -1.8 inches of water is a failure.

SUBTASK 21-27-00-720-081

(6) Measure rack E6-3 shelf pressure at the test port on the front of the rack using digital manometer, STD-11325. (Refer to Figure 501 for location of test port).

(a) Make sure that pressure is at or above (more negative than) the acceptable minimum -2.0 inches of water.

NOTE: For example, a pressure reading of -2.2 inches of water is acceptable and a reading of -1.8 inches of water is a failure.

SUBTASK 21-27-00-860-156

WARNING

MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE CLEAR OF CONTROL SURFACES AND LANDING GEAR DOOR AREAS BEFORE YOU SIMULATE THE AIRPLANE TO THE AIR MODE. THE SIMULATED AIR MODE CAN CAUSE MOVEMENT TO CONTROL SURFACES, LANDING GEAR, AND LANDING GEAR DOORS. IF YOU DO NOT OBEY THIS INSTRUCTION, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-27-00-860-163

(8) Put these switches on the P5-10 panel to the HIGH position:

(a) L PACK

(b) R PACK.

SUBTASK 21-27-00-860-164

(9) Put these switches on the P5-10 panel to the OFF position:

(a) L RECIRC FAN

(b) R RECIRC FAN.

SUBTASK 21-27-00-720-082

(10) Press and hold the P8-75 Cargo Fire Control Module Test switch.

(a) Make sure that the P8-75 Cargo Fire Control Module FWD light illuminates and fire bell alarms.

(11) Release the P8-75 Cargo Fire Control Module Test switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the P8-75 Cargo Fire Control Module FWD light turns off and the fire bell turns off.

(b) Make sure that the E6 rack cooling fan is not running.

SUBTASK 21-27-00-020-001

(12) Open these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A

SUBTASK 21-27-00-720-085

(13) Press and hold the P8-75 Cargo Fire Control Module Test switch.

(a) Make sure that the P8-75 Cargo Fire Control Module AFT light illuminates and fire bell alarms.

SUBTASK 21-27-00-720-086

(14) Release the P8-75 Cargo Fire Control Module Test switch.

(a) Make sure that the P8-75 Cargo Fire Control Module AFT light turns off and the fire bell turns off.

(b) Make sure that the E6 rack cooling fan is not running.

SUBTASK 21-27-00-410-010

(15) Close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	16	C01523	CARGO FIRE FWD DET B
C	17	C01522	CARGO FIRE FWD DET A

SUBTASK 21-27-00-860-157

(16) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-27-00-720-084

(17) Set the IFE/PASS SEAT switch on the P5-13 panel to OFF.

(a) Make sure that the E6 rack cooling fan is not running.

SUBTASK 21-27-00-860-174

(18) Open and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
D	1	C01444	A/C OVERBOARD EXHAUST VALVE CONT

SUBTASK 21-27-00-720-087

(19) Set the IFE/PASS SEAT switch on the P5-13 panel to ON.

(a) Make sure that the E6 rack cooling fan is running.

SUBTASK 21-27-00-720-088

(20) Set the IFE/PASS SEAT switch on the P5-13 panel to OFF.

H. Put the Airplane Back to its Usual Condition

SUBTASK 21-27-00-860-165

(1) Put these switches to the AUTO position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) L PACK

(b) R PACK

(c) L RECIRC FAN

(d) R RECIRC FAN.

SUBTASK 21-27-00-410-009

(2) Close the access panels on the E6 rack.

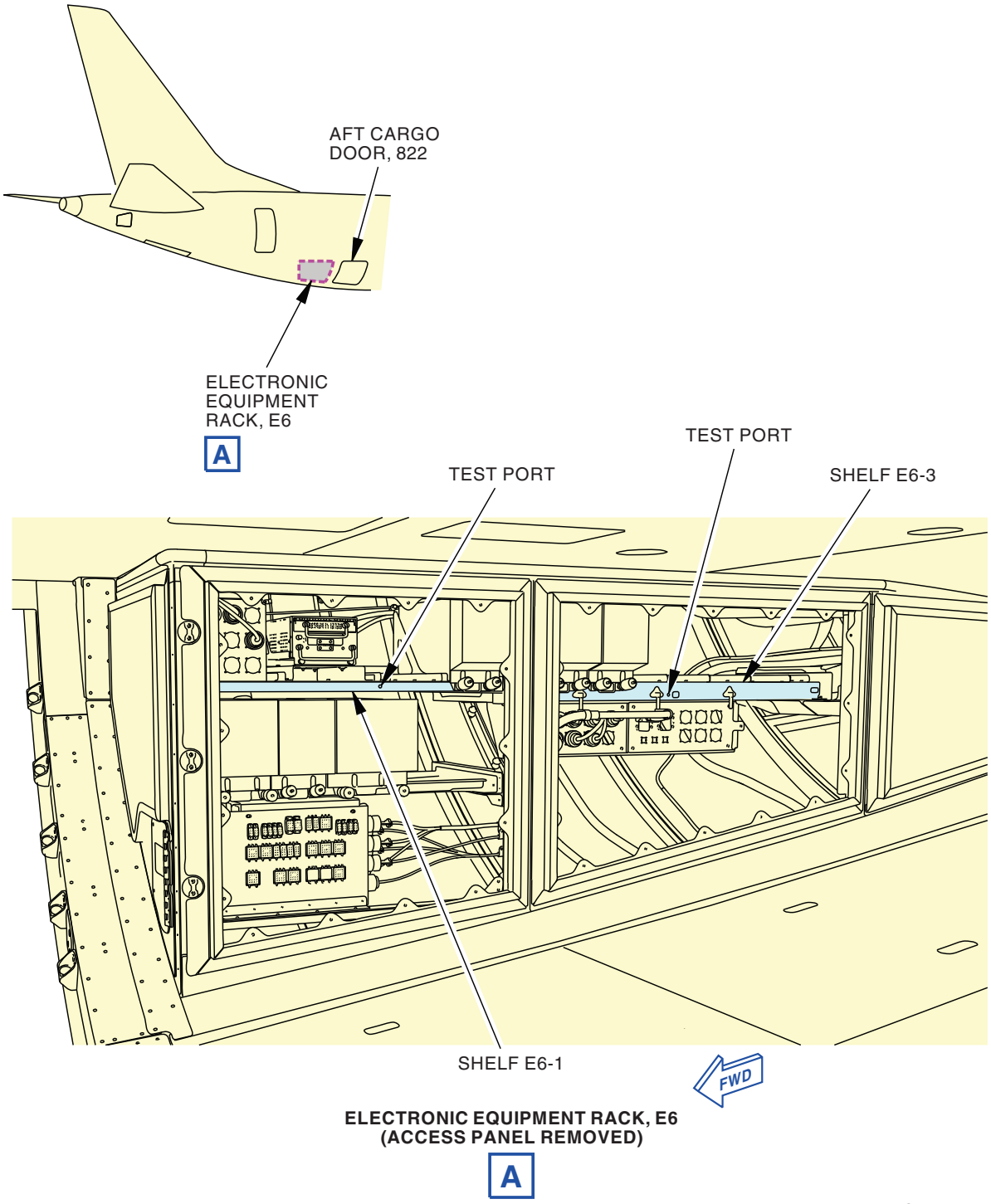
SUBTASK 21-27-00-420-002

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Equipment Rack Cooling Fan - Functional Test
Figure 501/21-27-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING AIR FILTER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the equipment cooling air filter

(2) An installation of the equipment cooling air filter.

TASK 21-27-01-000-801

2. Equipment Cooling Air Filter Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) You must remove electrical power from the airplane before you remove an equipment cooling air filter. This will make sure that the electrical/electronic equipment does not receive electrical power when the equipment cooling system is not in operation.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
50-11-01-000-801	Forward Cargo Compartment Forward Bulkhead Liner Removal (P/B 401)
52-31-00-580-801	Open the Cargo Door (P/B 201)
52-31-00-580-802	Close the Cargo Door (P/B 201)

C. Tools/Equipment

Reference	Description
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented
STD-6439	Coveralls - Hooded

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
B50280	Cleaner - Disinfectant - Zip-Chem Products Calla 1452	BAC5150
B50364	Cleaner - Disinfectant - Matrix #3	BAC5150
G00624	Bag - Plastic, General Purpose	
G50140	Gloves - Protective, Latex or Nitrile	

E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 21-27-01-860-001

CAUTION	MAKE SURE THAT YOU REMOVE ELECTRICAL POWER FROM THE AIRPLANE. IF YOU SUPPLY ELECTRICAL POWER TO THE ELECTRICAL/ELECTRONIC EQUIPMENT WHEN THE EQUIPMENT COOLING SYSTEM IS NOT IN OPERATION, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 21-27-01-010-001

(2) Open the forward-cargo compartment door (TASK 52-31-00-580-801).

SUBTASK 21-27-01-010-002

(3) Remove the forward right bulkhead liner in the forward cargo compartment (TASK 50-11-01-000-801).

SUBTASK 21-27-01-980-001

WARNING	PUT ON THE PERSONAL PROTECTIVE EQUIPMENT BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES AND INJURIES TO PERSONS.

(4) Before you touch the equipment cooling air filter [2] put on the following personal protective equipment listed below:

(a) hooded coveralls, STD-6439

(b) safety goggles, STD-6437

(c) particulate respirator, STD-6435

(d) protective gloves, G50140.

SUBTASK 21-27-01-500-001

(5) Select a plastic bag, G00624, larger than the filter and tape suitable for sealing the bag closed.

G. Equipment Cooling Air Filter Removal

SUBTASK 21-27-01-010-003

(1) If it is necessary, close the forward-cargo compartment door (TASK 52-31-00-580-802).

NOTE: The forward cargo compartment door can be lowered to provide access to remove the equipment cooling air filter.

SUBTASK 21-27-01-020-001

(2) Release the latch assemblies [3] that are on the top and the bottom of the filter housing assembly [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-01-020-002

WARNING DO NOT LET THE FILTER TOUCH YOUR SKIN. DO NOT SHAKE OR HIT THE FILTER. DO NOT LET THE FILTER FALL. DO NOT USE COMPRESSED AIR TO CLEAN THE FILTER OR FILTER HOUSING. THIS CAN CAUSE THE INFECTIOUS MATERIAL TO BECOME AIRBORNE. DISCARD THE FILTER IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(3) If possible, slide the plastic bag, G00624, over the filter when removing the equipment cooling air filter [2] from the filter housing assembly [1] to minimize contamination. Seal the bag closed with tape.

SUBTASK 21-27-01-980-002

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(4) Discard the bag with the filter. Refer to the airline, local, health, safety and regulatory procedures for disposal of material.

SUBTASK 21-27-01-010-004

(5) If necessary, open the forward-cargo compartment door (TASK 52-31-00-580-801).

H. Post-Filter Removal Cleanup

SUBTASK 21-27-01-670-002

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(1) Disinfect hard non-porous surfaces near the filter with a solution of 70% alcohol, B00130, Calla 1452 cleaner, B50280, or Matrix #3 cleaner, B50364.

NOTE: Follow the disinfectant manufacturer's instruction and refer to the Environmental Protection Agency (EPA) List N for applicable contact/dwell time.

NOTE: Refer to the EPA List N or local health authority guidance for required contact/dwell time for Isopropyl Alcohol to be used as a disinfectant.

SUBTASK 21-27-01-670-001

(2) Disinfect all tools used for filter replacement. Refer to the airline procedure and policies for the disinfecting of tools.

SUBTASK 21-27-01-020-005

WARNING DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(3) Remove the personal protective equipment and put them in a plastic bag, G00624. Seal the bag closed with tape.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(a) Discard the bag with the personal protective equipment. Refer to the airline, local health, safety and regulatory procedures for disposal of material.

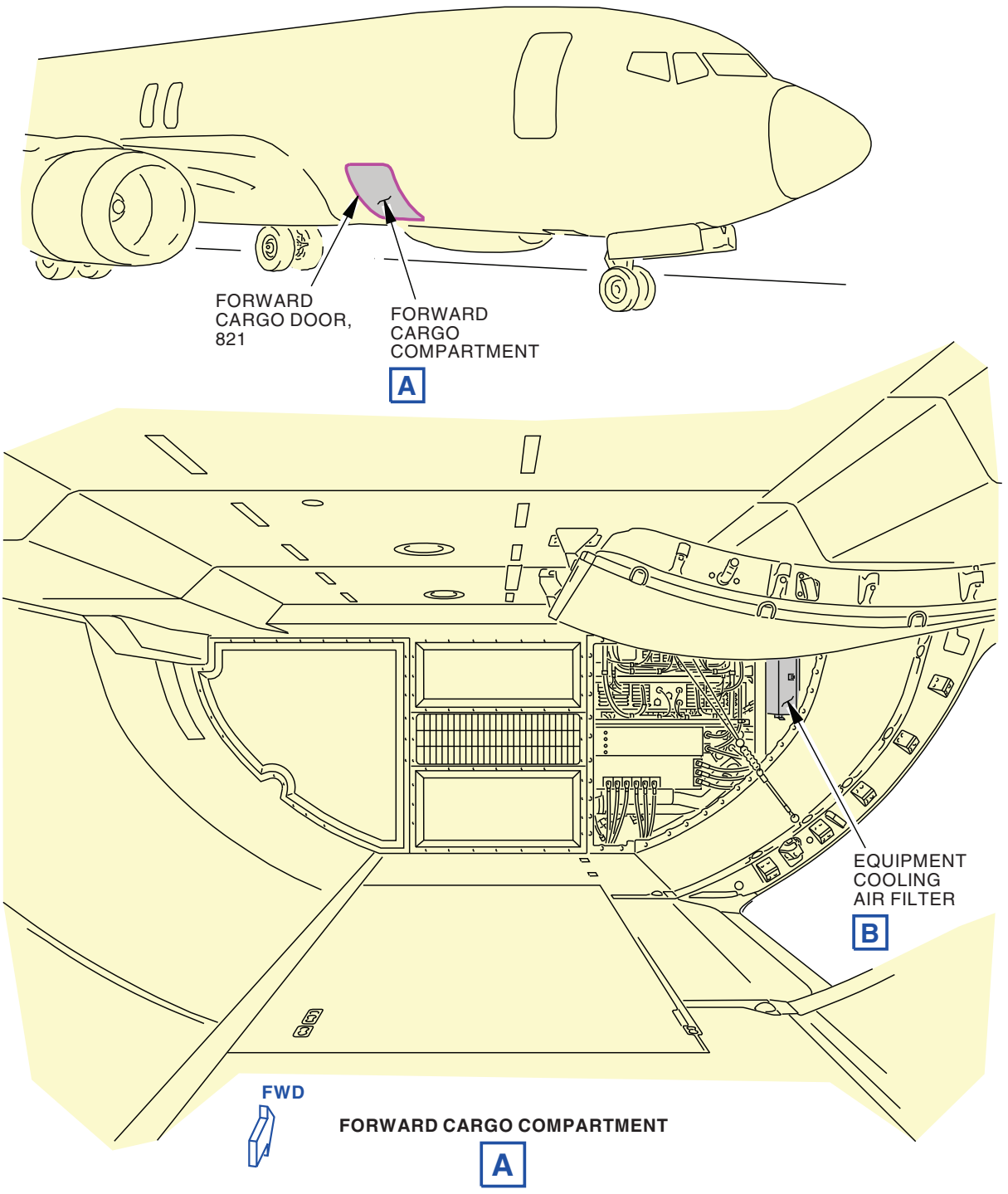
SUBTASK 21-27-01-110-001

WARNING CLEAN YOUR HANDS WITH SOAP AND FLUSH WITH HOT WATER. DIRTY HANDS WITH CONTAMINATION CAN CAUSE DISEASE AND ILLNESSES TO PERSONS.

(4) Clean your hands with soap and running hot water.

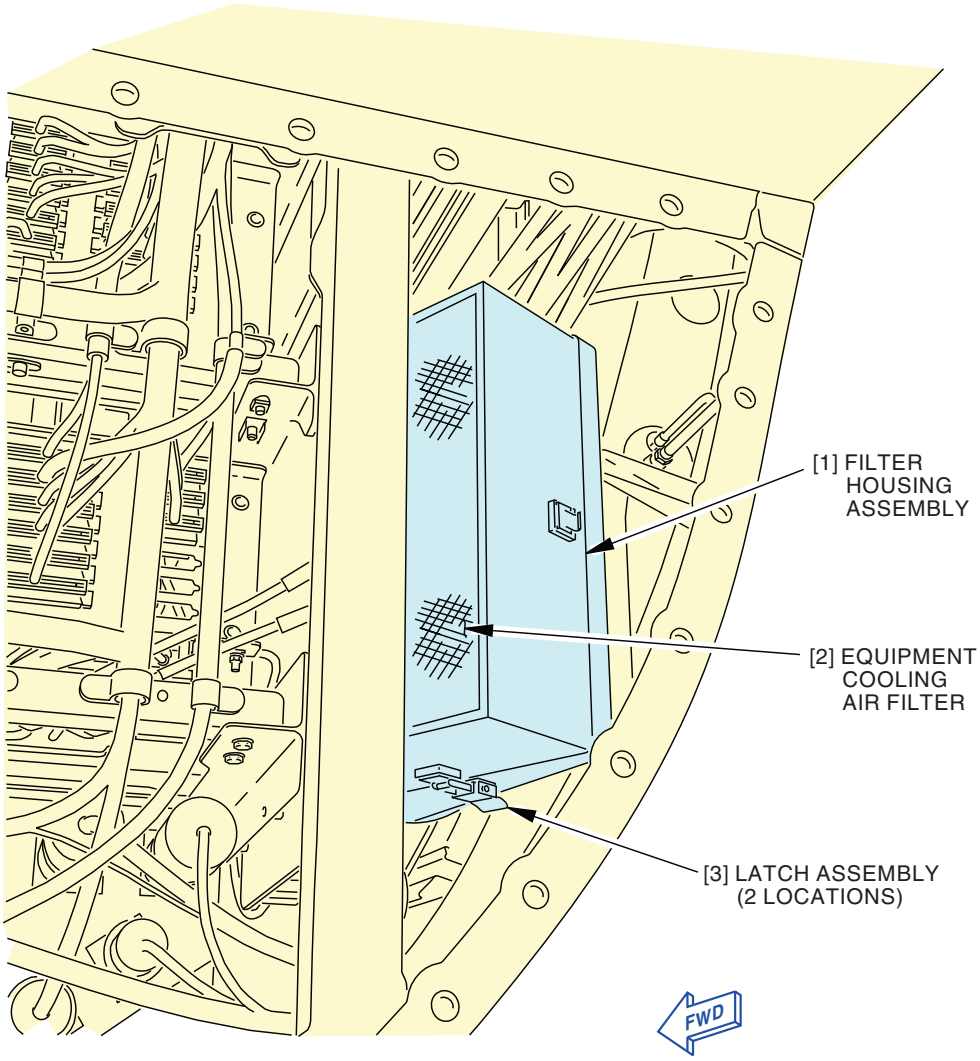
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Equipment Cooling Air Filter Installation
Figure 401/21-27-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EQUIPMENT COOLING AIR FILTER

B

2411313 S00061527591_V1

Equipment Cooling Air Filter Installation
Figure 401/21-27-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-27-01-400-801

3. Equipment Cooling Air Filter Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
50-11-01-400-801	Forward Cargo Compartment Forward Bulkhead Liner Installation (P/B 401)
52-31-00-580-801	Open the Cargo Door (P/B 201)
52-31-00-580-802	Close the Cargo Door (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Equipment cooling air filter	21-27-01-02-010	PTF ALL

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right

D. Prepare for the Installation

SUBTASK 21-27-01-010-005

(1) Open the forward-cargo compartment door (TASK 52-31-00-580-801).

E. Equipment Cooling Air Filter Installation

SUBTASK 21-27-01-400-001

(1) Get access to the filter housing assembly [1] at the forward right bulkhead in the cargo compartment.

SUBTASK 21-27-01-400-002

(2) If it is necessary, close the forward-cargo compartment door (TASK 52-31-00-580-802).

NOTE: The forward cargo compartment door can be lowered to provide access to install the equipment cooling air filter.

SUBTASK 21-27-01-020-003

(3) Put the equipment cooling air filter [2] in its position on the filter housing assembly [1].

SUBTASK 21-27-01-020-004

(4) Close the latch assemblies [3] that are on the top and the bottom of the filter housing assembly [1].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-01-410-001

(1) Install the forward right bulkhead liner in the forward cargo compartment (TASK 50-11-01-400-801).

SUBTASK 21-27-01-010-006

(2) Open the forward-cargo compartment door to allow technician to exit.

SUBTASK 21-27-01-410-002

(3) If it is necessary, close the forward-cargo compartment door.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING SUPPLY FAN - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the equipment cooling supply fans

(2) An installation of the equipment cooling supply fans.

B. There are two identical forward equipment cooling supply fans:

(1) The normal fan is installed in the top position

(2) The alternate fan is installed in the bottom position.

TASK 21-27-02-000-801

2. Forward Equipment Cooling Supply Fan Removal

(Figure 401)

A. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 21-27-02-860-004

CAUTION

MAKE SURE THAT YOU REMOVE ELECTRICAL POWER FROM THE AIRPLANE. IF YOU SUPPLY ELECTRICAL POWER TO THE ELECTRICAL/ELECTRONIC EQUIPMENT WHEN THE EQUIPMENT COOLING SYSTEM IS NOT IN OPERATION, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 21-27-02-010-001

(2) To get access to the equipment cooling supply fans [1], do this step:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Equipment Cooling Supply Fan Removal

SUBTASK 21-27-02-020-001

(1) Disconnect the electrical connector [6] from the equipment cooling supply fan [1].

(a) Make a record of the orientation of the electrical connector [6] so that you can install the equipment cooling supply fan [1] with the electrical connector [6] in the same position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-02-020-002

(2) For the normal equipment cooling supply fan [1], do these steps:

(a) Remove the nut [5], washer [3], washer [11], washer [10], and screw [2].

(b) Disconnect and stow the bonding jumper [4] and bonding jumper [9].

SUBTASK 21-27-02-020-008

(3) For the alternate equipment cooling supply fan [1], do these steps:

(a) Remove the nut [5], washer [3], washer [11], washer [10], and screw [2].

(b) Disconnect and stow the bonding jumper [9].

SUBTASK 21-27-02-020-003

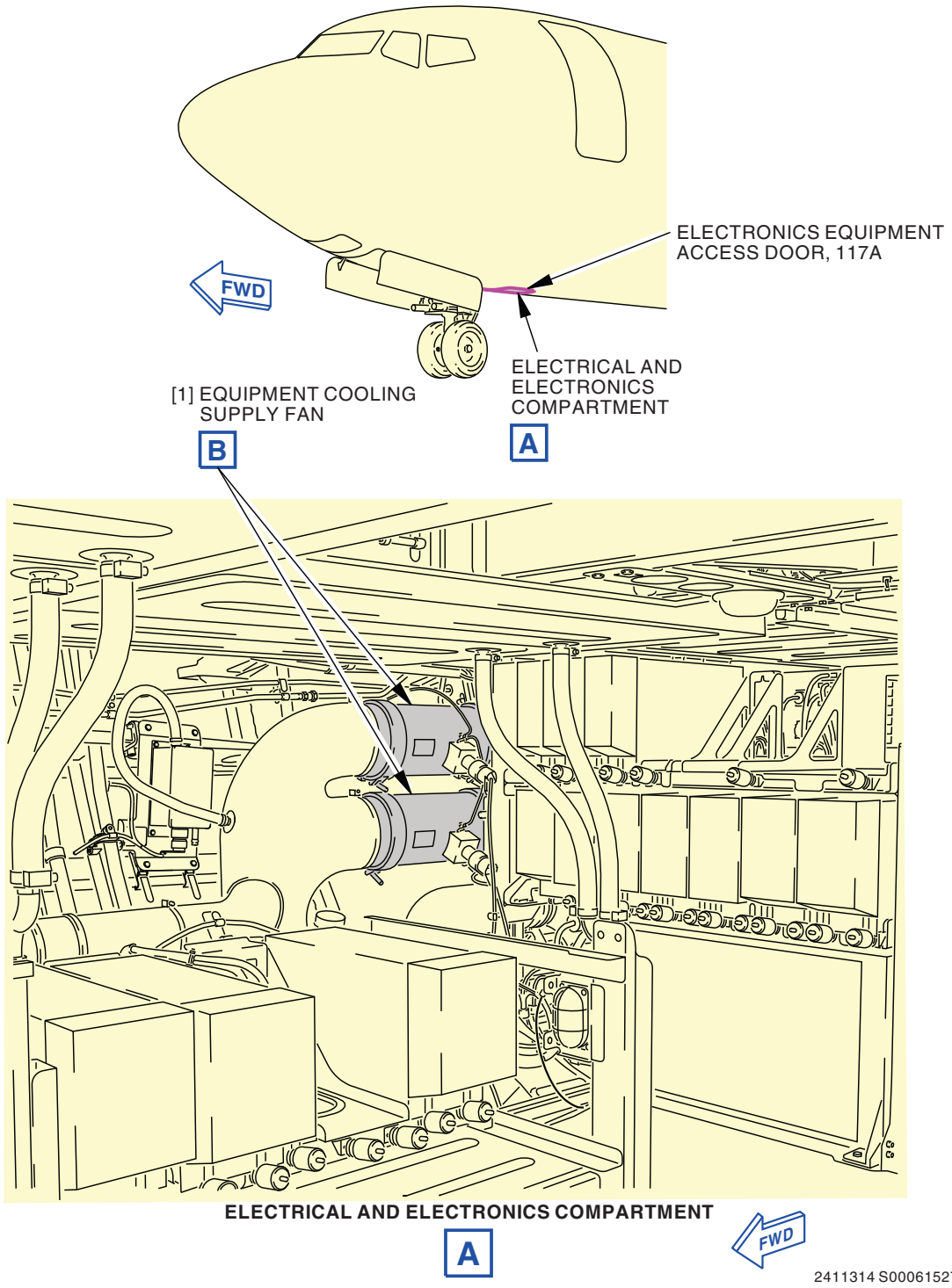
(4) Loosen the clamps [7].

SUBTASK 21-27-02-020-004

(5) Remove the equipment cooling supply fan [1] and the sleeves [8].

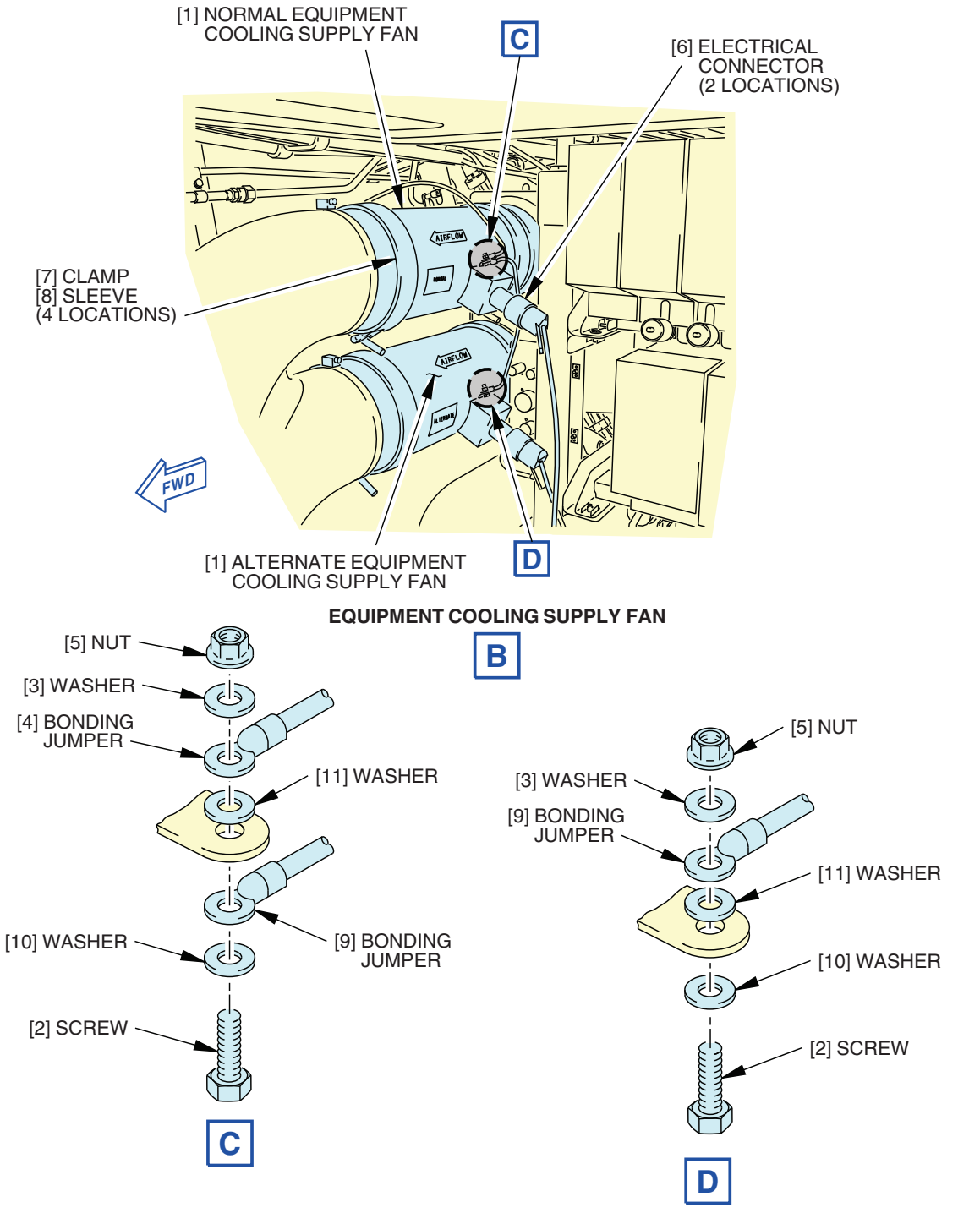
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Equipment Cooling Supply Fan Installation
Figure 401/21-27-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Equipment Cooling Supply Fan Installation
Figure 401/21-27-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-27-02-400-801

3. Forward Equipment Cooling Supply Fan Installation

(Figure 401)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Equipment cooling supply fan	21-27-02-01-130	PTF ALL

E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Equipment Cooling Supply Fan Installation

SUBTASK 21-27-02-420-001

(1) Put the equipment cooling supply fan [1] and the sleeves [8] in their position.

(a) Make sure that the fan impeller is aft of the fan check valve flappers until the flow of air is in the correct direction.

(b) Make sure that the orientation for the electrical connector [6] is oriented the same as it was on the removed equipment cooling supply fan [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-02-420-002

(2) Install the clamps [7].

(a) Tighten the clamps [7] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 21-27-02-420-003

(3) For the normal equipment cooling supply fan [1], do these steps:

(a) Use cotton wiper, G00034, moist in solvent, B00184, to final clean the mating surfaces of the airplane structure, washer [10], washer [11], bonding jumper [4], and bonding jumper [9].

(b) Install the screw [2], washer [10], bonding jumper [9], washer [11], bonding jumper [4], washer [3], and nut [5].

(c) Tighten the nut [5] to 35 in-lb (4.0 N·m) - 45 in-lb (5.1 N·m).

(d) Measure the electrical resistance between the bonding jumper [4] and airplane structure (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.01 ohm (10.0 milliohm) or less.

SUBTASK 21-27-02-420-008

(4) For the alternate equipment cooling supply fan [1], do these steps:

(a) Use cotton wiper, G00034, moist in solvent, B00184, to final clean the mating surfaces of the airplane structure, washer [10], washer [11], and bonding jumper [9].

(b) Install the screw [2], washer [10], washer [11], bonding jumper [9], washer [3], and nut [5].

(c) Tighten the nut [5] to 35 in-lb (4.0 N·m) - 45 in-lb (5.1 N·m).

(d) Measure the electrical resistance between the bonding jumper [4] and airplane structure (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.01 ohm (10.0 milliohm) or less.

SUBTASK 21-27-02-420-004

(5) Connect the electrical connector [6] to the equipment cooling supply fan [1].

H. Equipment Cooling Supply Fan Installation Test

SUBTASK 21-27-02-860-002

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-27-02-710-001

(2) Do these steps if you installed the normal equipment cooling supply fan [1]:

(a) Make sure that the EQUIP COOLING SUPPLY switch, on the equipment cooling panel, is in the NORMAL position.

(b) Make sure that the normal equipment cooling supply fan [1] operates.

NOTE: Put your hand on the equipment cooling supply fan to feel that it operates.

(c) Examine the normal equipment cooling supply fan [1] connections for leaks.

1) If a leak is found, repair the normal equipment cooling supply fan [1] connections.

SUBTASK 21-27-02-710-002

(3) Do these steps if you installed the alternate equipment cooling supply fan [1]:

(a) Make sure that the EQUIP COOLING SUPPLY switch, on the equipment cooling panel, is in the ALTERNATE position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the alternate equipment cooling supply fan [1] operates.

NOTE: Put your hand on the equipment cooling supply fan to feel that it operates.

(c) Examine the alternate equipment cooling supply fan [1] connections for leaks.

1) If a leak is found, repair the alternate equipment cooling supply fan [1] connections.

(d) Make sure that the EQUIP COOLING SUPPLY switch, on the equipment cooling panel, to the NORMAL position.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-02-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING LOW FLOW SENSORS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the equipment cooling supply or exhaust low flow sensor.

(2) An installation of the equipment cooling supply or exhaust low flow sensor.

(3) A cleaning of the equipment cooling supply or exhaust low flow sensor.

B. The equipment cooling low flow sensors are installed in the forward EE bay.

TASK 21-27-03-000-801

2. Equipment Cooling Low Flow Sensors Removal

(Figure 201)

A. General

(1) You must remove electrical power from the airplane before you remove one of the equipment cooling low flow sensors. This will make sure that the electrical/electronic equipment does not receive electrical power when the equipment cooling system is not in operation.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Prepare for the Removal

SUBTASK 21-27-03-860-001

(1) To get access to the equipment cooling supply or exhaust low flow sensor [4], do this step:

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 21-27-03-860-002

CAUTION

MAKE SURE YOU REMOVE ELECTRICAL POWER FROM THE AIRPLANE. IF YOU SUPPLY ELECTRICAL POWER TO THE ELECTRICAL/ELECTRONIC EQUIPMENT WHEN THE EQUIPMENT COOLING SYSTEM IS NOT IN OPERATION, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(2) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Equipment Cooling Low Flow Sensors Removal

SUBTASK 21-27-03-020-001

(1) Disconnect the electrical connector [3].

SUBTASK 21-27-03-020-002

(2) Remove the lockwire [2] (MS20995C32 lockwire, G01048).

SUBTASK 21-27-03-020-003

CAUTION

USE TWO WRENCHES WHEN YOU REMOVE THE SENSOR, ONE ON THE BOSS OF THE DUCT AND ONE ON THE SENSOR. IF YOU DO NOT USE TWO WRENCHES WHEN YOU REMOVE THE SENSOR, DAMAGE TO THE DUCT CAN OCCUR.

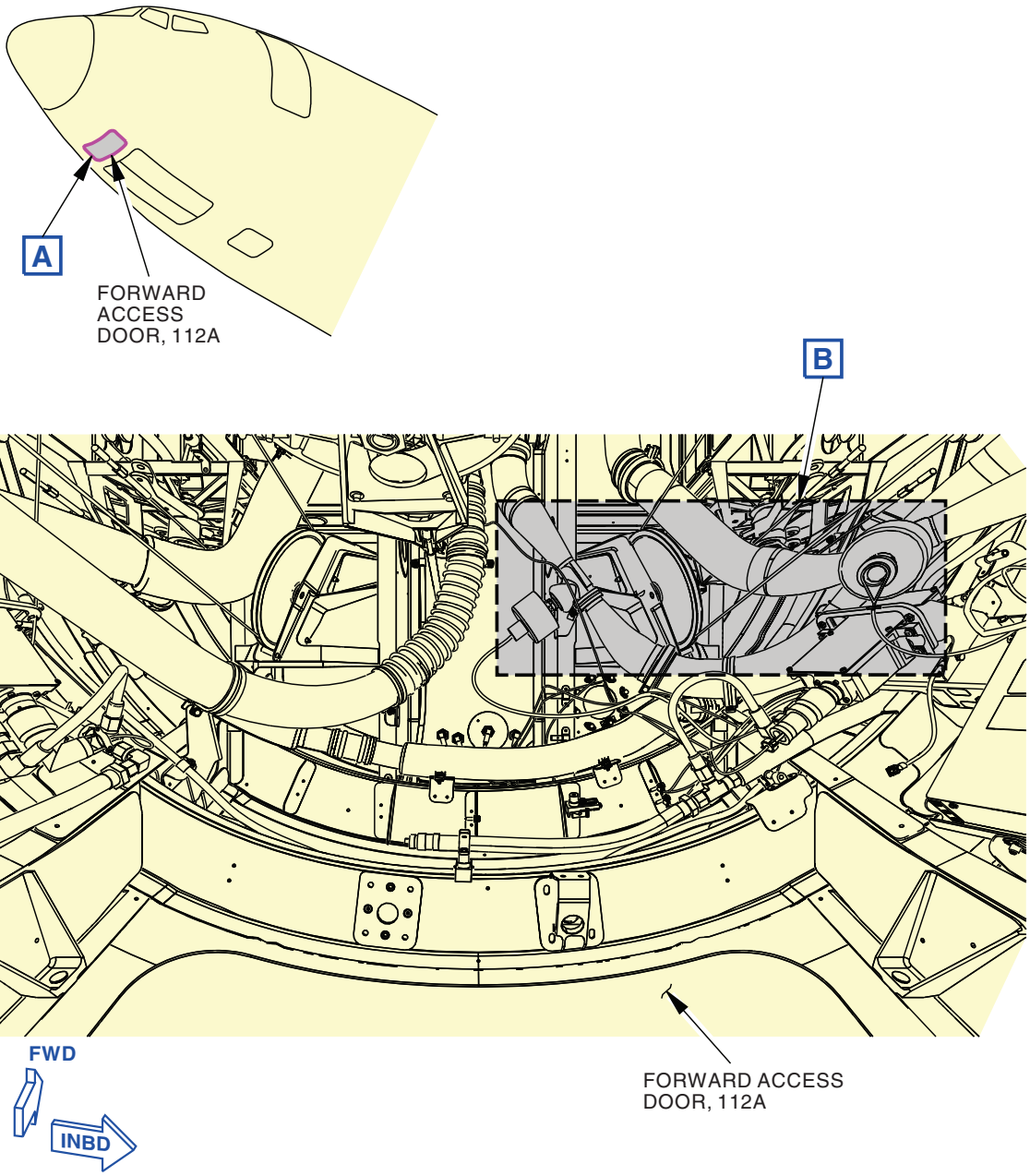
(3) Remove the equipment cooling low flow sensor [4].

SUBTASK 21-27-03-020-004

(4) Remove the packing [1] and discard.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



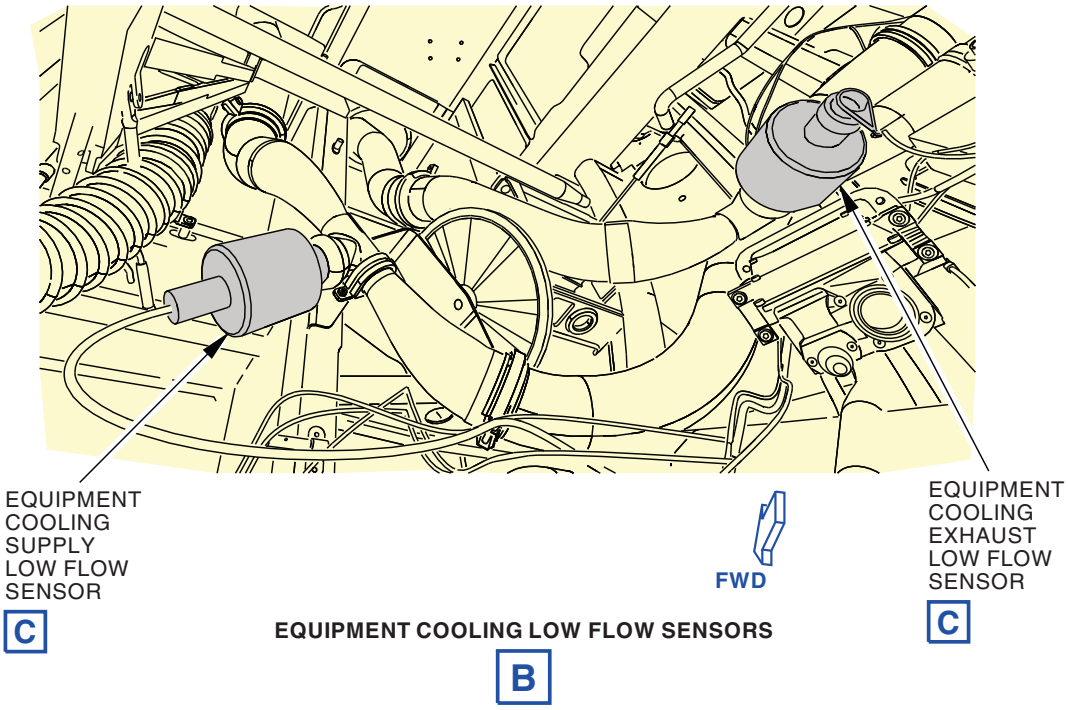
VIEW WHEN YOU ARE LOOKING UP THROUGH
THE FORWARD ACCESS DOOR, 112A

A

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Equipment Cooling Low Flow Sensors Installation
Figure 201/21-27-03-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Equipment Cooling Low Flow Sensors Installation
Figure 201/21-27-03-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-27-03-400-801

3. Equipment Cooling Low Flow Sensors Installation

(Figure 201)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51827	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 6 Inches Long	BACC13AT3K6

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Packing	21-27-03-01-010	PTF ALL
		21-27-03-02-020	PTF ALL
4	Low flow sensor	21-27-03-01-015	PTF ALL
		21-27-03-02-025	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

F. Equipment Cooling Low Flow Sensors Installation

SUBTASK 21-27-03-640-001

(1) Apply compound, D50004, to the threads of the equipment cooling low flow sensor [4].

SUBTASK 21-27-03-420-001

(2) Put a new packing [1] on the equipment cooling low flow sensor [4].

SUBTASK 21-27-03-420-002

CAUTION

USE TWO WRENCHES WHEN YOU TIGHTEN THE SENSOR, ONE ON THE BOSS OF THE DUCT AND ONE ON THE SENSOR. IF YOU DO NOT USE TWO WRENCHES WHEN YOU TIGHTEN THE SENSOR, DAMAGE TO THE DUCT CAN OCCUR.

(3) Install the equipment cooling low flow sensor [4] until the packing [1] makes contact with the duct, then tighten a maximum of 3/4 turn.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-03-420-003

(4) Install the lockwire [2] (MS20995C32 lockwire, G01048) with double twist method or safety cable kit, G51827.

SUBTASK 21-27-03-420-004

(5) Connect the electrical connector [3].

G. Equipment Cooling Low Flow Sensors Installation Test

SUBTASK 21-27-03-860-006

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-27-03-860-004

(2) Make sure that the equipment cooling supply and exhaust OFF lights on the equipment cooling Panel are not on.

SUBTASK 21-27-03-860-007

(3) Make sure that the EQUIP COOLING SUPPLY switch, on the equipment cooling panel, is in the NORM position.

SUBTASK 21-27-03-860-008

(4) Make sure that the EQUIP COOLING EXHAUST switch, on the equipment cooling panel, is in the NORM position.

SUBTASK 21-27-03-710-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

CAUTION

MAKE SURE THAT YOU CLOSE THE FAN CIRCUIT BREAKER IN LESS THAN 5 MINUTES AFTER YOU OPEN IT. IF YOU KEEP THE CIRCUIT BREAKER OPEN MORE THAN 5 MINUTES, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(5) Do these steps to do a check of the equipment cooling supply low flow sensor [4]:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

(c) Make sure that the equipment cooling supply OFF light comes on after approximately 20 to 60 seconds.

(d) Close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
D	10	C00934	EQPT CLG SPLY FAN PWR-NORM

(e) Make sure that the equipment cooling supply OFF light goes out within 20 seconds.

SUBTASK 21-27-03-710-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

CAUTION

MAKE SURE THAT YOU CLOSE THE FAN CIRCUIT BREAKER IN LESS THAN 5 MINUTES AFTER YOU OPEN IT. IF YOU KEEP THE CIRCUIT BREAKER OPEN MORE THAN 5 MINUTES, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(6) Do these steps to do a check of the equipment cooling exhaust low flow sensor [4]:

(a) Open this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

(b) Make sure that the equipment cooling exhaust OFF light comes on after approximately 20 to 60 seconds.

(c) Close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
E	1	C00836	EQPT CLG EXH FAN PWR-NORM

(d) Make sure that the equipment cooling exhaust OFF light goes out within 20 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-03-410-001

(1) Close these access panels:

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 21-27-03-400-802

4. Equipment Cooling Low Flow Sensors Cleaning

(Figure 201)

A. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

C. Equipment Cooling Low Flow Sensors Cleaning

SUBTASK 21-27-03-640-002

(1) Remove the equipment cooling low flow sensor [4]. To remove the equipment cooling low flow sensor [4], do this task: Equipment Cooling Low Flow Sensors Removal, TASK 21-27-03-000-801.

SUBTASK 21-27-03-420-005

(2) Clean the probe of the sensor with a cotton wiper, G00034 that is moist with alcohol, B00130.

SUBTASK 21-27-03-420-006

(3) Install the equipment cooling low flow sensor. To install the equipment cooling low flow sensor, do this task: Equipment Cooling Low Flow Sensors Installation, TASK 21-27-03-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING EXHAUST FAN - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the equipment cooling exhaust fans.

(2) An installation of the equipment cooling exhaust fans.

(3) A removal of the aft equipment cooling exhaust fan.

(4) An installation of the aft equipment cooling exhaust fan.

B. There are two identical equipment cooling exhaust fans installed in the electronic equipment center:

(1) The normal exhaust fan is installed in the forward position.

(2) The alternate exhaust fan is installed in the aft position.

TASK 21-27-04-000-801

2. Equipment Cooling Exhaust Fan Removal

(Figure 401)

A. General

(1) This task provides the instructions how to remove the normal and alternate equipment cooling exhaust fans from the electronic equipment center.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 21-27-04-860-001

CAUTION

MAKE SURE THAT YOU REMOVE ELECTRICAL POWER FROM THE AIRPLANE. IF YOU SUPPLY ELECTRICAL POWER TO THE ELECTRICAL/ELECTRONIC EQUIPMENT WHEN THE EQUIPMENT COOLING SYSTEM IS NOT IN OPERATION, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 21-27-04-010-001

(2) To get access to the equipment cooling exhaust fans, do this step:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-04-010-002

(3) Remove the access panel [11] on the top of the raised platform just aft and right of the aircraft center line as follows:

(a) Remove the screws [10] that attach the access panel [11] to the top of the raised platform.

1) Remove the access panel [11].

F. Equipment Cooling Exhaust Fan Removal

SUBTASK 21-27-04-020-001

(1) Disconnect the electrical connector [6] from the normal exhaust fan [1] or alternate exhaust fan [9].

SUBTASK 21-27-04-020-002

(2) Disconnect the bonding jumper [4] as follows:

(a) Remove the nut [5], washers [3], and screw [2].

(b) Disconnect and stow the bonding jumper [4].

NOTE: If you are removing the normal exhaust fan [1], also disconnect the bonding jumper [4] that connects the normal and the alternate exhaust fan [9] together.

SUBTASK 21-27-04-020-003

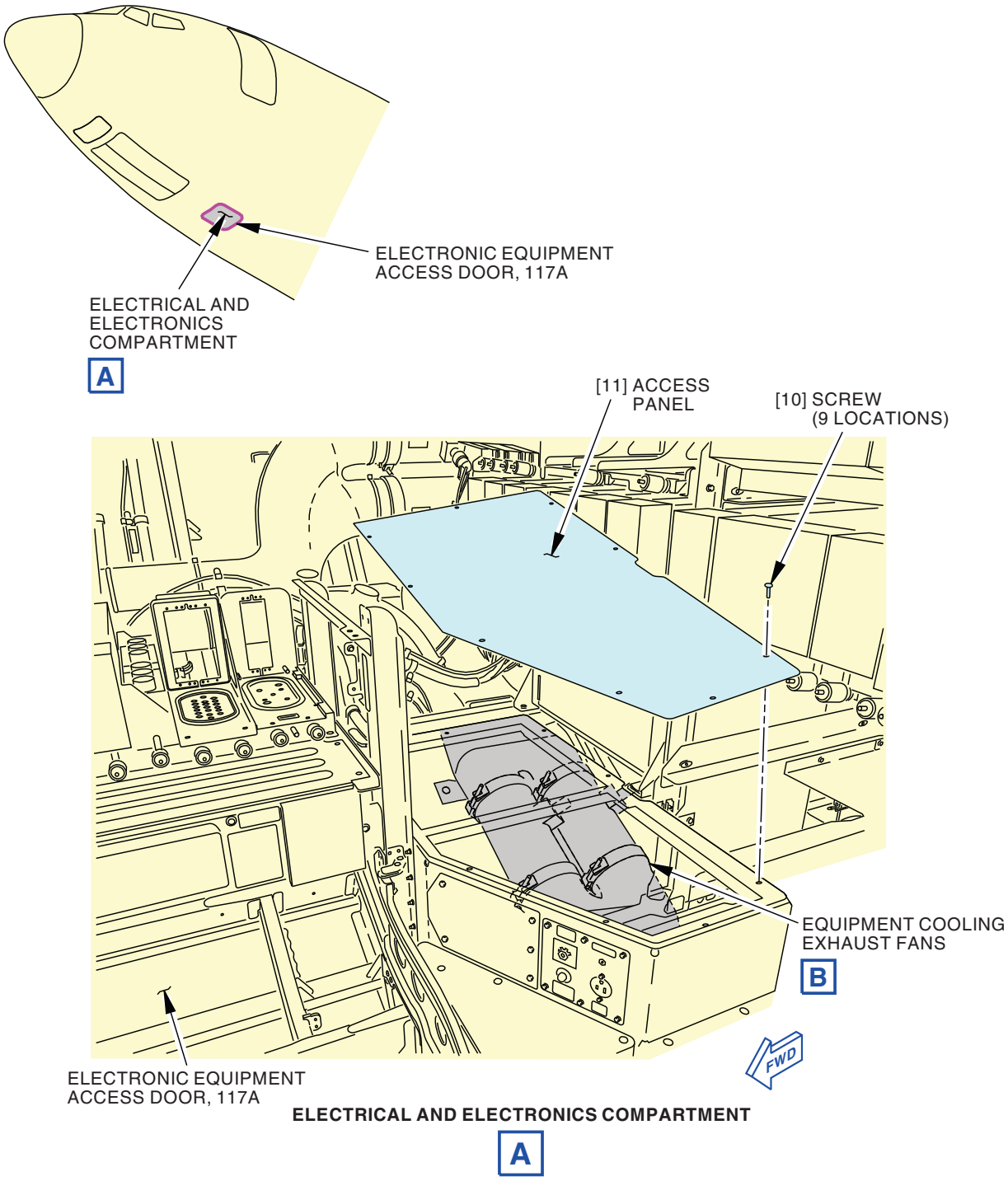
(3) Loosen the clamps [7].

SUBTASK 21-27-04-020-004

(4) Remove the normal exhaust fan [1] or alternate exhaust fan [9] and the sleeves [8].

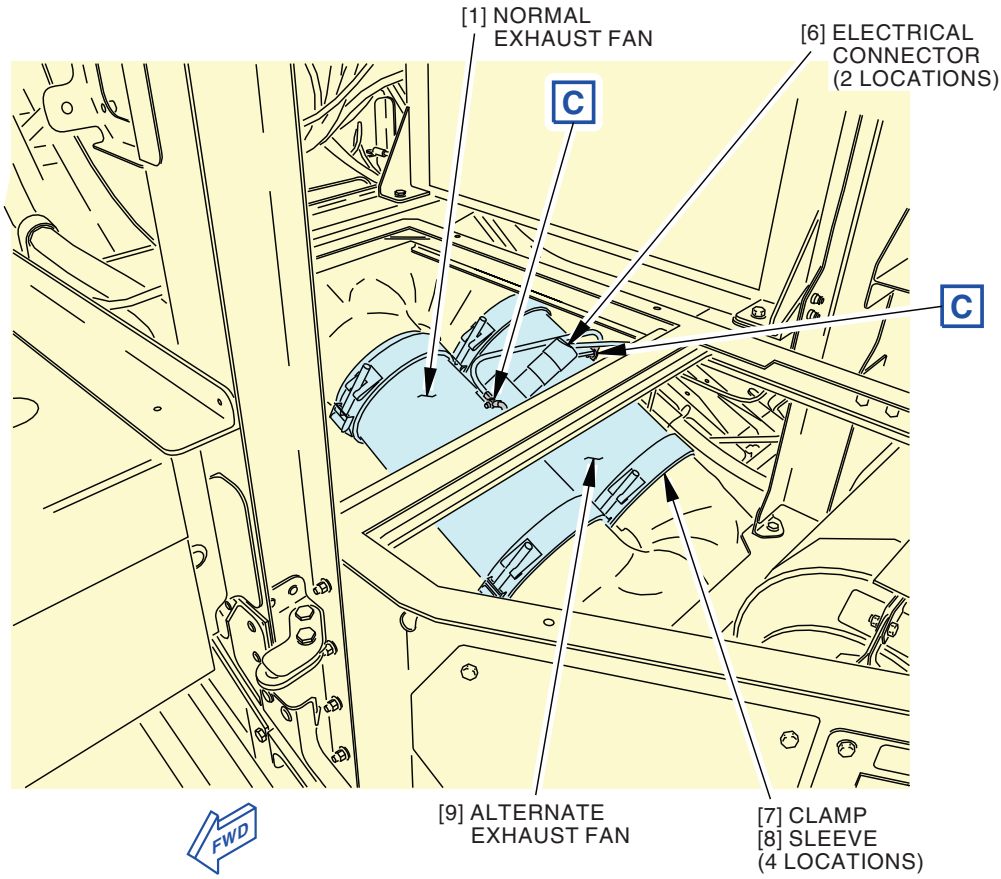
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



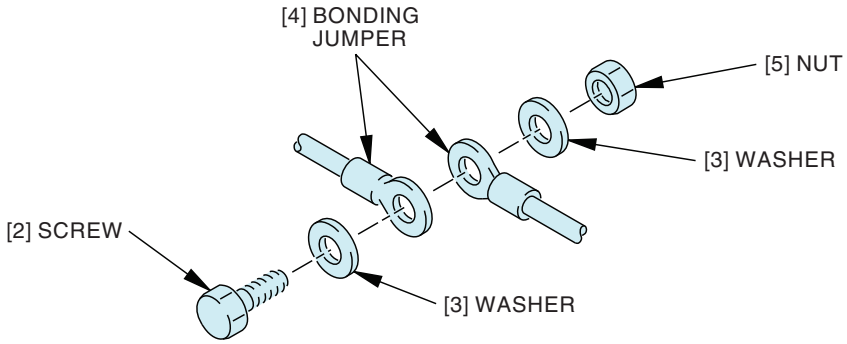
Equipment Cooling Exhaust Fan Installation
Figure 401/21-27-04-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EQUIPMENT COOLING EXHAUST FANS

B



NOTE:

EQUIPMENT COOLING EXHAUST SMOKE DETECTOR REMOVED FOR CLARITY

(EXAMPLE)

C

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Equipment Cooling Exhaust Fan Installation
Figure 401/21-27-04-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-27-04-400-801

3. Equipment Cooling Exhaust Fan Installation

(Figure 401)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Equipment Cooling Exhaust Fan Installation

SUBTASK 21-27-04-420-001

(1) Put the normal exhaust fan [1] or alternate exhaust fan [9] and the sleeves [8] in their position.

SUBTASK 21-27-04-420-002

(2) Install the clamps [7] and tighten to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 21-27-04-420-003

(3) Do the steps that follow to install the bonding jumper [4]:

(a) Make sure that the bonding jumper [4], two washers [3], nut [5], and screw [2] are clean.

(b) Put the bonding jumper [4] in its position.

(c) If you are installing the normal exhaust fan [1], also install the bonding jumper [4] that connects the normal and the alternate fans together.

(d) Install the nut [5], washers [3], and screw [2].

SUBTASK 21-27-04-420-004

(4) Connect the electrical connector [6] to the normal exhaust fan [1] or alternate exhaust fan [9].

E. Equipment Cooling Exhaust Fan Installation Test

SUBTASK 21-27-04-860-002

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-27-04-710-001

(2) Do these steps if you installed the normal exhaust fan [1]:

(a) Make sure that the EQUIP COOLING exhaust switch, on the Equipment Cooling Panel, is in the NORM position.

(b) Make sure that the normal exhaust fan [1] operates.

(c) Put your hand on the normal exhaust fan [1] to feel that it operates.

(d) Examine the fan connections for leaks.

1) If a leak is found, repair the normal exhaust fan [1] connections.

SUBTASK 21-27-04-710-002

(3) Do these steps if you installed the alternate exhaust fan [9]:

(a) On the P5 panel, make sure that the EQUIP COOLING exhaust switch, on the Equipment Cooling Panel, is in the ALTN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the alternate exhaust fan [9] operates.

(c) Put your hand on the fan to feel that it operates.

(d) Examine the fan connections for leaks.

1) If a leak is found, repair the alternate exhaust fan [9] connections.

(e) Put the EQUIP COOLING exhaust switch, on the Equipment Cooling Panel, to the NORM position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-04-410-001

(1) Install the access panel [11] that you removed on the top of the raised platform as follows:

(a) Install the access panel [11].

1) Install the screws [10] that attach the access panel [11] to the top of the raised platform.

SUBTASK 21-27-04-410-002

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 21-27-04-000-802

4. Aft Equipment Cooling Exhaust Fan Removal

(Figure 402)

A. References

Reference	Title
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
822	Aft Cargo Door

D. Prepare for the Removal

SUBTASK 21-27-04-040-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	8	C04069	EQUIPMENT COOLING E6 RACK FAN POWER
D	10	C04070	EQUIPMENT COOLING E6 RACK FAN CONTROL

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

SUBTASK 21-27-04-010-003

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 21-27-04-010-004

(3) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

SUBTASK 21-27-04-010-005

(4) Remove the electrical equipment rack, E6, aft access panel.

E. Aft Equipment Cooling Supply Fan Removal

SUBTASK 21-27-04-020-007

(1) Disconnect the electrical connector [13] from the aft equipment cooling exhaust fan [12].

SUBTASK 21-27-04-020-008

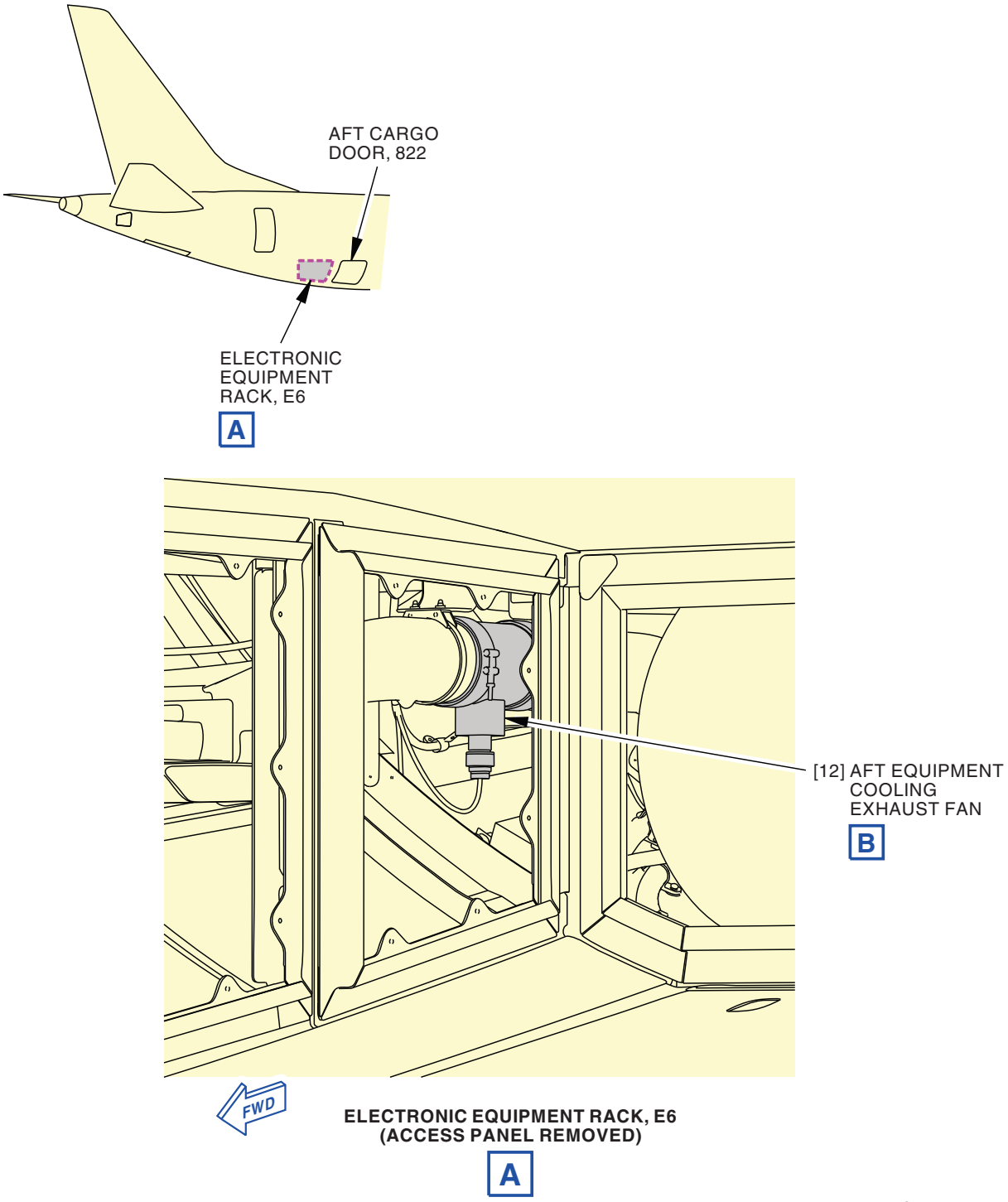
(2) Loosen clamps [14].

SUBTASK 21-27-04-020-009

(3) Remove the aft equipment cooling exhaust fan [12] and sleeves [15].

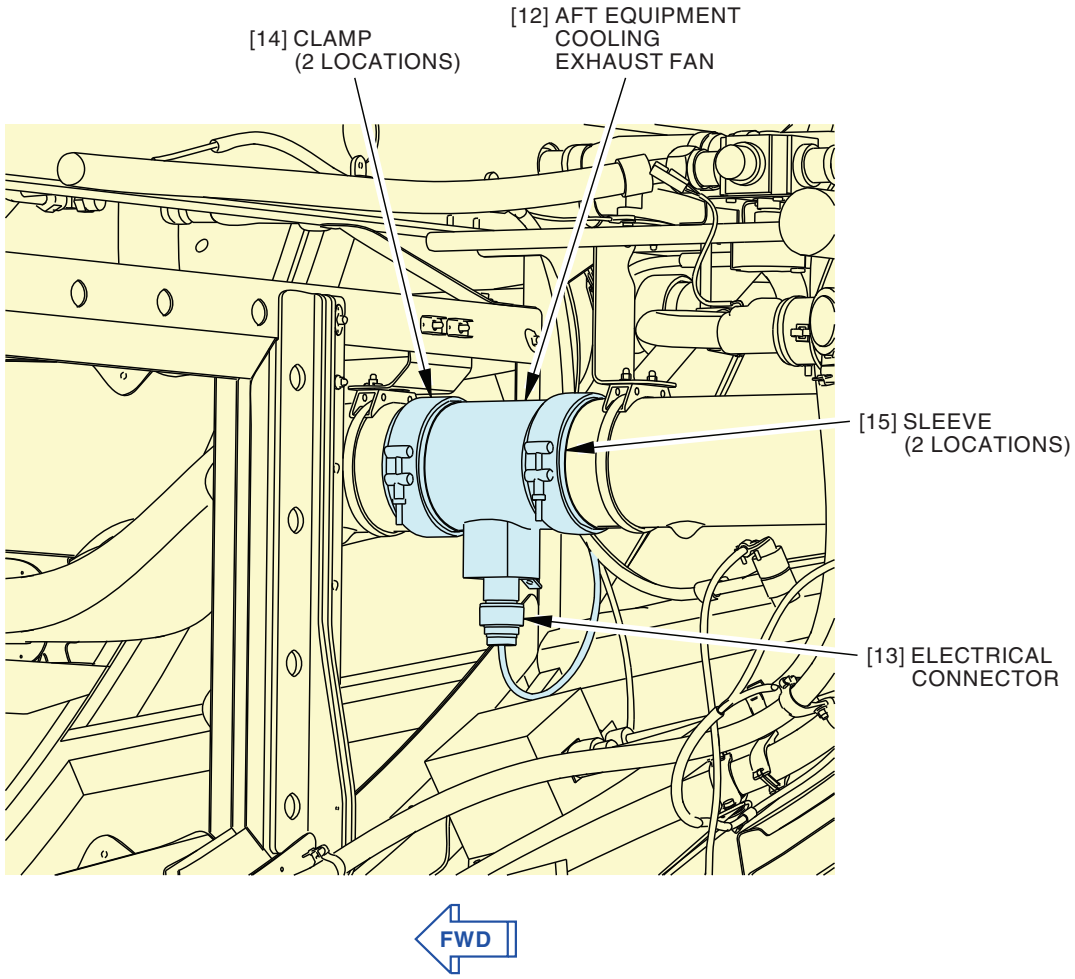
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Equipment Cooling Exhaust Fan Installation
Figure 402/21-27-04-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AFT EQUIPMENT COOLING EXHAUST FAN

B

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Aft Equipment Cooling Exhaust Fan Installation
Figure 402/21-27-04-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-27-04-400-802

5. Aft Equipment Cooling Exhaust Fan Installation

(Figure 402)

A. References

Reference	Title
21-27-00-700-808	Aft Equipment Cooling Fan - Functional Test (P/B 501)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
12	Aft equipment cooling exhaust fan	21-27-04-02-120	PTF ALL

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
822	Aft Cargo Door

E. Aft Equipment Cooling Supply Fan Installation

SUBTASK 21-27-04-420-009

(1) Put the aft equipment cooling exhaust fan [12] and sleeves [15] in their position.

(a) Align the electrical connector [13] with the protrusion of the aft duct.

SUBTASK 21-27-04-420-010

(2) Install the clamps [14].

(a) Tighten the clamps [14] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 21-27-04-420-011

(3) Connect the electrical connector [13] to the aft equipment cooling exhaust fan [12].

SUBTASK 21-27-04-440-001

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
C	7	C01537	VOICE RCDR RELAY
D	7	C00107	VOICE RCDR

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	2	C00857	COMMUNICATIONS HF 2

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	8	C04069	EQUIPMENT COOLING E6 RACK FAN POWER
D	10	C04070	EQUIPMENT COOLING E6 RACK FAN CONTROL

This circuit breaker is inoperative and should remain open:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
D	6	C01948	VOICE RCDR/RIPS (INOP)

F. Aft Equipment Cooling Supply Fan Installation Test

SUBTASK 21-27-04-710-003

(1) Do this task: Aft Equipment Cooling Fan - Functional Test, TASK 21-27-00-700-808.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-04-410-003

(1) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 21-27-04-410-004

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING EXHAUST FAN - CLEANING/PAINTING

1. General

A. This procedure has a task to clean the equipment cooling exhaust fans.

B. There are two identical equipment cooling exhaust fans installed in the electronic equipment center.

(1) The normal exhaust fan is installed in the forward position.

(2) The alternate exhaust fan is installed in the aft position.

TASK 21-27-04-110-801

2. Equipment Cooling Exhaust Fan Cleaning

A. General

(1) The equipment cooling exhaust fans may not operate correctly if they are dirty.

(2) This task provides instructions to clean only the impellers of the equipment cooling exhaust fans and the air passages.

(3) You must do an overhaul of the equipment cooling exhaust fans if it is necessary to disassemble them for a more thorough cleaning.

(4) This task is applicable for all of the fans in the equipment cooling system.

B. References

Reference	Title
21-27-04-000-801	Equipment Cooling Exhaust Fan Removal (P/B 401)
21-27-04-400-801	Equipment Cooling Exhaust Fan Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle

D. Consumable Materials

Reference	Description	Specification
B00039	Cleaner - Brulin Formula 815MX	
B00074	Solvent - Degreasing	MIL-PRF-680 (Supersedes P-D-680)
G50138	Cloth - Soft Cotton	

E. Prepare for the Cleaning

SUBTASK 21-27-04-020-005

(1) Remove the equipment cooling exhaust fan (TASK 21-27-04-000-801).

SUBTASK 21-27-04-930-001

(2) Make marks on the impellers so you can install them in the same position they were in when you removed them.

SUBTASK 21-27-04-030-001

(3) Remove the nut and washer at each impeller.

SUBTASK 21-27-04-020-006

(4) Move the impeller from side to side to remove it from the rotor shaft.

NOTE: You can lightly hit the rotor shaft with a rubber mallet while you pull on the impeller to make it easy to remove.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Equipment Cooling Exhaust Fan Cleaning

SUBTASK 21-27-04-160-001

(1) Soak the impellers in Brulin Formula 815MX cleaner, B00039, or solvent, B00074, until contamination loosens.

NOTE: Use the solvent, B00074, if the Brulin Formula 815MX cleaner, B00039, does not fully clean the impellers.

SUBTASK 21-27-04-160-002

(2) Remove the contamination with a soft bristle brush, STD-123.

SUBTASK 21-27-04-160-003

CAUTION

MAKE SURE TO DRY THE MOTOR FULLY WITH WARM AIR OR IN AN OVEN IF WATER GETS IN IT. IF THE MOTOR IS NOT FULLY DRIED BEFORE IT OPERATES, DAMAGE CAN OCCUR.

(3) Flush the impellers fully with clean water if you used Brulin Formula 815MX cleaner, B00039.

SUBTASK 21-27-04-160-004

(4) Dry the impellers with a soft cotton cloth, G50138, or with dry, compressed air.

SUBTASK 21-27-04-420-005

(5) Install the keys and the impellers on the motor shaft.

NOTE: Use the marks on the impellers to install them.

SUBTASK 21-27-04-420-006

(6) Make sure that the key is correctly engaged in each impeller.

SUBTASK 21-27-04-420-007

(7) Install the washers and self-locking nuts.

SUBTASK 21-27-04-430-001

(8) Tighten the nuts to 40 in-lb (5 N·m) to 50 in-lb (6 N·m).

SUBTASK 21-27-04-420-008

(9) Install the equipment cooling exhaust fan (TASK 21-27-04-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING MUFFLER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the equipment cooling muffler.

(2) An installation of the equipment cooling muffler.

B. There is an equipment cooling muffler installed behind galley No. 1. The muffler quiets the air being drawn from the P5 and P6 panels.

TASK 21-27-07-000-801

2. Equipment Cooling Muffler Removal

(Figure 401)

A. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)
25-31-11-000-801	Galley No. 1 - Removal (P/B 401)

B. Tools/Equipment

Reference	Description
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented
STD-6439	Coveralls - Hooded

C. Consumable Materials

Reference	Description	Specification
G50140	Gloves - Protective, Latex or Nitrile	

D. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

E. Prepare for the Removal

SUBTASK 21-27-07-860-001

CAUTION

MAKE SURE THAT YOU REMOVE ELECTRICAL POWER FROM THE AIRPLANE. IF YOU SUPPLY ELECTRICAL POWER TO THE ELECTRICAL/ELECTRONIC EQUIPMENT WHEN THE EQUIPMENT COOLING SYSTEM IS NOT IN OPERATION, THE ELECTRICAL/ELECTRONIC EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 21-27-07-010-001

(2) Remove the No. 1 galley (TASK 25-31-11-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-07-940-001

WARNING

PUT ON THE PERSONAL PROTECTIVE EQUIPMENT BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES AND INJURIES TO PERSONS.

(3) Before you touch the muffler/duct assembly, put on the personal protective equipment listed below:

(a) hooded coveralls, STD-6439

(b) safety goggles, STD-6437

(c) particulate respirator, STD-6435

(d) protective gloves, G50140

F. Equipment Cooling Muffler Removal

SUBTASK 21-27-07-010-002

(1) Do the steps that follow to remove the equipment cooling muffler [1]:

(a) Loosen the clamp [3].

(b) Slide the flex hose [2] away from the end of the muffler [1].

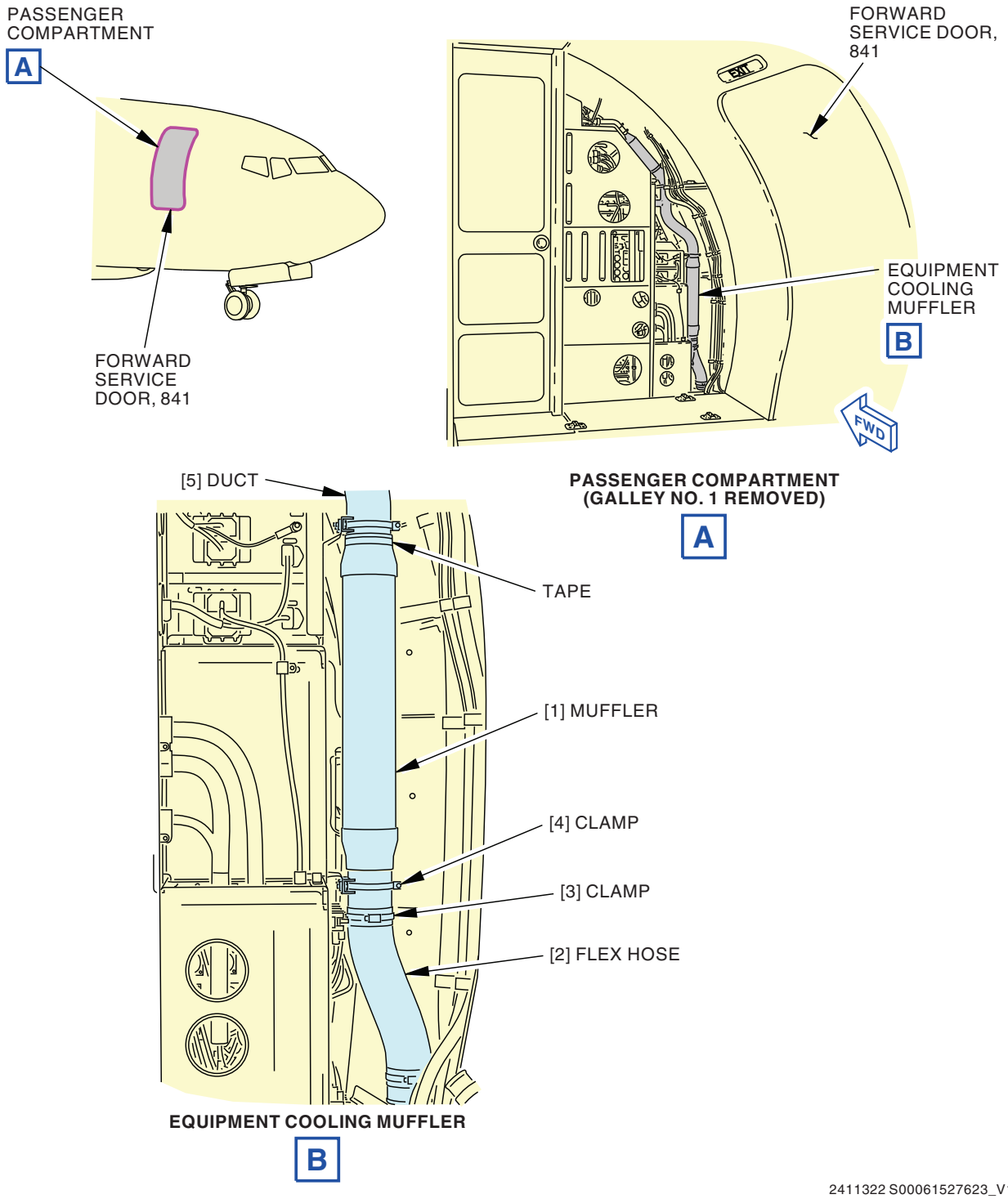
(c) Remove the clamp [4].

(d) Remove the tape between the muffler [1] and the duct [5].

(e) Move the muffler [1] down to remove it from the duct [5].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Equipment Cooling Muffler Installation
Figure 401/21-27-07-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-27-07-400-801

3. Equipment Cooling Muffler Installation

(Figure 401)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
25-31-11-400-801	Galley No. 1 - Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
A50464	Tape - Vinyl, Adhesive Coated	BMS5-179
B00090	Solvent - Inhibited Trichloroethane 1,1,1	MIL-T-81533

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Muffler	21-27-07-01-085	PTF ALL

D. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

E. Equipment Cooling Muffler Installation

SUBTASK 21-27-07-860-002

(1) Make sure that power has been removed from the airplane (Remove Electrical Power, TASK 24-22-00-860-802).

SUBTASK 21-27-07-110-001

(2) Clean the surfaces at the joint between the muffler [1] and the duct [5] with solvent, B00090.

SUBTASK 21-27-07-420-001

(3) Install the equipment cooling muffler [1] as follows:

(a) Slide the upper end of the muffler [1] over the end of the duct [5].

(b) Put the clamp [4] in its position and tighten to 3.00 in-lb (0.34 N·m) - 5.00 in-lb (0.56 N·m).

(c) Slide the flex hose [2] onto the lower end of the muffler [1].

(d) Put the clamp [3] in its position and tighten to 3.00 in-lb (0.34 N·m) - 5.00 in-lb (0.56 N·m).

1) Do a visual inspection of the muffler [1], make sure that it is not damaged.

(e) Install two layers of 2-inch wide tape, A50464 - Vinyl, Adhesive Coated BMS5-179, over the joint between the muffler [1] and the duct [5].

SUBTASK 21-27-07-860-003

(4) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-27-07-200-001

(5) Make sure that the equipment cooling fan operates.

SUBTASK 21-27-07-200-002

(6) Check for air leaks with your hand at all duct joints.

(a) Repair any leaks.

SUBTASK 21-27-07-410-001

(7) Install the No. 1 galley (TASK 25-31-11-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-27-07-020-001

WARNING DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(8) Remove all personal protective equipment and put them in a plastic bag, G00624. Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(a) Discard the bag with the personal protective equipment. Refer to the airline, local health, safety and regulatory procedures for disposal of material.

SUBTASK 21-27-07-110-002

WARNING CLEAN YOUR HANDS WITH SOAP AND FLUSH WITH HOT WATER. DIRTY HANDS WITH CONTAMINATION CAN CAUSE DISEASE AND ILLNESSES TO PERSONS.

(9) Clean your hands with soap and running hot water.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EQUIPMENT COOLING EXHAUST DUCT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the equipment cooling exhaust duct

(2) An installation of the equipment cooling overboard exhaust duct.

TASK 21-27-10-000-801

2. Equipment Cooling Exhaust Duct - Removal

(Figure 401)

A. General

(1) You must remove electrical power from the airplane before you remove an equipment cooling exhaust duct. This will make sure the electrical/electronic equipment does not receive electrical power when the equipment cooling system is not in operation.

B. References

Reference	Title
24-22-00-860-802	Remove Electrical Power (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 21-27-10-860-001

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 21-27-10-010-001

(2) To get access to the equipment cooling exhaust duct, do this step:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(3) Remove the access panel [1] and access panel [3] on the top of the raised platform.

(a) Remove the screws [2] that attach the access panel [1] and access panel [3] to the top of the raised platform.

F. Equipment Cooling Exhaust Duct Removal

SUBTASK 21-27-10-020-001

(1) Loosen the coupling [4].

SUBTASK 21-27-10-020-002

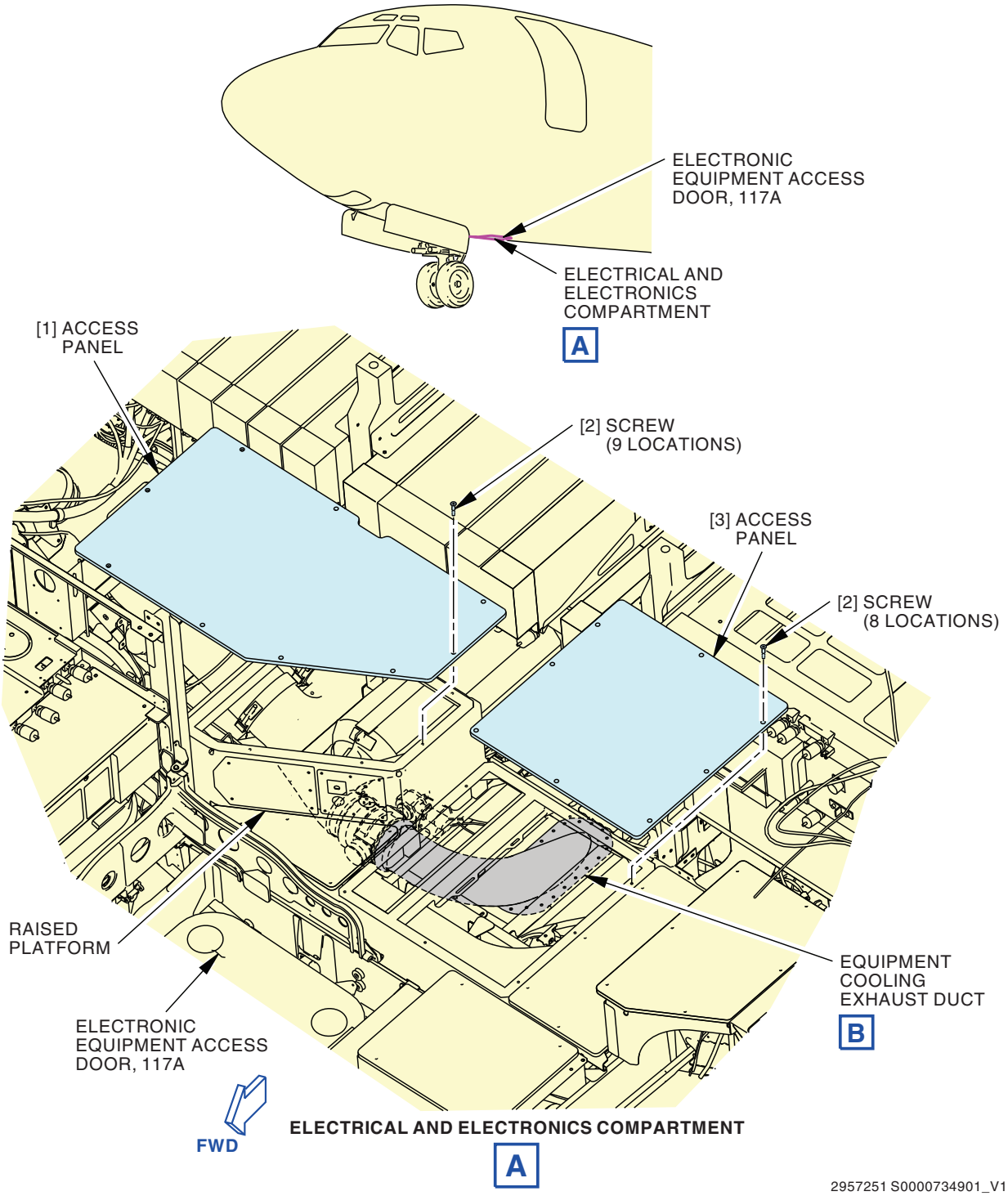
(2) Remove the bolts [6].

SUBTASK 21-27-10-020-003

(3) Remove the equipment cooling exhaust duct [5].

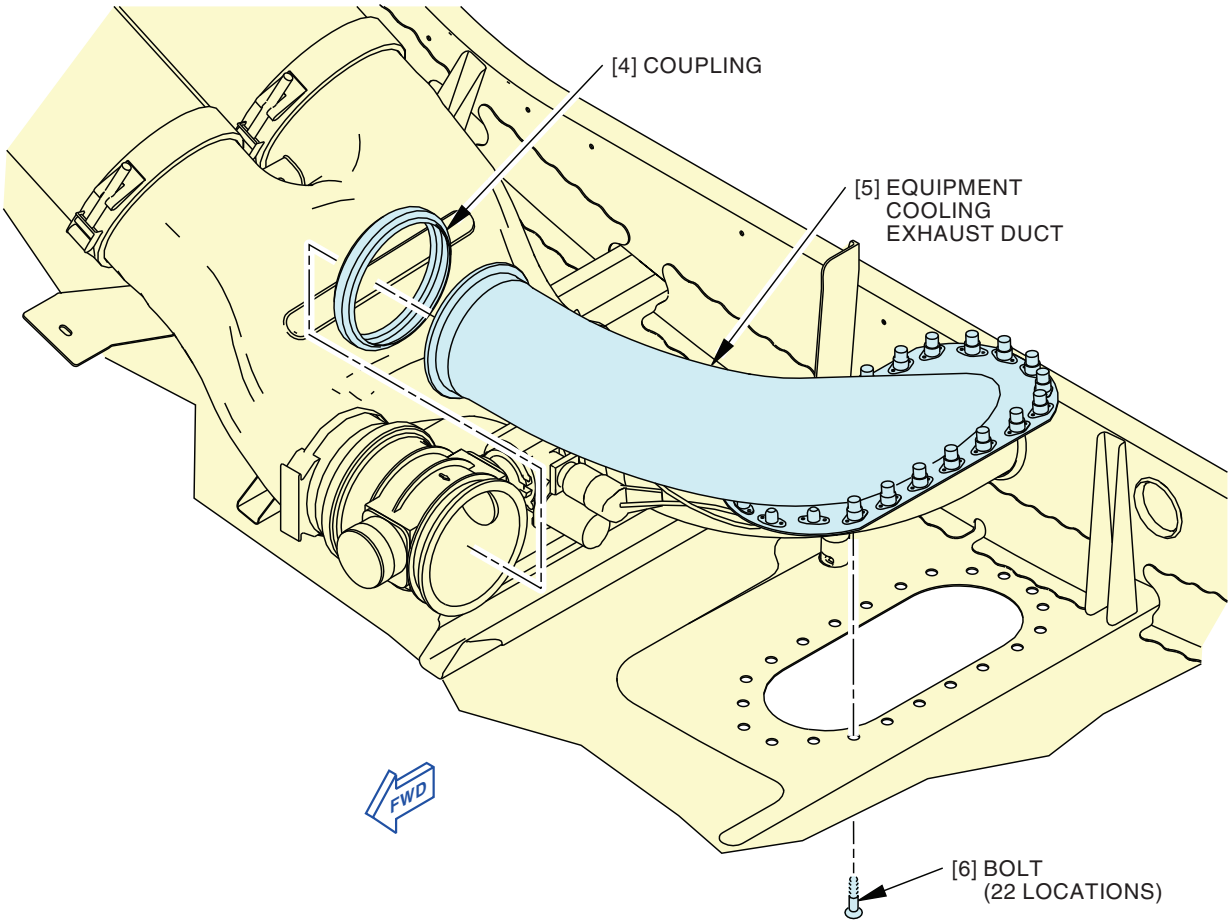
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Equipment Cooling Exhaust Duct Installation
Figure 401/21-27-10-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EQUIPMENT COOLING EXHAUST DUCT
(RAISED PLATFORM AND INSULATION ARE NOT SHOWN)

B

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Equipment Cooling Exhaust Duct Installation
Figure 401/21-27-10-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-27-10-400-801

3. Equipment Cooling Exhaust Duct - Installation

(Figure 401)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A50525	Sealant - P/S 870 Class C-80 Corrosion Inhibitive	BMS5-95 Class C-80
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Equipment Cooling Exhaust Duct Installation

SUBTASK 21-27-10-420-001

(1) Put the equipment cooling exhaust duct [5] in its position.

(a) Apply P/S 870 Class C-80 sealant, A50525, between surfaces of the skin and the equipment cooling exhaust duct [5].

SUBTASK 21-27-10-420-002

(2) Install the bolts [6] as follows:

(a) Apply compound, C00528, to the all surface of the hole.

(b) Immediately install the bolts [6] after you apply the compound, C00528.

SUBTASK 21-27-10-420-003

(3) Tighten the coupling [4].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-27-10-410-001

(1) Install the access panel [1] and access panel [3] on the top of the raised platform.

(a) Install the screws [2].

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-27-10-860-002

(3) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRESSURIZATION CONTROL SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A deactivation of the pressurization control system

(2) An activation of the pressurization control system.

TASK 21-31-00-800-801

2. Pressurization Control System - Deactivation

(Figure 201)

A. General

(1) This task will deactivate these components in the pressurization control system.

- Cabin Pressure Controller (CPC)
- Cabin Pressure Control Module
- Aft Outflow Valve

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
146	Aft Cargo Compartment Equipment Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Procedure

SUBTASK 21-31-00-010-001

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-31-00-860-001

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

F. Tryout

NOTE: This tryout is to make sure that the pressurization control system is in a zero energy state.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-31-00-860-002

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Set the mode selector switch on the P5-6 cabin pressure control module to the AUTO position.

SUBTASK 21-31-00-210-001

(2) Make sure that the P5-10 air conditioning panel amber AUTO FAIL LIGHT (L8) is illuminated.

SUBTASK 21-31-00-740-001

(3) Do these steps on the BITE module on the pressure controller No. 1:

(a) Push the ON/OFF button on the BITE module on the pressure controller No. 1.

(b) Push the MENU button.

(c) Push the down arrow button.

(d) Make sure that the display does not show “GROUND TEST”.

SUBTASK 21-31-00-740-002

(4) Do these steps on the BITE module on the pressure controller No. 2:

(a) Push the ON/OFF button on the BITE module on the pressure controller No. 2.

(b) Push the MENU button.

(c) Push the down arrow button.

(d) Make sure that the display does not show “GROUND TEST”.

SUBTASK 21-31-00-860-003

(5) Set the mode selector switch on the P5-6 cabin pressure control module to the MAN position.

SUBTASK 21-31-00-210-002

(6) Make sure that the P5-10 air conditioning panel green MANUAL LIGHT (L7) is illuminated.

(a) Position an assistant to monitor the cabin pressurization outflow valve for movement.

SUBTASK 21-31-00-860-004

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Push and hold the MANUAL MODE CLOSE/OPEN toggle switch (S2) on the P5-6 cabin pressure control module to the CLOSE position.

(a) Make sure that the cabin pressurization outflow valve gates do not move.

SUBTASK 21-31-00-860-005

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Push and hold the MANUAL MODE CLOSE/OPEN toggle switch (S2) on the P5-6 cabin pressure control module to the OPEN position.

(a) Make sure that the cabin pressurization outflow valve gates do not move.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-31-00-860-006

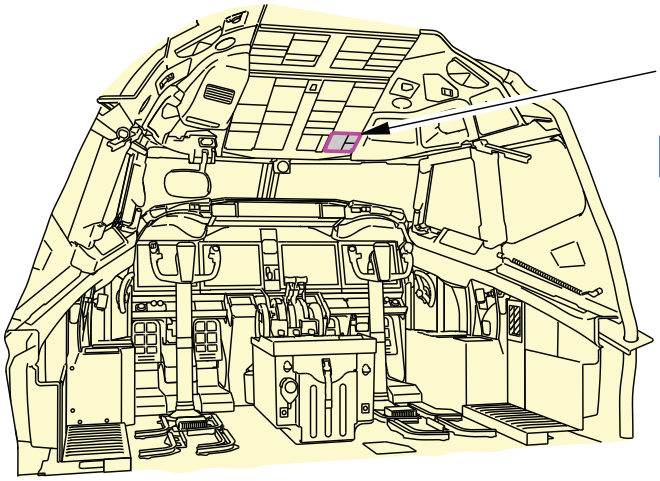
(9) Install DO NOT OPERATE tag, STD-858 on the following P5-6 cabin pressure control module switches:

(a) AUTO/ALTN/MAN select switch (S1).

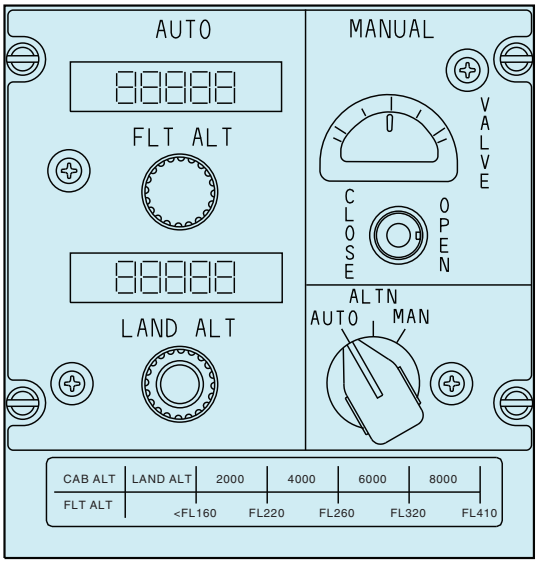
(b) MANUAL MODE CLOSE/OPEN toggle switch (S2).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



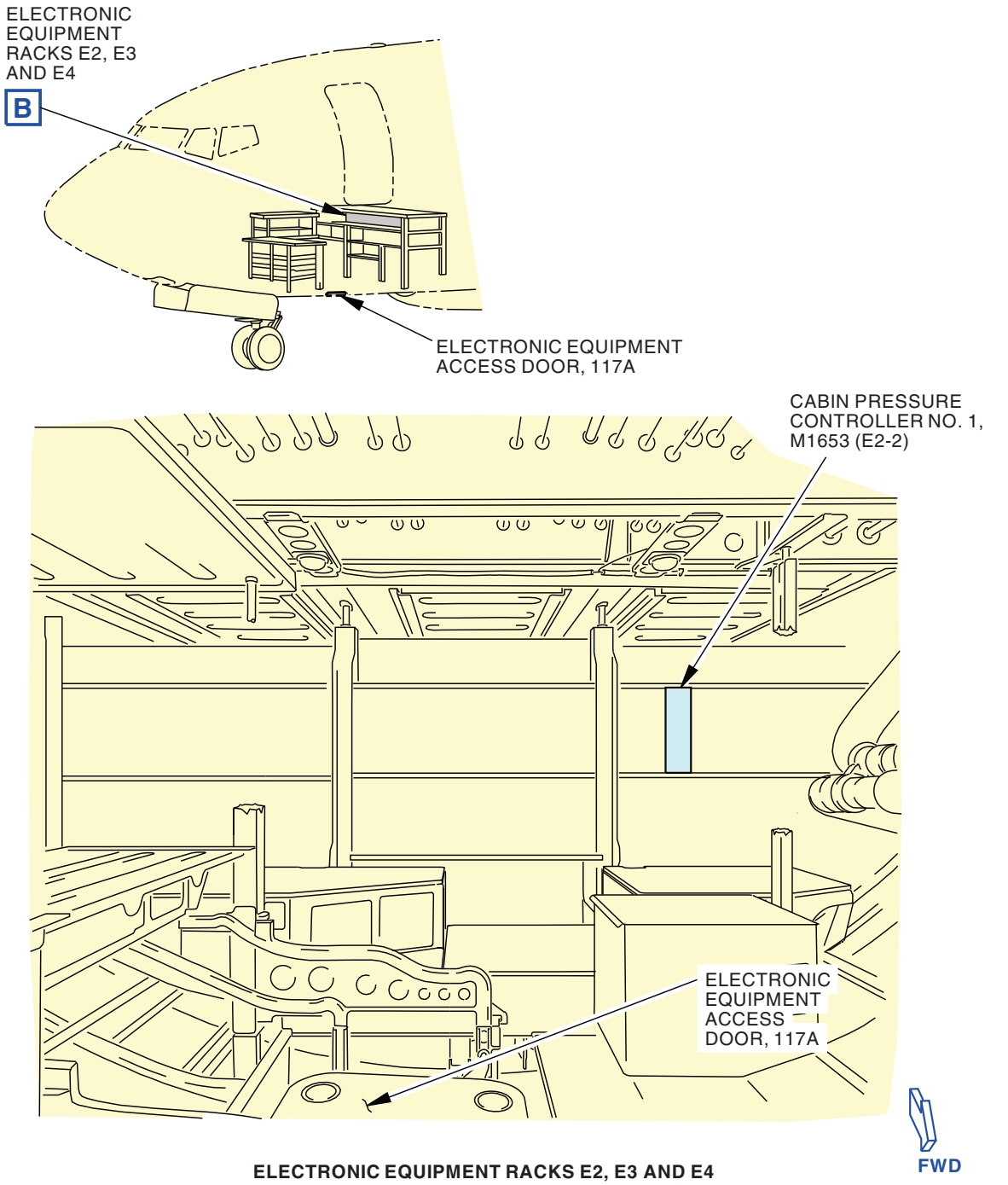
CABIN PRESSURE CONTROL MODULE

A

2411323 S00061527629_V1

Pressurization Control System
Figure 201/21-31-00-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



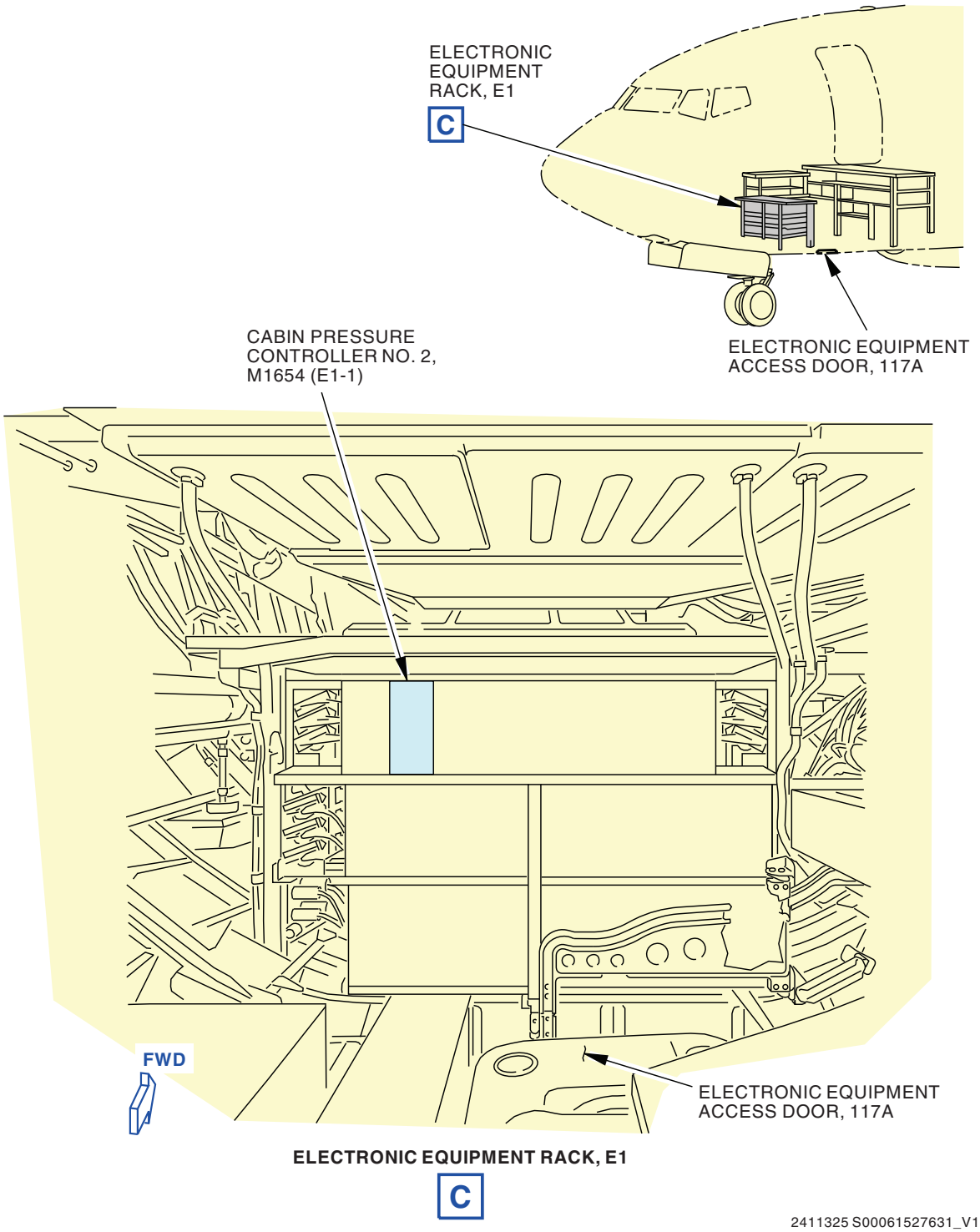
ELECTRONIC EQUIPMENT RACKS E2, E3 AND E4

B

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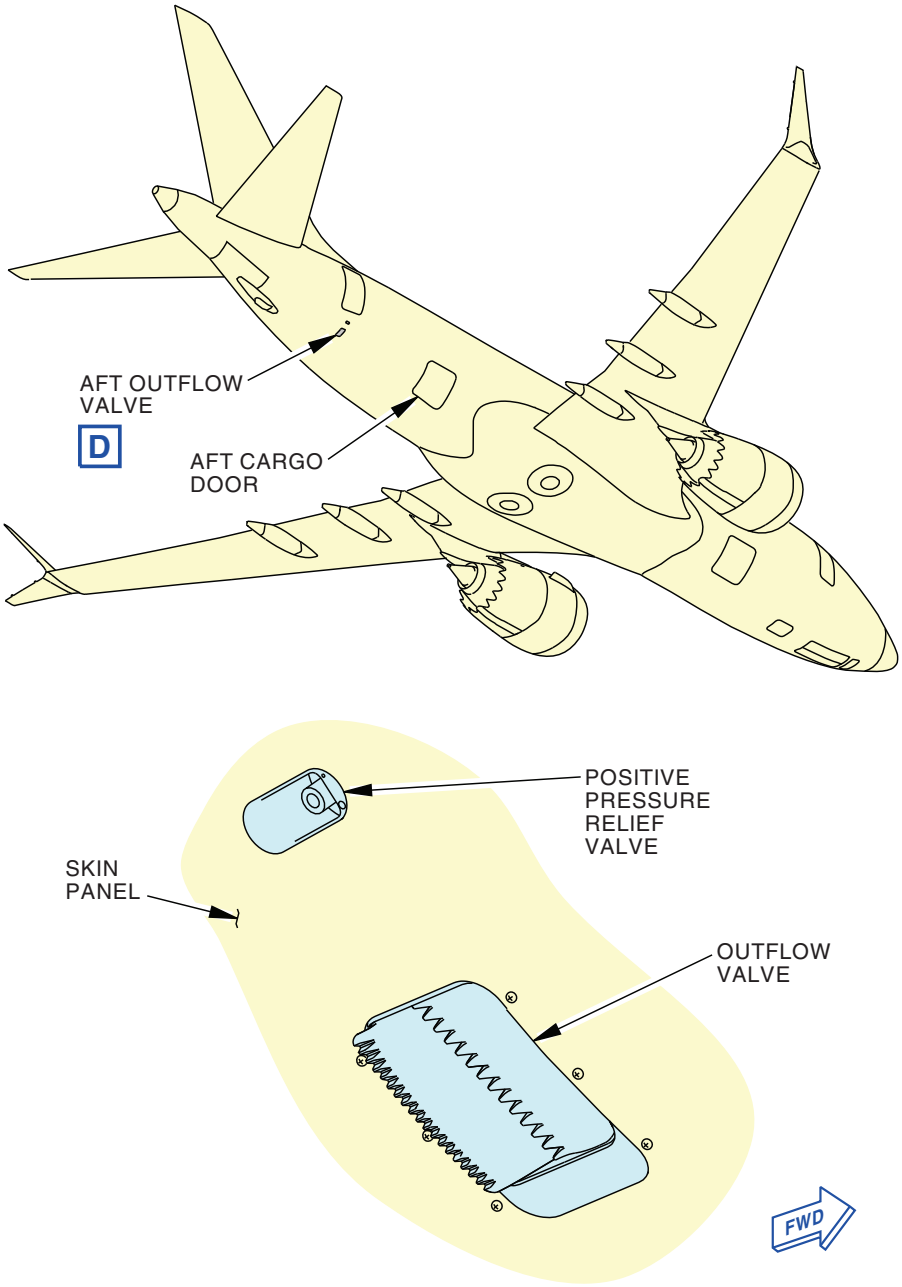
Pressurization Control System
Figure 201/21-31-00-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pressurization Control System
Figure 201/21-31-00-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTFLOW VALVE (EXTERIOR VIEW)

D

2411326 S00061527632_V1

Pressurization Control System
Figure 201/21-31-00-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-31-00-800-802

3. Pressurization Control System - Activation

(Figure 201)

A. General

(1) This task will activate these components in the pressurization control system.

- Cabin Pressure Controller (CPC)
- Cabin Pressure Control Module
- Aft Outflow Valve

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
146	Aft Cargo Compartment Equipment Bay - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Procedure

SUBTASK 21-31-00-860-007

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

SUBTASK 21-31-00-410-001

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRESSURE EQUALIZATION VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Pressure equalization valve removal.

(2) Pressure equalization valve installation.

TASK 21-31-00-000-801

2. Pressure Equalization Valve Removal

(Figure 401, Figure 402)

A. General

(1) This task includes the steps to remove the pressure equalization valve from the forward and aft cargo compartment.

(a) The forward cargo compartment has a pressure equalization valve on the aft bulkhead.

(b) The aft cargo compartment has a pressure equalization valve on the forward portion of the waste tank bulkhead.

B. References

Reference	Title
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
124	Forward Cargo Compartment - Right
142	Aft Cargo Compartment - Right

D. Prepare for the Removal

SUBTASK 21-31-00-020-002

(1) For the forward cargo compartment:

(a) Remove the forward cargo compartment access panel [5], do this task: Forward Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-11-02-000-801.

SUBTASK 21-31-00-020-003

(2) For the aft cargo compartment:

(a) Remove the waste tank access panel [5], do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

E. Cargo Compartment Pressure Equalization Valve Removal

SUBTASK 21-31-00-020-004

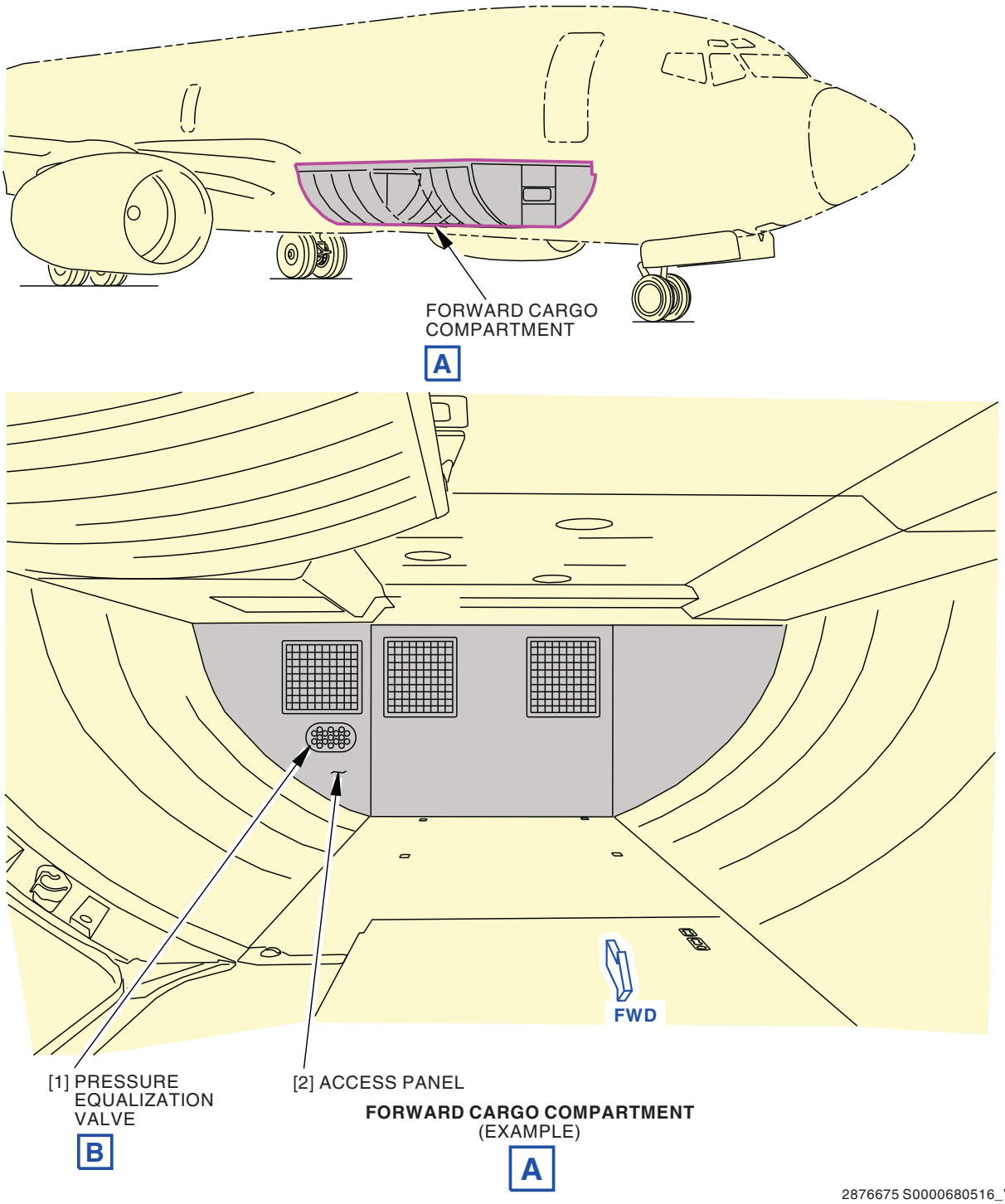
(1) Remove the bolts [4] and nuts [3] from the pressure equalization valve [1].

(a) Detach the pressure equalization valve [1] from the adapter [2].

NOTE: It is not required to remove the adapter [2] from the access panel [5] to replace the pressure equalization valve [1].

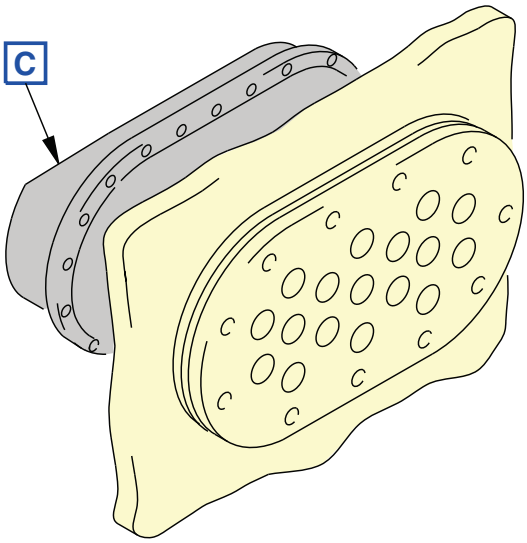
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

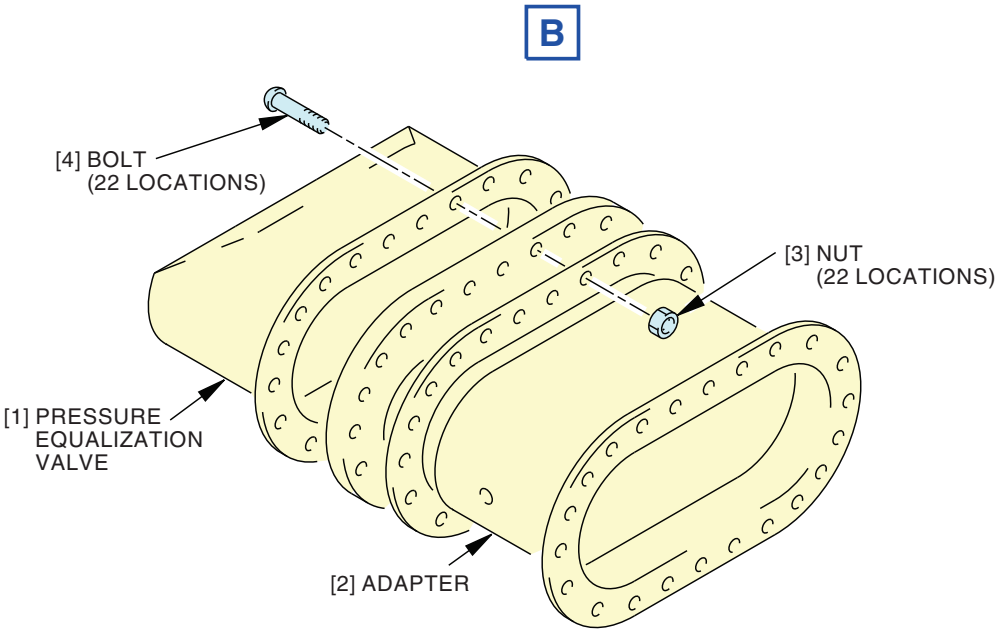


Forward Cargo Compartment Pressure Equalization Valve
Figure 401/21-31-00-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PRESSURE EQUALIZATION VALVE



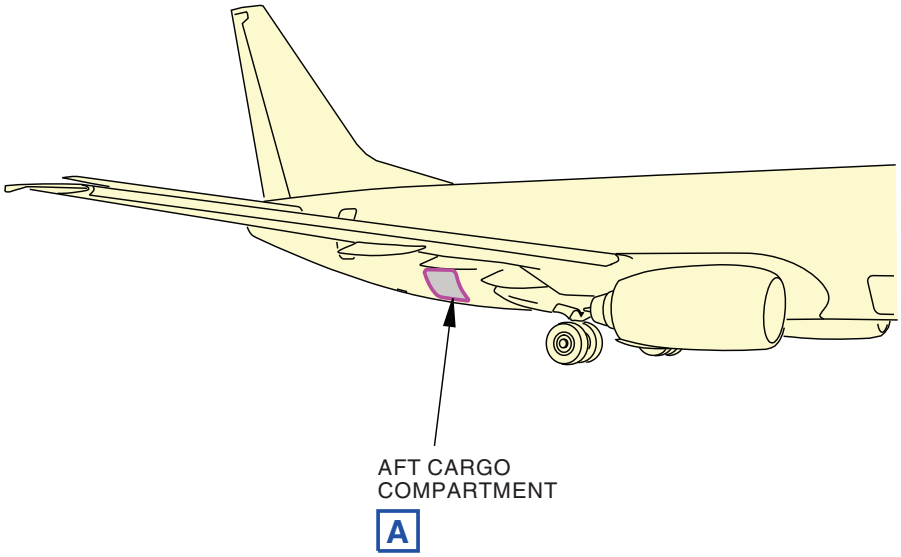
PRESSURE EQUALIZATION VALVE

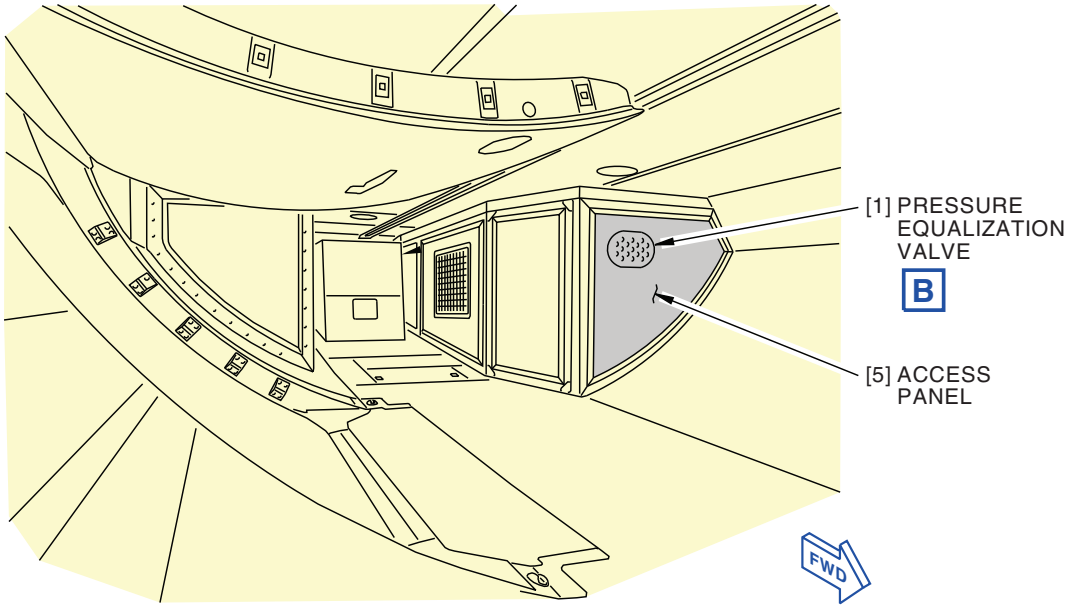
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Forward Cargo Compartment Pressure Equalization Valve
Figure 401/21-31-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





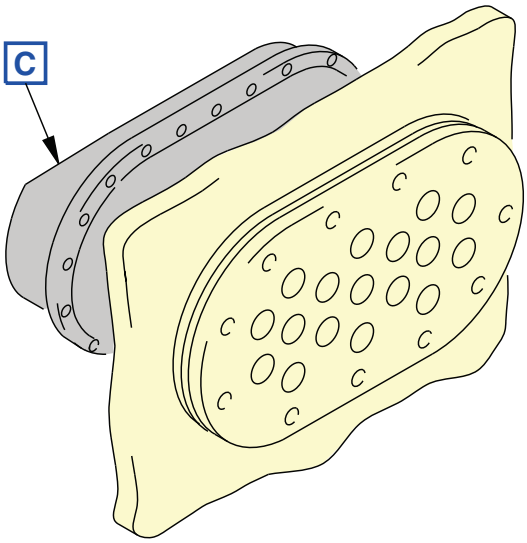
AFT CARGO COMPARTMENT

A

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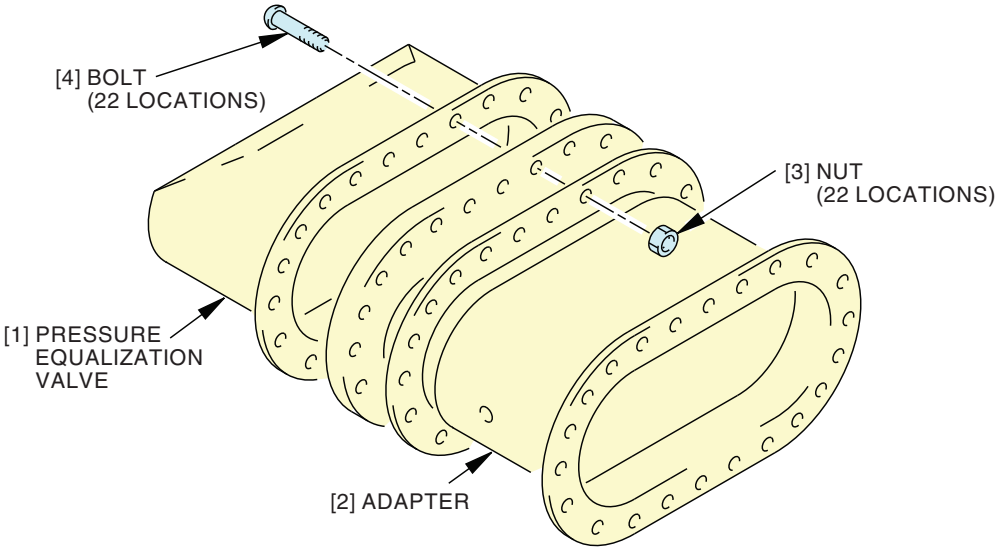
Aft Cargo Compartment Pressure Equalization Valve
Figure 402/21-31-00-990-809 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PRESSURE EQUALIZATION VALVE

B



PRESSURE EQUALIZATION VALVE

C

2876678 S0000680518_V1

Aft Cargo Compartment Pressure Equalization Valve
Figure 402/21-31-00-990-809 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-31-00-400-801

3. Pressure Equalization Valve Installation

(Figure 401, Figure 402)

A. General

(1) This task includes the steps to install the pressure equalization valve to the forward and aft cargo compartment.

(a) The forward cargo compartment has a pressure equalization valve on the aft bulkhead.

(b) The aft cargo compartment has a pressure equalization valve on the forward portion of the waste tank bulkhead.

B. References

Reference	Title
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Location Zones

Zone	Area
124	Forward Cargo Compartment - Right
142	Aft Cargo Compartment - Right

D. Cargo Compartment Pressure Equalization Valve Installation

SUBTASK 21-31-00-420-002

(1) Align the pressure equalization valve [1] with the adapter [2].

(2) Install the bolts [4] and nuts [3] that attach the pressure equalization valve [1] and the adapter [2].

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-31-00-420-004

(1) For the forward cargo compartment:

(a) Install the forward cargo compartment access panel [5], do this task: Forward Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-11-02-400-801.

SUBTASK 21-31-00-420-005

(2) For the aft cargo compartment:

(a) Install the aft cargo compartment access panel [5], do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

——— END OF TASK ———

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PRESSURIZATION CONTROL SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Pressurization system manual mode test

(2) 750 feet per minute descent with cargo fire mode test

(3) Master caution test

(4) Pressurization system ground test

(5) Cabin Pressure Controller (CPC) Built-In-Test Equipment (BITE) procedure.

TASK 21-31-00-710-801

2. Pressurization System Manual Mode Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task does an operational test of these components:

(a) The outflow valve manual mode motor

(b) The outflow valve selector panel

(c) The outflow valve position indicator

(d) The outflow valve feedback module for the outflow valve position indicator.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 21-31-00-860-012

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

D. Pressurization System Manual Mode Test

SUBTASK 21-31-00-860-013

CAUTION MAKE SURE THAT ONE OF THE EXTERNAL DOORS OR WINDOWS IS OPEN BEFORE YOU CONTINUE WITH THE TEST. UNCONTROLLED PRESSURIZATION OF THE CABIN CAN OCCUR WHEN THE OUTFLOW VALVE IS CLOSED. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Make sure that at least one exterior door or window is OPEN or that the L and R PACK switches, on the P5 forward overhead panel, are in the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-31-00-860-014

(2) Set the mode selector switch, on the P5-6 cabin pressure control module, to the MAN position.

SUBTASK 21-31-00-710-001

(3) Make sure that the MANUAL light, on the P5-10 air conditioning module, comes ON.

SUBTASK 21-31-00-860-015

(4) Set the LIGHTS switch, on the P2 center instrument panel, to the DIM position.

SUBTASK 21-31-00-710-002

(5) Make sure that the MANUAL light becomes dim.

SUBTASK 21-31-00-860-016

(6) Set the LIGHTS switch, on the P2 center instrument panel, to the BRT position.

SUBTASK 21-31-00-710-003

(7) Make sure that the MANUAL light becomes bright.

SUBTASK 21-31-00-710-004

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Push and hold the CLOSE/OPEN toggle switch, on the P5-6 cabin pressure control module, to the CLOSE position until the outflow valve is fully closed.

SUBTASK 21-31-00-710-005

(9) Make sure that the outflow valve is fully closed.

SUBTASK 21-31-00-710-006

(10) Make sure that the needle on the position indicator gage for the outflow valve is within approximately two needle widths of the fully closed mark.

SUBTASK 21-31-00-710-007

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(11) Push and hold the CLOSE/OPEN toggle switch, on the P5-6 cabin pressure control module, to the OPEN position until the outflow valve is fully open.

SUBTASK 21-31-00-710-008

(12) Make sure that the outflow valve is fully open.

SUBTASK 21-31-00-710-009

(13) Make sure that the needle on the position indicator gage for the outflow valve is within approximately two needle widths of the fully open mark.

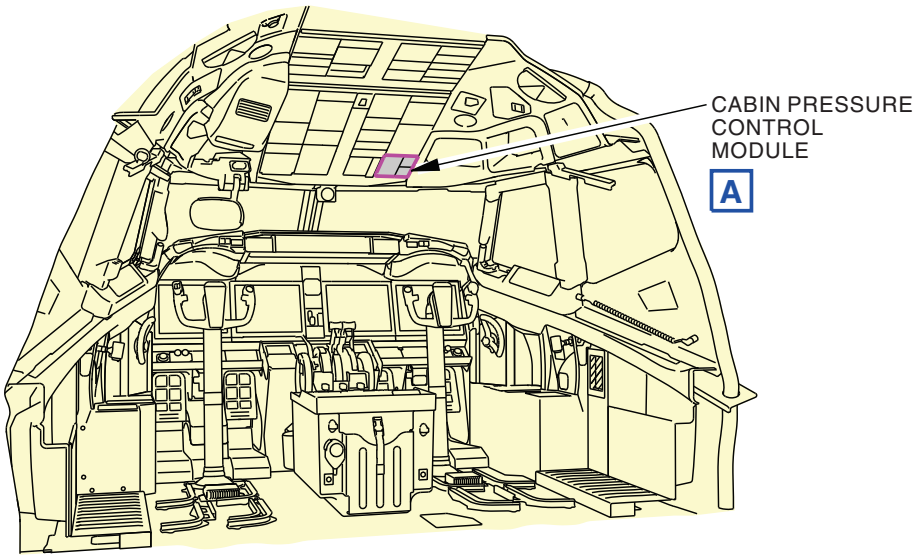
E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-31-00-860-017

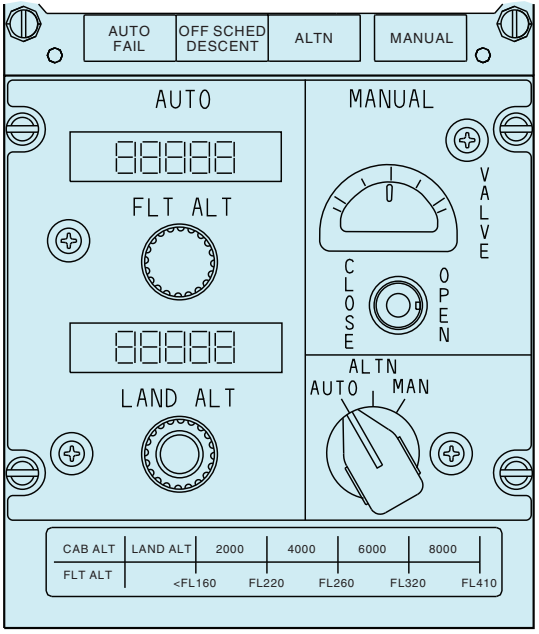
(1) Make sure that the mode selector switch, on the P5-6 cabin pressure control module, is returned to the AUTO position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



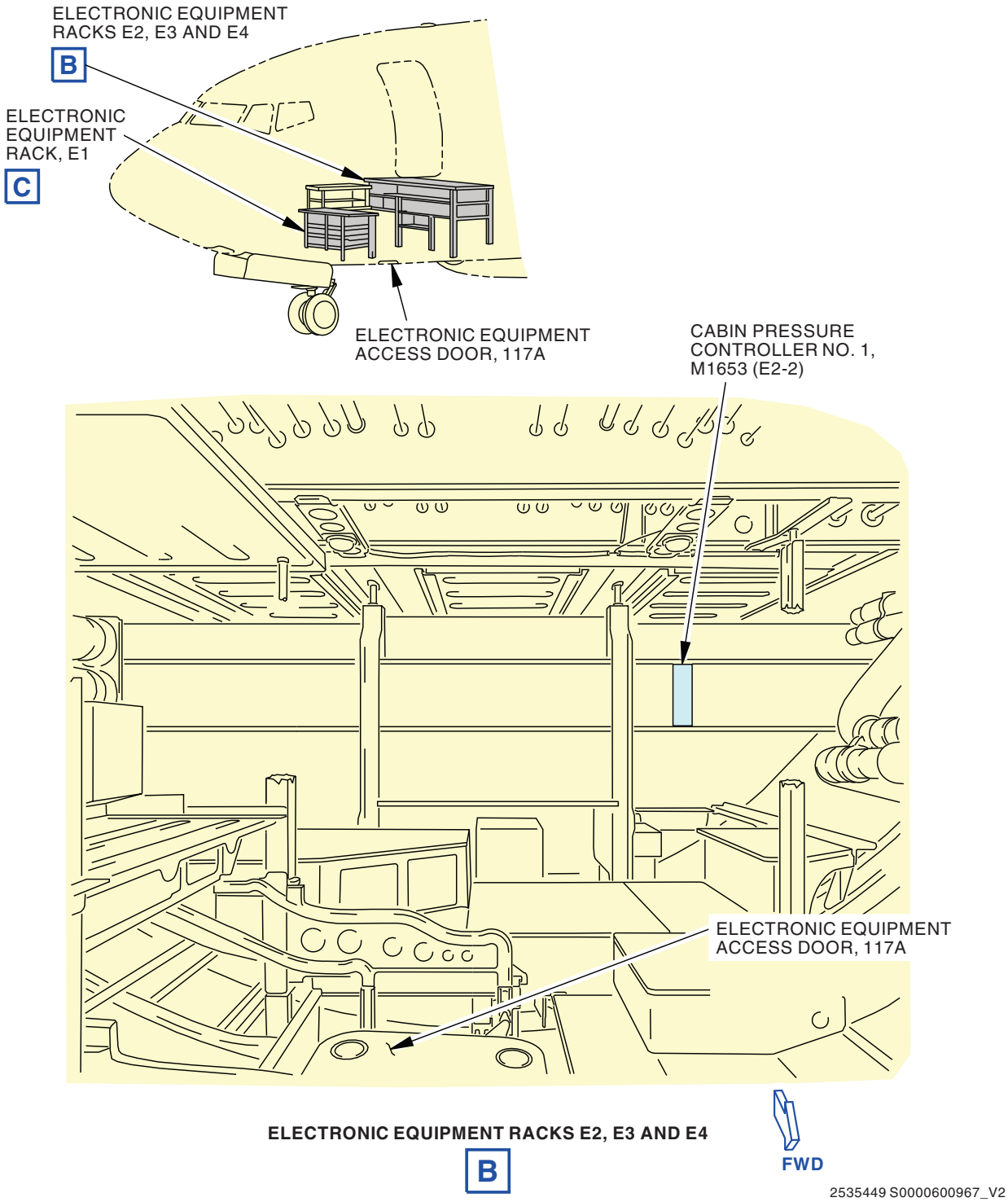
CABIN PRESSURE CONTROL MODULE

A

2535269 S0000600966_V1

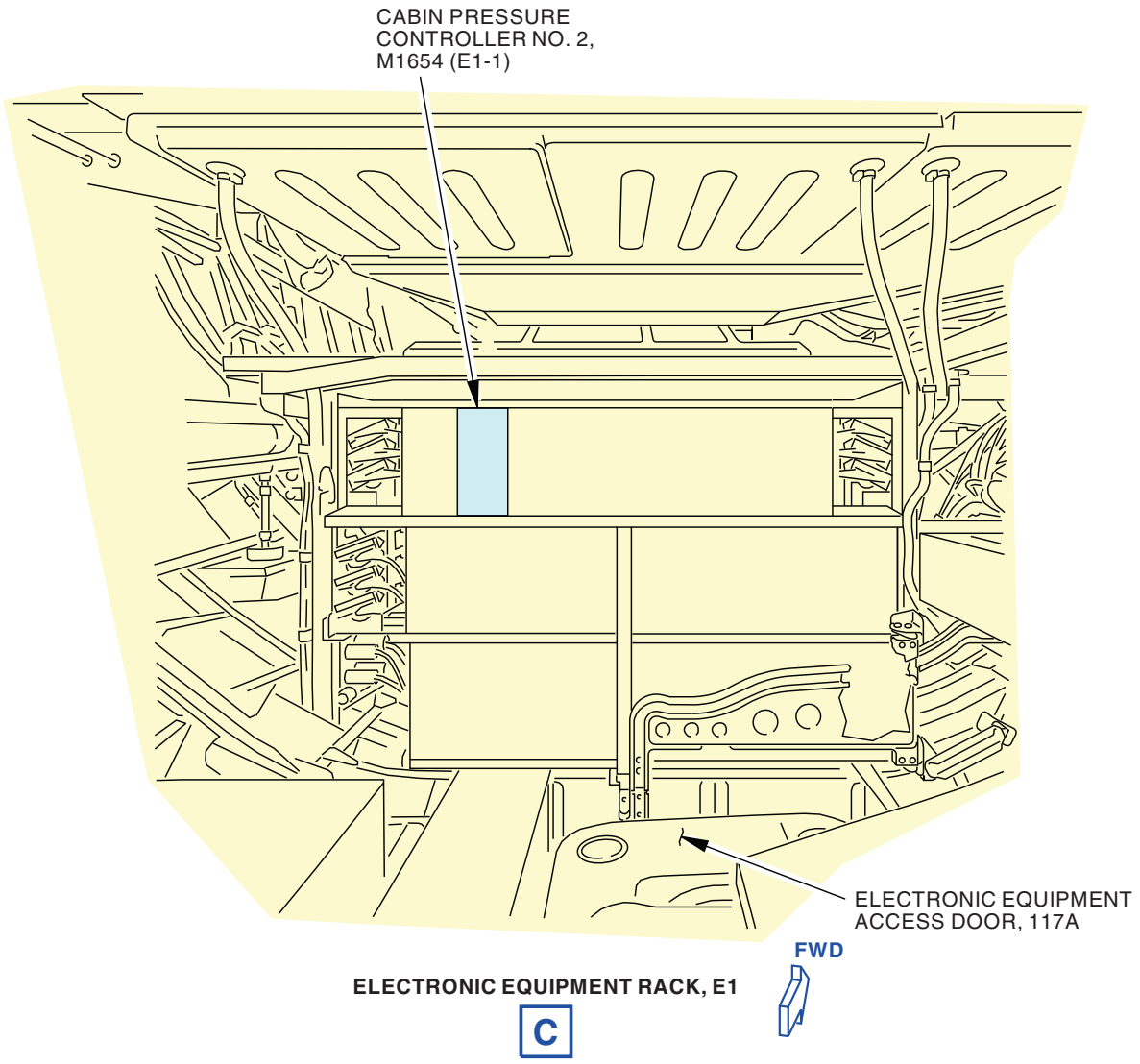
Cabin Pressure Control System Test
Figure 501/21-31-00-990-806 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Pressure Control System Test
Figure 501/21-31-00-990-806 (Sheet 2 of 4)

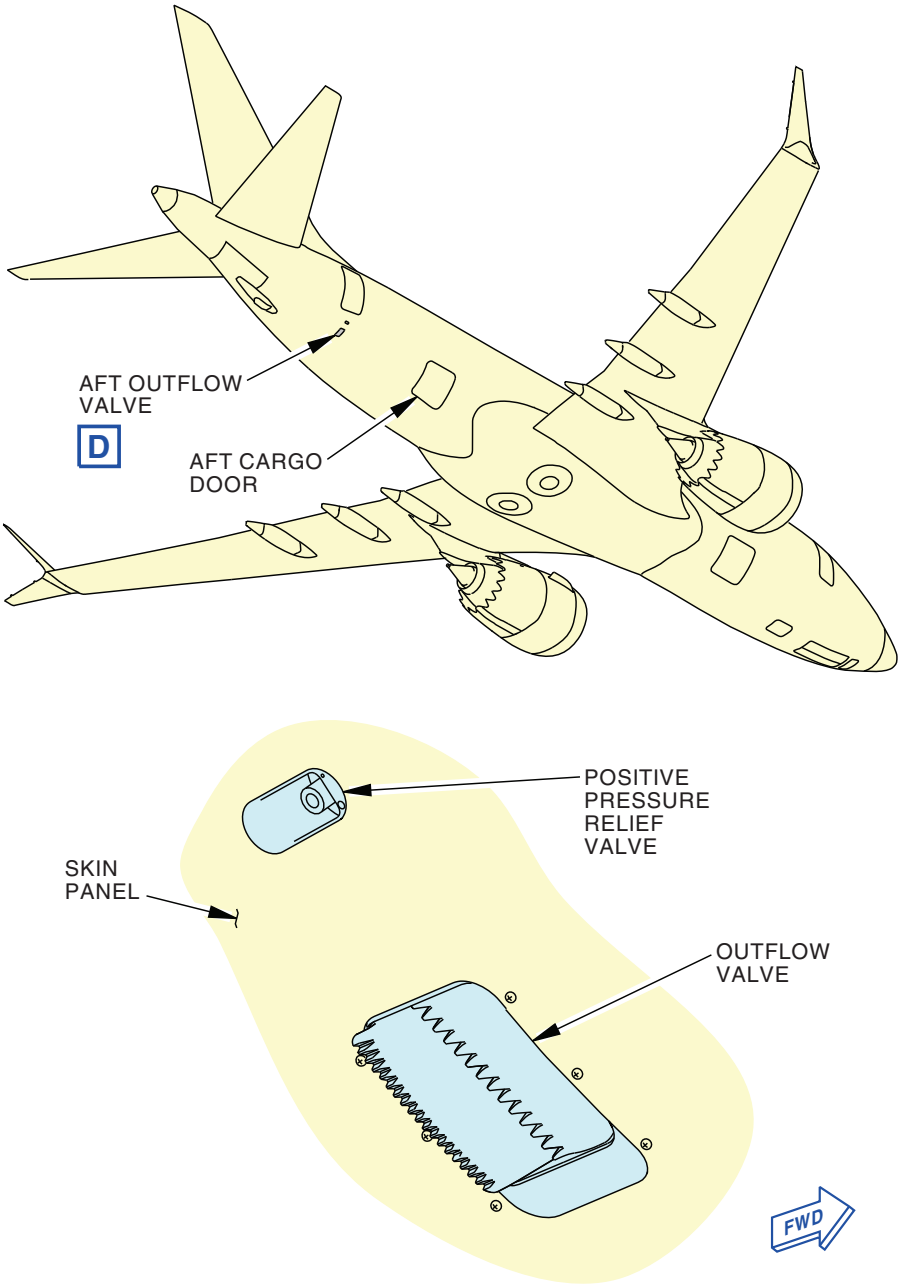
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3009740 S0000783413_V1

Cabin Pressure Control System Test
Figure 501/21-31-00-990-806 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTFLOW VALVE (EXTERIOR VIEW)

D

2411326 S00061527632_V1

Cabin Pressure Control System Test
Figure 501/21-31-00-990-806 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-31-00-710-802

3. 750 Feet Per Minute Descent With Cargo Fire Mode Test

(Figure 501)

A. General

(1) This task does an operational test of the 750 feet per minute descent rate with cargo fire mode.

B. References

Reference	Title
21-31-01-000-801	Cabin Pressure Controller (CPC) Removal (P/B 401)
21-31-01-400-801	Cabin Pressure Controller (CPC) Installation (P/B 401)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Test

SUBTASK 21-31-00-860-018

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL
F	7	C01272	PRESSURIZATION CONTROL IND

F. 750 Feet Per Minute Descent With Cargo Fire Mode Test

SUBTASK 21-31-00-860-019

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-31-00-020-001

(2) Remove the cabin pressure controller 1 and cabin pressure controller 2, do this task: Cabin Pressure Controller (CPC) Removal, TASK 21-31-01-000-801.

SUBTASK 21-31-00-860-020

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

SUBTASK 21-31-00-710-010

(4) Push and hold the TEST switch on the P8 Cargo Fire Control Panel.

SUBTASK 21-31-00-710-011

(5) Connect a digital/analog multimeter, COM-1793, between the E2-2 shelf connector D10738B electrical contacts D7, B10 and airplane ground.

NOTE: Pressure controller No. 1 is installed on the E2-2 shelf.

(a) Make sure that there is less than 4.0 ohms between electrical contacts D7, B10 and airplane ground.

SUBTASK 21-31-00-710-012

(6) Disconnect the digital/analog multimeter, COM-1793, from electrical contact D7 or B10 of E2-2 shelf connector D10738B.

SUBTASK 21-31-00-710-013

(7) Connect the digital/analog multimeter, COM-1793, between the E1-1 shelf connector D10740B electrical contacts D7, B10 and airplane ground.

NOTE: Pressure controller No. 2 is installed on the E1-1 shelf.

(a) Make sure that there is less than 4.0 ohms between electrical contacts D7, B10 and airplane ground.

SUBTASK 21-31-00-710-020

(8) Disconnect the digital/analog multimeter, COM-1793, from electrical contact D7 or B10 of E1-1 shelf connector D10740B.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-31-00-710-014

(9) Release the TEST switch on the P8 Cargo Fire Control Panel.

SUBTASK 21-31-00-860-021

(10) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-31-00-860-022

(1) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-31-00-410-003

(2) Install the cabin pressure controller 1 and cabin pressure controller 2, do this task: Cabin Pressure Controller (CPC) Installation, TASK 21-31-01-400-801.

——— END OF TASK ———

TASK 21-31-00-700-801

4. Master Caution Test

(Figure 501)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Test

SUBTASK 21-31-00-860-023

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	13	C01277	MASTER CAUTION ANNUNCIATOR CONT 3
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
E	11	C00313	INDICATOR MASTER DIM SECT 1
E	12	C00314	INDICATOR MASTER DIM SECT 2
E	13	C00315	INDICATOR MASTER DIM SECT 3
E	14	C00316	INDICATOR MASTER DIM SECT 4
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	12	C00318	INDICATOR MASTER DIM SECT 6

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

C. Master Caution Test

SUBTASK 21-31-00-860-024

(1) Put the mode selector switch on the P5-6 pressure control module to the AUTO position.

SUBTASK 21-31-00-860-025

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1

SUBTASK 21-31-00-710-015

(3) Make sure that the AUTO FAIL and ALTN lights on the P5-10 air conditioning module come ON in less than 5 seconds.

SUBTASK 21-31-00-860-026

(4) Set the mode selector switch on the P5-6 pressure control module to the ALTN position.

SUBTASK 21-31-00-710-016

(5) Make sure the AUTO FAIL light on the P5-10 air conditioning module goes OFF and the ALTN light stays ON.

SUBTASK 21-31-00-860-027

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

SUBTASK 21-31-00-710-017

(7) Make sure that the AUTO FAIL light comes ON and the ALTN light goes OFF in less than 5 seconds.

SUBTASK 21-31-00-860-028

(8) Disconnect electrical connector D10724 from the P5-6 pressure control module:

(a) Get access to the back of the P5-6 pressure control panel as follows:

1) Loosen the 1/4-turn fasteners and lower the forward end of the P5 forward overhead panel.

SUBTASK 21-31-00-860-029

(9) Make sure the AUTO FAIL light is ON.

SUBTASK 21-31-00-860-030

(10) Connect electrical connector D10724 to the P5-6 pressure control module.

(a) Push the P5 forward overhead panel back into its installed position and tighten the 1/4-turn fasteners to hold the panel.

SUBTASK 21-31-00-860-031

(11) Set the mode selector switch on the pressure control module to the MAN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-31-00-860-032

(12) Make sure the AUTO FAIL light goes OFF and the MANUAL light comes ON.

SUBTASK 21-31-00-860-033

(13) Set the mode selector switch on the pressure control module to the AUTO position.

SUBTASK 21-31-00-860-034

(14) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

SUBTASK 21-31-00-860-035

(15) Make sure the AUTO FAIL, ALTN, and the MANUAL lights are OFF.

——— END OF TASK ———

TASK 21-31-00-700-802

5. Pressurization System Ground Test

(Figure 501, Figure 502, Figure 503)

A. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Pressurization System Ground Test (No. 1 Pressure Controller)

SUBTASK 21-31-00-860-036

(1) Put the mode selector switch on the P5-6 pressure control module to AUTO.

SUBTASK 21-31-00-710-018

CAUTION MAKE SURE THAT ONE OF THE EXTERNAL DOORS OR WINDOWS IS OPEN BEFORE YOU CONTINUE WITH THE TEST. UNCONTROLLED PRESSURIZATION OF THE CABIN CAN OCCUR WHEN THE OUTFLOW VALVE IS CLOSED. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Make sure at least one external door or window is open or that the L and R PACK switches on the P5 forward overhead panel are off.

SUBTASK 21-31-00-740-003

(3) Do these steps on the BITE module on the pressure controller No. 1:

(a) Push the ON/OFF button on the BITE module on the pressure controller No. 1.

(b) Push the MENU button.

(c) Continue to push the button with the down arrow label until the display shows this message:

Table 501/21-31-00-993-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GROUND TEST

(d) Push the YES button.

(e) Make sure this message is shown on the display:

Table 502/21-31-00-993-802

DISPLAY TEST

(f) Continue to push the button with the down arrow label until the display shows this message:

Table 503/21-31-00-993-803

SYSTEM TEST

(g) Push the YES button to start the system test.

(h) Make sure this message is shown on the display:

Table 504/21-31-00-993-804

DOORS OPEN?

(i) Make sure that at least one external airplane door is open and then push the YES button.

(j) Make sure this message is shown on the display:

Table 505/21-31-00-993-805

DADC ON?

NOTE: The DADC is referring to the Air Data Inertial Reference Units (ADIRU) that provides the air data to the pressure controller.

(k) Make sure that both ADIRU L and R are ON.

1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	8	C00425	ADIRU LEFT EXC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
C	14	C01008	ADIRU RIGHT AC
C	15	C00426	ADIRU RIGHT EXC
C	17	C01010	ADIRU RIGHT DC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Push the YES button.

(m) Make sure this message is shown on the display:

Table 506/21-31-00-993-806

SMC ON?

NOTE: SMC is the Stall Management Yaw Damper (SMYD) computer that provides data to the pressure controller.

(n) Make sure that both SMYD-1 and SMYD -2 are ON.

(o) Push the YES button on the pressure controller No. 1.

(p) Make sure this message is shown on the display:

Table 507/21-31-00-993-807

OFV CLEAR?

(q) Do these steps in less than one minute or the test will fail:

NOTE: You will try to do a system test on pressure controller No. 2 while the test for pressure controller No. 1 is in progress. The system test cannot be done on the two controllers at the same time. This step makes sure that if you try to do the system test on pressure controller No. 2 when the test is in progress on pressure controller No. 1 that the test on pressure controller No. 2 will be aborted.

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

1) Push the YES button on the BCM on pressure controller No. 1 after you make sure the outflow valve is not blocked.

2) Make sure this message is shown on the display:

NOTE: Five or eight digits will come on below the word "TESTING" at intervals of approximately 12 seconds. Older controllers will show eight digits. Newer controllers will show 5 digits.

Table 508/21-31-00-993-808

TESTING

3) Do these steps on the BITE module on pressure controller No. 2:

a) Push the ON/OFF button on the front of the pressure controller No. 2.

b) Push the MENU button.

c) Continue to push the button with the down arrow label until the display shows this message:

Table 509/21-31-00-993-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GROUND TEST

d) Push the YES button.

e) Make sure this message is shown on the display:

Table 510/21-31-00-993-810

DISPLAY TEST

f) Continue to push the button with the down arrow label until the display shows this message:

Table 511/21-31-00-993-811

SYSTEM TEST

g) Push the YES button to start the system test.

h) Make sure the display on the BITE module of pressure controller No. 2 shows these messages:

NOTE: The messages will show one after the other with a two second delay between the first and the second message. This shows that the test on pressure controller No. 2 is aborted.

Table 512/21-31-00-993-812

BOTH SYS
IN IBIT
IBIT
ABORTED

i) Push the ON/OFF button on the BITE module of pressure controller No. 2 (M1654).

(r) Make sure that five or eight digits are on for pressure controller No. 1.

NOTE: Five or eight digits will come on below the word "TESTING" at intervals of approximately 12 seconds. Older controllers will show eight digits. Newer controllers will show 5 digits.

(s) Make sure the display on the BITE module of pressure controller No. 1 shows these messages:

NOTE: The messages will show one after the other with a two second delay between the first and the second message.

Table 513/21-31-00-993-813

SYSTEM
OK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 513/21-31-00-993-813 (Continued)

SYSTEM TEST
& CLEAR

(t) Push the YES button on the BITE module on pressure controller No. 1 (M1653).

(u) Push the ON/OFF button on the BITE module on pressure controller No. 1 (M1653).

C. Pressurization System Ground Test (No. 2 Pressure Controller)

SUBTASK 21-31-00-860-037

(1) Put the mode selector switch on the P5-6 pressure control module to AUTO.

SUBTASK 21-31-00-710-019

CAUTION MAKE SURE THAT ONE OF THE EXTERNAL DOORS OR WINDOWS IS OPEN BEFORE YOU CONTINUE WITH THE TEST. UNCONTROLLED PRESSURIZATION OF THE CABIN CAN OCCUR WHEN THE OUTFLOW VALVE IS CLOSED. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Make sure at least one external door or window is open or that the L and R PACK switches on the P5 forward overhead panel are off.

SUBTASK 21-31-00-740-004

(3) Do these steps on the BITE module on the pressure controller No. 2:

(a) Push the ON/OFF button on the BITE module on the pressure controller No. 2.

(b) Push the MENU button.

(c) Continue to push the button with the down arrow label until the display shows this message:

Table 514/21-31-00-993-814

GROUND TEST

(d) Push the YES button.

(e) Make sure this message is shown on the display:

Table 515/21-31-00-993-815

DISPLAY TEST

(f) Continue to push the button with the down arrow label until the display shows this message:

Table 516/21-31-00-993-816

SYSTEM TEST

(g) Push the YES button to start the system test.

(h) Make sure this message is shown on the display:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 517/21-31-00-993-817

DOORS OPEN?

(i) Make sure that at least one external airplane door is open and then push the YES button.

(j) Make sure this message is shown on the display:

Table 518/21-31-00-993-818

DADC ON?

NOTE: The DADC is referring to the Air Data Inertial Reference Units (ADIRU) that provides the air data to the pressure controller.

(k) Make sure that both ADIRU L and R are ON.

(l) Push the YES button.

(m) Make sure this message is shown on the display:

Table 519/21-31-00-993-819

SMC ON?

NOTE: SMC is the Stall Management Yaw Damper (SMYD) computer that provides data to the pressure controller.

(n) Make sure that both SMYD-1 and SMYD -2 are ON.

(o) Push the YES button on the pressure controller No. 2.

(p) Make sure this message is shown on the display:

Table 520/21-31-00-993-820

OFV CLEAR?

(q) Do these steps in less than one minute or the test will fail:

NOTE: You will try to do a system test on pressure control No. 1 while the test for pressure controller No. 2 is in progress. The system test cannot be done on the two pressure controllers at the same time. This test makes sure that if you try to do the system test on pressure controller No. 1 when the test is in progress on pressure controller No. 2 that the test on pressure controller No. 1 will be aborted.

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

1) Push the YES button on pressure controller No. 2 after you make sure the outflow valve is not blocked.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure this message is shown on the display:

NOTE: Five or eight digits will come on below the word "TESTING" at intervals of approximately 12 seconds. Older controllers will show eight digits. Newer controllers will show 5 digits.

Table 521/21-31-00-993-821

TESTING

3) Do these steps on the BITE module on pressure controller No. 1:

a) Push the ON/OFF button on the front of pressure controller No. 1 (M1653).

b) Push the MENU button.

c) Continue to push the button with the down arrow label until the display shows this message:

Table 522/21-31-00-993-822

GROUND TEST

d) Push the YES button.

e) Make sure this message is shown on the display:

Table 523/21-31-00-993-823

DISPLAY TEST

f) Continue to push the button with the down arrow label until the display shows this message:

Table 524/21-31-00-993-824

SYSTEM TEST

g) Push the YES button to start the system test.

h) Make sure the display on the BITE module of pressure controller No. 1 shows these messages:

NOTE: The messages will show one after the other with a two second delay between the first and the second message. This shows that the test on pressure controller No. 1 is aborted.

Table 525/21-31-00-993-825

BOTH SYS
IN IBIT
IBIT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 525/21-31-00-993-825 (Continued)

ABORTED

i) Push the ON/OFF button on the BITE module of pressure controller No. 1 (M1653).

(r) Make sure that five or eight digits on pressure controller No. 2 are on.

NOTE: Five or eight digits will come on below the word "TESTING" at intervals of approximately 12 seconds. Older controllers will show eight digits. Newer controllers will show 5 digits.

(s) Make sure the display on the BITE module of pressure controller No. 2 shows these messages:

NOTE: The messages will show one after the other with a two second delay between the first and the second message.

Table 526/21-31-00-993-826

SYSTEM
OK
SYSTEM TEST
& CLEAR

(t) Push the YES button on the BITE module on pressure controller No. 2 (M1654).

(u) Push the ON/OFF button on the BITE module on pressure controller No. 2 (M1654).

——— END OF TASK ———

TASK 21-31-00-740-801

6. Cabin Pressure Controller (CPC) BITE Procedure

A. General

(1) The Cabin Pressure Controller (CPC) is a part of the Cabin Pressure Control System. It automatically controls with related aircraft systems, the cabin pressure and the rate of pressure change.

(2) There are two the same CPC.

(a) CPC No. 1 is on the E2-2 shelf in the electronic equipment bay.

(b) CPC No. 2 is on the E1-1 shelf in the electronic equipment bay.

(3) The steps that follow give a summary of the CPC button functions.

(a) ON/OFF button activates or de-activates the Built-In-Test Equipment (BITE) functions in the CPC.

(b) MENU button is used to show the BITE menus or move up one level in the BITE menus.

(c) YES button is used to respond to questions.

(d) NO button is used to respond to questions.

(e) Up arrow button is used to scroll up through the menu or the results of a BITE test.

(f) Down arrow button is used to scroll down through the menu or the results of a BITE test.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the BITE Procedure

SUBTASK 21-31-00-040-001

(1) Do the steps below to prepare for the applicable BITE procedures:

(a) Push the ON/OFF button on the CPC front panel.

(b) Put the airplane in the ground mode (TASK 32-09-00-860-802).

E. BITE Procedure for EXISTING FAULTS

SUBTASK 21-31-00-740-005

(1) Do these steps on the CPC:

(a) Make sure that the CPC front panel display shows EXISTING FAULTS.

(b) Push the YES button on the CPC front panel.

(c) The CPC front panel display shows the first fault (maintenance message) in the existing fault list.

NOTE: When the cabin pressure controller shows a maintenance message that requires maintenance action, there is an equivalent maintenance message that shows on the Onboard Maintenance Function (OMF).

(d) Find and record the existing faults:

1) Make a record of the existing fault (maintenance message).

2) Push the YES button, on the CPC front panel, to see the fault details (fault numbers) for the maintenance message.

3) Make a record of the fault detail (fault number) for the maintenance message.

4) Push the down arrow button, on the CPC front panel, to look for the next fault detail (fault number).

5) Make a record of this fault detail (fault number) also.

6) Do the above two steps again until the CPC front panel display shows BOTTOM OF LIST for two seconds, then shows the maintenance message again.

7) Push the MENU button, on the CPC front panel, to go back to see the maintenance message in the existing fault list.

8) Push the down arrow button, on the CPC front panel, to see the next maintenance message in the EXISTING FAULTS list.

NOTE: If there are no more faults, the display will show "BOTTOM OF LIST" for two seconds, then the previous maintenance message will show.

9) Do the eight steps above until the CPC front panel display shows BOTTOM OF LIST for two seconds so that all maintenance messages and associated fault numbers have been recorded.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Push the MENU button on the CPC front panel.

(f) Go to the IFIM and do the applicable procedure(s) for each maintenance message.

(g) If the CPC front panel display is not at a main menu selection, push the MENU button until you are in the main menu.

F. BITE Procedure for FAULT HISTORY

SUBTASK 21-31-00-740-006

(1) Do these steps on the CPC:

(a) Make sure that the CPC front panel display shows FAULT HISTORY.

(b) Push the YES button on the CPC front panel.

(c) The CPC front panel display shows FLIGHT LEG XX.

NOTE: XX is the last flight leg that faults were recorded in history.

(d) If there are no FLIGHT LEGS shown on the CPC front panel display, go to the IFIM and do the applicable procedure(s) for each maintenance message.

NOTE: This shows that there are no faults recorded in FAULT HISTORY.

(e) Push the YES button on the CPC front panel.

(f) The CPC front panel display shows the first fault (maintenance message) in the flight leg fault list.

(g) Find and record the faults in the flight leg fault list as follows:

1) Make a record of the fault (maintenance message).

2) Push the YES button, on the CPC front panel, to see the fault details (fault numbers) for the maintenance message.

NOTE: When the CPC shows a maintenance message that requires maintenance action, there is an equivalent maintenance message that shows on the OMF.

3) Make a record of the fault detail (fault number) for the maintenance message.

4) Push the down arrow button, on the CPC front panel, to look for the next fault detail (fault number) for the maintenance message.

5) Make a record of this fault detail (fault number) also.

6) Do the above two steps again until the CPC front panel display shows BOTTOM OF LIST for two seconds, then returns to the maintenance message in the fight leg fault list.

7) Push the down arrow button, on the CPC front panel, to look for the next fault (maintenance message) for the flight leg.

8) Do the seven steps above until all fault entries in the flight leg fault list have been examined and recorded for the flight leg.

(h) Push the MENU button, on the CPC front panel, until the display shows FLIGHT LEG XX.

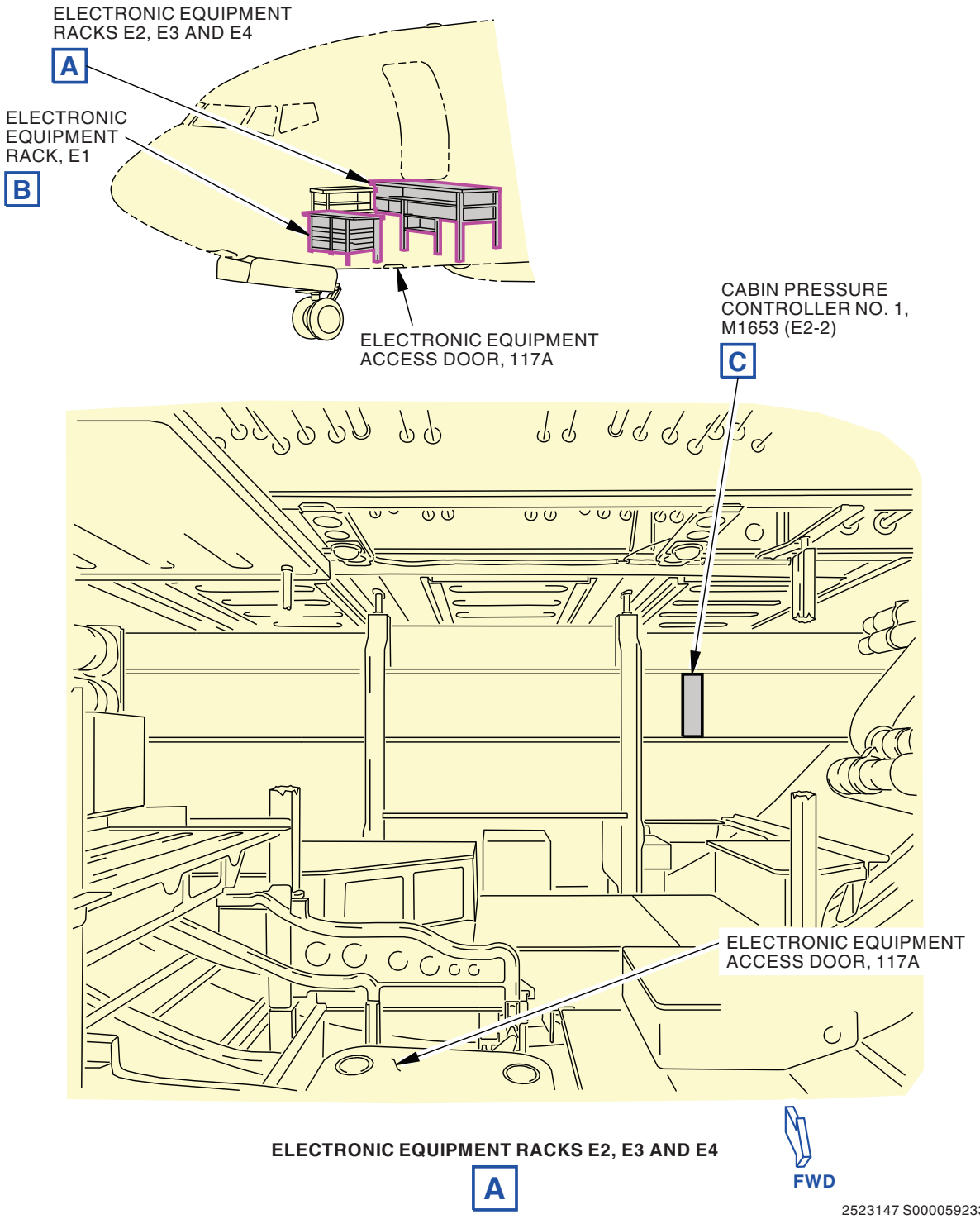
(i) Push the down arrow button, on the CPC front panel, to look for the next flight leg that recorded faults in FAULT HISTORY.

(j) Repeat this procedure until all flight legs with faults in FAULT HISTORY have been examined and recorded.

(k) If the CPC front panel display is not at a main menu selection, push the MENU button until you are in the main menu.

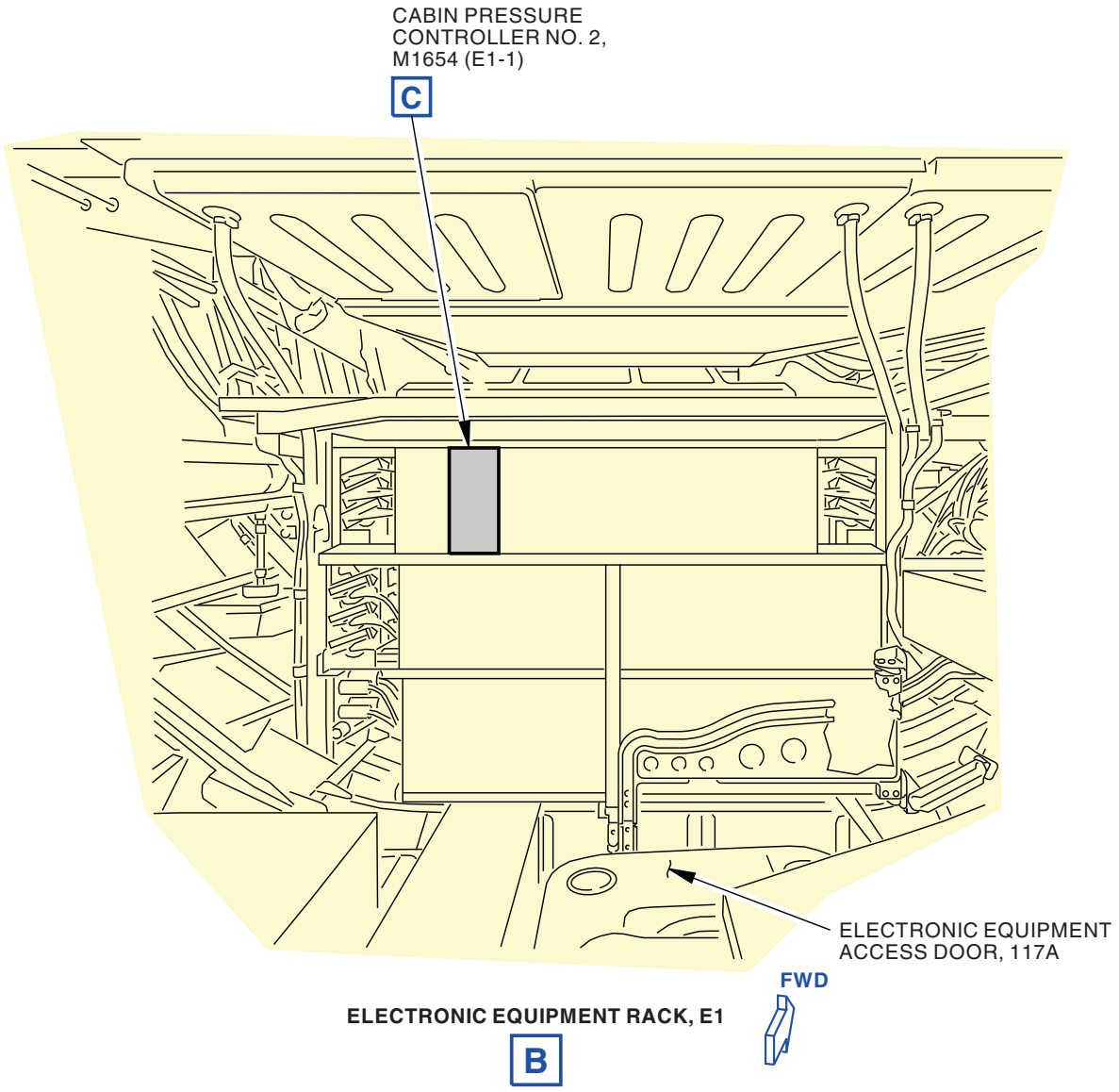
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Pressure Control System
Figure 502/21-31-00-990-804 (Sheet 1 of 3)

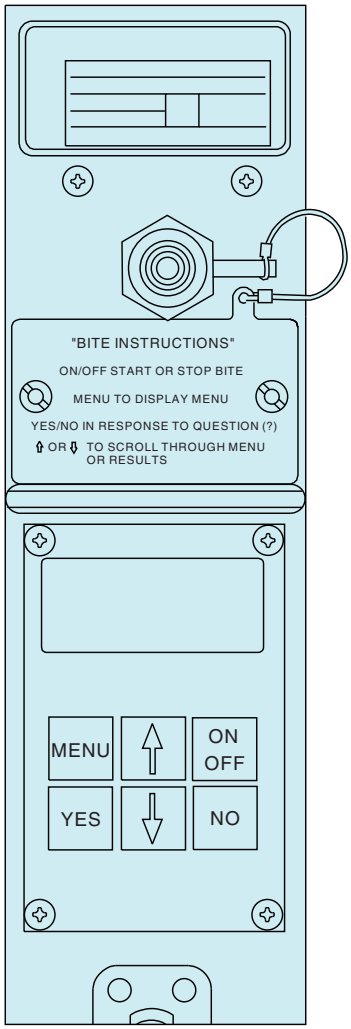
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3009742 S0000783415_V1

Cabin Pressure Control System
Figure 502/21-31-00-990-804 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



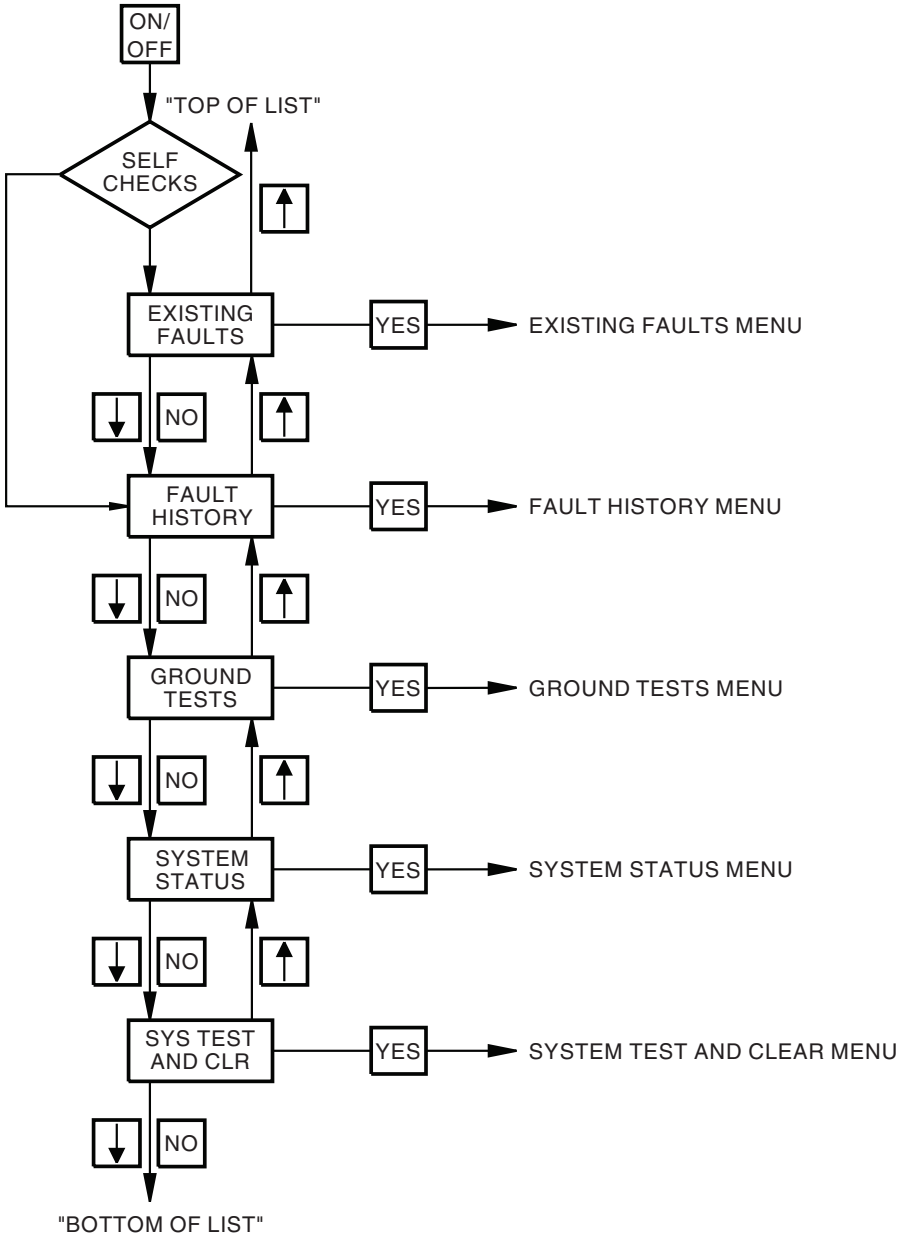
CABIN PRESSURE CONTROLLER (EXAMPLE)

C

2523159 S0000592335_V2

Cabin Pressure Control System
Figure 502/21-31-00-990-804 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

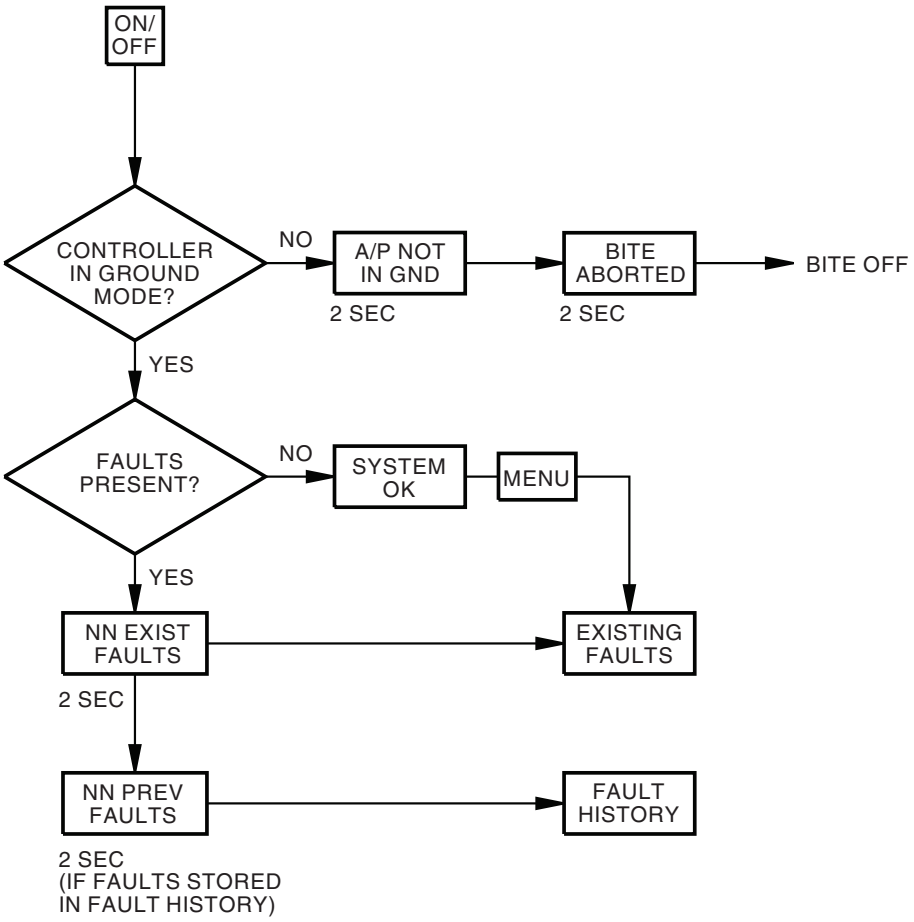


CPC BITE MAIN MENUS

2523161 S0000592344_V1

Cabin Pressure Controller BITE Procedure
Figure 503/21-31-00-990-805 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

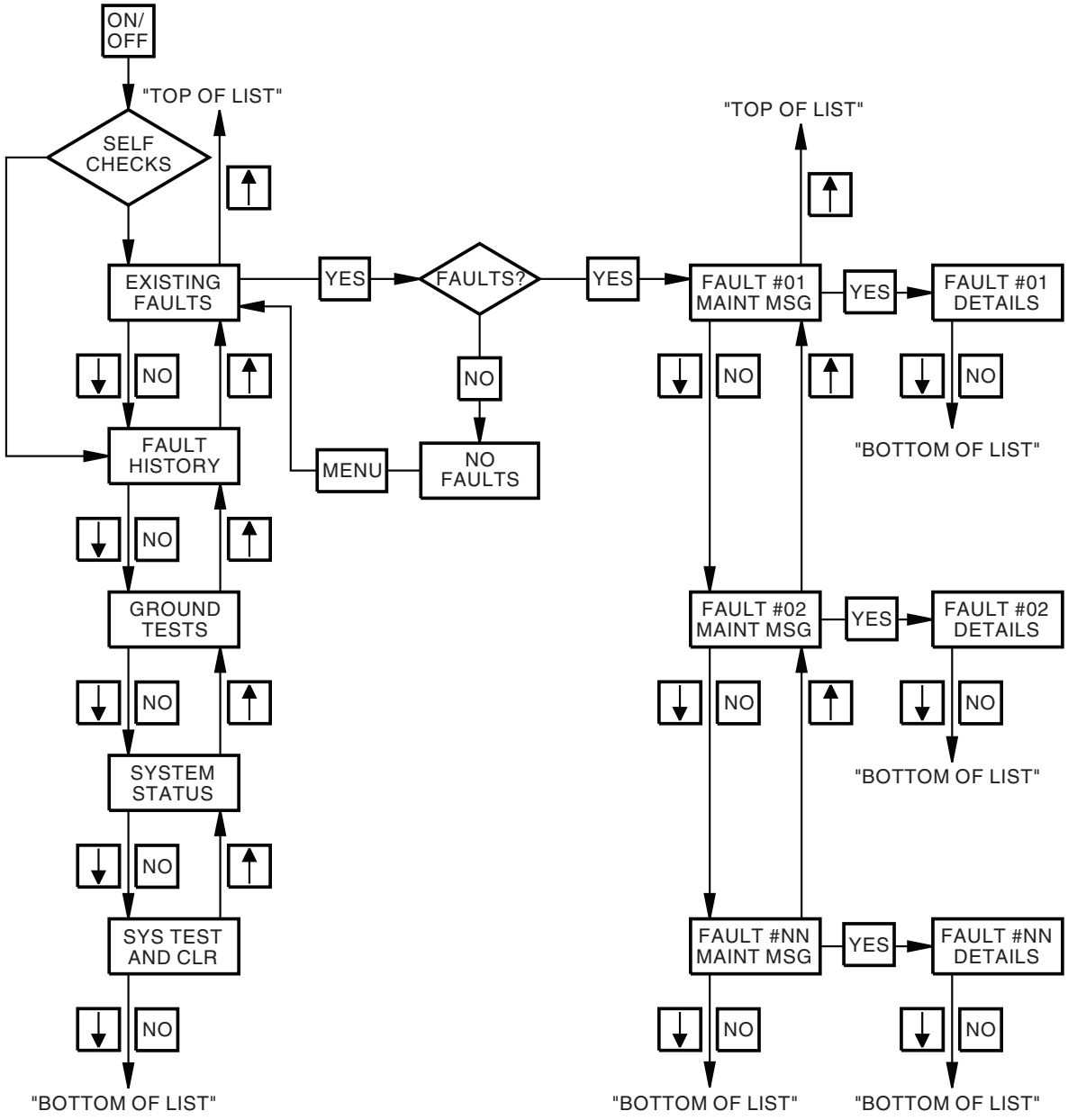


BITE ACCESS

2523188 S0000592347_V1

Cabin Pressure Controller BITE Procedure
Figure 503/21-31-00-990-805 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

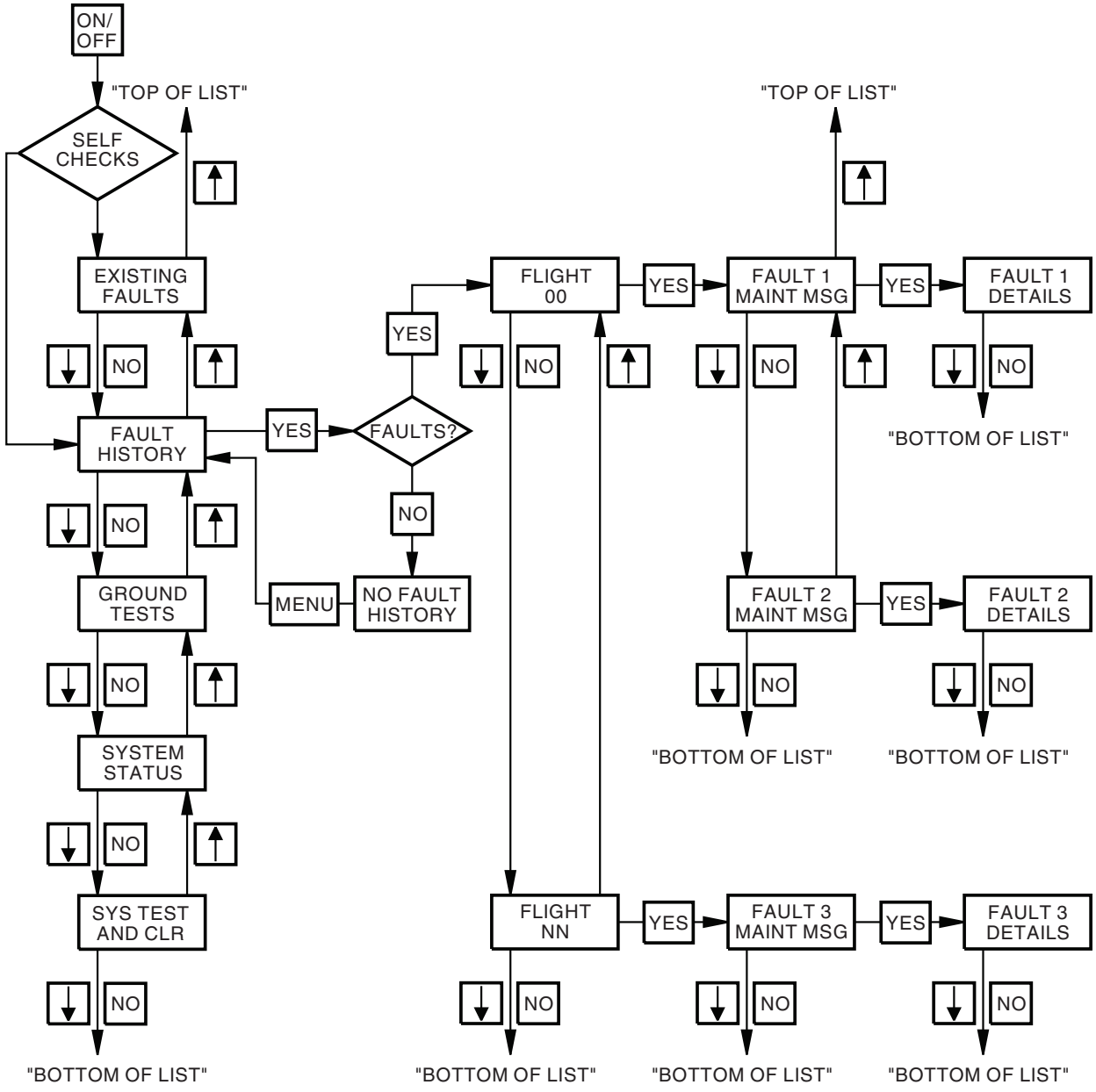


EXISTING FAULTS MENU

2523214 S0000592348_V1

Cabin Pressure Controller BITE Procedure
Figure 503/21-31-00-990-805 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

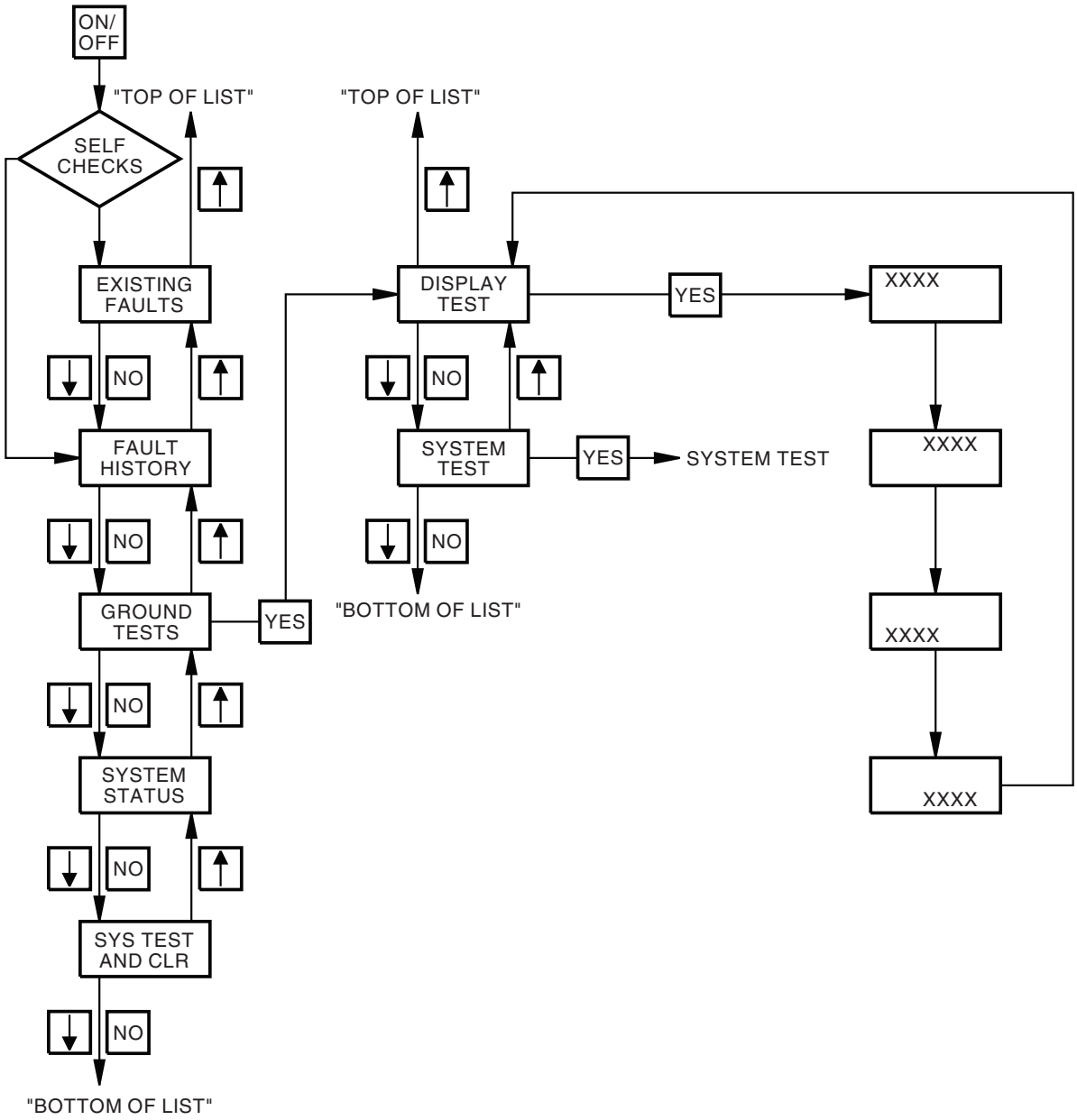


FAULT HISTORY MENU

2523222 S0000592349_V1

Cabin Pressure Controller BITE Procedure
Figure 503/21-31-00-990-805 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

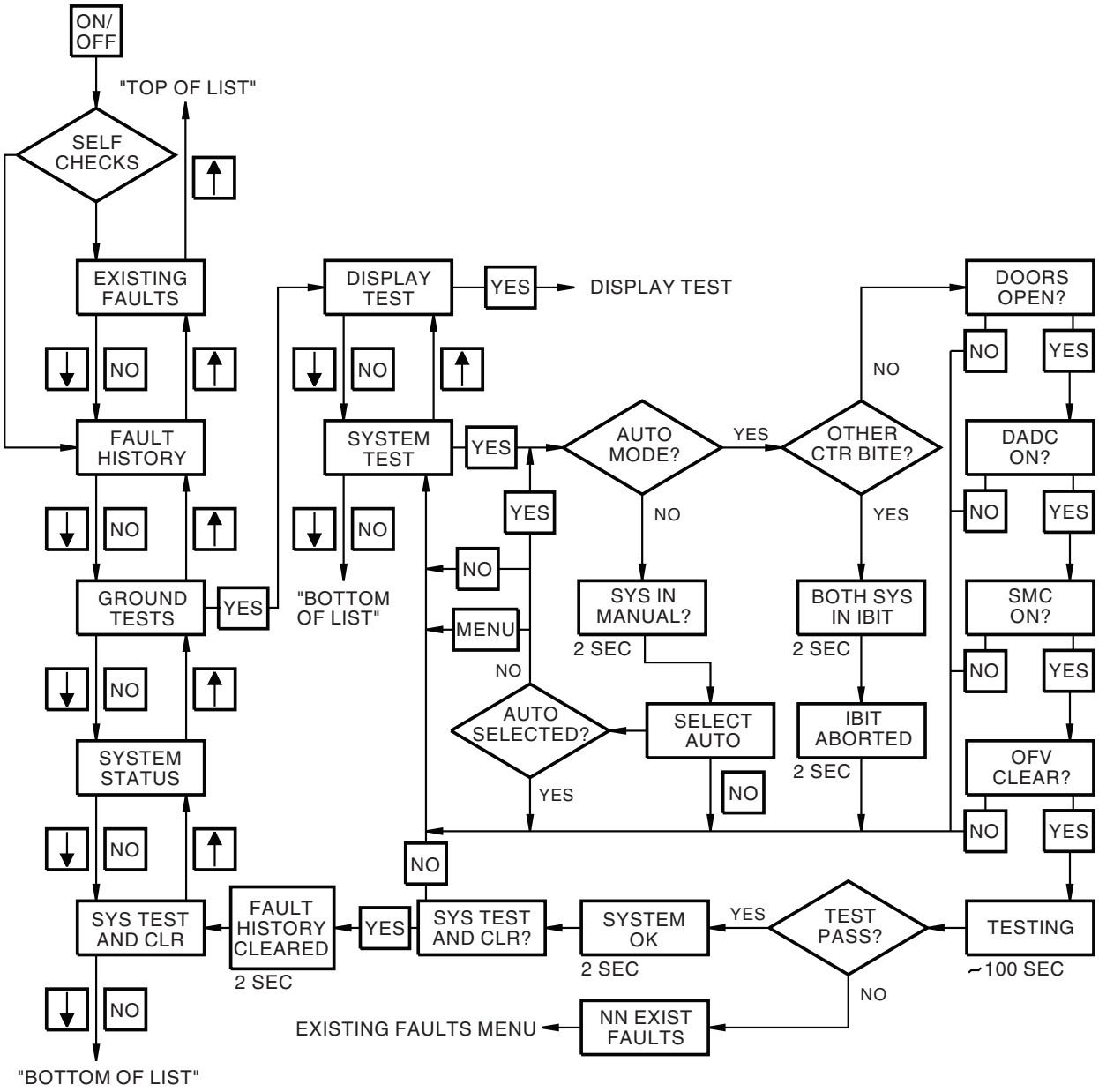


GROUND TESTS MENUS - DISPLAYS

2523227 S0000592351_V1

Cabin Pressure Controller BITE Procedure
Figure 503/21-31-00-990-805 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



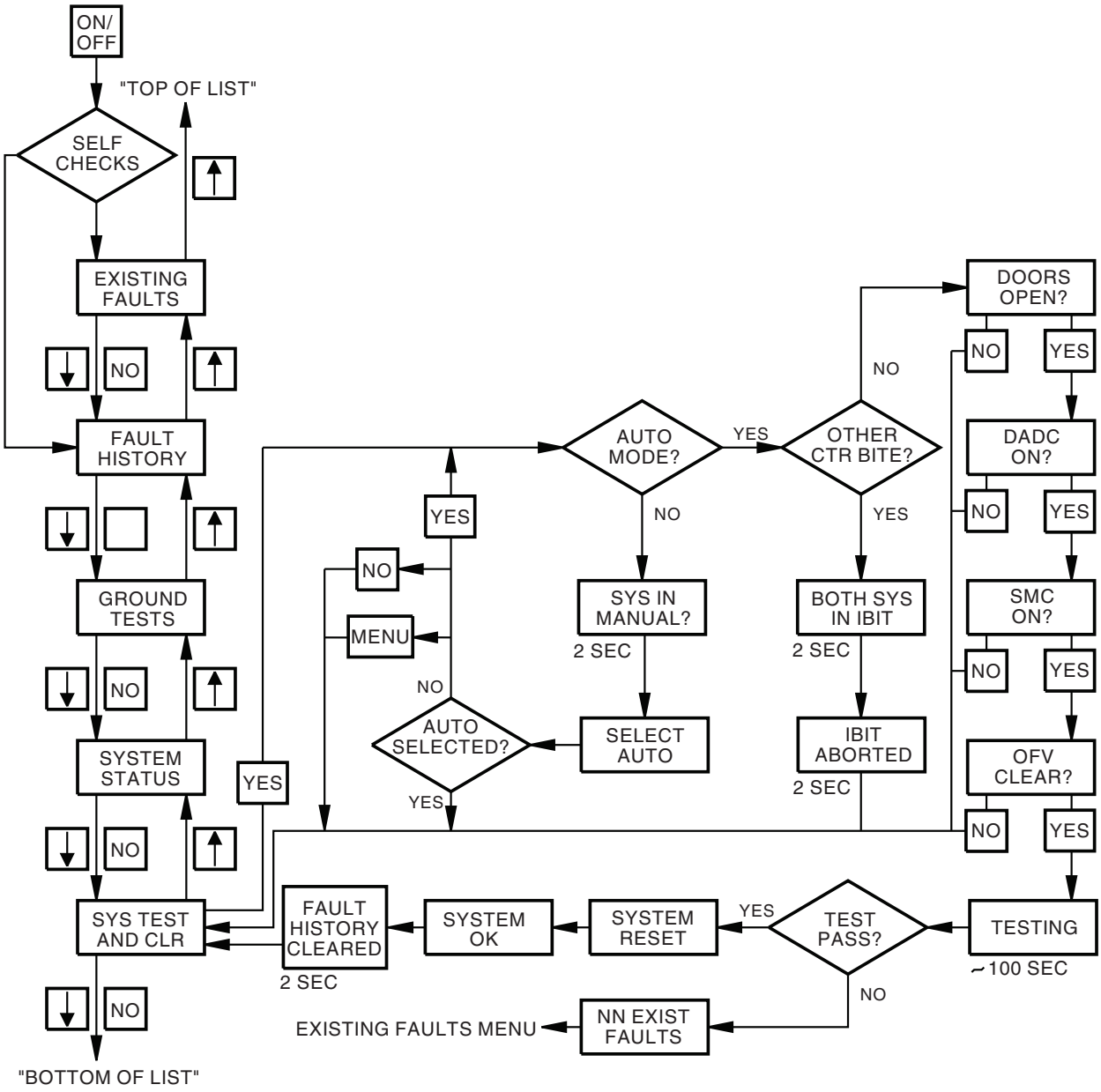
GROUND TESTS MENUS - SYSTEMS TEST

NOTE: DADC REFERS TO THE AIR DATA INERTIAL REFERENCE UNIT (ADIRU).
SMC REFERS TO THE STALL MANAGEMENT YAW DAMPENER (SMYD) COMPUTER.

2523655 S0000592352_V2

Cabin Pressure Controller BITE Procedure
Figure 503/21-31-00-990-805 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SYSTEM TEST & CLEAR MENUS

NOTE: DADC REFERS TO THE AIR DATA INERTIAL REFERENCE UNIT (ADIRU).
SMC REFERS TO THE STALL MANAGEMENT YAW DAMPENER (SMYD) COMPUTER.

2523665 S0000592354_V2

Cabin Pressure Controller BITE Procedure
Figure 503/21-31-00-990-805 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN PRESSURE CONTROLLER (CPC) - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cabin pressure controller (CPC).

(2) An installation of the cabin pressure controller (CPC).

TASK 21-31-01-000-801

2. Cabin Pressure Controller (CPC) Removal

(Figure 401)

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Procedure

SUBTASK 21-31-01-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

SUBTASK 21-31-01-010-001

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-31-01-020-001

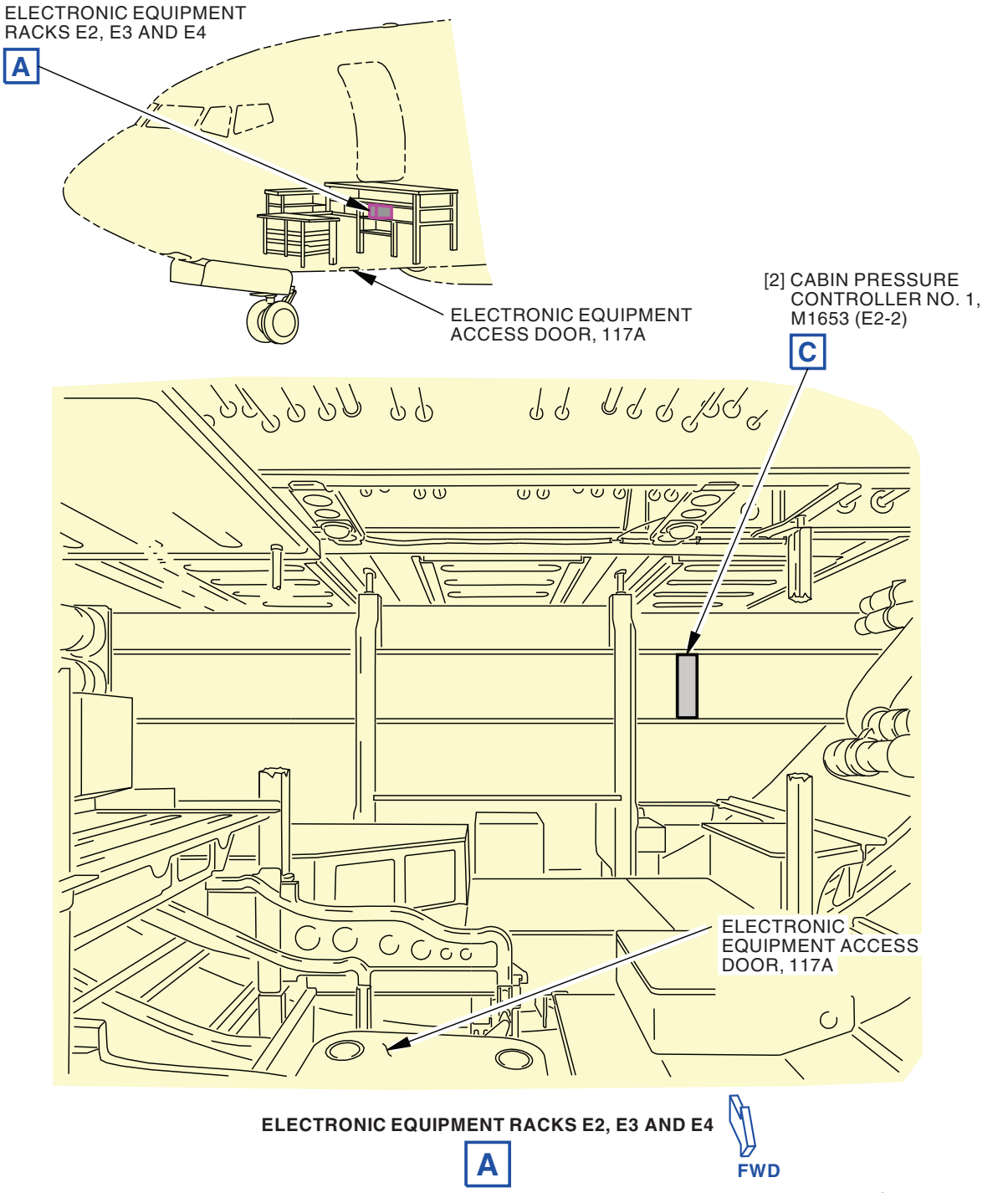
CAUTION

REMOVE AND INSTALL THE PRESSURE CONTROLLER CAREFULLY TO PREVENT DAMAGE TO THE COPPER TAPE ON THE FRONT OF THE PRESSURE CONTROLLER. THE COPPER TAPE PREVENTS ELECTROMAGNETIC INTERFERENCE. IF YOU ARE NOT CAREFUL, DAMAGE TO THE COPPER TAPE WILL OCCUR.

(3) Remove the cabin pressure controller No. 2 [1] or the cabin pressure controller No. 1 [2]. To remove the controller, do this task: E/E Box Removal, TASK 20-10-07-000-801.

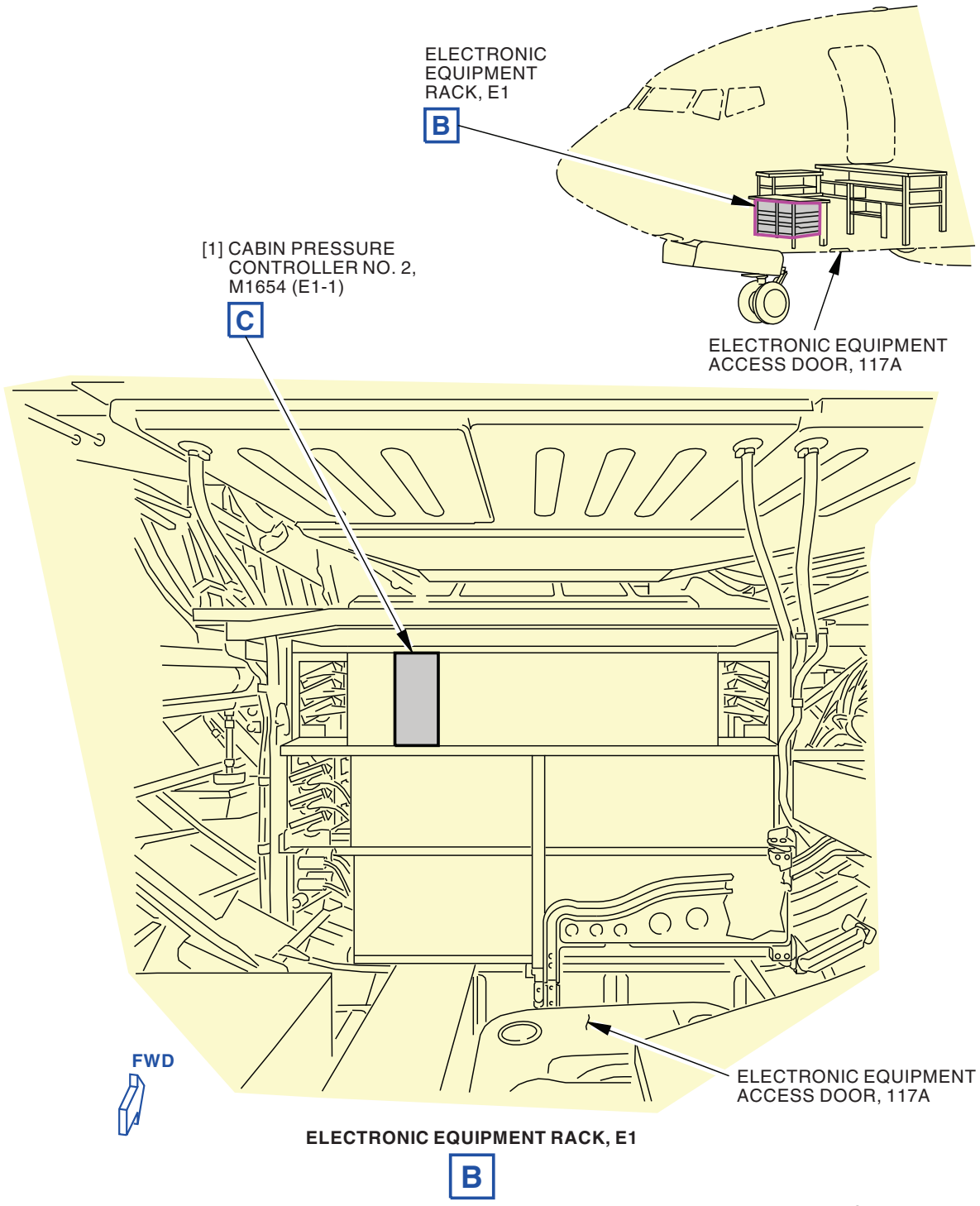
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



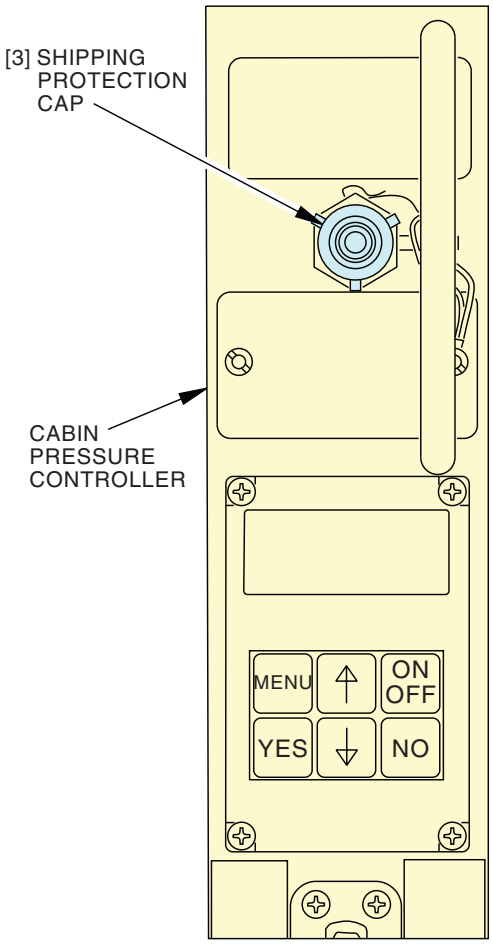
Cabin Pressure Controller Installation
Figure 401/21-31-01-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Pressure Controller Installation
Figure 401/21-31-01-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN PRESSURE CONTROLLER (EXAMPLE)

C

2479871 S0000582552_V1

Cabin Pressure Controller Installation
Figure 401/21-31-01-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-31-01-400-801

3. Cabin Pressure Controller (CPC) Installation

(Figure 401)

A. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
21-31-00-700-802	Pressurization System Ground Test (P/B 501)
24-22-00-860-801	Supply Electrical Power (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Pressure controller No. 2	21-31-01-03-010	PTF ALL

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Cabin Pressure Controller (CPC) Installation

SUBTASK 21-31-01-420-001

(1) Make sure the shipping protection cap [3] is removed from the cabin pressure port on the faceplate of the controller.

SUBTASK 21-31-01-420-002

(2) Install the cabin pressure controller No. 2 [1] or the cabin pressure controller No. 1 [2]. To install the controller, do this task: E/E Box Installation, TASK 20-10-07-400-801.

SUBTASK 21-31-01-860-002

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2

F. Cabin Pressure Controller (CPC) Installation Test

SUBTASK 21-31-01-860-003

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-31-01-740-001

(2) Do a pressurization system ground test on the controller. To do the ground test, do this task: Pressurization System Ground Test, TASK 21-31-00-700-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-31-01-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN PRESSURE CONTROL MODULE - REMOVAL/INSTALLATION

1. General

A. The pressure control module is installed on the lower right corner of the P5 forward overhead panel in the flight deck.

B. This procedure has these tasks:

(1) A removal of the cabin pressure control module.

(2) An installation of the cabin pressure control module.

TASK 21-31-02-000-801

2. Cabin Pressure Control Module Removal

(Figure 401)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Cabin Pressure Control Module Removal

SUBTASK 21-31-02-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	7	C01272	PRESSURIZATION CONTROL IND

SUBTASK 21-31-02-010-001

(2) Get access to the electrical connectors on the back of the P5-6 cabin pressure control module in the P5 forward overhead panel.

(a) Loosen the 1/4-turn fasteners that hold the P5 forward overhead panel in position and let the panel move downward.

SUBTASK 21-31-02-020-001

(3) Remove the P5-6 cabin pressure controlmodule [1] from the P5 forward overhead panel as follows:

(a) Disconnect the electrical connectors at the back of the module.

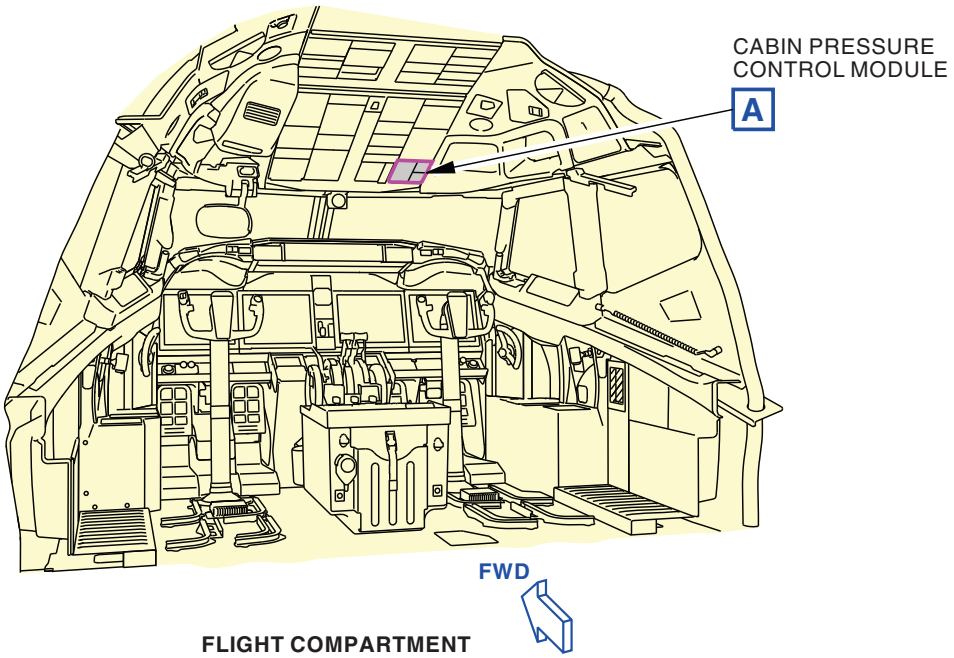
WARNING HOLD THE PRESSURE CONTROL MODULE WHEN YOU LOOSEN THE 1/4-TURN FASTENERS. IF THE MODULE FALLS, INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Hold the cabin pressure control module [1] while you loosen the 1/4-turn fasteners that attach the cabin pressure control module [1] to the P5 forward overhead panel.

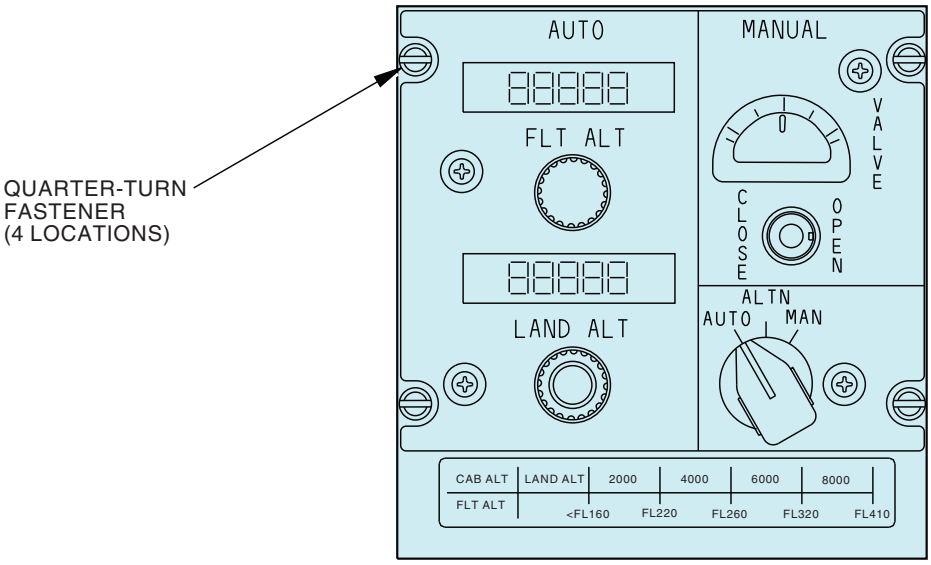
(c) Remove the cabin pressure control module [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



[1] CABIN PRESSURE CONTROL MODULE

A

2411332 S00061527652_V1

Cabin Pressure Control Module Installation
Figure 401/21-31-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-31-02-400-801

3. Cabin Pressure Control Module Installation

(Figure 401)

A. References

Reference	Title
21-31-00-700-802	Pressurization System Ground Test (P/B 501)
21-31-00-710-801	Pressurization System Manual Mode Test (P/B 501)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Cabin Pressure Control Module Installation

SUBTASK 21-31-02-860-002

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	7	C01272	PRESSURIZATION CONTROL IND

SUBTASK 21-31-02-010-002

(2) If it has not already been done, lower the P5 forward overhead panel.

SUBTASK 21-31-02-420-001

(3) Install the P5-6 cabin pressure control module [1] into the P5 forward overhead panel as follows:

(a) Put the module [1] into the P5 forward overhead panel and hold it in position.

(b) Turn the 1/4-turn fasteners on the cabin pressure control module [1] to hold the module to the P5 forward overhead panel.

(c) Connect the two electrical connectors to the module.

SUBTASK 21-31-02-010-003

(4) Lift the P5 forward overhead panel to the closed position and turn the 1/4-turn fasteners.

SUBTASK 21-31-02-860-003

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	7	C01272	PRESSURIZATION CONTROL IND

D. Cabin Pressure Control Module Installation Test

SUBTASK 21-31-02-860-004

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-31-02-860-005

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	1	C01273	PRESSURIZATION CONTROL LCD LTG
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-31-02-710-001

(3) Do a cabin pressurization system ground test. To do the ground test, do this task: Pressurization System Ground Test, TASK 21-31-00-700-802.

SUBTASK 21-31-02-710-002

(4) Do a Pressurization System Manual Mode Test. To do the manual mode test, do this task: Pressurization System Manual Mode Test, TASK 21-31-00-710-801.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-31-02-410-001

(1) Remove electrical power if it is not necessary. To remove electrical power, do this task: Remove Electrical Power, TASK 24-22-00-860-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AFT OUTFLOW VALVE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure gives instructions for the removal and installation of the aft outflow valve assembly.

B. The aft outflow valve assembly is found on the bottom of the airplane near the aft service door.

TASK 21-31-03-000-801

2. Aft Outflow Valve Assembly Removal

(Figure 401 and Figure 402)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14208	Extension Cable - Outflow Valve Installation, Aft Fuselage Part #: C21008-1 Supplier: 81205

B. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

C. Access Panels

Number	Name/Location
822	Aft Cargo Door

D. Prepare for the Removal

SUBTASK 21-31-03-860-001

WARNING

DO NOT OPERATE THE OUTFLOW VALVE. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT THE VALVE GATES STAY FULLY CLOSED DURING TRANSPORTATION, REPAIR OR OVERHAUL. DAMAGE CAN OCCUR TO THE SEALS AND MOVABLE PARTS IF NOT PREPARED CORRECTLY.

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-31-03-860-009

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-31-03-000-002

(3) Remove the E6 electrical equipment rack, aft access panel.

E. Aft Outflow Valve Assembly Removal

SUBTASK 21-31-03-020-001

CAUTION

BE CAREFUL WHEN YOU REACH THROUGH THE OPENING OF THE OUTFLOW VALVE. DAMAGE CAN OCCUR TO THE ANTISKATING FOIL ON THE FORWARD AND AFT GATES OF THE OUTFLOW VALVE.

(1) Reach through the opening of the outflow valve [1] to disconnect electrical connectors or other components (Figure 402).

NOTE: The function of the antiskating foil is to reduce noise made by the air going through the outflow valve.

SUBTASK 21-31-03-020-002

(2) Disconnect these electrical connectors from the outflow valve [1]:

NOTE: You can get access to the electrical connectors through the valve door opening.

(a) ELACT 1 electrical connector [3]

(b) ELACT 2 electrical connector [2]

(c) feedback module connector 1 [5]

(d) feedback module connector 2 [6].

SUBTASK 21-31-03-020-003

(3) Remove the nut [7] and washer [8] that attach the bonding jumper [9] to the outflow valve [1].

SUBTASK 21-31-03-020-004

(4) Disconnect the bonding jumper [9] from the outflow valve [1].

SUBTASK 21-31-03-020-008

(5) Loosen the three aft screws [12] until the three aft lugs [11] can be turned.

SUBTASK 21-31-03-020-005

(6) Turn the three aft lugs [11] until they point inboard-to-outboard.

SUBTASK 21-31-03-860-002

WARNING

DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN THE VALVE IS IN OPERATION. IF YOU PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN IT OPERATES YOU CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(7) Do these steps to move the outflow valve [1] gates until the outflow valve [1] can be moved through the fuselage opening:

(a) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(b) Put the mode select switch, on the P5-6 cabin pressure control module on the overhead P5 panel, to the MAN position.

(c) Set the outflow valve [1] in position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Use the service interphone to make sure that the technician at the outflow valve [1] location can hear and speak with the technician in the flight compartment.

2) Push and hold the CLOSE/OPEN toggle switch, on the P5-6 pressure control module, to the CLOSE position until the technician at the outflow valve [1] tells you that the outflow valve [1] is approximately in the specified position (View B-B, Figure 401).

SUBTASK 21-31-03-860-003

(8) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-31-03-020-006

(9) Do these steps to remove the outflow valve [1]:

CAUTION DO NOT APPLY TOO MUCH FORCE TO FORWARD ROW SCREWS AT THE AFT OUTFLOW VALVE AFTER THEY STOPPED. TOO MUCH FORCE APPLIED CAN CAUSE ROTATION OF THE LATCH LUGS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRCRAFT STRUCTURE CAN OCCUR.

(a) Turn the forward three screws [12] in the counterclockwise direction until they stop to release the forward lugs [11] without their rotation.

(b) Slowly push up and aft against the aft end of the outflow valve [1].

(c) Turn the outflow valve [1] until you can get to the manual motor electrical connector [4].

(d) Disconnect the manual motor electrical connector [4].

(e) Lift and turn the outflow valve [1] until it is diagonal to the skin opening of the airplane and until the valve motors are up.

(f) Carefully remove the outflow valve [1] through the skin opening of the airplane.

NOTE: Remove the outflow valve so that the valve motors come out last.

(g) Remove the gasket [10].

SUBTASK 21-31-03-210-001

(10) Visually examine the condition of the antiskating foil on the forward and aft gates of the outflow valve [1] (Figure 402):

(a) If the antiskating foil is missing, do one of these actions:

1) Remove all adhesive or dirt that remains on the valve gates and do not replace the foil as the outflow valve [1] is functionally serviceable with the antiskating foil missing.

2) Repair the antiskating foil per the vendor component maintenance manual at the next convenient maintenance opportunity.

(b) If the antiskating foil is damaged or peeled away from the gate surface, trim the damaged or loose portion of the foil from the gate.

F. Prepare for the Transportation of the Outflow Valve Assembly

SUBTASK 21-31-03-210-002

(1) The outflow valve [1] must be in the fully closed position for transportation.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-31-03-860-006

(2) If you do not have outflow valve installation extension cable, SPL-14208, available, do these steps to close the outflow valve gates for transportation:

(a) Lift the outflow valve [1] sufficiently so that you can connect the manual motor electrical connector [4] on the outflow valve [1].

(b) Connect the manual motor electrical connector [4] on the outflow valve [1].

(c) Put the outflow valve [1] on a suitable support so that the valve doors will be free to move.

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN THE VALVE IS IN OPERATION. IF YOU PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN IT OPERATES YOU CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(e) Put the mode select switch, on the P5-6 cabin pressure control module on the overhead P5 panel, to the MAN position.

(f) Position the outflow valve [1] as follows:

1) Use the service interphone system to make sure that there is communication between the technician at the aft outflow valve location and technician in the flight deck.

2) Push and hold the CLOSE/OPEN toggle switch, on the P5-6 pressure control module, to the CLOSE position until the aft outflow valve is fully closed.

(g) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(h) Disconnect the manual motor electrical connector [4] from the outflow valve [1].

SUBTASK 21-31-03-210-003

(3) The outflow valve [1] must be in the fully closed position for transportation.

SUBTASK 21-31-03-860-005

(4) If the outflow valve installation extension cable, SPL-14208, is available, do these steps to close the outflow valve gates for transportation:

(a) Position the outflow valve [1] so that the valve gates are free to move and are not obstructed.

(b) Connect J1 connector of the outflow valve installation extension cable, SPL-14208, to connector D10730 on the ship's wiring inside the outflow valve opening in the fuselage.

(c) Connect P1 of the outflow valve installation extension cable, SPL-14208, to the manual motor electrical connector [4] (J5) on the aft outflow valve DC motor.

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WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(e) Put the mode select switch, on the P5-6 cabin pressure control module, to the MAN position.

(f) Position the outflow valve [1] as follows:

1) Use the service interphone system to make sure there is communication between the technician at the aft outflow valve location and the technician in the flight deck.

2) Push and hold the CLOSE/OPEN toggle switch, on the P5-6 pressure control module, to the CLOSE position until the outflow valve [1] is fully closed.

(g) Open this circuit breaker and install safety tag:

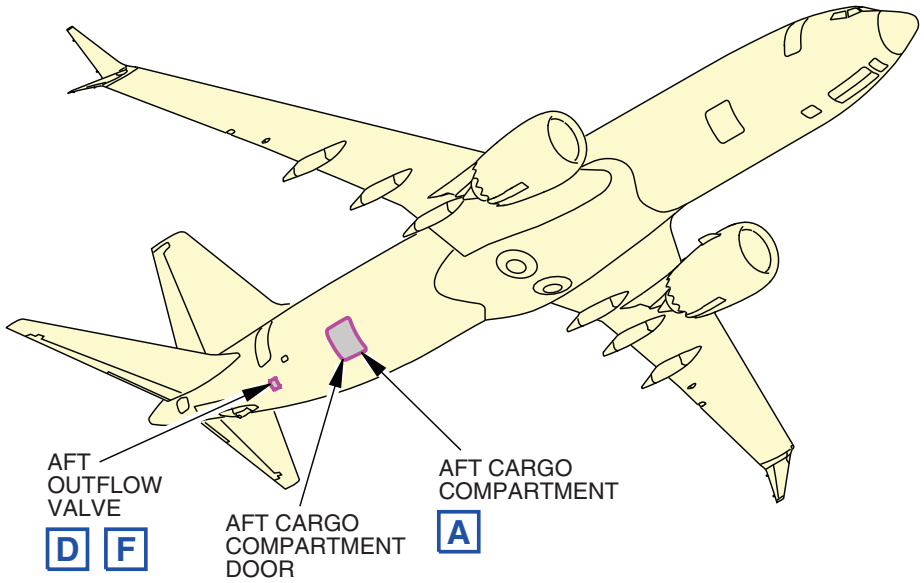
F/O Electrical System Panel, P6-4

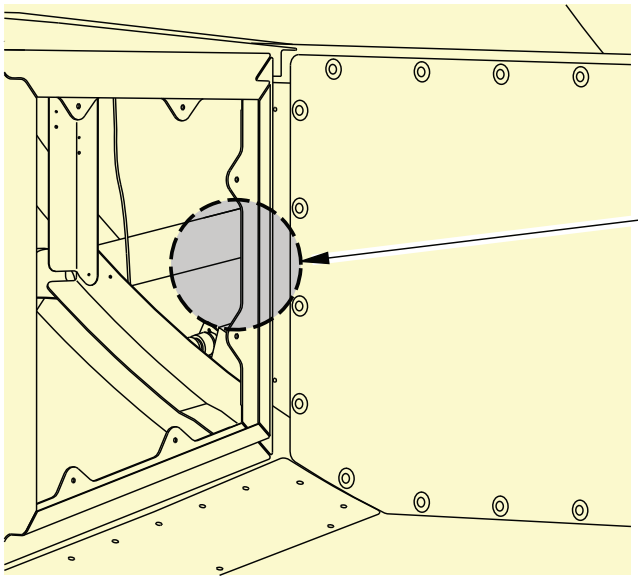
Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(h) Disconnect the outflow valve installation extension cable, SPL-14208, from the aft outflow valve connector J5 and connector D10730 on the ship's wiring.

——— END OF TASK ———

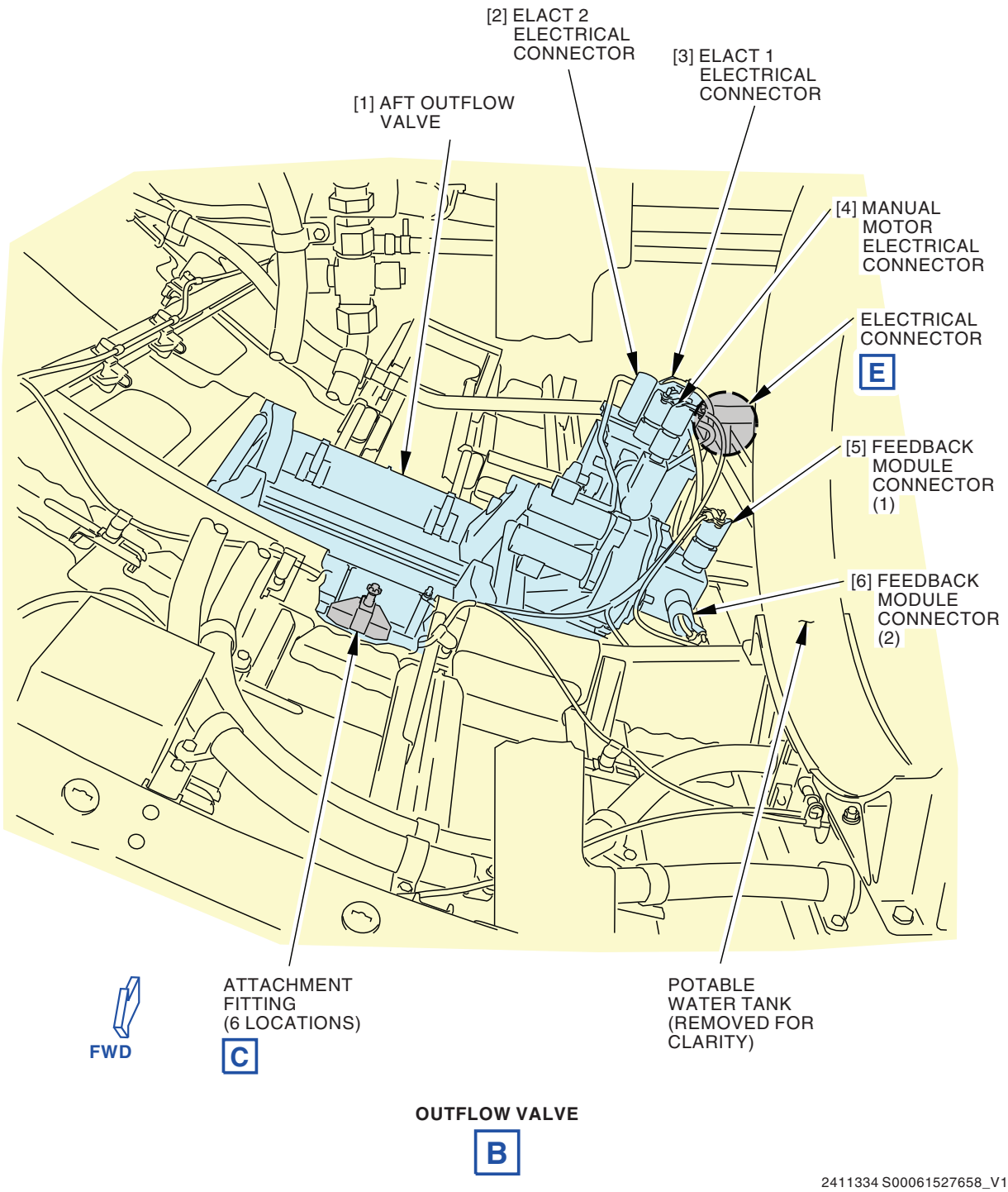
ECCN 9E991 BOEING PROPRIETARY - See title page for details





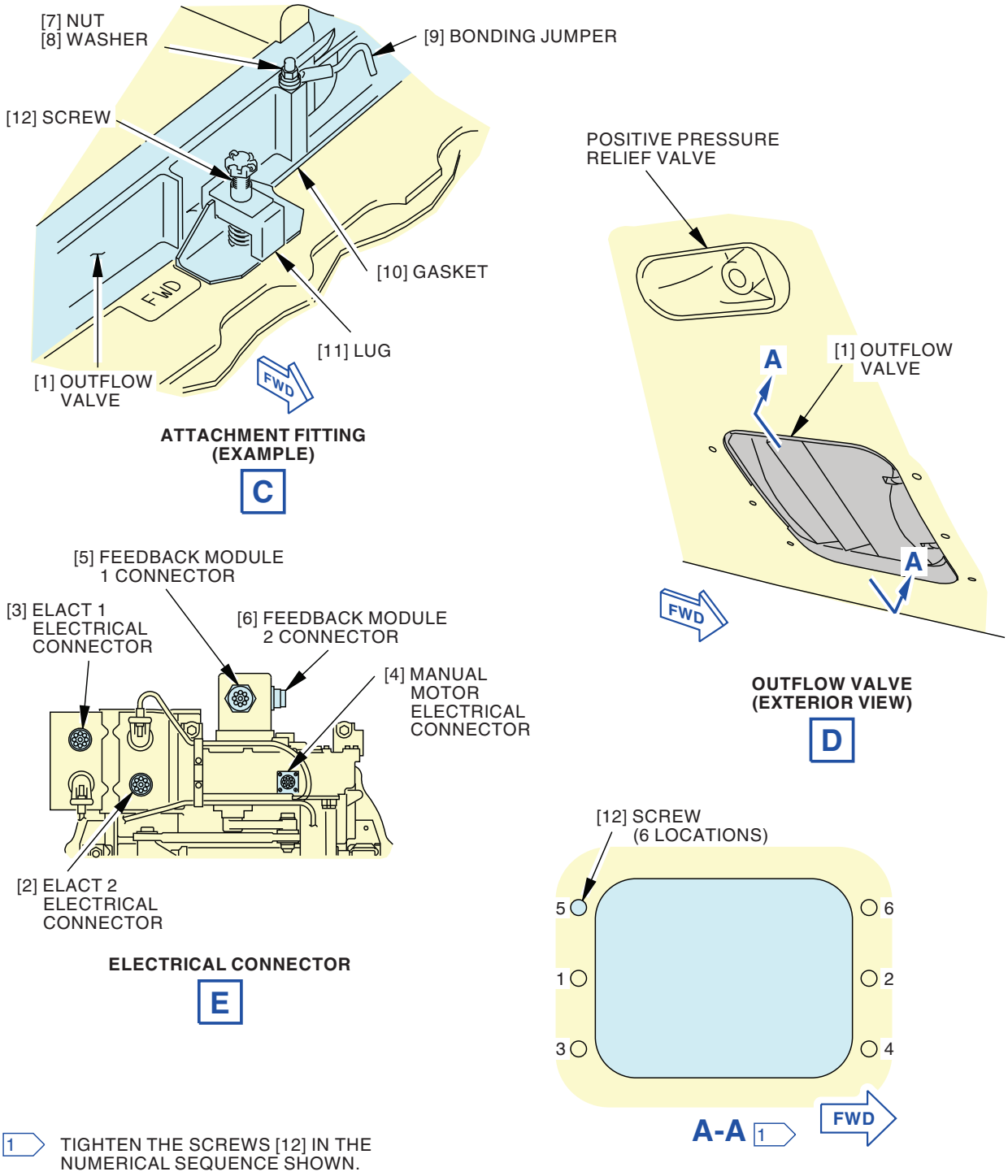
Aft Outflow Valve Installation
Figure 401/21-31-03-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Outflow Valve Installation
Figure 401/21-31-03-990-801 (Sheet 2 of 4)

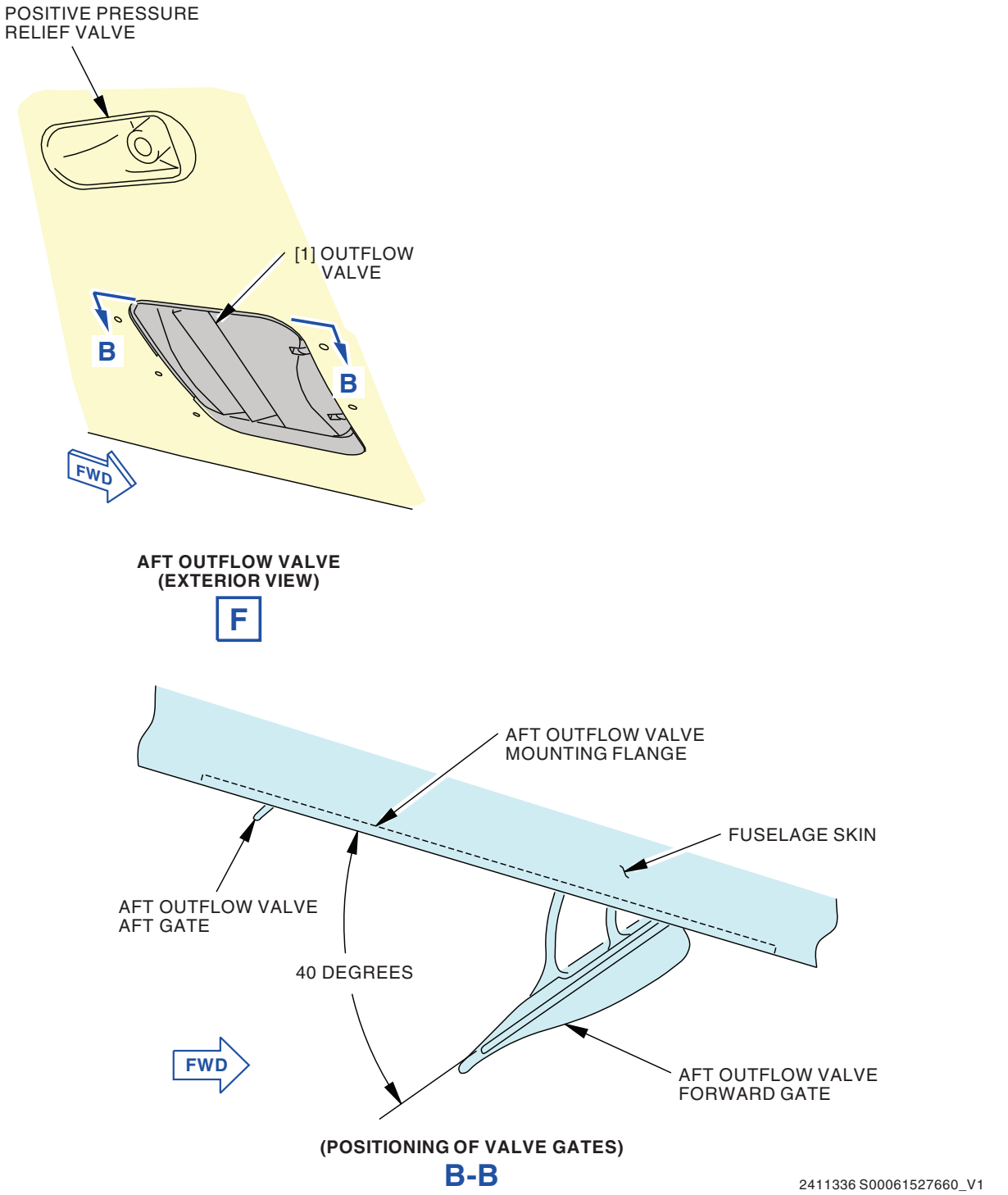
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411335 S00061527659_V1

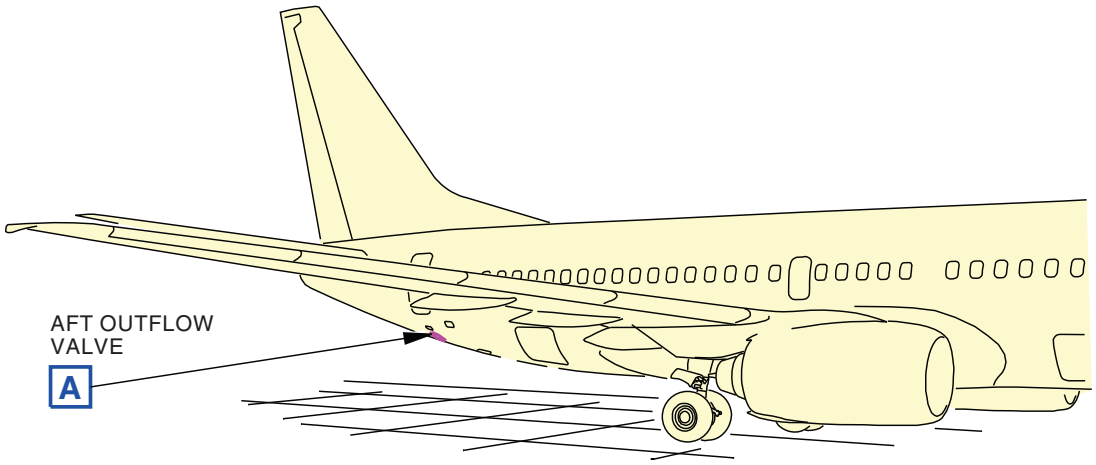
Aft Outflow Valve Installation
Figure 401/21-31-03-990-801 (Sheet 3 of 4)

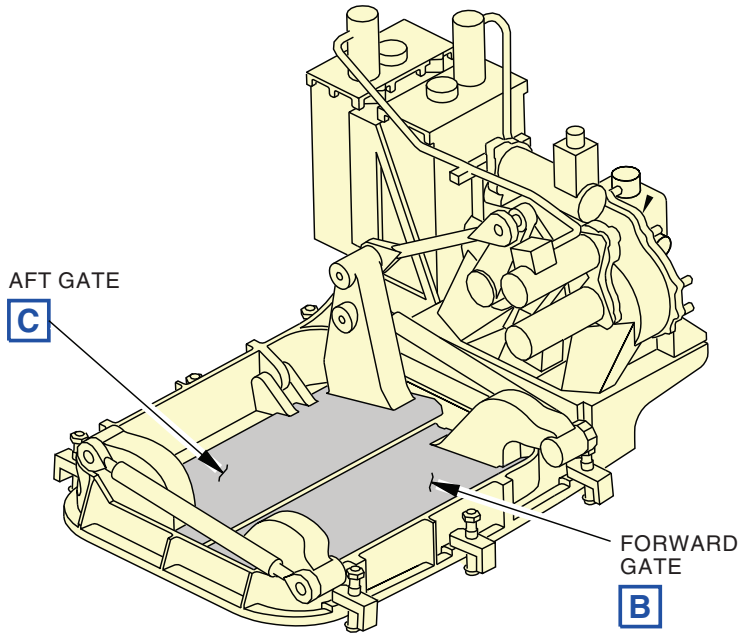
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Outflow Valve Installation
Figure 401/21-31-03-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



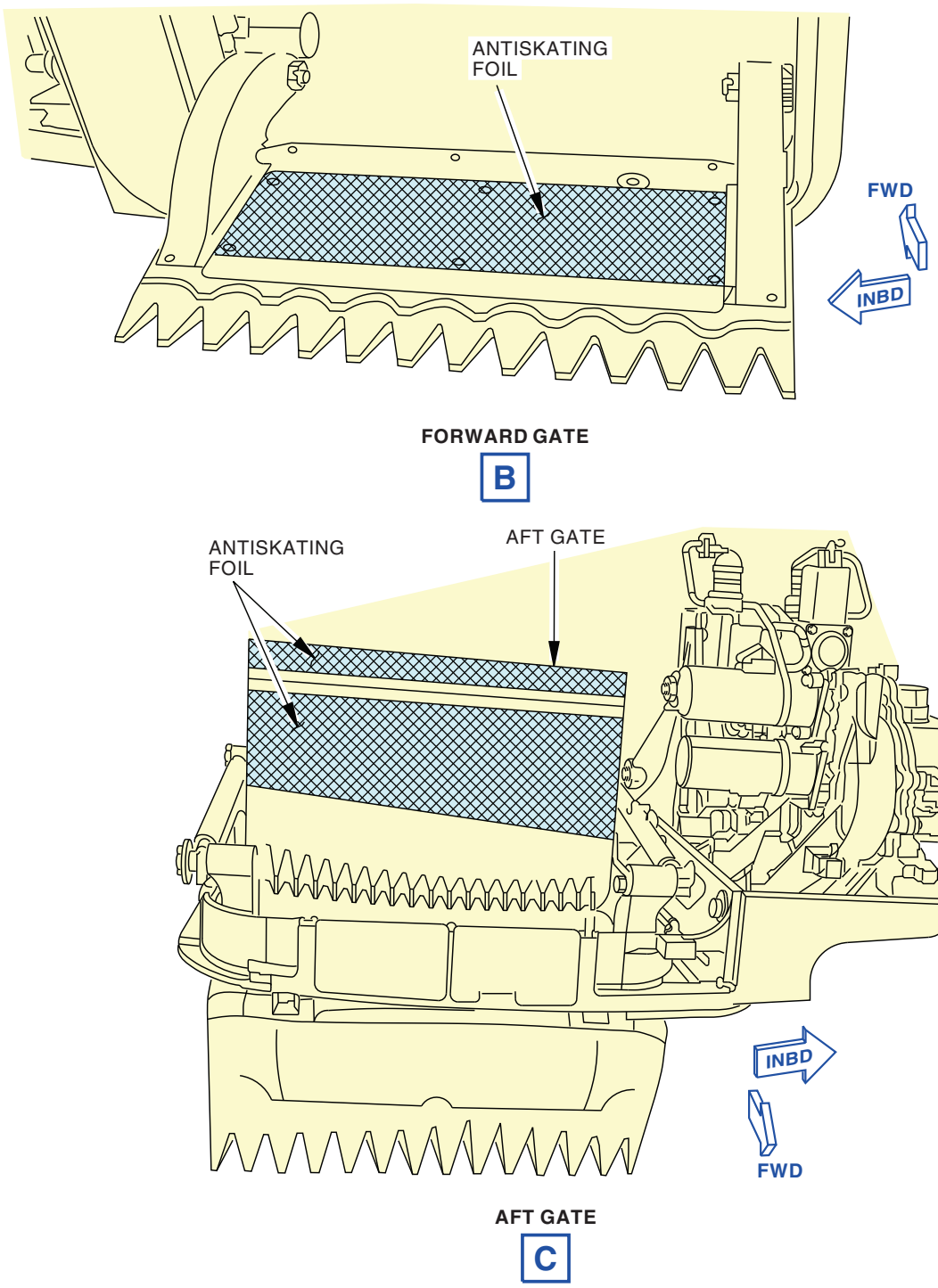


AFT OUTFLOW VALVE
A

2411337 S00061527661_V1

Location of Antiskating Foil on Forward and Aft Gates
Figure 402/21-31-03-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Location of Antiskating Foil on Forward and Aft Gates
Figure 402/21-31-03-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-31-03-400-801

3. Aft Outflow Valve Assembly Installation

(Figure 401 and Figure 402)

A. References

Reference	Title
21-31-00-700-802	Pressurization System Ground Test (P/B 501)
21-31-00-710-801	Pressurization System Manual Mode Test (P/B 501)
24-22-00-860-801	Supply Electrical Power (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
COM-10996	Bit - Screwdriver, Torq-set Part #: 212-5/16A Supplier: 03705 Part #: TS212-5/16B Supplier: 67179
SPL-14208	Extension Cable - Outflow Valve Installation, Aft Fuselage Part #: C21008-1 Supplier: 81205
STD-10994	Screwdriver - Torque, 0-100 in-lbs, +/-5% accuracy

C. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Outflow valve	21-31-03-01-197	PTF ALL
10	Gasket	21-31-03-01-080	PTF ALL

E. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

F. Access Panels

Number	Name/Location
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Aft Outflow Valve Assembly Installation

SUBTASK 21-31-03-860-007

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-31-03-860-008

(2) Make sure that the electrical power is on (TASK 24-22-00-860-801).

SUBTASK 21-31-03-860-010

(3) Make sure that the BAT switch, on the P5-13 electrical meters, battery, and galley power module, is in the ON position.

SUBTASK 21-31-03-420-001

(4) Make sure that all shipping plugs are removed from the vent ports of the two electronic actuators.

NOTE: The vent ports are below the electronic actuators.

SUBTASK 21-31-03-640-001

(5) If a screw [12] is replaced for any reason or if a screw [12] was difficult to loosen, apply a light and even application of compound, D50004, to the threads and shank of the screw [12].

SUBTASK 21-31-03-420-002

CAUTION

BE CAREFUL WHEN YOU REACH THROUGH THE OPENING OF THE OUTFLOW VALVE. DAMAGE CAN OCCUR TO THE ANTISKATING FOIL ON THE FORWARD AND AFT GATES OF THE OUTFLOW VALVE.

(6) Reach through the opening of the outflow valve [1] to connect electrical connectors or other components (Figure 402).

(a) If you install the same outflow valve [1] that you removed and the foil is damaged or peeled away from the gate, the foil can be removed or cut away until all loose foil has been removed.

SUBTASK 21-31-03-980-001

(7) If the outflow valve [1] to be installed is in the fully open position or closed position and you do not have outflow valve installation extension cable, SPL-14208, available, do these steps to put the outflow valve gates in position so that the outflow valve [1] can be moved through the fuselage opening:

NOTE: If you are re-installing the same outflow valve that you removed, then you do not need to change the position of the valve gates.

(a) Lift the outflow valve [1] sufficiently so that you can connect the manual motor electrical connector [4] on the outflow valve [1].

(b) Connect the manual motor electrical connector [4] (J5) on the outflow valve [1].

(c) Put the outflow valve [1] on a suitable support so that the outflow valve [1] doors will be free to move.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN THE VALVE IS IN OPERATION. IF YOU PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN IT OPERATES YOU CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(d) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(e) Put the mode select switch, on the P5-6 cabin pressure control module on the overhead P5 panel, to the MAN position.

(f) Set the outflow valve [1] in its position:

1) Use the service interphone system to make sure that the technician at the outflow valve [1] location can hear and speak with the technician in the flight compartment.

2) Push and hold the CLOSE/OPEN toggle switch, on the P5-6 pressure control module, to the OPEN position until the technician at the outflow valve [1] location tells you that the outflow valve [1] is approximately in the specified position (View B-B, Figure 401).

(g) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(h) Disconnect the manual motor electrical connector [4] from the outflow valve [1].

SUBTASK 21-31-03-480-001

(8) If the outflow valve [1] to be installed is in the fully open position or closed position and outflow valve installation extension cable, SPL-14208, is available, do these steps to position the outflow valve gates so that the outflow valve [1] can be moved through the fuselage opening:

NOTE: If you are re-installing the same outflow valve that you removed, then you do not need to change the position of the valve gates.

(a) Set the outflow valve [1] so that the valve gates are free to move and are not blocked.

(b) Connect the J1 connector of the outflow valve installation extension cable, SPL-14208, to connector D10730 on the ship's wiring inside the outflow valve [1] opening in the fuselage.

(c) Connect P1 of the outflow valve installation extension cable, SPL-14208, to the manual motor electrical connector [4] (J5) on the outflow valve [1] DC motor.

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTLET OF THE OUTFLOW VALVE. THE VALVE IS MOTOR-OPERATED. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(d) Close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Put the mode select switch, on the P5-6 cabin pressure control module to the MAN position.

(f) Set the outflow valve [1] in position:

1) Use the service interphone system to make sure the technician at the outflow valve location can hear and speak with the technician in the flight deck.

2) Push and hold the CLOSE/OPEN toggle switch, on the P5-6 pressure control module, to the OPEN position until the technician at the outflow valve [1] location tells you that the valve is approximately in the specified position (View B-B, Figure 401).

(g) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(h) Disconnect the outflow valve installation extension cable, SPL-14208, from the outflow valve [1] connector J5 and connector D10730 on the ship's wiring.

SUBTASK 21-31-03-420-003

(9) Do these steps to install the outflow valve [1]:

(a) Make sure that the aft three lugs [11] are turned until they point inboard-to-outboard.

NOTE: The lugs will be 1/4 turn away from the installed position.

(b) Put the outflow valve [1] through the skin opening on the airplane so that the motors go in first with the outflow valve [1] diagonal to the skin opening.

NOTE: The motors will be on the inboard side of the skin opening.

(c) Set the outflow valve [1] down on the inside of the airplane so that you can connect the manual motor electrical connector [4].

(d) Connect the manual motor electrical connector [4] to the outflow valve [1].

(e) Install the gasket [10] in its position on the airplane skin opening with the thinner rubber side against the airplane (down).

(f) Do these steps to put the outflow valve [1] in its position:

1) Put the outflow valve [1] above the skin opening with the motors facing inboard.

2) Tilt the outflow valve [1] to put the forward end of the outflow valve [1] below the three forward lugs [11], then lower the aft end of the outflow valve [1].

3) Turn the three aft lugs [11] 90 degrees in the counterclockwise direction so that they face aft to forward.

(g) Do these steps to tighten the screws [12] to 60 ±5 in-lb (6.78 ±0.57 N·m):

1) Tighten the screws [12] no more than one turn at a time in sequence (1 through 6) as shown (View D, Figure 401).

2) Continue to tighten the screws [12] until all of the latches touch the valve.

3) Use a torque screwdriver, STD-10994, set at 60 ±5 in-lb (6.78 ±0.57 N·m) with a bit, COM-10996, to tighten the screws no more than 1/2 turn at a time in the sequence shown in (View D, Figure 401).

4) Continue to tighten the screws [12] in sequence until all latch screws [12] are tightened to 60 ±5 in-lb (6.78 ±0.57 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) If a screw [12] does not turn when the final application of torque is applied, back that screw [12] off one turn and tighten it to the final torque value.

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN THE VALVE IS IN OPERATION. IF YOU PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN IT OPERATES YOU CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(h) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

(i) Put the mode select switch, on the P5-6 cabin pressure control module on the overhead P5 panel, to the MAN position.

(j) Use the service interphone system so that the technician at the outflow valve [1] location can communicate the position of the outflow valve [1] to the technician in the flight compartment.

(k) Push and hold the manual control switch to the OPEN position until the outflow valve [1] is in the full open position.

(l) Open this circuit breaker:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-31-03-420-004

(10) Connect these electrical connectors to the outflow valve [1]:

NOTE: You can get access the electrical connectors through the valve door opening.

(a) ELACT 1 electrical connector [3]

(b) ELACT 2 electrical connector [2]

(c) feedback module connector 1 [5]

(d) feedback module connector 2 [6].

SUBTASK 21-31-03-020-007

(11) Do these steps to install the bonding jumper [9] on the outflow valve [1].

(a) Use the solvent cleaning method to clean these parts:

1) The surfaces of the outflow valve [1] where the bonding jumper [9] will touch

2) The stud

3) The terminal of the bonding jumper [9]

4) The washer [8]

5) The nut [7].

(b) Put the bonding jumper [9] in its position on the stud of the outflow valve [1] and install the washer [8] and nut [7].

(c) Measure the bonding resistance between the bonding jumper [9] and outflow valve [1] (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure that the bonding resistance is 0.001 ohm (1 milliohm) or less.

SUBTASK 21-31-03-860-004

(12) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

H. Aft Outflow Valve Assembly Installation Test

SUBTASK 21-31-03-710-001

(1) Do a test of the manual mode (TASK 21-31-00-710-801).

SUBTASK 21-31-03-710-002

(2) Do a cabin pressure control system ground test (TASK 21-31-00-700-802).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 21-31-03-400-002

(1) Install the E6 electrical equipment rack, aft access panel.

SUBTASK 21-31-03-860-011

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

POSITIVE PRESSURE RELIEF VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Positive pressure relief valve removal.

(2) Positive pressure relief valve installation.

B. Two positive pressure relief valves are installed in the area aft of the aft cargo compartment behind the aft bulkhead liner.

TASK 21-32-01-000-801

2. Positive Pressure Relief Valve Removal

(Figure 401)

A. References

Reference	Title
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)
38-11-01-000-801	Water Tank Removal (P/B 401)

B. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

C. Prepare for the Removal

SUBTASK 21-32-01-010-001

(1) Do one of the tasks that follow to get access to the positive pressure relief valves [1]:

(a) Remove the potable water tank (TASK 38-11-01-000-801).

(b) Remove the aft outflow valve (TASK 21-31-03-000-801).

SUBTASK 21-32-01-860-005

(2) If the potable water tank is removed and the aft outflow valve stays installed, open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

D. Positive Pressure Relief Valve Removal

SUBTASK 21-32-01-020-001

(1) Loosen the clamp [3] at the bottom of the positive pressure relief valve [1].

(a) Slide the clamp [3] down over the pedestal [4].

SUBTASK 21-32-01-020-002

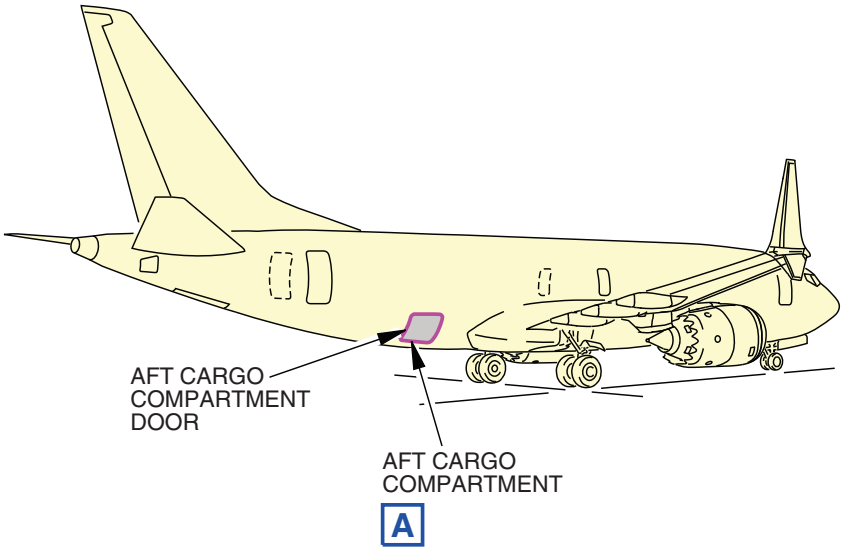
(2) Remove the positive pressure relief valve [1].

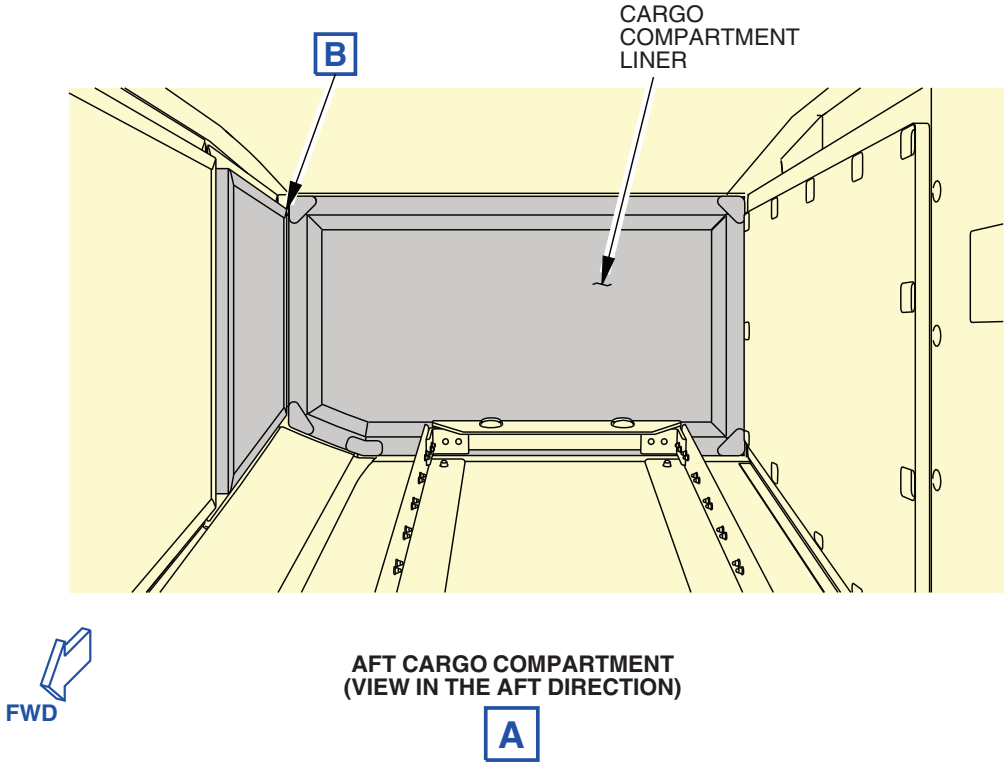
SUBTASK 21-32-01-020-003

(3) Remove and discard the gasket [2].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

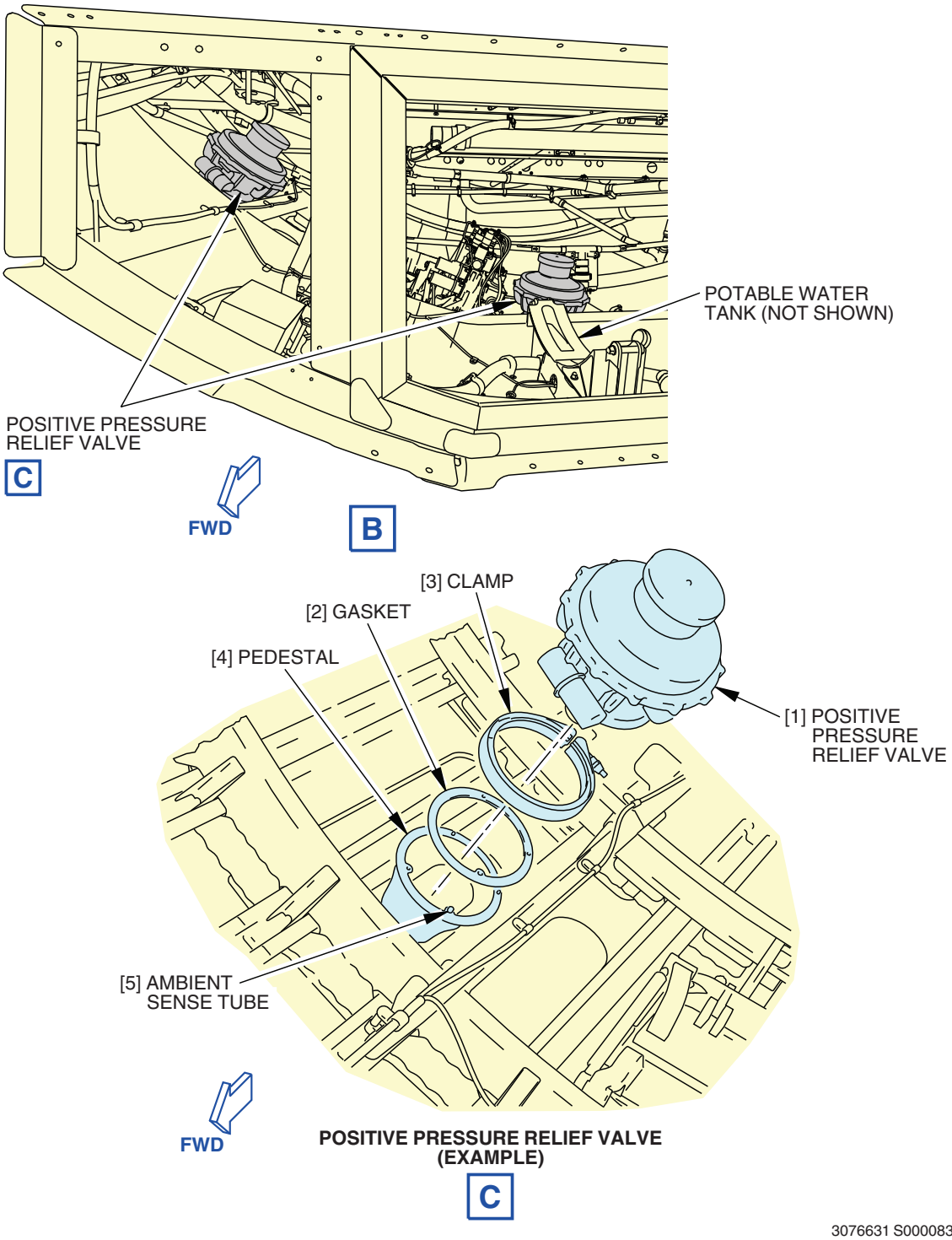




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Positive Pressure Relief Valve Installation
Figure 401/21-32-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Positive Pressure Relief Valve Installation
Figure 401/21-32-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-32-01-400-801

3. Positive Pressure Relief Valve Installation

(Figure 401)

A. References

Reference	Title
21-31-03-400-801	Aft Outflow Valve Assembly Installation (P/B 401)
21-32-01-700-801	Positive Pressure Relief Valve - System Test with the Use of Hamilton Sundstrand Test Equipment (P/B 501)
21-32-01-700-802	Positive Pressure Relief Valve - System Test with the Use of Boeing Test Equipment (P/B 501)
38-11-01-400-801	Water Tank Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
G50063	Lubricant - PTFE Release Agent - Miller-Stephenson MS-122XD (Replaces MS-122DF)	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Positive pressure relief valve	21-32-01-01-010	PTF ALL
2	Gasket	21-32-01-01-030	PTF ALL

D. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

E. Positive Pressure Relief Valve Installation

SUBTASK 21-32-01-640-001

(1) Apply MS-122XD release agent, G50063, to the two sides of the gasket [2].

SUBTASK 21-32-01-420-001

CAUTION

MAKE SURE THAT YOU DO NOT GET UNWANTED MATERIALS IN THE AMBIENT SENSE TUBE WHEN YOU INSTALL THE GASKET. IF YOU GET UNWANTED MATERIALS IN THE AMBIENT SENSE TUBE, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(2) Install the gasket [2] on the pedestal [4].

SUBTASK 21-32-01-420-002

CAUTION

BE CAREFUL WHEN YOU INSTALL THE POSITIVE PRESSURE RELIEF VALVE ON THE PEDESTAL. DAMAGE COULD OCCUR TO THE AMBIENT SENSE TUBE.

(3) Put the positive pressure relief valve [1] in its position on the pedestal [4].

SUBTASK 21-32-01-420-003

(4) Put the clamp [3] in its position around the top of the pedestal [4] and the bottom of the positive pressure relief valve [1].

(a) Tighten the clamp [3] to 65 +5 / -0 in-lb (7.3 +0.6 / -0.0 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-32-01-710-001

(5) To do the required post-installation test of the positive pressure relief valve [1], do one of these tasks:

(a) Positive Pressure Relief Valve - System Test with the Use of Boeing Test Equipment, TASK 21-32-01-700-802

(b) Positive Pressure Relief Valve - System Test with the Use of Hamilton Sundstrand Test Equipment, TASK 21-32-01-700-801.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-32-01-410-001

(1) If it was removed, install the potable water tank (TASK 38-11-01-400-801).

SUBTASK 21-32-01-410-002

(2) If it was removed, install the aft outflow valve (TASK 21-31-03-400-801).

SUBTASK 21-32-01-860-006

(3) If these circuit breakers were opened, remove the safety tags and close them:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

POSITIVE PRESSURE RELIEF VALVE - ADJUSTMENT/TEST

1. General

A. This procedure has two tasks. Each task does a test to find the differential pressure when the positive pressure relief valve opens. The first task uses a Hamilton Sundstrand valve test fixture. The second task uses Boeing test equipment.

B. The two positive pressure relief valves are installed on the lower exterior part of the airplane near the aft entry door. One of the valves is above the aft outflow valve and one is below the aft outflow valve.

C. Each task is accomplished with the positive pressure relief valves installed on the aircraft. The task will make sure that the positive pressure relief valves can operate correctly at specific differential pressures if the pressure inside the airplane becomes too high as compared to the pressure outside the airplane. To accomplish this task, the differential pressure at which the valve is designed to relieve is simulated with ground test equipment.

TASK 21-32-01-700-801

2. Positive Pressure Relief Valve - System Test with the Use of Hamilton Sundstrand Test Equipment

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-6692	Vacuum Tank - Pressure Relief Valve Tester Part #: A21010-203 Supplier: 81205 Part #: GS15047-1 Supplier: 99167 Opt Part #: A21010-187 Supplier: 81205 Opt Part #: A21010-188 Supplier: 81205 Opt Part #: A21010-189 Supplier: 81205 Opt Part #: A21010-190 Supplier: 81205
STD-1114	Air Source - Regulated, Dry, Filtered 0-150 PSIG with Pressure Gauge Range 0-100 PSIG, 1 PSIG Increment, and +/-1 PSI Minimum Accuracy

B. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

C. Prepare for the Test

SUBTASK 21-32-01-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-32-01-840-001

(2) Make sure that there is no blockage or unwanted materials at these locations:

(a) The fuselage skin in the area of the relief valves.

(b) The overboard vent tube opening in the center of each relief valve.

(c) The ambient sense tube in the fuselage pedestal mount.

SUBTASK 21-32-01-840-002

(3) Close the pressure regulator on the pressure relief valve tester, SPL-6692.

SUBTASK 21-32-01-840-003

(4) Connect the Regulated Dry Filtered Air Source, STD-1114, to the pressure regulator of the pressure relief valve tester, SPL-6692.

D. Positive Pressure Relief Valve System Test

NOTE: This test can only be done on one relief valve at a time. Do the test on the outboard relief valve, then do the test again on the inboard relief valve.

SUBTASK 21-32-01-730-001

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN THE VALVE IS IN OPERATION. IF YOU PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN IT OPERATES YOU CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do these steps to do a test of the positive pressure relief valve:

(a) Put the pressure relief valve tester, SPL-6692, on the relief valve on the fuselage.

(b) Put the locating crossbar on the pressure relief valve tester, SPL-6692, forward of the deflector tube on the relief valve and hold the fixture in position.

NOTE: The fixture is installed to make sure that the test fixture gasket is not on the opening in the valve.

(c) Open the air pressure regulator on the pressure relief valve tester, SPL-6692, to start the flow of air.

(d) Adjust the air pressure regulator so that the indication on the vacuum gage increases 2.21 psi/minute (15.24 kPa/minute) until the vacuum gage shows 7.76 psig (53.50 kPa).

WARNING MANUALLY HOLD THE TEST FIXTURE AGAINST THE FUSELAGE. WHEN THE VALVE OPENS, IT CAN CAUSE THE TEST FIXTURE TO FALL. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(e) When the vacuum gage shows 7.76 psig (53.50 kPa), manually hold the test fixture against the airplane fuselage.

(f) When the vacuum gage shows 7.76 psig (53.50 kPa), adjust the air pressure regulator to increase the vacuum in 0.05 psi (0.34 kPa) increments.

(g) Stop the increase of vacuum for 10 seconds between each increase of 0.05 psi (0.34 kPa).

(h) Continue the intermittent vacuum increases of 0.05 psi (0.34 kPa) until the valve opens as noted by these indications:

1) The cracking point pressure of the valve occurs when the vacuum gauge needle stops increasing, then suddenly decreases slightly, then increases slightly again and then stabilizes.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Monitor the indication on the vacuum gage when the valve opens.

(j) Make sure that the valve opens to 8.69 psig (59.92 kPa) - 9.09 psig (62.67 kPa) (17.7 - 18.5 inches Hg).

(k) If the pressure relief valve cracks open at a vacuum gauge indication greater than 9.09 psig (62.67 kPa) (18.5 inches of Hg), make sure that there are no leaks anywhere on the valve test fixture.

(l) Adjust the vacuum regulator valve to decrease the differential pressure to approximately 7.5 psid (51.7 kPa) - 7.0 psid (48.3 kPa) (15.04 - 14.02 inches Hg) at a rate not more than 2.00 psig/minute (13.79 kPa/minute) (4.07 inches Hg/minute).

1) Make sure that the relief valve closes.

NOTE: The indication that the relief valve has closed is seen on the differential pressure gauge when the gauge stops, and then starts to increase and then becomes stable.

(m) Adjust the air pressure regulator to reduce the indication on the vacuum gage to 0 psig (0 kPa) (0 inches Hg) at a rate of 4.0 psig/minute (27.58 kPa/minute) (8.14 inches Hg/minute) or slower.

(n) Close the pressure regulator on the pressure relief valve tester, SPL-6692.

(o) Remove the pressure relief valve tester, SPL-6692, from the airplane.

(p) Disconnect the Regulated Dry Filtered Air Source, STD-1114, from the pressure regulator.

SUBTASK 21-32-01-730-002

(2) Do the test for the other relief valve.

SUBTASK 21-32-01-860-002

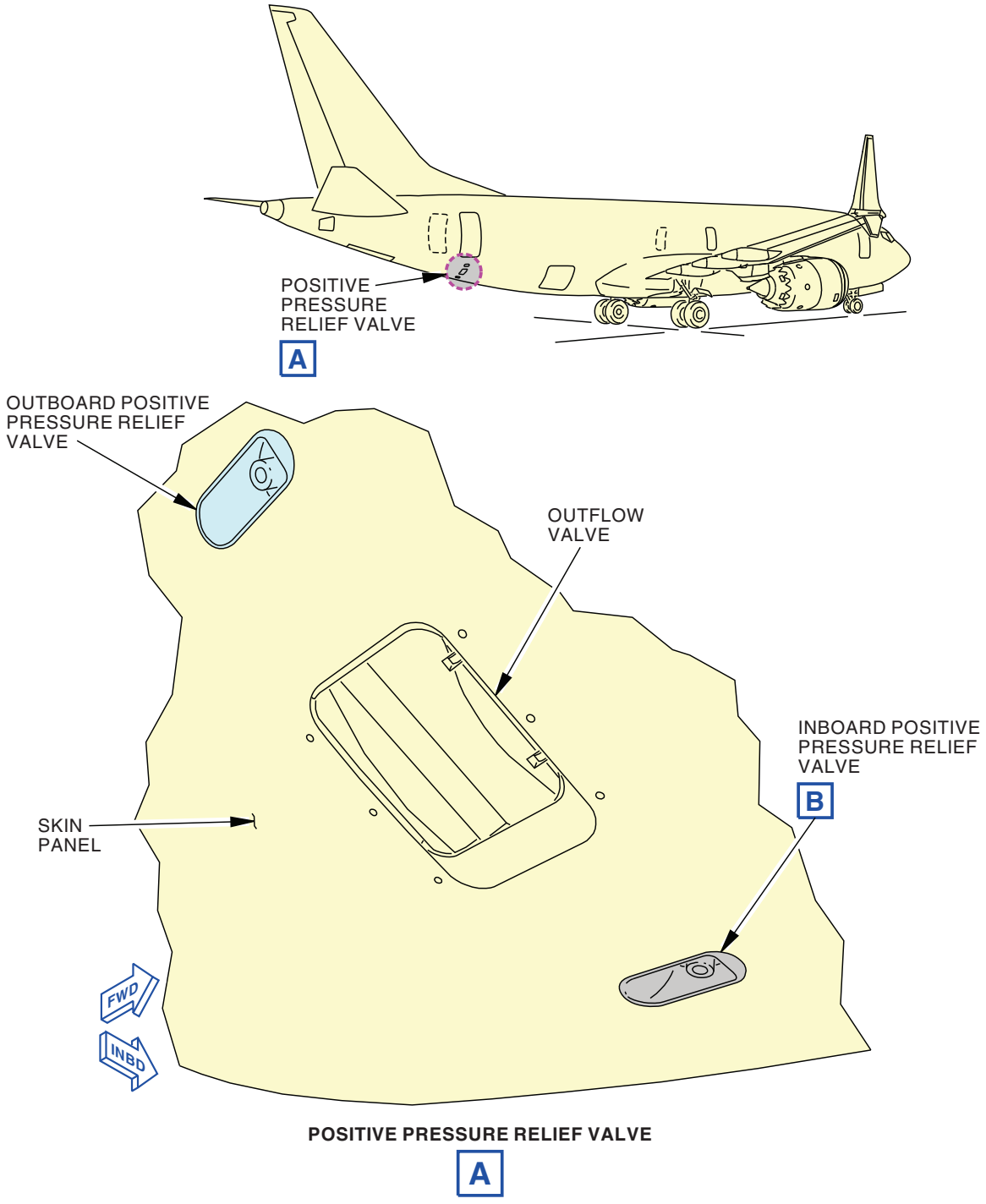
(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

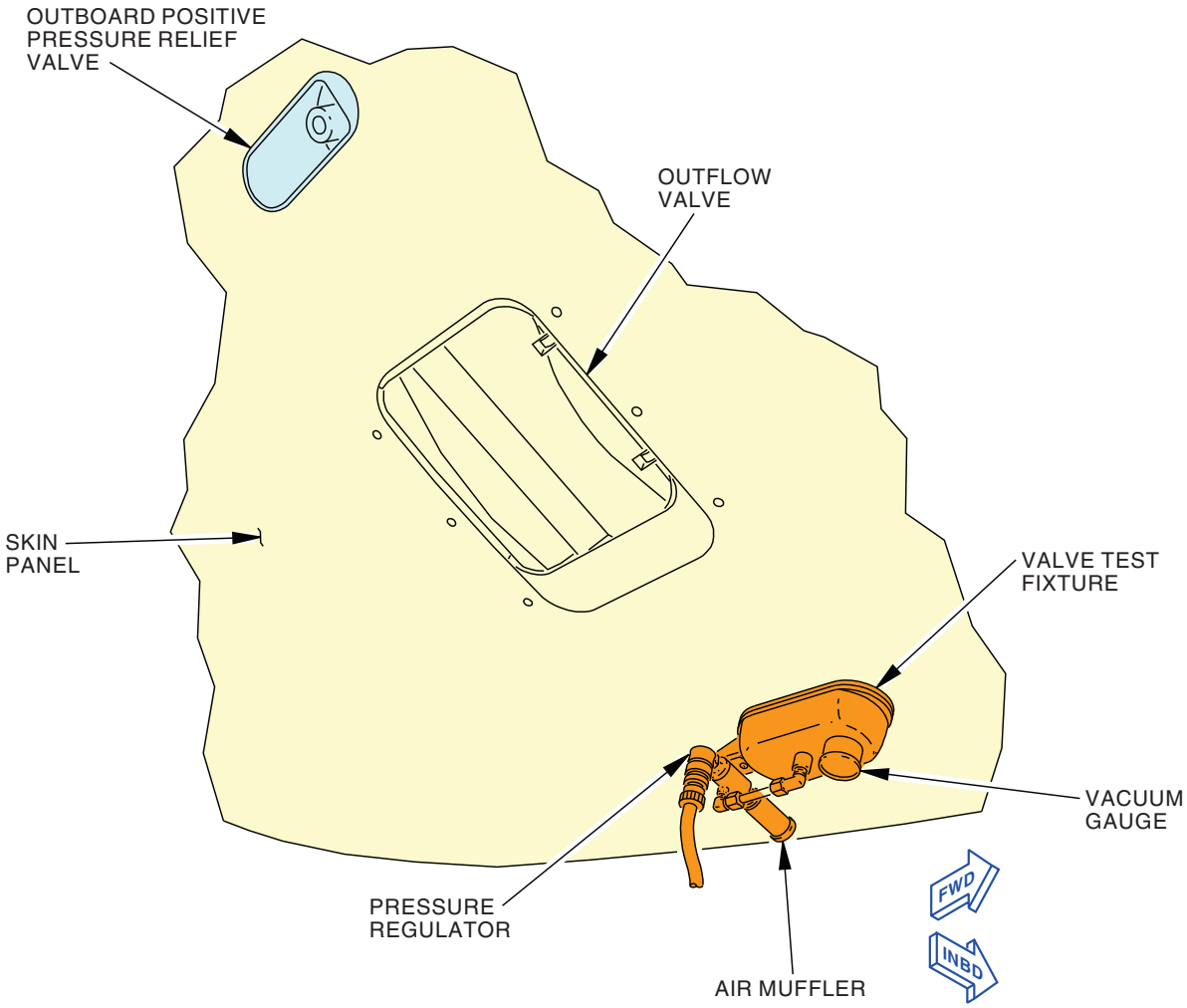
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Positive Pressure Relief Valve Test Setup
Figure 501/21-32-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GS15047-1 POSITIVE RELIEF VALVE TEST FIXTURE

B

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Positive Pressure Relief Valve Test Setup
Figure 501/21-32-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-32-01-700-802

3. Positive Pressure Relief Valve - System Test with the Use of Boeing Test Equipment

(Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-6692	Vacuum Tank - Pressure Relief Valve Tester Part #: A21010-203 Supplier: 81205 Part #: GS15047-1 Supplier: 99167 Opt Part #: A21010-187 Supplier: 81205 Opt Part #: A21010-188 Supplier: 81205 Opt Part #: A21010-189 Supplier: 81205 Opt Part #: A21010-190 Supplier: 81205
STD-1114	Air Source - Regulated, Dry, Filtered 0-150 PSIG with Pressure Gauge Range 0-100 PSIG, 1 PSIG Increment, and +/-1 PSI Minimum Accuracy

B. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

C. Prepare for the Test

SUBTASK 21-32-01-860-003

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

SUBTASK 21-32-01-210-001

(2) Make sure that there is no blockage or unwanted materials at these locations:

(a) The fuselage skin in the area of the relief valve.

(b) The overboard vent tube opening in the center of each relief valve.

(c) The ambient sense tube in the fuselage pedestal mount.

D. Positive Pressure Relief Valve System Test

NOTE: This test can only be done on one relief valve at a time. Do the test on the outboard relief valve, then do the test again on the inboard relief valve.

SUBTASK 21-32-01-480-001

WARNING DO NOT PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN THE VALVE IS IN OPERATION. IF YOU PUT YOUR HANDS OR TOOLS IN THE OUTFLOW VALVE WHEN IT OPERATES YOU CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Do these steps to connect the pressure relief valve tester, SPL-6692, to the relief valve:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that all the valves on the pressure relief valve tester, SPL-6692, are closed.

(b) Connect the Regulated Dry Filtered Air Source, STD-1114, to the quick-disconnect connection on the pressure relief valve tester, SPL-6692.

(c) Adjust the air control valve on the pressure relief valve tester, SPL-6692, until the indication on the pressure gage is 60 psi (414 kPa) - 80 psi (552 kPa).

(d) Adjust the vacuum regulator valve until the differential pressure is approximately 2.0 psi (13.8 kPa) (4.07 in/Hg).

NOTE: Turn the vacuum regulator valve counter-clockwise to increase the vacuum (differential pressure). The arrow on the vacuum regulator valve indicates an increase of pressure not vacuum.

(e) Put and hold the vacuum head assembly over the relief valve, and open the vacuum head assembly valve.

1) Make sure that the vacuum head assembly is sealed to the airplane skin.

SUBTASK 21-32-01-730-003

(2) Do these steps to find the differential pressure when the relief valve opens:

(a) Adjust the vacuum regulator valve so the indication on the vacuum rate gage does not exceed 2.4 psi/minute (16.5 kPa/minute).

WARNING

YOU MUST HOLD THE VACUUM HEAD ASSEMBLY MANUALLY AGAINST THE AIRPLANE FUSELAGE WHEN THE DIFFERENTIAL PRESSURE GAGE INDICATION IS 8.25 PSID (56.88 KPA). WHEN THE VALVE OPENS, THE VACUUM HEAD ASSEMBLY CAN FALL FROM THE FUSELAGE SURFACE. INJURY TO PERSONNEL OR DAMAGE TO EQUIPMENT CAN OCCUR.

(b) When the differential pressure gage shows 8.25 psid (56.88 kPa), manually hold the vacuum head assembly against the fuselage.

(c) When the differential pressure gage shows 8.25 psid (56.88 kPa), adjust the vacuum regulator valve to increase the differential pressure in 0.05 psi (0.34 kPa) increments.

(d) Stop the increase of vacuum for 10 seconds between each increase of 0.05 psid (0.34 kPa).

(e) Continue the intermittent vacuum increases of 0.05 psid (0.34 kPa) until the relief valve opens.

(f) Monitor the indication on the differential pressure gage when the relief valve opens.

NOTE: The indication that the relief valve has opened is seen on the differential pressure gage when the pressure starts to decrease.

(g) Make sure that the valve opens at a differential pressure of 8.69 psid (59.92 kPa) - 9.09 psid (62.67 kPa) (17.7 in/Hg - 18.5 in/Hg).

NOTE: When the valve opens the differential pressure will decrease to near 0 psid.

(h) If the relief valve opens at a differential pressure more than 9.09 psid (62.67 kPa) (18.5 in/Hg), make sure that there are no leaks at any of the tubing connections on the vacuum tank assembly.

(i) Adjust the vacuum regulator valve to decrease the differential pressure to approximately 7.5 psig (51.7 kPa) - 7.0 psig (48.3 kPa) (15.04 in/Hg - 14.02 in/Hg) at a rate not more than 2.00 psig/minute (13.79 kPa/minute) (4.07 inches Hg/minute).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the relief valve closes.

NOTE: The indication that the relief valve has closed is seen on the differential pressure gage when the gage stops, and then starts to increase and then becomes stable.

SUBTASK 21-32-01-080-001

(3) Adjust the vacuum regulator valve fully clockwise and reduce the differential pressure to 0 psig (0 kPa) (0 in/Hg), at a rate of 4.0 psig/minute (27.58 kPa/minute) (8.14 inches Hg/minute) or slower.

SUBTASK 21-32-01-080-002

(4) Close the vacuum head assembly valve and remove the vacuum head assembly.

SUBTASK 21-32-01-730-004

(5) Do the test for the other relief valve.

SUBTASK 21-32-01-860-004

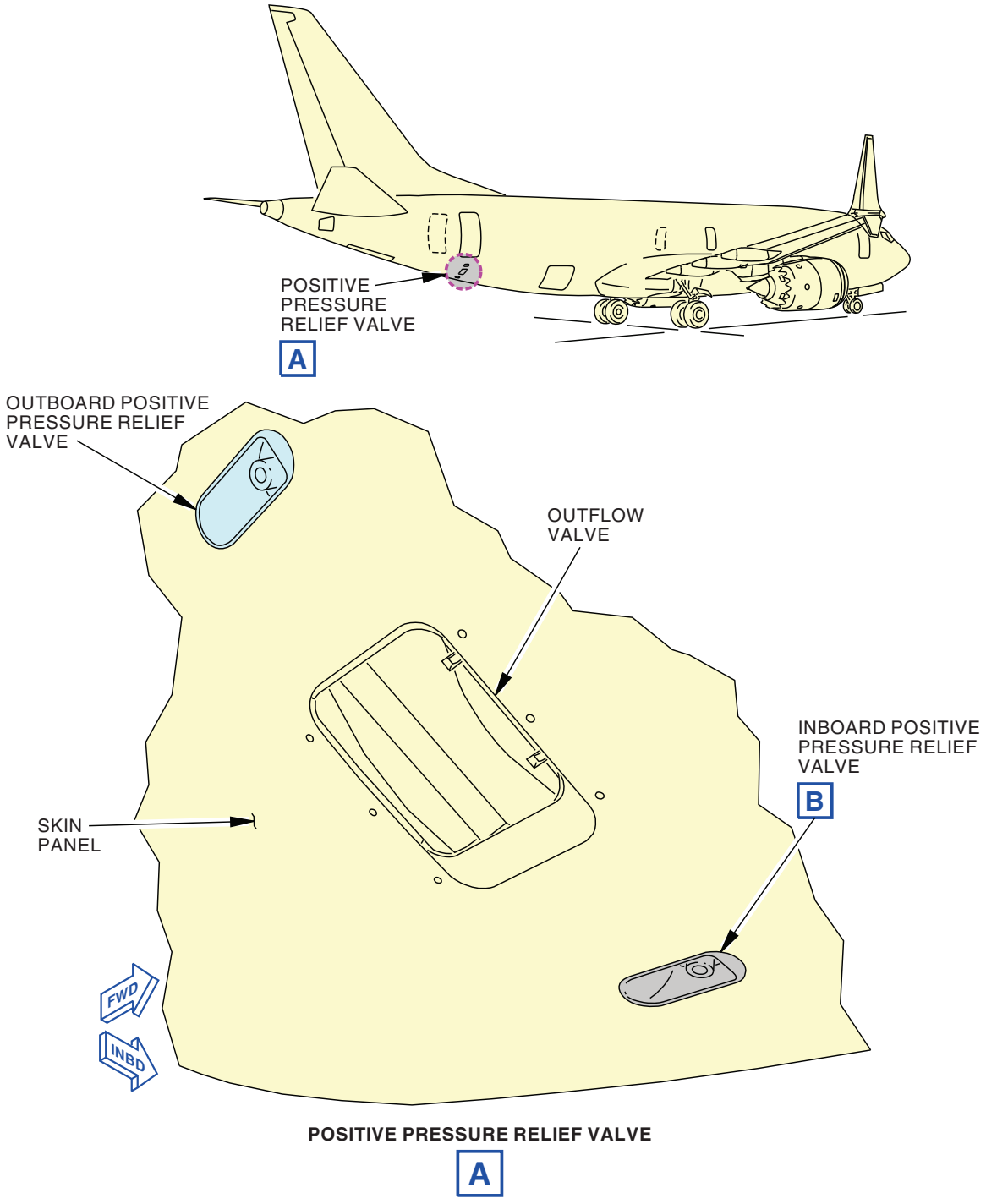
(6) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	3	C01270	PRESSURIZATION CONTROL AUTO 1
F	5	C01271	PRESSURIZATION CONTROL AUTO 2
F	6	C01269	PRESSURIZATION CONTROL MANUAL

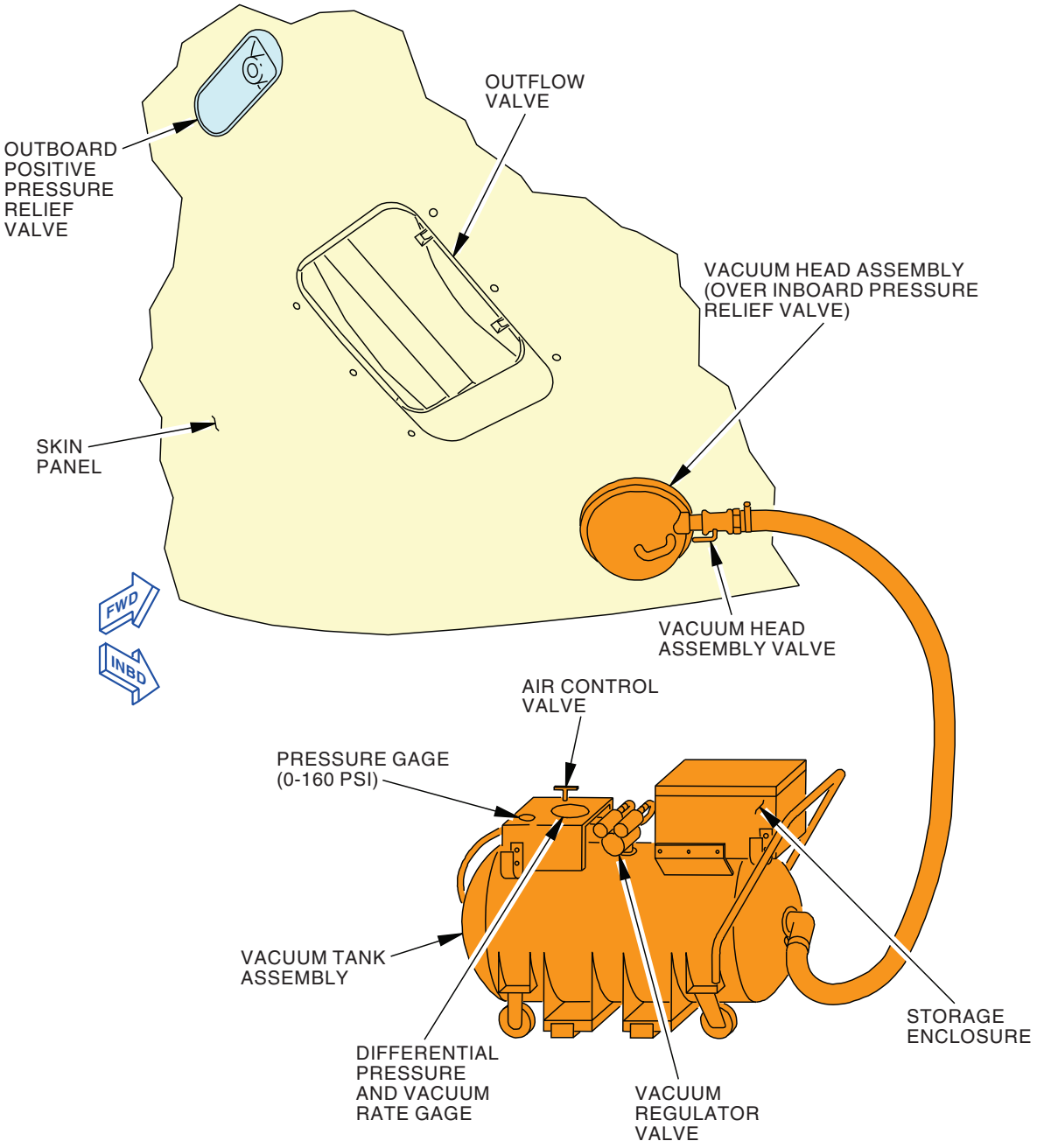
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Positive Pressure Relief Valve Test Setup
Figure 502/21-32-01-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A21010 VACUUM TANK - RELIEF VALVE TEST

B

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Positive Pressure Relief Valve Test Setup
Figure 502/21-32-01-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

POSITIVE PRESSURE RELIEF VALVE FILTER - REMOVAL/INSTALLATION

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has these tasks:

(1) Positive pressure relief valve filter removal.

(2) Positive pressure relief valve filter installation.

C. Two positive pressure relief valves are installed in the aft cargo compartment behind just inboard and outboard of the aft outflow valve. One filter is installed on the top side of each positive pressure relief valve.

TASK 21-32-02-000-801

2. Positive Pressure Relief Valve Filter Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)
38-11-01-000-801	Water Tank Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

B. Tools/Equipment

Reference	Description
STD-6435	Respirator - Particulate (Meets NIOSH 42 CFR 84 Class N95, EN149 Class FFP2, or Better)
STD-6437	Goggles - Safety, Non Vented
STD-6439	Coveralls - Hooded

C. Consumable Materials

Reference	Description	Specification
B50073	Alcohol - Isopropyl	ASTM D 770
G00624	Bag - Plastic, General Purpose	
G50140	Gloves - Protective, Latex or Nitrile	

D. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

E. Prepare for the Removal

SUBTASK 21-32-02-010-002

(1) Do one of these steps to get access to the positive pressure relief valve [1]:

(a) Remove the potable water tank (TASK 38-11-01-000-801).

(b) Remove the aft outflow valve (TASK 21-31-03-000-801).

(c) Remove the applicable floor panels above the positive pressure relief valve [1] (TASK 53-21-00-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-32-02-940-001

WARNING PUT ON THE PERSONAL PROTECTIVE EQUIPMENT BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES AND INJURIES TO PERSONS.

(2) Before you touch the positive pressure relief valve filter [3], put on the personal protective equipment listed below:

(a) hooded coveralls, STD-6439

(b) safety goggles, STD-6437

(c) particulate respirator, STD-6435

(d) protective gloves, G50140.

SUBTASK 21-32-02-500-001

(3) Select a plastic bag, G00624, larger than the filters and tape suitable for sealing the bag closed.

F. Positive Pressure Relief Valve Filter Removal

SUBTASK 21-32-02-020-001

(1) Remove the cover [4] from the top of the positive pressure relief valve [1].

SUBTASK 21-32-02-020-002

(2) Remove the filter [3] and packing [2] from the positive pressure relief valve [1].

SUBTASK 21-32-02-020-003

WARNING DO NOT LET THE FILTER TOUCH YOUR SKIN. DO NOT SHAKE OR HIT THE FILTER. DO NOT LET THE FILTER FALL. DO NOT USE COMPRESSED AIR TO CLEAN THE FILTER OR FILTER HOUSING. THIS CAN CAUSE THE INFECTIOUS MATERIAL TO BECOME AIRBORNE. DISCARD THE FILTER IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(3) If it is possible, slide plastic bag, G00624, over the filter [3] with the packing [2] when removing filter from the positive pressure relief valve [1] to minimize contamination.

(a) Seal the bag closed with a tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(b) Discard the bag with the filter.

NOTE: Refer to the airline, local health, safety and regulatory procedures for the disposal of material.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Post-Filter Removal Cleanup

SUBTASK 21-32-02-670-002

WARNING ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(1) Disinfect the filter cover, filter housing and exterior surfaces of the valve with alcohol, B50073.

NOTE: Refer to the Environmental Protection Agency (EPA) List N or local health authority guidance for required contact/dwell time for Isopropyl Alcohol to be used as a disinfectant.

SUBTASK 21-32-02-670-001

(2) Disinfect all tools used for filter replacement. Refer to the airline procedure and policies for the disinfecting of tools.

SUBTASK 21-32-02-020-004

WARNING DISCARD ALL PERSONAL PROTECTIVE EQUIPMENT AFTER YOU USE IT ONE TIME. DO NOT TRY TO CLEAN IT. DISCARD THE EQUIPMENT IN A PLASTIC DISPOSAL BAG. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(3) Remove the personal protective equipment and put them in plastic bag, G00624.

(a) Seal the bag closed with tape.

WARNING OBEY THE AIRLINE POLICY, LOCAL HEALTH DEPARTMENT, AND LAW ENFORCEMENT REGULATIONS WHEN YOU DISCARD THIS MATERIAL. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL.

(b) Discard the bag with the personal protective equipment.

NOTE: Refer to the airline, local health, safety and regulatory procedures for disposal of material.

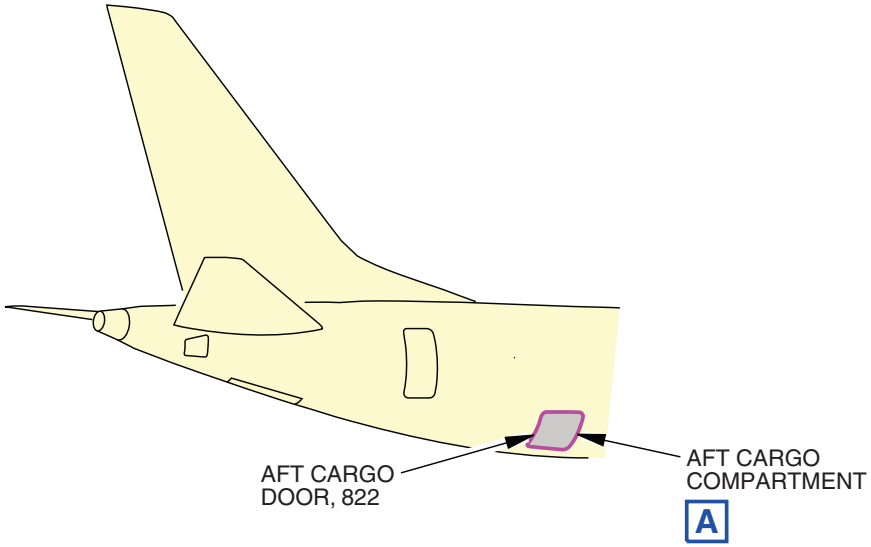
SUBTASK 21-32-02-110-001

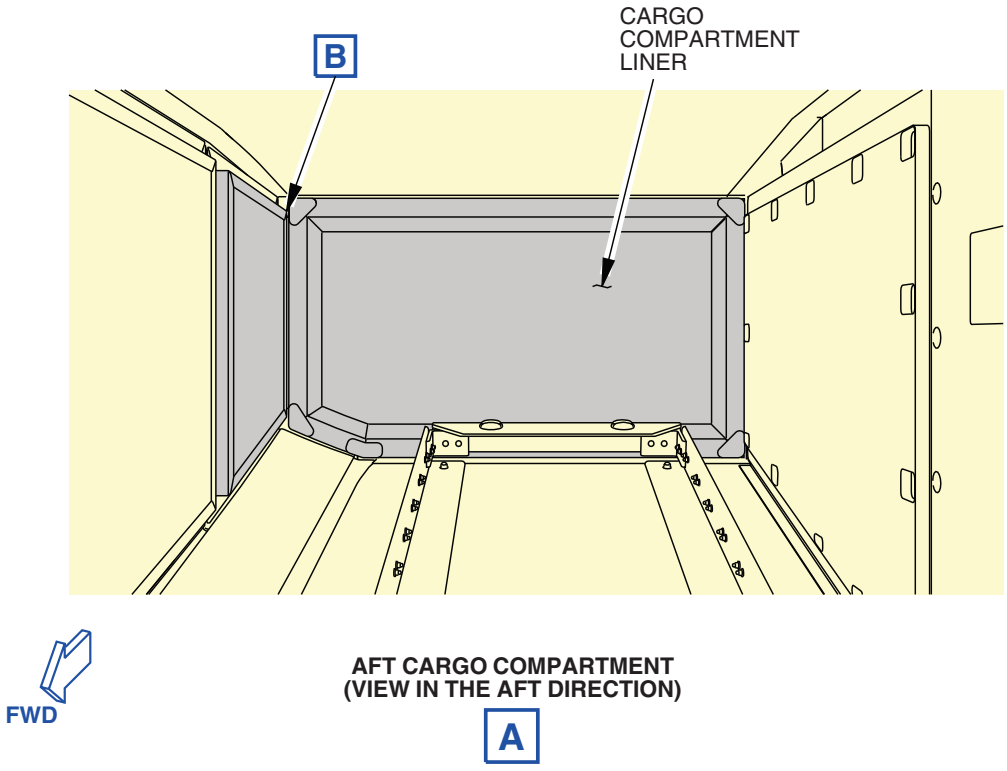
WARNING CLEAN YOUR HANDS WITH SOAP AND FLUSH WITH HOT WATER. DIRTY HANDS WITH CONTAMINATION CAN CAUSE DISEASE AND ILLNESSES TO PERSONS.

(4) Clean your hands with soap and running hot water.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

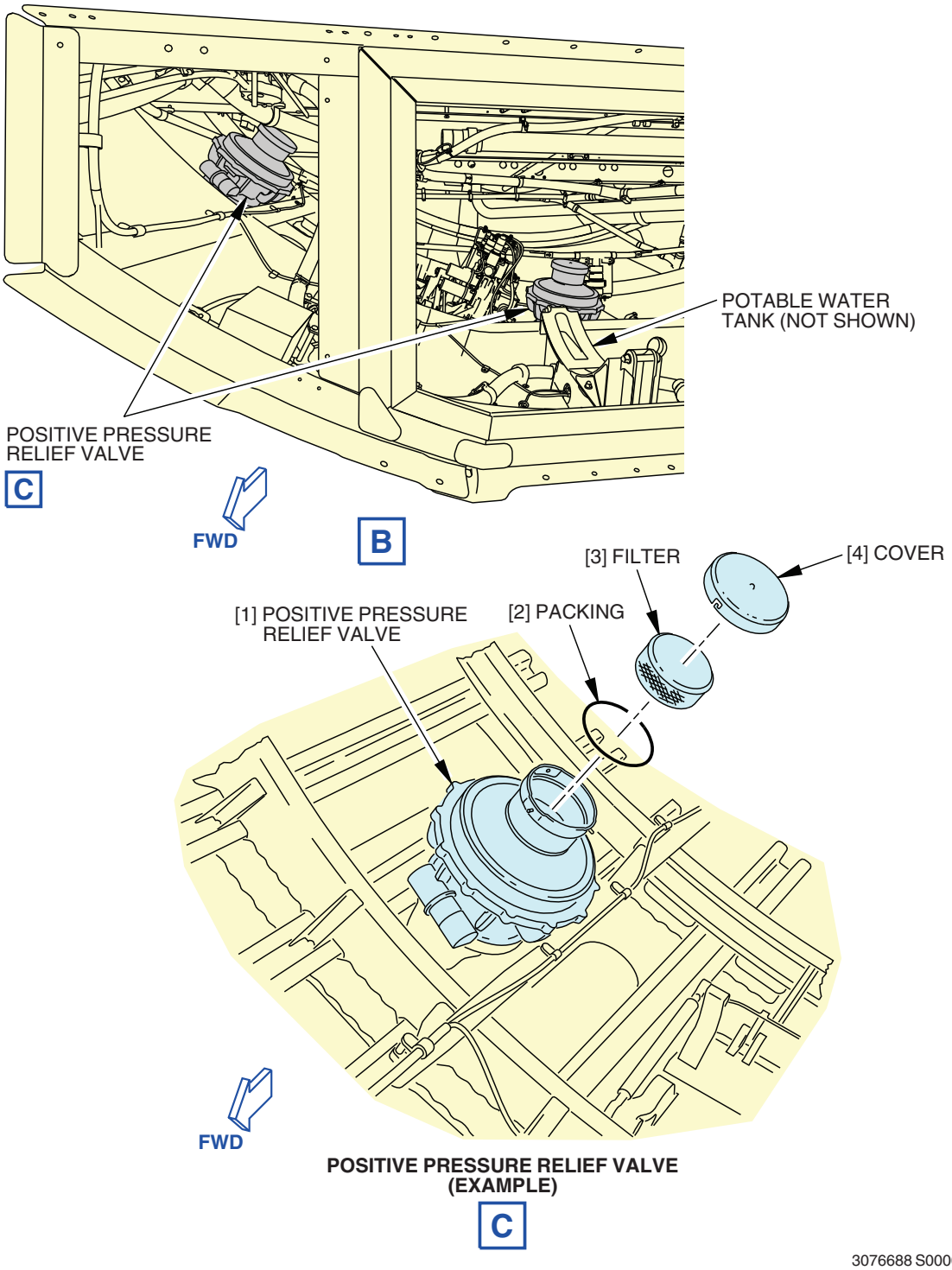




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Positive Pressure Relief Valve Filter Installation
Figure 401/21-32-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Positive Pressure Relief Valve Filter Installation
Figure 401/21-32-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-32-02-400-801

3. Positive Pressure Relief Valve Filter Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
21-31-03-400-801	Aft Outflow Valve Assembly Installation (P/B 401)
38-11-01-400-801	Water Tank Installation (P/B 401)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Packing	21-32-02-01-015	PTF ALL
3	Filter	21-32-02-01-020	PTF ALL

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

D. Positive Pressure Relief Valve Filter Installation

SUBTASK 21-32-02-420-001

(1) Install new filter [3] and new packing [2] in the top of the positive pressure relief valve [1].

SUBTASK 21-32-02-420-002

(2) Install and turn the cover [4] until it snaps into position.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-32-02-410-001

(1) If it is removed, install the potable water tank (TASK 38-11-01-400-801).

SUBTASK 21-32-02-410-003

(2) If it is removed, install the aft outflow valve (TASK 21-31-03-400-801).

SUBTASK 21-32-02-410-004

(3) If it is removed, install the floor panels above the positive pressure relief valve [1] (TASK 53-21-00-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NEGATIVE PRESSURE RELIEF VALVE - MAINTENANCE PRACTICES

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has these tasks:

(1) Negative pressure relief valve removal

(2) Negative pressure relief valve installation

(3) Negative pressure relief valve adjustment/test.

TASK 21-32-03-000-801

2. Negative Pressure Relief Valve Removal

(Figure 201)

A. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

B. Prepare for the Removal

SUBTASK 21-32-03-010-001

(1) Open the aft cargo compartment door.

SUBTASK 21-32-03-010-003

(2) Remove the E6 electrical equipment rack, aft access panel.

C. Negative Pressure Relief Valve Removal

SUBTASK 21-32-03-020-001

(1) Do these steps to remove the door assembly [6] from the frame:

(a) Remove the nut [5] and washer [4] from the end of the bolt [1].

(b) Remove the bolt [1] and washer [2].

(c) Remove the bushing [3] from the frame.

(d) Carefully lower the door assembly [6] and disengage the end of the spring [7] from the hole in the frame.

1) Hold the end of the spring [7] as you disengage it from the hole in the frame so that it does not move suddenly when it is free from the frame hole.

SUBTASK 21-32-03-020-002

(2) Do these steps to remove the spring [7] from the door assembly [6]:

(a) Remove the nut [14] and washer [13] from the end of the bolt [10].

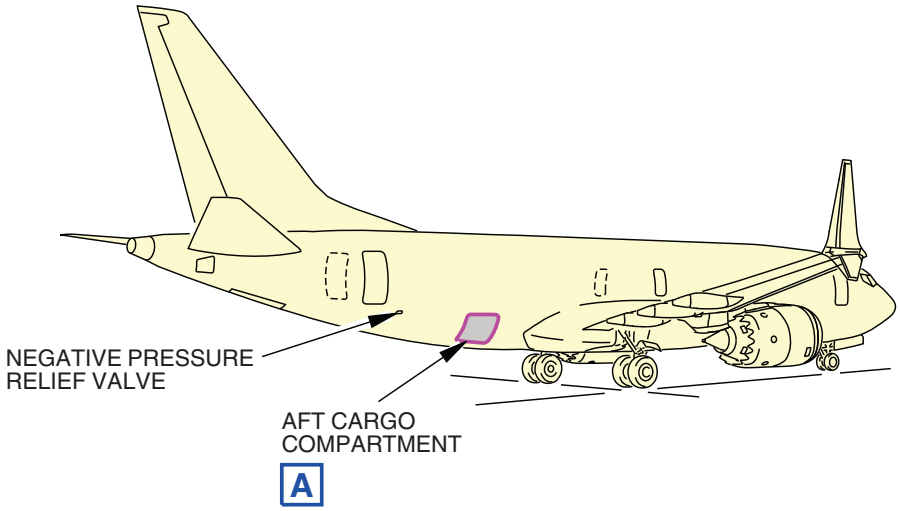
(b) Remove the bolt [10] and washer [11].

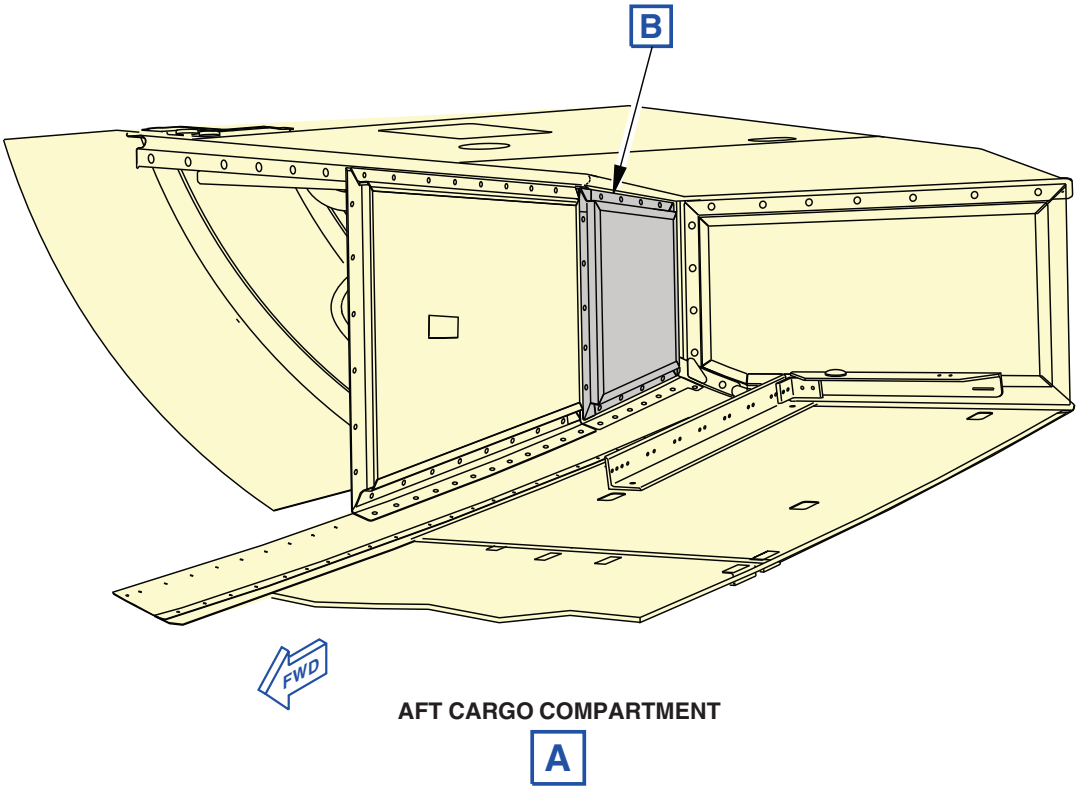
(c) Remove the retainer [9] and shim [12].

(d) Remove the spring support [8] and spring [7].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

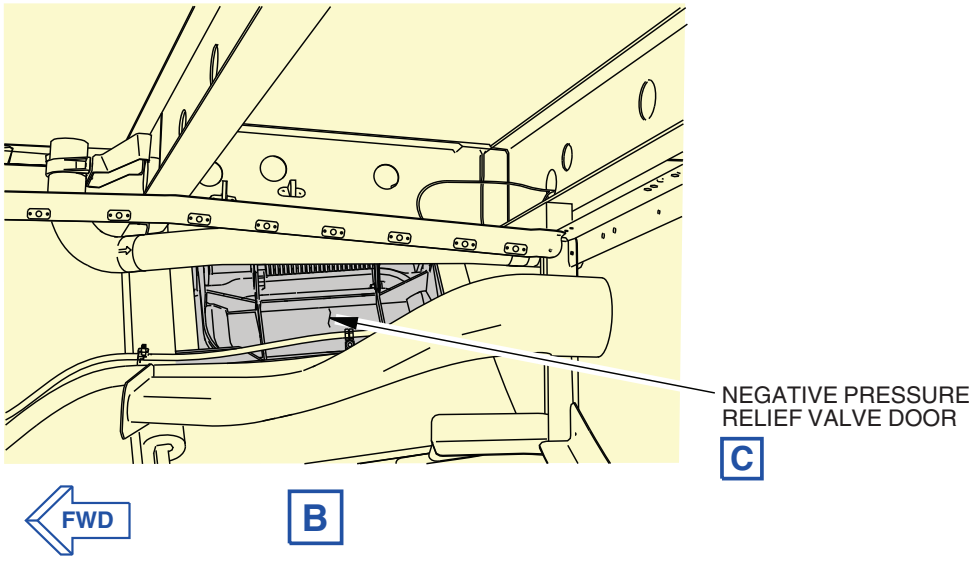


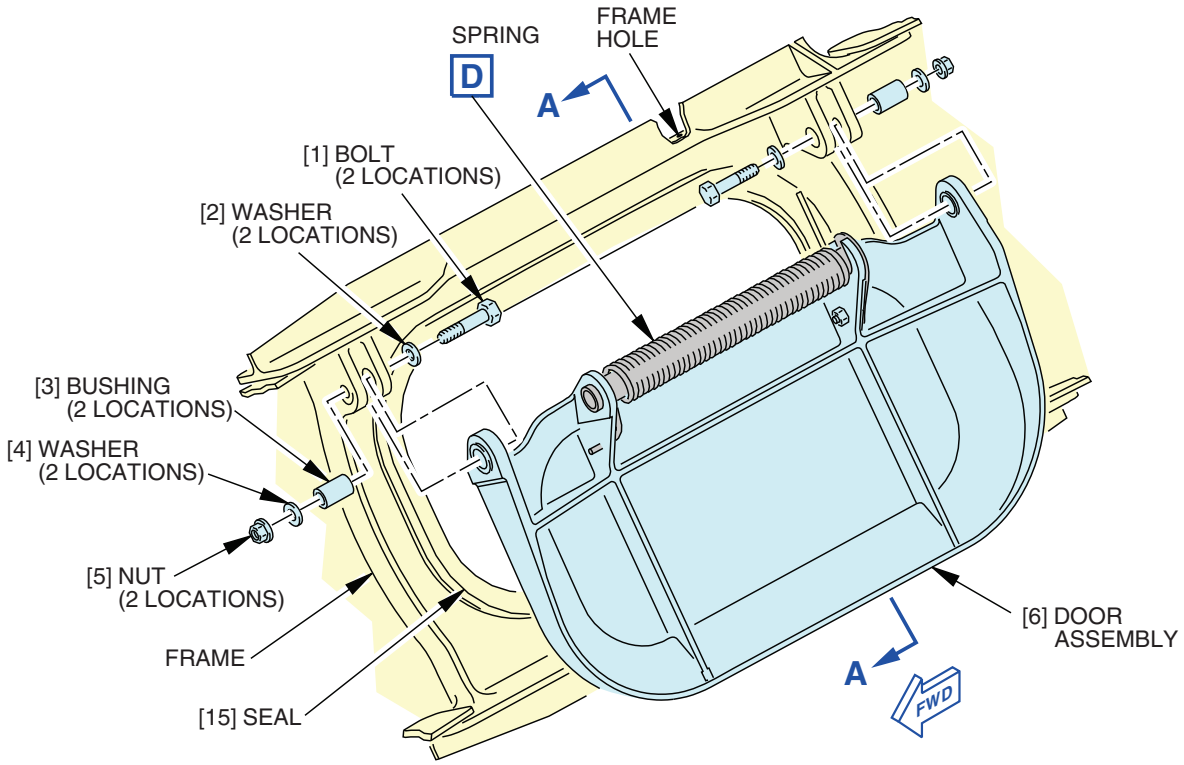


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Negative Pressure Relief Valve Installation
Figure 201/21-32-03-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





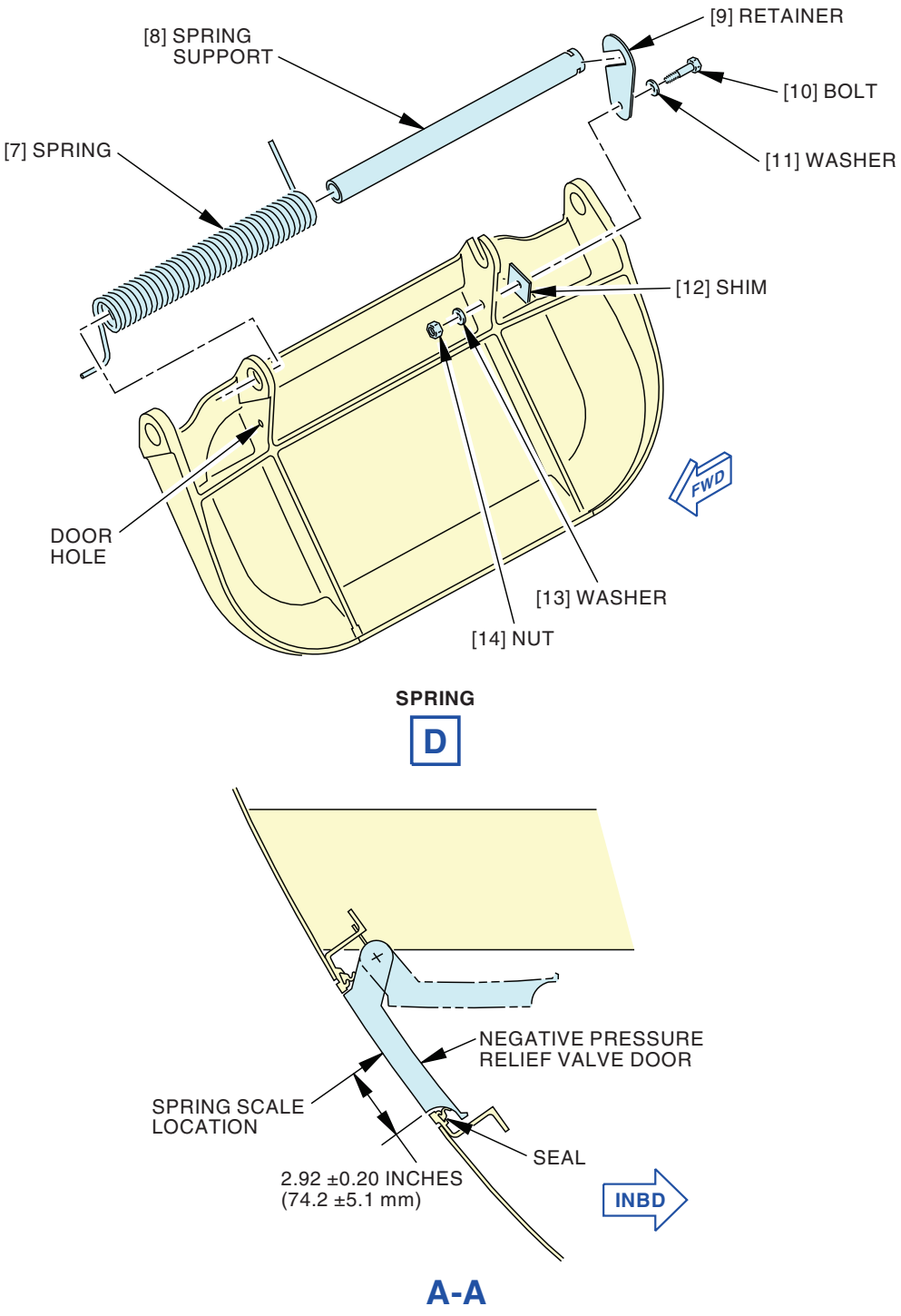
NEGATIVE PRESSURE RELIEF VALVE DOOR

C

3076827 S0000834447_V2

Negative Pressure Relief Valve Installation
Figure 201/21-32-03-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Negative Pressure Relief Valve Installation
Figure 201/21-32-03-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-32-03-400-801

3. Negative Pressure Relief Valve Installation

(Figure 201)

A. Consumable Materials

Reference	Description	Specification
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
6	Door assembly	21-32-03-01-100	PTF ALL

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

D. Negative Pressure Relief Valve Installation

SUBTASK 21-32-03-420-001

(1) Do the steps which follow to install the spring [7] on the door assembly [6]:

(a) Put the spring [7] on the spring support [8].

(b) Put the end of the spring [7] in the door hole.

(c) Put the end of the spring support [8] in the large hole in the door.

(d) Turn the spring [7] approximately one full turn and hold it in that position.

(e) Put the free end of the spring support [8] in the slot of the door assembly [6].

(f) Slowly release the free end of the spring [7] until it rests against the edge of the door assembly [6].

(g) Put the shim [12] and the retainer [9] in their position.

(h) Install the bolt [10] and washer [11].

(i) Install the washer [13] and nut [14].

1) Tighten the nut [14] to 20 in-lb (2.3 N·m) – 25 in-lb (2.8 N·m).

SUBTASK 21-32-03-210-001

(2) Examine the condition of the seal [15] and replace it if it is damaged.

SUBTASK 21-32-03-420-002

(3) Do the steps which follow to install the door assembly [6] in the frame:

(a) Lubricate the bolt [1] with the grease, D00013.

(b) Put the door assembly [6] in its position.

1) Hold the door assembly [6] in an approximately 45° open position while you turn the free end of the spring [7] and insert it into the frame hole.

(c) Install the bushing [3].

(d) Install the bolt [1] and washer [2].

(e) Install the washer [4] and nut [5].

1) Tighten the nut [5] to 60 in-lb (6.8 N·m) – 70 in-lb (7.9 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-32-03-720-001

(4) Do this task: Negative Pressure Relief Valve Adjustment/Test, TASK 21-32-03-700-801.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-32-03-410-002

(1) Install the electrical equipment rack, E6, aft access panel.

——— END OF TASK ———

TASK 21-32-03-700-801

4. Negative Pressure Relief Valve Adjustment/Test

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. Tools/Equipment

Reference	Description
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

C. Procedure

SUBTASK 21-32-03-480-001

(1) On the external surface of the airplane, hold the push/pull scale 0-25 pound (0-11 kilogram), STD-753, against the relief valve (View A-A, Figure 201).

(a) Put the push/pull scale 0-25 pound (0-11 kilogram), STD-753, at approximately the center of the relief valve door and 2.92 ±0.20 in. (74.2 ±5.1 mm) above the lower edge of the relief valve door.

SUBTASK 21-32-03-720-002

(2) Slowly push the push/pull scale 0-25 pound (0-11 kilogram), STD-753, against the relief valve door.

(a) Keep the push/pull scale 0-25 pound (0-11 kilogram), STD-753, at approximately a 90° angle to the relief valve door.

SUBTASK 21-32-03-720-003

(3) Make sure the relief valve door starts to open when the push/pull scale 0-25 pound (0-11 kilogram), STD-753, shows between 4 lbf (17.8 N) and 8 lbf (35.6 N).

SUBTASK 21-32-03-720-004

(4) Continue to push the relief valve door to the full open position.

(a) Make sure the door moves smoothly without any binding.

SUBTASK 21-32-03-720-005

(5) Make sure the push/pull scale 0-25 pound (0-11 kilogram), STD-753, shows between 8 lbf (35.6 N) and 10.5 lbf (46.7 N) when the relief valve door is in the full open position.

NOTE: The door will be at an 80° angle in the full open position.

SUBTASK 21-32-03-720-006

(6) Slowly release the relief valve door until it is in the closed position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-32-03-080-001

(7) Remove the push/pull scale 0-25 pound (0-11 kilogram), STD-753.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO COMPARTMENT BLOWOUT PANELS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cargo compartment ceiling blowout panels.

(2) An installation of the cargo compartment ceiling blowout panels.

(3) A removal of the cargo compartment bulkhead blowout panels.

(4) An installation of the cargo compartment bulkhead blowout panels.

B. There are three ceiling blowout panels, two in the aft cargo compartment and one in the forward cargo compartment.

C. There are seven bulkhead blowout panels, three in the forward cargo forward bulkhead, three in the forward cargo aft bulkhead and one in the aft cargo compartment waste tank enclosure.

TASK 21-32-05-000-801

2. Ceiling Blowout Panel Removal

(Figure 401)

A. Location Zones

Zone	Area
122	Forward Cargo Compartment - Right
142	Aft Cargo Compartment - Right

B. Prepare for the Removal

SUBTASK 21-32-05-010-001

(1) Do this step to access the blowout panels [1] in the forward cargo compartment:

(a) Open the forward cargo door.

SUBTASK 21-32-05-010-002

(2) Do this step to access the blowout panels [1] in the aft cargo compartment:

(a) Open the aft cargo door.

C. Ceiling Blowout Panel Removal

SUBTASK 21-32-05-020-001

(1) Remove the screws [3] on one of the cap strips [4].

SUBTASK 21-32-05-020-002

(2) Remove the cap strip [4].

SUBTASK 21-32-05-020-003

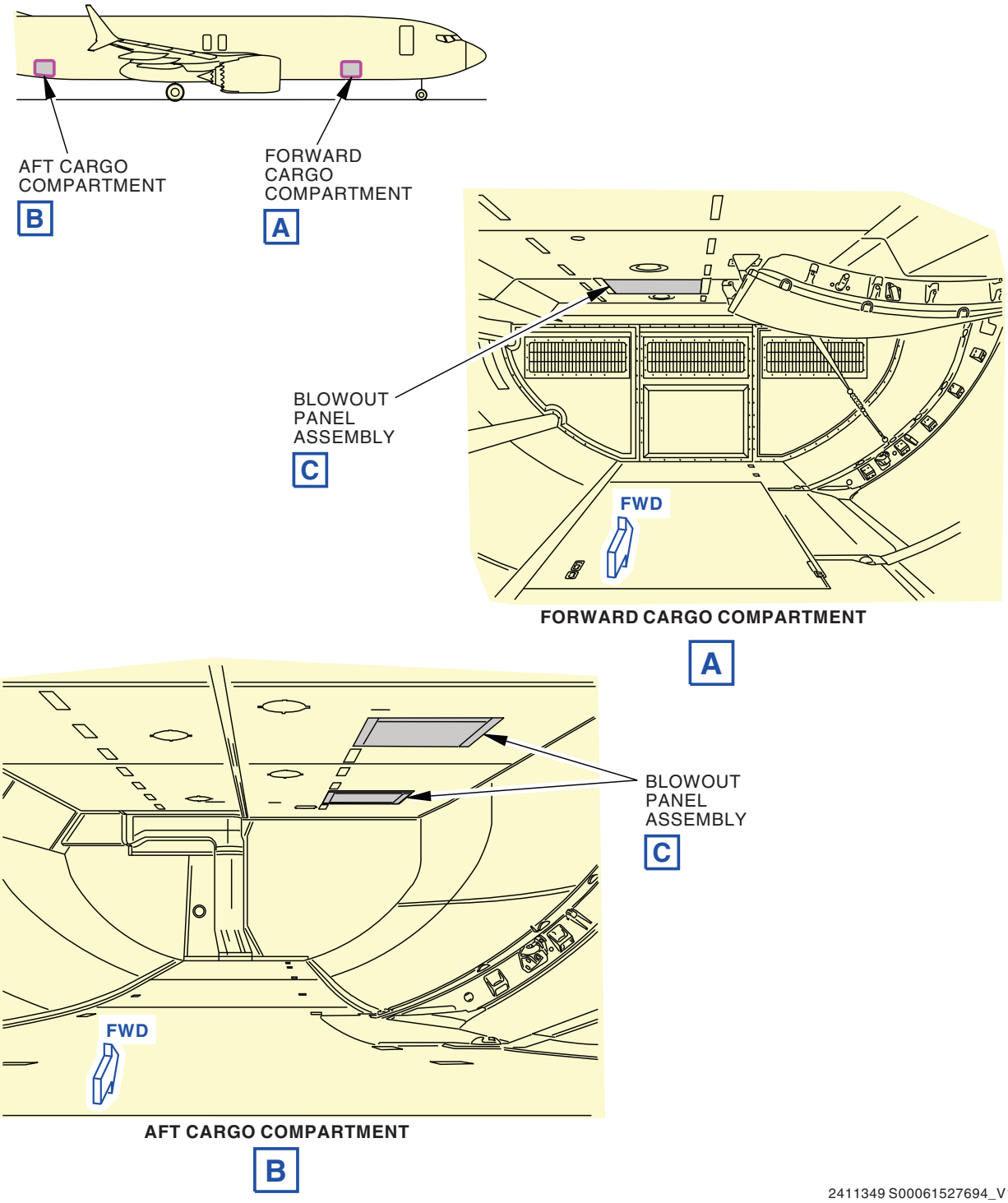
(3) Loosen the screws [2] on the remaining three cap strips [4].

SUBTASK 21-32-05-020-004

(4) Slide out the blowout panel [1].

——— END OF TASK ———

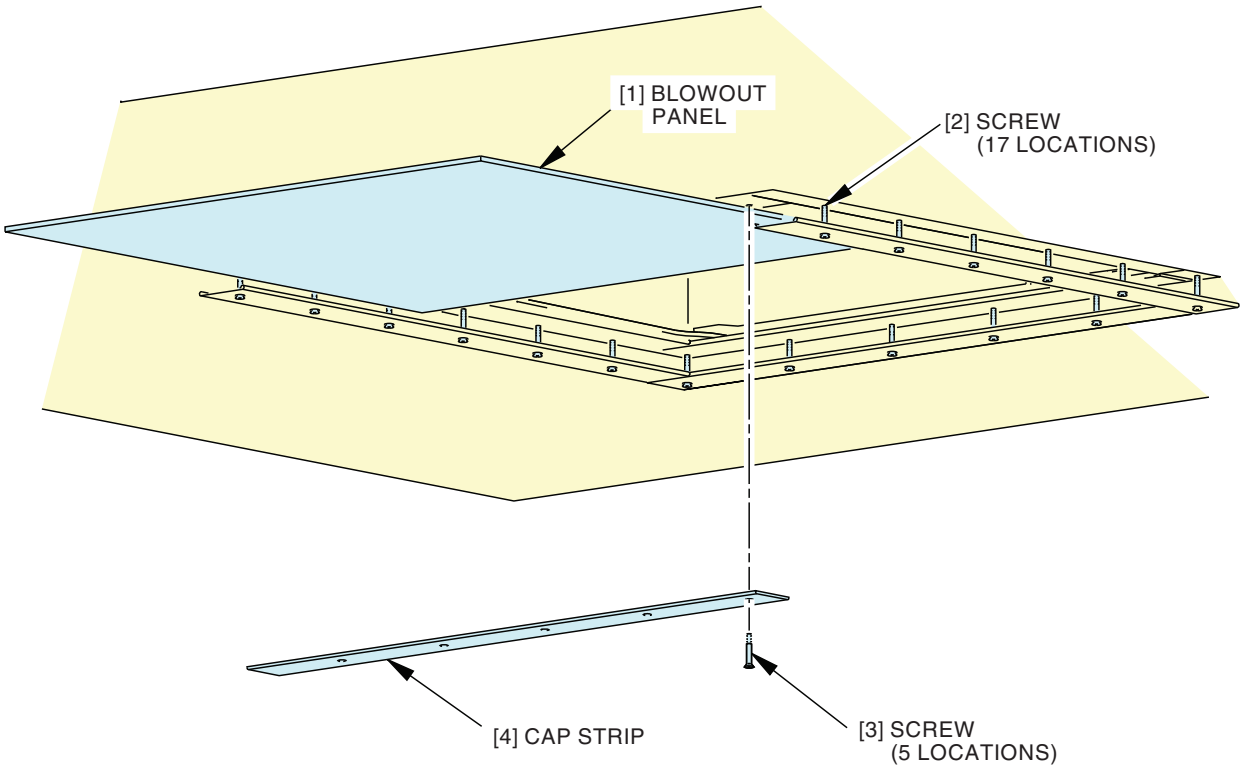
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Cargo Compartment Ceiling Blowout Panels Installation
Figure 401/21-32-05-990-801 (Sheet 1 of 2)

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BLOWOUT PANEL ASSEMBLY

C

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Cargo Compartment Ceiling Blowout Panels Installation
Figure 401/21-32-05-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-32-05-400-801

3. Ceiling Blowout Panel Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Blowout panel	21-32-05-30-050	PTF ALL
		21-32-05-40-065	PTF ALL

B. Location Zones

Zone	Area
122	Forward Cargo Compartment - Right
142	Aft Cargo Compartment - Right

C. Ceiling Blowout Panel Installation

SUBTASK 21-32-05-420-001

(1) Tighten the screws [2] on the three cap strips [4] on the ceiling until there is about a 0.125 in. (3.18 mm) gap in the cap strips [4].

SUBTASK 21-32-05-420-002

(2) Slide the blowout panel [1] into its position.

(a) Engage the blowout panel [1] edges to expose the black edge band 0.00 to 0.10 in. (2.5 mm).

SUBTASK 21-32-05-420-003

(3) Put the cap strip [4] that was removed in its position.

SUBTASK 21-32-05-420-004

(4) Install the remaining screws [3] loosely in the cap strip [4].

SUBTASK 21-32-05-420-005

WARNING

DO NOT APPLY TAPE ON THE BLOWOUT PANELS OR CAP STRIPS ON THE CEILING LINER. IN THE AFT CARGO COMPARTMENT, DO NOT APPLY TAPE BETWEEN THE CEILING LINER AND CAP STRIP ALONG THE CEILING CENTERLINE. THE JOINT GIVES A VENT PATH FOR AIR FLOW IF A DECOMPRESSION OCCURS. IF THE PRESSURE RELIEF FUNCTION OPERATES INCORRECTLY, THIS CAN CAUSE INJURY TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(5) Tighten the screws [3] to 20 in-lb (2.3 N·m) – 25 in-lb (2.8 N·m).

NOTE: Do not apply tape over the cap strips [4].

D. Put the Airplane Back to Its Usual Condition

SUBTASK 21-32-05-010-003

(1) Do this step if you installed the blowout panel [1] in the forward cargo compartment:

(a) Close the forward cargo door.

SUBTASK 21-32-05-010-004

(2) Do this step if you installed the blowout panels [1] in the aft cargo compartment:

(a) Close the aft cargo door.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-32-05-000-802

4. Cargo Compartment Bulkhead Blowout Panel Removal

(Figure 402)

A. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
821	Forward Cargo Door
822	Aft Cargo Door

B. Prepare for the Removal

SUBTASK 21-32-05-010-005

(1) Open the applicable forward or aft cargo door.

C. Cargo Compartment Bulkhead Blowout Panel Removal

SUBTASK 21-32-05-020-005

(1) Do these steps to remove the blowout panel [15] for the forward bulkhead:

(a) Loosen the quarter-turn fasteners [8] on the blowout panel assembly [7].

(b) Remove the blowout panel assembly [7].

(c) Remove the bolts [4] and washers [5] that attach the grill [1].

(d) Remove the grill [1]. Make sure the seals [6] do not detach.

(e) Remove the bolts [2] that attach the bellmouth [3].

(f) Remove the bellmouth [3].

(g) Remove the nuts [10], washers [11], washers [12], and bolts [13].

(h) Remove the retaining ring [14].

CAUTION

MAKE SURE THAT YOU DO NOT TRY TO REMOVE THE INDEX PINS. IF YOU TRY TO REMOVE THE INDEX PINS, YOU CAN DAMAGE THEM.

(i) Remove the blowout panel [15].

SUBTASK 21-32-05-020-006

(2) Do these steps to remove the blowout panel [20] for the aft bulkhead or waste tank enclosure:

(a) Remove the screws [17] that attach the decompression assembly [19] to the bulkhead.

(b) Remove the decompression assembly [19]].

(c) Remove the screws [24] and the nuts [25], at 6 locations.

(d) Remove the grill [22].

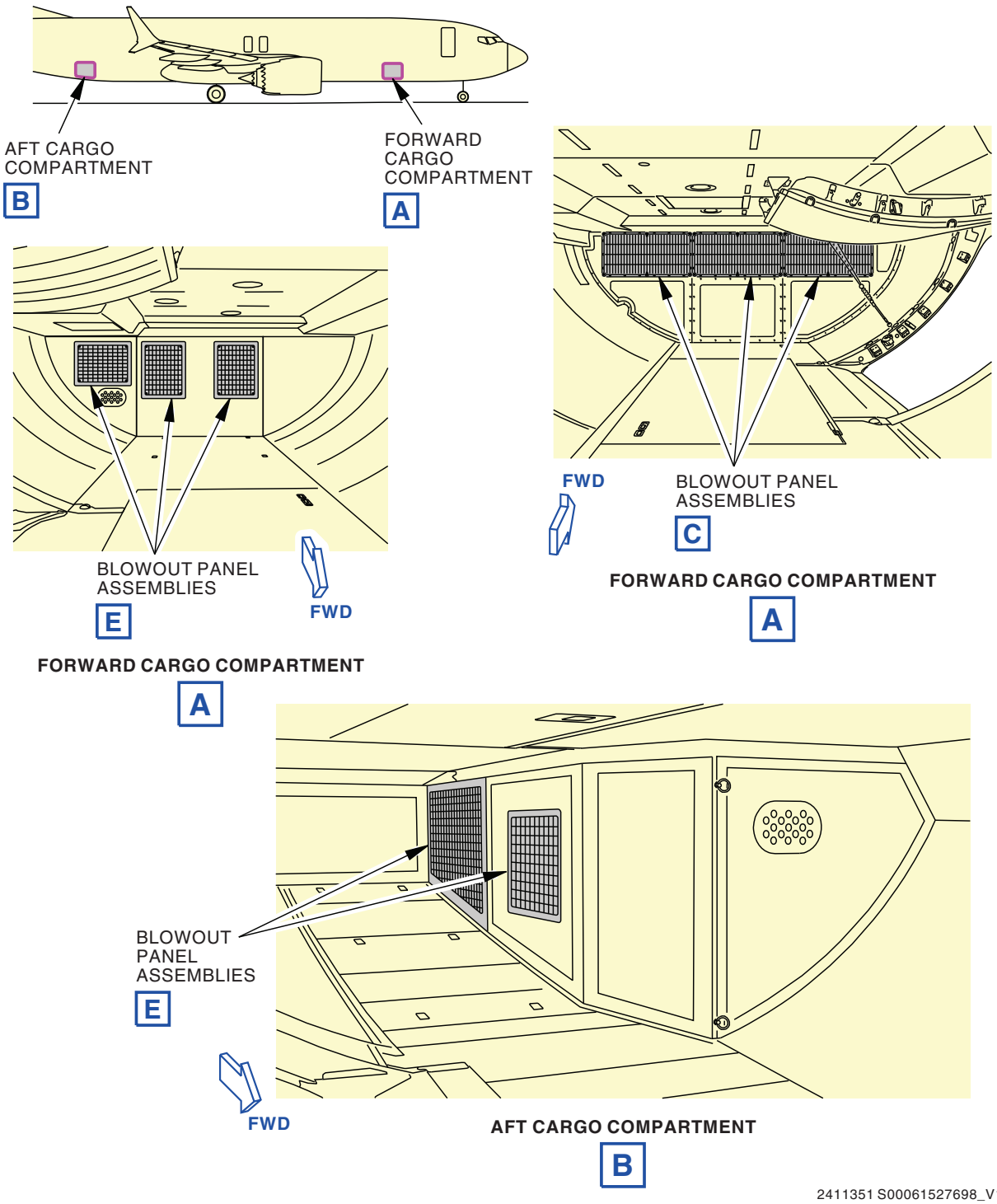
(e) Remove the screws [23] and the nuts [16], at 26 locations.

(f) Remove the cap strip [21] assemblies.

(g) Remove the blowout panel [20].

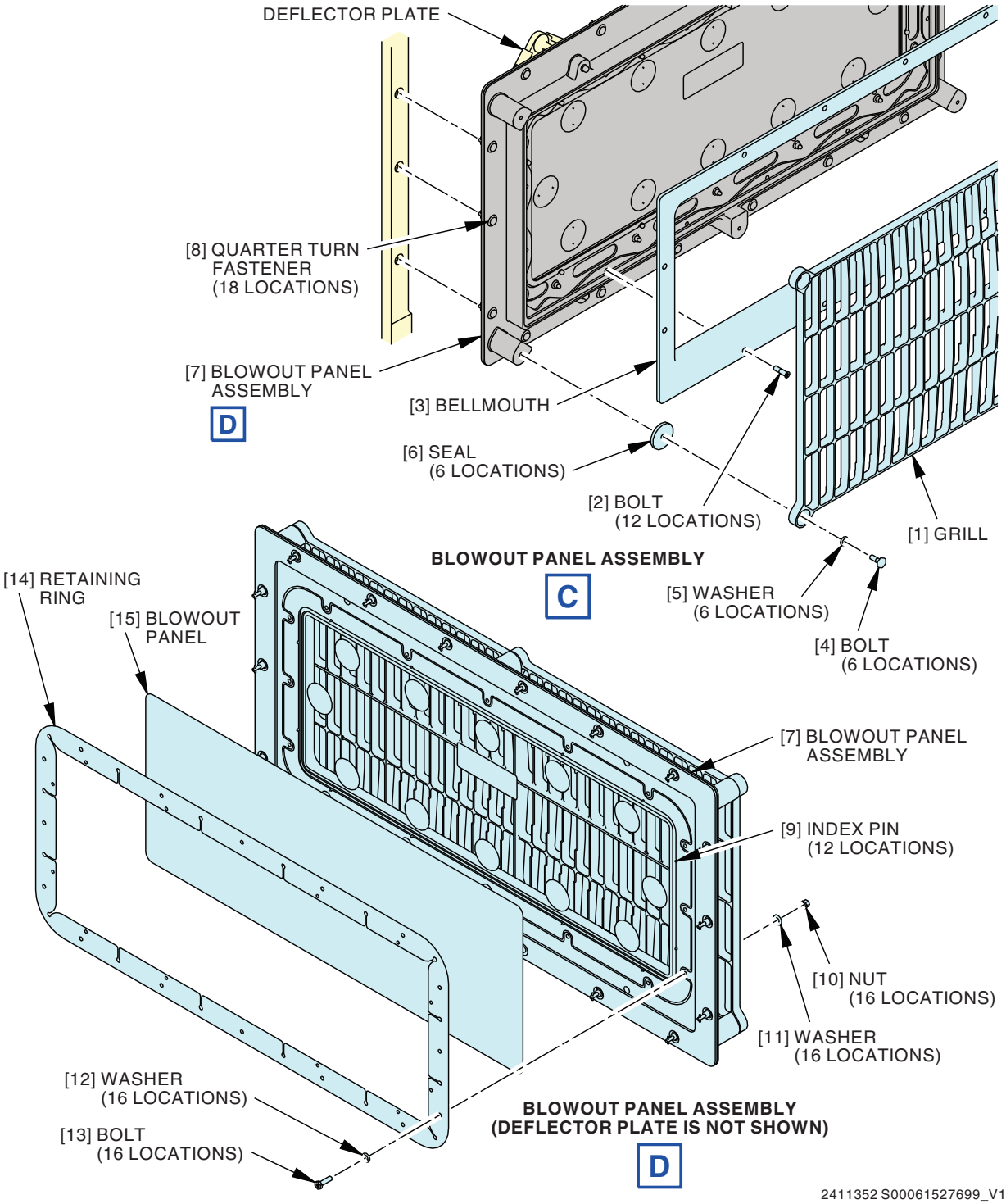
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



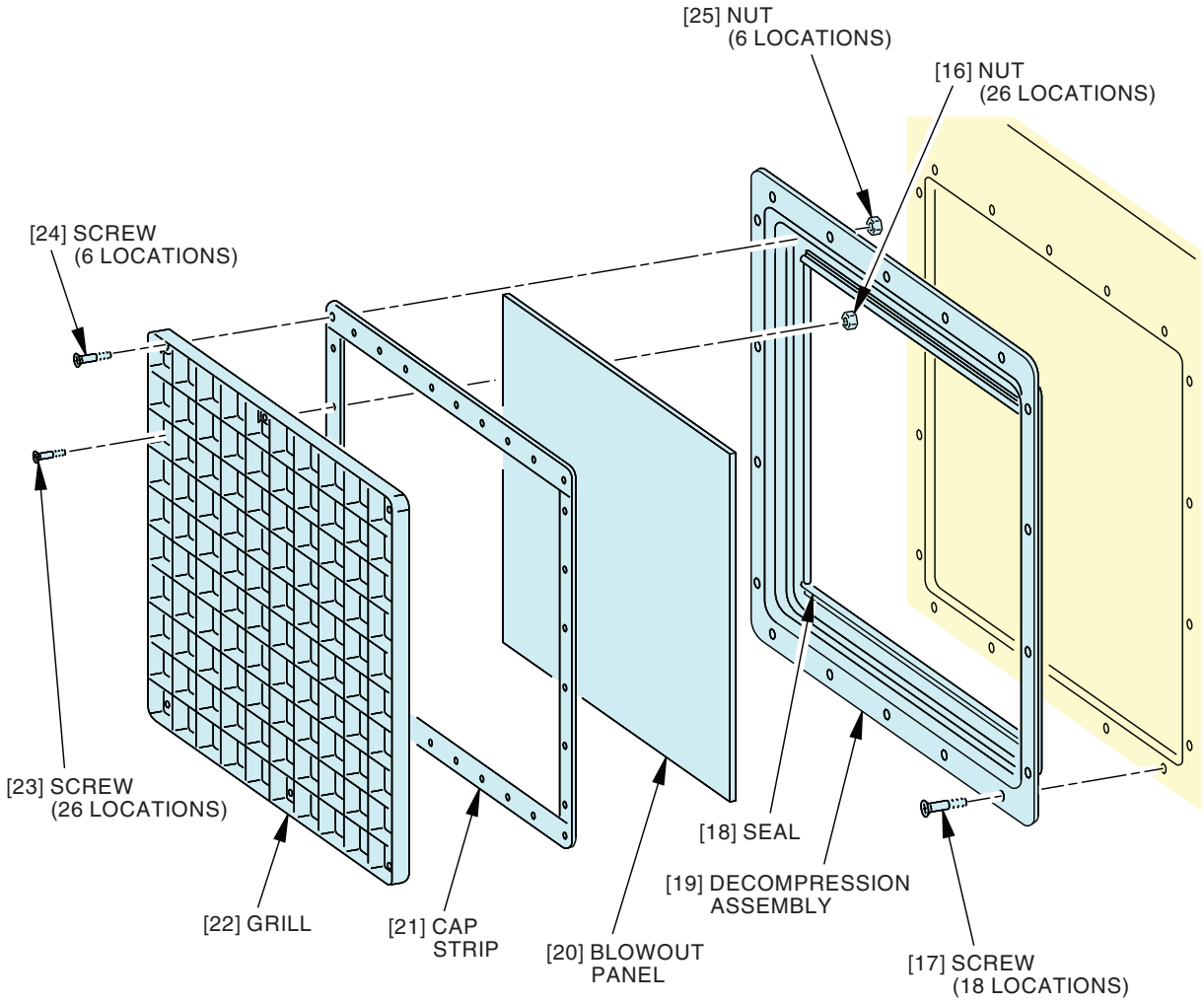
Cargo Compartment Bulkhead Blowout Panels Installation
Figure 402/21-32-05-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cargo Compartment Bulkhead Blowout Panels Installation
Figure 402/21-32-05-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BLOWOUT PANEL ASSEMBLY
(EXAMPLE)

E

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Cargo Compartment Bulkhead Blowout Panels Installation
Figure 402/21-32-05-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-32-05-400-802

5. Cargo Compartment Bulkhead Blowout Panel Installation

(Figure 402)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
15	Blowout panel	21-32-05-01-205	PTF ALL
20	Blowout panel	21-32-05-10-155	PTF ALL
		21-32-05-15-065	PTF ALL
		21-32-05-20-060	PTF ALL

B. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right
821	Forward Cargo Door
822	Aft Cargo Door

C. Cargo Compartment Bulkhead Blowout Panel Installation

SUBTASK 21-32-05-420-006

(1) Do these steps to install the blowout panel [15] for the forward bulkhead:

(a) Put the retaining ring [14] so you can slide the blowout panel [15] into its position.

(b) Put the blowout panel [15] in its position.

NOTE: The panel will be held in its position by the index pins [9].

(c) Install the nuts [10], washers [11], washers [12], and bolts [13] at 16 locations.

(d) Put the bellmouth [3] in its position.

(e) Install the bolts [2] at 12 locations.

(f) Make sure that the seals [6] are in their positions.

(g) Put the grill [1] in its position.

(h) Install the bolts [4] and washers [5] at 6 locations.

(i) Make sure that the deflector plate is attached to the blowout panel assembly [7].

(j) Put the blowout panel assembly [7] in its position and tighten the quarter-turn fasteners [8] at 18 locations.

SUBTASK 21-32-05-420-007

(2) Do these steps to install the blowout panel [20] for the aft bulkhead or waste tank enclosure:

(a) Position the decompression assembly [19] on a flat surface so that the nutplates are facing down.

(b) Put the blowout panel [20] in its position over the seal [18].

(c) Install the cap strip [21] assemblies.

(d) Install the screws [23] and the nuts [16] at 26 locations.

(e) Install the grill [22].

(f) Install the screws [24] and the nuts [25] at 6 locations.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Put the decompression assembly [19] in its position on the bulkhead.

(h) Install the screws [17] at 18 locations.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 21-32-05-010-006

(1) Close the applicable forward or aft cargo door.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN PRESSURE ALTITUDE WARNING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has one task:

(1) Cabin pressure altitude warning switch functional test.

31-CMR-05

TASK 21-33-00-700-802

2. Cabin Altitude Warning Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is for the functional test of the cabin pressure altitude warning system.

(2) For 737-7 airplanes, this task satisfies Certification Maintenance Requirement (CMR) Item Number 31-CMR-05.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197

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C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Prepare for the Test

SUBTASK 21-33-00-860-002

(1) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN

SUBTASK 21-33-00-860-004

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	12	C00318	INDICATOR MASTER DIM SECT 6

SUBTASK 21-33-00-010-001

(3) Open this access panel:

Number	Name/Location
112A	Forward Access Door

F. Cabin Altitude Warning Functional Test

SUBTASK 21-33-00-720-001

(1) Do the test of the 10000 feet cabin pressure altitude warning as follows:

NOTE: There are two identical pressure switches (S128 and S1153) for the 10000 feet cabin altitude warning. These switches are set to alert the crew when the cabin pressure altitude goes above 10000 feet. One switch can trip the cabin altitude warning. The second switch is a backup. The test that you will do one at a time for each switch is the same.

(a) Remove the screen port assembly from the cabin pressure altitude warning switch, S128, or S1153.

(b) Install an adapter fitting and matching hose on the cabin pressure altitude warning switch, S128, or S1153.

NOTE: The female threads on the cabin altitude warning switch are 7/16-20-UNJF-3A with a full thread depth of 0.438 in. (1.113 cm) minimum. The total port depth is 0.541 in. (1.374 cm) - 0.551 in. (1.400 cm). Any mating adapter should have threads having a full thread depth of no greater than 0.438 in. (1.113 cm) and an overall length that is less than 0.500 in. (1.270 cm) to avoid false test results.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Use one of the following adapters, or equivalent adapter, and matching hose to connect to the cabin altitude warning switch:

NOTE: Adapters are considered to be equivalent as long as the mating side with the switch is:

- Greater than or equal to 0.265 in. (0.673 cm) X 7/16-20-UNJF-3A and less than or equal to 0.438 in. (1.113 cm) X 7/16-20-UNJF-3A for the flareless end, or
- Less than or equal to 0.500 in. (1.270 cm) total with greater than or equal to 0.265 in. (0.673 cm) X 7/16-20-UNJF-3A thread for AN4 flared end.

a) SAE J514 P/N 070220 90 Degree Straight Thread Elbow and appropriate sized O-ring (Preferred).

<1> Use a Barfield Pitot Hose, or equivalent 25 ft (7.62 m) to 40 ft (12.19 m) long hose, with #4 AN fitting to the adapter and quick disconnect (if applicable) to the air data test set.

<2> Make sure that the flat side of the adapter is connected with the cabin altitude warning switch.

NOTE: Do not connect the flared side of the adapter with the cabin altitude warning switch. Connecting the flared side of the adapter with the cabin altitude warning switch can bottom out the cabin altitude warning switch, resulting in false test results.

b) SAE J514 P/N 070320 45 Degree Straight Thread Elbow and appropriate sized O-ring (Preferred).

<1> Use a Barfield Pitot Hose, or equivalent 25 ft (7.62 m) to 40 ft (12.19 m) long hose, with #4 AN fitting to the adapter and quick disconnect (if applicable) to the air data test set.

<2> Make sure that the flat side of the adapter is connected with the cabin altitude warning switch.

NOTE: Do not connect the flared side of the adapter with the cabin altitude warning switch. Connecting the flared side of the adapter with the cabin altitude warning switch can bottom out the cabin altitude warning switch, resulting in false test results.

c) SAE J514 P/N 070120 Straight Thread Connector Short and appropriate sized O-ring (Preferred).

<1> Use a Barfield Pitot Hose, or equivalent 25 ft (7.62 m) to 40 ft (12.19 m) long hose, with #4 AN fitting to the adapter and quick disconnect (if applicable) to the air data test set.

<2> Make sure that the flat side of the adapter is connected with the cabin altitude warning switch.

NOTE: Do not connect the flared side of the adapter with the cabin altitude warning switch. Connecting the flared side of the adapter with the cabin altitude warning switch can bottom out the cabin altitude warning switch, resulting in false test results.

d) AS21900-4 (or MS21900-4) Flareless Tube to Flared Tube Adapter and appropriate sized O-ring (Preferred).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> Use a Barfield Pitot Hose, or equivalent 25 ft (7.62 m) to 40 ft (12.19 m) long hose, with #4 AN fitting to the adapter and quick disconnect (if applicable) to the air data test set.

<2> Make sure that the flat side of the adapter is connected with the cabin altitude warning switch.

NOTE: Do not connect the flared side of the adapter with the cabin altitude warning switch. Connecting the flared side of the adapter with the cabin altitude warning switch can bottom out the cabin altitude warning switch, resulting in false test results.

e) P/N JUD321 Hose Fitting with MS28778-4 O-ring (Eaton Aerospace LLC, Bethel, CT 02750) (Preferred).

<1> Use a Barfield Pitot Hose, or equivalent 25 ft (7.62 m) to 40 ft (12.19 m) long hose, with #4 AN fitting to the adapter and quick disconnect (if applicable) to the air data test set.

f) AN807-4D (or AS5180D04 or AS5180W04) Tube to Hose Adapter, AN924-4 nut and appropriate sized O-ring (on the mating side with the switch) and spacer or washers (Alternate).

NOTE: This adapter can be used if the steps below are carefully followed. This adapter is not preferred because if the AN924-4 nut is not connected carefully as recommended below, this can bottom out the cabin altitude warning switch, resulting in false test results.

<1> Use a Barfield Pitot Hose, or equivalent 25 ft (7.62 m) to 40 ft (12.19 m) long hose, with quick disconnect (if applicable) to the air data test set.

<2> Make sure that the thread length, including fitting end after the installation of AN924-4 nut and appropriate sized 7/16 spacer or washers, is less than 0.500 in. (1.270 cm) to avoid false test results.

(c) Connect the air data model test set, COM-1914, or equivalent, to the adapter fitting.

1) Make sure that the quick disconnect (if applicable) is fully engaged.

(d) Slowly increase the cabin altitude at a rate of not more than 4000 ft/minute.

(e) Make sure that the aural warning signal comes on at an altitude of 10000 ± 1000 feet.

(f) Make a record of the altitude that the aural warning sounds.

(g) Make sure that the red CABIN ALTITUDE indicators, on the captain's and first officer's instrument panels, come on.

(h) Press the ALT HORN CUTOUT switch, on the P5-16 Cabin Pressure Control Panel (CPCP).

(i) Make sure that the aural warning signal goes off.

(j) Make sure that the red CABIN ALTITUDE indicators, on the captain's and first officer's instrument panels, stay on.

(k) Decrease the cabin altitude to approximately 500 to 1500 feet below the cabin altitude the aural warning came on.

NOTE: Decrease the cabin altitude at a rate of not more than 4000ft/minute.

(l) Make sure that the CABIN ALTITUDE indicators, on the captain's and first officer's instrument panels, go off.

(m) Increase the cabin altitude again at a rate of not more than 4000 ft/minute.

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(n) Make sure that the aural and visual warnings for the cabin altitude come on at an altitude of 10000 ±1000 feet.

(o) Decrease the cabin altitude to ground level at a rate of not more than 4000 ft/minute.

(p) Make sure that the aural warning signal goes off.

(q) Make sure that the CABIN ALTITUDE indicators, on the captain's and first officer's instrument panels, go off.

(r) Remove the air data model test set, COM-1914, or equivalent, and adapter fitting from the cabin pressure altitude warning switch.

(s) Install the screen port assembly on the cabin pressure altitude warning switch.

SUBTASK 21-33-00-720-004

(2) Do the steps again to test the second cabin pressure altitude warning switch, S128, or S1153.

G. Put the Airplane Back to Its Usual Condition

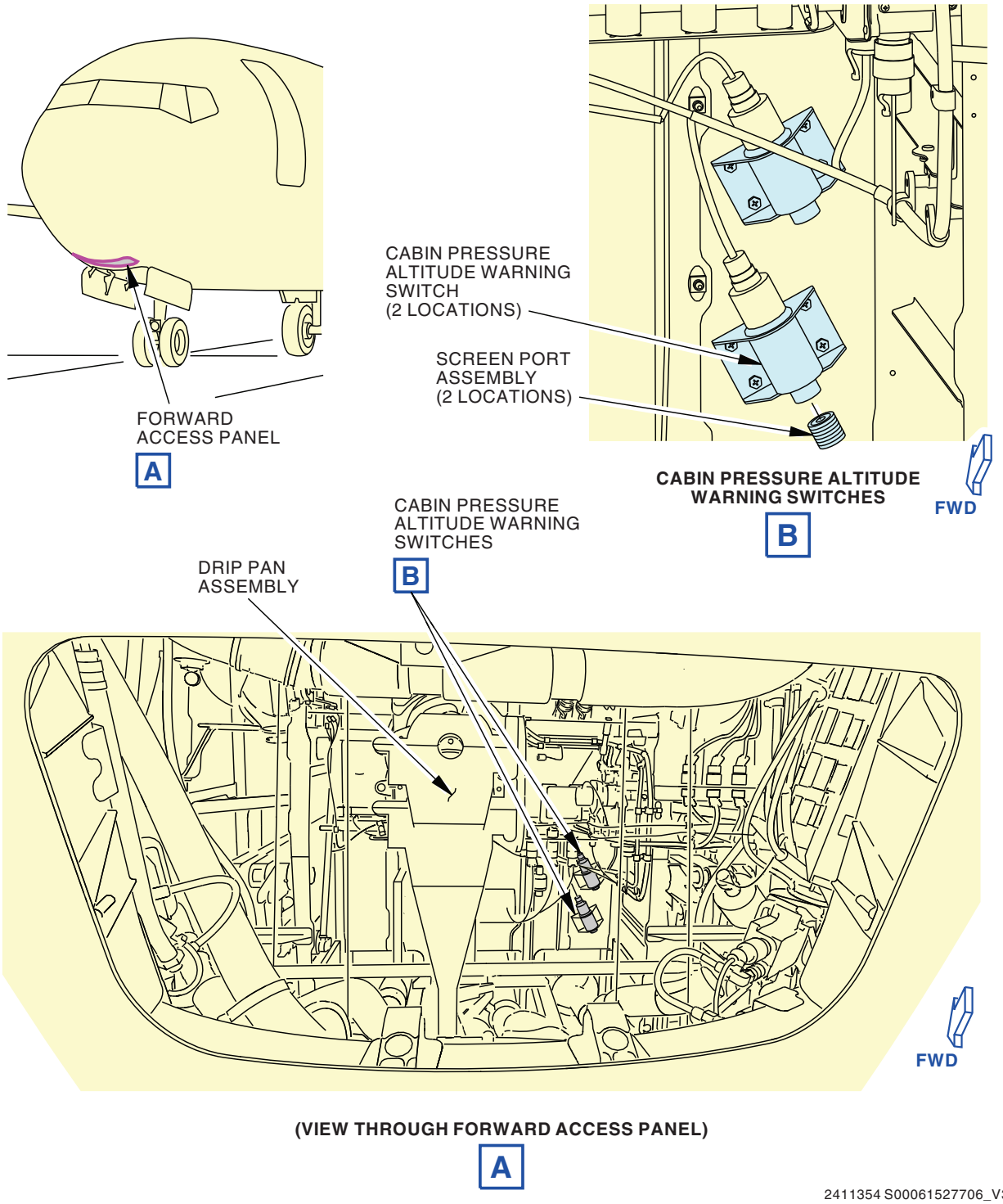
SUBTASK 21-33-00-410-001

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Altitude Warning Test
Figure 501/21-33-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-33-00-700-801

3. Cabin Rate of Climb Indicator Check

(Figure 502f)

A. General

(1) This task is for the check of the cabin rate of climb indicator.

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Cabin Rate of Climb Indicator Check

SUBTASK 21-33-00-710-001

(1) Do a check of the cabin rate of climb indicator as follows:

(a) Make sure that the cabin entry door is open.

NOTE: This will make sure that there is no pressurization on the cabin.

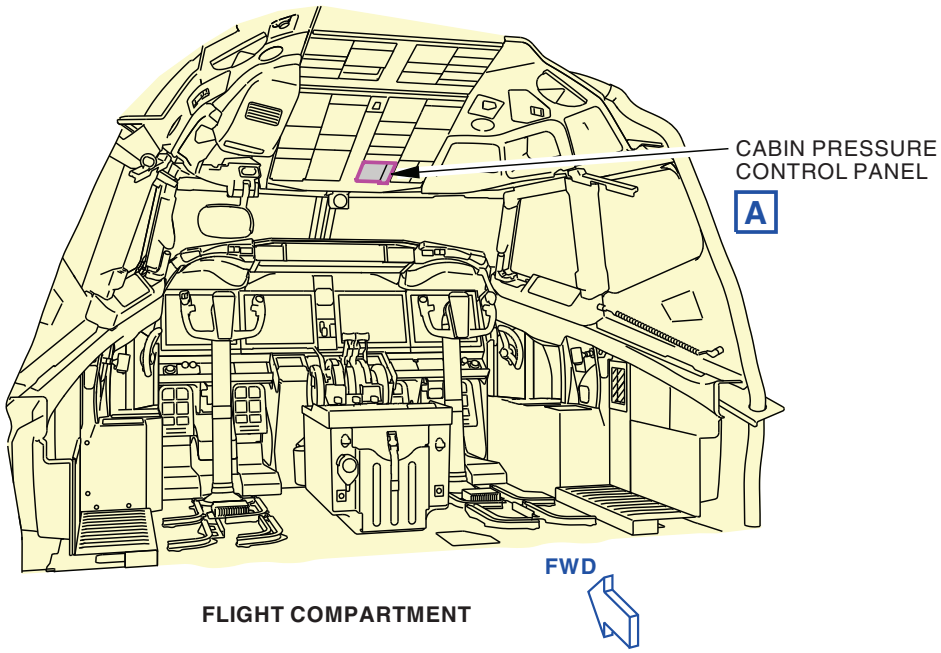
(b) Turn the zero adjustment screw on the lower left corner of the gauge as follows:

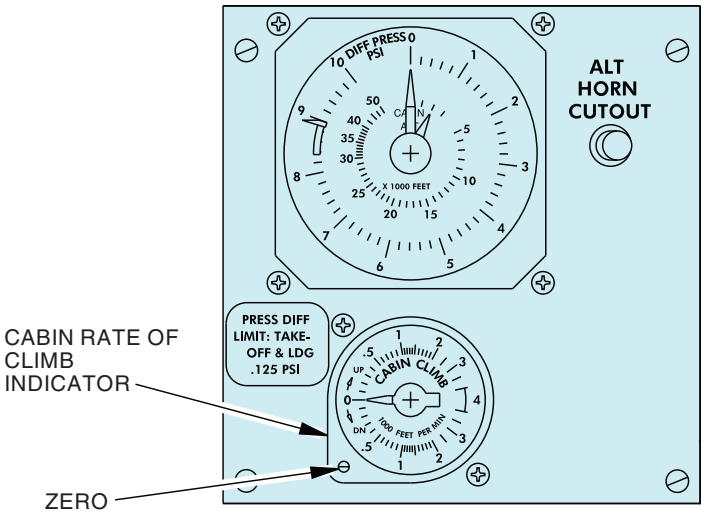
1) Make sure that you can adjust the pointer a minimum of 400 ft/min on both sides of the zero graduation.

(c) Reset the pointer to zero.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





CABIN PRESSURE CONTROL PANEL
(P5-16)
(EXAMPLE)

A

2411355 S00061527708_V4

Cabin Rate of Climb Indicator Check
Figure 502/21-33-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN RATE OF CLIMB INDICATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Cabin rate of climb indicator removal.

(2) Cabin rate of climb indicator installation.

TASK 21-33-01-000-801

2. Cabin Rate of Climb Indicator Removal

(Figure 401)

A. General

(1) This task is for the removal of the cabin rate of climb indicator.

B. Location Zones

Zone	Area
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 21-33-01-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	10	C00328	OVERHEAD 5V SECONDARY

D. Cabin Rate of Climb Indicator Removal

SUBTASK 21-33-01-020-001

CAUTION

HOLD THE INDICATOR WHEN YOU LOOSEN THE CLAMP. THE INDICATOR CAN FALL WHEN THE CLAMP IS LOOSE. DAMAGE CAN OCCUR TO THE INDICATOR IF IT FALLS.

(1) Loosen the clamp tension screws [3].

SUBTASK 21-33-01-020-002

(2) Carefully pull the cabin rate of climb indicator [1] out of the P5-16 panel [4].

SUBTASK 21-33-01-020-003

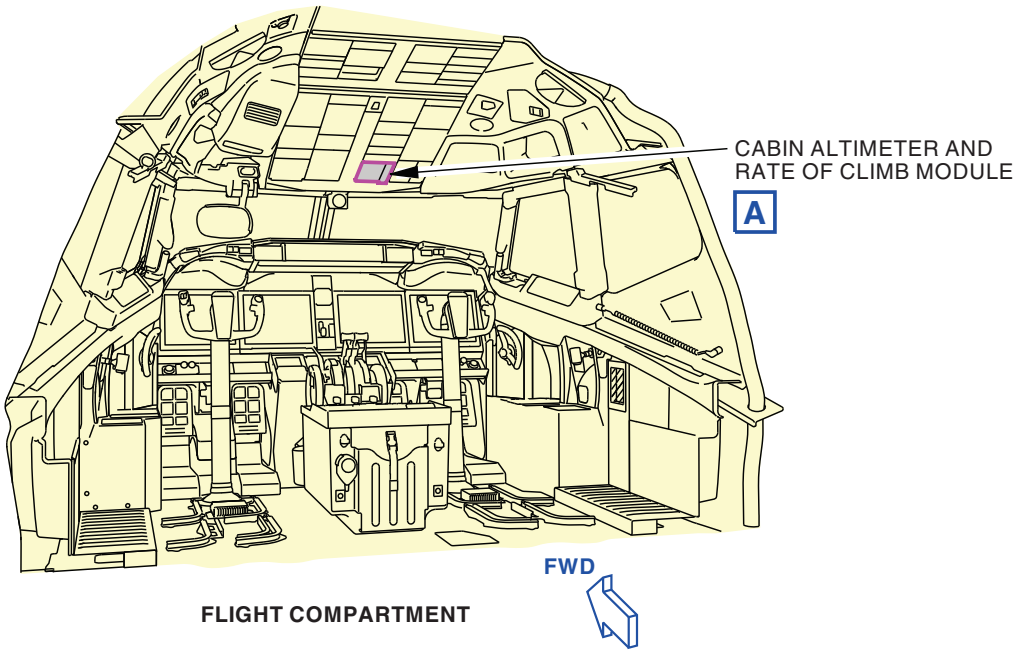
(3) Disconnect the electrical connector [2] from the cabin rate of climb indicator [1].

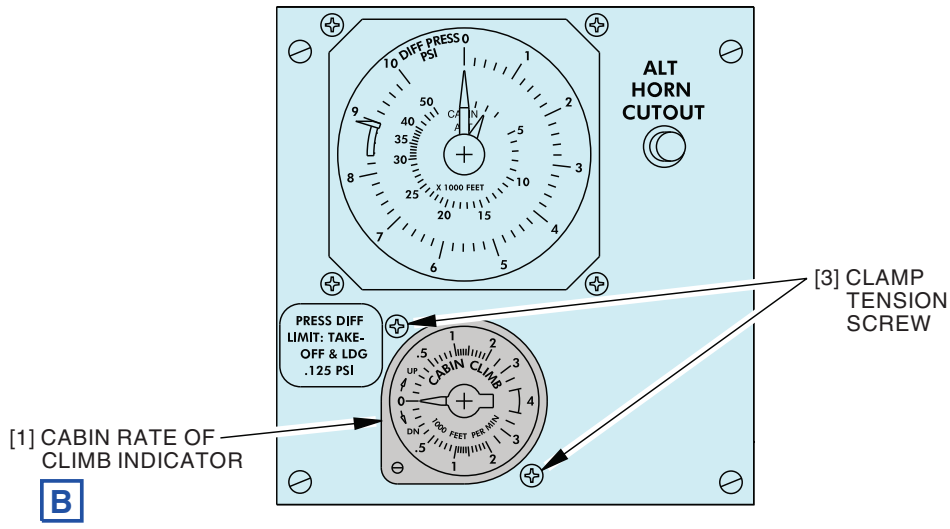
SUBTASK 21-33-01-480-001

(4) Put a dust cap on the cabin pressure port on the back of the cabin rate of climb indicator [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





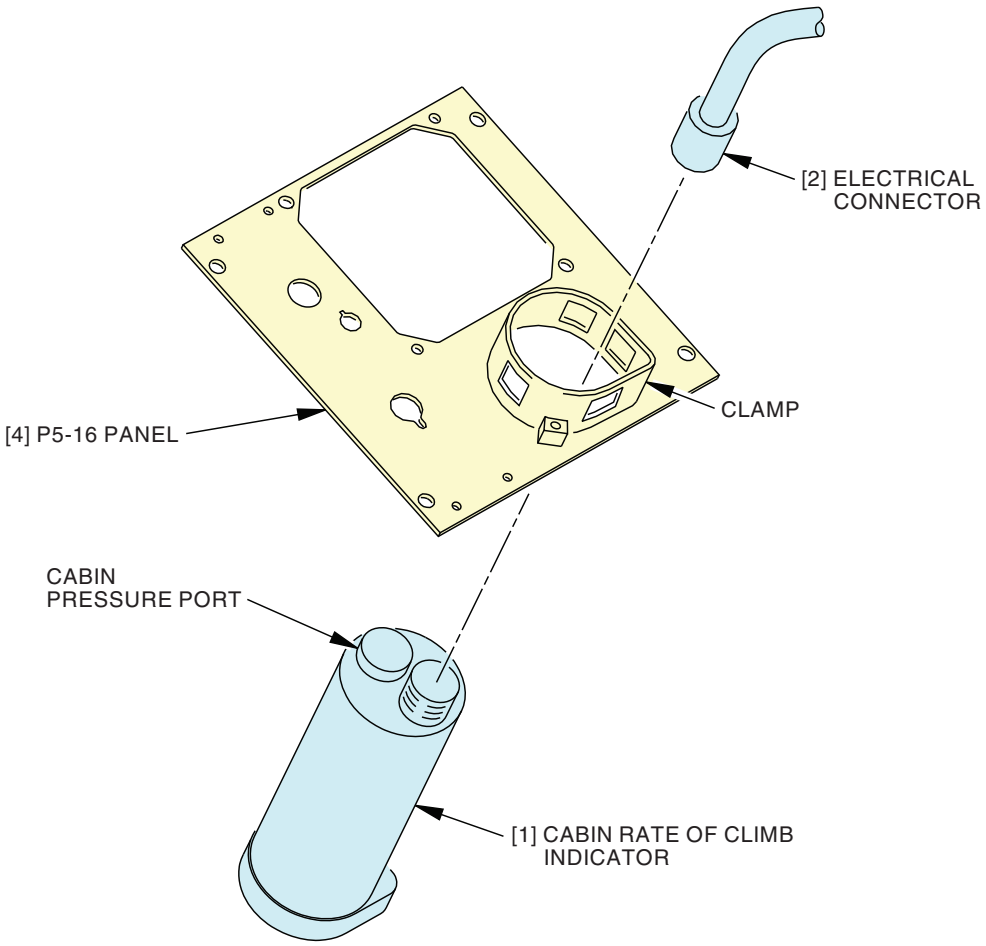
**CABIN ALTIMETER AND RATE OF CLIMB MODULE (P5-16)
(EXAMPLE)**

A

2411356 S00061527712_V4

**Cabin Rate of Climb Indicator Installation
Figure 401/21-33-01-990-801 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN RATE OF CLIMB INDICATOR

B

2411357 S00061527713_V1

Cabin Rate of Climb Indicator Installation
Figure 401/21-33-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-33-01-400-801

3. Cabin Rate of Climb Indicator Installation

(Figure 401)

A. General

(1) This task is for the installation of the cabin rate of climb indicator.

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cabin rate of climb indicator	21-33-01-01-050	PTF ALL
		31-11-94-01-025	PTF ALL

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Cabin Rate Indicator Installation

SUBTASK 21-33-01-080-001

(1) Remove the dust cap from the cabin pressure port on the back of the cabin rate of climb indicator [1].

SUBTASK 21-33-01-420-001

(2) Connect the electrical connector [2] to the cabin rate of climb indicator [1].

SUBTASK 21-33-01-820-001

(3) Carefully push the cabin rate of climb indicator [1] into its position in the P5-16 panel [4].

SUBTASK 21-33-01-420-002

(4) Hold the cabin rate of climb indicator [1] against the P5-16 panel [4].

(a) Tighten the clamp tension screws [3] 5.0 in-lb (0.6 N·m) to 8.0 in-lb (0.9 N·m).

E. Put the Airplane Back to Its Usual Condition

SUBTASK 21-33-01-860-002

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	10	C00328	OVERHEAD 5V SECONDARY

F. Cabin Rate of Climb Indicator Installation Test

SUBTASK 21-33-01-700-001

(1) Do this steps to do a check of the cabin rate of climb indicator [1]:

(a) Make sure that the cabin is not pressurized.

(b) Turn the zero adjustment screw on the lower left corner of the gauge.

(c) Make sure that the pointer can be adjusted to a minimum of 400 ft/min on both sides of the zero graduation.

(d) Reset the pointer to zero.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN ALTITUDE AND DIFFERENTIAL PRESSURE INDICATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Cabin altitude and differential pressure indicator removal

(2) Cabin altitude and differential pressure indicator installation.

TASK 21-33-02-000-801

2. Cabin Altitude and Differential Pressure Indicator Removal

(Figure 401)

A. General

(1) This task is for the removal of the cabin altitude and differential pressure indicator.

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cabin altitude and differential pressure indicator	21-33-02-01-050	PTF ALL
		31-11-94-01-020	PTF ALL

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 21-33-02-860-003

(1) Open this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	10	C00328	OVERHEAD 5V SECONDARY

E. Cabin Altitude and Differential Pressure Indicator Removal

SUBTASK 21-33-02-020-001

CAUTION

HOLD THE INDICATOR WHEN YOU LOOSEN THE CLAMP. THE INDICATOR CAN FALL WHEN THE CLAMP IS LOOSE. DAMAGE CAN OCCUR TO THE INDICATOR IF IT FALLS.

(1) Loosen the clamp tension screws [6].

SUBTASK 21-33-02-020-002

(2) Carefully pull the cabin altitude and differential pressure indicator [1] out of the P5-16 panel [7].

SUBTASK 21-33-02-020-003

(3) Disconnect the electrical connector [5] from the cabin altitude and differential pressure indicator [1].

SUBTASK 21-33-02-020-004

(4) Disconnect the static pressure hose [4] from the male quick-disconnect [2].

SUBTASK 21-33-02-020-005

(5) If you will replace the cabin altitude and differential pressure indicator [1], do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the male quick-disconnect [2] and the packing [3] from the cabin altitude and differential pressure indicator [1].

(b) Discard the packing [3].

SUBTASK 21-33-02-800-001

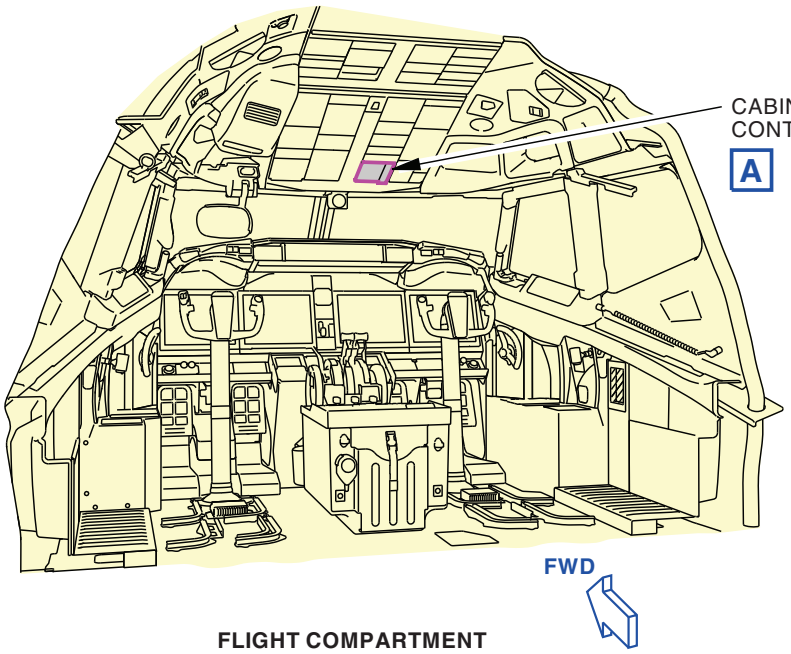
(6) If you will install a serviceable cabin altitude and differential pressure indicator [1], keep the male quick-disconnect [2] and packing [3] in place.

SUBTASK 21-33-02-480-001

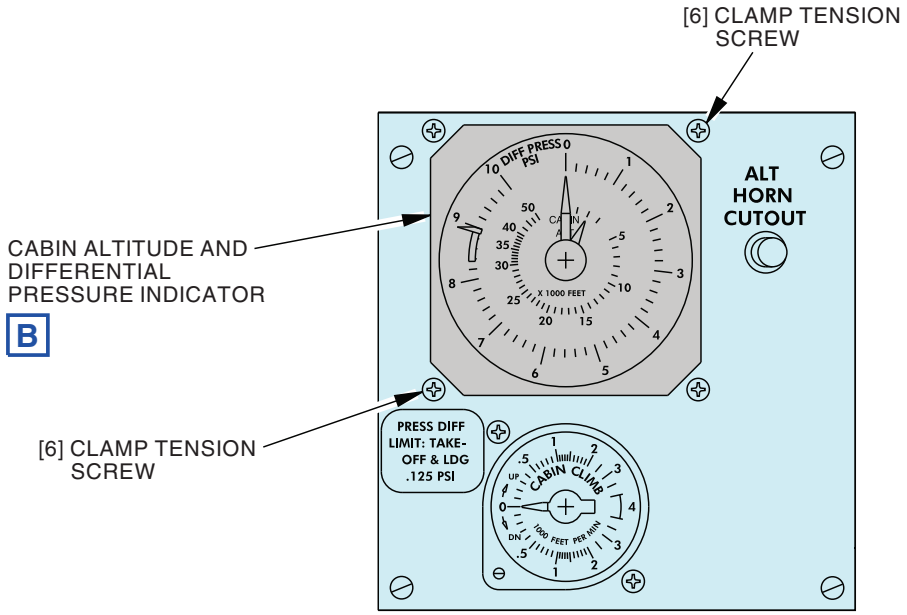
(7) Put a dust cap on the cabin pressure port.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



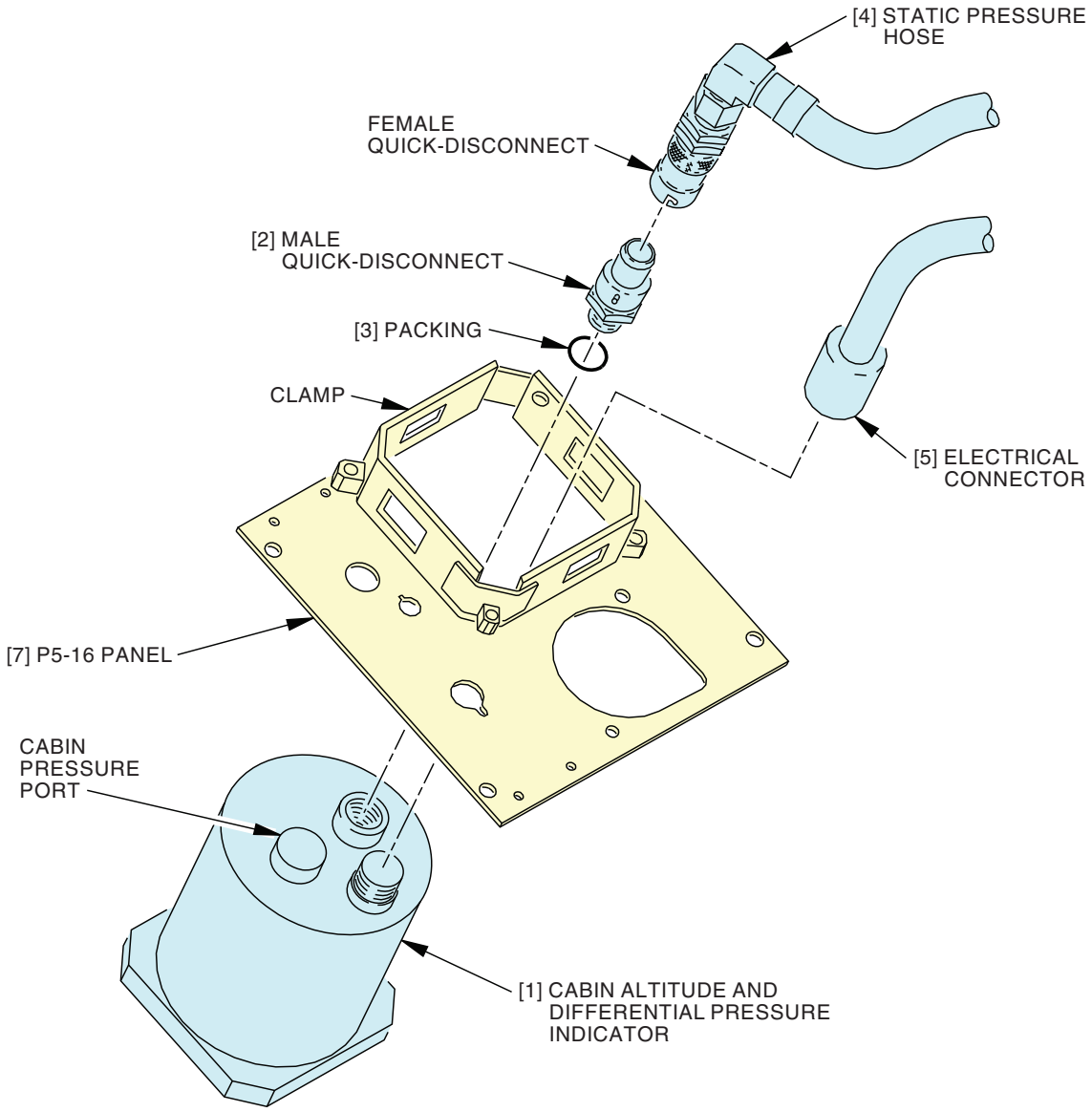
CABIN PRESSURE CONTROL PANEL (P5-16)
(EXAMPLE)

A

2411358 S00061527718_V4

Cabin Altitude and Differential Pressure Indicator Installation
Figure 401/21-33-02-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABIN ALTITUDE AND DIFFERENTIAL PRESSURE INDICATOR

B

2411359 S00061527719_V1

Cabin Altitude and Differential Pressure Indicator Installation
Figure 401/21-33-02-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-33-02-400-801

3. Cabin Altitude and Differential Pressure Indicator Installation

(Figure 401)

A. General

(1) This task is for the installation of the cabin altitude and differential pressure indicator.

B. References

Reference	Title
34-11-00-790-805	Alternate Static System Low-range Leak Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
D50004	Compound - Antiseize	BMS3-28

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cabin altitude and differential pressure indicator	21-33-02-01-050	PTF ALL
		31-11-94-01-020	PTF ALL

E. Location Zones

Zone	Area
212	Flight Compartment - Right

F. Cabin Altitude and Differential Pressure Indicator Installation

SUBTASK 21-33-02-080-001

(1) Remove the dust cap from the cabin pressure port on the rear of the indicator.

SUBTASK 21-33-02-420-001

(2) If it is necessary to install the male quick-disconnect [2], do these steps:

(a) Put a new packing [3] on the male quick-disconnect [2].

(b) Apply anti-seize compound, D50004 to the threads of the male quick-disconnect [2].

(c) Install the male quick-disconnect [2] on the cabin altitude and differential pressure indicator [1].

(d) Torque the male quick-disconnect [2] to 120 ±12 in-lb (13.56 ±1.36 N·m).

SUBTASK 21-33-02-020-006

(3) Connect the static pressure hose [4] to the male quick-disconnect [2] on the cabin altitude and differential pressure indicator [1].

SUBTASK 21-33-02-420-002

(4) Connect the electrical connector [5] to the cabin altitude and differential pressure indicator [1].

SUBTASK 21-33-02-820-001

(5) Carefully push the cabin altitude and differential pressure indicator [1] into its position in the P5-16 panel [7].

SUBTASK 21-33-02-420-003

(6) Hold the cabin altitude and differential pressure indicator [1] against the P5-16 panel [7].

(a) Torque the clamp tension screws [6] to 5 in-lb (0.6 N·m) - 8 in-lb (0.9 N·m).

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SUBTASK 21-33-02-790-001

(7) If you installed the male quick-disconnect [2] on the cabin altitude and differential pressure indicator [1], do this task: Alternate Static System Low-range Leak Test, TASK 34-11-00-790-805.

NOTE: If the replacement indicator came with the quick-disconnect installed or if the quick-disconnect was not removed from the indicator, it is not necessary to do the leak test.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-33-02-860-002

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	10	C00328	OVERHEAD 5V SECONDARY

H. Cabin Altitude and Differential Pressure Indicator Installation

SUBTASK 21-33-02-710-001

(1) Do these steps to do a check of the cabin altimeter gauge:

(a) Make sure that a main entry door is open.

(b) Set the barometric pressure for the pilot's altimeter to 29.92 inches of Hg.

(c) Compare the altitude on the pilot's altimeter to the altitude on the cabin altimeter.

(d) If the altitude on the cabin altimeter is within 1000 feet of the altitude on the pilot's altimeter, the cabin altimeter is satisfactory.

(e) If the altitude on the cabin altimeter is not within 1000 feet of the altitude on the pilot's altimeter, replace the cabin alltitude and differential pressure indicator (TASK 21-33-02-000-801 and TASK 21-33-02-400-801).

SUBTASK 21-33-02-710-002

(2) Do these steps to do a check of the cabin differential pressure gauge:

(a) Observe the differential pressure on the DIFF PRESS gauge.

(b) If the differential pressure is within 0 (± 0.2) psid, the cabin differential pressure gauge is satisfactory.

(c) If the differential pressure is not within 0 (± 0.2) psid, replace the cabin altitude and differential pressure indicator (TASK 21-33-02-000-801 and TASK 21-33-02-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN PRESSURE SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the cabin pressure switch

(2) An installation of the cabin pressure switch.

TASK 21-33-04-000-801

2. Cabin Pressure Warning Switch Removal

(Figure 401)

A. General

(1) This task is for the removal of the cabin pressure warning switch.

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 21-33-04-010-001

(1) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 21-33-04-862-001

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN

E. Cabin Pressure Warning Switch Removal

SUBTASK 21-33-04-020-001

(1) Remove the cabin pressure warning switch [1] as follows:

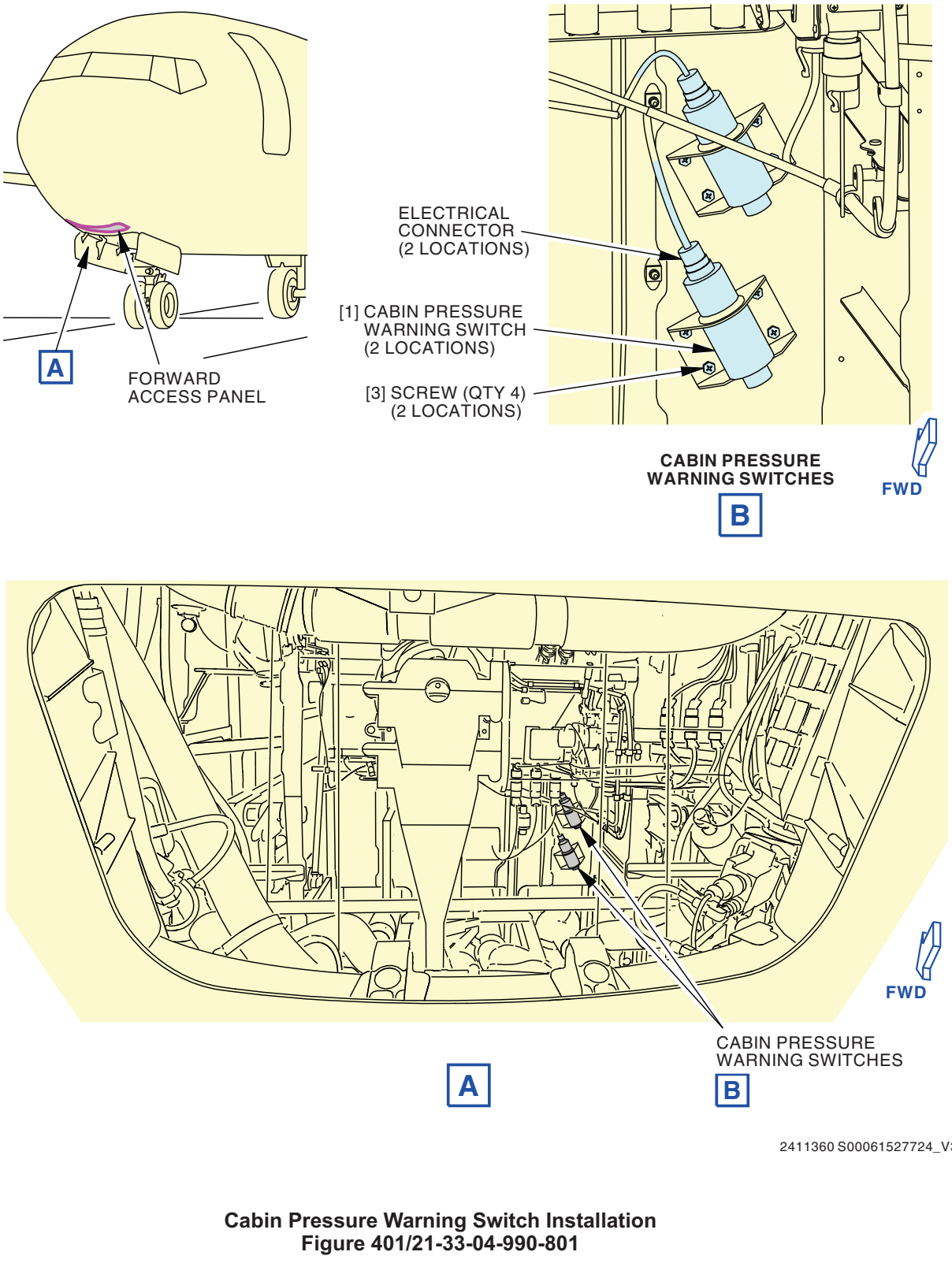
(a) Disconnect the electrical connector from the cabin pressure warning switch [1].

(b) Remove the screws [3] from the cabin pressure warning switch [1].

(c) Remove the cabin pressure warning switch [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cabin Pressure Warning Switch Installation
Figure 401/21-33-04-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-33-04-400-801

3. Cabin Pressure Warning Switch Installation

(Figure 401)

A. General

(1) This task is for the installation of the cabin pressure warning switch.

B. References

Reference	Title
20-50-11-910-801	Standard Torque Values (P/B 201)
21-33-00-700-802	Cabin Altitude Warning Test (P/B 501)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cabin pressure warning switch	21-33-04-01-015	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Cabin Pressure Warning Switch Installation

SUBTASK 21-33-04-820-001

(1) Install the cabin pressure warning switch [1] as follows:

(a) Put the cabin pressure warning switch [1] in its position.

(b) Install the screws [3] that attach the cabin pressure warning switch [1].

(c) Tighten the screws [3] to specifications (TASK 20-50-11-910-801).

(d) Connect the electrical connector to the cabin pressure warning switch [1].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-33-04-410-001

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 21-33-04-865-001

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN

H. Cabin Pressure Warning Switch Installation Test

SUBTASK 21-33-04-720-001

(1) Do this task: Cabin Altitude Warning Test, TASK 21-33-00-700-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABIN ALTIMETER AND RATE OF CLIMB MODULE - REMOVAL AND INSTALLATION

1. General

A. This procedure has these tasks:

(1) Cabin altimeter and rate of climb module removal

(2) Cabin altimeter and rate of climb module installation.

TASK 21-33-07-000-801

2. Cabin Altimeter and Rate of Climb Module Removal

Figure 401

A. General

(1) This task is for the removal of the cabin altimeter and rate of climb module.

B. References

Reference	Title
21-33-02-000-801	Cabin Altitude and Differential Pressure Indicator Removal (P/B 401)

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 21-33-07-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	10	C00328	OVERHEAD 5V SECONDARY

SUBTASK 21-33-07-000-001

(2) Loosen the 1/4-turn fasteners that hold the P5 Forward Overhead Panel in its position and let the panel rotate down.

E. Cabin Altimeter and Rate of Climb Module Removal

SUBTASK 21-33-07-020-001

(1) Remove the P5-16 Cabin Altimeter and Rate of Climb Module from the P5 Forward Overhead Panel as follows:

(a) Disconnect the electrical connector in the back of the cabin altimeter and rate of climb module.

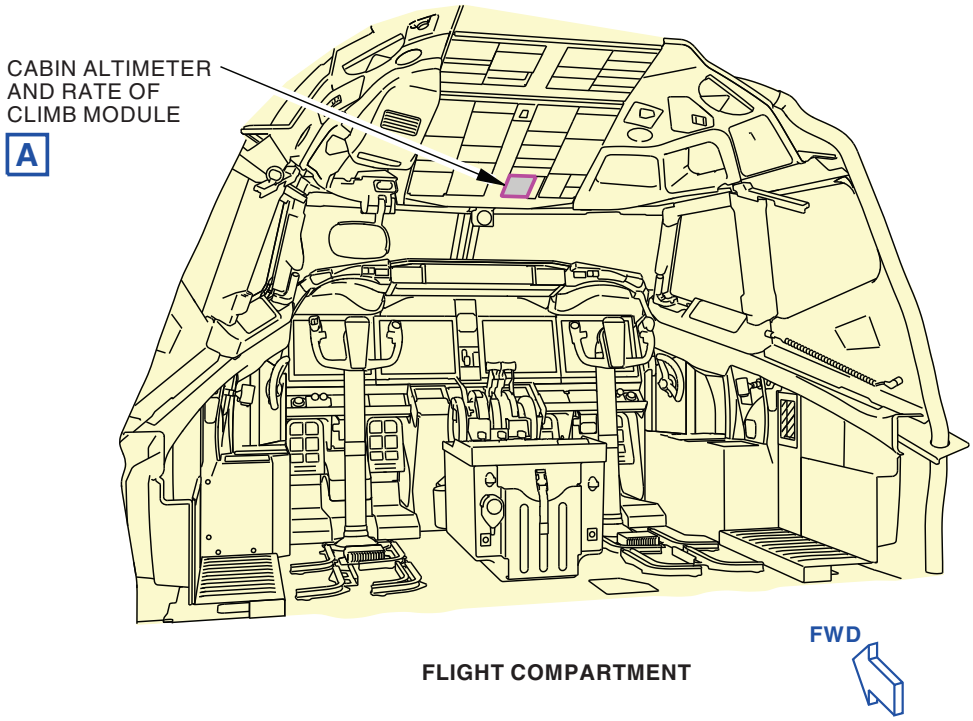
(b) Disconnect the static pressure hose from the male quick-disconnect on the back of the cabin altitude and differential pressure indicator (TASK 21-33-02-000-801).

(c) Hold the cabin altimeter and rate of climb module [1] in its position and loosen the 1/4-turn fasteners that hold the module to the P5 Forward Overhead Panel.

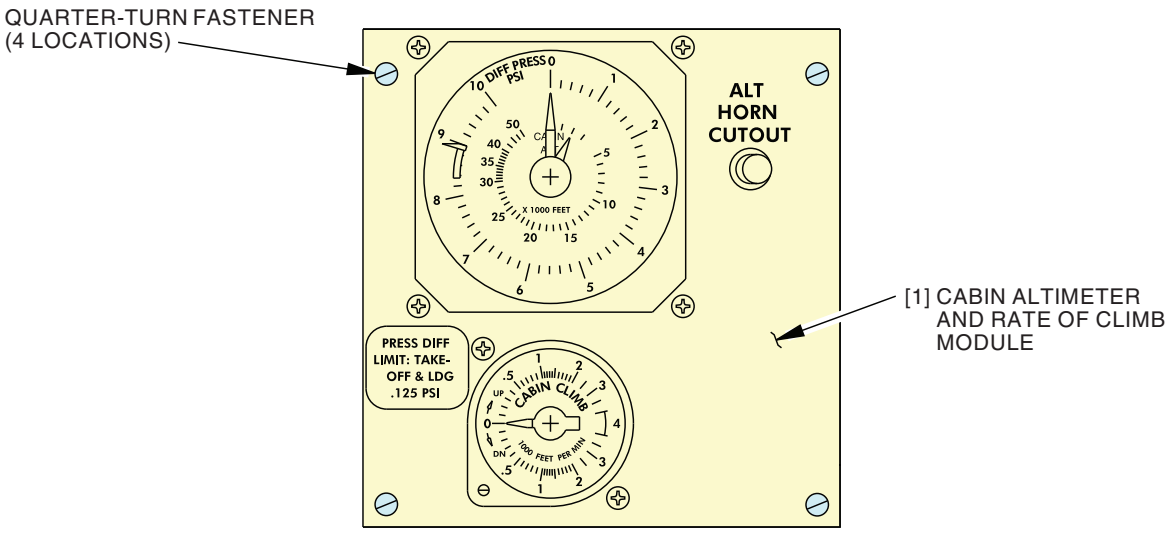
(d) Remove the cabin altimeter and rate of climb module [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



CABIN ALTIMETER AND RATE OF CLIMB MODULE
(EXAMPLE)

A

2411361 S00061527729_V4

Cabin Altimeter and Rate of Climb Module Installation
Figure 401/21-33-07-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-33-07-400-801

3. Cabin Altimeter and Rate of Climb Module Installation

Figure 401

A. References

Reference	Title
34-11-00-790-805	Alternate Static System Low-range Leak Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cabin altimeter and rate of climb module	21-33-07-01-050	PTF ALL

C. Location Zones

Zone	Area
212	Flight Compartment - Right

D. Procedure

SUBTASK 21-33-07-020-002

(1) If it has not already been done, lower the P5 Forward Overhead Panel.

SUBTASK 21-33-07-420-001

(2) Install the P5-16 Cabin Altimeter and Rate of Climb Module into the P5 Forward Overhead Panel as follows:

(a) Put the cabin altimeter and rate of climb module [1] into the P5 Forward Overhead Panel.

(b) Turn the 1/4-turn fasteners on the cabin altimeter and rate of climb module [1] to hold the module to the P5 Forward Overhead Panel.

(c) Connect the static pressure hose to the quick-disconnect at the back of the cabin altitude and differential pressure indicator.

(d) Connect the electrical connector to the cabin rate of climb indicator and the cabin altitude and differential pressure indicator at the back of the module.

SUBTASK 21-33-07-410-001

(3) Lift the P5 Forward Overhead Panel to the closed position and turn the 1/4-turn fasteners.

SUBTASK 21-33-07-720-001

(4) Do this task: Alternate Static System Low-range Leak Test, TASK 34-11-00-790-805.

NOTE: It is not necessary to do the (Alternate Static System Low-range Leak Test, TASK 34-11-00-790-805) if the replacement module has the quick-disconnect already installed on the cabin altitude and differential pressure indicator.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OVERBOARD EXHAUST VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Equipment Cooling Overboard Exhaust Valve Removal.

(2) Equipment Cooling Overboard Exhaust Valve Installation.

B. The equipment cooling overboard exhaust valve is installed in the electronic equipment center.

TASK 21-43-01-000-801

2. Overboard Exhaust Valve Removal

(Figure 401)

A. General

(1) This task provides the instructions to remove the overboard exhaust valve.

B. References

Reference	Title
12-15-21-600-804	Crew Oxygen Cylinder Replacement (P/B 301)
24-22-00-860-802	Remove Electrical Power (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 21-43-01-860-001

CAUTION

MAKE SURE YOU REMOVE ELECTRICAL POWER FROM THE AIRPLANE. IF YOU SUPPLY ELECTRICAL POWER TO THE ELECTRICAL/ELECTRONIC EQUIPMENT WHEN THE AFT EQUIPMENT COOLING SYSTEM IS NOT IN OPERATION, THE EQUIPMENT CAN BECOME TOO HOT. THIS CAN CAUSE DAMAGE TO THE ELECTRICAL/ELECTRONIC EQUIPMENT.

(1) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 21-43-01-010-001

(2) To get access to the overboard exhaust valve [1], do this step:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-43-01-010-002

(3) Remove the access panel [5] on the top of the raised platform just aft and right of the aircraft center line as follows (Figure 401):

(a) Remove the screws [4] that attach the access panel [5] to the top of the raised platform.

(b) Remove the access panel [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-43-01-010-003

(4) Remove the crew oxygen system cylinder (TASK 12-15-21-600-804).

F. Equipment Cooling Overboard Exhaust Valve Removal

SUBTASK 21-43-01-020-001

(1) Disconnect the electrical connector [2] from the overboard exhaust valve [1] (Figure 401).

SUBTASK 21-43-01-020-002

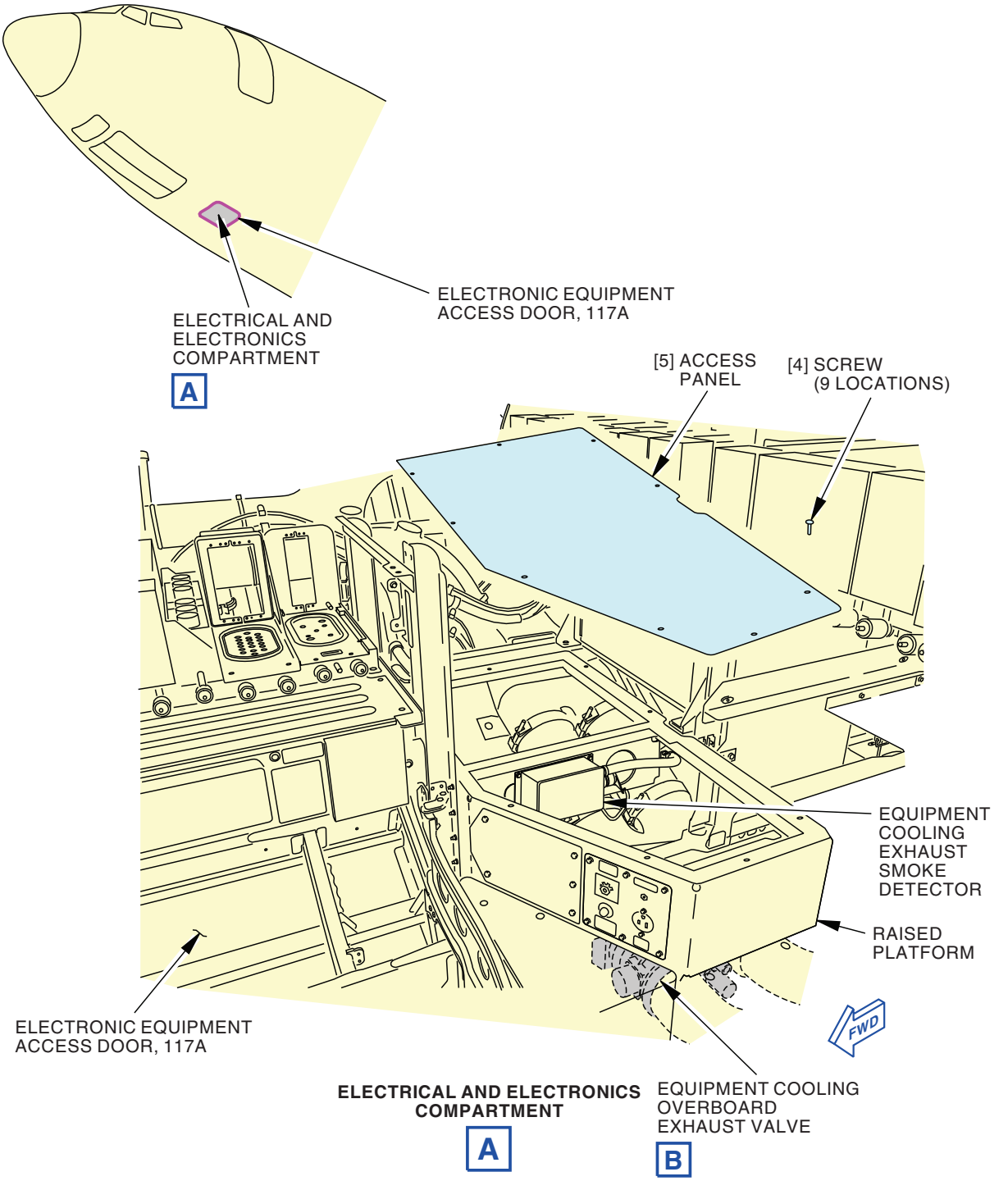
(2) Loosen the couplings [3].

SUBTASK 21-43-01-020-003

(3) Remove the overboard exhaust valve [1].

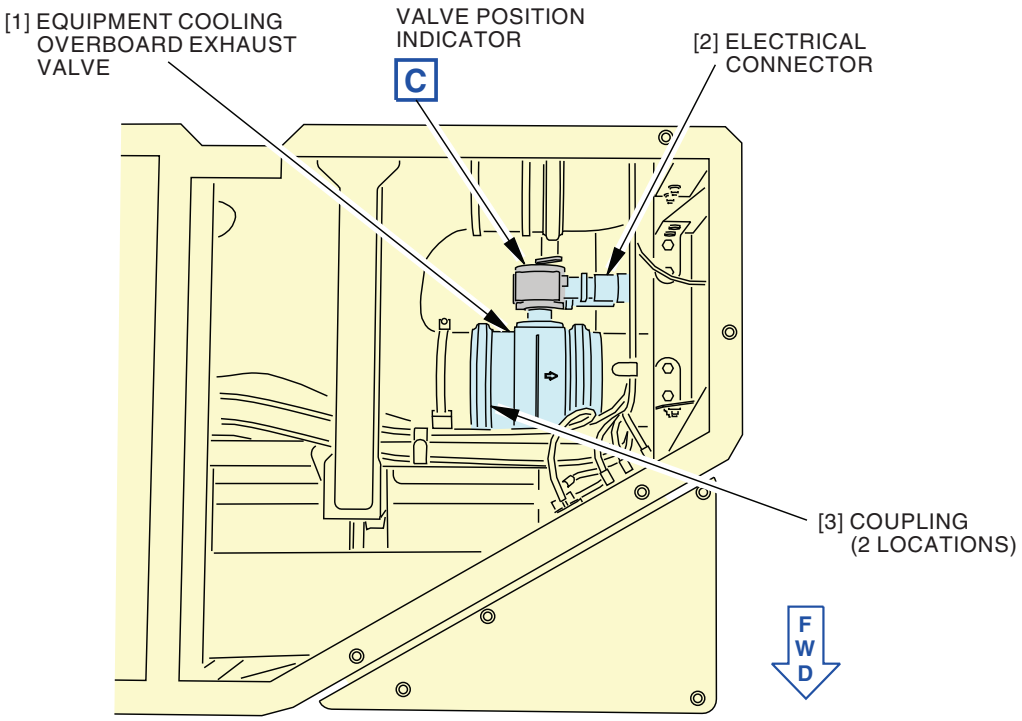
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



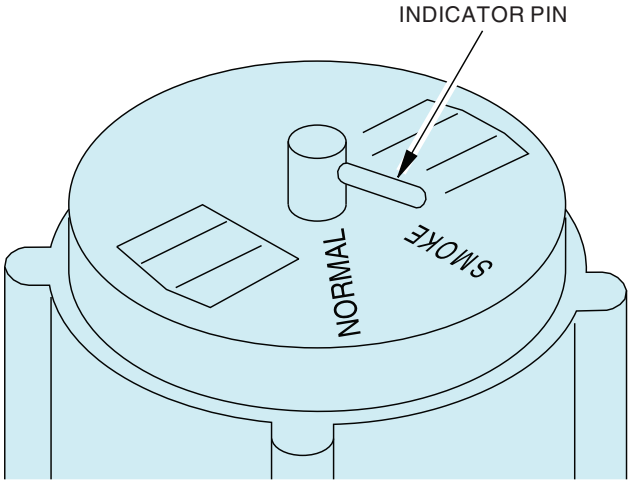
Equipment Cooling Overboard Exhaust Valve Installation
Figure 401/21-43-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EQUIPMENT COOLING OVERBOARD EXHAUST VALVE (OXYGEN BOTTLE INSTALLATION NOT SHOWN)

B



VALVE POSITION INDICATOR

C

NOTE:

EQUIPMENT COOLING EXHAUST SMOKE DETECTOR REMOVED FOR CLARITY

2411321 S00061527618_V2

Equipment Cooling Overboard Exhaust Valve Installation
Figure 401/21-43-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-43-01-400-801

3. Overboard Exhaust Valve Installation

(Figure 401)

A. References

Reference	Title
12-15-21-600-804	Crew Oxygen Cylinder Replacement (P/B 301)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Overboard exhaust valve	21-27-05-01-015	PTF ALL

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Exhaust Valve Installation

SUBTASK 21-43-01-420-001

(1) Put the overboard exhaust valve [1] in its position.

SUBTASK 21-43-01-420-002

(2) Tighten the couplings [3].

SUBTASK 21-43-01-420-003

(3) Connect the electrical connector [2] to the overboard exhaust valve [1].

F. Equipment Cooling Overboard Exhaust Valve Installation Test

SUBTASK 21-43-01-860-002

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 21-43-01-860-003

(2) Operate the air conditioning packs (TASK 21-00-00-800-803).

SUBTASK 21-43-01-860-004

(3) Make sure that these switches, on the P5 Forward Overhead Panel, are in the AUTO position:

(a) L PACK

(b) R PACK

(c) L RECIRC FAN

(d) R RECIRC FAN

SUBTASK 21-43-01-860-005

(4) Make sure that these switches, on the P5 Forward Overhead Panel, are in the NORMAL position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) EQUIP COOLING, EXHAUST

(b) EQUIP COOLING, SUPPLY

SUBTASK 21-43-01-790-001

(5) Make sure that there is no air leakage at thecouplings [3].

SUBTASK 21-43-01-860-006

(6) Make sure that the overboard exhaust valve [1] is in the NORMAL position (Figure 401).

NOTE: Look at the indicator on the valve actuator to see the position of the valve.

SUBTASK 21-43-01-860-007

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(7) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-43-01-860-008

(8) Put these switches, on the P5 Forward Overhead Panel, to the HIGH position:

(a) L PACK

(b) R PACK

SUBTASK 21-43-01-860-009

(9) Put the R RECIRC FAN switch, on the P5 Forward Overhead Panel, to the OFF position.

SUBTASK 21-43-01-860-010

(10) Make sure that the overboard exhaust valve [1] is in the SMOKE position (Figure 401).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 21-43-01-860-011

(1) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-43-01-860-012

(2) Shutdown the air conditioning packs (TASK 21-00-00-800-804).

SUBTASK 21-43-01-860-013

(3) Put these switches, on the P5 Forward Overhead Panel, in the AUTO position:

(a) L PACK

(b) R PACK

(c) L RECIRC FAN

(d) R RECIRC FAN

SUBTASK 21-43-01-420-004

(4) Install the crew oxygen system cylinder (TASK 12-15-21-600-804).

SUBTASK 21-43-01-410-001

(5) Install the access panel [5] on the top of the raised platform that you removed as follows:

(a) Install the access panel [5].

(b) Install the screws [4] that attach the access panel [5] to the top of the raised platform.

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SUBTASK 21-43-01-410-002

(6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DOOR AREA HEATING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An operational test of the door area heaters.

(2) An operational test of the emergency exit hatch heaters.

NOTE: The emergency exit hatch heaters have over-temperature thermostats. The thermostats will turn the heaters off very shortly after they come on when the airplane is on the ground, in the air mode, with an air conditioning pack on. You must have a minimum of two persons to do this test correctly.

B. The door area and emergency exit hatch heaters operate when these conditions are met:

(1) The airplane is in the air.

(2) One or both of the air conditioning packs are on.

C. There are five heater blankets associated with each emergency exit hatch. The heaters are installed as follows:

(1) There are two heaters behind the lining of the emergency exit hatch.

(2) There are two heaters for the emergency exit hatch doorway. There is a heater behind the forward and the aft surround panels.

(3) There is a heater on the outboard side of the closeout panel of emergency exit hatch.

TASK 21-45-00-700-801

2. Door Area Heater Operational Test

A. General

(1) This task will test the operation of the door area heater.

B. References

Reference	Title
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

D. Prepare for the Test

SUBTASK 21-45-00-860-012

(1) Make sure that the CAB/UTIL switch, on the P5-13 electrical meters, battery and galley power panel, is set to the ON position.

SUBTASK 21-45-00-860-013

(2) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

E. Door Area Heater Operational Test

SUBTASK 21-45-00-200-005

(1) Feel the airflow from the door heater outlet.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the airflow from the door heater outlet is cool.

SUBTASK 21-45-00-860-015

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-45-00-860-041

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	11	C01274	AIR CONDITIONING DOOR AREA HEAT CONT

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	7	C01279	DOOR AREA HTR-AFT
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 21-45-00-200-006

(4) Feel the airflow from the door heater outlet.

(a) Make sure that the air that exits the outlet becomes warmer.

SUBTASK 21-45-00-860-016

(5) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-45-00-840-001

(1) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

——— END OF TASK ———

TASK 21-45-00-700-802

3. Emergency Exit Hatch Heaters Operational Test

A. References

Reference	Title
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
21-00-00-800-804	Remove Conditioned Air Supplied by a Cooling Pack (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

B. Consumable Materials

Reference	Description	Specification
G02319	Spray - Freeze - Miller-Stephenson MS-242N	

C. Location Zones

Zone	Area
212	Flight Compartment - Right
832	Left Forward Emergency Exit

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
833	Left Emergency Exit (STA 627.5)
842	Right Forward Emergency Exit
843	Right Emergency Exit (STA 627.5)

D. Prepare for the Test

NOTE: The emergency exit hatch heaters have over-temperature thermostats. The thermostats will turn the heaters off very shortly after they come on when the airplane is on the ground, in the air mode, with an air conditioning pack on. The exit hatch heaters will not operate when the temperature is more than approximately 70 ±5°F (21 ±3°C) (when the overheat thermostat is closed) or 50 ±7°F (10 ±4°C) (when the overheat thermostat is open due to a previous overheat condition). In warm temperatures, the heater thermostats may need to be cooled before the heaters will operate.

SUBTASK 21-45-00-860-018

(1) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

SUBTASK 21-45-00-200-015

(2) Make sure that the temperature in the area of the associated thermostats is low enough to make the heaters operate during the test (Figure 501).

- 70°F (21°C) or less for the hatch heaters and trim heaters.
- 80°F (27°C) or less for the closeout panel heater.

(a) If necessary, remove any lining required for access to cool the thermostats during the heater test.

E. Emergency Exit Hatch Heater Operational Test

SUBTASK 21-45-00-200-016

(1) For the applicable heater(s), check the adjacent surface temperature as follows (Figure 501):

(a) Feel the surface of the emergency exit hatch lining along the vertical edges.

1) Make sure that the temperature of the lining is approximately the same as the ambient air temperature.

(b) Feel the surface of the trim strips of the emergency exit hatch doorway.

1) Make sure that the temperature of the trim strips is approximately the same as the ambient air temperature.

(c) Feel the surface of the closeout panel.

1) Make sure that the temperature of the closeout panel is approximately the same as the ambient air temperature.

SUBTASK 21-45-00-860-020

(2) Do this task: Remove Conditioned Air Supplied by a Cooling Pack, TASK 21-00-00-800-804.

SUBTASK 21-45-00-860-021

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-45-00-860-042

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	11	C01274	AIR CONDITIONING DOOR AREA HEAT CONT

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 21-45-00-860-022

(5) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

(a) Have one person turn the air conditioning pack on. Have one or more persons check the operation of the emergency exit hatch heaters.

SUBTASK 21-45-00-200-013

(6) If necessary, cool the associated control thermostat and overheat thermostat to make the heater operate.

- 70°F (21°C) or less for the hatch heaters and trim heaters.
- 80°F (27°C) or less for the closeout panel heater.

NOTE: The time to cool the thermostats and the amount of cooling required will vary depending on the ambient temperature.

(a) Check the heater quickly after cooling the thermostats.

NOTE: The thermostats will shut the hatch and trim heaters down when the heaters reach 80°F (27°C). The closeout heater will shut down at 100°F (38°C).

(b) Cool air source MS-242N spray, G02319 or equivalent can be used to cool the thermostats.

SUBTASK 21-45-00-200-014

(7) Make sure that the applicable heaters operate as follows (Figure 501):

(a) Feel the surface of the emergency exit hatch lining along the vertical edges.

1) Make sure that you feel the warmth from the heater.

(b) Feel the surface of the trim strips for the emergency exit hatch doorway.

1) Make sure that you feel the warmth from the heater.

(c) Feel the surface of the closeout panel.

1) Make sure that you feel the warmth from the heater.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-45-00-860-023

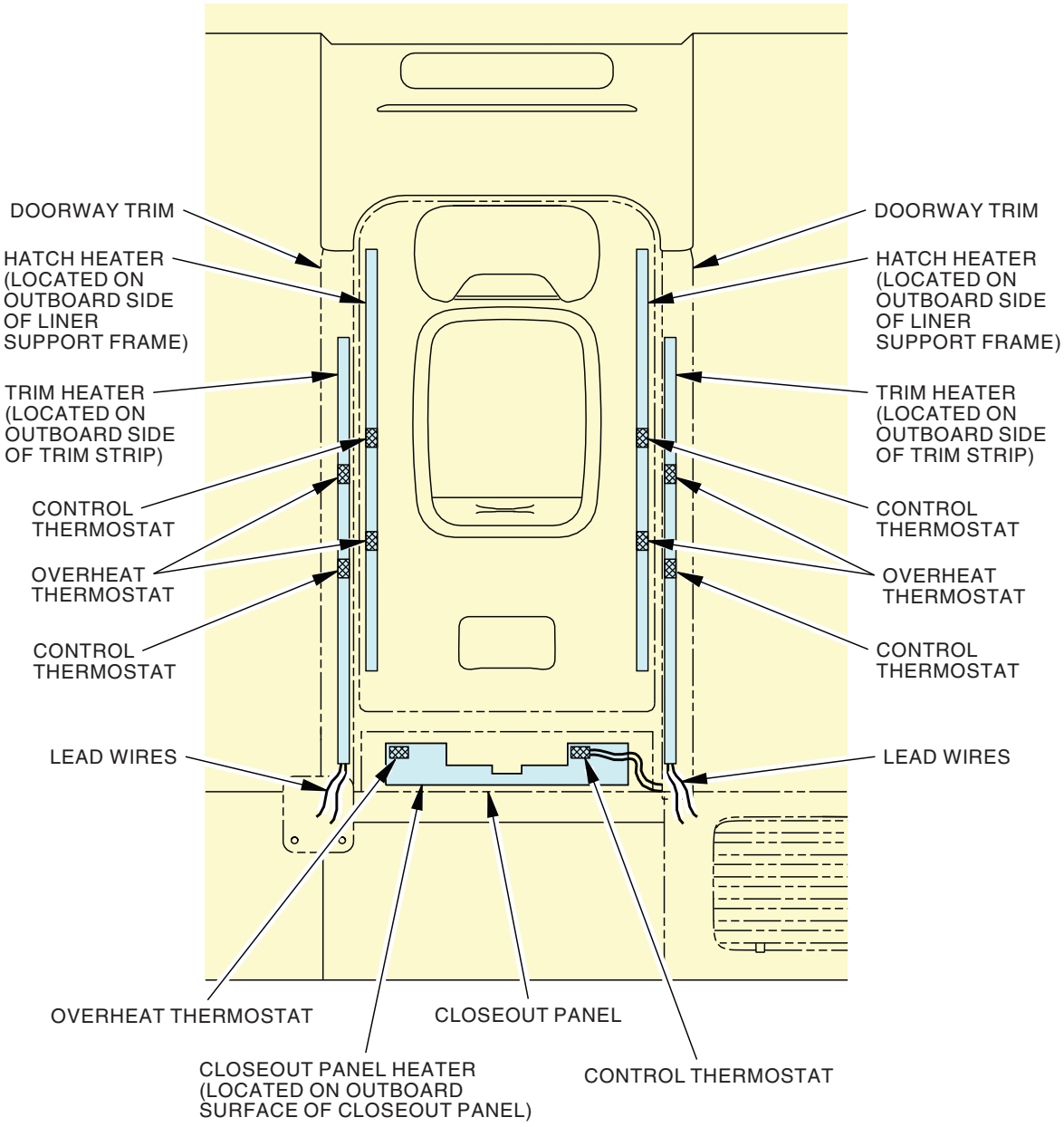
(1) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-45-00-840-003

(2) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Exit Hatch Heater Locations
Figure 501/21-45-00-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DOOR AREA HEATER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the door area heater

(2) An installation of the door area heater.

B. The door area heaters operate when these conditions are met:

(1) The airplane is in the air.

(2) One or two of the air conditioning packs are on.

TASK 21-45-01-000-801

2. Door Area Heater Removal

(Figure 401 or Figure 402)

A. General

(1) This task is for the removal of the forward or mid or aft door area heater.

B. References

Reference	Title
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
117	Electrical and Electronics Compartment - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
113BW	Forward Nose Wheel Well Panel
117A	Electronic Equipment Access Door

F. Prepare for the Removal

SUBTASK 21-45-01-840-001

(1) Do these steps to make sure that conditioned air is not supplied to the passenger compartment:

NOTE: This will make sure that there is no airflow in the ducts when the door area heater is removed.

(a) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

NOTE: Do not operate a ground conditioned air source.

(b) Make sure that the L RECIRC FAN and R RECIRC FAN switches, on the P5-10 air conditioning panel, are set to the OFF position.

SUBTASK 21-45-01-010-001

(2) If the forward door area heater [1] will be removed, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) To get access to the P91 and P92 panels, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(b) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

(c) Open this access panel:

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

SUBTASK 21-45-01-010-002

(3) If the aft door area heater [1B] will be removed, do these steps:

(a) To get access to the P91 and P92 panels, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

> **WARNING** DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(b) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	7	C01279	DOOR AREA HTR-AFT

(c) At the forward attendant panel, set the switch for the entry lights to the OFF position and attach the DO NOT OPERATE tag, STD-858.

> **WARNING** DO NOT TOUCH THE SOCKETS OR THE METAL ENDS OF THE LAMP. SHOCK CAN OCCUR WHEN YOU SUPPLY ELECTRICAL POWER TO THE LIGHT.

(d) Lower the aft side of the aft entry light lens assembly.

G. Door Area Heater Removal

SUBTASK 21-45-01-020-008

(1) If the forward door area heater [1] will be removed, do these steps:

(a) Disconnect the electrical connector [2] from the forward door area heater [1].

(b) Loosen the clamps [3] and clamp [5].

(c) Move the clamps [3] and clamp [5] away from the forward door area heater [1] and onto the flex ducts.

(d) Remove and discard the tape from the insulation [4].

(e) Remove the insulation [4] from the forward door area heater [1].

NOTE: If there is no damage, keep the insulation for re-installation.

(f) Remove the forward door area heater [1] from the flex ducts.

SUBTASK 21-45-01-020-002

(2) If the aft door area heater [1B] will be removed, do these steps:

(a) Disconnect the electrical connector [2] from the aft door area heater [1B].

(b) Remove the nut [6], washer [7], and wire bundle mounting plate [8].

(c) Loosen the duct support clamps [9] from around the insulation [4].

(d) Remove and discard the tape from the insulation [4].

(e) Remove the insulation [4] from the aft door area heater [1B].

NOTE: If there is no damage, keep the insulation for re-installation.

(f) Loosen the clamp [3A] and clamp [5].

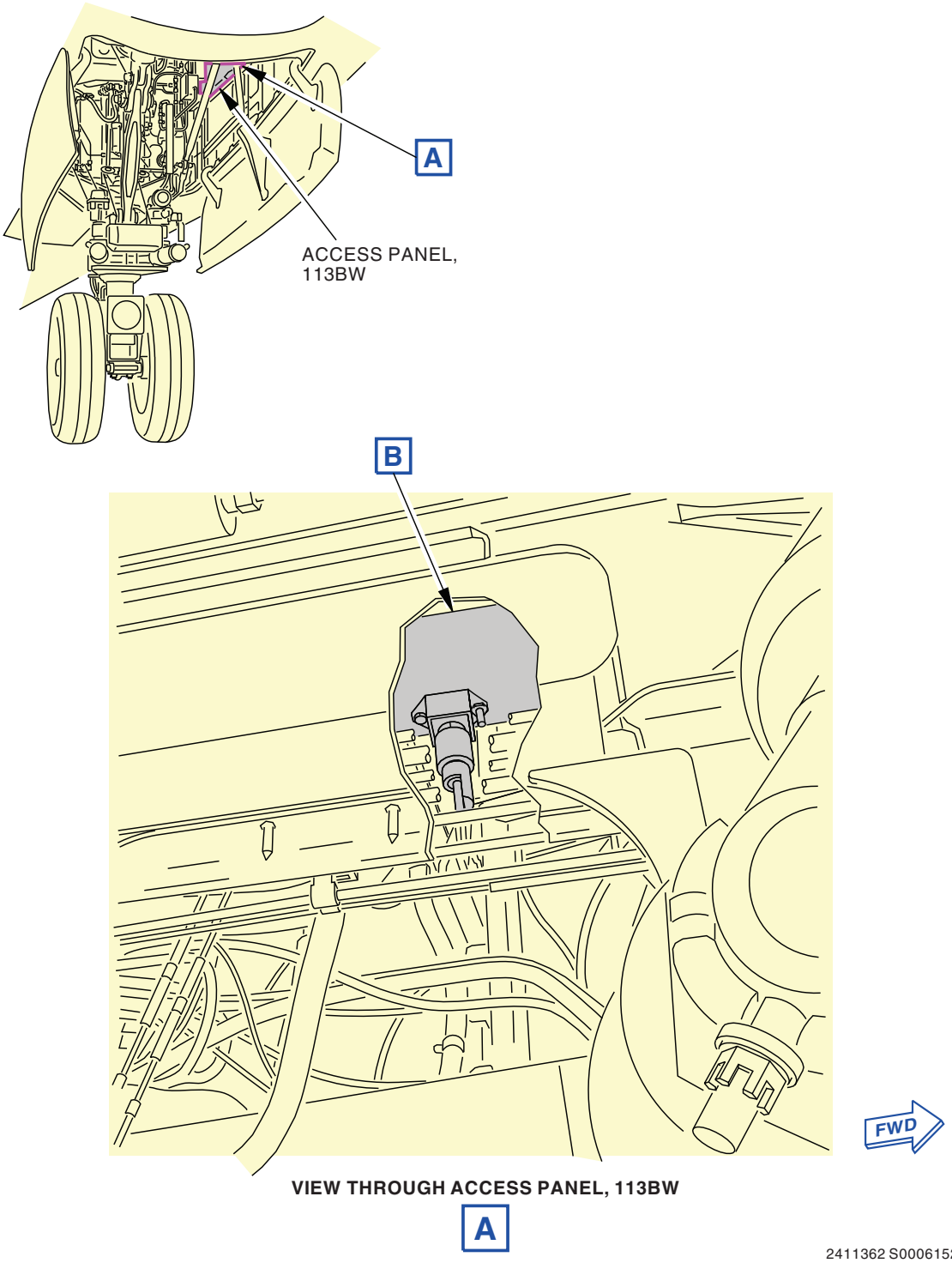
(g) Move the clamp [3A] and clamp [5] away from the aft door area heater [1B] and onto the flex ducts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Remove the aft door area heater [1B] from the flex ducts.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



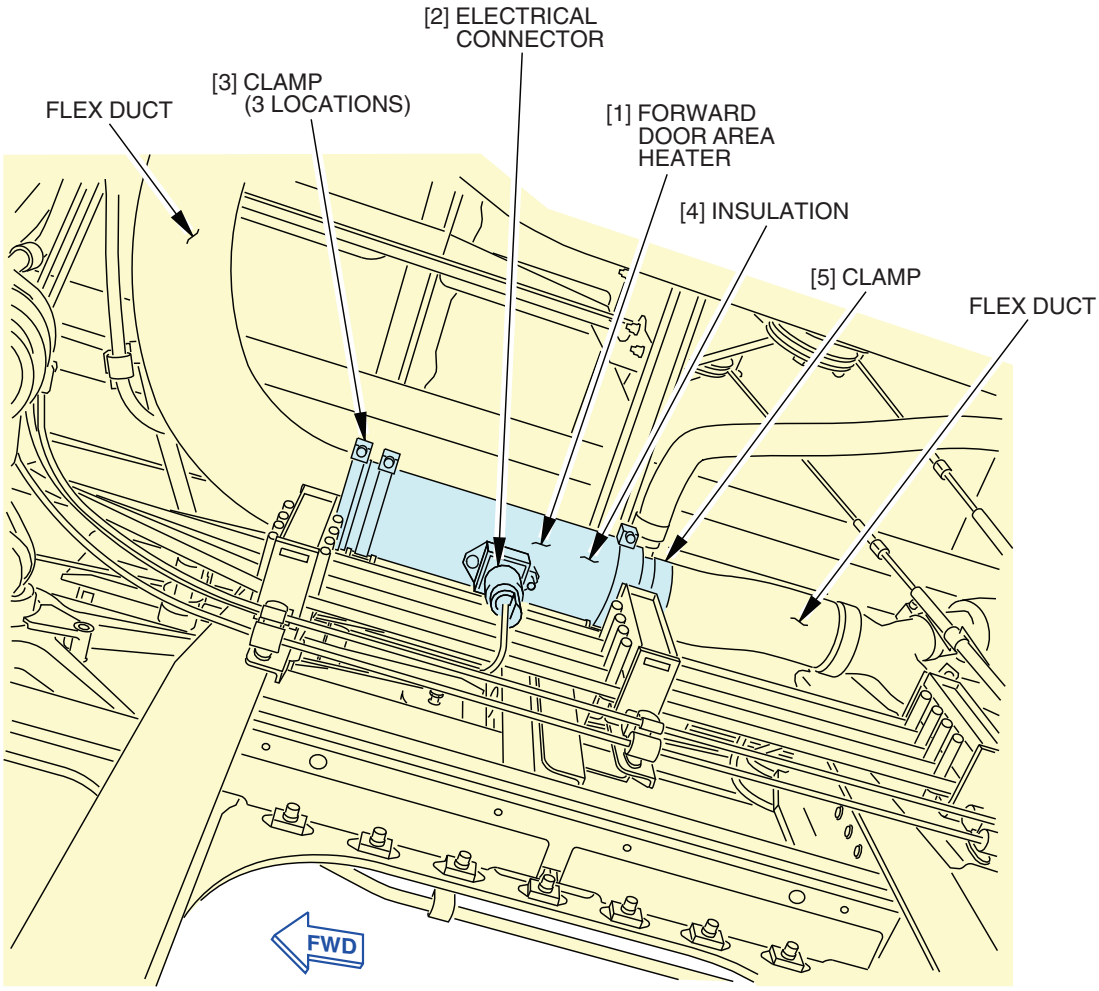
VIEW THROUGH ACCESS PANEL, 113BW

A

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Forward Door Area Heater Installation
Figure 401/21-45-01-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



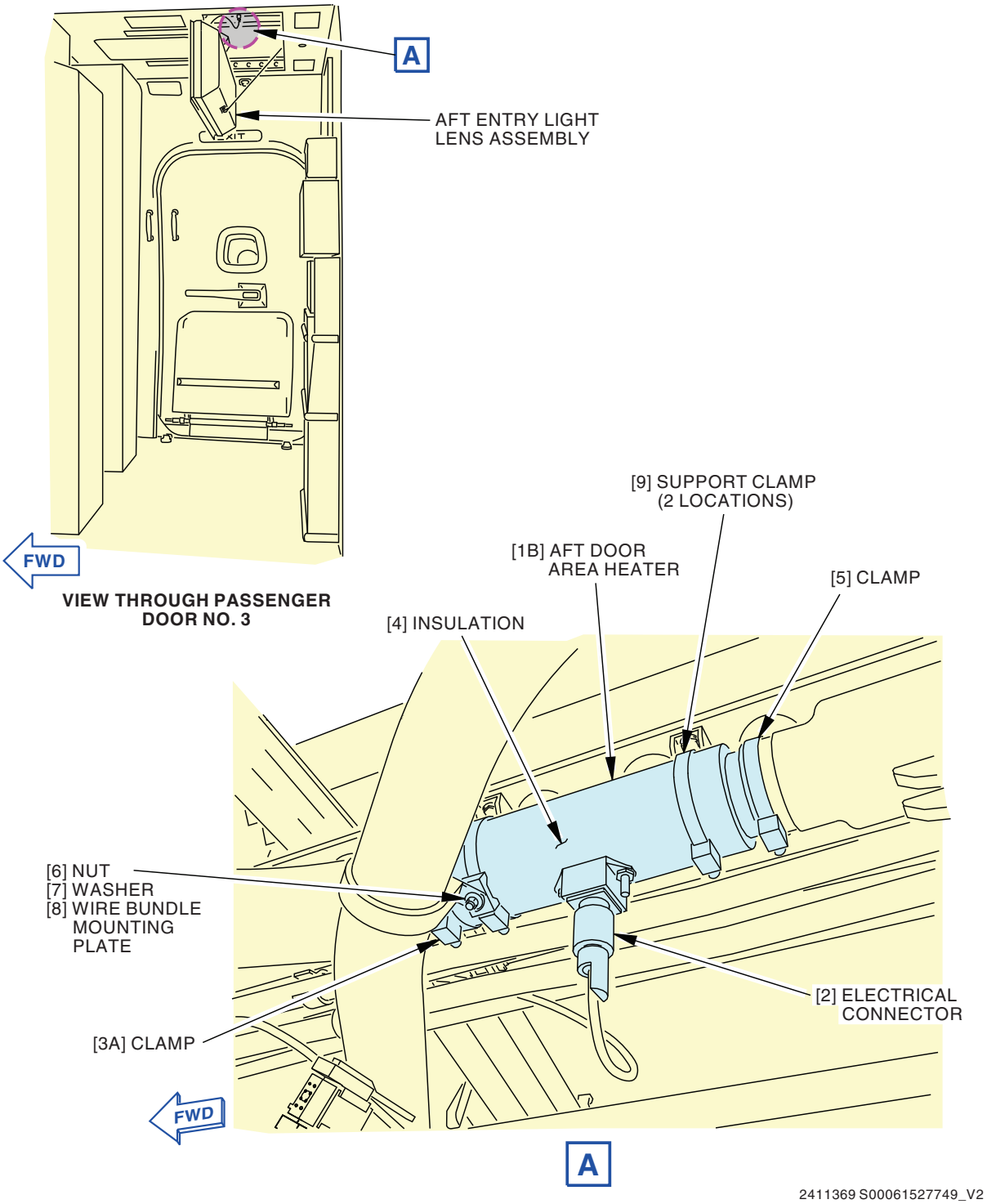
VIEW THROUGH ACCESS PANEL, 113BW

B

2411368 S00061527748_V2

Forward Door Area Heater Installation
Figure 401/21-45-01-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Door Area Heater Installation
Figure 402/21-45-01-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-45-01-400-801

3. Door Area Heater Installation

(Figure 401 or Figure 402)

A. General

(1) Some of the ducts and equipment in the air distribution system are wrapped with insulation material. CFR 14 Part 121.312 mandates that insulation material installed in the fuselage as a replacement after September 2, 2005 must meet the flame propagation requirements of CFR Part 25.856, effective September 2, 2003.

(2) If the existing insulation material is not damaged, it can be removed and re-installed. Replacement insulation material must meet the flame propagation requirement.

(3) Melamine foam insulation, G50449, BMS8-385, is the preferred foam insulation material that meets the flame propagation requirement. BMS8-385 Type 4 is the preferred foam insulation material that meets the flame propagation requirement. BMS 8-385 Type 6 and BMS 8-300 Type VIII insulation is optional to BMS8-385 Type 4.

(4) Air conditioning ducts with polyimide foam insulation, G02470, must be wrapped with tape, G50327, tape, G51202, or tape, G50789, to meet the flammability/flame propagation requirements. Wrap the tape, G50327, tape, G51202, or tape, G50789, across 100% of the brown hypalon insulation surface on the polyimide foam insulation, G02470.

(5) Do not use BMS8-39 polyurethane foam insulation for repairs due to the degradation in flammability properties over time.

(6) When insulation material is removed and re-installed, any existing tape that does not meet the flame propagation requirement must be removed or completely covered with tape that does meet the requirement.

(a) Tape specifications BMS 5-157, Type I, Class 1, Grade B, composition MPVF and BMS 5-157, Type I, Class 2, Grade C, composition MPEEK are qualified tapes used to secure insulation to ducts.

B. References

Reference	Title
21-00-00-800-802	Remove Conditioned Air from the Airplane (P/B 201)
21-00-00-800-803	Supply Conditioned Air with a Cooling Pack (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G02470	Foam - Flexible Polyimide	BMS8-300 Type I
G50327	Tape - Advanced Insulation Blanket (use until stock depleted)	BMS5-157 Type I Class 1 Grade B Composition MPVF
G50449	Foam - Flexible Melamine	BMS8-385 Type 4 Grade 1
G50789	Tape - Adhesive Insulation Blanket	BMS5-157 Type I Class 2 Grade C Composition MPEEK
G51202	Tape - Advanced Insulation Blanket	BMS5-157 Type I Class 1 Grade A Composition MPVF

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward door area heater	21-45-01-01-150	PTF ALL
1B	Aft door area heater	21-45-01-07-045	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
117	Electrical and Electronics Compartment - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

F. Access Panels

Number	Name/Location
113BW	Forward Nose Wheel Well Panel
117A	Electronic Equipment Access Door

G. Door Area Heater Installation

SUBTASK 21-45-01-420-001

(1) Install the forward door area heater [1] between the flex ducts as follows:

(a) Put the forward door area heater [1] in its position with the electrical connector [2] pointed down.

NOTE: Point the flow arrow on the heater in the same direction as the flow arrow on the support structure.

(b) Install the free ends of the forward and aft flex ducts on the forward door area heater [1] as follows:

NOTE: Route the flex ducts such that the duct length that is not necessary is taken up.

1) Use large radius bends to prevent a kink or turn of the ducts.

2) Compress the flex duct between the support clamps.

(c) Make sure that the flex ducts overlap the ends of the heater duct 1.25 ±0.13 in. (31.75 ±3.30 mm).

(d) Move the clamps [3] and clamp [5] into their positions 0.38 ±0.13 in. (9.65 ±3.30 mm) from the ends of the flex ducts.

(e) Tighten the clamps [3] and clamp [5] to 5 in-lb (0.6 N·m) - 6 in-lb (0.7 N·m).

(f) Close this access panel:

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

SUBTASK 21-45-01-420-002

(2) Install the aft door area heater [1B] between the flex ducts as follows:

(a) Put the aft door area heater [1B] in its position with the electrical connector [2] pointed down.

NOTE: Point the flow arrow on the heater in the same direction as the flow arrow on the support structure.

(b) Install the free ends of the forward and aft flex ducts on the aft door area heater [1B] until the ends of the flex ducts overlap the heater duct by 1.25 ±0.13 in. (31.75 ±3.30 mm).

(c) Move the clamp [3A] and clamp [5] into their positions 0.38 ±0.13 in. (9.65 ±3.30 mm) from the ends of the heater duct.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Tighten the clamp [3A] and clamp [5] to 5 in-lb (0.6 N·m) - 6 in-lb (0.7 N·m).

(e) Install new tape, G50327, tape, G51202, or tape, G50789, on the seam of the insulation [4].

(f) Install the insulation [4] on the aft door area heater [1B].

(g) Install the duct support clamps [9] around the insulation [4] on the aft door area heater [1B].

(h) Connect the electrical connector [2] to the aft door area heater [1B].

(i) Close the aft entry light lens assembly.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 21-45-01-840-002

(1) Close the applicable circuit breaker as follows:

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(a) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	7	C01279	DOOR AREA HTR-AFT
D	9	C01280	DOOR AREA HTR-FWD

(b) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-45-01-860-006

(2) Make sure that the L RECIRC FAN and R RECIRC FAN switches, on the P5-10 bleed control air panel, are set to the AUTO position.

I. Door Area Heater Installation Test

SUBTASK 21-45-01-860-002

(1) Make sure that the CAB/UTIL switch, on the P5-13 electrical meters, battery and galley power panel, is set to the ON position.

SUBTASK 21-45-01-860-003

(2) Do this task: Supply Conditioned Air with a Cooling Pack, TASK 21-00-00-800-803.

SUBTASK 21-45-01-700-001

(3) Make sure that the airflow from the door heater outlet is cool.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-45-01-860-004

(4) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

SUBTASK 21-45-01-700-002

(5) Make sure that the airflow from the door heater outlet becomes warmer.

SUBTASK 21-45-01-860-005

(6) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 21-45-01-840-003

(7) Do this task: Remove Conditioned Air from the Airplane, TASK 21-00-00-800-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMERGENCY EXIT DOOR HEATER SYSTEM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the emergency exit door heater blanket

(2) An installation of the emergency exit door heater blanket.

(3) A removal of the door surround panel heater blanket of the emergency exit door

(4) An installation of the door surround panel heater blanket of the emergency exit door.

(5) A removal of the closeout panel heater blanket of the emergency exit door

(6) A installation of the closeout panel heater blanket of the emergency exit door.

B. There are two heater blankets in each emergency exit door. There is one heater blanket behind the closeout panel of each emergency exit door doorway. There is a heater blanket behind the forward and aft door surround panel of each emergency exit door.

C. The emergency exit door heater blankets operate when these conditions are met:

(1) The airplane is in the air

(2) One or two of the air conditioning packs are on.

TASK 21-45-02-000-801

2. Emergency Exit Door Heater Blanket Removal

(Figure 401)

A. References

Reference	Title
25-21-20-000-801	Emergency Exit Doorway Lining Removal (P/B 401)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
212	Flight Compartment - Right
832	Left Forward Emergency Exit
833	Left Emergency Exit (STA 627.5)
842	Right Forward Emergency Exit
843	Right Emergency Exit (STA 627.5)

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 21-45-02-010-014

(1) To get access to the P91 and P92 panels, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-45-02-840-001

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 21-45-02-010-001

(3) Open the emergency exit door.

E. Emergency Exit Door Heater Blanket Removal

SUBTASK 21-45-02-010-002

CAUTION DO NOT APPLY TENSION TO THE WIRE BUNDLE BEHIND THE LINING. DAMAGE TO THE WIRE BUNDLE CAN OCCUR.

(1) Remove the lining on the emergency exit door (TASK 25-21-20-000-801).

SUBTASK 21-45-02-020-001

(2) Disconnect the electrical connectors from the heater blanket [1].

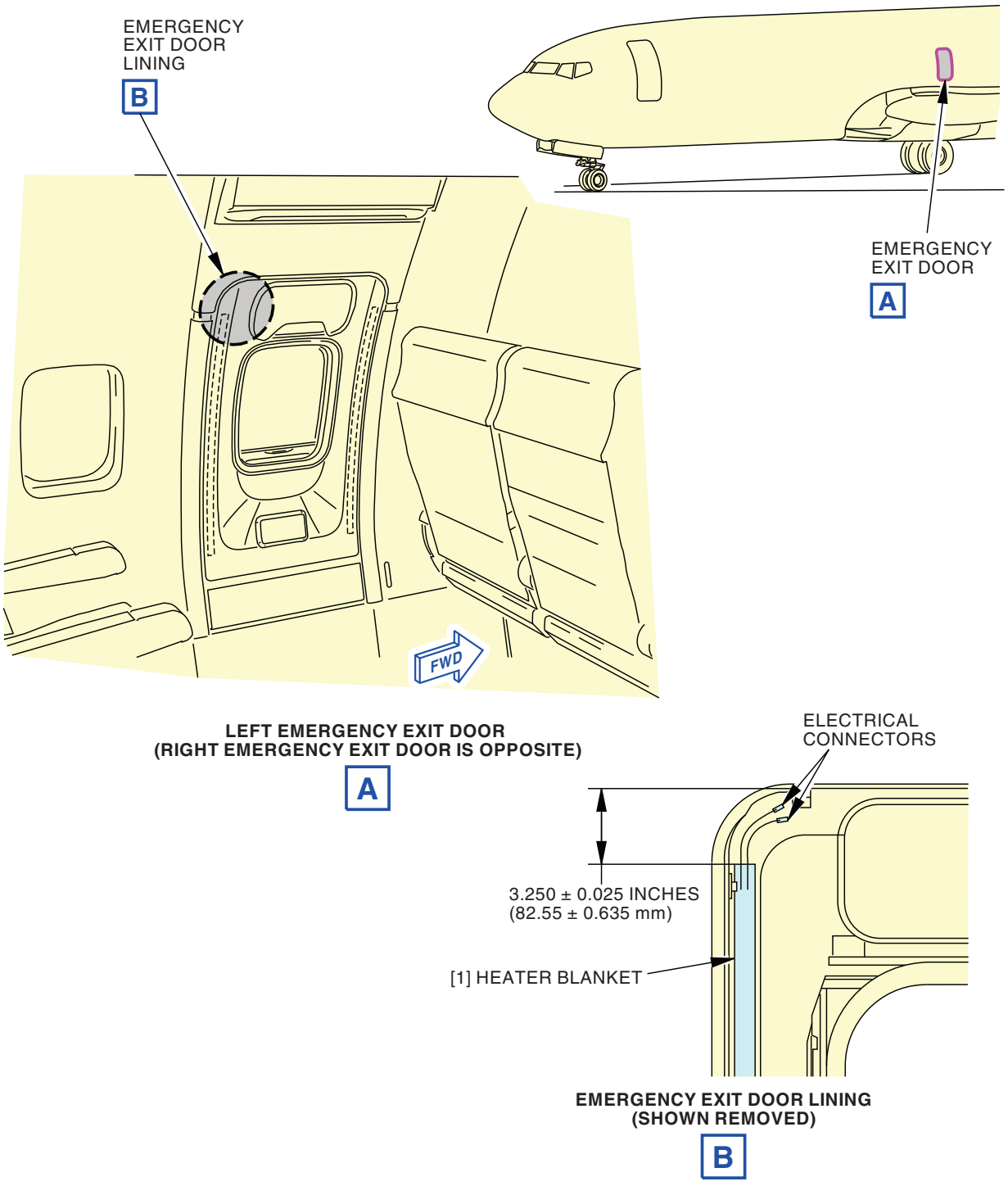
SUBTASK 21-45-02-020-002

(3) Remove the heater blanket [1].

NOTE: The heater blanket is attached to the lining with adhesive.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Emergency Exit Door Heater Blanket Installation
Figure 401/21-45-02-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-45-02-400-801

3. Emergency Exit Door Heater Blanket Installation

(Figure 401)

A. References

Reference	Title
21-45-00-700-802	Emergency Exit Hatch Heaters Operational Test (P/B 501)
25-21-20-400-801	Emergency Exit Doorway Lining Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Heater blanket	52-22-51-01-165	PTF ALL
		52-22-51-03-257	PTF ALL
		52-22-51-03-280	PTF ALL
		52-22-51-04-260	PTF ALL
		52-22-51-04-300	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
212	Flight Compartment - Right
832	Left Forward Emergency Exit
833	Left Emergency Exit (STA 627.5)
842	Right Forward Emergency Exit
843	Right Emergency Exit (STA 627.5)

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Emergency Exit Door Heater Blanket Installation

SUBTASK 21-45-02-110-001

(1) Clean the lining assembly with solvent, B00083, to remove all the adhesive.

SUBTASK 21-45-02-420-001

(2) Install the heater blanket [1] 3.250 ±0.025 in. (82.6 ±0.7 mm) from the top of the lining assembly.

(a) Make sure that there are no air bubbles between the heater and lining assembly.

SUBTASK 21-45-02-020-003

(3) Connect the electrical connectors for the heater blanket [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-45-02-010-003

CAUTION

DO NOT APPLY TENSION TO THE WIRE BUNDLE BEHIND THE LINING. DAMAGE TO THE WIRE BUNDLE CAN OCCUR.

(4) Install the lining on the emergency exit door (TASK 25-21-20-400-801).

SUBTASK 21-45-02-010-004

(5) Close the emergency exit door.

SUBTASK 21-45-02-840-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 21-45-02-410-001

(7) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-45-02-710-001

(8) Do an operational test of the emergency exit hatch heaters (TASK 21-45-00-700-802).

——— END OF TASK ———

TASK 21-45-02-000-802

4. Emergency Exit Door Surround Panel Heater Blanket Removal

(Figure 402)

A. References

Reference	Title
25-21-20-000-801	Emergency Exit Doorway Lining Removal (P/B 401)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
832	Left Forward Emergency Exit
833	Left Emergency Exit (STA 627.5)
842	Right Forward Emergency Exit
843	Right Emergency Exit (STA 627.5)

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 21-45-02-010-013

(1) To get access to the P91 and P92 panels, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-45-02-840-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 21-45-02-010-005

(3) Open the emergency exit door.

E. Emergency Exit Door Surround Panel Heater Blanket Removal

SUBTASK 21-45-02-010-006

CAUTION

DO NOT APPLY TENSION TO THE WIRE BUNDLE BEHIND THE TRIM STRIP. DAMAGE TO THE WIRE BUNDLE CAN OCCUR.

(1) Remove the door surround panel from the emergency exit door doorway (TASK 25-21-20-000-801).

SUBTASK 21-45-02-020-004

(2) Disconnect the electrical connectors from the heater blanket [21].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

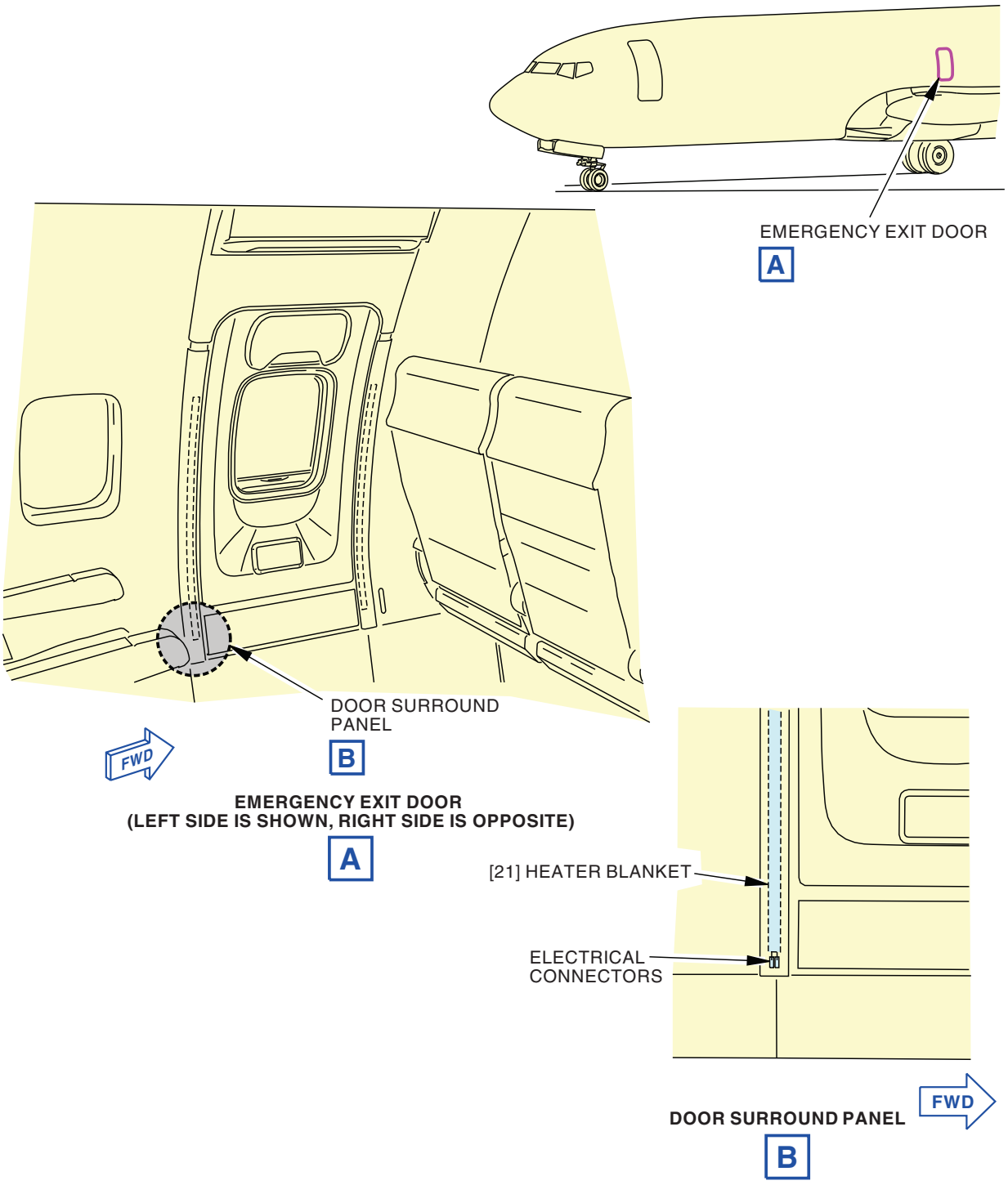
SUBTASK 21-45-02-020-005

(3) Remove the heater blanket [21].

NOTE: The heater blanket is attached to the trim strip with adhesive.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Emergency Exit Door Surround Panel Heater Blanket Installation
Figure 402/21-45-02-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-45-02-400-802

5. Emergency Exit Door Surround Panel Heater Blanket Installation

(Figure 402)

A. References

Reference	Title
21-45-00-700-802	Emergency Exit Hatch Heaters Operational Test (P/B 501)
25-21-20-400-801	Emergency Exit Doorway Lining Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Heater blanket	52-22-51-01-165	PTF ALL
		52-22-51-03-257	PTF ALL
		52-22-51-03-280	PTF ALL
		52-22-51-04-260	PTF ALL
		52-22-51-04-300	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
212	Flight Compartment - Right
832	Left Forward Emergency Exit
833	Left Emergency Exit (STA 627.5)
842	Right Forward Emergency Exit
843	Right Emergency Exit (STA 627.5)

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Emergency Exit Door Surround Heater Blanket Installation

SUBTASK 21-45-02-110-002

(1) Clean the aluminum part of the surround panel assembly with solvent, B00083, to remove all the adhesive.

SUBTASK 21-45-02-420-002

(2) Install the heater blanket [21] in the center of the aluminum part of the surround panel assembly.

(a) Make sure that there are no air bubbles between the heater and lining assembly.

SUBTASK 21-45-02-020-006

(3) Connect the electrical connectors for the heater blanket [21].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-45-02-010-007

CAUTION

DO NOT APPLY TENSION TO THE WIRE BUNDLE BEHIND THE TRIM STRIP. DAMAGE TO THE WIRE BUNDLE CAN OCCUR.

(4) Install the surround panel in the emergency exit doorway (TASK 25-21-20-400-801).

SUBTASK 21-45-02-010-008

(5) Close the emergency exit door.

SUBTASK 21-45-02-840-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 21-45-02-410-002

(7) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-45-02-710-002

(8) Do an operational test of the emergency exit hatch heaters (TASK 21-45-00-700-802).

——— END OF TASK ———

TASK 21-45-02-000-803

6. Emergency Exit Door Closeout Panel Heater Blanket Removal

(Figure 403)

A. References

Reference	Title
21-23-03-000-801	Return Air Grille Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
832	Left Forward Emergency Exit
833	Left Emergency Exit (STA 627.5)
842	Right Forward Emergency Exit
843	Right Emergency Exit (STA 627.5)

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 21-45-02-010-015

(1) To get access to the P91 and P92 panels, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-45-02-840-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 21-45-02-010-009

(3) Open the emergency exit door.

SUBTASK 21-45-02-010-016

(4) To get access to the heater blanket [41] electrical lead splices, do this task: Return Air Grille Removal, TASK 21-23-03-000-801.

E. Emergency Exit Door Closeout Panel Heater Blanket Removal

SUBTASK 21-45-02-010-010

(1) Push the prop [42] on the support assembly to release the closeout panel.

SUBTASK 21-45-02-010-011

(2) Hold the closeout panel in the vertical position to remove the heater blanket [41].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-45-02-020-007

(3) Cut and remove the two splices to disconnect the heater blanket [41].

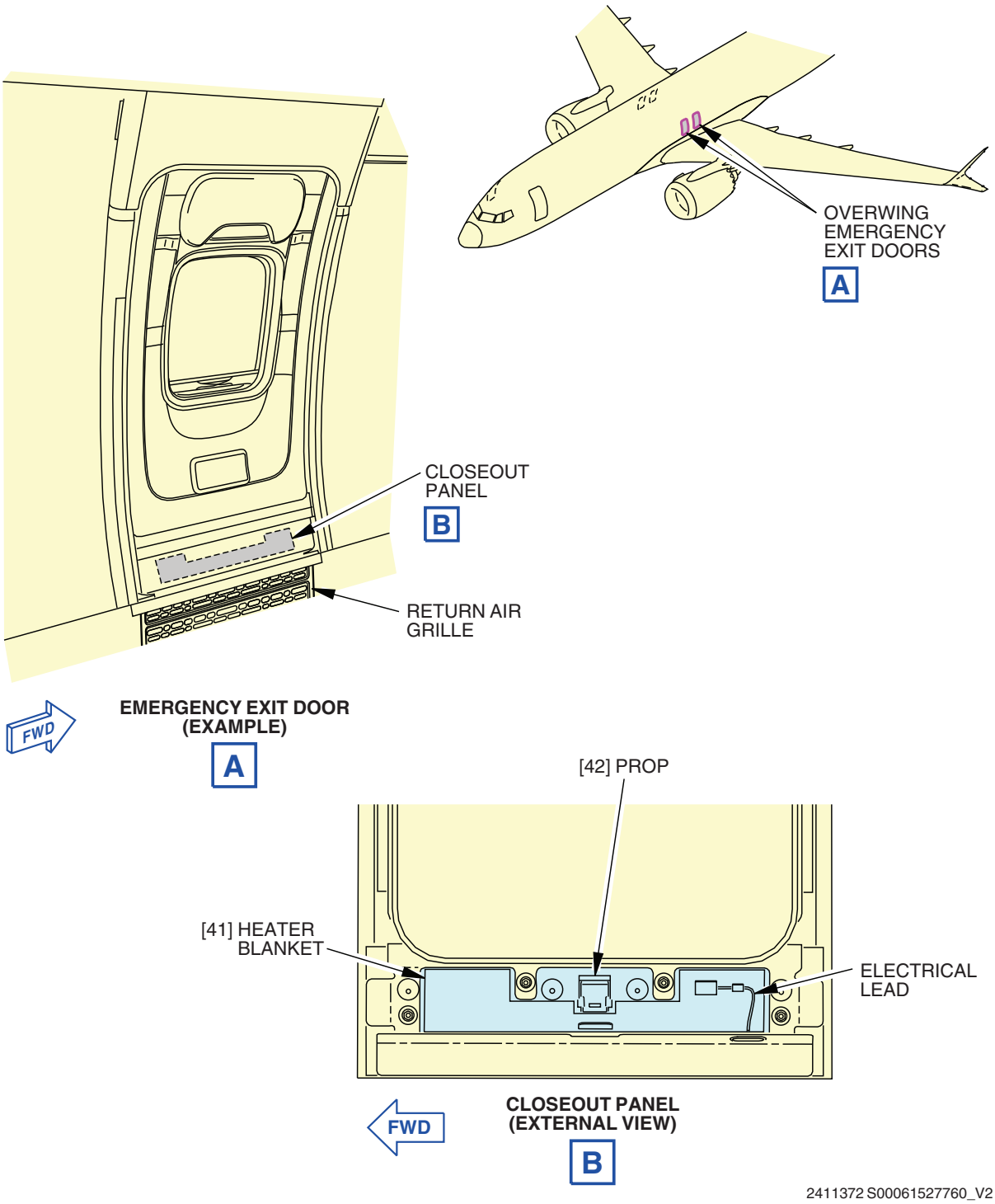
SUBTASK 21-45-02-020-008

(4) Remove the heater blanket [41].

NOTE: The heater blanket is attached to the panel with adhesive.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Emergency Exit Door Closeout Panel Heater Blanket Installation
Figure 403/21-45-02-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-45-02-400-803

7. Emergency Exit Door Closeout Panel Heater Blanket Installation

(Figure 403)

A. References

Reference	Title
21-23-03-400-802	Return Air Grille Installation (P/B 201)
21-45-00-700-802	Emergency Exit Hatch Heaters Operational Test (P/B 501)
SWPM 20-30-12	Assembly of Splices

B. Consumable Materials

Reference	Description	Specification
A00188	Adhesive - Urethane, Two Component	BMS5-105 Type V
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
41	Heater blanket	52-22-51-03-410	PTF ALL
		52-22-51-04-415	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
212	Flight Compartment - Right
832	Left Forward Emergency Exit
833	Left Emergency Exit (STA 627.5)
842	Right Forward Emergency Exit
843	Right Emergency Exit (STA 627.5)

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Emergency Exit Door Closeout Panel Heater Blanket Installation

SUBTASK 21-45-02-110-003

(1) Clean the closeout panel with solvent, B00083, to remove all the adhesive.

SUBTASK 21-45-02-420-003

(2) Apply a layer of adhesive, A00188, to the closeout panel where the heater blanket and closeout panel touch.

SUBTASK 21-45-02-420-004

(3) Apply a layer of adhesive, A00188, to the heater blanket [41].

SUBTASK 21-45-02-420-005

(4) Put the heater blanket [41] in its position on the closeout panel and apply pressure.

SUBTASK 21-45-02-420-006

(5) Assemble the two electrical lead splices to connect the heater blanket [41] (SWPM 20-30-12).

SUBTASK 21-45-02-420-007

(6) Let the adhesive dry, refer to the manufacturers specifications.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-45-02-410-004

(7) Do this task: Return Air Grille Installation, TASK 21-23-03-400-802.

SUBTASK 21-45-02-010-012

(8) Close the emergency exit door.

SUBTASK 21-45-02-840-006

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(9) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	9	C01280	DOOR AREA HTR-FWD

SUBTASK 21-45-02-410-003

(10) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 21-45-02-710-003

(11) Do an operational test of the emergency exit hatch heaters (TASK 21-45-00-700-802).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK FLOW CONTROL AND PACK COOLING SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Air Cycle Air Conditioning and Temperature Control System - Deactivation

(2) Air Cycle Air Conditioning and Temperature Control System - Activation

(3) Ram Air System - Deactivation

(4) Ram Air System - Activation

TASK 21-51-00-800-801

2. Air Cycle Air Conditioning and Temperature Control System - Deactivation

A. General

(1) This task will deactivate these components in the air cycle air conditioning and temperature control system.

- Flow Control and Shutoff Valve (FCSOV)
- Heat Exchanger and Plenum/Diffuser Assembly
- Air Cycle Machine (ACM)
- Conditioned Air Check Valve
- Temperature Control Valve (TCV)
- Standby Temperature Control Valve (STCV)
- Condenser
- Reheater
- Water Extractor
- Water Extractor Duct
- Water Spray Nozzle
- Compressor Discharge Overheat Switch
- Turbine Inlet Overheat Switch
- Pack Discharge Duct Overheat Switch
- Ram Air Control Temperature Sensor
- Mix Manifold Temperature Sensor
- Pack Temperature Sensor
- Pack Temperature Bulb
- Air Conditioning Module.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Air Cycle Air Conditioning and Temperature Control System Deactivation

SUBTASK 21-51-00-860-001

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM BEFORE YOU DO MAINTENANCE. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the pneumatic system is off (TASK 36-00-00-860-806).

SUBTASK 21-51-00-860-002

(2) Do these steps, on the P5-10 air conditioning panel:

(a) Put the L and R PACK switches to the OFF position.

1) Attach the DO NOT OPERATE tags, STD-858, on the switches.

(b) Put the ISOLATION VALVE switch to the OPEN position.

1) Attach the DO NOT OPERATE tag, STD-858, on the switch.

(c) Put the BLEED 1 and 2 switches to the OFF position.

1) Attach the DO NOT OPERATE tags, STD-858, on the switches.

(d) Put the BLEED APU switch to the OFF position.

1) Attach the DO NOT OPERATE tag, STD-858, on the switch.

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(e) If an external ground air source was used, disconnect the external ground air source from the ground pneumatic service connector.

SUBTASK 21-51-00-860-003

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	5	C00259	AIR CONDITIONING BLEED AIR VALVE ISLN
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	2	C01169	A/C ZONE TEMP DUCT OVHT AFT PASS
C	3	C01166	A/C ZONE TEMP DUCT OVHT FWD PASS
C	4	C01164	A/C ZONE TEMP DUCT OVHT FLT DECK
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT
D	3	C01613	AIR COND PACK CONT VALVES L ALT
D	4	C01614	AIR COND PACK CONT VALVES R ALT
D	8	C00076	AIR CONDITIONING TEMP IND
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

F. System Check

NOTE: This check is to make sure that the air cycle air conditioning and temperature control system is in a zero energy state.

SUBTASK 21-51-00-210-001

(1) Make sure that the duct temperature indicator, on the P5-17 cabin temperature control panel, shows less than 200°F (93°C).

SUBTASK 21-51-00-210-002

(2) Make sure that both duct pressure needles on the dual duct pressure indicator, on the P5-10 air conditioning panel, show 0 psi (0 kPa).

——— END OF TASK ———

TASK 21-51-00-800-802

3. Air Cycle Air Conditioning and Temperature Control System - Activation

A. General

(1) This task will activate these components in the air cycle air conditioning and temperature control system.

- Flow Control and Shutoff Valve (FCSOV)
- Heat Exchanger and Plenum/Diffuser Assembly
- Air Cycle Machine (ACM)
- Conditioned Air Check Valve
- Temperature Control Valve (TCV)
- Standby Temperature Control Valve (STCV)
- Condenser
- Reheater
- Water Extractor
- Water Extractor Duct

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Water Spray Nozzle
- Compressor Discharge Overheat Switch
- Turbine Inlet Overheat Switch
- Pack Discharge Duct Overheat Switch
- Ram Air Control Temperature Sensor
- Mix Manifold Temperature Sensor
- Pack Temperature Sensor
- Pack Temperature Bulb
- Air Conditioning Module.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Air Cycle Air Conditioning and Temperature Control System Activation

SUBTASK 21-51-00-860-049

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM BEFORE YOU DO MAINTENANCE. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the pneumatic system is off (TASK 36-00-00-860-806).

SUBTASK 21-51-00-860-004

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	1	C01157	A/C TEMP CONT VALVE CLOSE LEFT
A	3	C01170	A/C ZONE TEMP VALVE/FAN CONT AFT PASS
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	5	C00259	AIR CONDITIONING BLEED AIR VALVE ISLN
A	7	C00796	AIR CONDITIONING BLEED AIR VALVES L
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	1	C01160	A/C TEMP CONT VALVE CLOSE RIGHT
B	2	C01167	A/C ZONE TEMP VALVE/FAN CONT FWD PASS
B	3	C01163	A/C ZONE TEMP VALVE/FAN CONT FLT DECK
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	7	C00797	AIR CONDITIONING BLEED AIR VALVES R

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC
C	2	C01169	A/C ZONE TEMP DUCT OVHT AFT PASS
C	3	C01166	A/C ZONE TEMP DUCT OVHT FWD PASS
C	4	C01164	A/C ZONE TEMP DUCT OVHT FLT DECK
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT
D	3	C01613	AIR COND PACK CONT VALVES L ALT
D	4	C01614	AIR COND PACK CONT VALVES R ALT
D	8	C00076	AIR CONDITIONING TEMP IND
D	9	C01172	AIR CONDITIONING TRIM AIR PRESS

SUBTASK 21-51-00-840-002

(3) Do these steps, on the P5-10 air conditioning panel:

(a) Remove the DO NOT OPERATE tags, STD-858, from the L and R PACK switches.

1) Put the L and R PACK switches to the AUTO position.

(b) Remove the DO NOT OPERATE tag, STD-858, from the ISOLATION VALVE switch.

1) Put the ISOLATION VALVE switch to the AUTO position.

(c) Remove the DO NOT OPERATE tags, STD-858, from the BLEED 1 and 2 switches.

1) Put the BLEED 1 and 2 switches to the ON position.

(d) Remove the DO NOT OPERATE tag, STD-858, from the BLEED APU switch.

1) Put the BLEED APU switch to the ON position.

——— END OF TASK ———

TASK 21-51-00-800-803

4. Ram Air System - Deactivation

A. General

(1) This task will deactivate these components in the ram air system:

- Ram Air Inlet Actuator
- Ram Air Inlet Deflector Door
- Ram Air Inlet Modulation Panels
- Ram Air Exhaust Actuator.

B. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Ram Air System Deactivation

SUBTASK 21-51-00-040-001

(1) When the ram air components for left cooling pack will be deactivated, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-00-040-002

(2) When the ram air components for the right cooling pack will be deactivated, do this step:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

——— END OF TASK ———

TASK 21-51-00-800-804

5. Ram Air System - Activation

A. General

(1) This task will activate these components in the ram air system:

- Ram Air Inlet Actuator
- Ram Air Inlet Deflector Door
- Ram Air Inlet Modulation Panels
- Ram Air Exhaust Actuator.

B. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Ram Air System Activation

SUBTASK 21-51-00-440-001

(1) If the ram air inlet door and ram air exhaust duct louvers for the left cooling pack were deactivated, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MOVE ALL PERSONNEL AWAY FROM THE PANELS, DOORS, VANES, AND LINKAGES FOR RAM AIR. THESE CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC

SUBTASK 21-51-00-440-002

(2) If the ram air inlet door and ram air exhaust duct louvers for the right cooling pack were deactivated, do this step:

WARNING MOVE ALL PERSONNEL AWAY FROM THE PANELS, DOORS, VANES, AND LINKAGES FOR RAM AIR. THESE CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK FLOW CONTROL AND PACK COOLING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Pack Flow Control Valve Operational Test.

(2) Ram Air System Operational Test.

(3) Ground Pack Run Test - APU AUTO Flow Mode.

TASK 21-51-00-700-802

2. Pack Flow Control Valve Operational Test

A. General

(1) This test does a check of the pack flow control and shutoff valve AUTO and APU high flow modes.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
36-00-00-860-803	Supply Pressure to the Pneumatic System with the APU (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

E. Pack Flow Control Valve Operational Test

SUBTASK 21-51-00-010-003

(1) Make sure that this access panel is open:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-00-010-013

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(2) To make sure that the right ECS access door is open do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(TASK 06-41-00-800-801)

(a) Make sure that this access panel is open:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Make sure that this access panel is open:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-00-860-013

(3) Do this task: Supply Pressure to the Pneumatic System with the APU, TASK 36-00-00-860-803.

SUBTASK 21-51-00-860-014

(4) Do these steps to make sure the left pack and right pack flow control valves are closed with the pack switches in the OFF position:

(a) Put the L PACK switch and the R PACK switch, on the P5-10 air conditioning panel, to the OFF position.

(b) Make sure the position indicators on the left and right flow control and shutoff valves are in the closed position.

1) Use a mirror to verify the valve position.

(c) Make sure that the air conditioning maintenance data page indicates that the pack flow is 0 lb/min (0 kg/min).

SUBTASK 21-51-00-860-015

(5) Do these steps to make sure the left and right flow control valves are partially open with the pack switches in the AUTO position:

(a) Make sure that either the BLEED 1 or BLEED 2 switch, on the P5-10 air conditioning panel, is set to the ON position.

(b) Set the CONT CAB, FWD CAB and AFT CAB temperature selectors, on the P5-17 cabin temperature panel, to the AUTO position.

(c) Put the L PACK switch and the R PACK switch in the AUTO position.

NOTE: The packs will operate.

(d) Make sure that the air conditioning maintenance data page indicates that both packs have a pack flow greater than 0 lb/min (0 kg/min).

SUBTASK 21-51-00-710-001

(6) Do these steps to make sure that a pack in operation will go to the engine high position when the other pack shuts down:

(a) Put the airplane in the air mode (TASK 32-09-00-860-801).

(b) Retract the trailing edge flaps to the fully up position (TASK 27-51-00-860-804).

(c) Position the L PACK switch in the AUTO position.

(d) Position the R PACK switch to the OFF position.

(e) Make sure that the air conditioning maintenance data page indicates that pack flow increases on the L PACK.

(f) Position the R PACK switch to the AUTO position.

(g) Position the L PACK switch to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Make sure that the air conditioning maintenance data page indicates that pack flow increases on the R PACK.

(i) Put the L PACK switch and the R PACK switch, on the P5-10 air conditioning panel, to the OFF position.

SUBTASK 21-51-00-860-016

(7) Return the airplane to the ground mode (TASK 32-09-00-860-802).

SUBTASK 21-51-00-860-018

(8) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-00-410-006

(9) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-00-410-009

(10) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

——— END OF TASK ———

TASK 21-51-00-730-801

3. Ram Air System Operational Test

(Figure 501)

A. General

(1) This procedure does a test of the ram air system actuators and associated components.

B. References

Reference	Title
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Operational Test

SUBTASK 21-51-00-860-023

(1) Set the switches, on the P5-10 Air Conditioning Module, to these positions:

(a) BLEED 1 and BLEED 2 switches to OFF

(b) APU BLEED switch to OFF

(c) L PACK and R PACK switches to AUTO

SUBTASK 21-51-00-860-024

(2) Set the switches, on the P5-17 Cabin Temperature Selector, to these positions:

(a) TRIM AIR switch to ON

(b) CONT CAB selector switch to AUTO

(c) FWD CAB selector switch to AUTO

(d) AFT CAB selector switch to AUTO

E. Ram Air Modulation Operational Test

SUBTASK 21-51-00-730-001

(1) Do this ground test: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

NOTE: There are two system test procedures within the task referenced above: one for the left system and one for the right system.

(a) When the test is completed, make sure that PASS shows adjacent to TEST CONDITION.

(b) If FAILED shows, refer to the IFIM and do the applicable procedure(s).

SUBTASK 21-51-00-860-025

(2) Make sure that the aircraft is in the ground mode (TASK 32-09-00-860-802).

SUBTASK 21-51-00-710-002

(3) Make sure that the left and right ram air inlet deflector doors are extended (View B, Figure 501).

SUBTASK 21-51-00-860-026

(4) Extend the trailing edge flaps to position 1 (TASK 27-51-00-860-803).

SUBTASK 21-51-00-210-003

(5) Make sure that the left and right ram air inlet deflector doors are still extended.

SUBTASK 21-51-00-860-027

(6) Put the airplane in the air mode (TASK 32-09-00-860-801).

SUBTASK 21-51-00-710-006

(7) Make sure that the left and right ram air inlet deflectors door move to the stowed position (View A, Figure 501).

SUBTASK 21-51-00-860-028

(8) Retract the trailing edge flaps to the fully up position (TASK 27-51-00-860-804).

SUBTASK 21-51-00-710-009

(9) Make sure that the left and right exhaust deflector doors move to the partially stowed position (View C, Figure 501).

SUBTASK 21-51-00-860-029

(10) Return the airplane to the ground mode (TASK 32-09-00-860-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-00-860-030

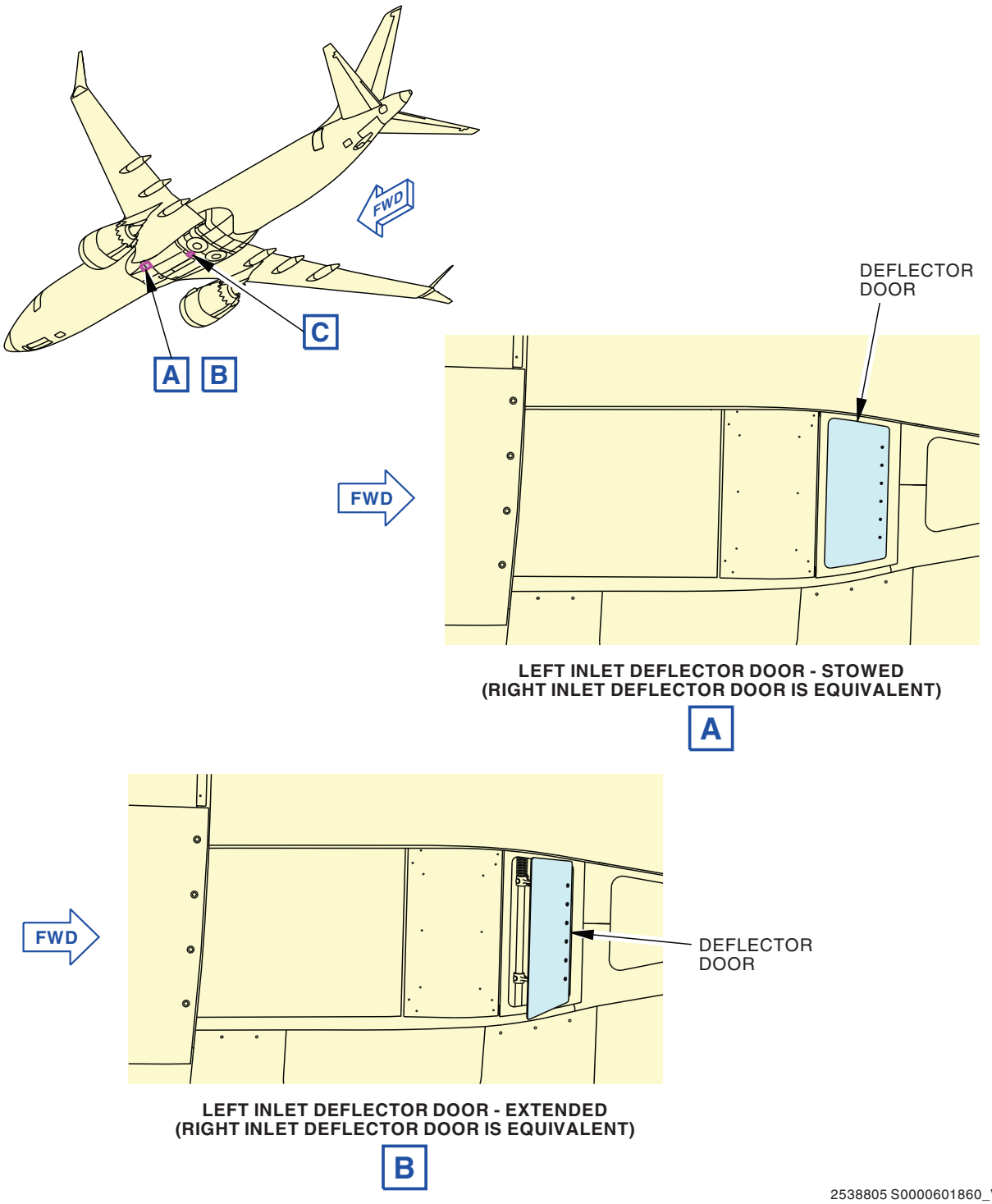
(11) Make sure that the left and right ram air inlet deflector doors are extended (View B, Figure 501).

SUBTASK 21-51-00-860-031

(12) Put the switches back to the usual condition.

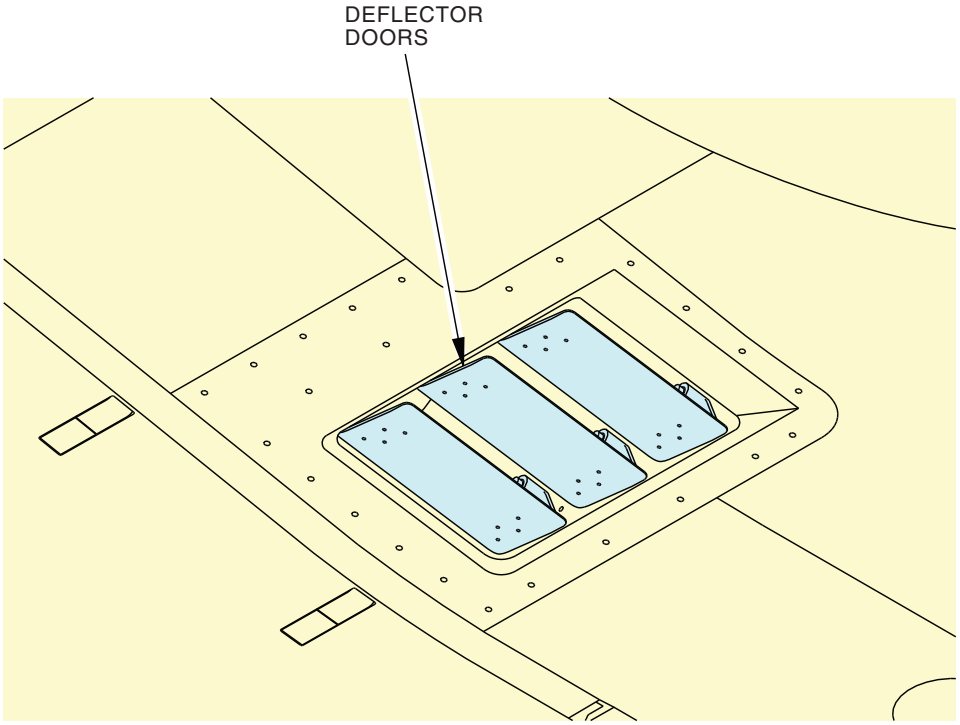
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RAM Air System Operational Test
Figure 501/21-51-00-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT EXHAUST DEFLECTOR DOORS - PARTIALLY OPEN
(RIGHT EXHAUST DEFLECTOR DOORS ARE EQUIVALENT)

C

2538817 S0000601861_V2

RAM Air System Operational Test
Figure 501/21-51-00-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 21-51-00-700-804

4. Ground Pack Run Test - APU AUTO Flow Mode

A. General

(1) The objective of this task is to monitor the temperatures on the Multi Function Display (MFD) air conditioning maintenance data pages while the packs are operating on the ground. The data collected from this procedure can be utilized to trend the health of the system and troubleshoot temperature control issues. The data collected over time can also help operators set the threshold at which maintenance is performed on the pack/system. For this exercise, if the Outside Air Temperature (OAT) is 75°F (24°C) or higher, set the L/R packs to the full COLD setting. If the OAT is below 75°F (24°C), maintain the packs at the 12 o'clock AUTO position for data collection.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
31-65-00-970-801	System Maintenance Data Pages Function (P/B 201)
49-11-00-860-801	APU Starting and Operation (P/B 201)
49-11-00-860-802	APU Usual Shutdown (P/B 201)

C. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

D. Preparation for the Ground Pack Run Test

SUBTASK 21-51-00-010-014

(1) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-00-010-015

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(2) Open the access panels in the sequence that follows (TASK 06-41-00-800-801):

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 21-51-00-710-011

(3) Set the LIGHTS switch on the P2-1 center instrument panel momentarily to the TEST position and make sure these lights come on and then go off:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) MASTER CAUTION, on the P7 panel.

(b) AIR COND annunciator lights, on the P7 panel.

(c) PACK TRIP OFF lights, on the P5-10 air conditioning panel.

SUBTASK 21-51-00-860-043

(4) Do these steps on the Flight Management Computer (FMC) to display the Flight Management Computer System (FMCS) ANALOG page 1/4:

(a) Press the INIT REF function key on the Control Display Unit (CDU) front panel.

(b) Push the LSK adjacent to <INDEX (LSK 6L) to show the INIT REF INDEX 1/1 page.

(c) Push the LSK adjacent to MAINT> (LSK 6R) to show the MAINT BITE INDEX page.

(d) Push the LSK adjacent to <FMCS (LSK 1L) to show the FMCS BITE 1/1 page.

(e) Push the LSK adjacent to <DISCRETES (LSK 4L) to show the FMCS ANALOG DISC 1/4 page.

(f) Use the Environmental Control System (ECS) PACK H/L (high flow, low flow) line entry on the ANALOG DISC 1/4 page.

(g) The following information can be displayed on the FMC:

1) ECS PACK ON/OFF (LEFT and RIGHT)

a) Indicates if the left or right pack is on or off.

2) ECS PACK H/L LO HI (LEFT and RIGHT)

a) Indicates the pack is in the low flow mode or if it has been commanded to the high flow mode.

NOTE: A pack can be commanded to the high flow mode even if it is off.

3) ISOLATION VALVE OPEN (CLOSED)

a) Indicates if the valve is open or closed.

SUBTASK 21-51-00-860-044

(5) Set these switches, on the P5-10 air conditioning panel, as follows:

(a) Set the BLEED 1 and 2 switches to the OFF position.

(b) Set the APU BLEED switch to the OFF position.

(c) Set the L and R PACK switches the OFF position.

(d) Set the ISOLATION VALVE switch to the CLOSE position.

(e) Set the L and R RECIRC FAN switches to the OFF position.

E. Ground Pack Run Test - APU AUTO Flow Mode

SUBTASK 21-51-00-860-045

(1) Do this task to start the APU: APU Starting and Operation, TASK 49-11-00-860-801.

SUBTASK 21-51-00-970-001

(2) Make a record of the OAT and dew point (humidity) on the data collection sheet: Table 501.

Table 501/21-51-00-993-801 Visual Inspection Data Sheet

Outside Air Temperature (OAT):	Visual Inspection			
Time	t= 0 minute	t= 5 minutes	t= 10 minutes	t= 15 minutes
LEFT STCV POSITION				

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Table 501/21-51-00-993-801 Visual Inspection Data Sheet (Continued)

Outside Air Temperature (OAT):	Visual Inspection			
Time	**t= 0 minute**	**t= 5 minutes**	**t= 10 minutes**	**t= 15 minutes**
LEFT TCV POSITION				
CONTROL CABIN TAV POSITION				
IS THERE LEAKAGE FROM THE LEFT HEAT EXCHANGER?				
IS THERE LEAKAGE FROM THE LEFT CONDENSER?				
IS THERE ICE ON THE LEFT CONDITIONED AIR CHECK VALVE?				
RIGHT STCV POSITION				
RIGHT TCV POSITION				
FORWARD TAV POSITION				
AFT TAV POSITION				
IS THERE LEAKAGE FROM THE RIGHT HEAT EXCHANGER?				
IS THERE LEAKAGE FROM THE RIGHT CONDENSER?				
IS THERE ICE ON THE RIGHT CONDITIONED AIR CHECK VALVE?				

SUBTASK 21-51-00-710-012

(3) Make sure that all doors are closed.

SUBTASK 21-51-00-860-046

(4) Set these switches, on the P5-10 air conditioning panel, to reflect the APU AUTO flow mode:

NOTE: The engines should be OFF when recording temperature readings in APU AUTO flow mode.

(a) Set the BLEED 1 switch to the ON position.

(b) Set the APU BLEED switch to the ON position.

(c) Set the ISOLATION VALVE switch to the AUTO position.

(d) Set the L and R RECIRC FAN switches to the AUTO position.

(e) Set the L and R PACK switches the AUTO position.

(f) Set the TRIM AIR switch to the ON Position.

(g) Make sure that the correct FLOW MODE configuration is displayed on the FMC.

SUBTASK 21-51-00-970-002

(5) Take a snapshot of the temperatures at t= 0 minute on the MFD (ATA 21 - Air Conditioning Maintenance Page) (TASK 31-65-00-970-801):

(a) Make sure that the following temperatures are on the Air Conditioning Maintenance Page snapshot:

1) LEFT/RIGHT PACK SENSOR TEMP

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) CONT CAB/FORWARD/AFT ZONE TEMP

3) CONT CAB/FORWARD/AFT ZONE DUCT TEMP

SUBTASK 21-51-00-970-006

(6) Do a visual inspection to record if the position is OPEN, MODULATING, or CLOSED on the data sheet of Table 501 for the following:

(a) Left and right Standby Temperature Control Valve (STCV)

(b) Left and right Temperature Control Valve (TCV)

(c) Trim Air Valve (TAV)

1) Control Cabin

2) Forward Cabin

3) Aft Cabin

SUBTASK 21-51-00-970-003

(7) Repeat the snapshot of the temperatures and recording of the physical positions at t= 5 minutes.

SUBTASK 21-51-00-970-004

(8) Repeat the snapshot of the temperatures and recording of the physical positions at t= 10 minutes.

SUBTASK 21-51-00-970-005

(9) Repeat the snapshot of the temperatures and recording of the physical positions at t= 15 minutes.

SUBTASK 21-51-00-200-001

(10) Make sure that there are no leakages from the Left/Right Heat Exchanger.

(a) Make a record on the data sheet of Table 501.

SUBTASK 21-51-00-200-002

(11) Make sure that there are no leakages from the Left/Right Condenser.

(a) Make a record on the data sheet of Table 501.

SUBTASK 21-51-00-200-003

(12) Do a check to see if there is ice on the Left/Right Conditioned Air Check Valve.

NOTE: Ice is an indicator that the pack is performing well.

(a) Make a record on the data sheet of Table 501.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-00-860-047

(1) Set these switches, on the P5-10 air conditioning panel, as follows:

(a) Set the BLEED 1 switch to the OFF position.

(b) Set the APU BLEED switch to the OFF position.

(c) Set the L and R PACK switches the OFF position.

1) Make sure that the L and R PACK indicate a duct pressure of zero.

2) Make sure that the PACK configuration on the FMC is correct.

(d) Set the ISOLATION VALVE switch to the CLOSE position.

(e) Set the L and R RECIRC FAN switches to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 21-51-00-860-048

(2) Do this task: APU Usual Shutdown, TASK 49-11-00-860-802.

SUBTASK 21-51-00-410-010

(3) Close the access panels in the sequence that follows (TASK 06-41-00-800-801):

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

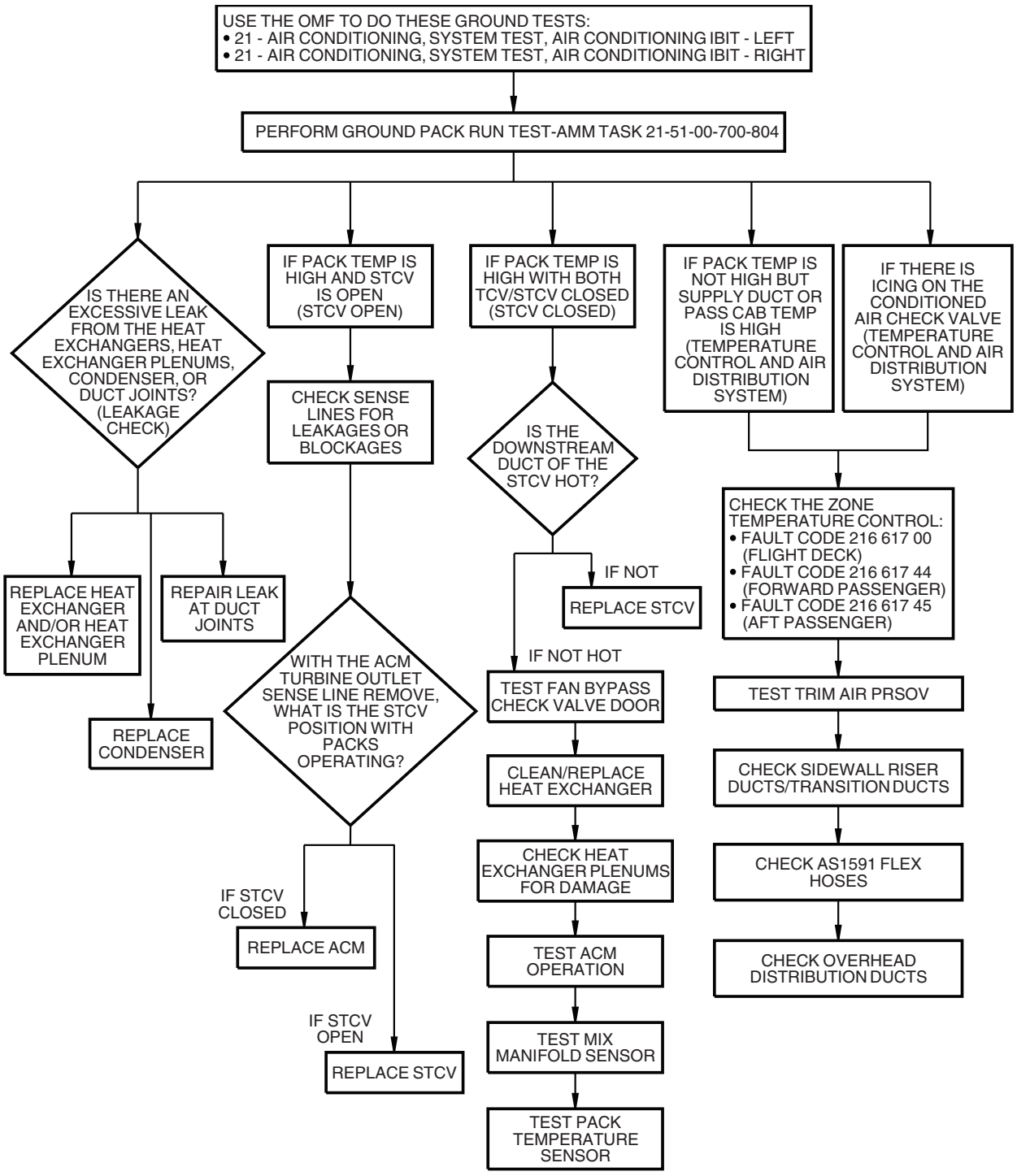
SUBTASK 21-51-00-410-011

(4) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Ground Pack Run Test
Figure 502/21-51-00-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PACK FLOW CONTROL AND PACK COOLING SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) A static breakaway torque check of the Air Cycle Machine (ACM).

TASK 21-51-00-700-803

2. Air Cycle Machine Static Breakaway Torque Check

(Figure 601)

A. General

(1) This task gives instructions to do the static breakaway torque check the Air Cycle Machine (ACM).

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
21-00-00-730-801	Air Conditioning, System Test, Air Conditioning IBIT (P/B 501)
21-51-04-000-801	Air Cycle Machine Removal (P/B 401)
21-51-04-400-801	Air Cycle Machine Installation (P/B 401)
21-51-17-000-801	High Pressure Water Separator Mix Muff Removal (P/B 401)
21-51-17-400-801	High Pressure Water Separator Mix Muff Installation (P/B 401)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00006	Compound - Antiseize, Pure Nickel Special - Never-Seez NSBT-8N/-16N	
D00504	Grease - Petrolatum	VV-P-236
D50104	Grease - Molykote 44 High Temperature Bearing Grease (Medium) (Formerly Dow Corning DC-44)	
D50320	Lubricant - Moly-50 Anti-Seize	MIL-PRF-83483
G01061	Water - Distilled	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Air cycle machine	21-51-04-01-025	PTF ALL
		21-51-04-01-510	PTF ALL
5	Packing	21-51-03-02-040	PTF ALL
		21-51-03-02-200	PTF ALL
9	Packing	21-51-03-02-055	PTF ALL
		21-51-03-02-210	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Flexible duct	21-51-08-05-130	PTF ALL
		21-51-17-02-020	PTF ALL

F. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box
212	Flight Compartment - Right

G. Access Panels

Number	Name/Location
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

H. Prepare for the Check

SUBTASK 21-51-00-860-032

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that there is no pressure in the pneumatic system.

(a) If it is necessary, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 21-51-00-860-033

(2) Do these steps on the P5-10 air conditioning panel (located on the P5 forward overhead panel):

(a) Set the L PACK and R PACK switches to the OFF position.

1) Attach the DO NOT OPERATE tags, STD-858.

(b) Set the BLEED 1 and BLEED 2 switches to the OFF position.

1) Attach the DO NOT OPERATE tags, STD-858.

(c) Set the BLEED APU switch to the OFF position.

1) Attach the DO NOT OPERATE tag, STD-858.

SUBTASK 21-51-00-860-039

(3) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-00-010-010

(4) When the air cycle machine for the left pack will be removed, open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-00-010-011

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) When the air cycle machine for the right pack will be removed, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access panel:

Number	Name/Location
192CR	ECS Access Door

I. Air Cycle Machine Static Breakaway Torque Check

SUBTASK 21-51-00-010-012

WARNING

DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

(1) To remove the compressor outlet duct, do these steps:

(a) Disconnect the flex hose from the compressor outlet duct [12].

(b) Remove the electrical connector [3] from the ram air control sensor.

(c) Remove the electrical connector [6] from the compressor discharge overheat switch.

(d) Remove the nut [10] and washer [11] that hold the bonding jumper to the clamp on the duct.

(e) Remove the clamshell clamps [7] at the air cycle machine [1].

1) Lift the latch pawls at the same time to release the clamps [7].

(f) Move the sleeves [8] to the compressor outlet duct [12].

(g) Remove the coupling [4] at the heat exchanger.

(h) Remove the compressor outlet duct [12].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Remove and discard the packing [5] at the heat exchanger.

(j) Remove and discard the packings [9].

SUBTASK 21-51-00-000-001

(2) Remove the HPWS mix muff [2], do this task: High Pressure Water Separator Mix Muff Removal, TASK 21-51-17-000-801.

SUBTASK 21-51-00-210-006

(3) Check the flexible duct [13] that is connected to the high pressure water separator condenser for any of the conditions listed below:

(a) Fraying

(b) Discoloration

(c) Rips

(d) Splits

(e) Tears

(f) Rupture defects.

SUBTASK 21-51-00-210-007

(4) If any of the above conditions are found, remove the flexible duct [13].

(a) Loosen the clamp [14] that attach the flexible duct [13] to the water separator condenser.

(b) Remove the flexible duct [13] from the water separator condenser.

(c) Discard the flexible duct [13].

SUBTASK 21-51-00-720-001

(5) Do the torque check of the air cycle machine [1] as follows:

CAUTION

TURN THE ROTATING COMPONENTS OF THE ACM ONLY IN THE DIRECTION THAT THEY TURN WHEN IN OPERATION. IF YOU TURN THE ACM ROTATING COMPONENTS IN THE INCORRECT DIRECTION, YOU CAN CAUSE DAMAGE TO THE FOIL AIR BEARINGS.

(a) Use a torque wrench on the nut to turn the turbine wheel in the direction of normal rotation.

NOTE: Normal rotation is clockwise when viewed from the turbine end and counterclockwise when viewed from the fan end.

(b) The torque necessary to turn the nut must not be more than 20 in-lb (2 N·m).

(c) If the torque is more than specified, replace of the ACM (TASK 21-51-04-000-801 and TASK 21-51-04-400-801).

NOTE: Wheel rub, endplay is too tight, dirt on the bearings or shaft, components not seated correctly, shaft surface finish, journal bearing set preload is too high, or housing assembly misalignment are probable cause for incorrect torque.

SUBTASK 21-51-00-420-001

(6) Install the new flexible duct [13] as follows:

(a) Loosely position the clamps [14] onto the flexible duct [13].

(b) Install the flexible duct [13] onto the water separator condenser.

(c) Position the forward clamp [14] to retain the flexible duct [13] onto the water separator condenser.

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(d) Tighten the forward clamp [14] to 13 in-lb (1.5 N·m) - 17 in-lb (1.9 N·m).

NOTE: The aft clamp is connected in the installation of the HPWS mix muff in the next step.

SUBTASK 21-51-00-400-001

(7) Do this task: High Pressure Water Separator Mix Muff Installation, TASK 21-51-17-400-801.

SUBTASK 21-51-00-420-002

(8) To install the compressor outlet duct, do these steps:

(a) Apply a light coat of distilled water, G01061, or grease, D00504, or Molykote 44 grease, medium, D50104, to the new packing [5] and new packings [9].

(b) Install the packings [9] on the duct ends.

(c) Install the packing [5] at the heat exchanger.

(d) Put the compressor outlet duct [12] into its position.

(e) Apply compound, D50320, to the bolt threads of the coupling [4].

(f) Install the coupling [4] at the heat exchanger.

1) Do not tighten the coupling [4] fully.

(g) Move the sleeve [8] over the outlet duct connection.

(h) Install the clamshell clamp [7] over the outlet duct connection.

(i) Tighten the nut on the coupling [4] to 45 in-lb (5.1 N·m) - 50 in-lb (5.6 N·m).

(j) With hand force, gently push the compressor outlet duct [12] in the direction of the adjacent mix muff duct until the compressor outlet duct [12] reaches the maximum flexed position.

(k) Make sure that there is a minimum clearance of 0.20 in. (5.1 mm) to the adjacent mix muff duct when the compressor outlet duct [12] is in the flexed position.

(l) Apply a light coat of Pure Nickel Special compound, D00006, to the threads of the flex hose.

(m) Connect the flex hose to the compressor outlet duct [12].

(n) Install the electrical connector [3] to the ram air control sensor.

(o) Install the electrical connector [6] to the compressor discharge overheat switch.

(p) Install the nut [10] and washer [11] that hold the bonding jumper to the clamp on the duct.

J. ACM Installation Test

SUBTASK 21-51-00-860-042

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	4	C01950	AIR CONDITIONING RAM AIR MOD LEFT
A	8	C00555	AIR CONDITIONING MISC LOADS LEFT
A	9	C01158	AIR CONDITIONING PACK CONTROL LEFT DC
A	11	C01159	AIR CONDITIONING PACK CONTROL LEFT AC
B	4	C01951	AIR CONDITIONING RAM AIR MOD RIGHT
B	8	C01178	AIR CONDITIONING MISC LOADS RIGHT
B	9	C01161	AIR CONDITIONING PACK CONT RIGHT DC
B	11	C01162	AIR CONDITIONING PACK CONT RIGHT AC

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(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	5	C00263	AIR CONDITIONING PACK CONT VALVES R
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L
C	7	C01177	A/C PACK/ENGINE BLEED AIR OVHT RIGHT
C	8	C01176	A/C PACK/ENGINE BLEED AIR OVHT LEFT

SUBTASK 21-51-00-860-034

(2) Do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 21-51-00-860-036

(3) Remove the DO NOT OPERATE tags, STD-858, from the L PACK, R PACK, BLEED 1, BLEED 2, and BLEED APU switches, on the P5-10 air conditioning panel.

SUBTASK 21-51-00-860-035

(4) Put the applicable pack L PACK or R PACK switch, on the P5-10 air conditioning panel, to the AUTO position.

SUBTASK 21-51-00-790-001

(5) Do a soap bubble test of all the duct joints at the HPWS mix muff [2] and at the air cycle machine [1].

NOTE: No air leakage is permitted.

(a) If there is leakage, do these steps:

1) Put the L PACK and R PACK switches, on the P5-10 air conditioning panel, to the OFF position.

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. THE COOLING PACK DUCTS CAN BE HOT AND CAN CAUSE INJURY TO PERSONS.

2) Loosen the clamp [14] and clamp [15].

3) Make sure that the HPWS mix muff [2] and air cycle machine [1] are aligned at the joints.

4) Tighten the clamp [14] to 13 in-lb (1.5 N·m) - 17 in-lb (1.9 N·m).

5) Tighten the clamp [15] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

6) Put the L PACK and R PACK switches to the AUTO position.

7) Make sure that the leak has been repaired.

SUBTASK 21-51-00-740-004

(6) Do a IBIT test on the left and right pack/zone temperature controllers (M2957 and M2939).

NOTE: This will clear any fault introduced during maintenance when power is interrupted.

(a) Do this task: Air Conditioning, System Test, Air Conditioning IBIT, TASK 21-00-00-730-801.

K. Put the Airplane Back to Its Usual Condition

SUBTASK 21-51-00-860-037

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

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SUBTASK 21-51-00-410-007

(2) Close this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-00-410-008

(3) To close the right ECS access doors, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access panel:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access panel:

Number	Name/Location
192DR	ECS High Pressure Access Door

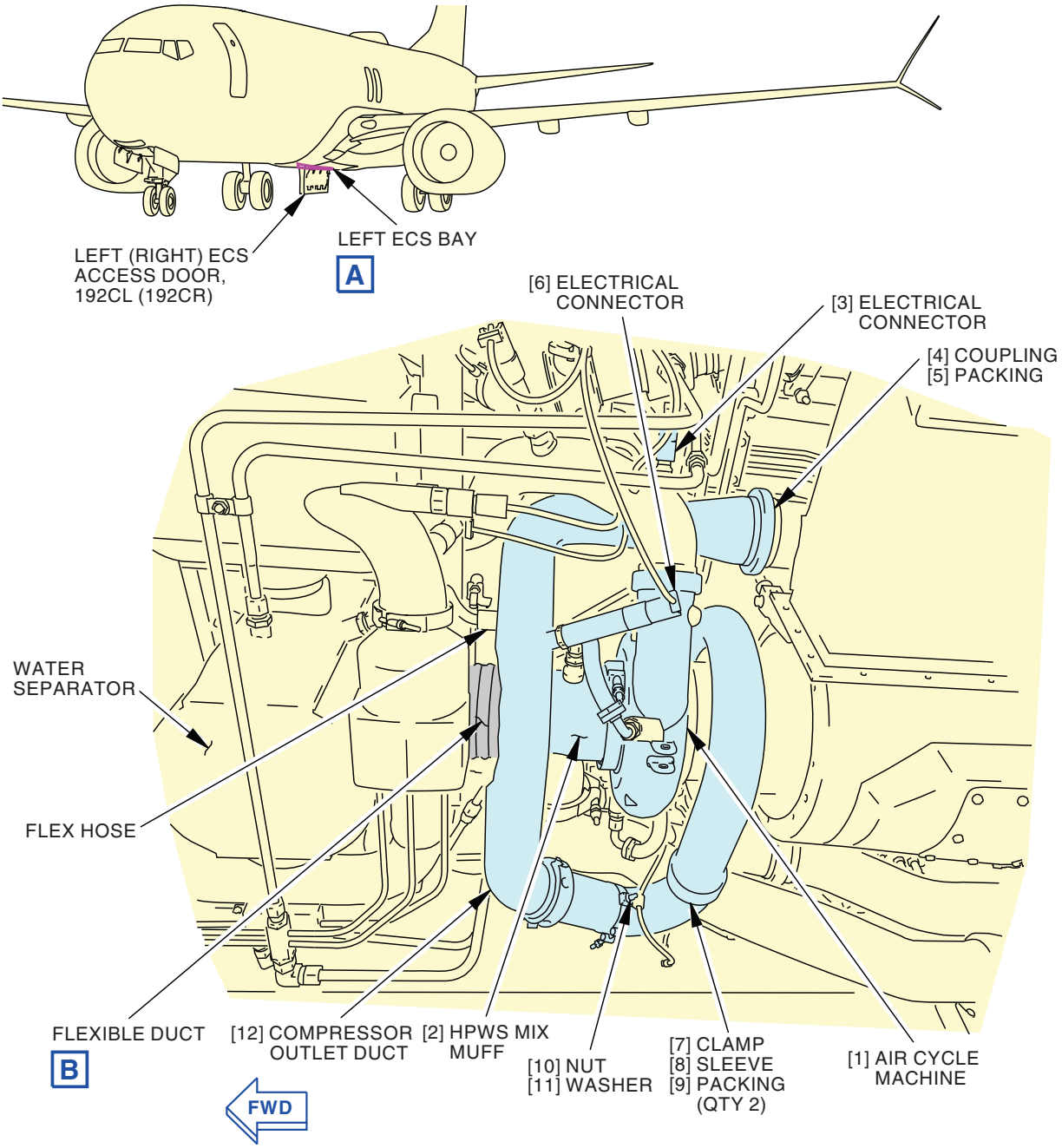
NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 21-51-00-860-038

(4) Put the L PACK and R PACK switches, on the P5-10 air conditioning panel, to the OFF position.

——— END OF TASK ———

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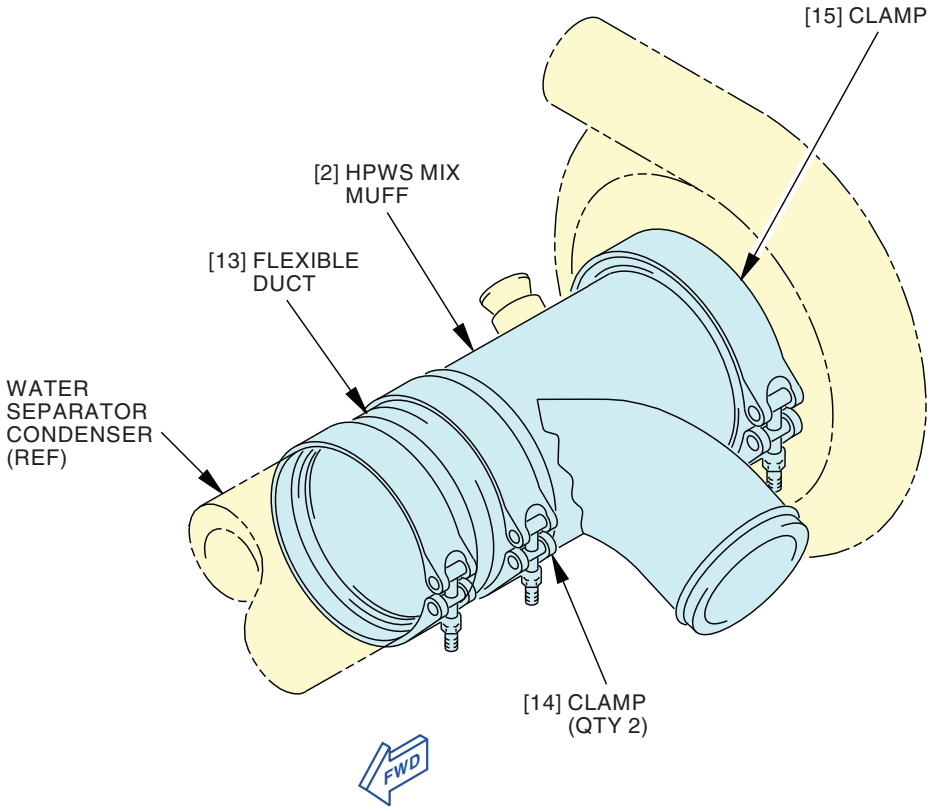
LEFT ECS BAY
(RIGHT ECS BAY IS EQUIVALENT)

A

3008686 S0000782529_V1

Air Cycle Machine Static Breakaway Torque Check
Figure 601/21-51-00-990-807 (Sheet 1 of 2)

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FLEXIBLE DUCT
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

B

3008687 S0000782530_V1

Air Cycle Machine Static Breakaway Torque Check
Figure 601/21-51-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLOW CONTROL AND SHUTOFF VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Left Flow Control and Shutoff Valve Removal

(2) Left Flow Control and Shutoff Valve Installation

(3) Right Flow Control and Shutoff Valve Removal

(4) Right Flow Control and Shutoff Valve Installation.

TASK 21-51-01-000-801

2. Left Flow Control and Shutoff Valve Removal

(Figure 401)

A. General

(1) The high pressure water separator mix muff is referred to as HPWS mix muff in this task.

B. References

Reference	Title
20-10-51-000-801	Flareless Tubing Assembly Removal (P/B 401)
21-51-17-000-801	High Pressure Water Separator Mix Muff Removal (P/B 401)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192CL	ECS Access Door

F. Prepare for the Removal

SUBTASK 21-51-01-860-008

WARNING REMOVE THE PRESSURE FROM THE PNEUMATIC SYSTEM. PRESSURE IN THE PNEUMATIC SYSTEM CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove pressure from the pneumatic system (TASK 36-00-00-860-806).

SUBTASK 21-51-01-860-009

(2) On the P5-10 Bleed Air Control Module, set the switches and attach DO NOT OPERATE tags, STD-858, as follows:

(a) Set the L PACK and R PACK switches to the OFF position.

(b) Set the BLEED 1 and 2 switches to the OFF position.

(c) Set the BLEED APU switch to the OFF position.

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SUBTASK 21-51-01-010-003

(3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
C	6	C00262	AIR CONDITIONING PACK CONT VALVES L

SUBTASK 21-51-01-010-004

(4) Open this access panel:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 21-51-01-020-041

(5) Remove the electrical connector [1] from the ram air control sensor.

SUBTASK 21-51-01-020-042

(6) Remove the electrical connector [15] from the compressor discharge overheat switch.

SUBTASK 21-51-01-010-005

WARNING DO NOT TOUCH THE COOLING PACK DUCTS WHEN THEY ARE HOT. HOT DUCTS CAN CAUSE INJURIES TO PERSONNEL.

(7) Remove the compressor outlet duct [11], as follows:

(a) Disconnect the flexhose [2] from the compressor outlet duct [11] (TASK 20-10-51-000-801).

(b) Remove and discard the O-ring [3].

(c) Remove the coupling [12] between the compressor outlet duct [11] and the air cycle machine.

NOTE: Lift the three latch pawls at the same time to release the coupling [12].

(d) Move the sleeve [13] to the adjacent duct.

(e) Remove the clamp coupling [16] between the compressor outlet duct [11] and the heat exchanger.

(f) Remove the compressor outlet duct [11].

(g) Remove and discard the packing [17].

(h) Remove and discard the O-rings [14].

SUBTASK 21-51-01-020-014

(8) Remove the mix muff duct [5], as follows:

(a) Remove the screw [6], washers [7], nut [8] that holds the bonding jumper [9] and bonding jumper [10] to the bonding tab on the mix muff duct [5].

(b) Hold the mix muff duct [5].

(c) Remove the coupling clamps [4].

(d) Remove the mix muff duct [5].

SUBTASK 21-51-01-010-006

(9) Remove the HPWS mix muff (TASK 21-51-17-000-801).

SUBTASK 21-51-01-010-007

(10) Remove the compressor inlet duct [21], as follows:

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(a) Remove the screw [25] and plain washer [26] that holds the ground clip [27] to the air cycle machine.

(b) Move the jumper [28] from the air cycle machine.

NOTE: Keep the ground clip [27] on the jumper [28].

(c) Remove the screws [22] and plain washers [23] that holds the compressor inlet duct [21] to the air cycle machine.

(d) Remove and discard the O-ring [24].

(e) Remove the self-bonding coupling [18].

NOTE: Lift the three latch pawls at the same time to release the clamp.

(f) Move the sleeve [19] to the compressor inlet duct [21].

(g) Remove and discard the O-rings [20].

G. Left Flow Control and Shutoff Valve Removal

SUBTASK 21-51-01-020-045

(1) Remove the screw [30], washers [78], and nut [31] from the clamps [32] to disconnect the wiring from the flow control and shutoff valve [35].

SUBTASK 21-51-01-020-043

(2) Disconnect the electrical connector [29] from the flow control and shutoff valve [35].

SUBTASK 21-51-01-020-015

(3) Disconnect the electrical connector [33] from the flow control and shutoff valve [35].

SUBTASK 21-51-01-020-016

(4) Disconnect the electrical connector [34] from the flow control and shutoff valve [35].

SUBTASK 21-51-01-020-017

(5) Remove the screw [37], plain washers [38], and nut [40] that connects the jumper [39] to the bonding tab on the flow control and shutoff valve [35].

SUBTASK 21-51-01-020-018

(6) Hold the flow control and shutoff valve [35].

SUBTASK 21-51-01-020-044

(7) Remove the coupling clamp [36] and coupling clamp [42].

SUBTASK 21-51-01-020-019

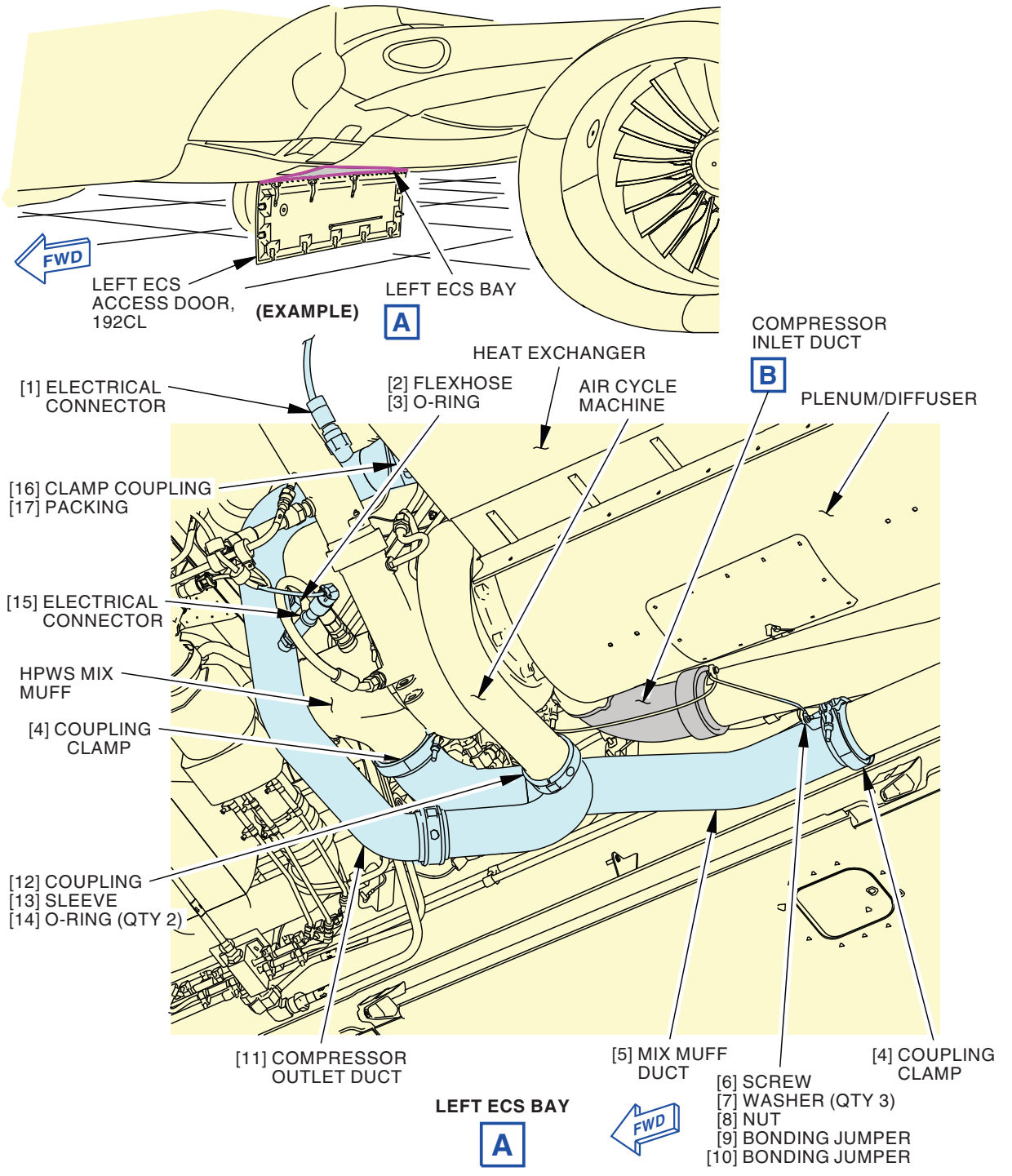
(8) Remove the flow control and shutoff valve [35].

SUBTASK 21-51-01-410-010

(9) Put covers on the duct openings to keep out unwanted materials.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Left Flow Control and Shutoff Valve Installation
Figure 401/21-51-01-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details